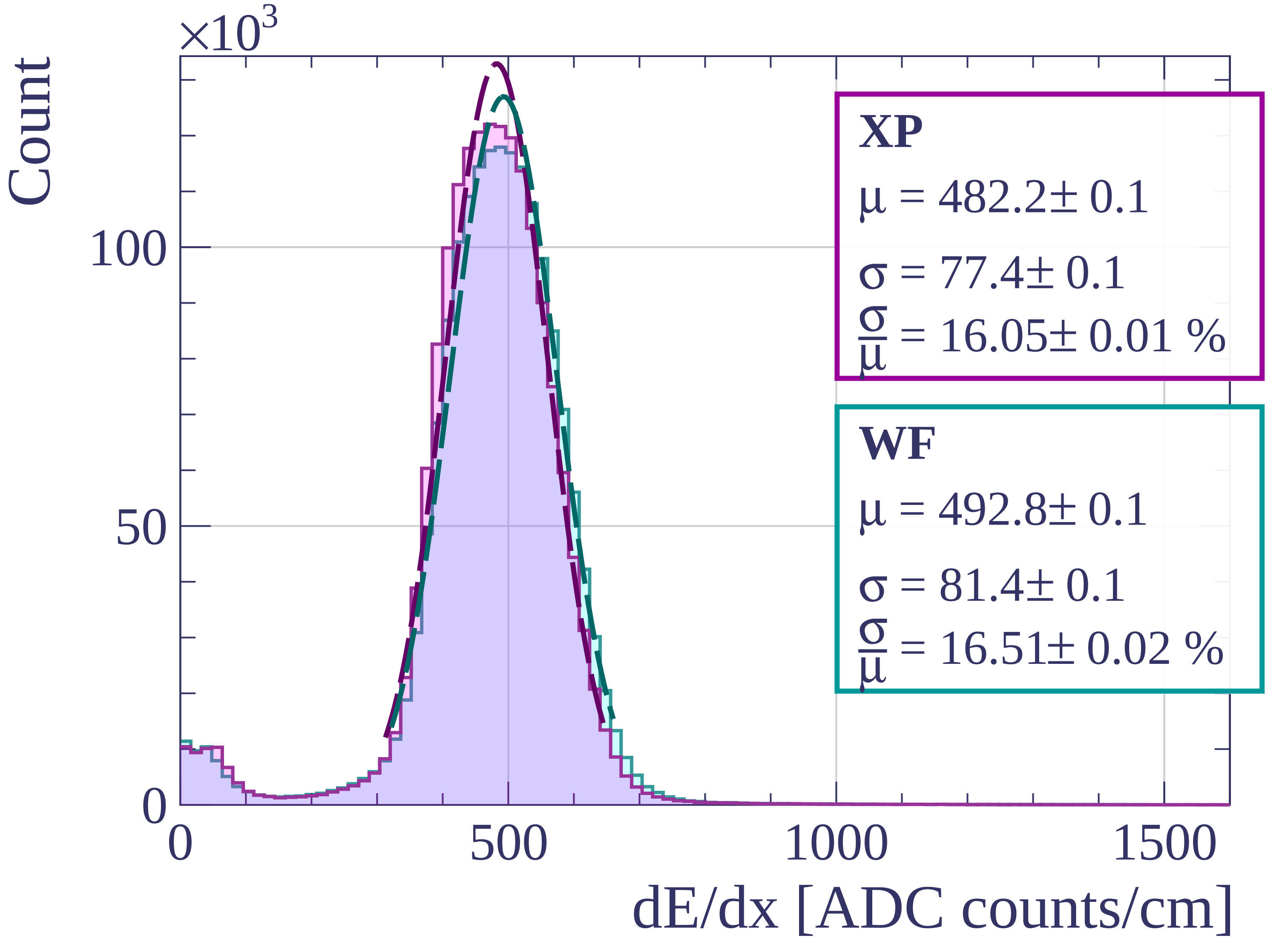
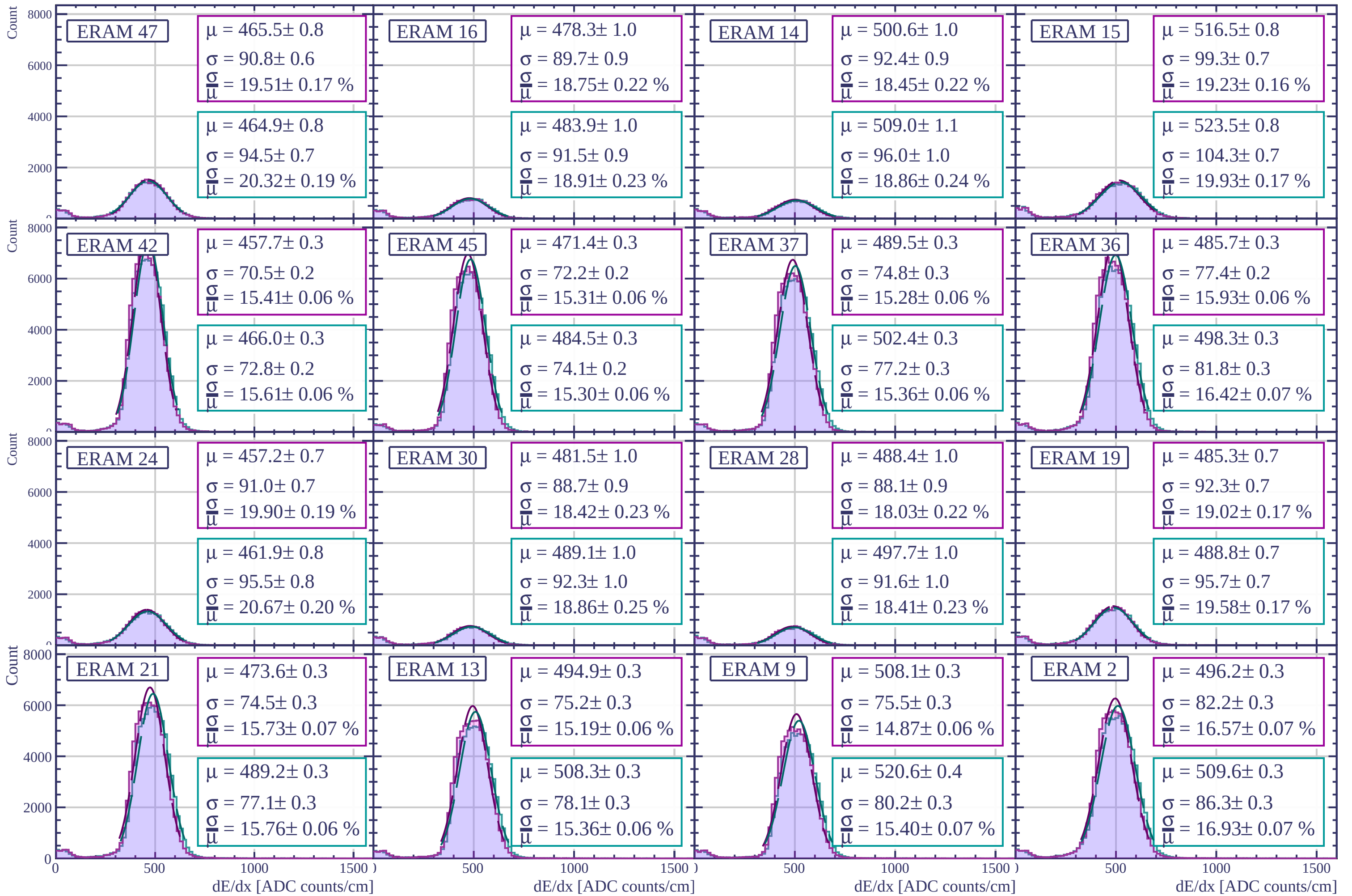
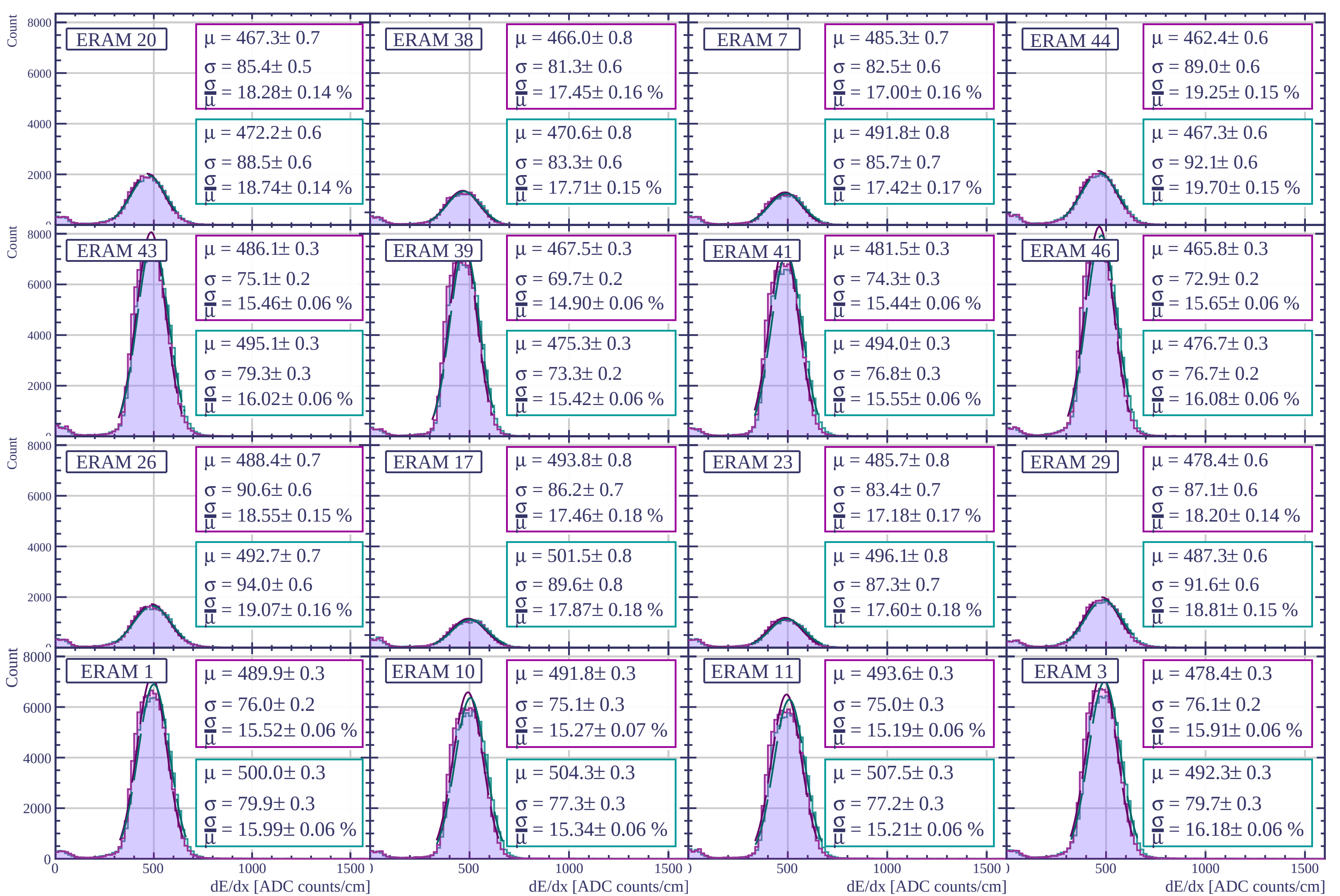
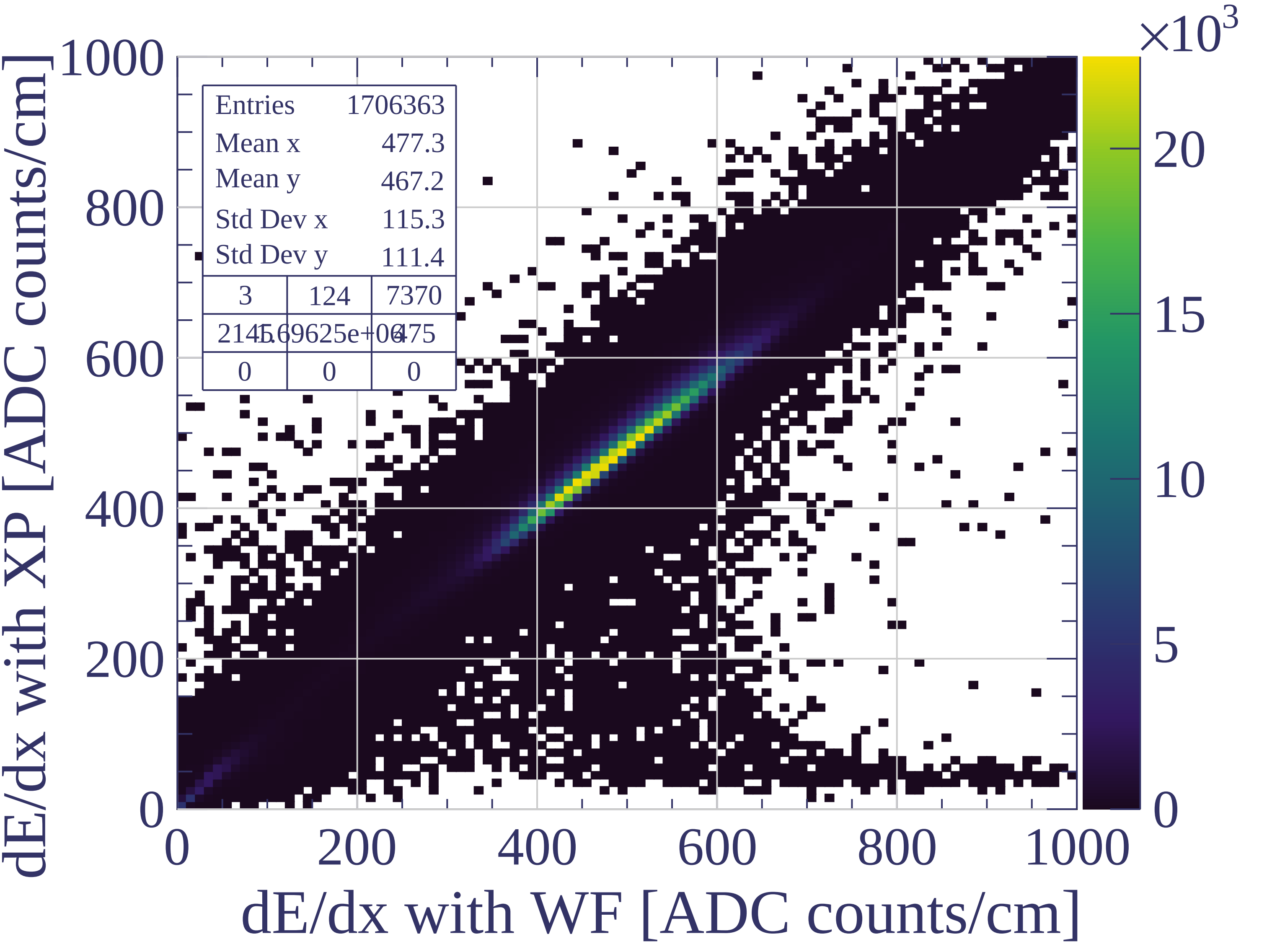


×10^3
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 482.2± 0.1
σ = 77.4± 0.1
σ/μ = 16.05± 0.01 %
WF
μ = 492.8± 0.1
σ = 81.4± 0.1
σ/μ = 16.51± 0.02 %

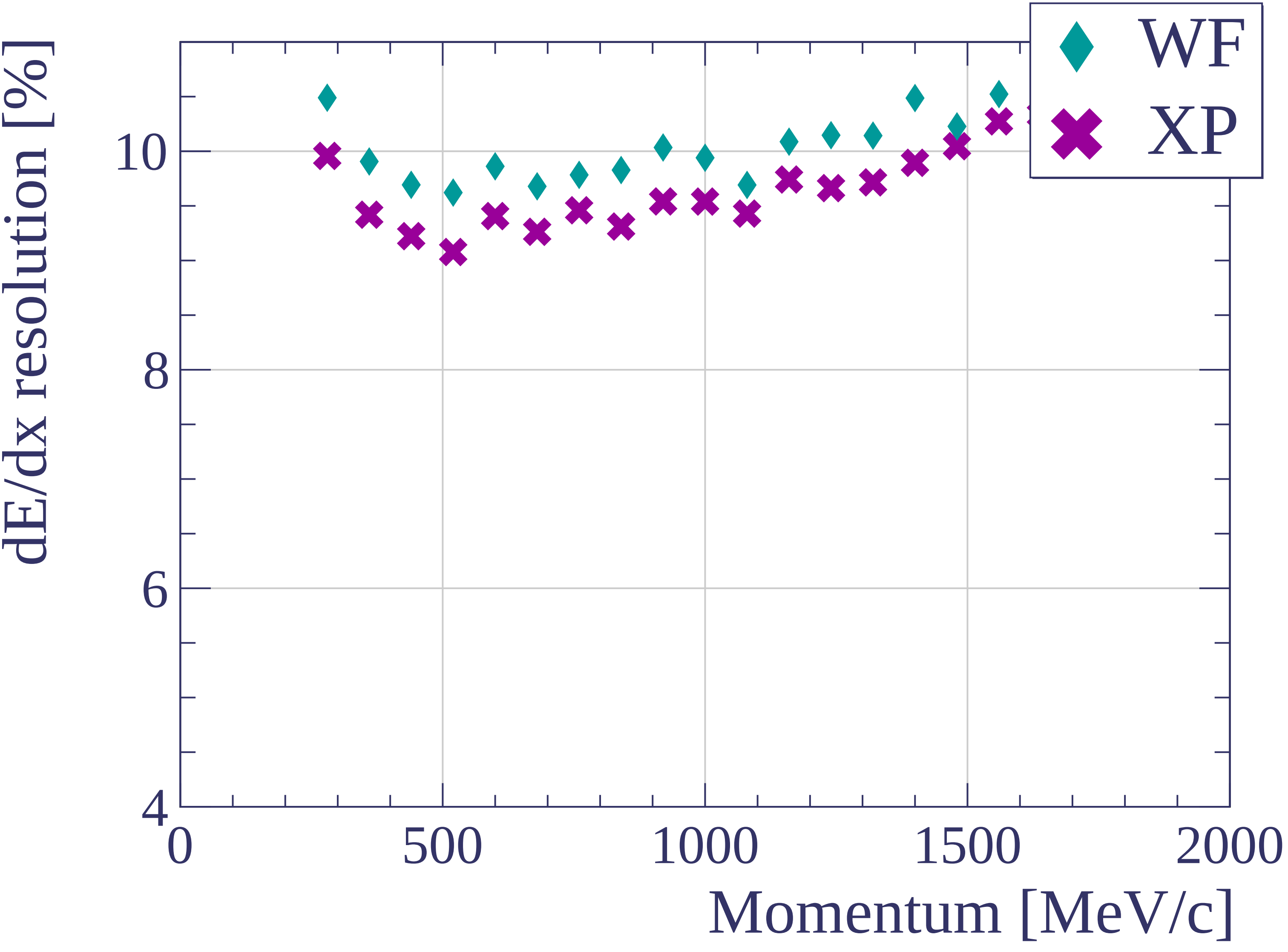


ERAM 47
μ = 465.5± 0.8
σ = 90.8± 0.6
σ/μ = 19.51± 0.17 %
μ = 464.9± 0.8
σ = 94.5± 0.7
σ/μ = 20.32± 0.19 %
ERAM 16
μ = 478.3± 1.0
σ = 89.7± 0.9
σ/μ = 18.75± 0.22 %
μ = 483.9± 1.0
σ = 91.5± 0.9
σ/μ = 18.91± 0.23 %
ERAM 14
μ = 500.6± 1.0
σ = 92.4± 0.9
σ/μ = 18.45± 0.22 %
μ = 509.0± 1.1
σ = 96.0± 1.0
σ/μ = 18.86± 0.24 %
ERAM 15
μ = 516.5± 0.8
σ = 99.3± 0.7
σ/μ = 19.23± 0.16 %
μ = 523.5± 0.8
σ = 104.3± 0.7
σ/μ = 19.93± 0.17 %
ERAM 42
μ = 457.7± 0.3
σ = 70.5± 0.2
σ/μ = 15.41± 0.06 %
μ = 466.0± 0.3
σ = 72.8± 0.2
σ/μ = 15.61± 0.06 %
ERAM 45
μ = 471.4± 0.3
σ = 72.2± 0.2
σ/μ = 15.31± 0.06 %
μ = 484.5± 0.3
σ = 74.1± 0.2
σ/μ = 15.30± 0.06 %
ERAM 37
μ = 489.5± 0.3
σ = 74.8± 0.3
σ/μ = 15.28± 0.06 %
μ = 502.4± 0.3
σ = 77.2± 0.3
σ/μ = 15.36± 0.06 %
ERAM 36
μ = 485.7± 0.3
σ = 77.4± 0.2
σ/μ = 15.93± 0.06 %
μ = 498.3± 0.3
σ = 81.8± 0.3
σ/μ = 16.42± 0.07 %
ERAM 24
μ = 457.2± 0.7
σ = 91.0± 0.7
σ/μ = 19.90± 0.19 %
μ = 461.9± 0.8
σ = 95.5± 0.8
σ/μ = 20.67± 0.20 %
ERAM 30
μ = 481.5± 1.0
σ = 88.7± 0.9
σ/μ = 18.42± 0.23 %
μ = 489.1± 1.0
σ = 92.3± 1.0
σ/μ = 18.86± 0.25 %
ERAM 28
μ = 488.4± 1.0
σ = 88.1± 0.9
σ/μ = 18.03± 0.22 %
μ = 497.7± 1.0
σ = 91.6± 1.0
σ/μ = 18.41± 0.23 %
ERAM 19
μ = 485.3± 0.7
σ = 92.3± 0.7
σ/μ = 19.02± 0.17 %
μ = 488.8± 0.7
σ = 95.7± 0.7
σ/μ = 19.58± 0.17 %
ERAM 21
μ = 473.6± 0.3
σ = 74.5± 0.3
σ/μ = 15.73± 0.07 %
μ = 489.2± 0.3
σ = 77.1± 0.3
σ/μ = 15.76± 0.06 %
ERAM 13
μ = 494.9± 0.3
σ = 75.2± 0.3
σ/μ = 15.19± 0.06 %
μ = 508.3± 0.3
σ = 78.1± 0.3
σ/μ = 15.36± 0.06 %
ERAM 9
μ = 508.1± 0.3
σ = 75.5± 0.3
σ/μ = 14.87± 0.06 %
μ = 520.6± 0.4
σ = 80.2± 0.3
σ/μ = 15.40± 0.07 %
ERAM 2
μ = 496.2± 0.3
σ = 82.2± 0.3
σ/μ = 16.57± 0.07 %
μ = 509.6± 0.3
σ = 86.3± 0.3
σ/μ = 16.93± 0.07 %
Count
8000
6000
4000
2000
0
500
1000
1500
dE/dx [ADC counts/cm]

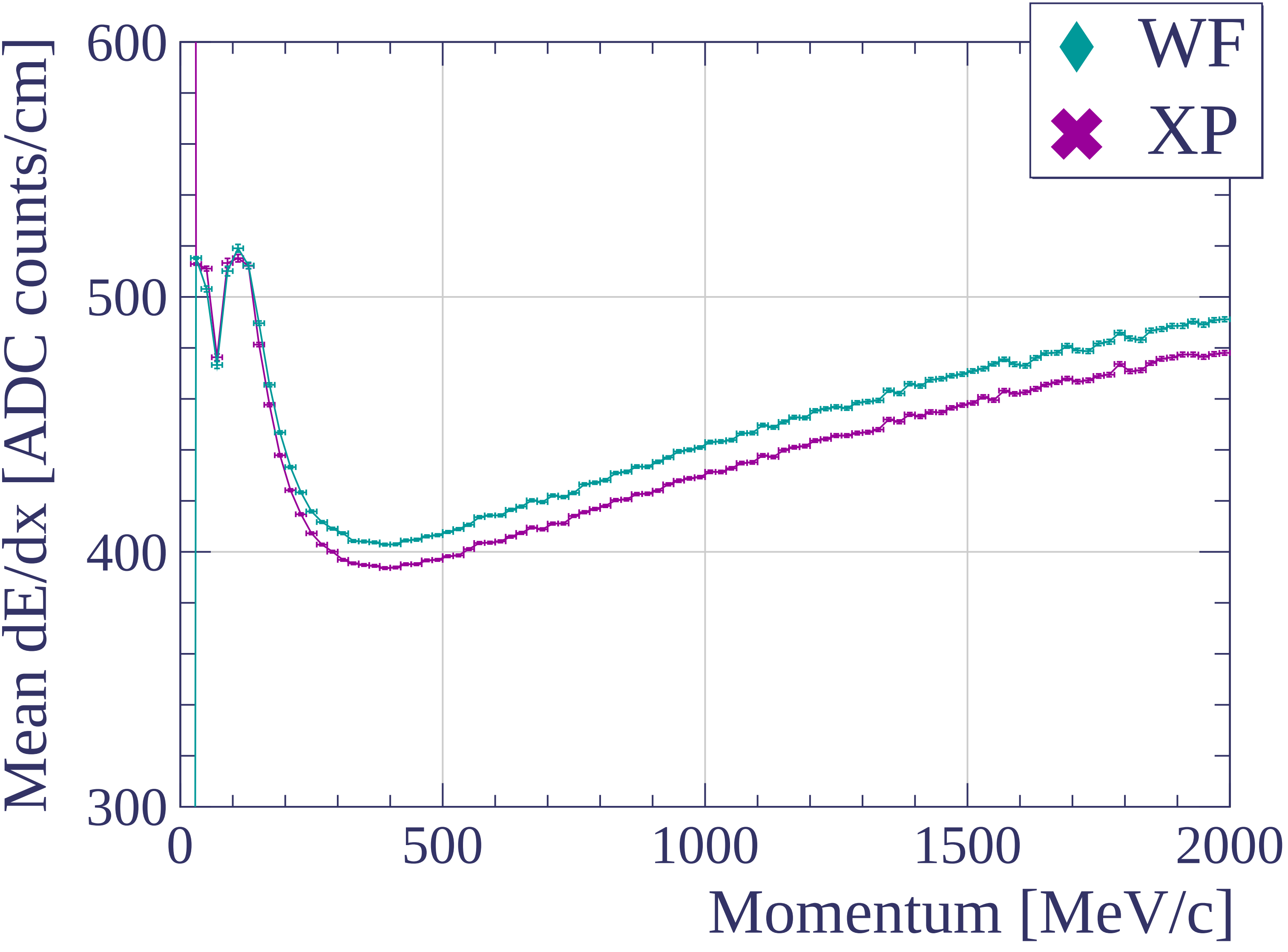

Count
dE/dx [ADC counts/cm]
ERAM 20
μ = 467.3± 0.7
σ = 85.4± 0.5
σ/μ = 18.28± 0.14 %
μ = 472.2± 0.6
σ = 88.5± 0.6
σ/μ = 18.74± 0.14 %
ERAM 38
μ = 466.0± 0.8
σ = 81.3± 0.6
σ/μ = 17.45± 0.16 %
μ = 470.6± 0.8
σ = 83.3± 0.6
σ/μ = 17.71± 0.15 %
ERAM 7
μ = 485.3± 0.7
σ = 82.5± 0.6
σ/μ = 17.00± 0.16 %
μ = 491.8± 0.8
σ = 85.7± 0.7
σ/μ = 17.42± 0.17 %
ERAM 44
μ = 462.4± 0.6
σ = 89.0± 0.6
σ/μ = 19.25± 0.15 %
μ = 467.3± 0.6
σ = 92.1± 0.6
σ/μ = 19.70± 0.15 %
ERAM 43
μ = 486.1± 0.3
σ = 75.1± 0.2
σ/μ = 15.46± 0.06 %
μ = 495.1± 0.3
σ = 79.3± 0.3
σ/μ = 16.02± 0.06 %
ERAM 39
μ = 467.5± 0.3
σ = 69.7± 0.2
σ/μ = 14.90± 0.06 %
μ = 475.3± 0.3
σ = 73.3± 0.2
σ/μ = 15.42± 0.06 %
ERAM 41
μ = 481.5± 0.3
σ = 74.3± 0.3
σ/μ = 15.44± 0.06 %
μ = 494.0± 0.3
σ = 76.8± 0.3
σ/μ = 15.55± 0.06 %
ERAM 46
μ = 465.8± 0.3
σ = 72.9± 0.2
σ/μ = 15.65± 0.06 %
μ = 476.7± 0.3
σ = 76.7± 0.2
σ/μ = 16.08± 0.06 %
ERAM 26
μ = 488.4± 0.7
σ = 90.6± 0.6
σ/μ = 18.55± 0.15 %
μ = 492.7± 0.7
σ = 94.0± 0.6
σ/μ = 19.07± 0.16 %
ERAM 17
μ = 493.8± 0.8
σ = 86.2± 0.7
σ/μ = 17.46± 0.18 %
μ = 501.5± 0.8
σ = 89.6± 0.8
σ/μ = 17.87± 0.18 %
ERAM 23
μ = 485.7± 0.8
σ = 83.4± 0.7
σ/μ = 17.18± 0.17 %
μ = 496.1± 0.8
σ = 87.3± 0.7
σ/μ = 17.60± 0.18 %
ERAM 29
μ = 478.4± 0.6
σ = 87.1± 0.6
σ/μ = 18.20± 0.14 %
μ = 487.3± 0.6
σ = 91.6± 0.6
σ/μ = 18.81± 0.15 %
ERAM 1
μ = 489.9± 0.3
σ = 76.0± 0.2
σ/μ = 15.52± 0.06 %
μ = 500.0± 0.3
σ = 79.9± 0.3
σ/μ = 15.99± 0.06 %
ERAM 10
μ = 491.8± 0.3
σ = 75.1± 0.3
σ/μ = 15.27± 0.07 %
μ = 504.3± 0.3
σ = 77.3± 0.3
σ/μ = 15.34± 0.06 %
ERAM 11
μ = 493.6± 0.3
σ = 75.0± 0.3
σ/μ = 15.19± 0.06 %
μ = 507.5± 0.3
σ = 77.2± 0.3
σ/μ = 15.21± 0.06 %
ERAM 3
μ = 478.4± 0.3
σ = 76.1± 0.2
σ/μ = 15.91± 0.06 %
μ = 492.3± 0.3
σ = 79.7± 0.3
σ/μ = 16.18± 0.06 %



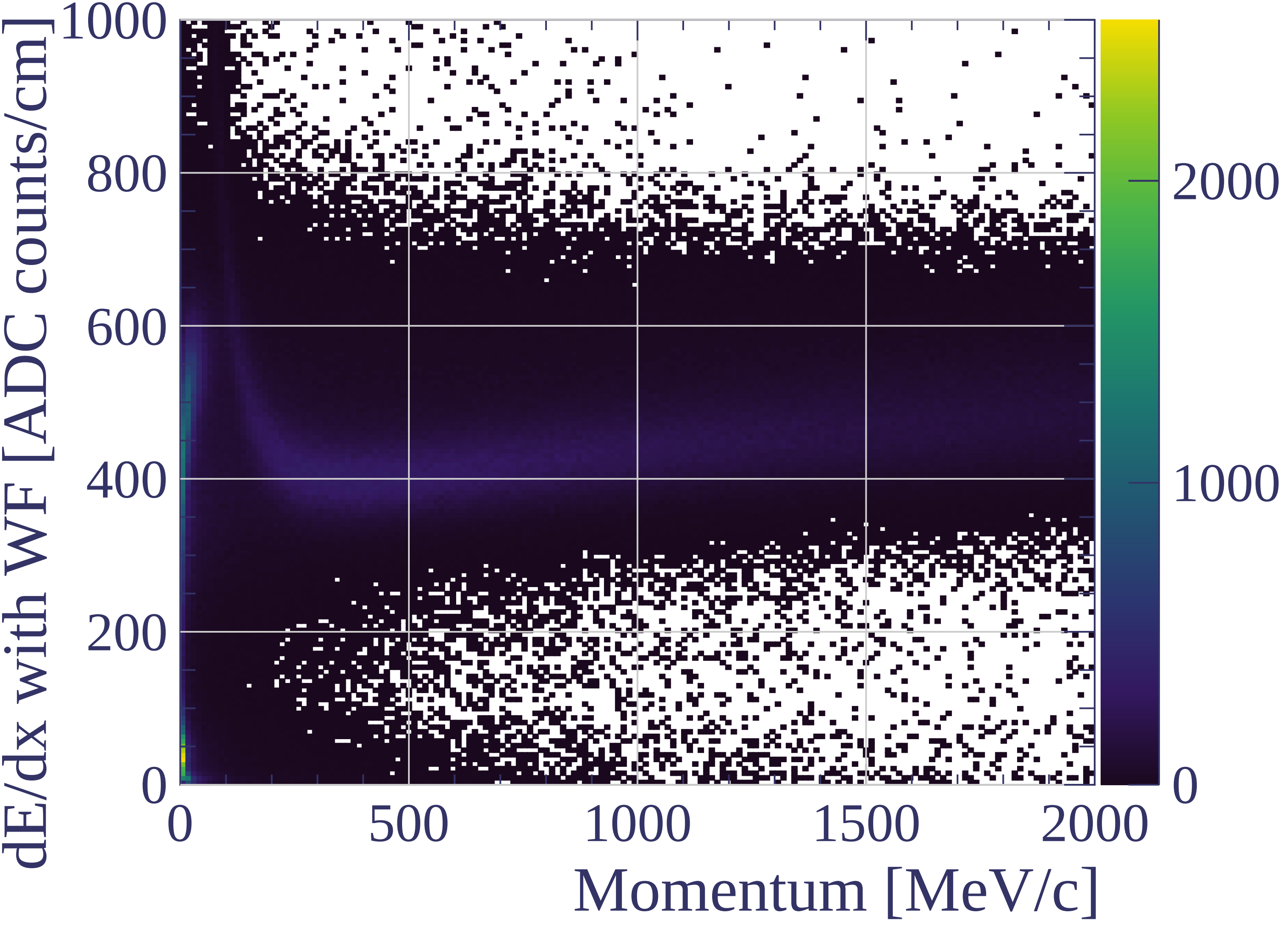
Entries 1706363
Mean x 477.3
Mean y 467.2
Std Dev x 115.3
Std Dev y 111.4
3
124
7370
0
0
0
×10^3
0
5
10
15
20
0
200
400
600
800
1000
0
200
400
600
800
1000
dE/dx with XP [ADC counts/cm]
dE/dx with WF [ADC counts/cm]



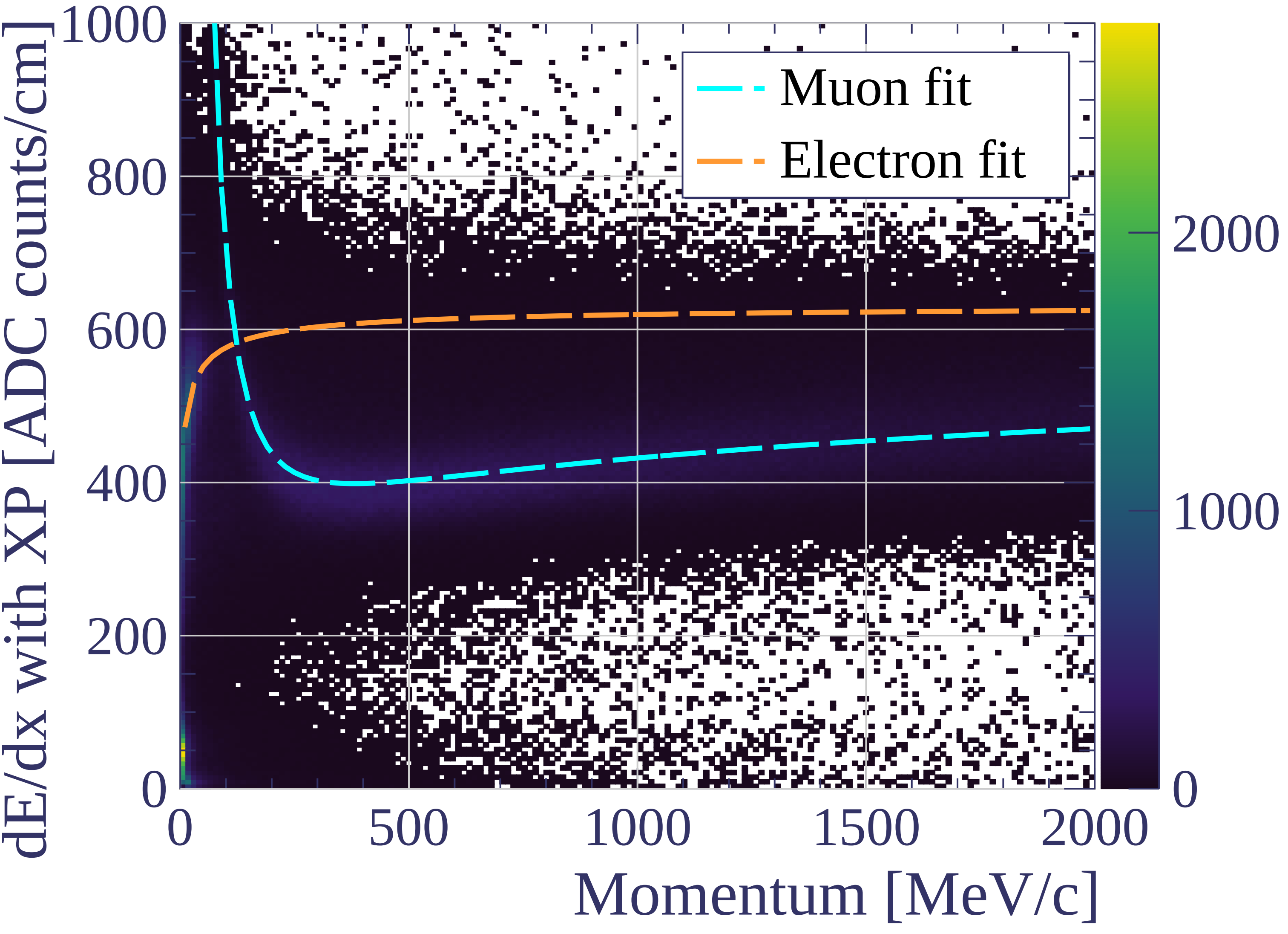
dE/dx resolution [%]
10
8
6
4
0
500
1000
1500
2000
Momentum [MeV/c]
WF
XP



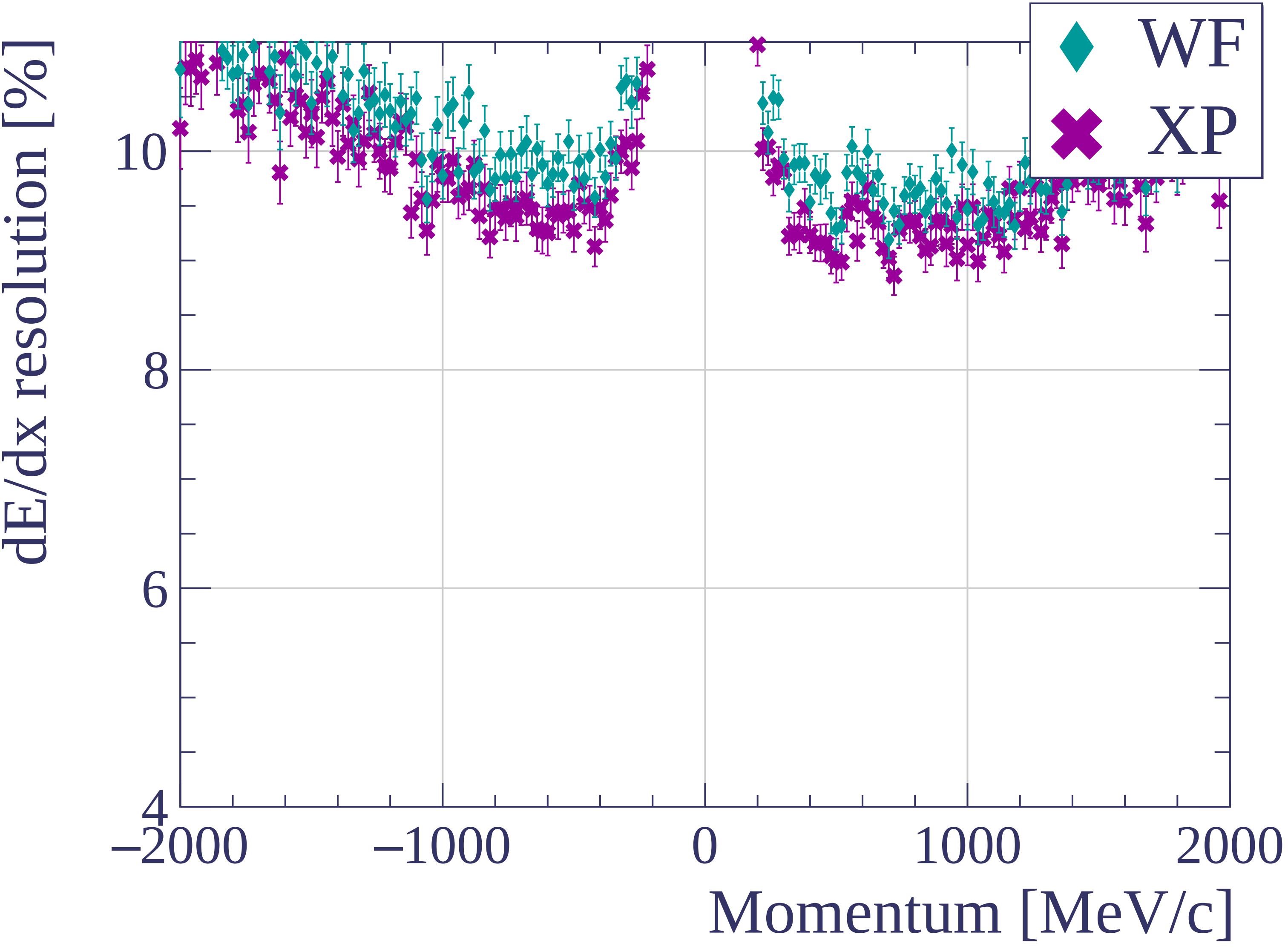
WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
0
500
1000
1500
2000
Momentum [MeV/c]



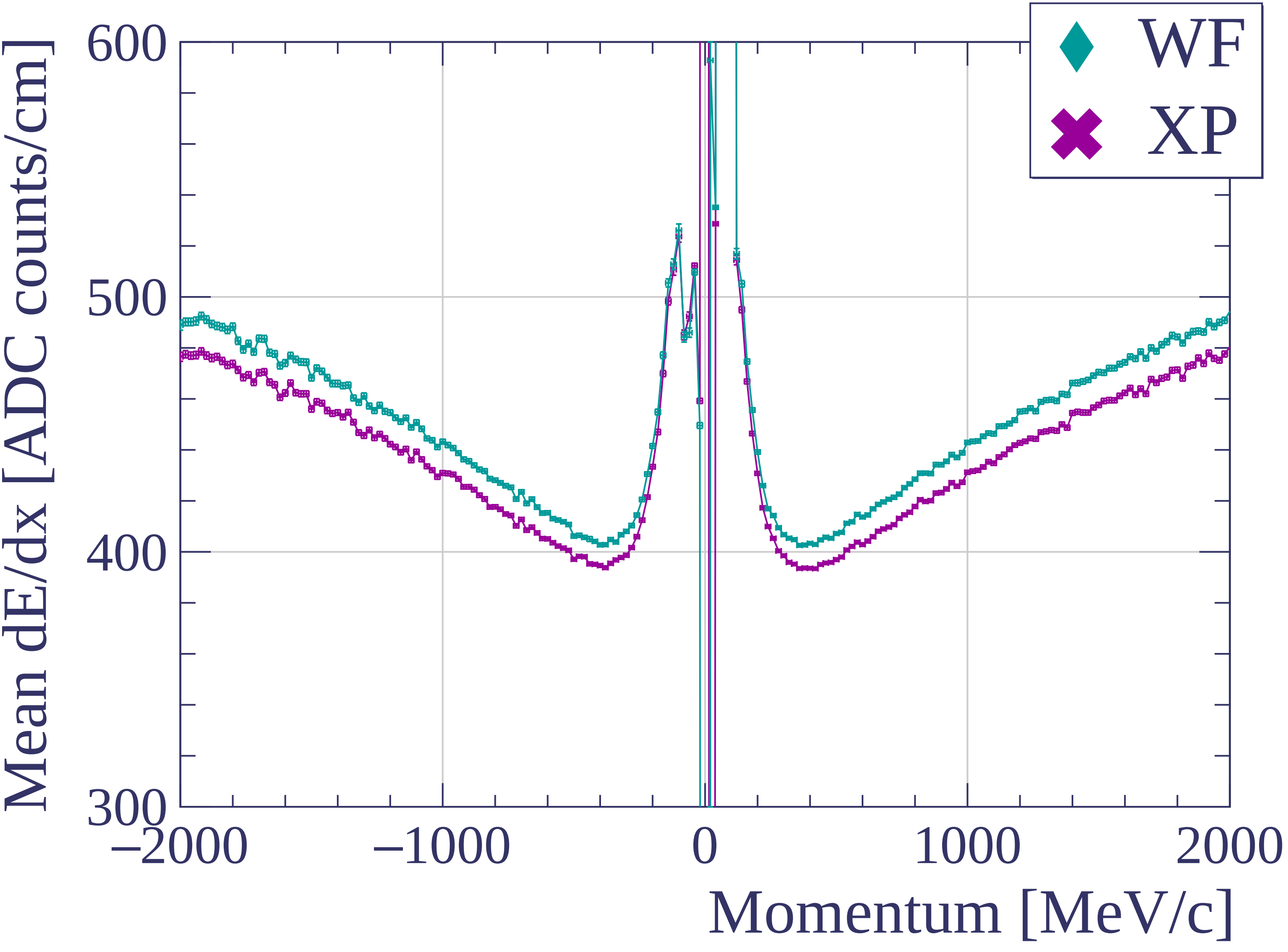
dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
0
500
1000
1500
2000
Momentum [MeV/c]
2000
1000
0

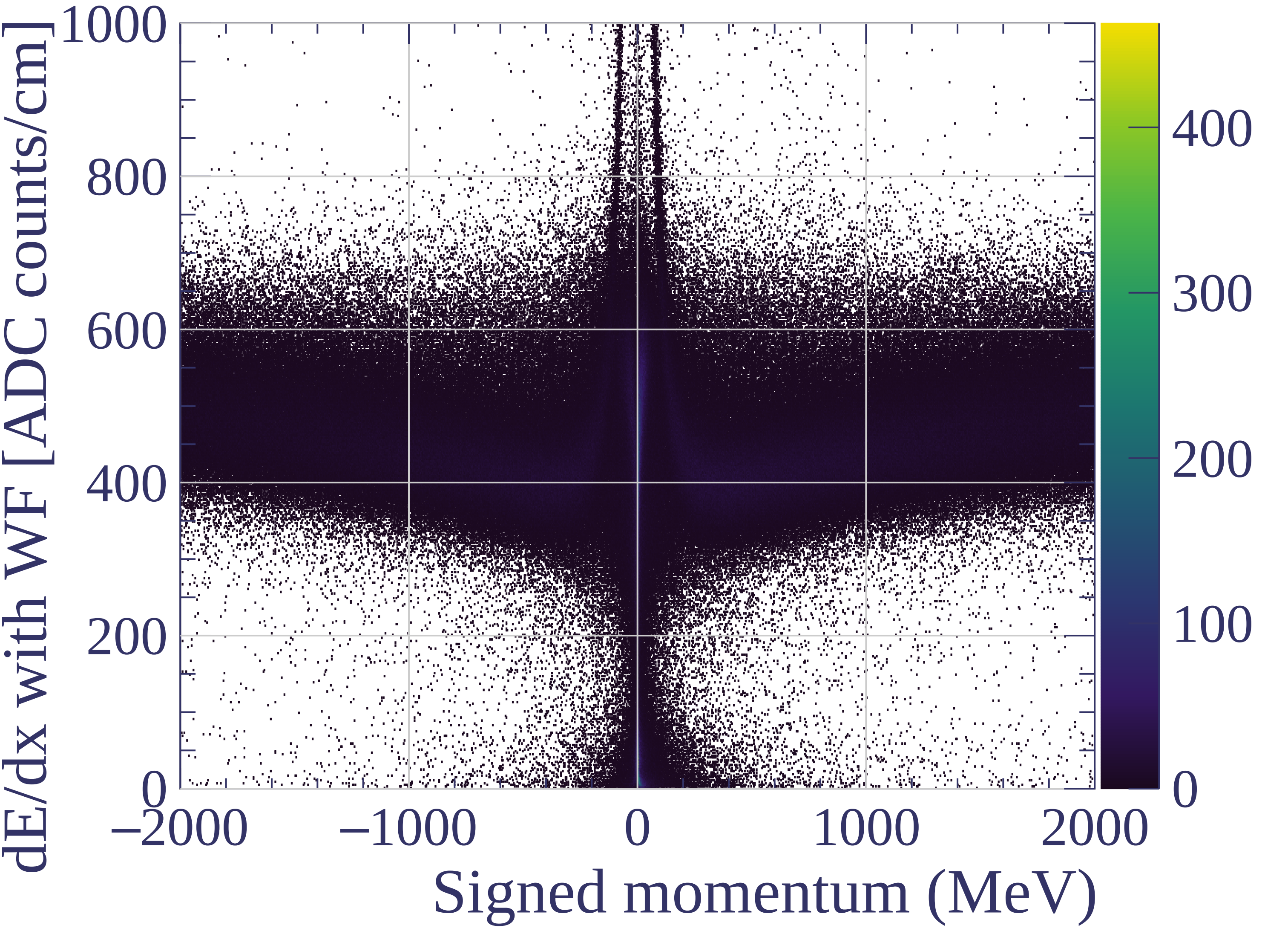


Muon fit
Electron fit
dE/dx with XP [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
500
1500
2000

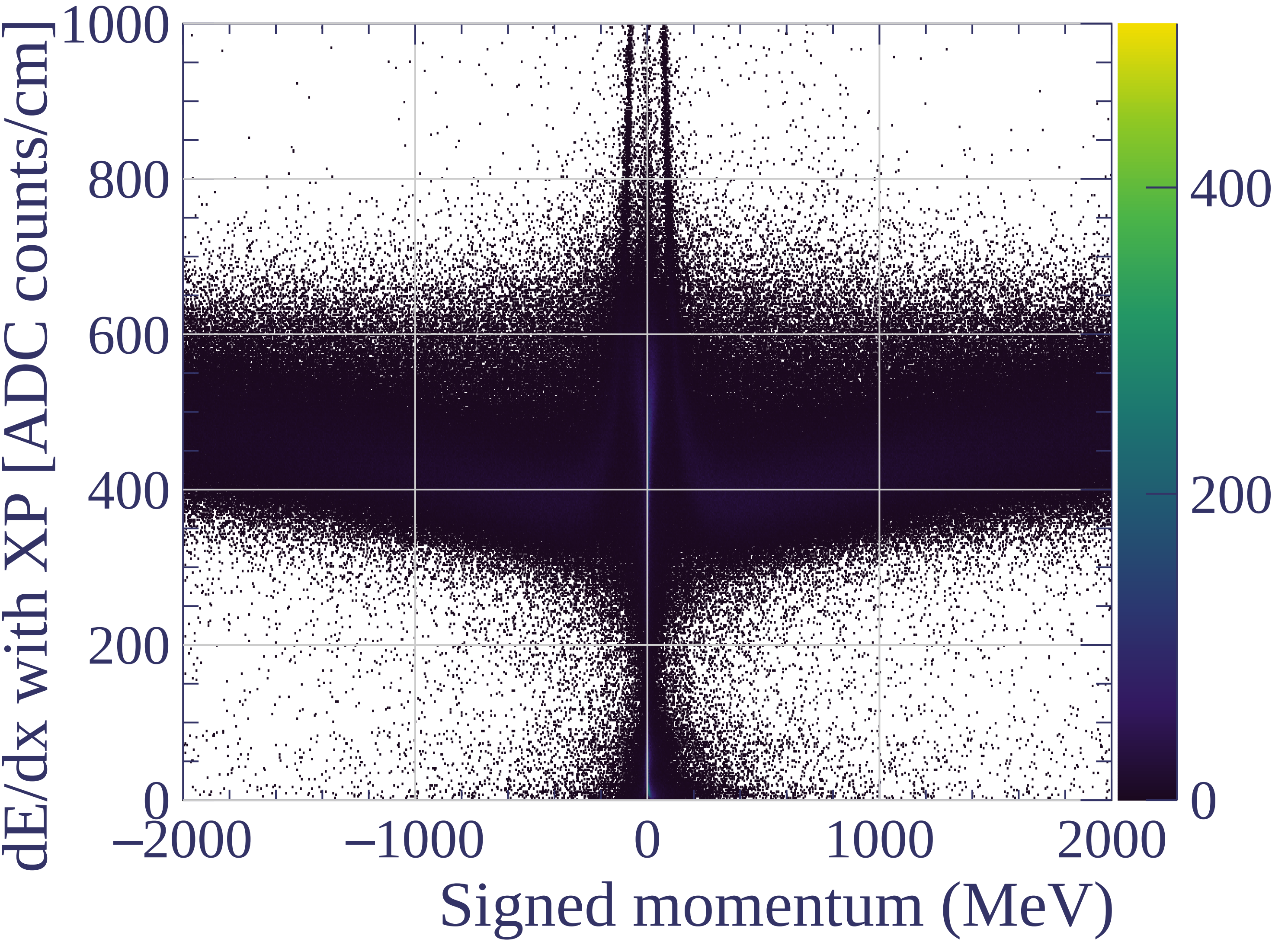


WF
XP
dE/dx resolution [%]
10
8
6
4
−2000
−1000
0
1000
2000
Momentum [MeV/c]

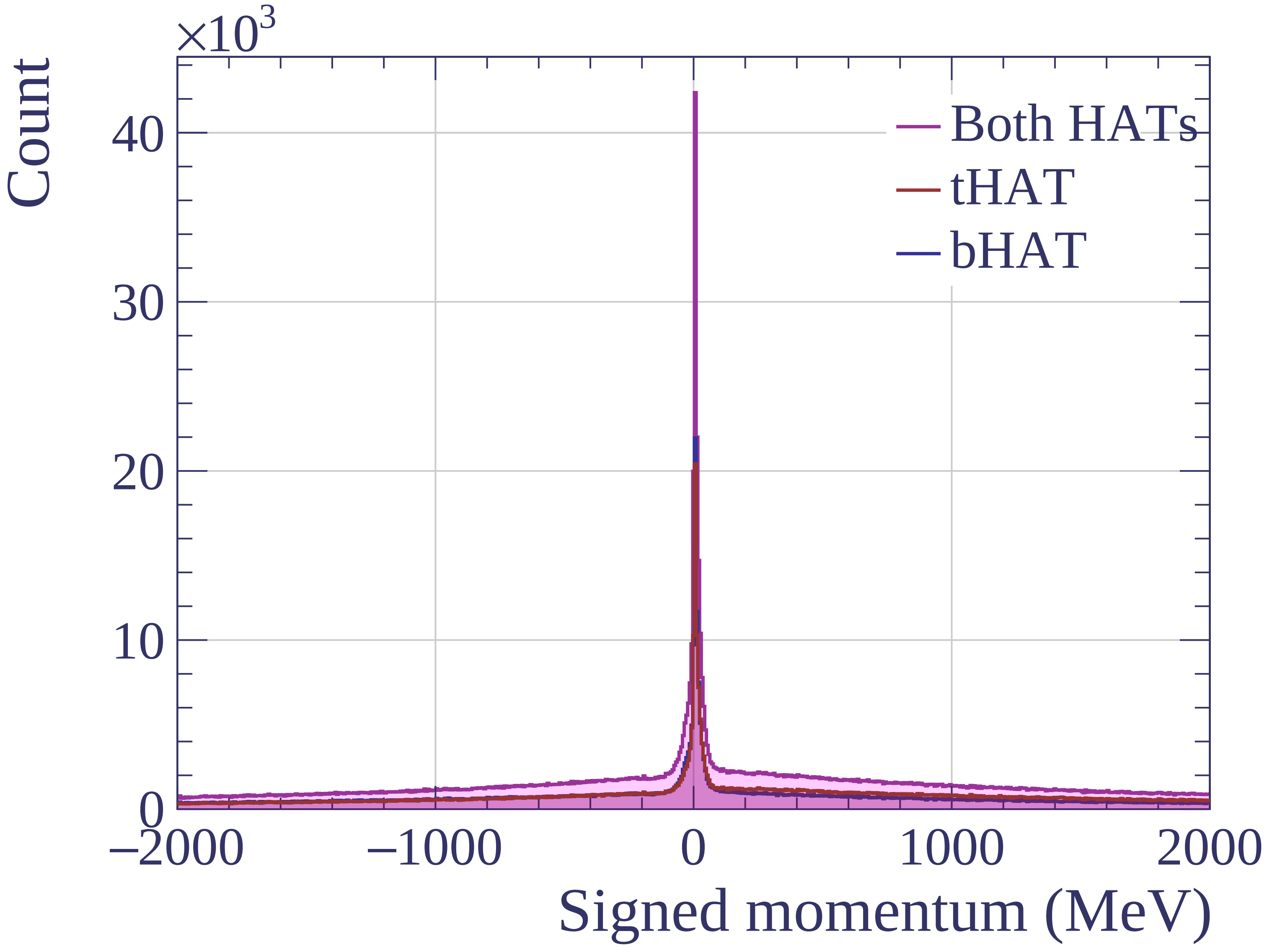

WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
−2000
−1000
0
1000
2000
Momentum [MeV/c]



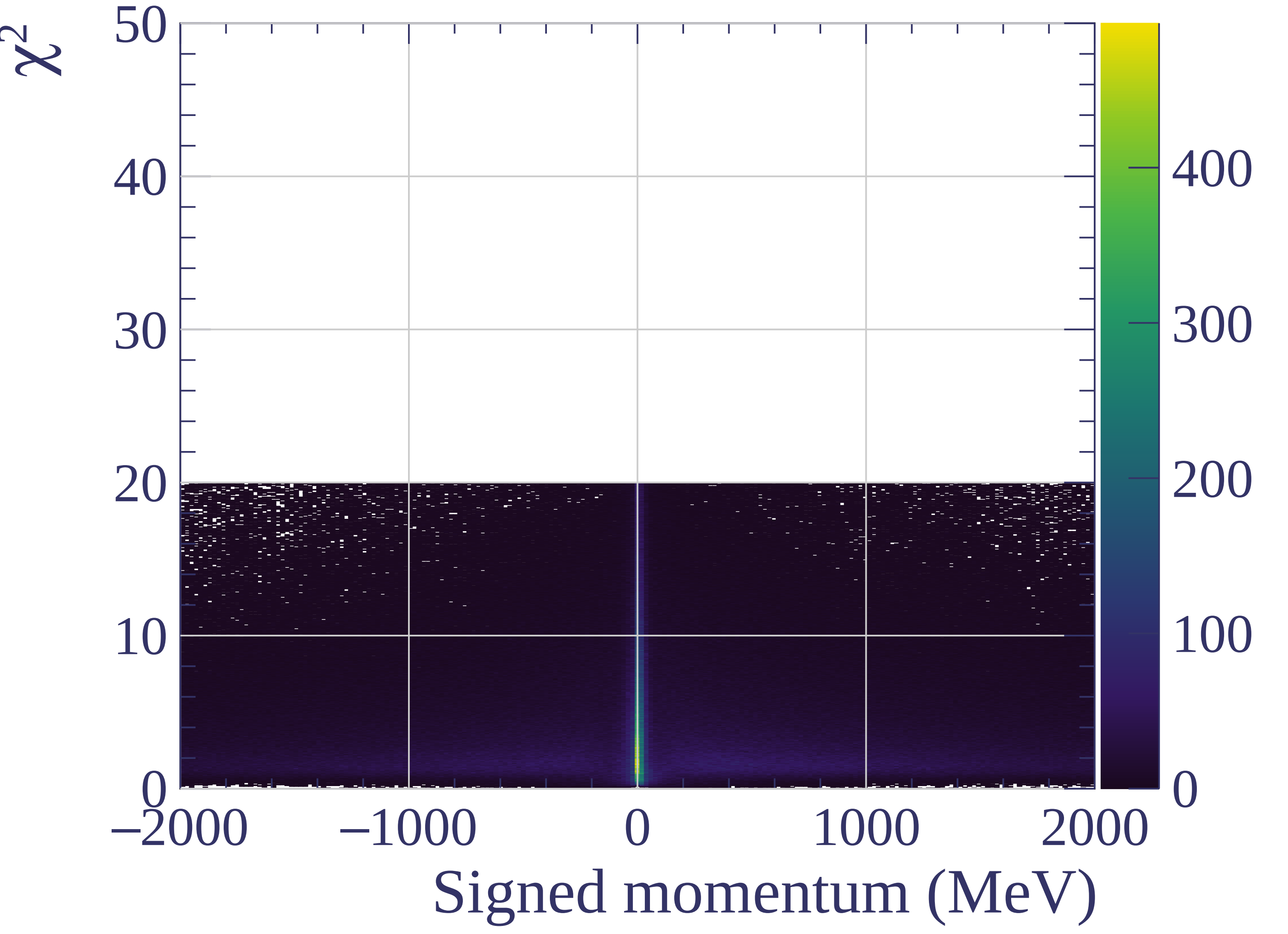
dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
400
300
200
100
0



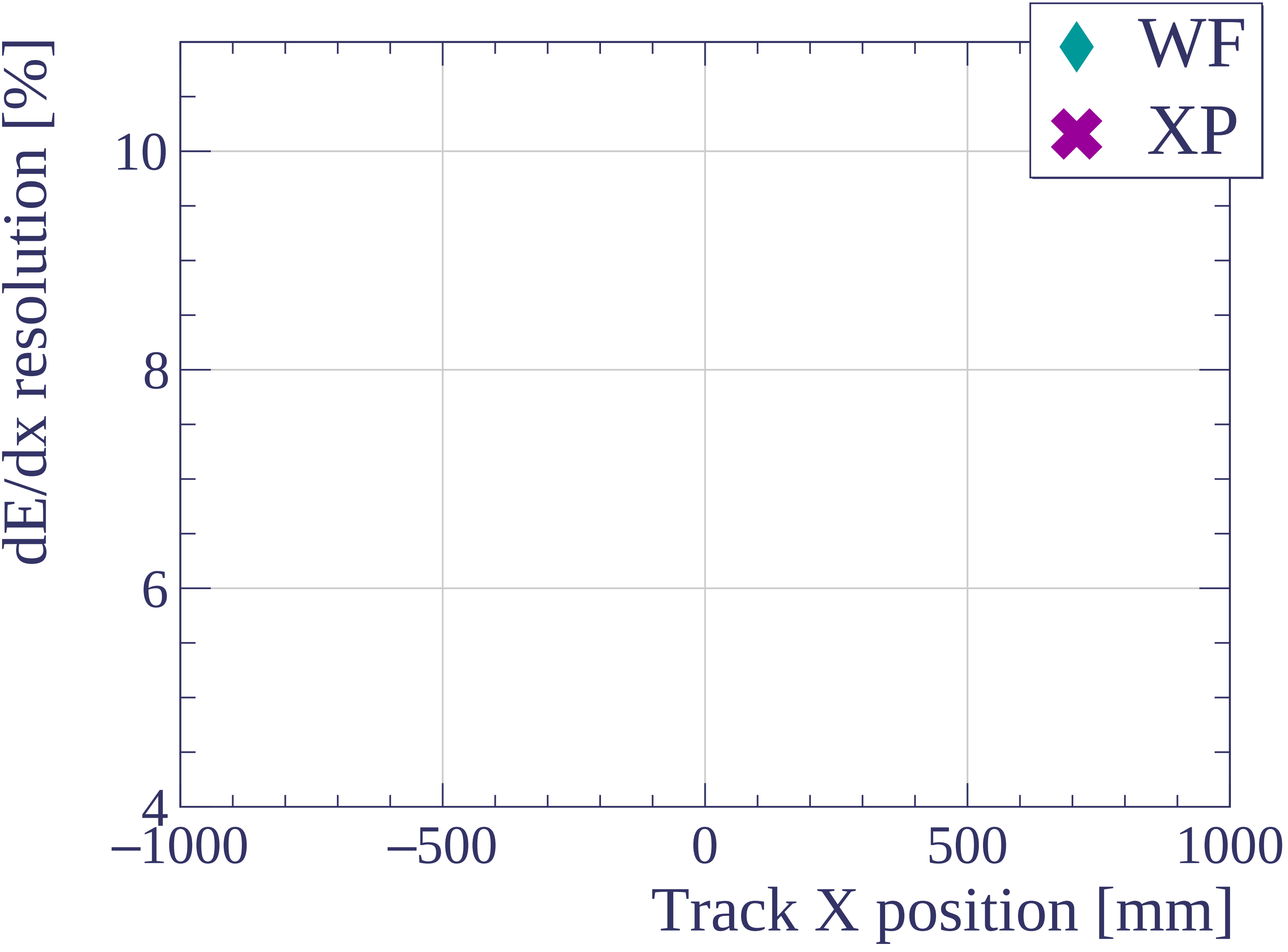
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
400
200
0



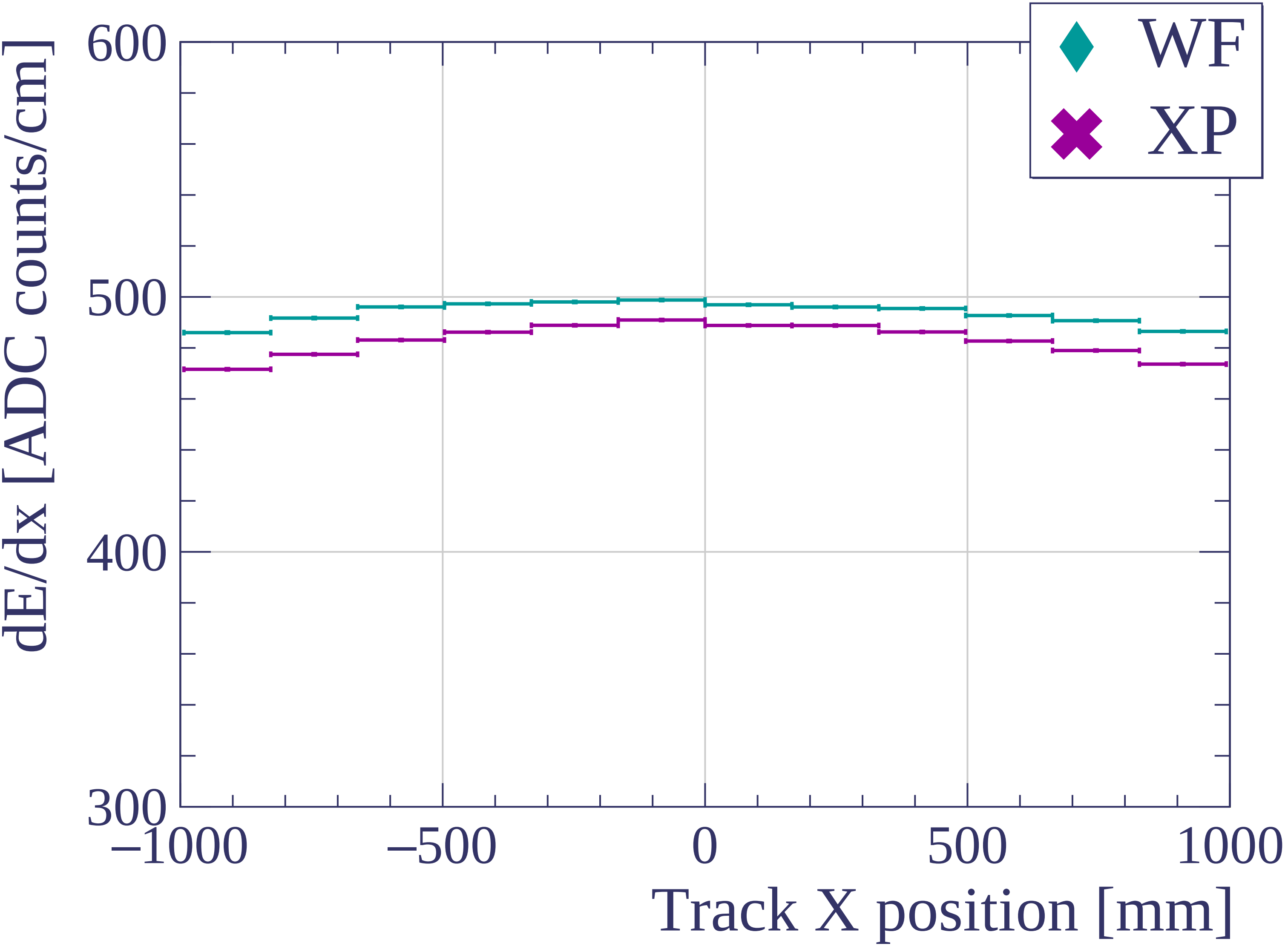
×10³
Count
Both HATs
tHAT
bHAT
0
10
20
30
40
−2000
−1000
0
1000
2000
Signed momentum (MeV)



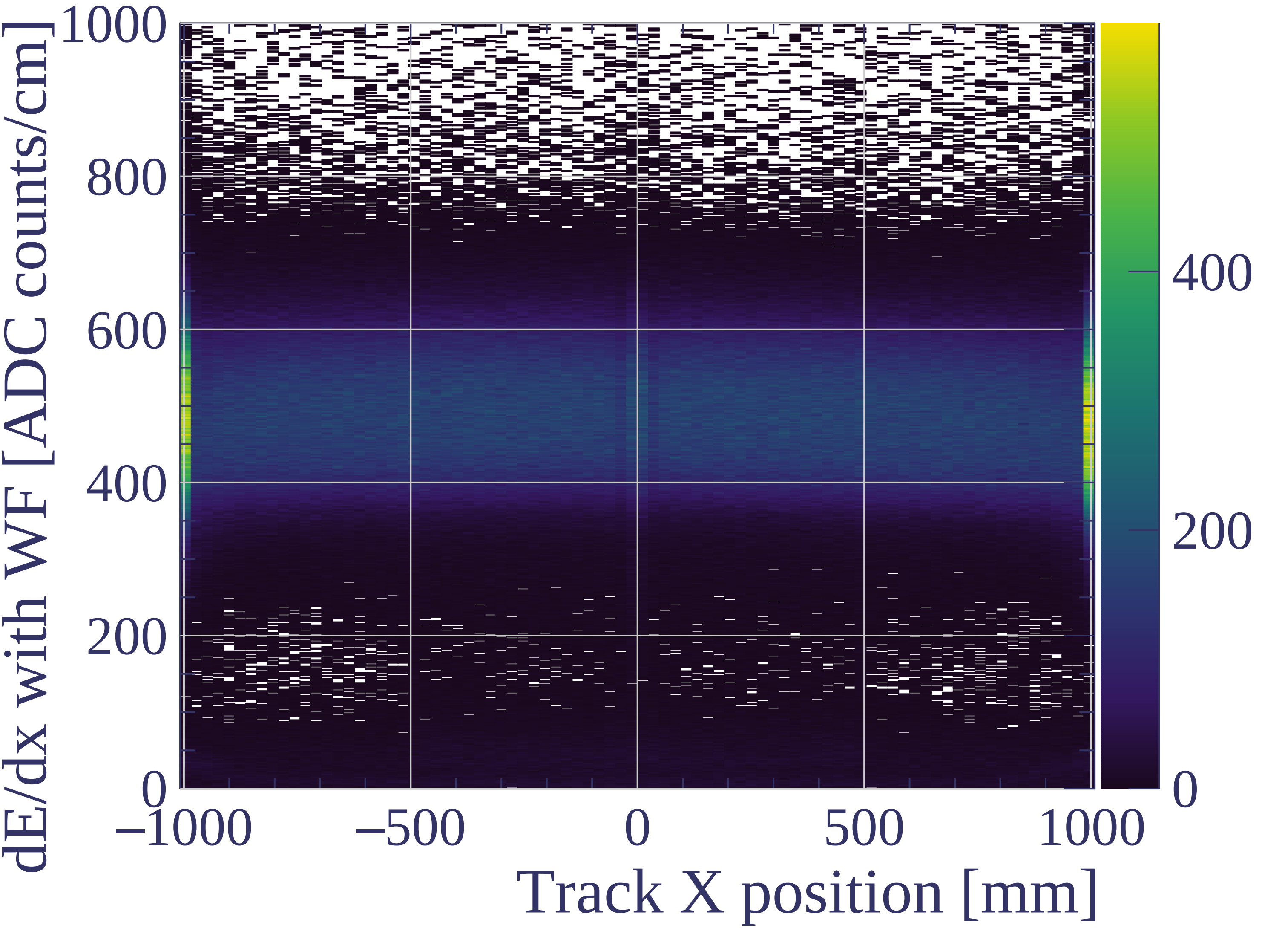
χ^2
50
40
30
20
10
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
400
300
200
100
0

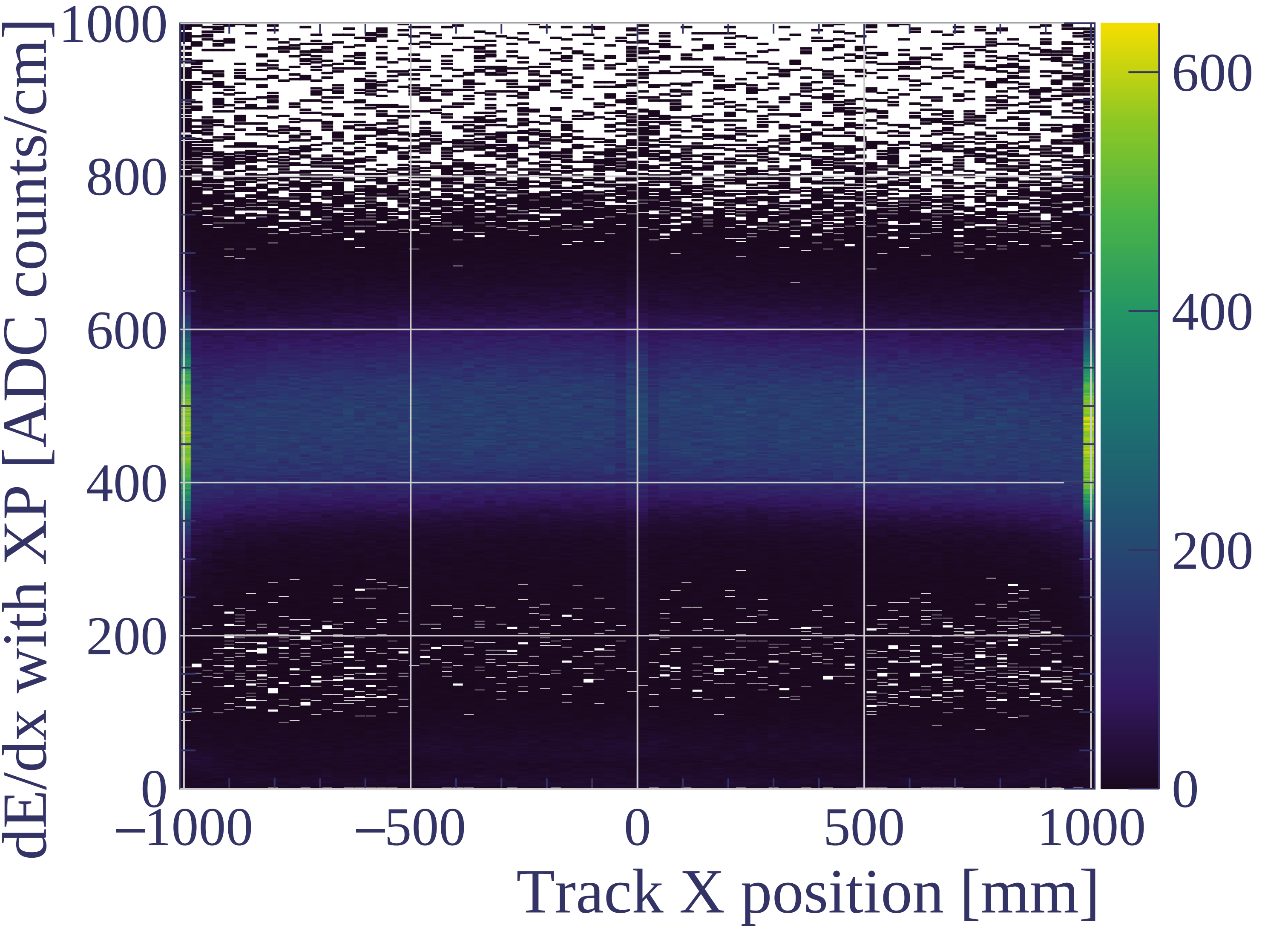


WF
XP
dE/dx resolution [%]
10
8
6
4
−1000
−500
0
500
1000
Track X position [mm]

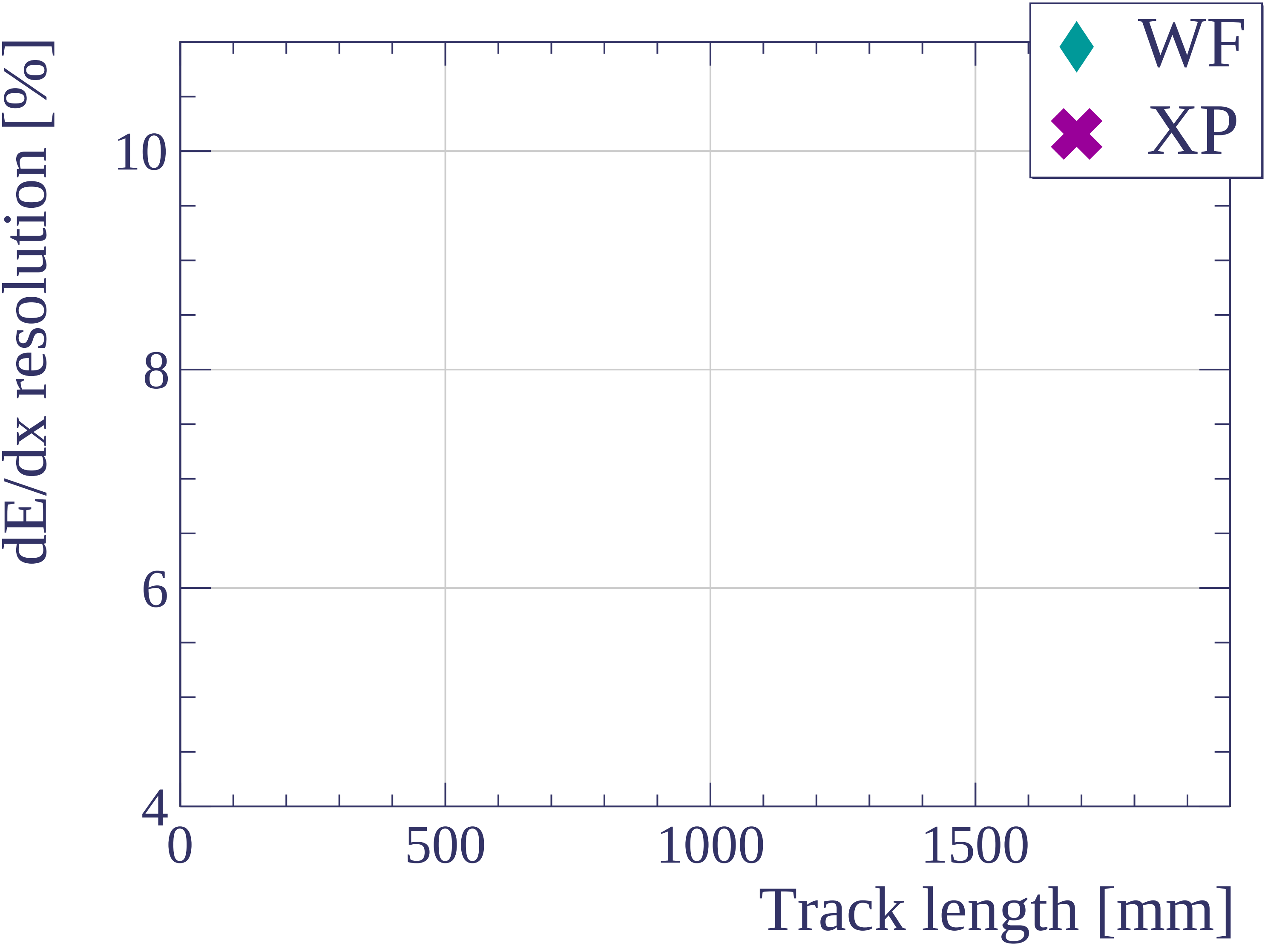


WF
XP
600
500
400
300
−1000
−500
0
500
1000
dE/dx [ADC counts/cm]
Track X position [mm]

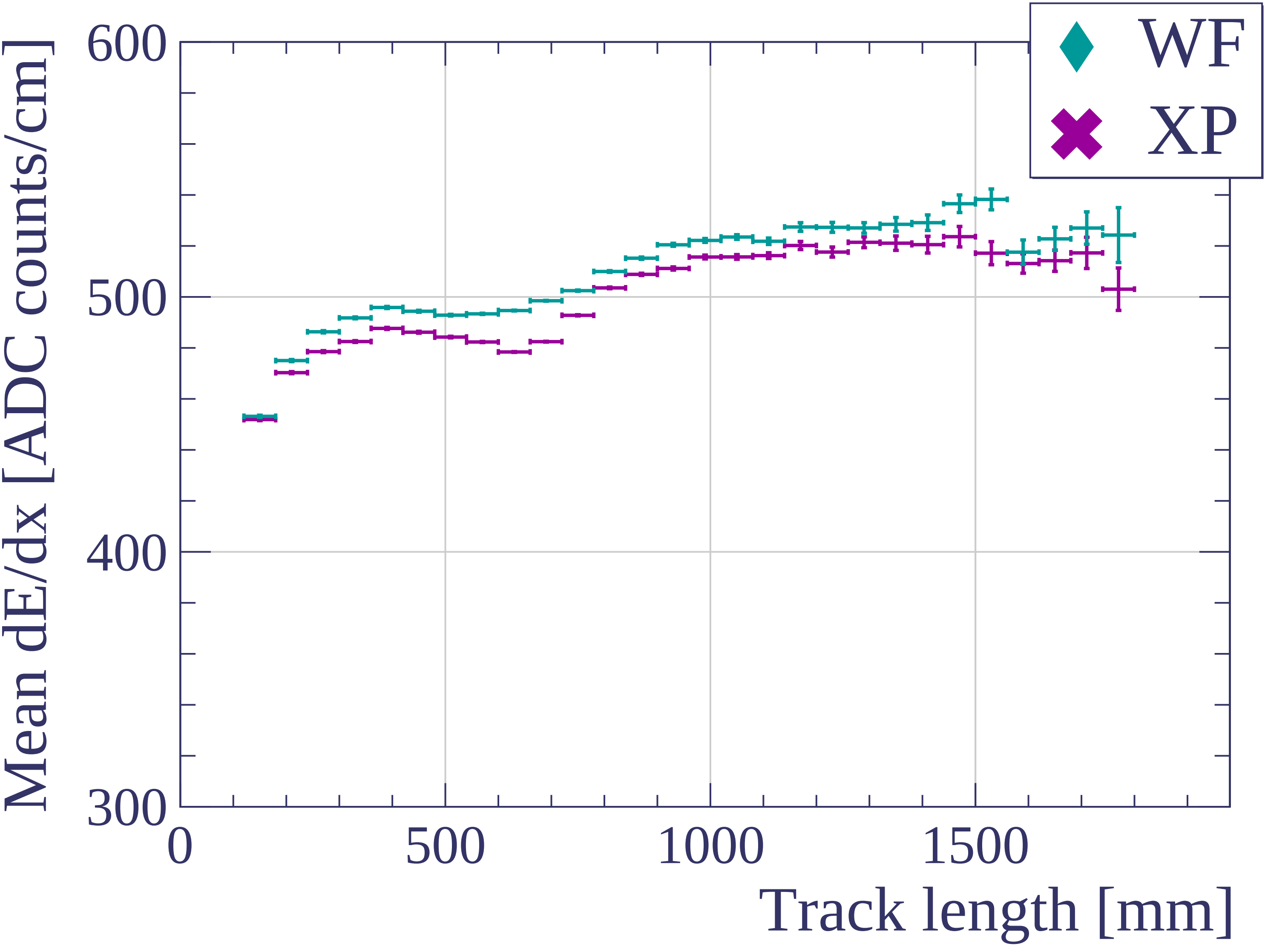

dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
400
200
0

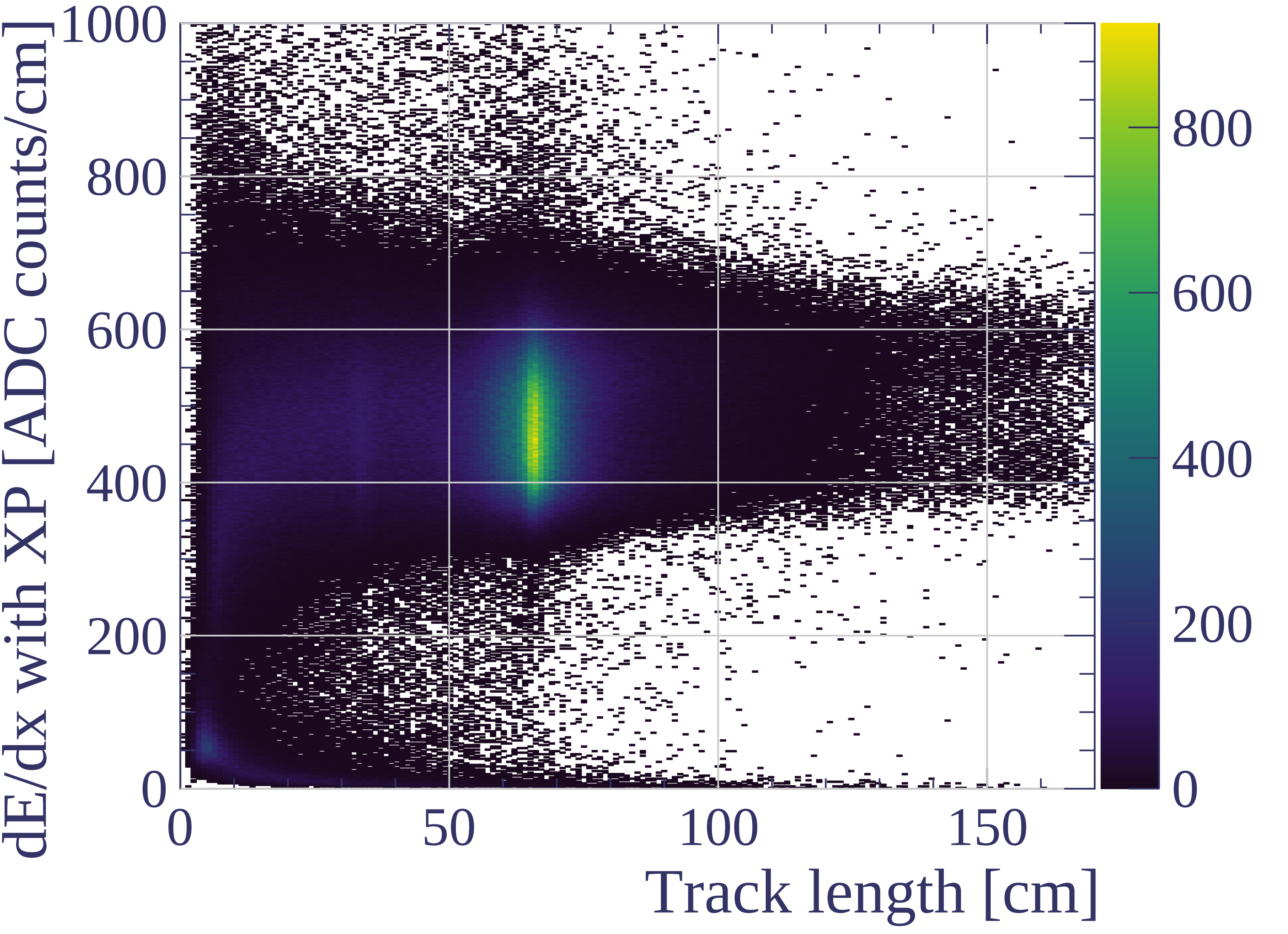


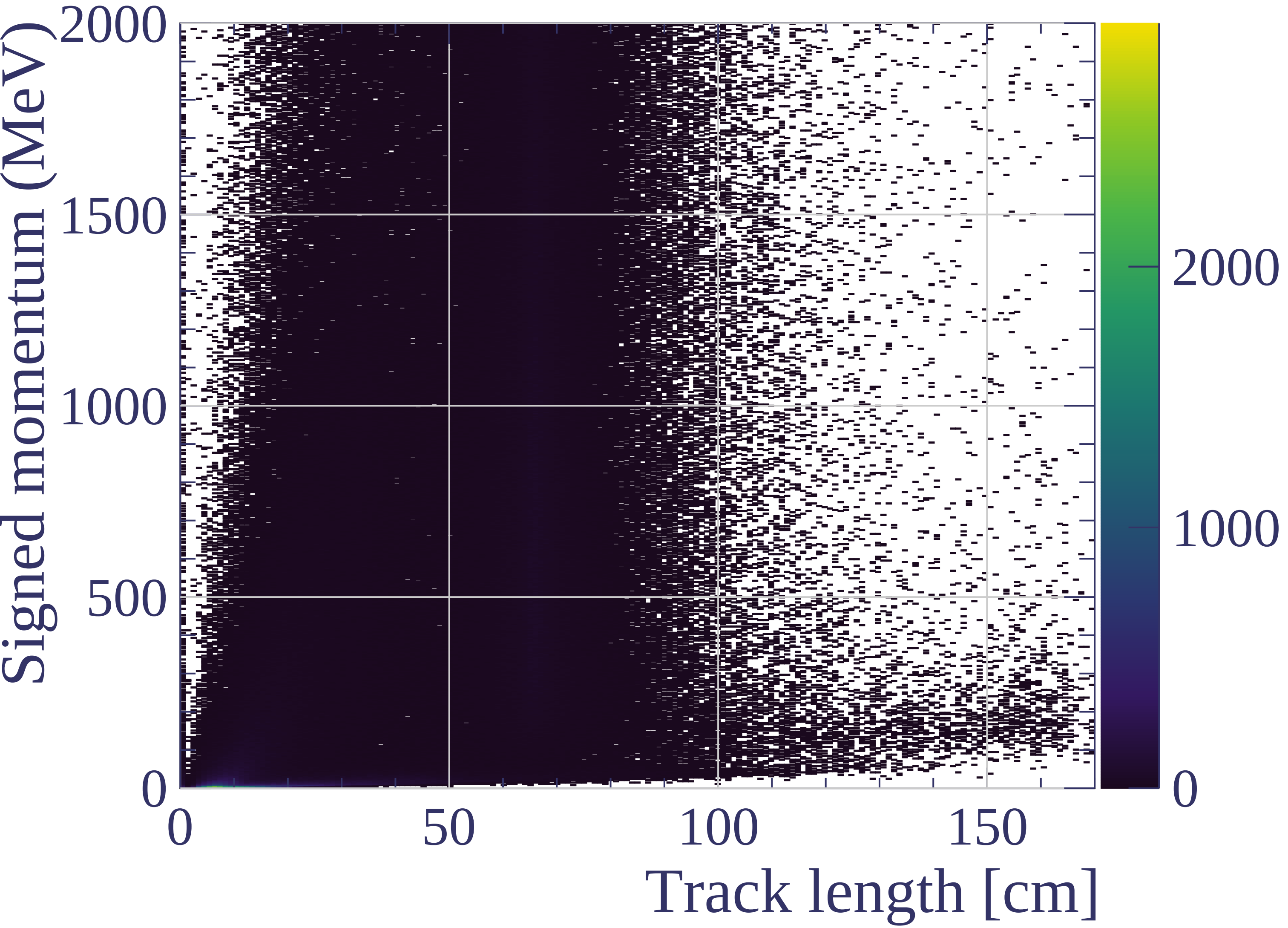
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
600
400
200
0



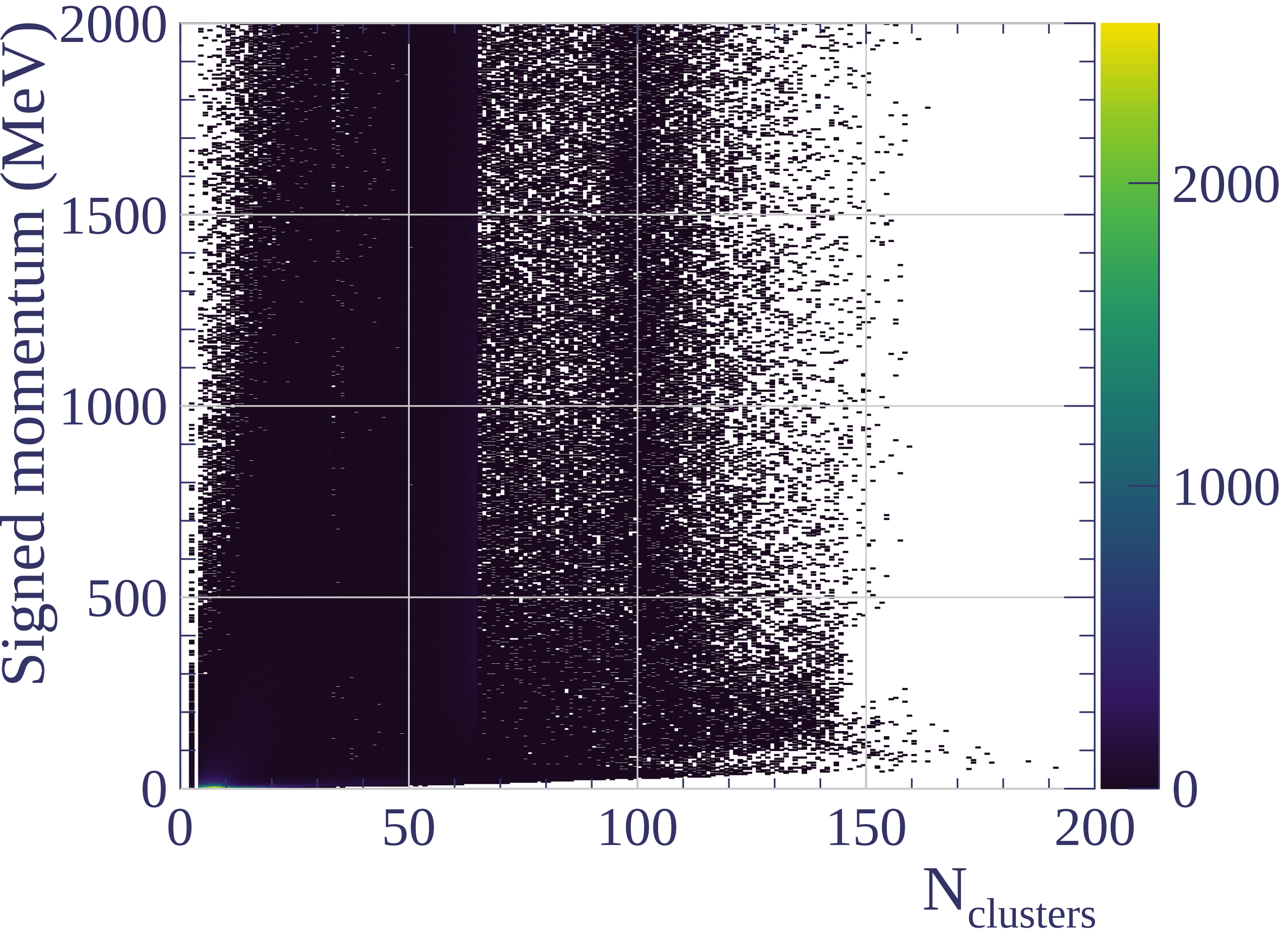
WF
XP
dE/dx resolution [%]
10
8
6
4
0
500
1000
1500
Track length [mm]

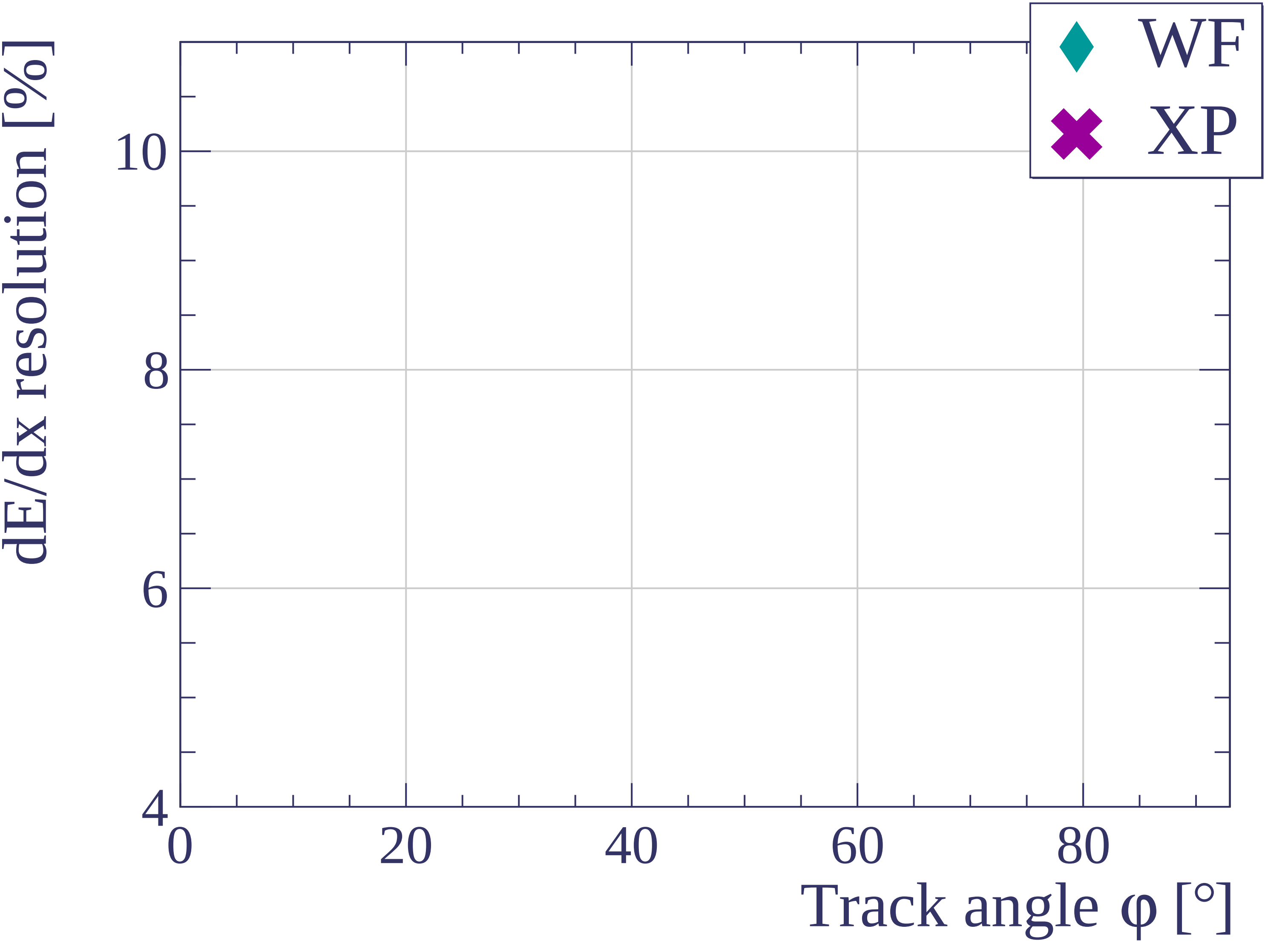

WF
XP
Mean dE/dx [ADC counts/cm]
Track length [mm]
600
500
400
300
0
500
1000
1500


dE/dx with XP [ADC counts/cm]
Track length [cm]
0
50
100
150
0
200
400
600
800
1000
800
600
400
200
0

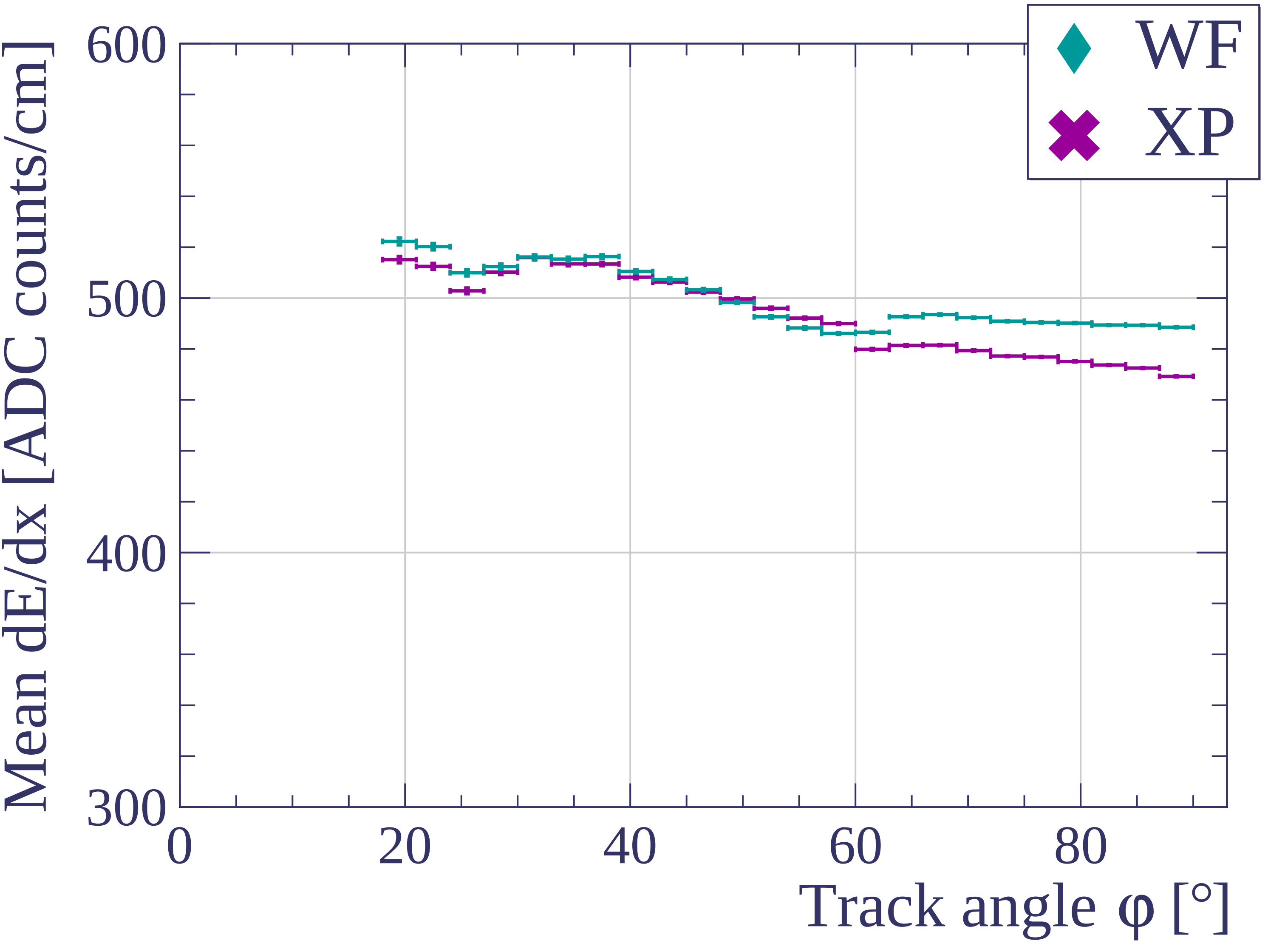


Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
Track length [cm]
2000
1000
0

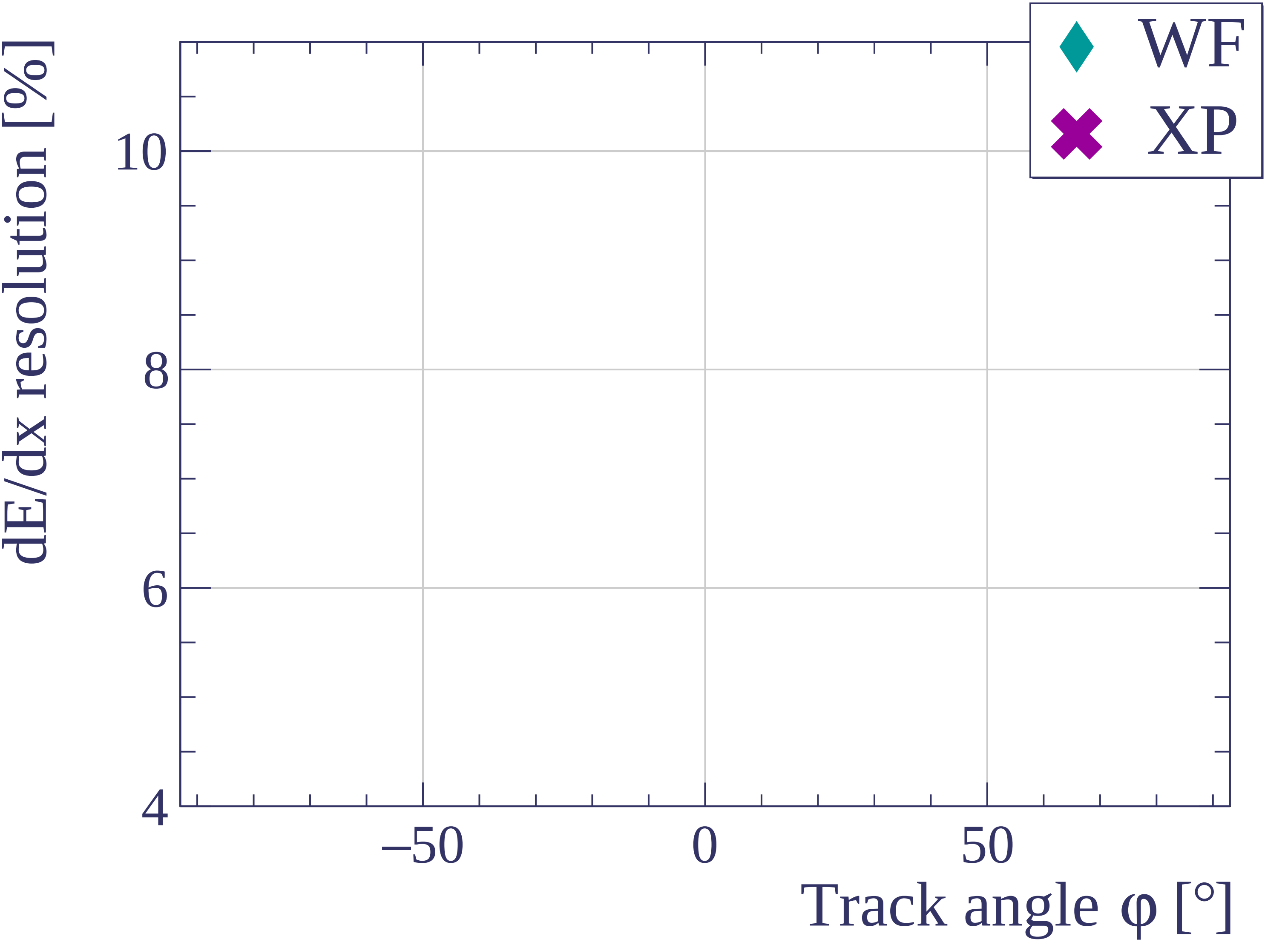

Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
200
$N_{clusters}$
2000
1000
0

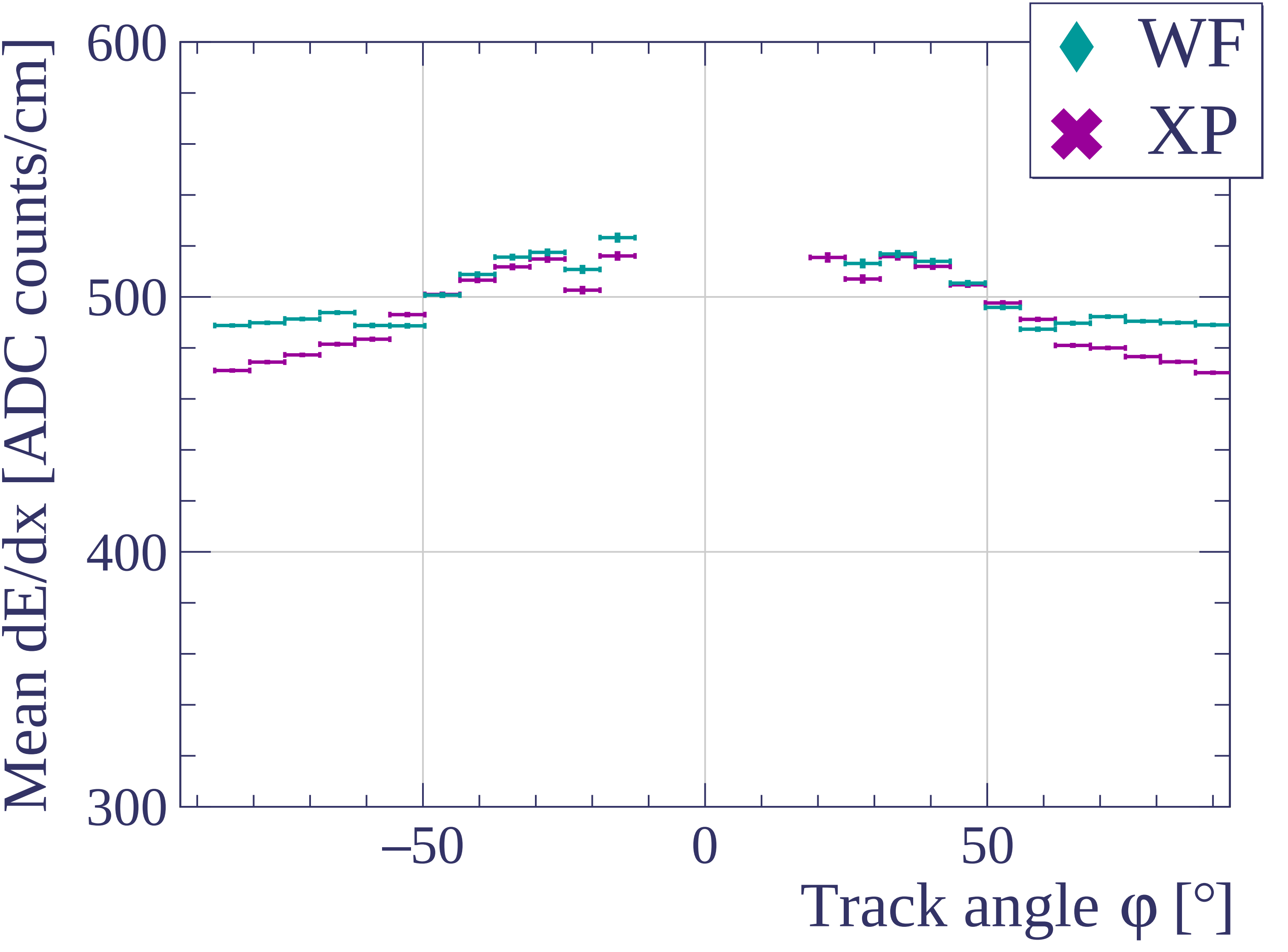


WF
XP
dE/dx resolution [%]
Track angle φ [°]
4
6
8
10
0
20
40
60
80

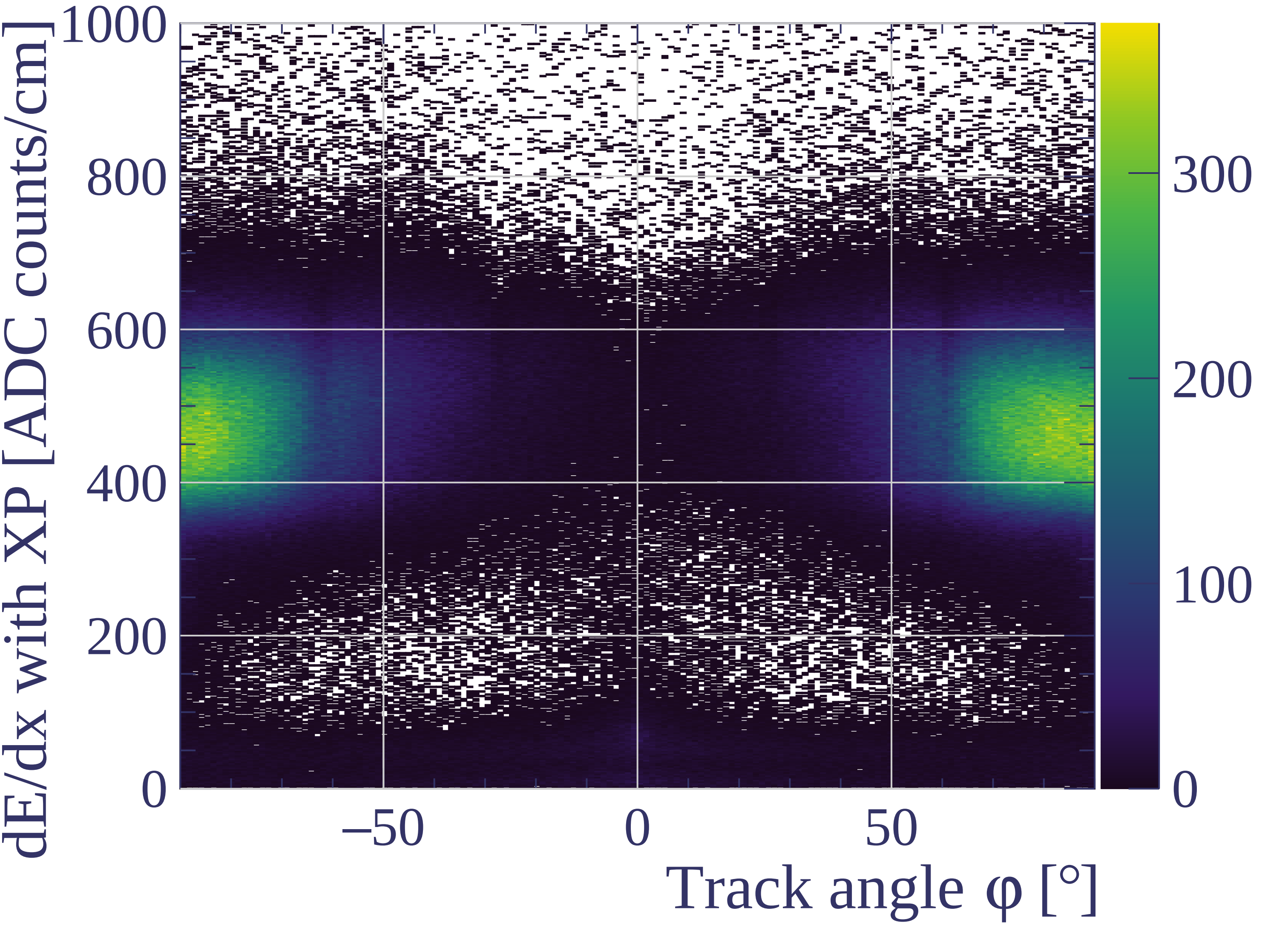


WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
0
20
40
60
80
Track angle φ [°]

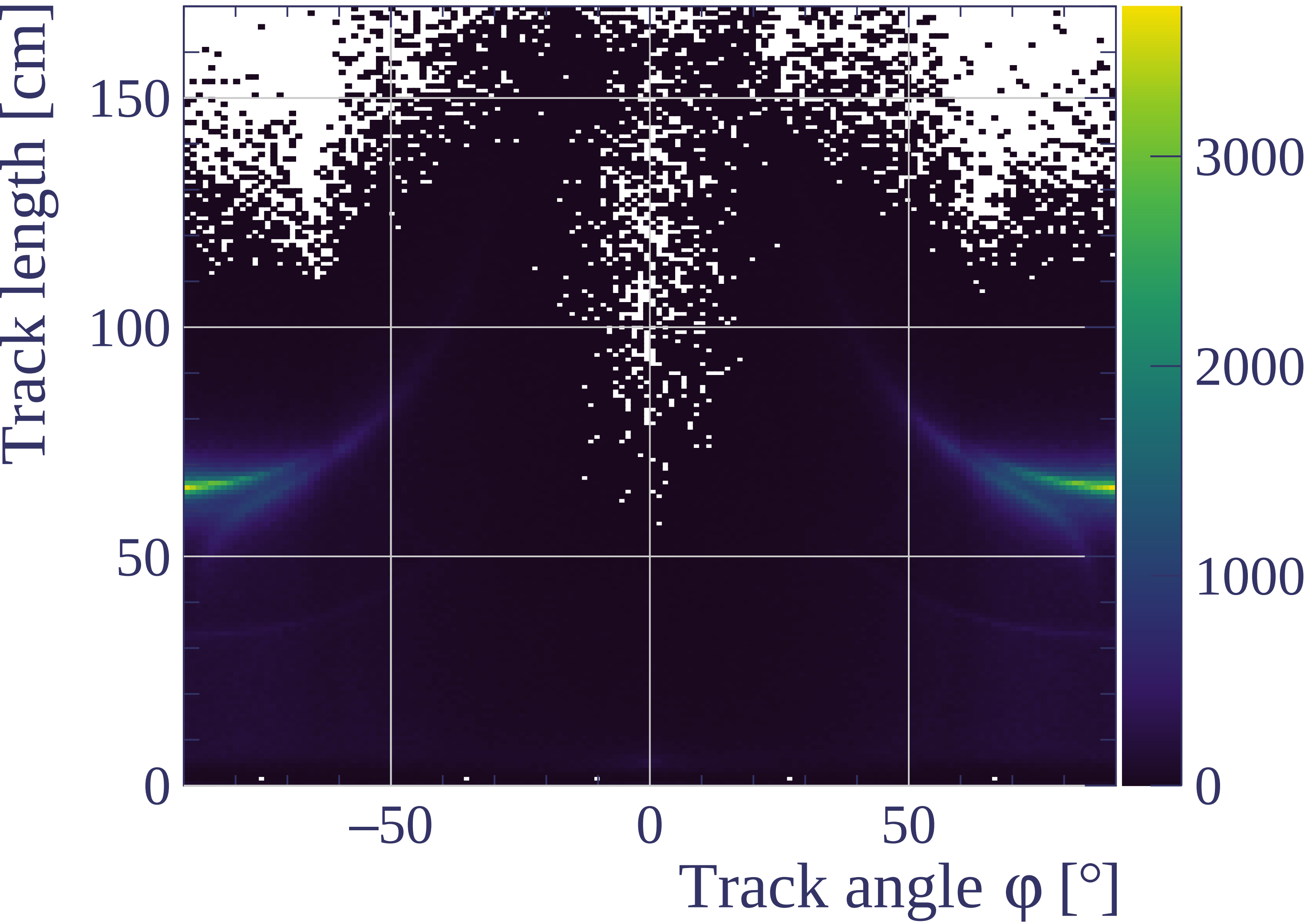

WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle φ [°]

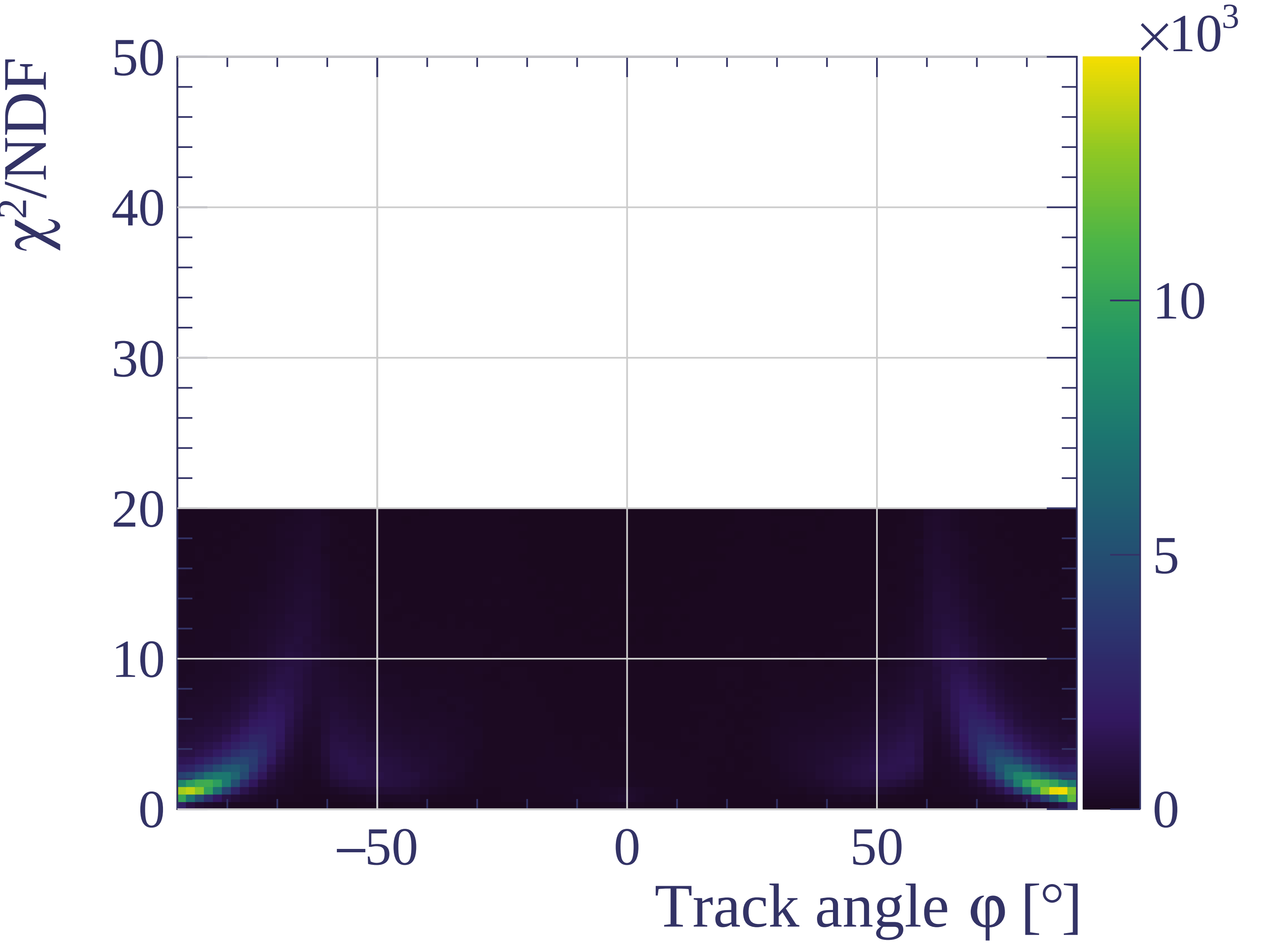


WF
XP
600
500
400
300
−50
0
50
Mean dE/dx [ADC counts/cm]
Track angle φ [°]

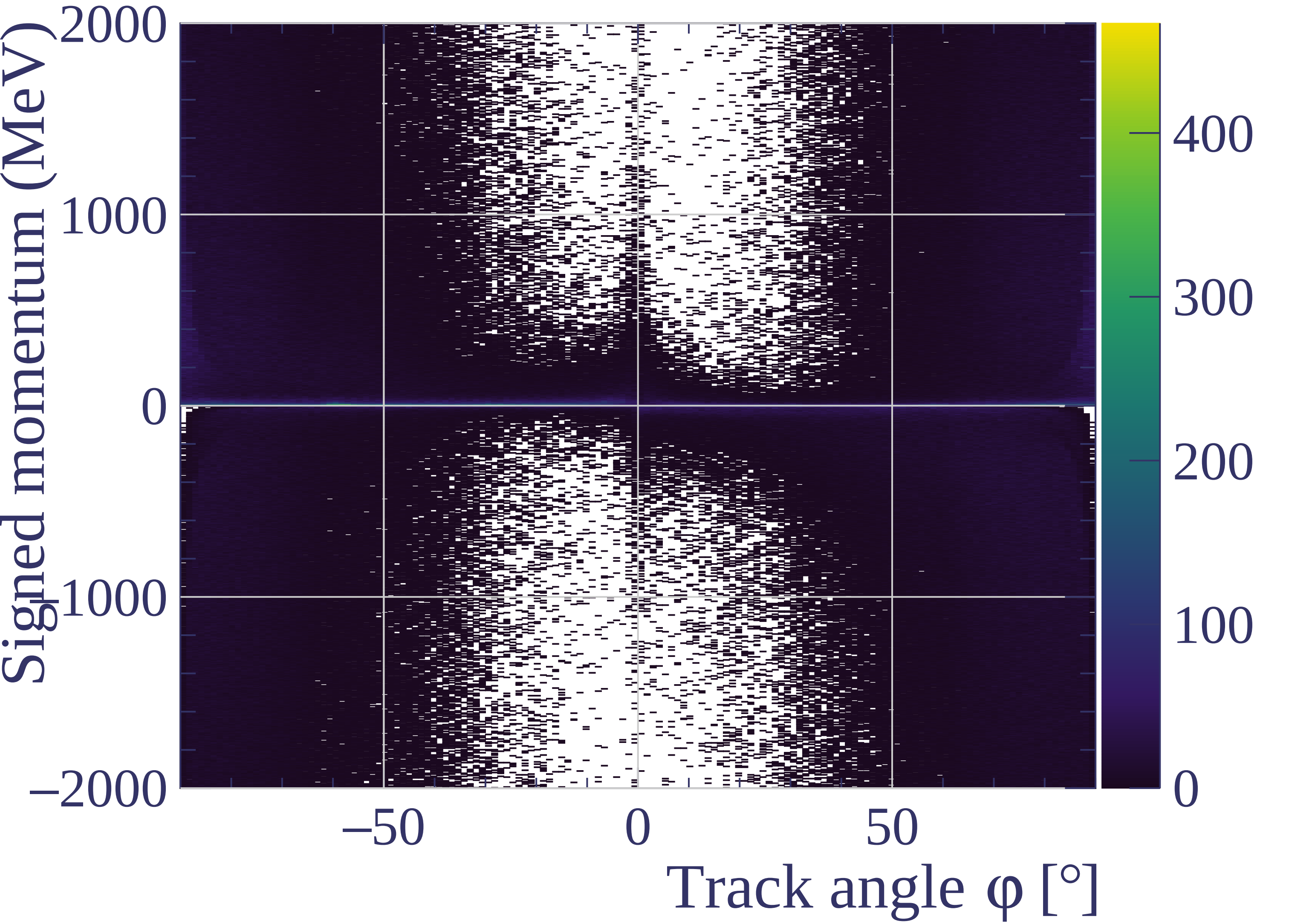


dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle φ [°]
300
200
100
0

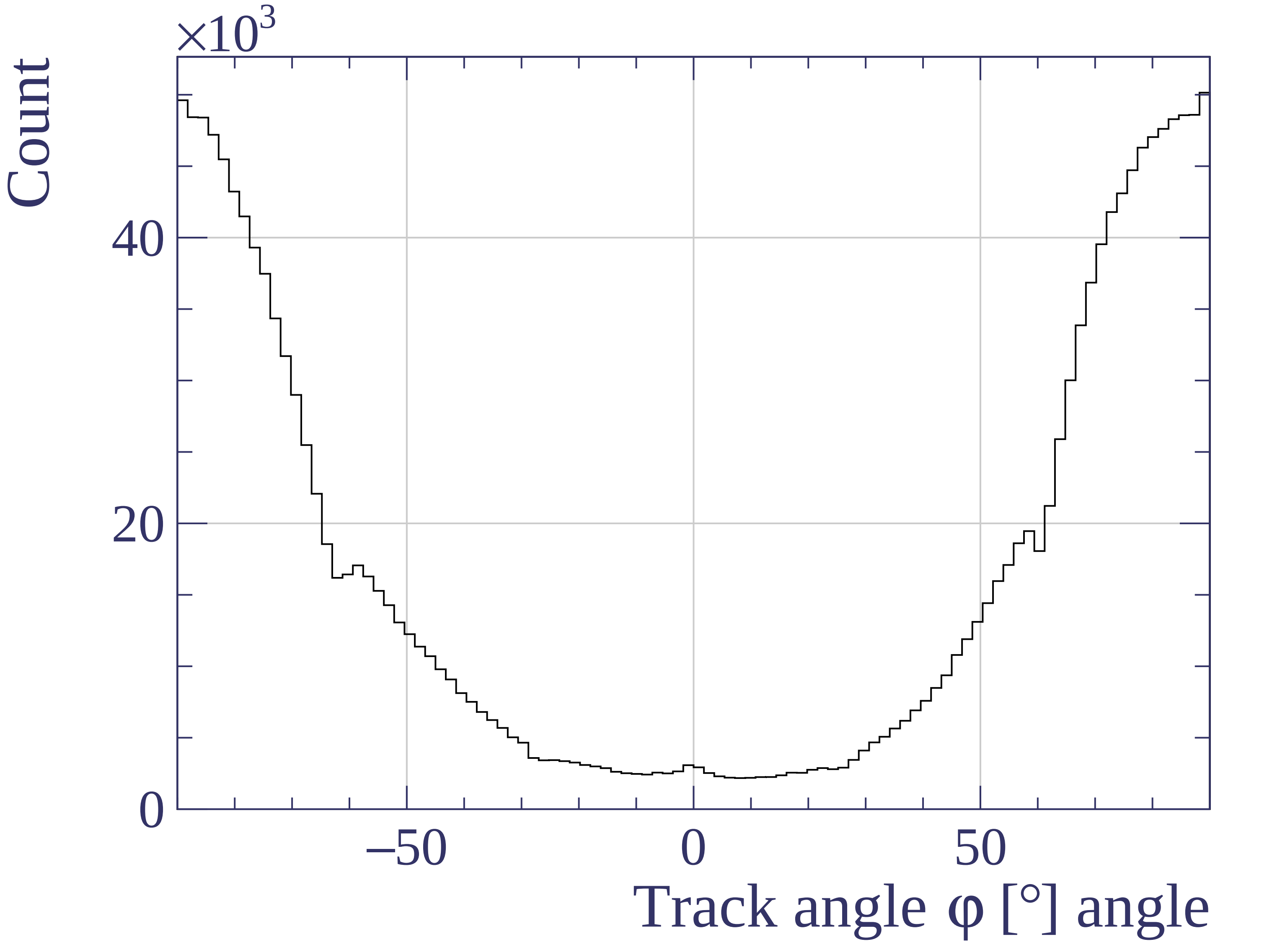

Track length [cm]
150
100
50
0
−50
0
50
Track angle φ [°]
3000
2000
1000
0

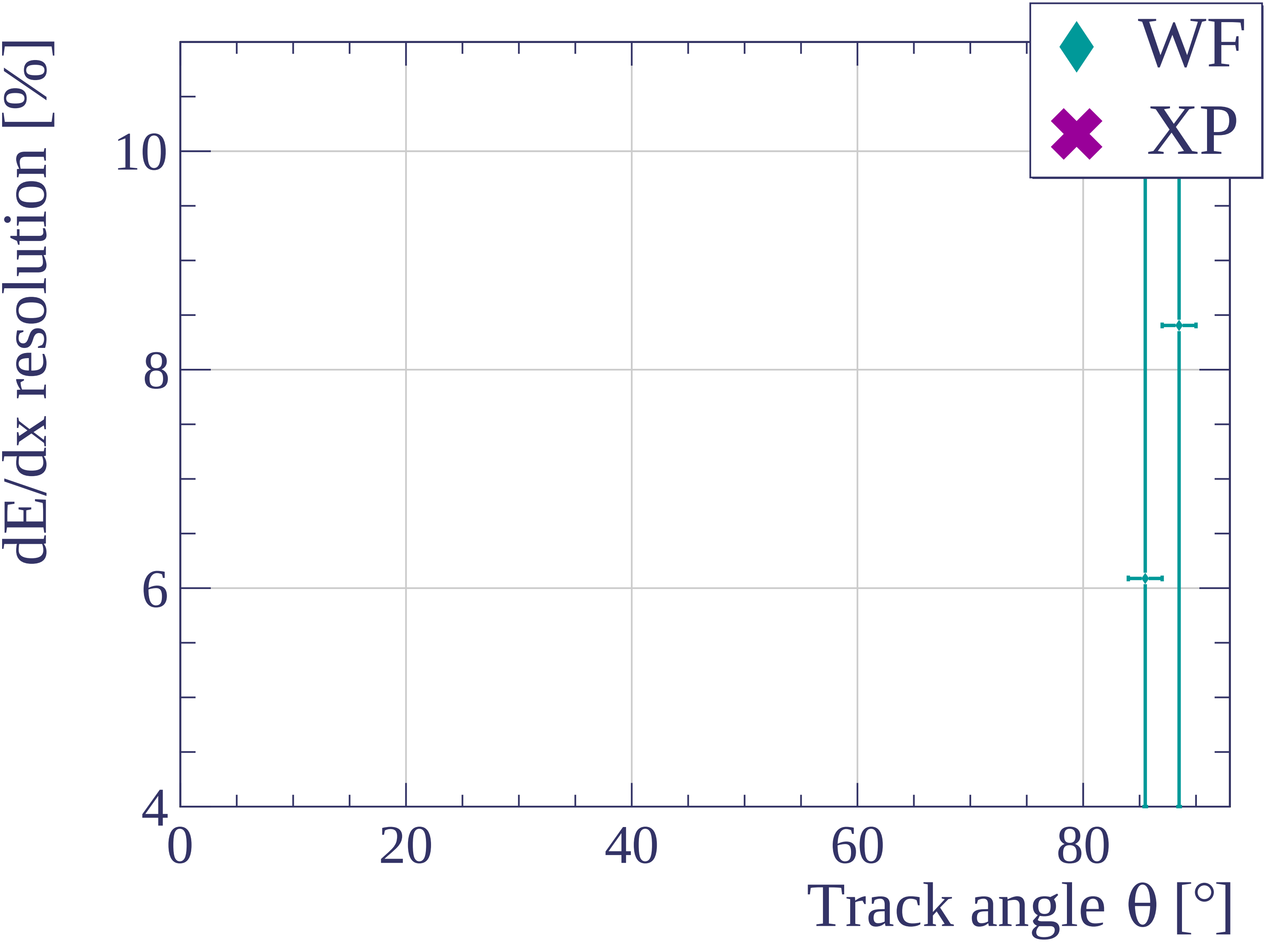


χ^2/NDF
50
40
30
20
10
0
−50
0
50
Track angle φ [°]
×10³
10
5
0

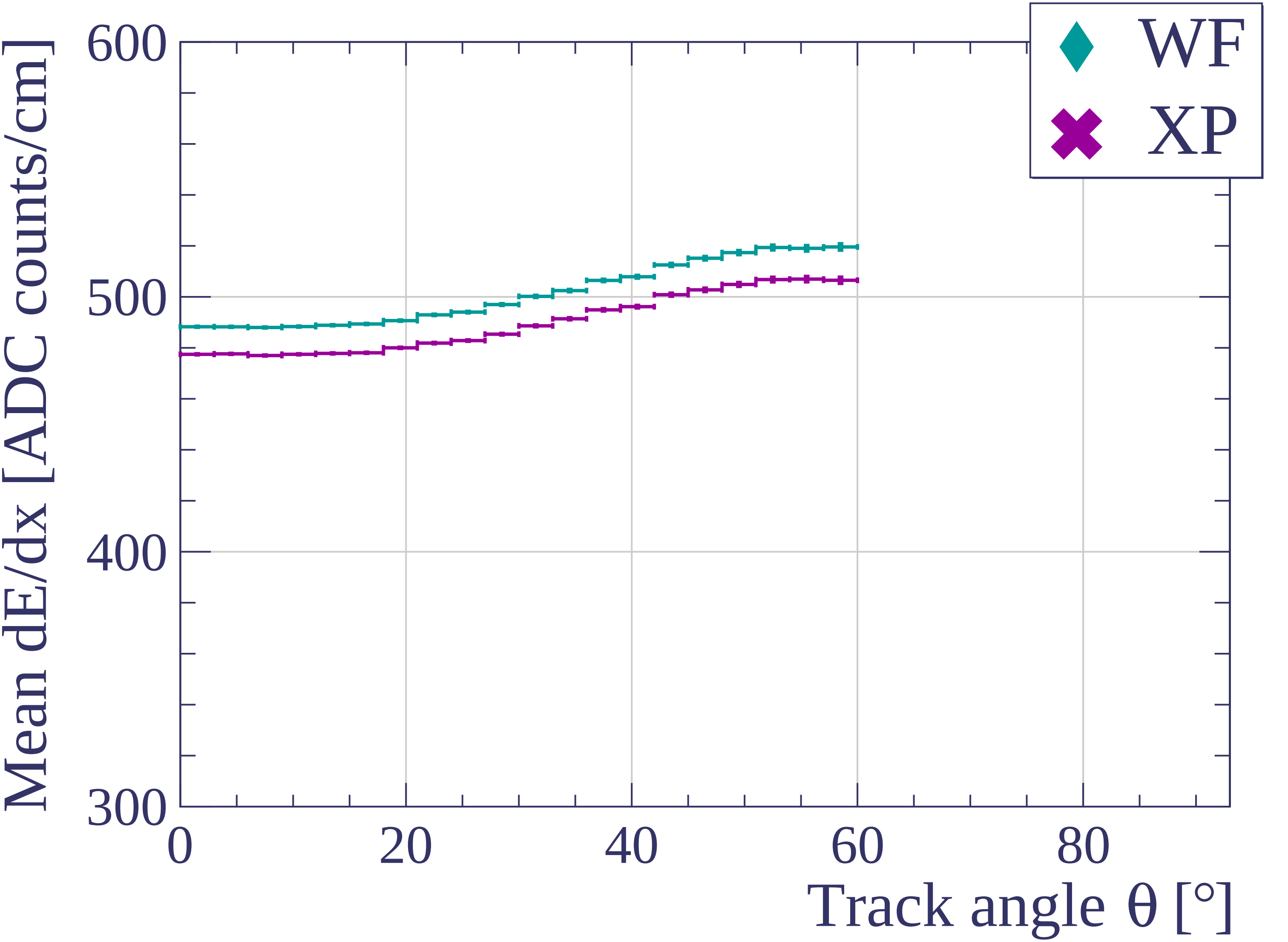


Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle φ [°]
400
300
200
100
0

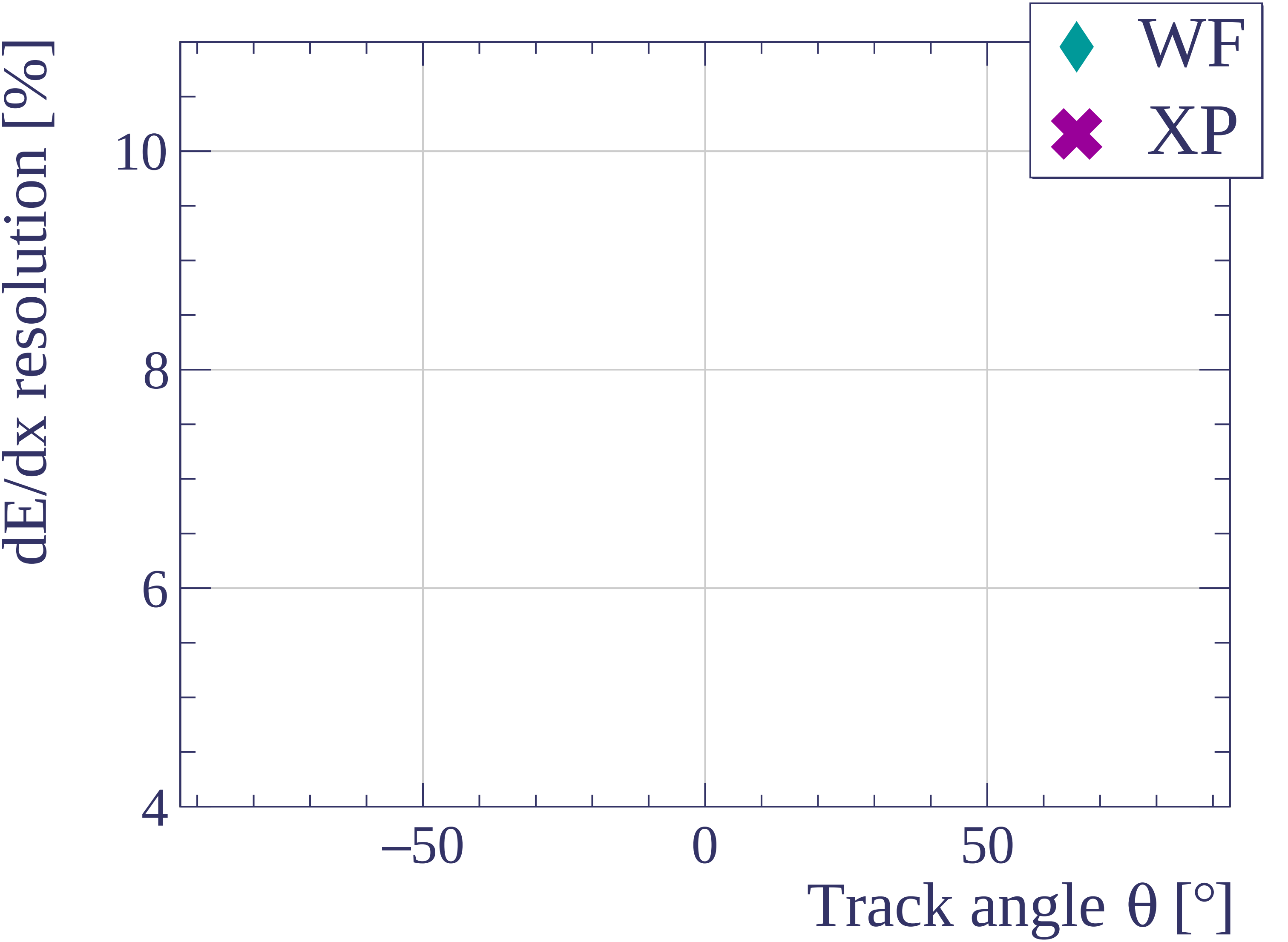

×10^3
Count
0
20
40
−50
0
50
Track angle φ [°] angle

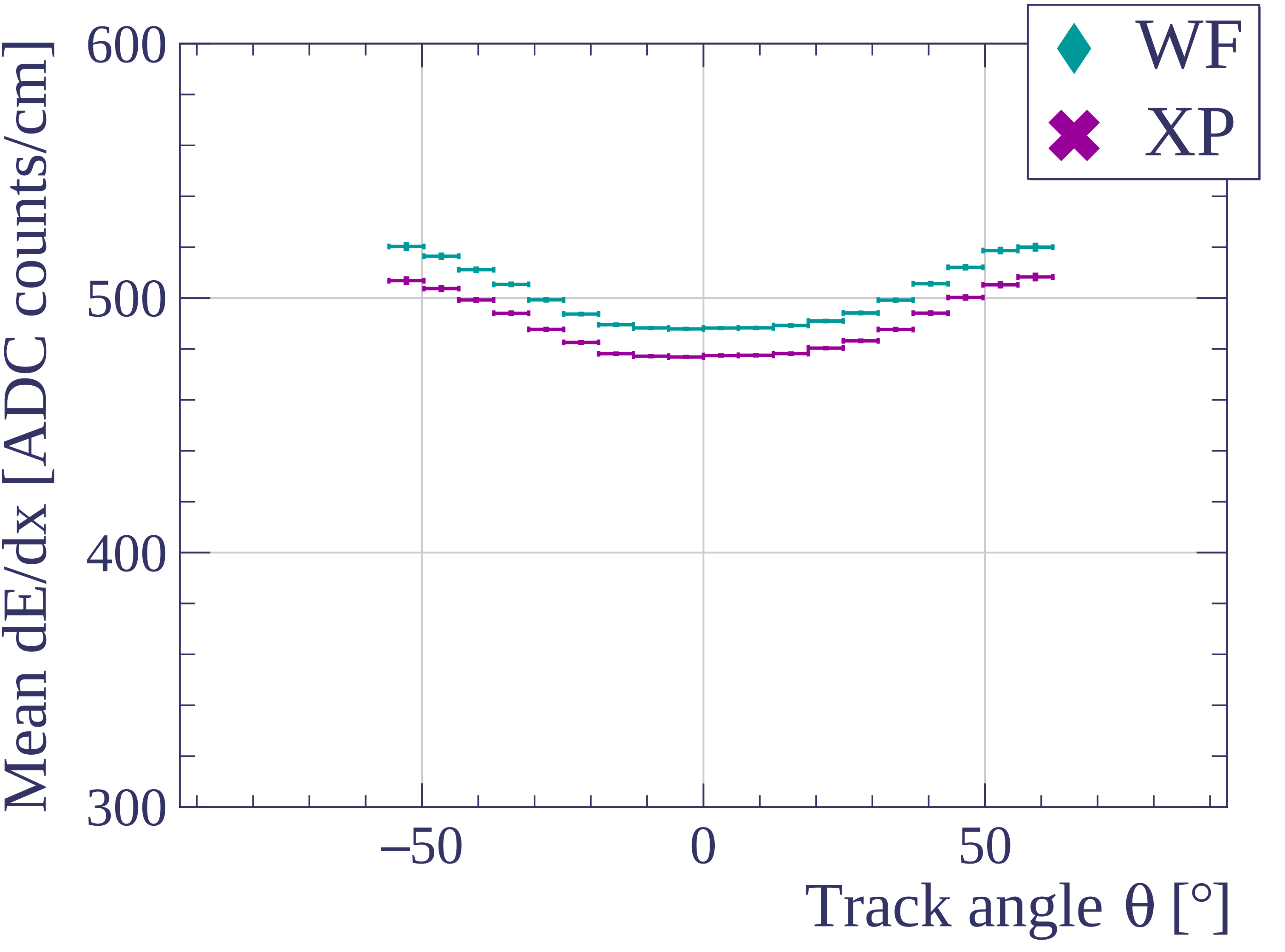


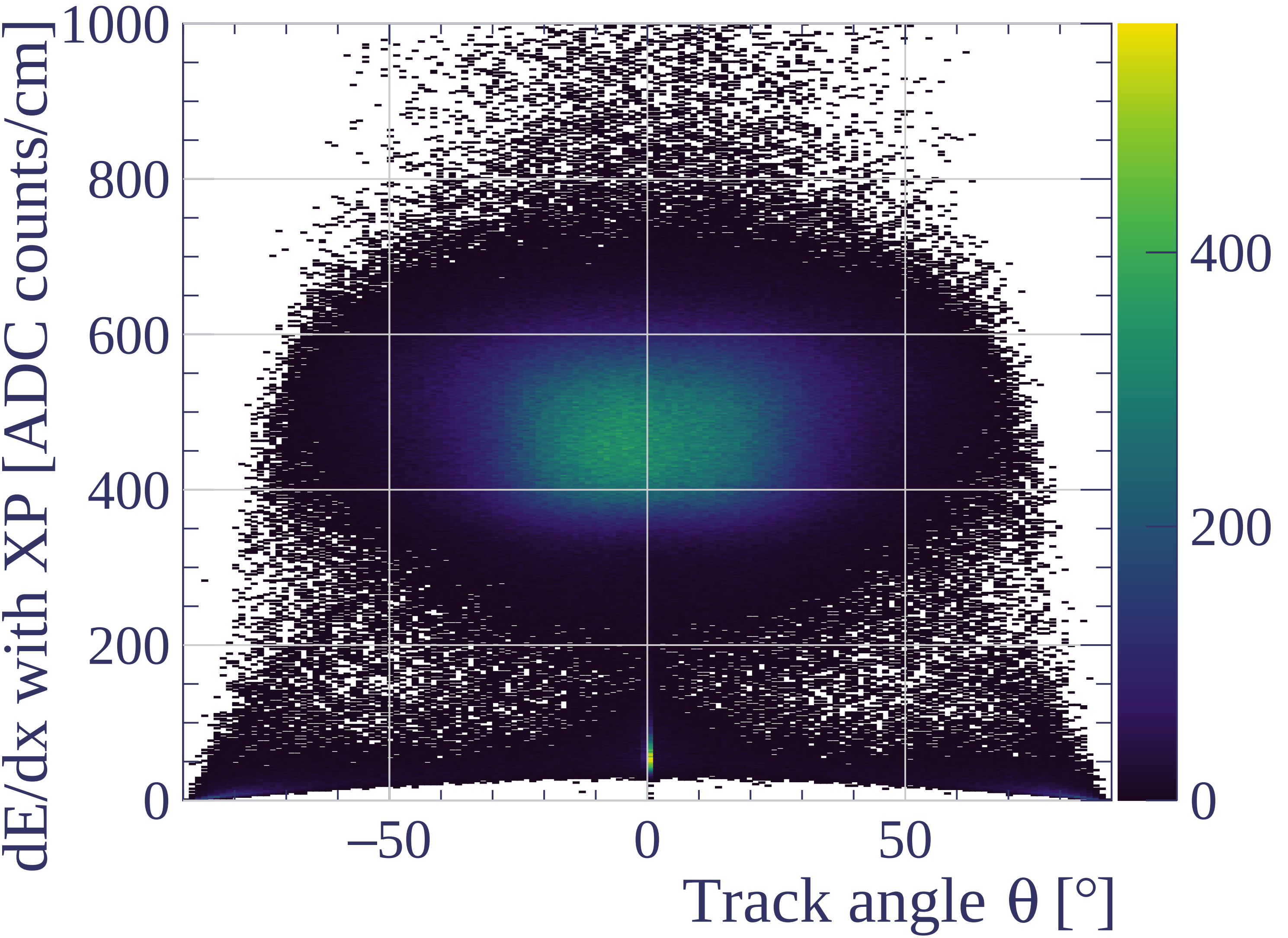
WF
XP
dE/dx resolution [%]
10
8
6
4
0
20
40
60
80
Track angle θ [°]



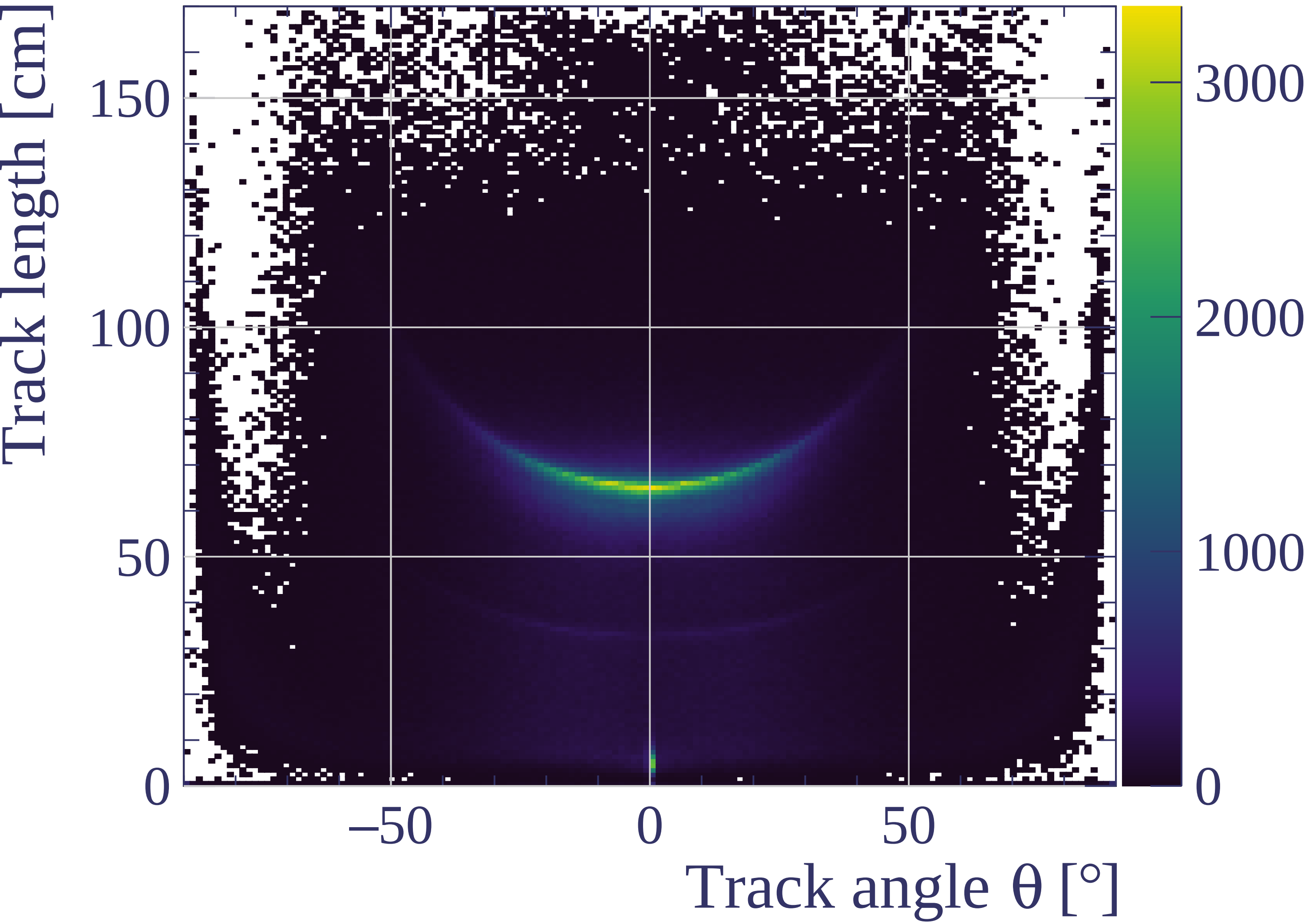
WF
XP
Mean dE/dx [ADC counts/cm]
Track angle θ [°]
300
400
500
600
0
20
40
60
80


WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle θ [°]

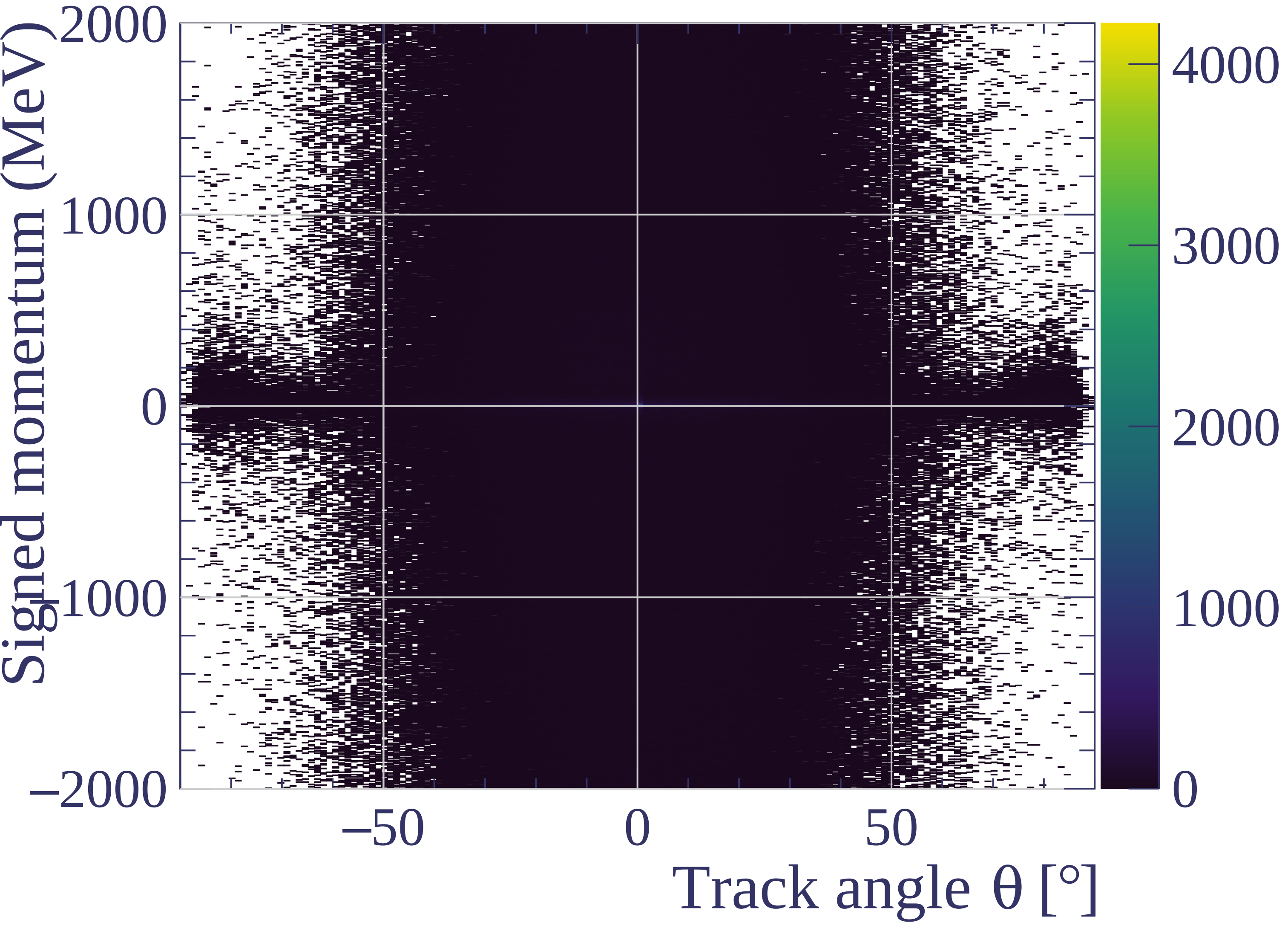

WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
−50
0
50
Track angle θ [°]

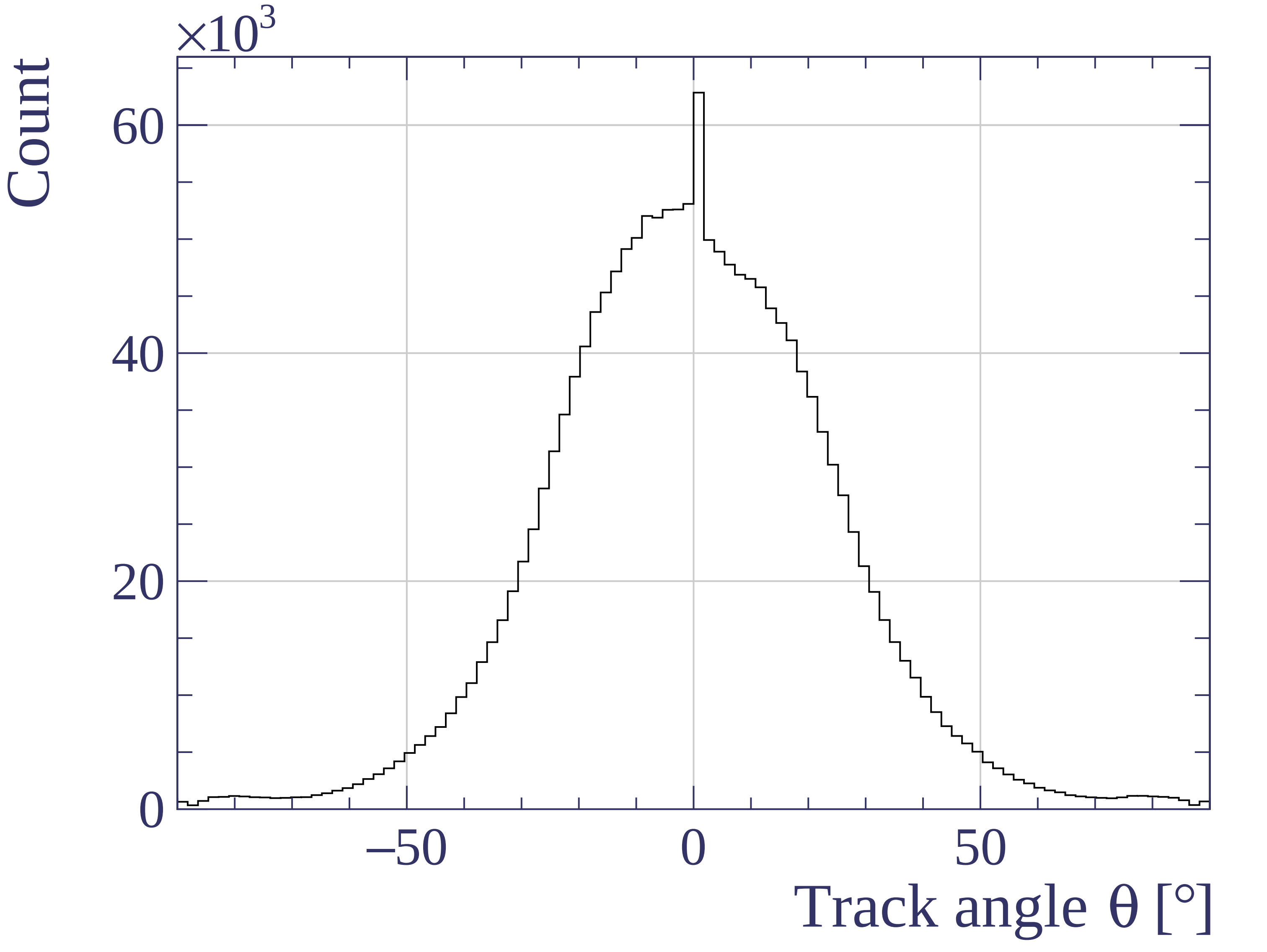


dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle θ [°]
400
200
0

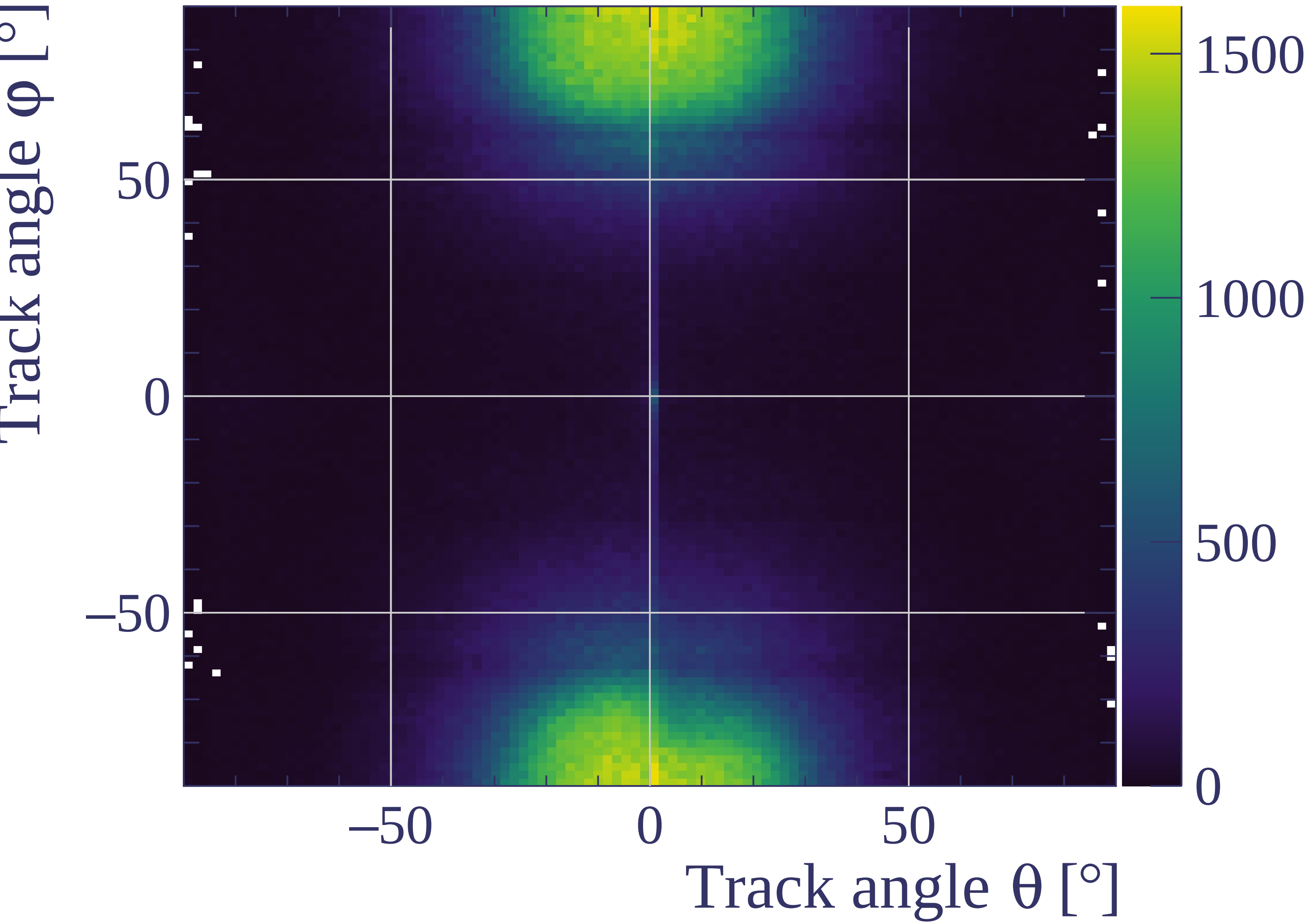


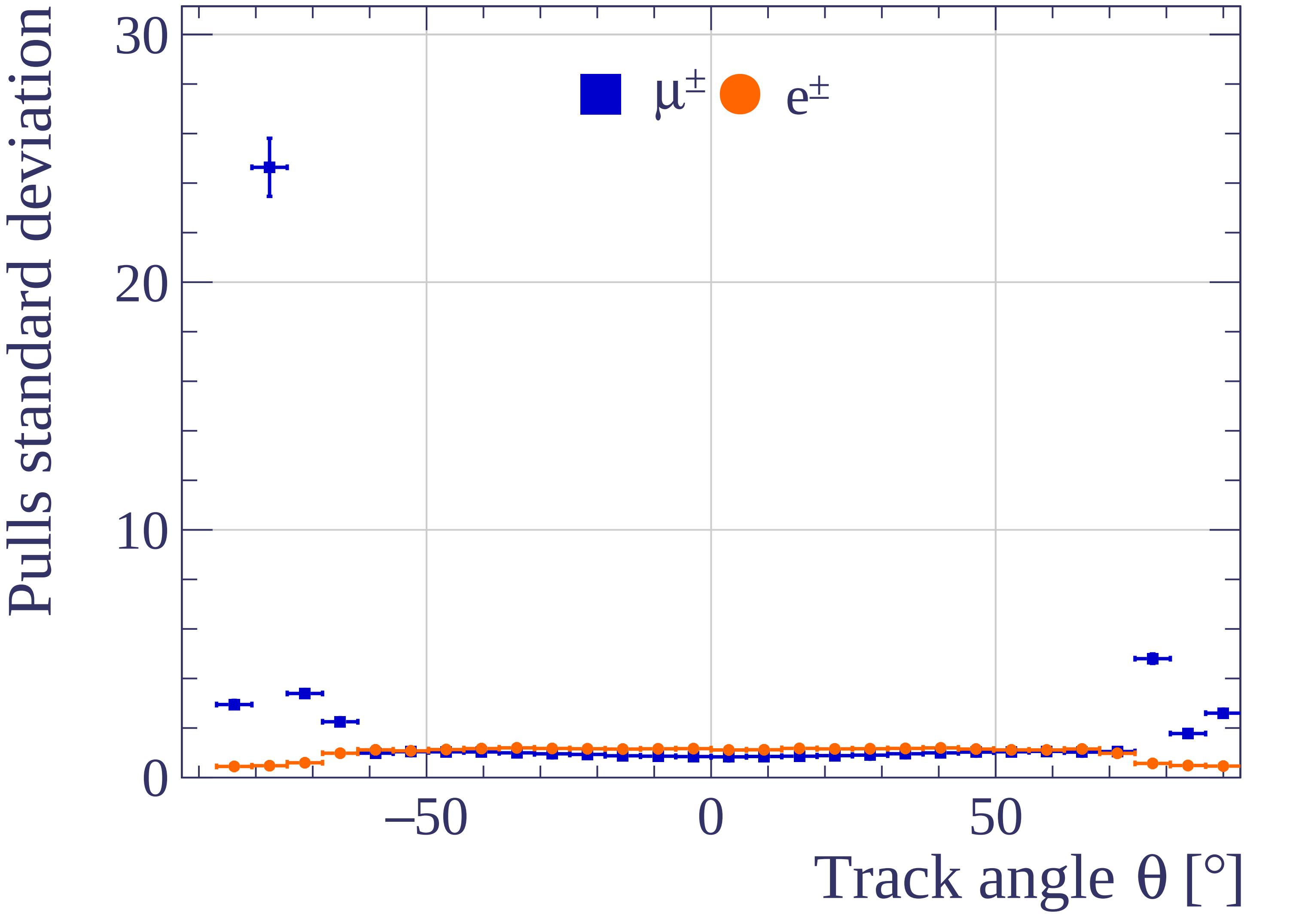
Track length [cm]
150
100
50
0
−50
0
50
Track angle θ [°]
3000
2000
1000
0


Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle θ [°]
4000
3000
2000
1000
0

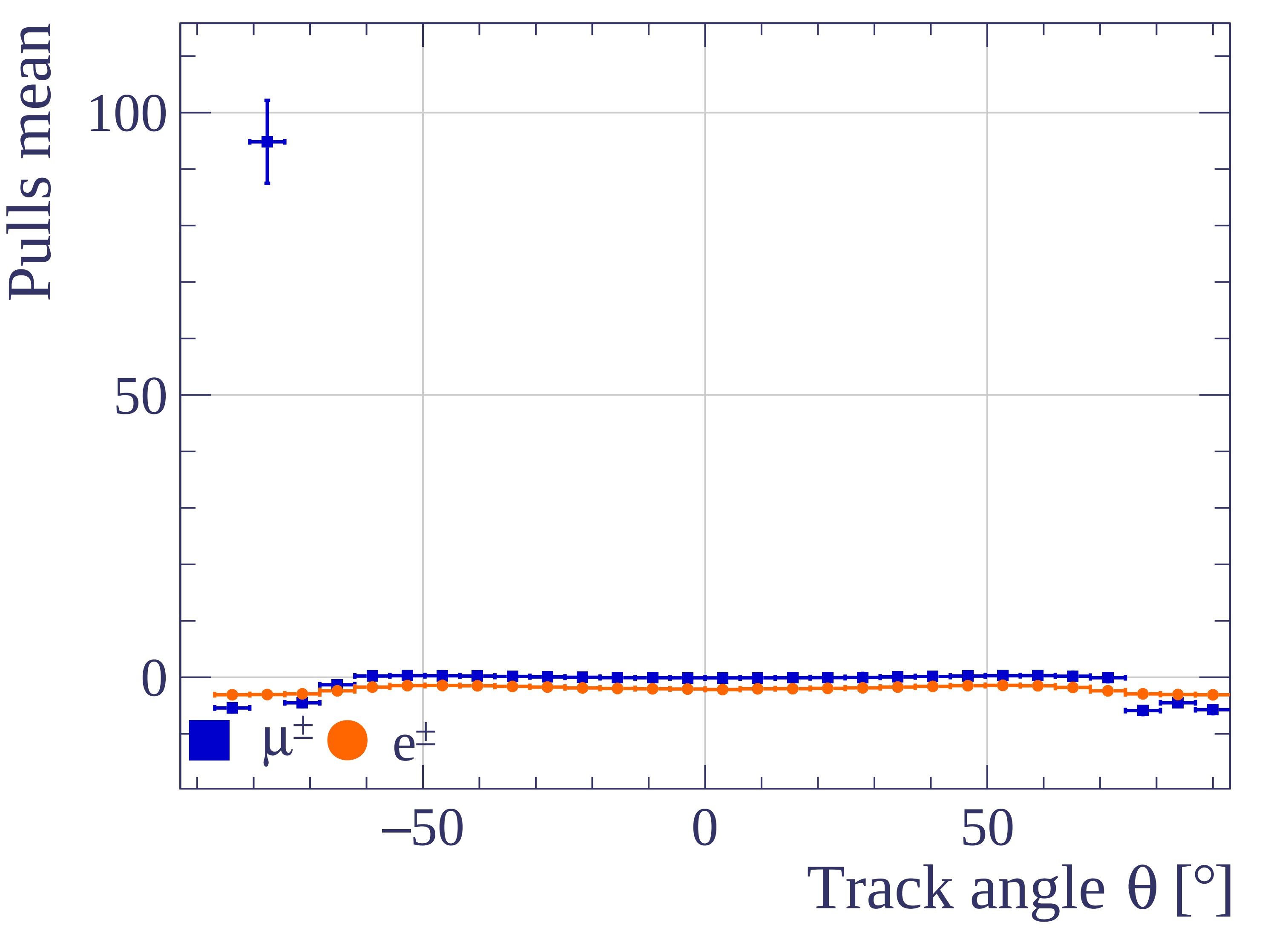


×10^3
Count
60
40
20
0
−50
0
50
Track angle θ [°]

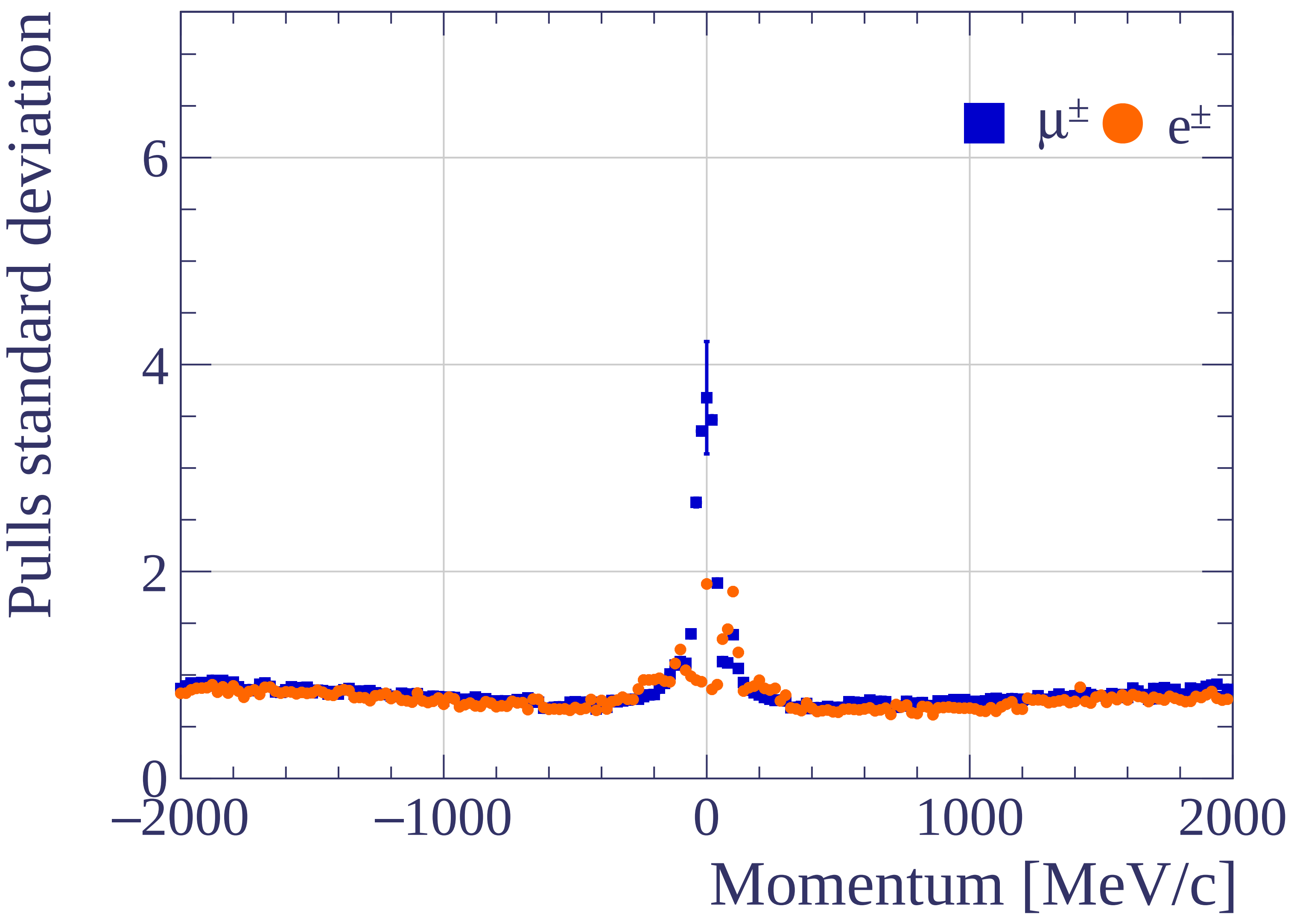

Track angle φ [°]
50
0
−50
−50
0
50
Track angle θ [°]
1500
1000
500
0



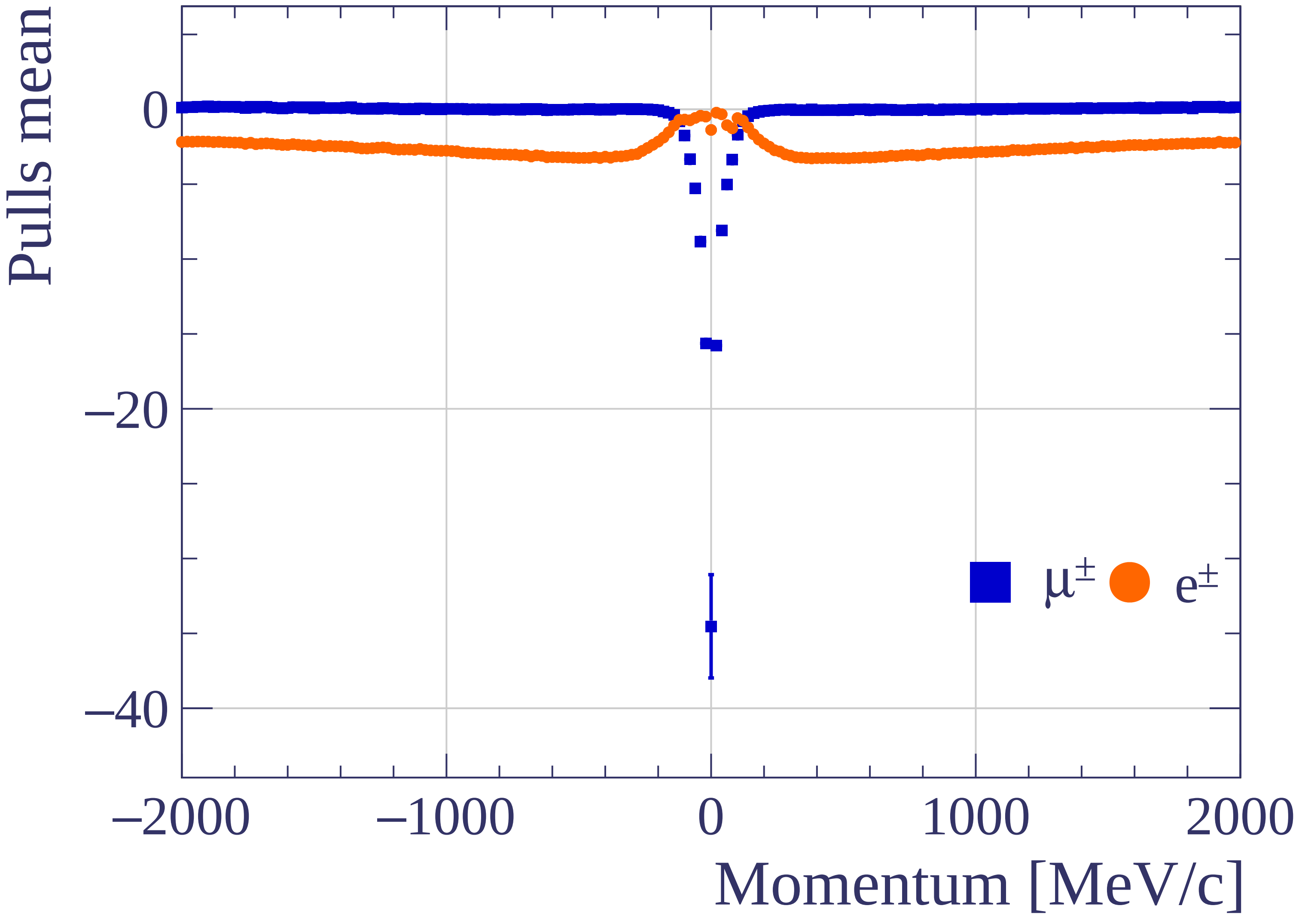
Pulls standard deviation
30
20
10
0
−50
0
50
Track angle θ [°]
μ±
e±

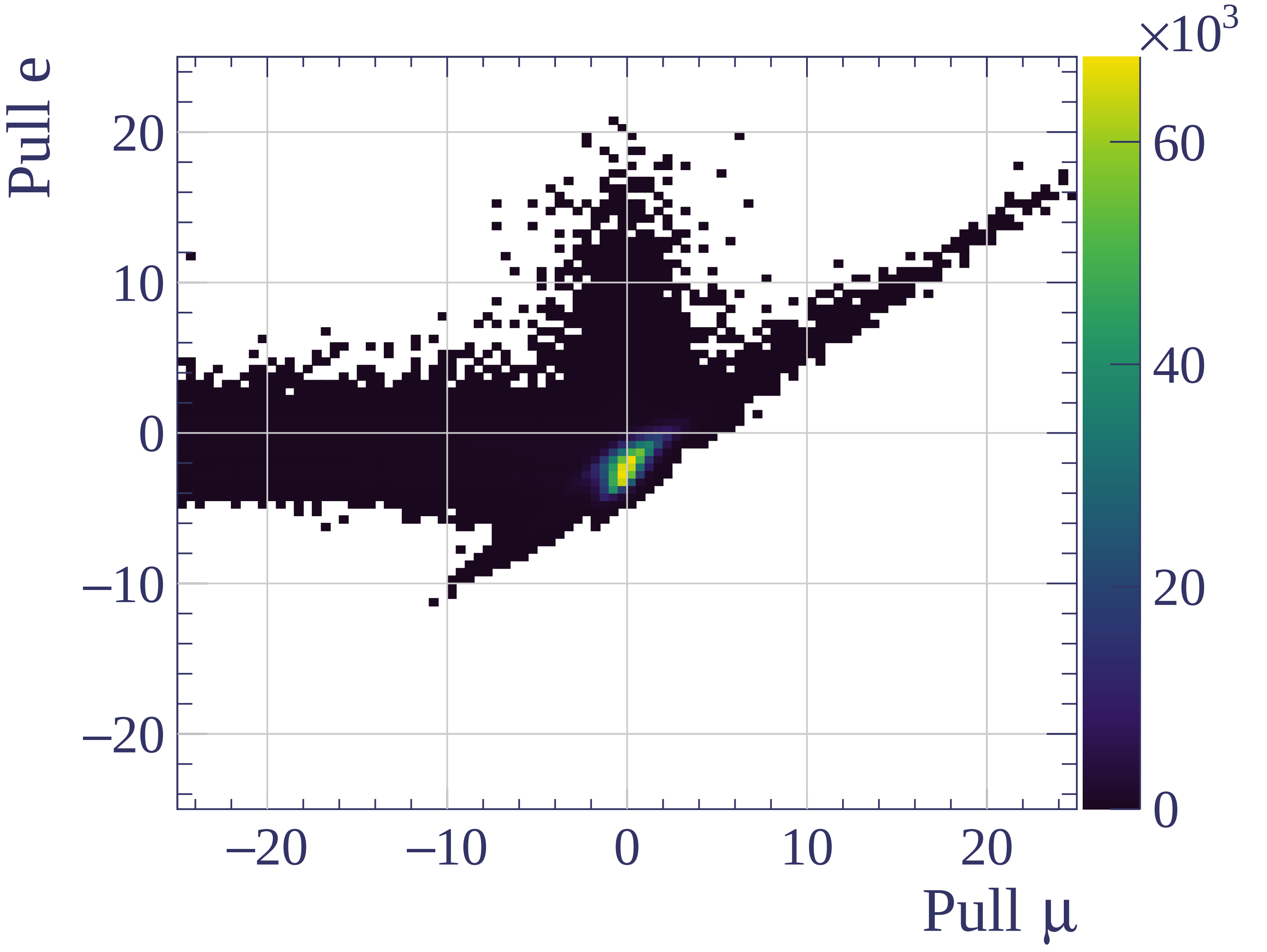


Pulls mean
100
50
0
μ±
e±
−50
0
50
Track angle θ [°]

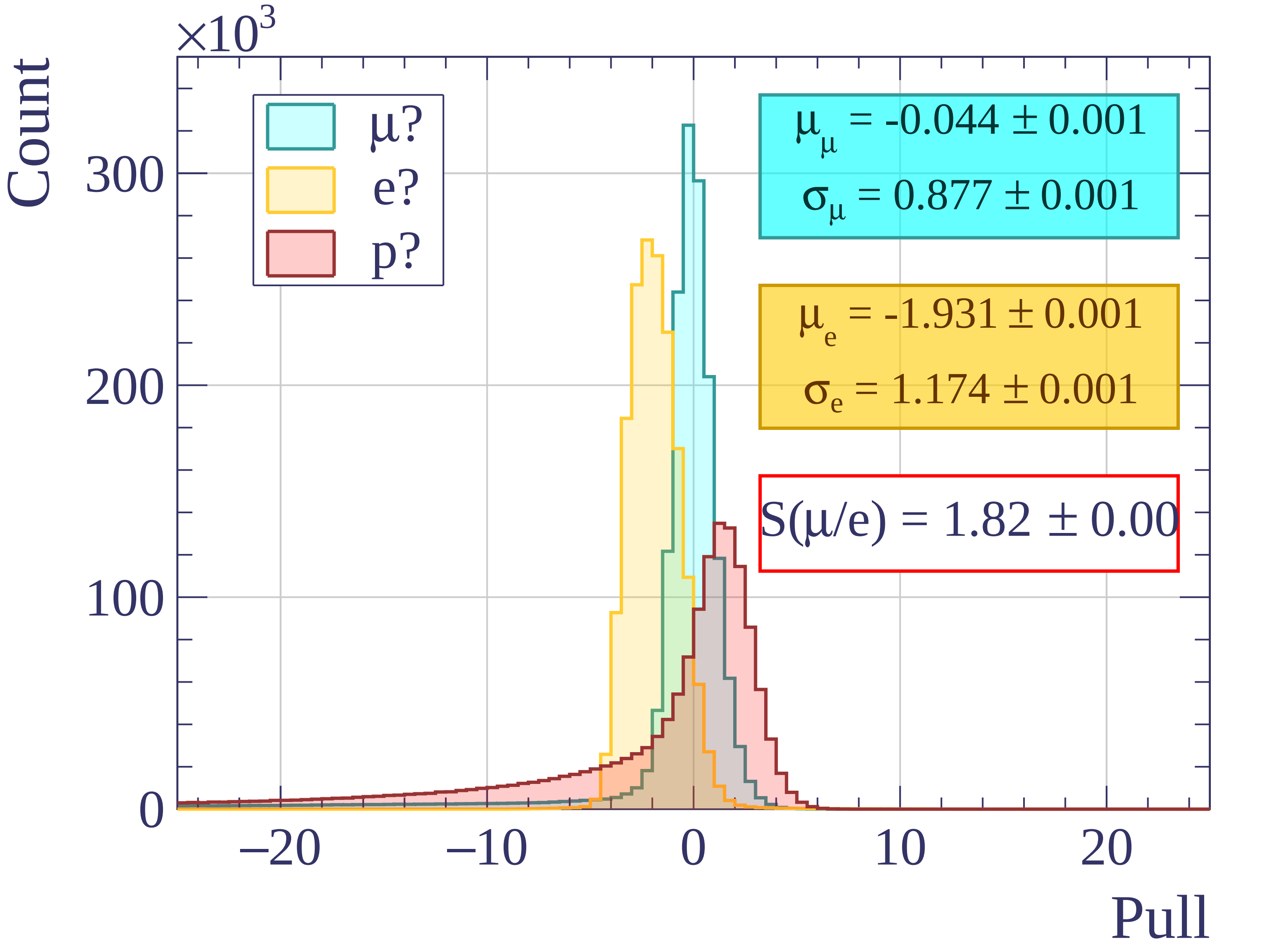


Pulls standard deviation
Momentum [MeV/c]
0
2
4
6
−2000
−1000
0
1000
2000
$\mu^{\pm}$
$e^{\pm}$

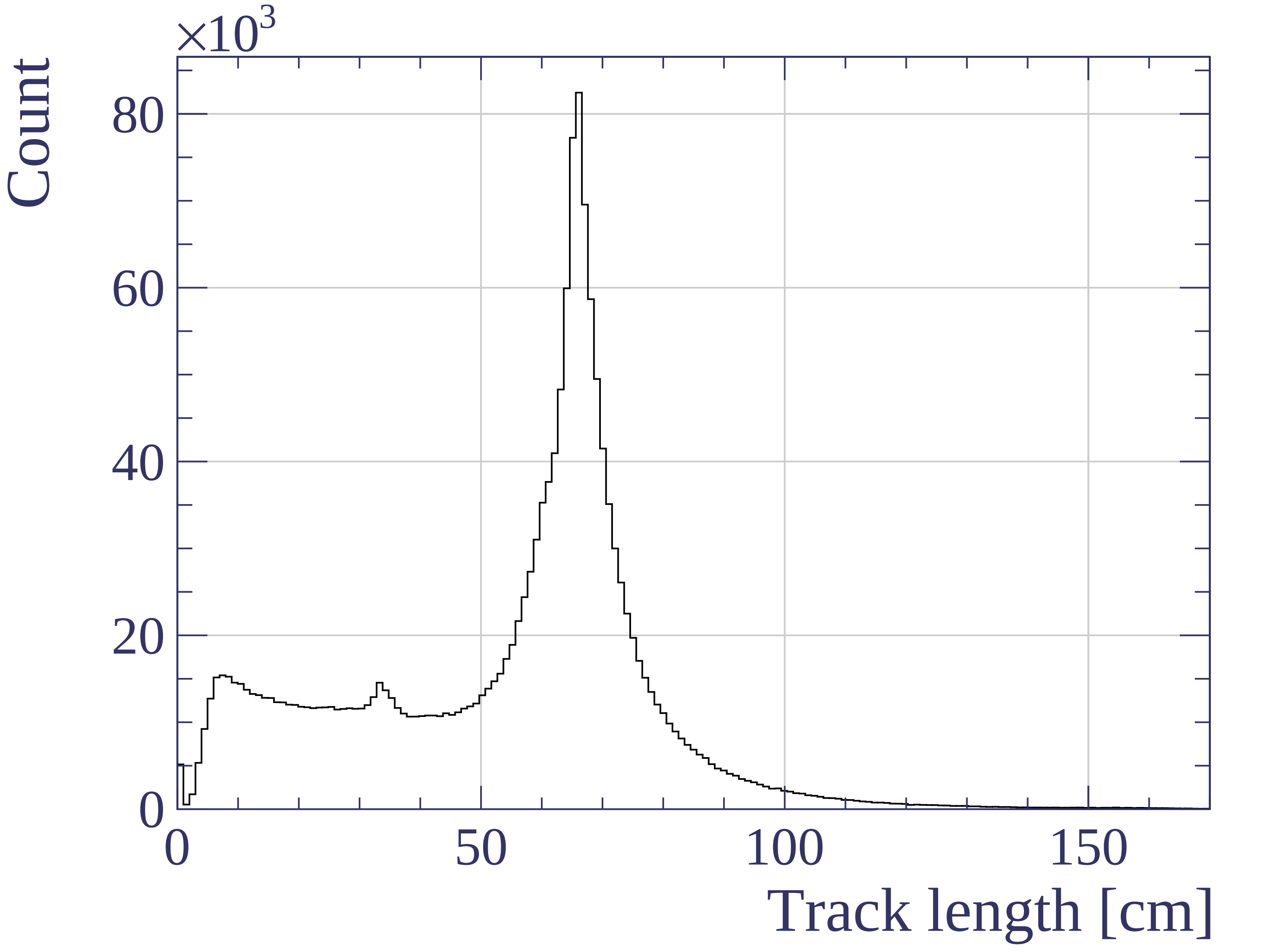

Pulls mean
0
−20
−40
−2000
−1000
0
1000
2000
Momentum [MeV/c]
$\mu^{\pm}$
$e^{\pm}$



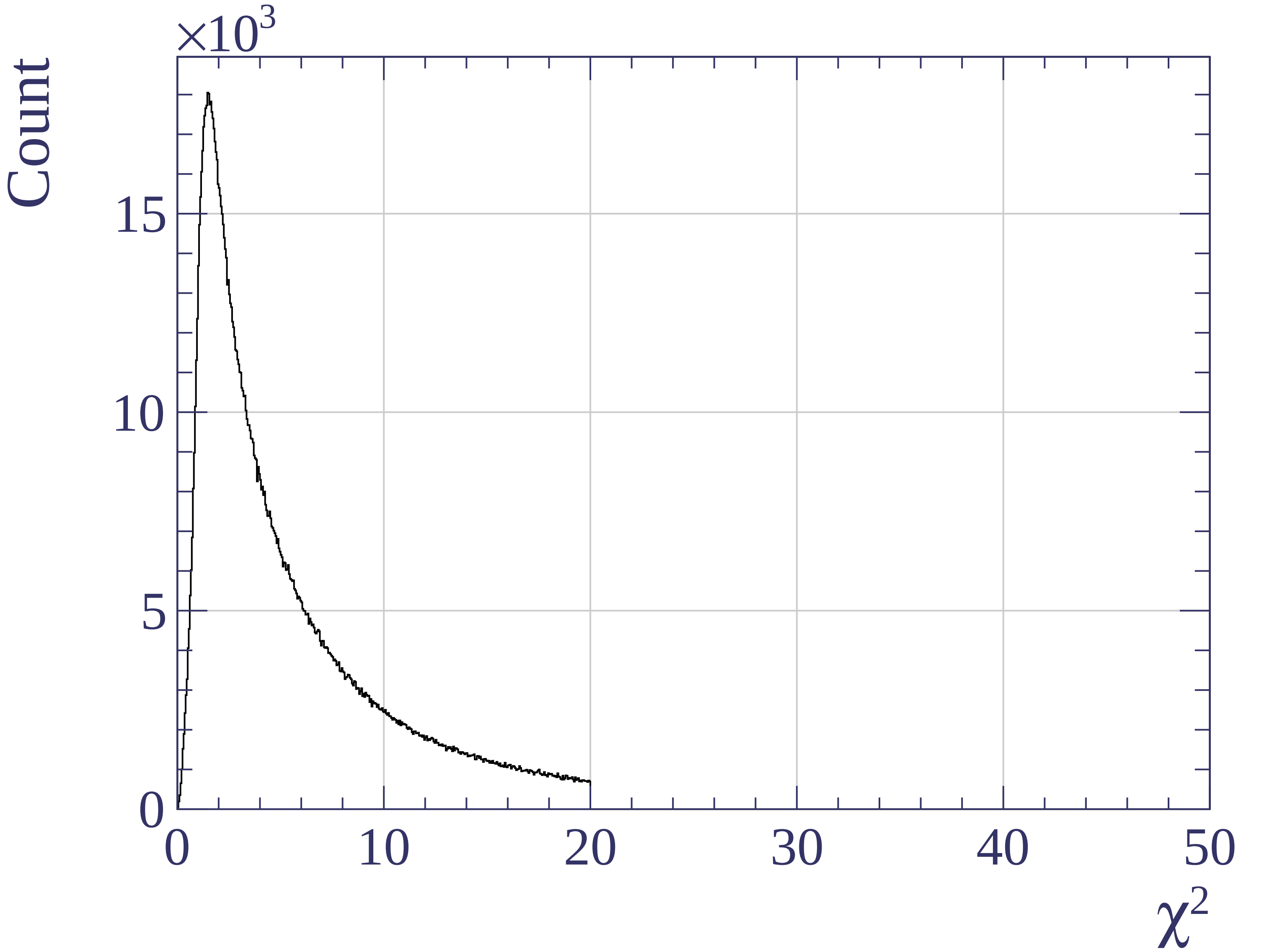
Pull e
Pull μ
×10³
20
10
0
−10
−20
−20
−10
0
10
20
60
40
20
0



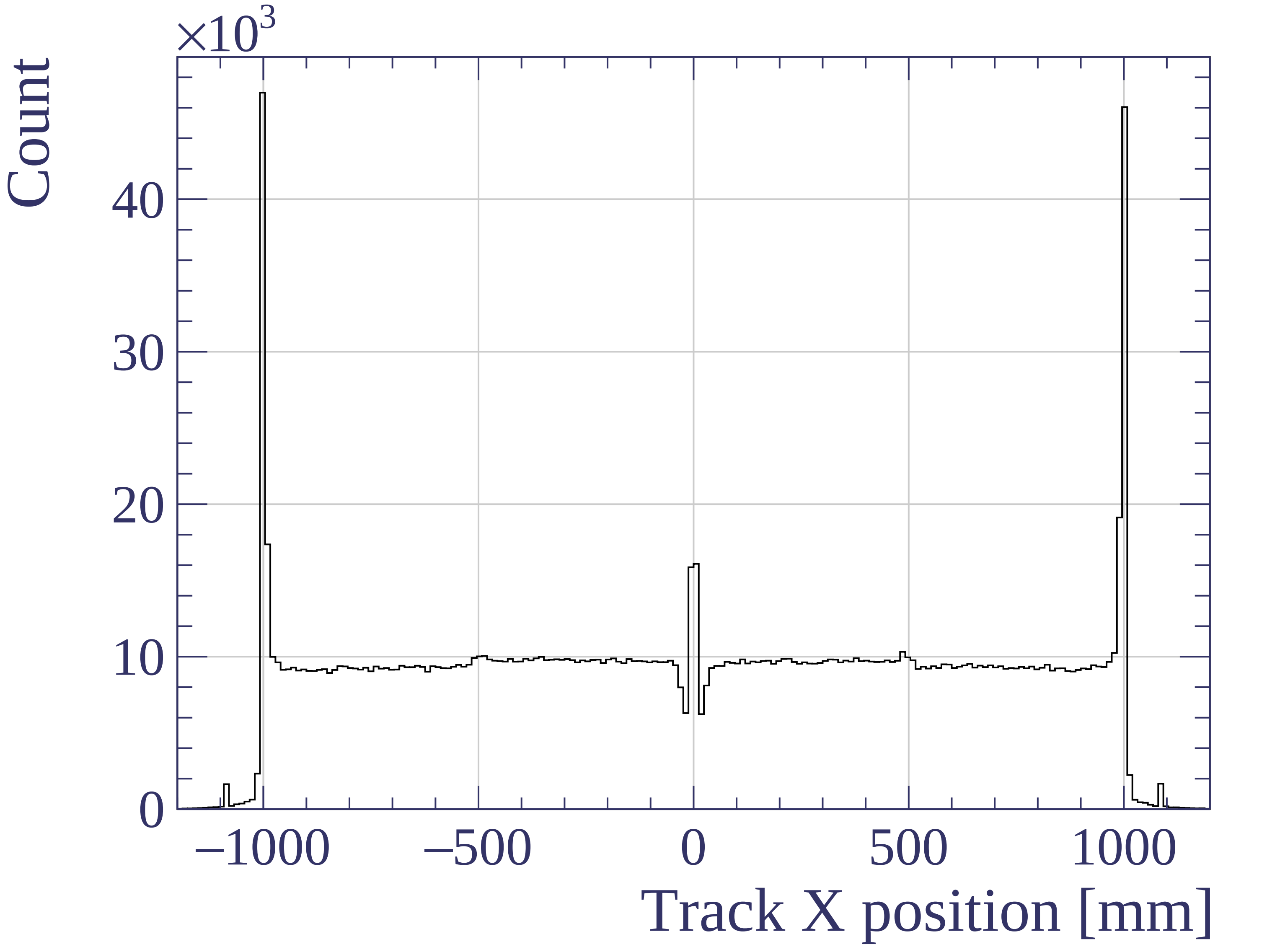
×10³
Count
300
200
100
0
μ?
e?
p?
μ_μ = -0.044 ± 0.001
σ_μ = 0.877 ± 0.001
μ_e = -1.931 ± 0.001
σ_e = 1.174 ± 0.001
S(μ/e) = 1.82 ± 0.00
−20
−10
0
10
20
Pull



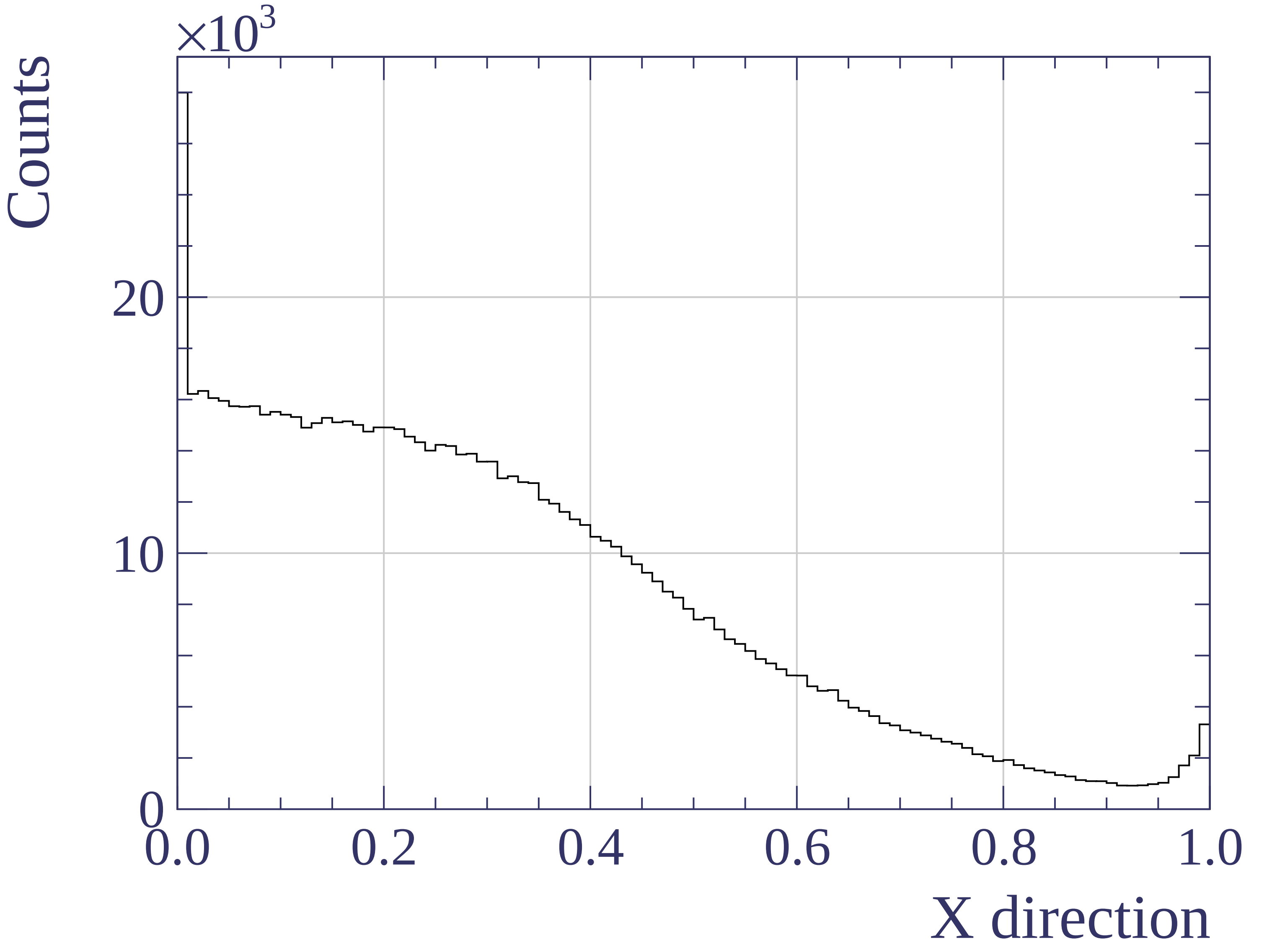
×10^3
Count
80
60
40
20
0
0
50
100
150
Track length [cm]



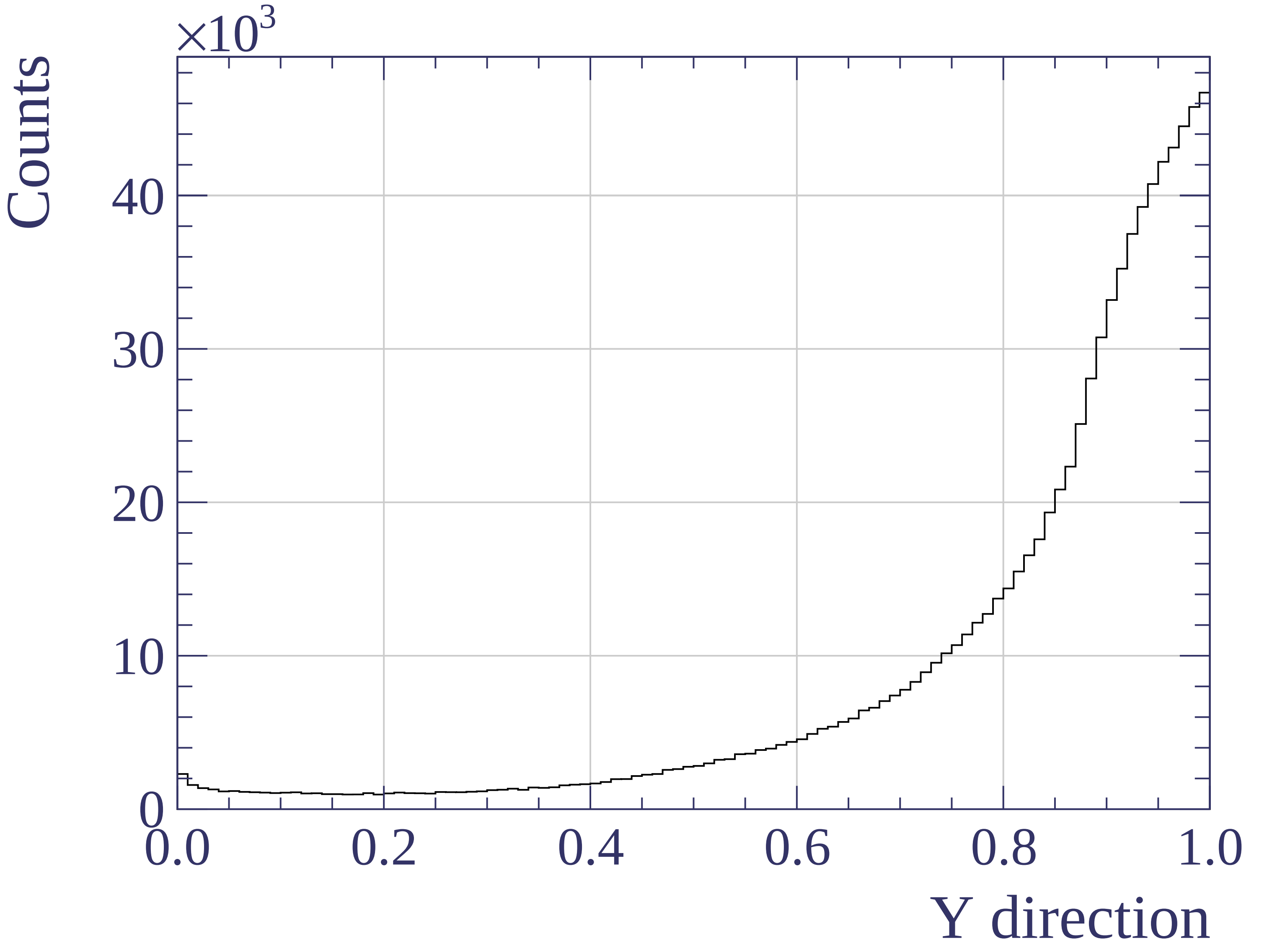
×10^3
Count
0
5
10
15
0
10
20
30
40
50
χ^2



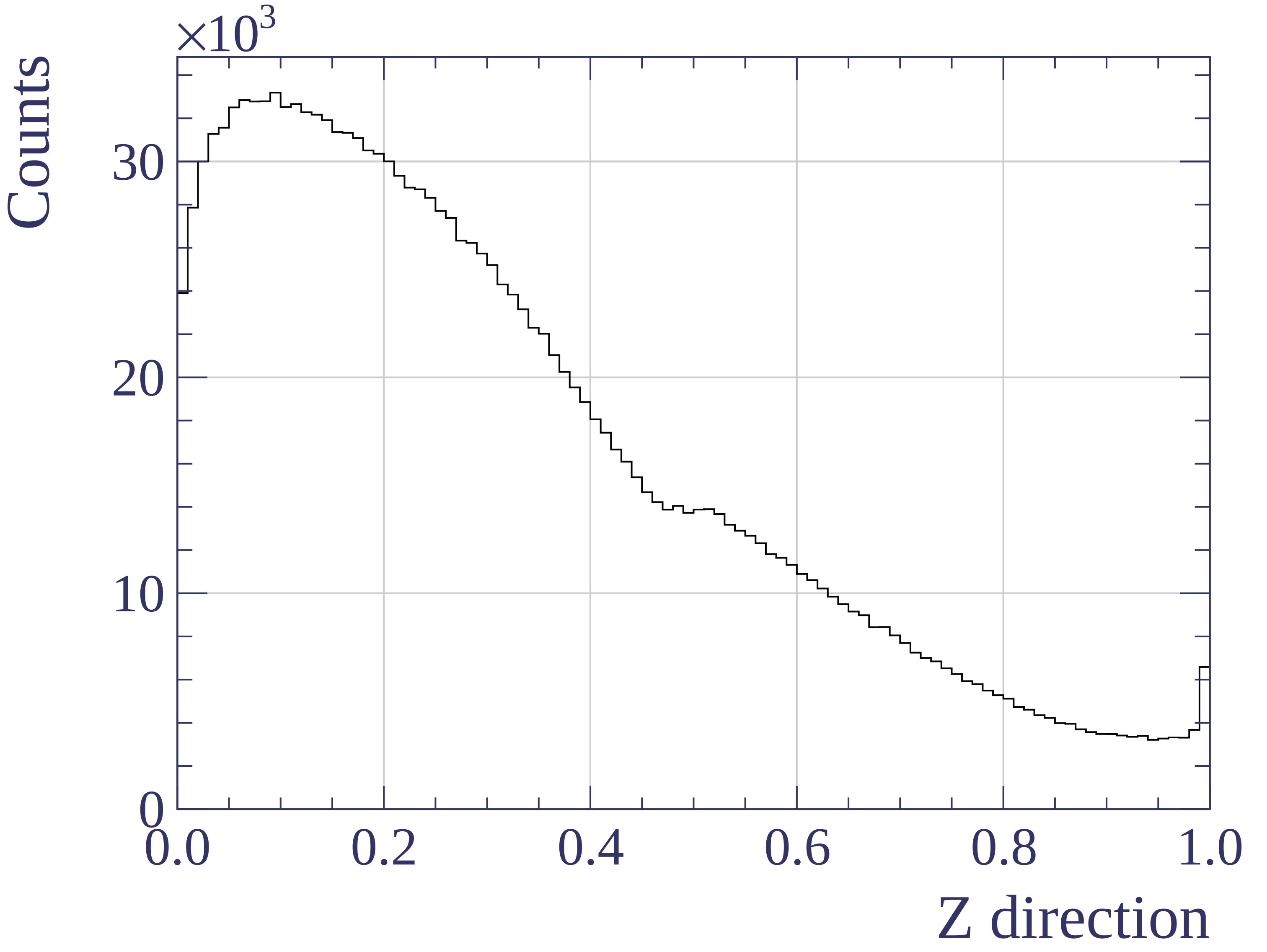
$\times 10^3$
Count
0
10
20
30
40
−1000
−500
0
500
1000
Track X position [mm]



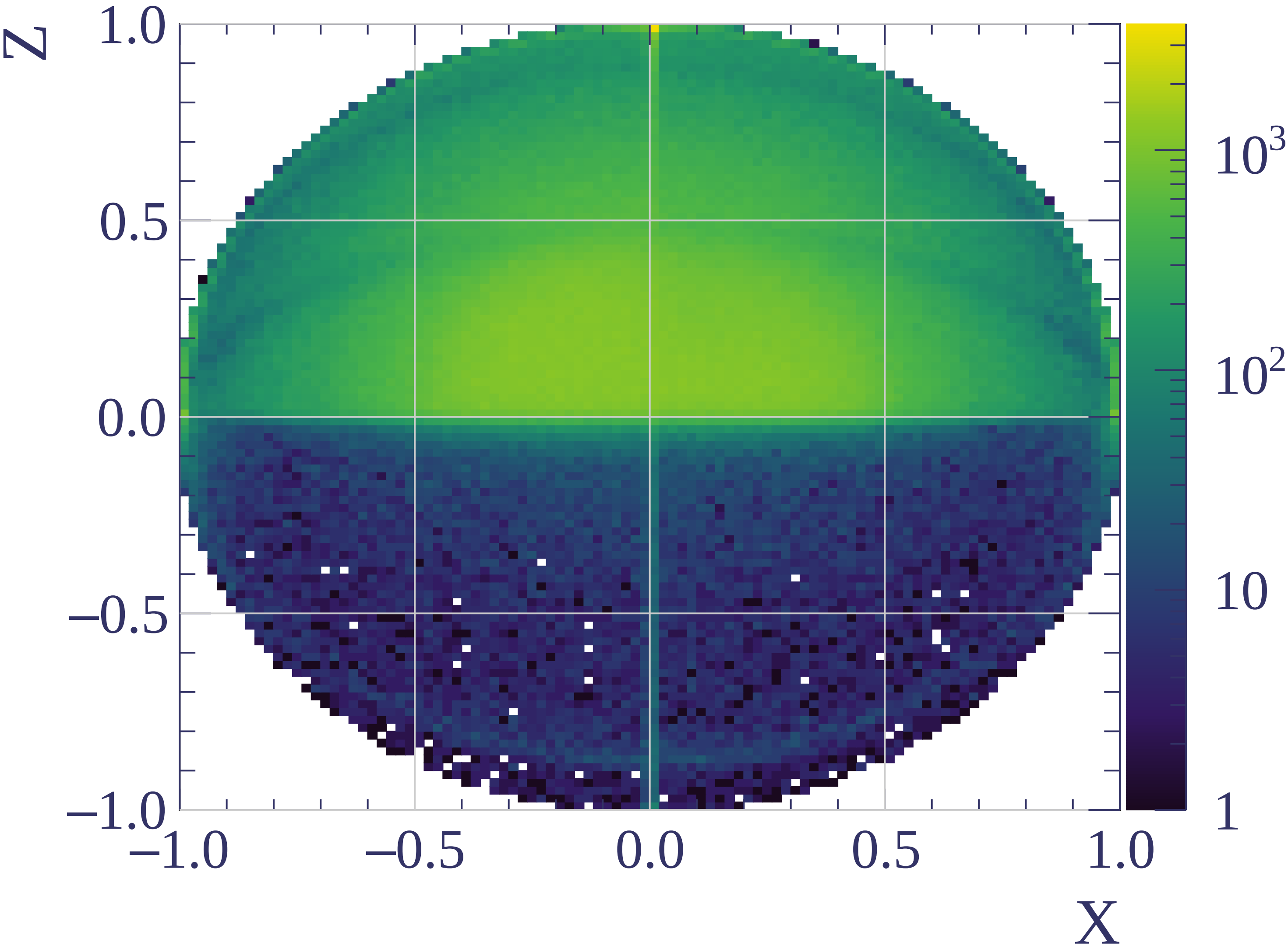
×10^3
Counts
20
10
0
0.0
0.2
0.4
0.6
0.8
1.0
X direction

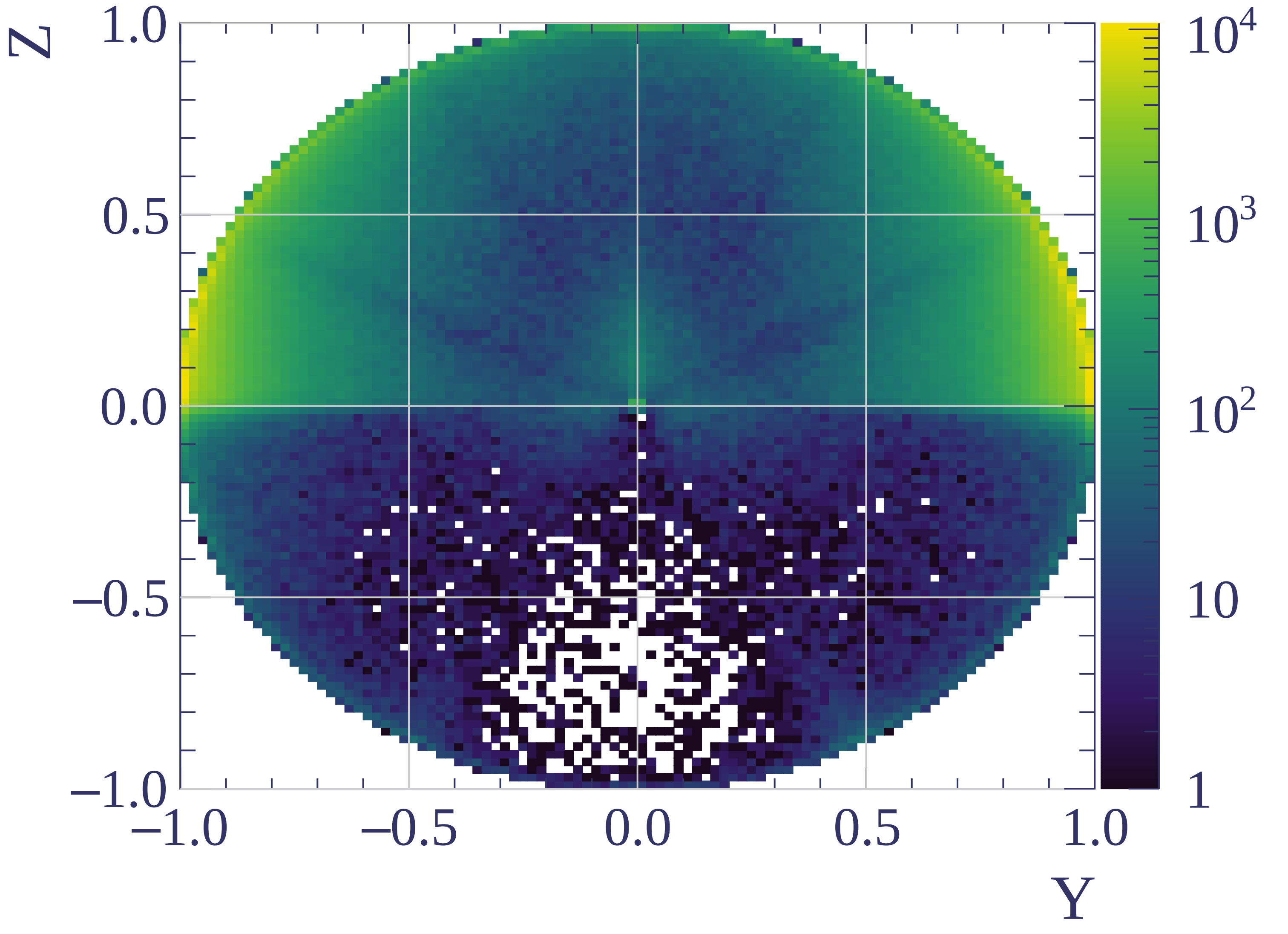


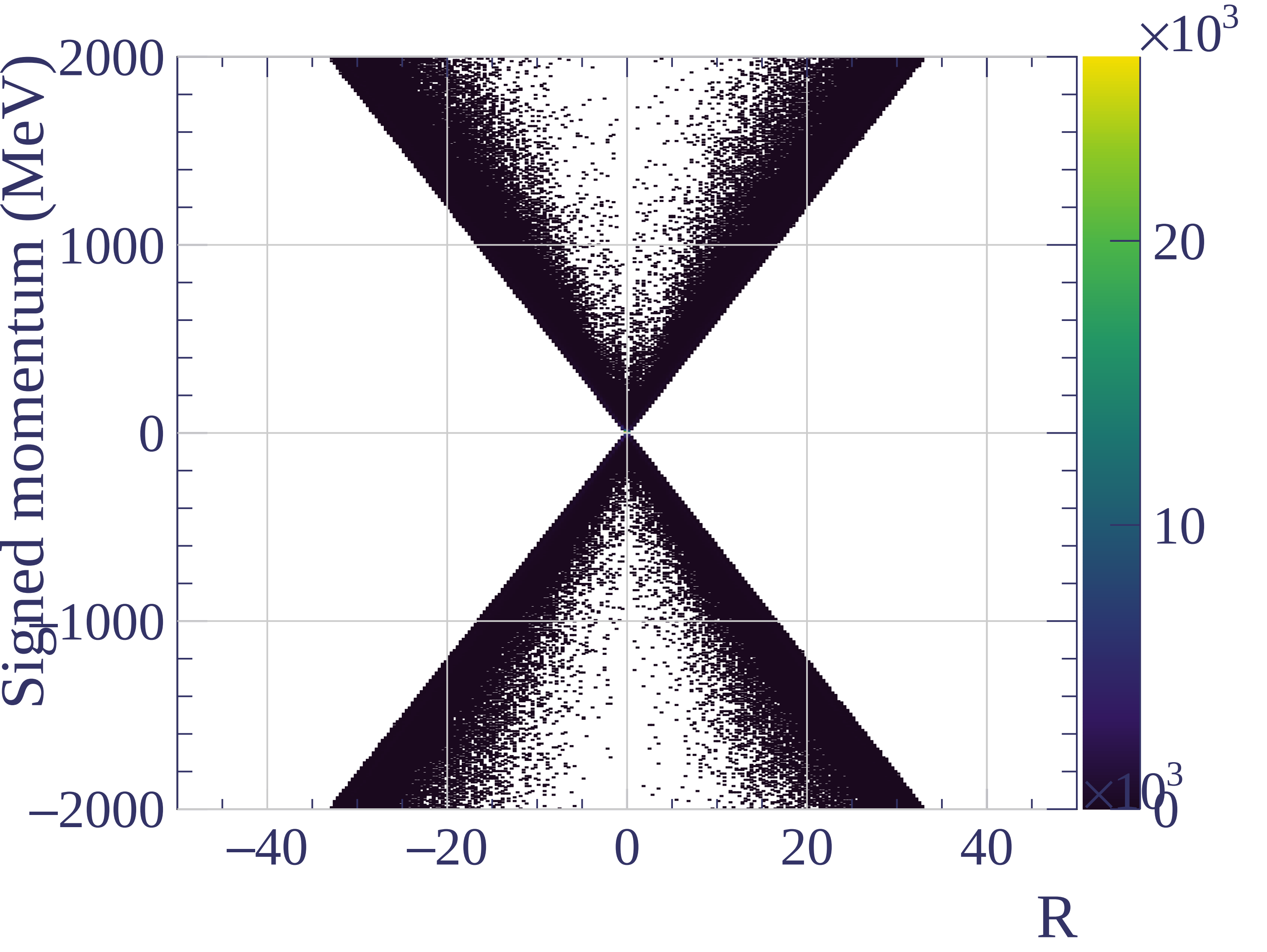
×10³
Counts
40
30
20
10
0
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



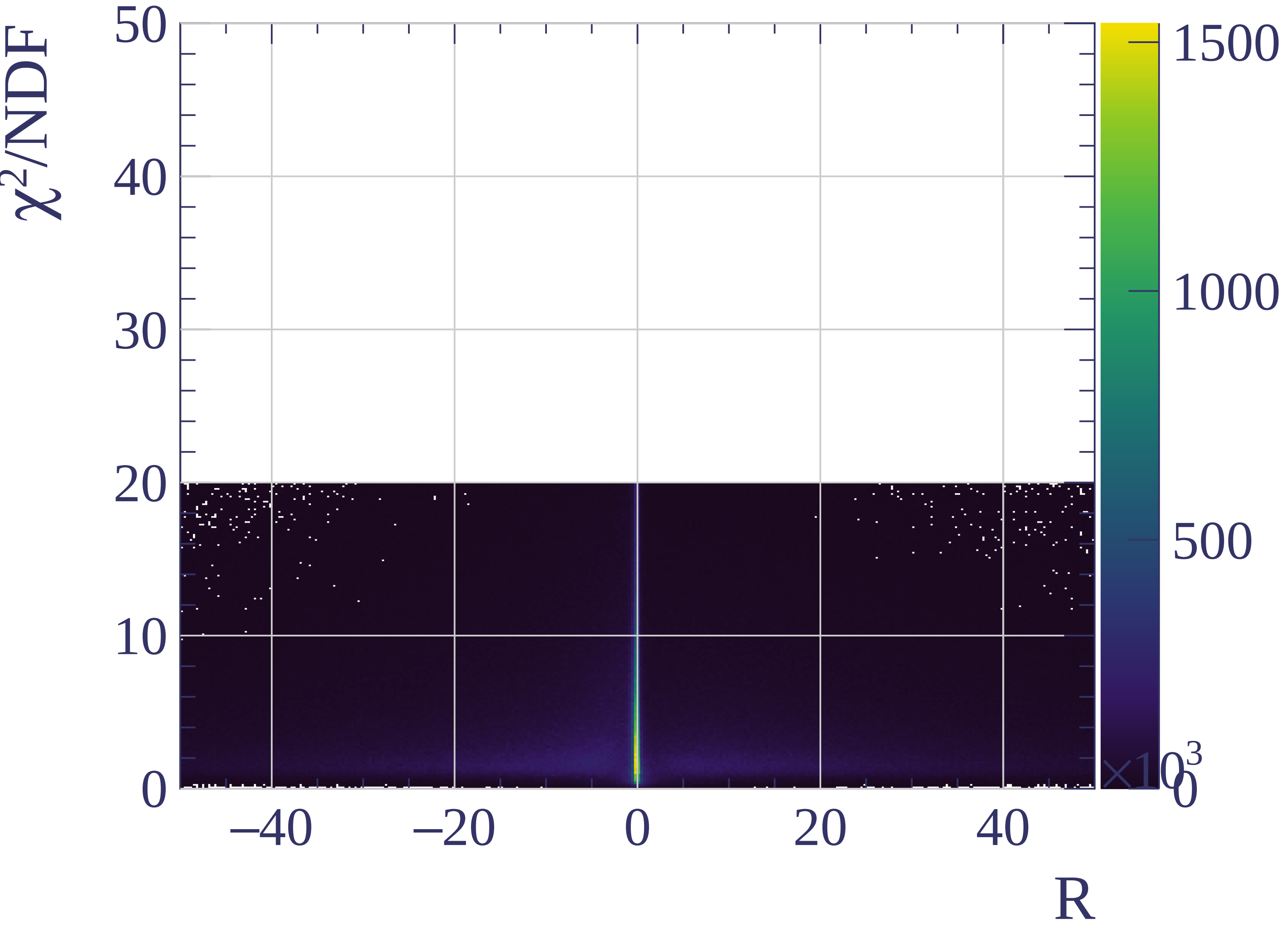
×10^3
Counts
30
20
10
0
0.0
0.2
0.4
0.6
0.8
1.0
Z direction


Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
X
10^3
10^2
10
1

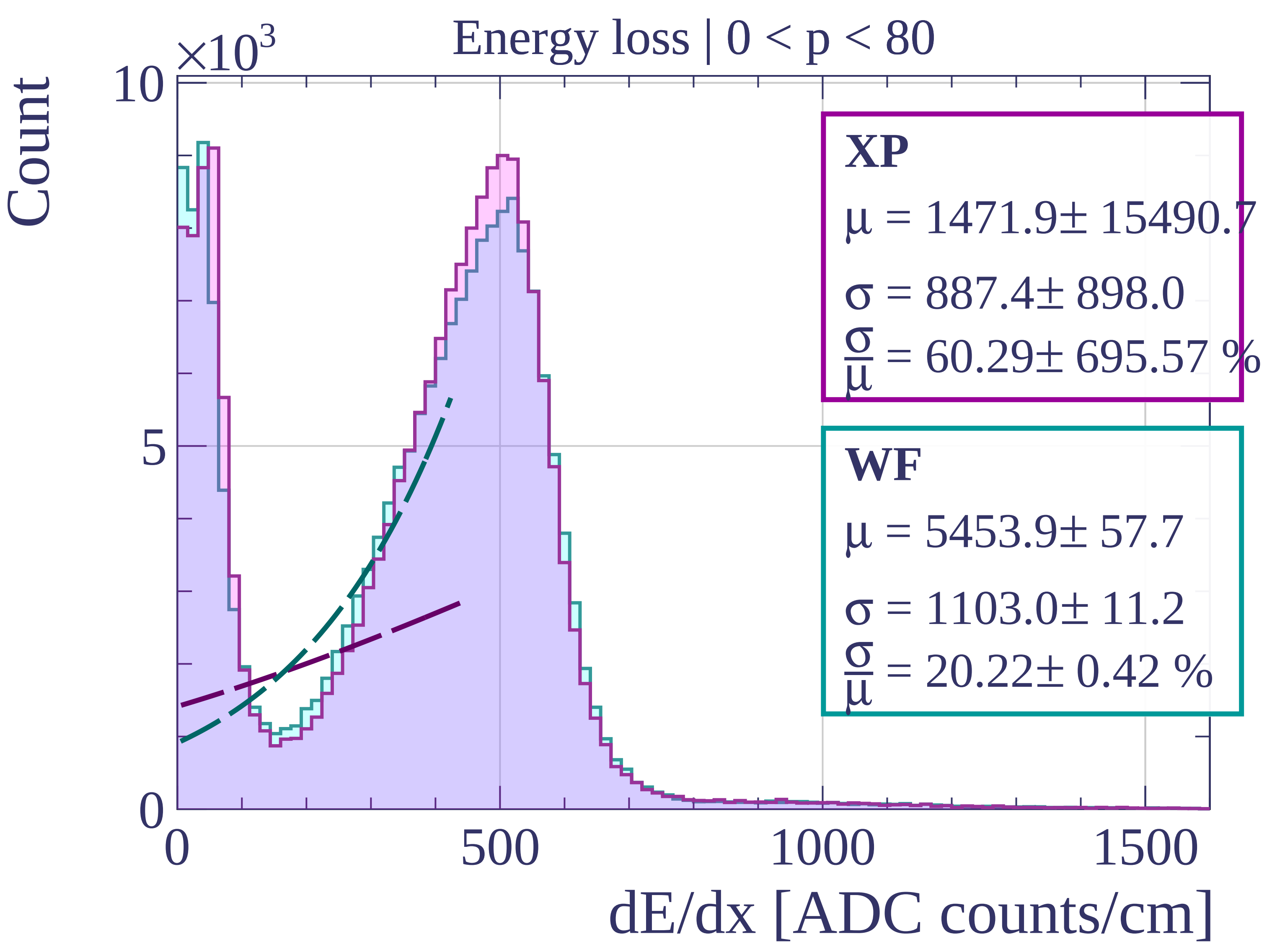

Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Y
10^4
10^3
10^2
10
1

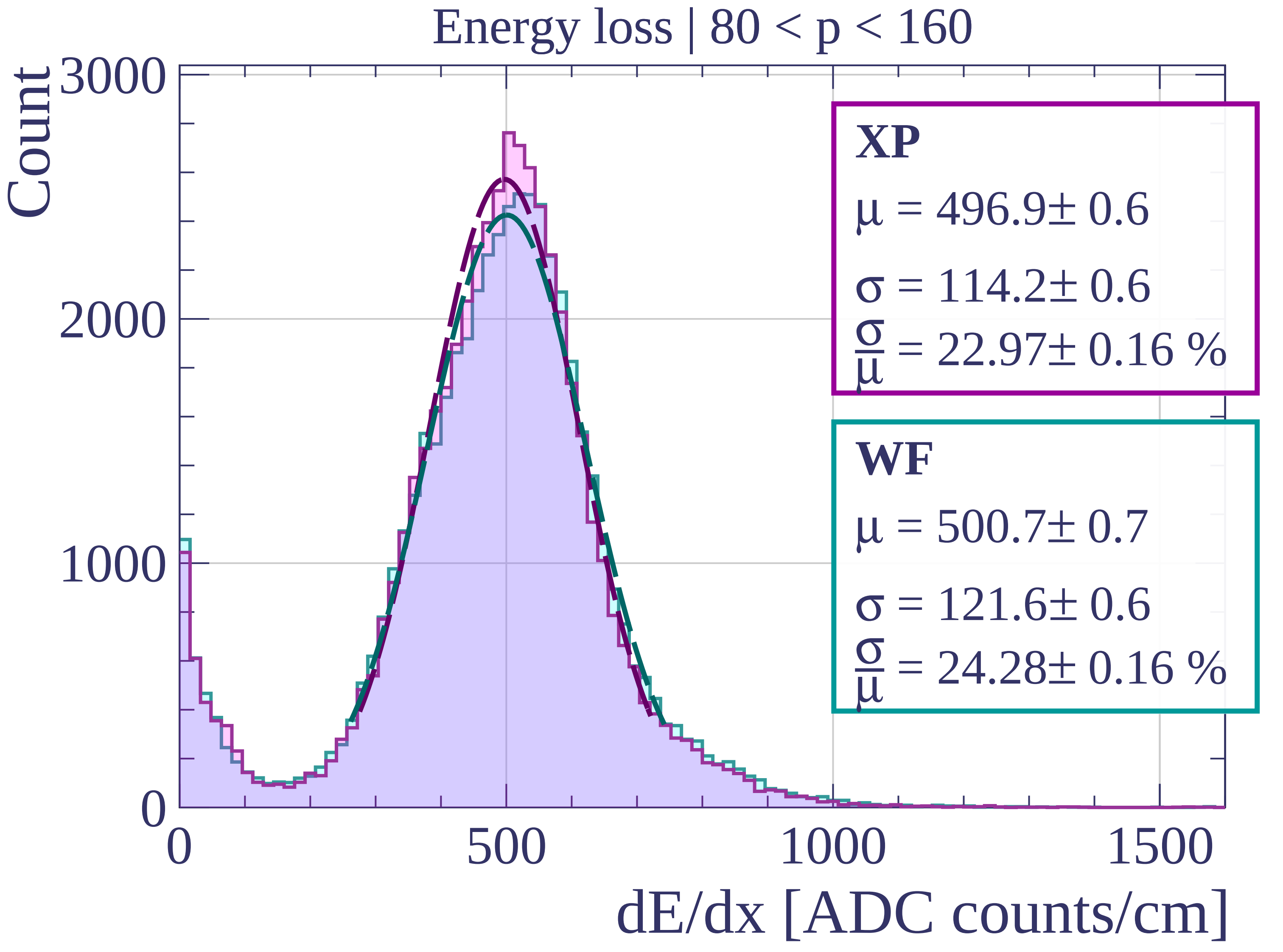


Signed momentum (MeV)
2000
1000
0
−1000
−2000
−40
−20
0
20
40
R
×10³
20
10
0

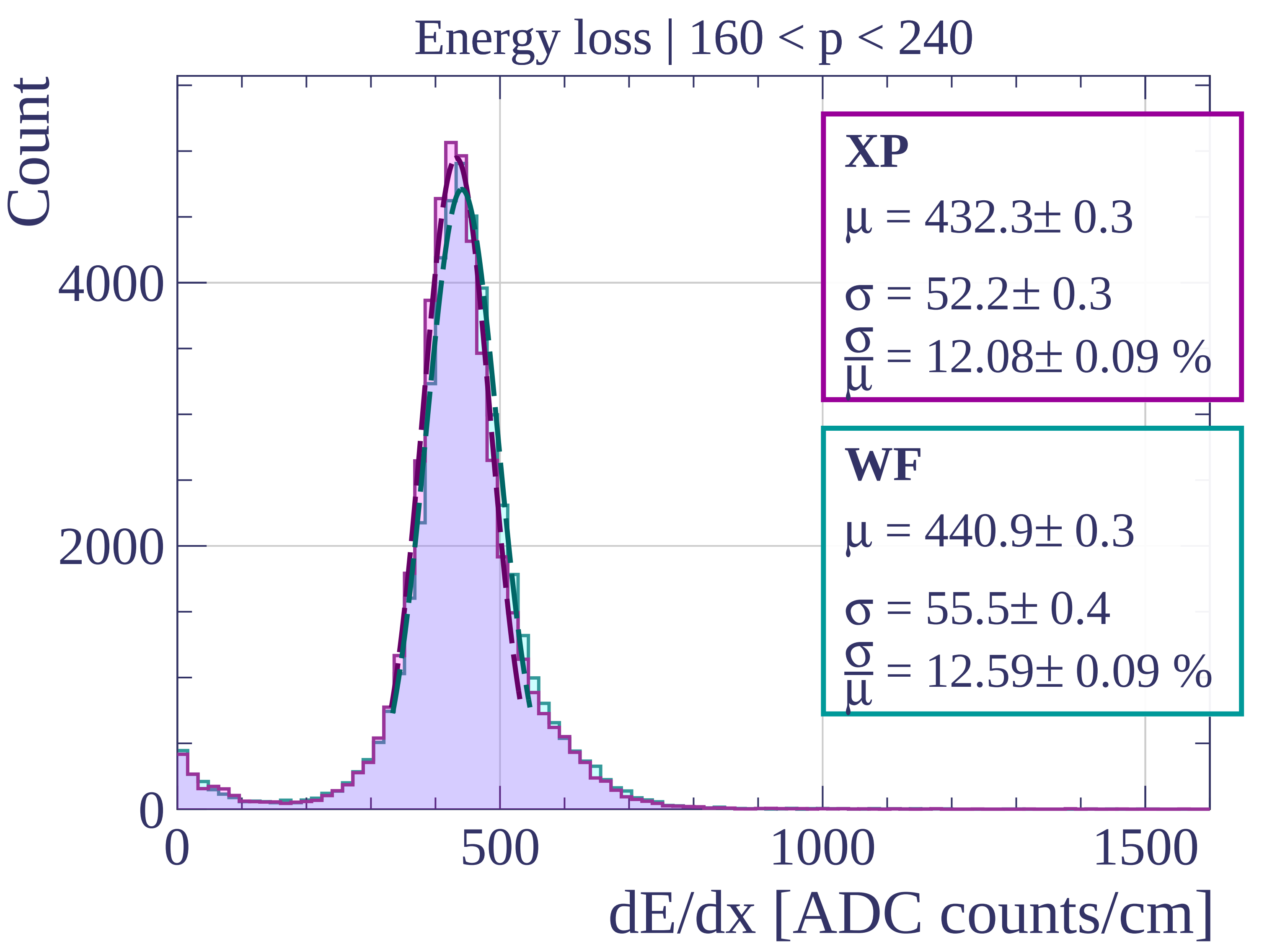


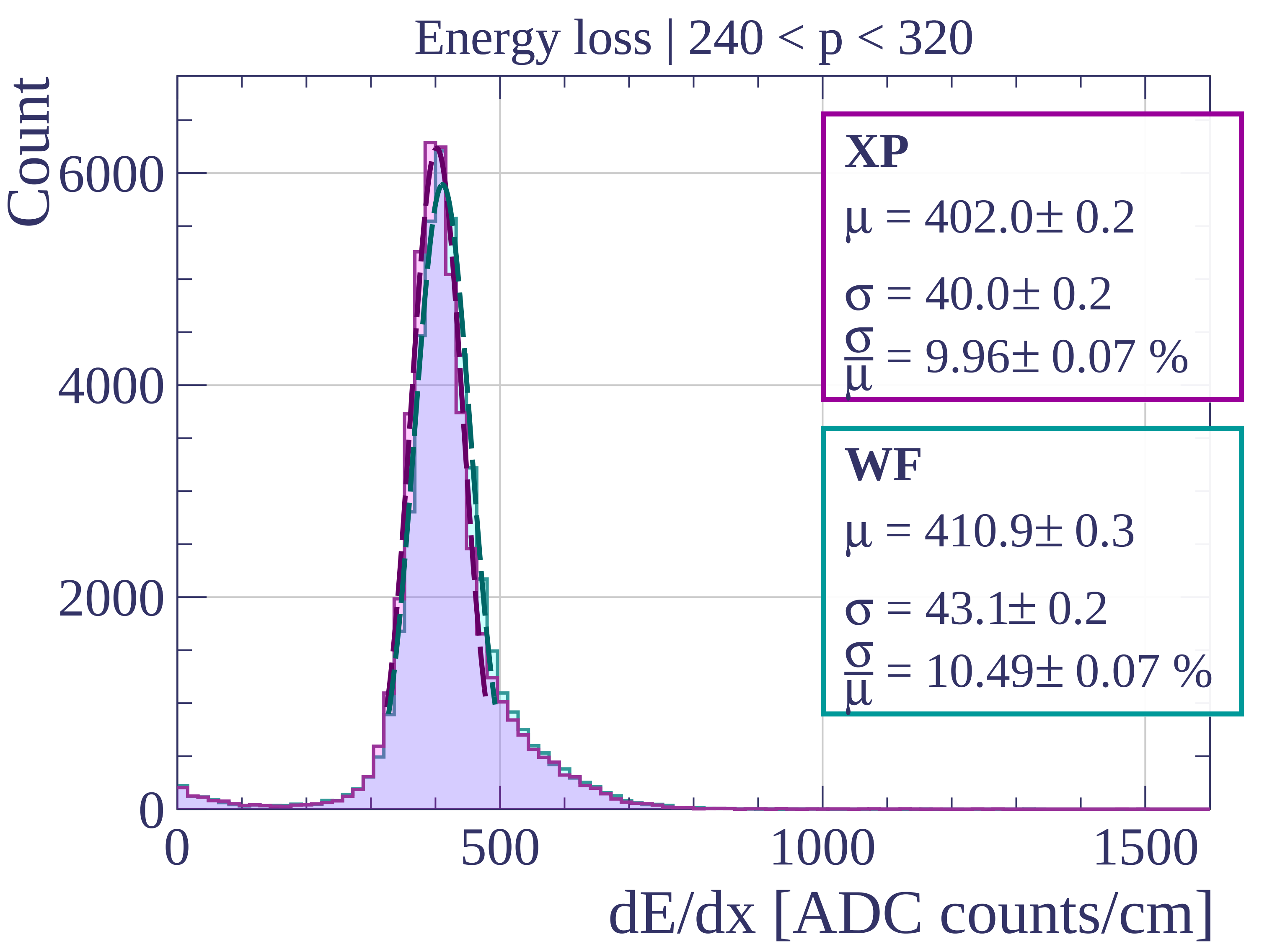
χ^2/NDF
50
40
30
20
10
0
−40
−20
0
20
40
R
1500
1000
500
0
$\times10^3$


Energy loss | 0 < p < 80
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 1471.9± 15490.7
σ = 887.4± 898.0
σ/μ = 60.29± 695.57 %
WF
μ = 5453.9± 57.7
σ = 1103.0± 11.2
σ/μ = 20.22± 0.42 %

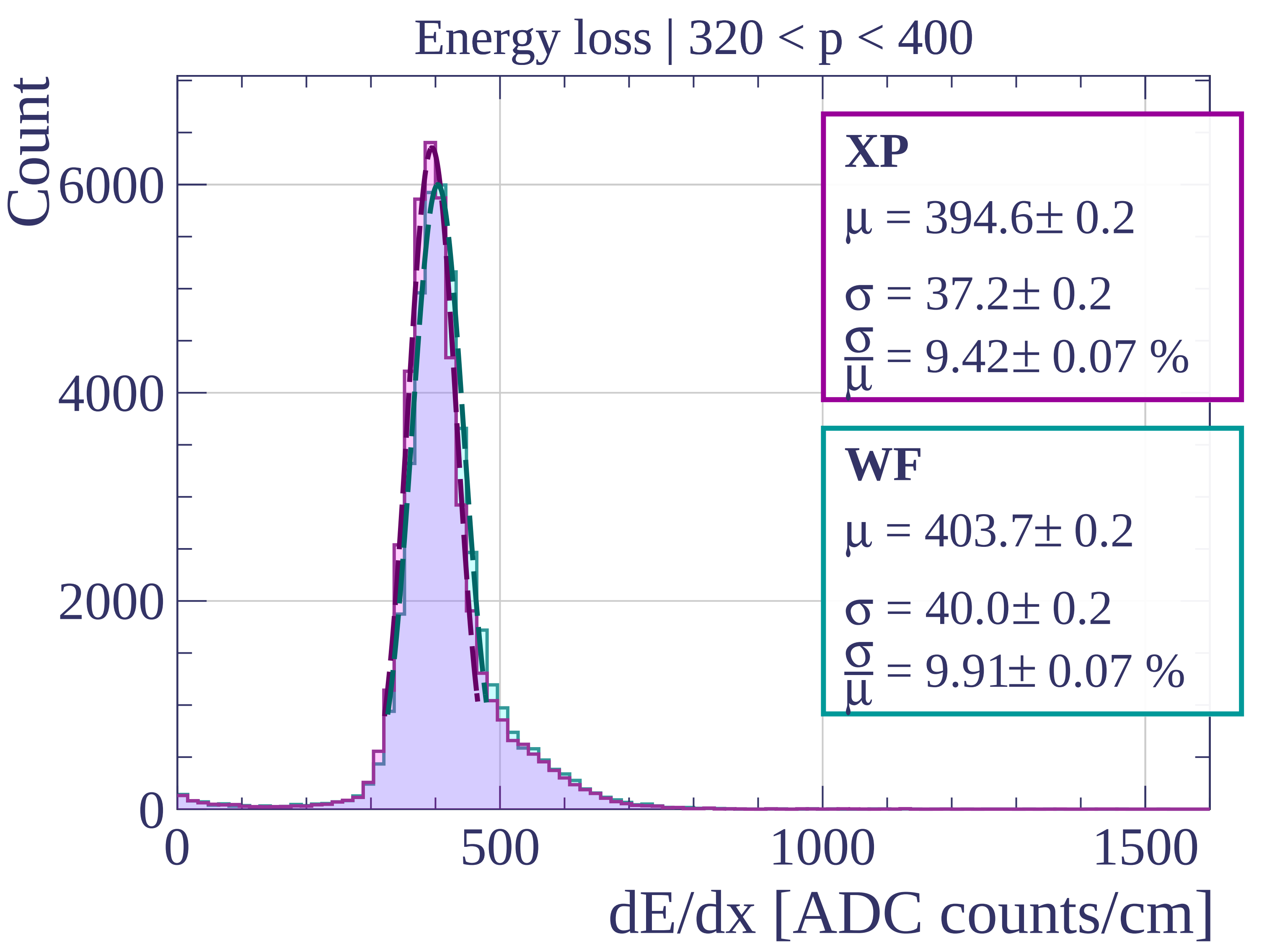


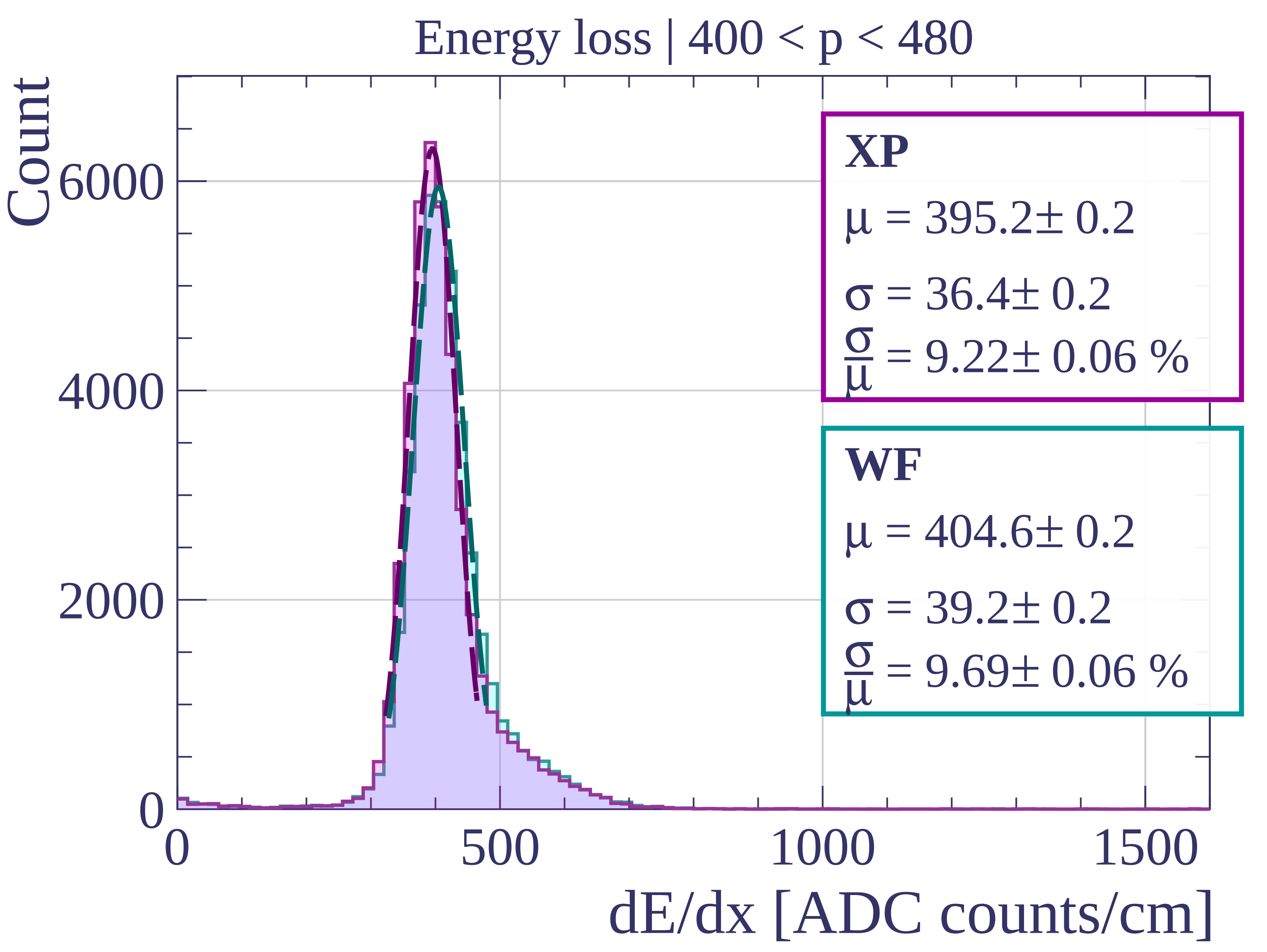
Energy loss | 80 < p < 160
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
1000
2000
3000
XP
μ = 496.9± 0.6
σ = 114.2± 0.6
σ/μ = 22.97± 0.16 %
WF
μ = 500.7± 0.7
σ = 121.6± 0.6
σ/μ = 24.28± 0.16 %


Energy loss | 160 < p < 240
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
4000
XP
μ = 432.3± 0.3
σ = 52.2± 0.3
σ/μ = 12.08± 0.09 %
WF
μ = 440.9± 0.3
σ = 55.5± 0.4
σ/μ = 12.59± 0.09 %

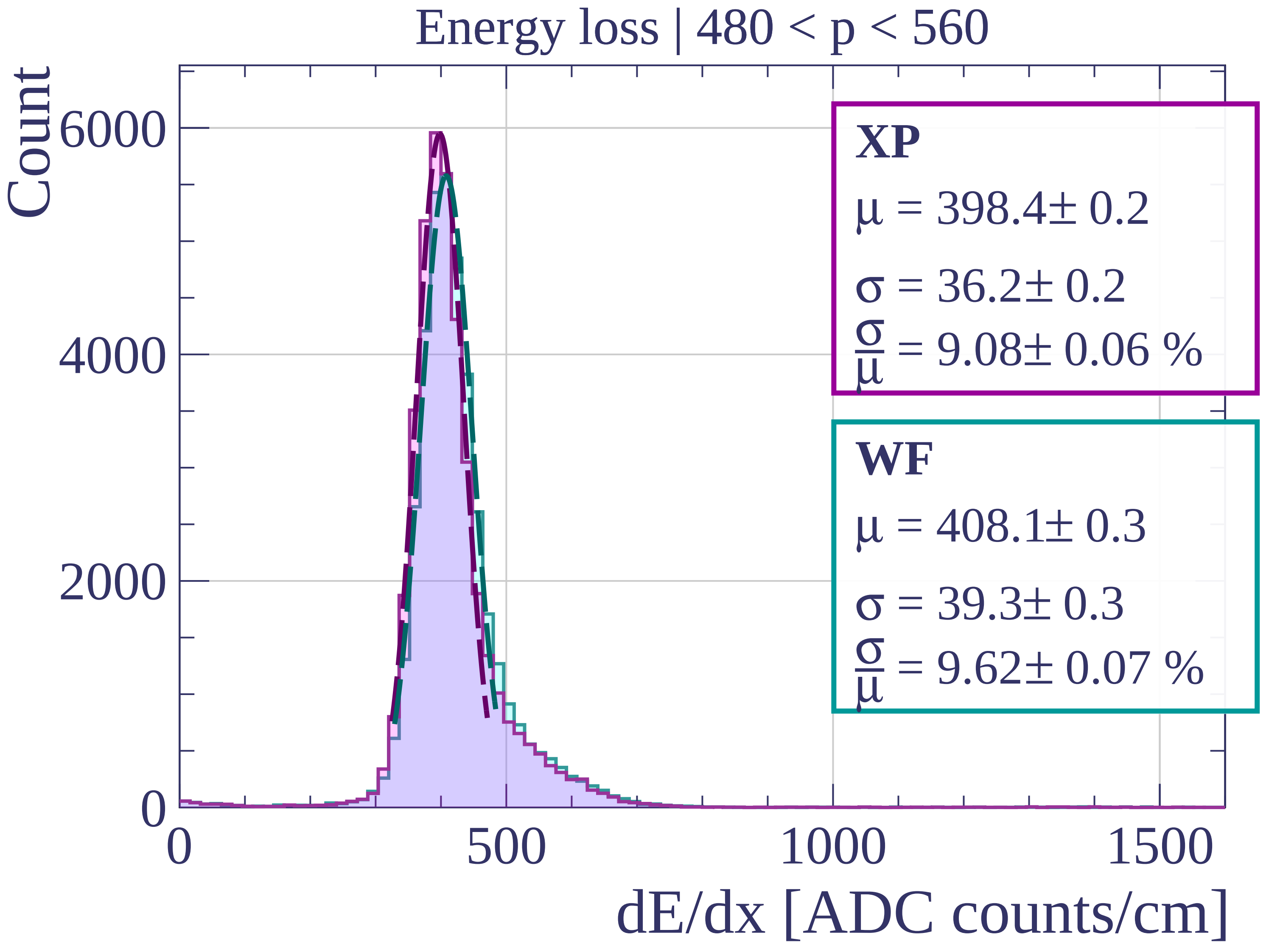


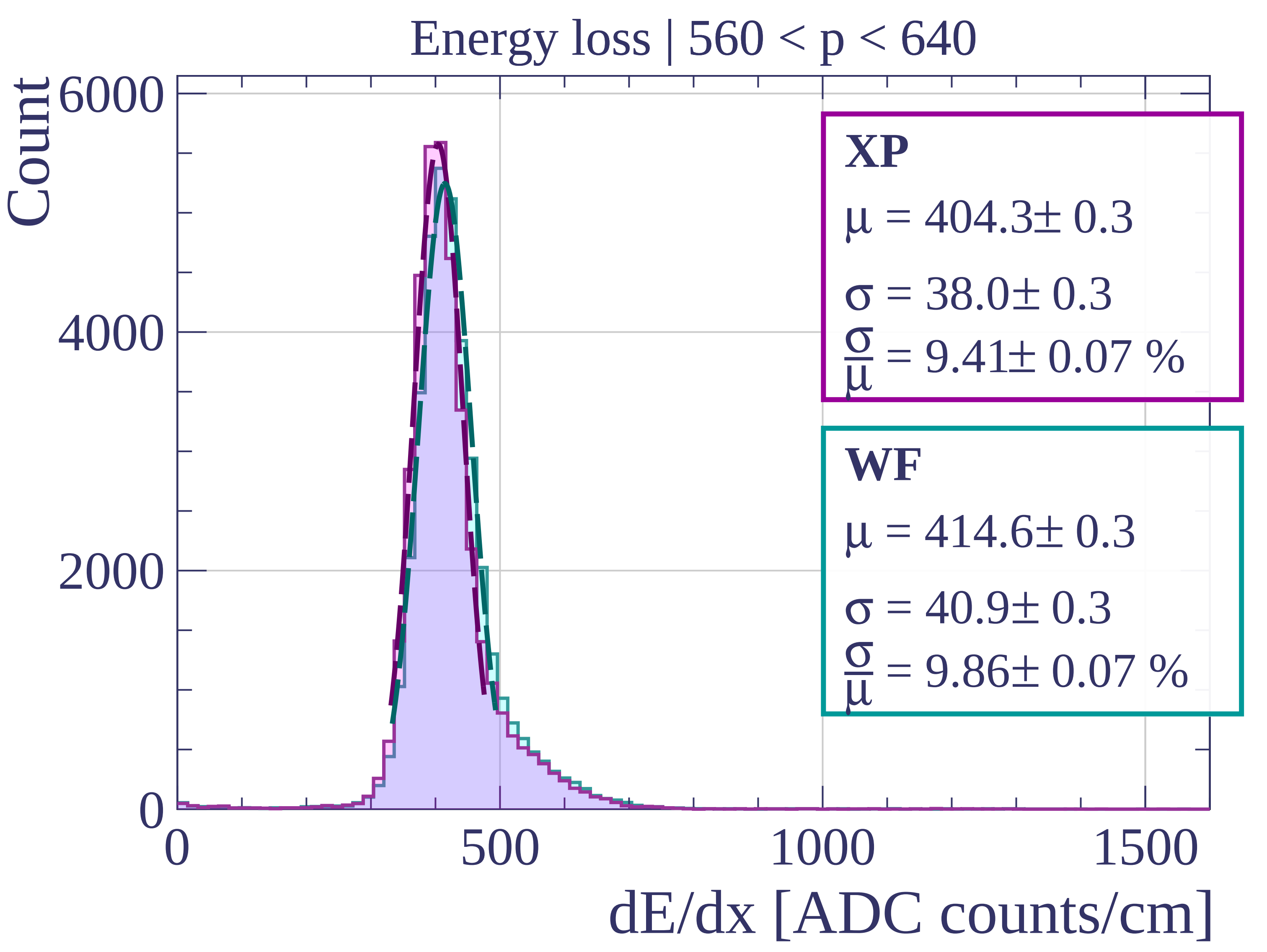
Energy loss | 240 < p < 320
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 402.0± 0.2
σ = 40.0± 0.2
σ/μ = 9.96± 0.07 %
WF
μ = 410.9± 0.3
σ = 43.1± 0.2
σ/μ = 10.49± 0.07 %

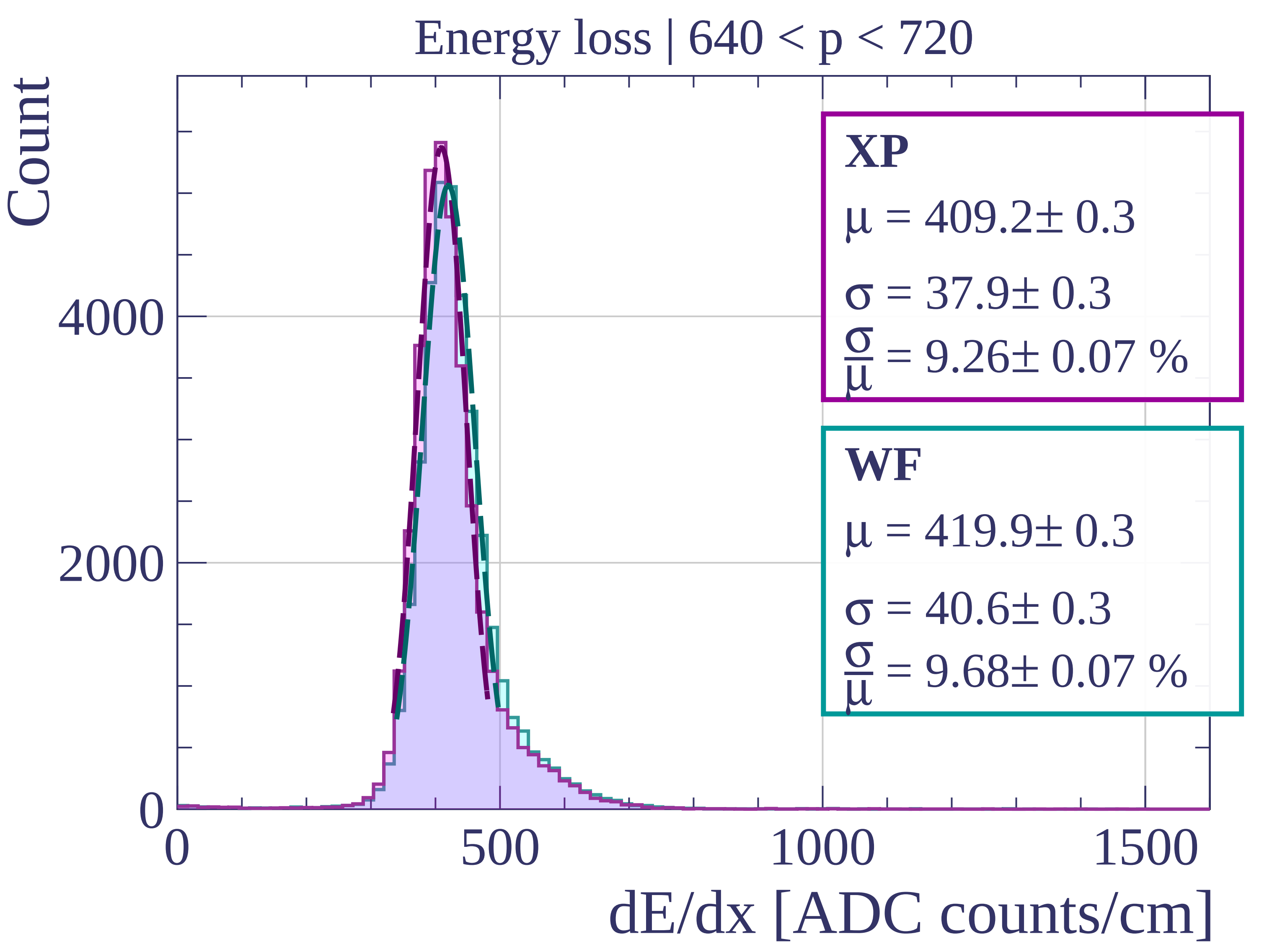

Energy loss | 320 < p < 400
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 394.6± 0.2
σ = 37.2± 0.2
σ/μ = 9.42± 0.07 %
WF
μ = 403.7± 0.2
σ = 40.0± 0.2
σ/μ = 9.91± 0.07 %



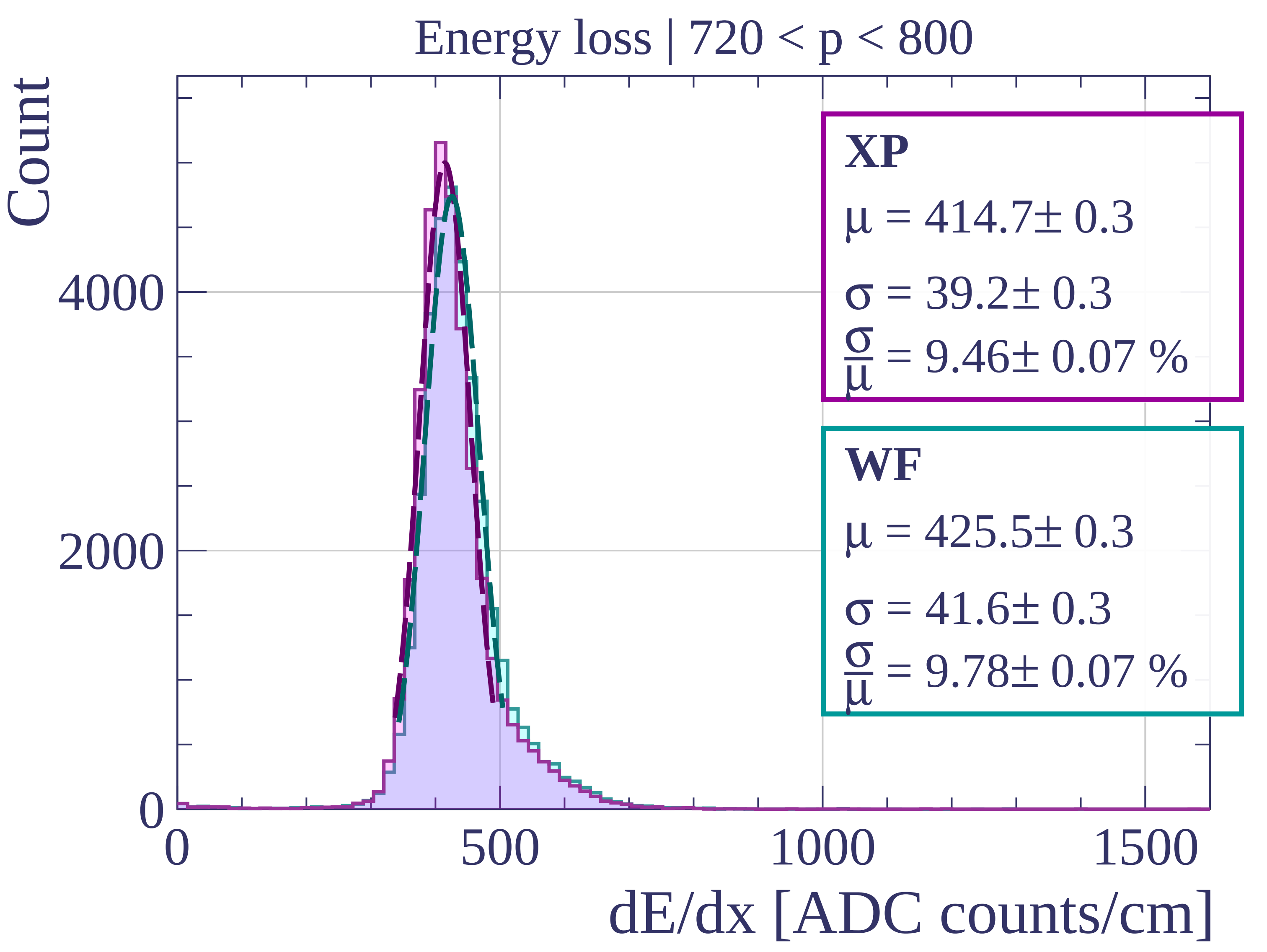
Energy loss | 400 < p < 480
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 395.2± 0.2
σ = 36.4± 0.2
σ/μ = 9.22± 0.06 %
WF
μ = 404.6± 0.2
σ = 39.2± 0.2
σ/μ = 9.69± 0.06 %


Energy loss | 480 < p < 560
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 398.4± 0.2
σ = 36.2± 0.2
σ/μ = 9.08± 0.06 %
WF
μ = 408.1± 0.3
σ = 39.3± 0.3
σ/μ = 9.62± 0.07 %

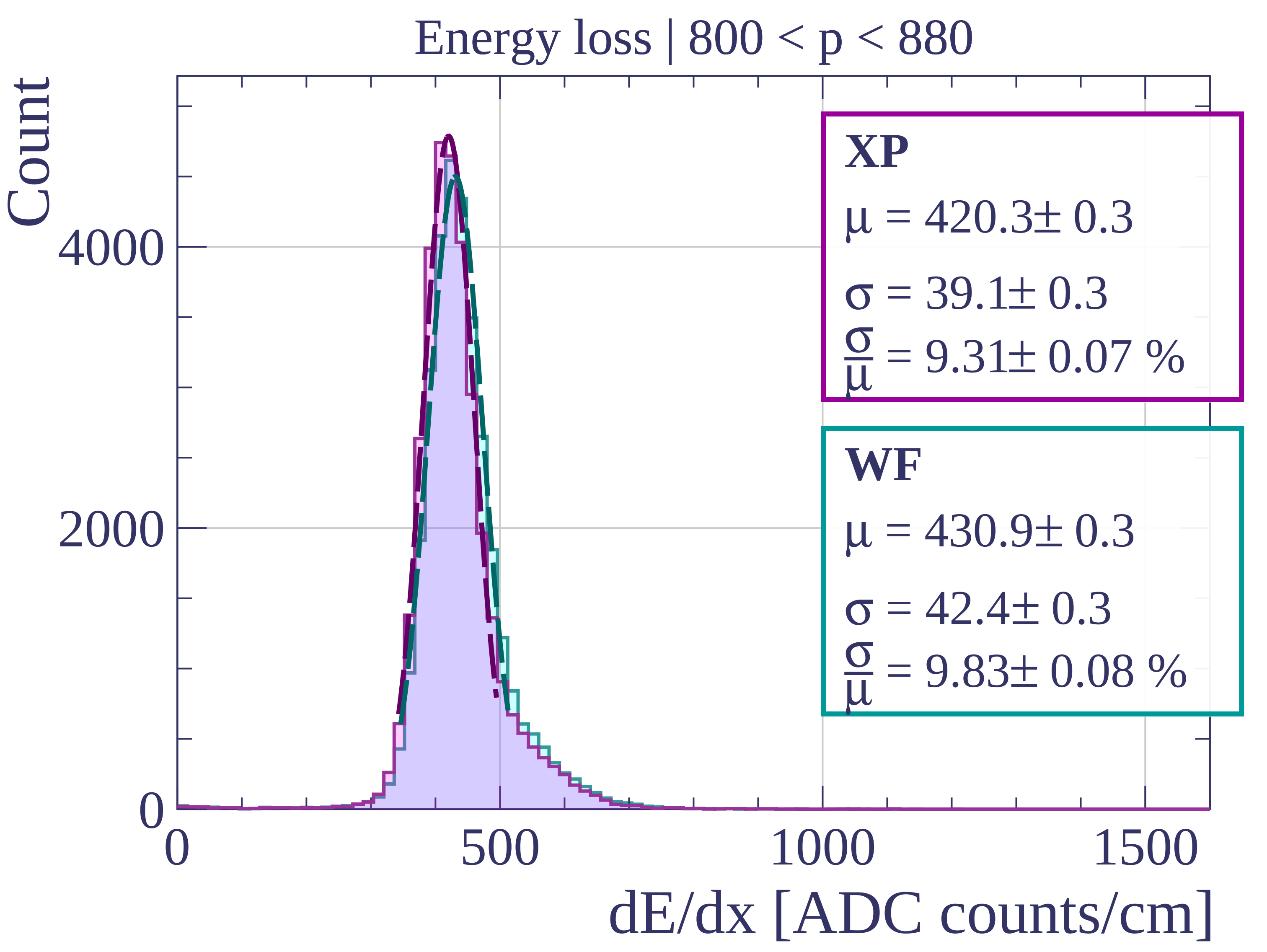

Energy loss | 560 < p < 640
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 404.3± 0.3
σ = 38.0± 0.3
σ/μ = 9.41± 0.07 %
WF
μ = 414.6± 0.3
σ = 40.9± 0.3
σ/μ = 9.86± 0.07 %

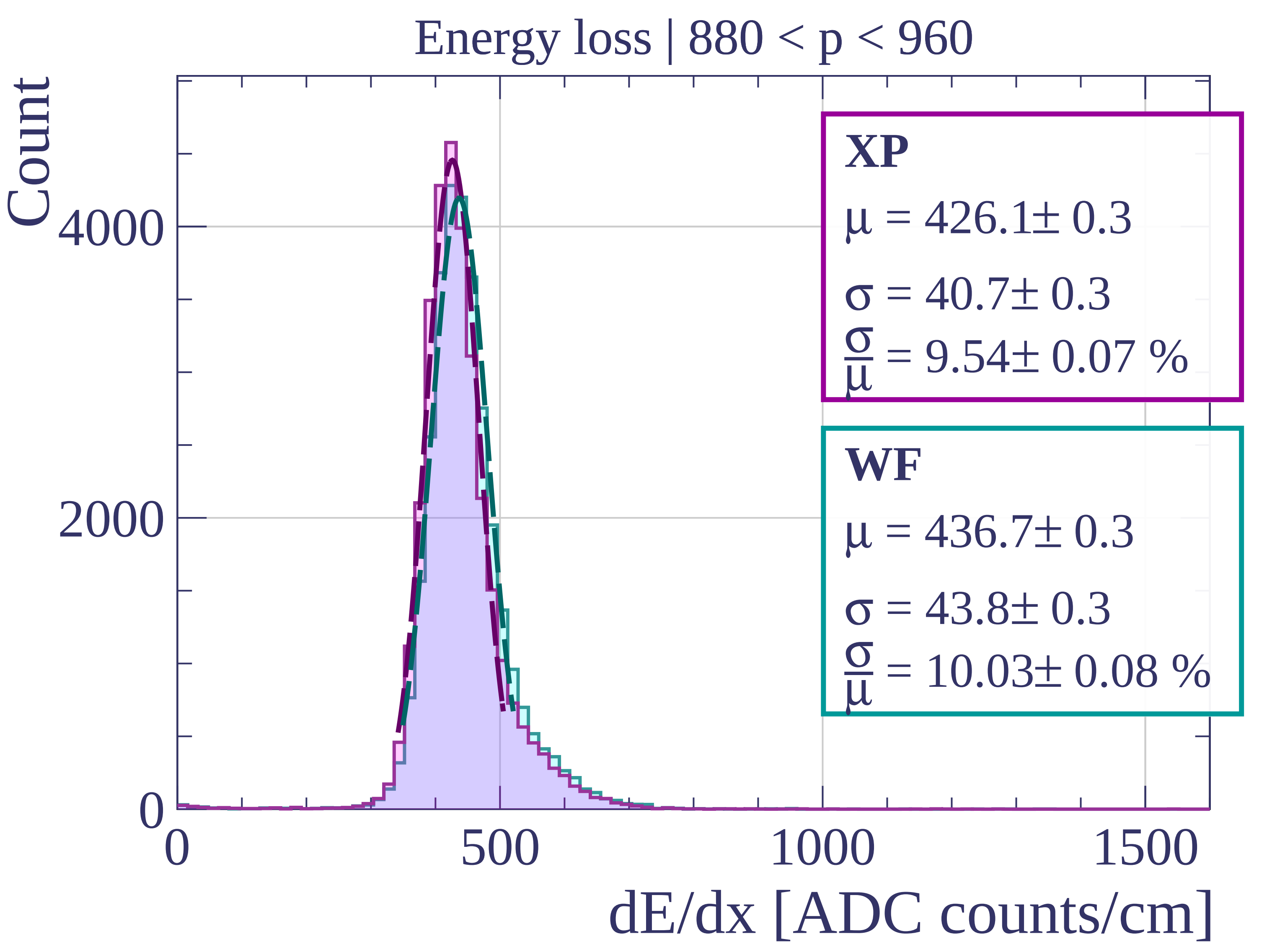


Energy loss | 640 < p < 720
Count
dE/dx [ADC counts/cm]
0
2000
4000
0
500
1000
1500
XP
μ = 409.2± 0.3
σ = 37.9± 0.3
σ/μ = 9.26± 0.07 %
WF
μ = 419.9± 0.3
σ = 40.6± 0.3
σ/μ = 9.68± 0.07 %

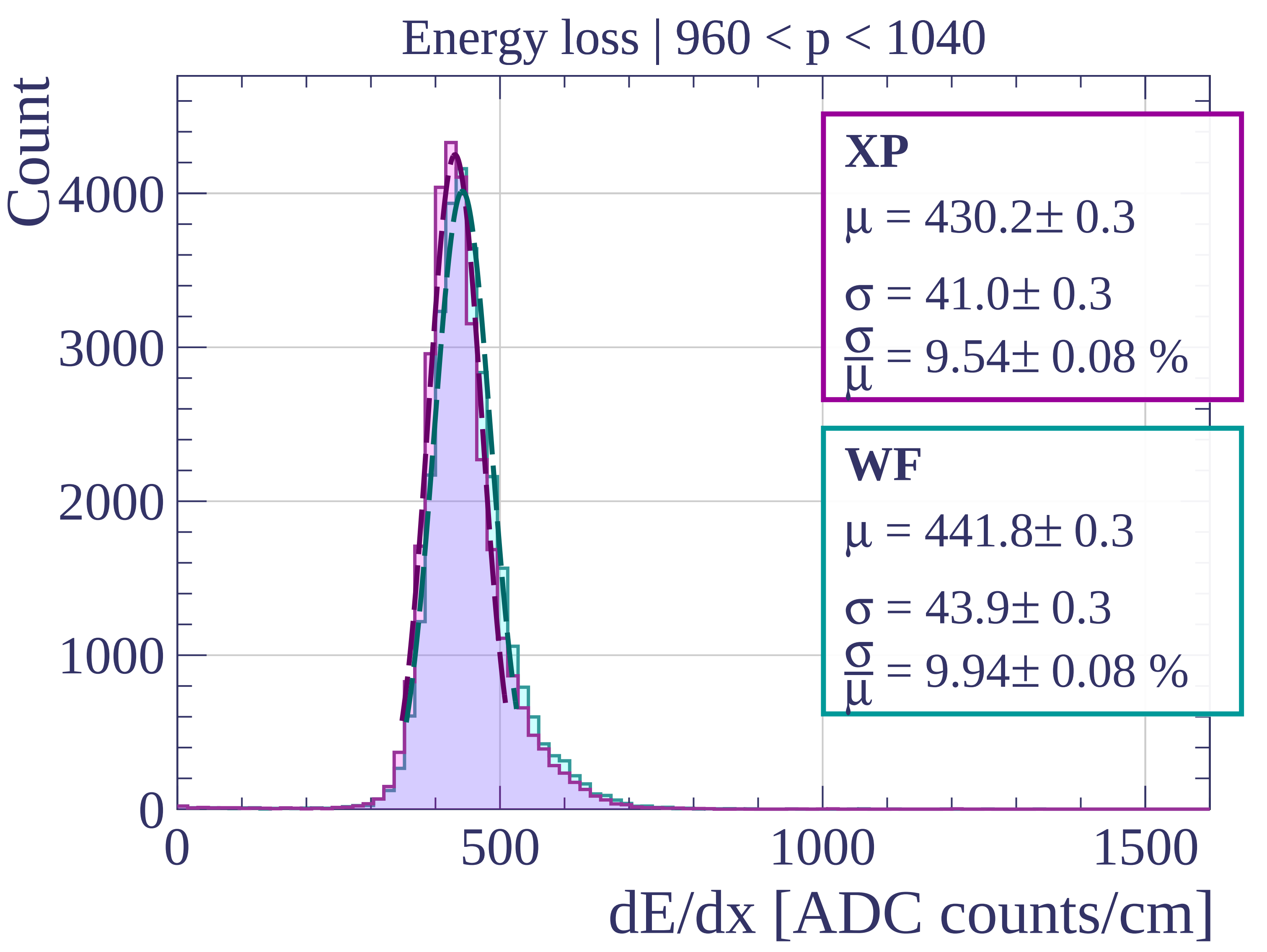


Energy loss | 720 < p < 800
Count
dE/dx [ADC counts/cm]
0
2000
4000
0
500
1000
1500
XP
μ = 414.7± 0.3
σ = 39.2± 0.3
σ/μ = 9.46± 0.07 %
WF
μ = 425.5± 0.3
σ = 41.6± 0.3
σ/μ = 9.78± 0.07 %

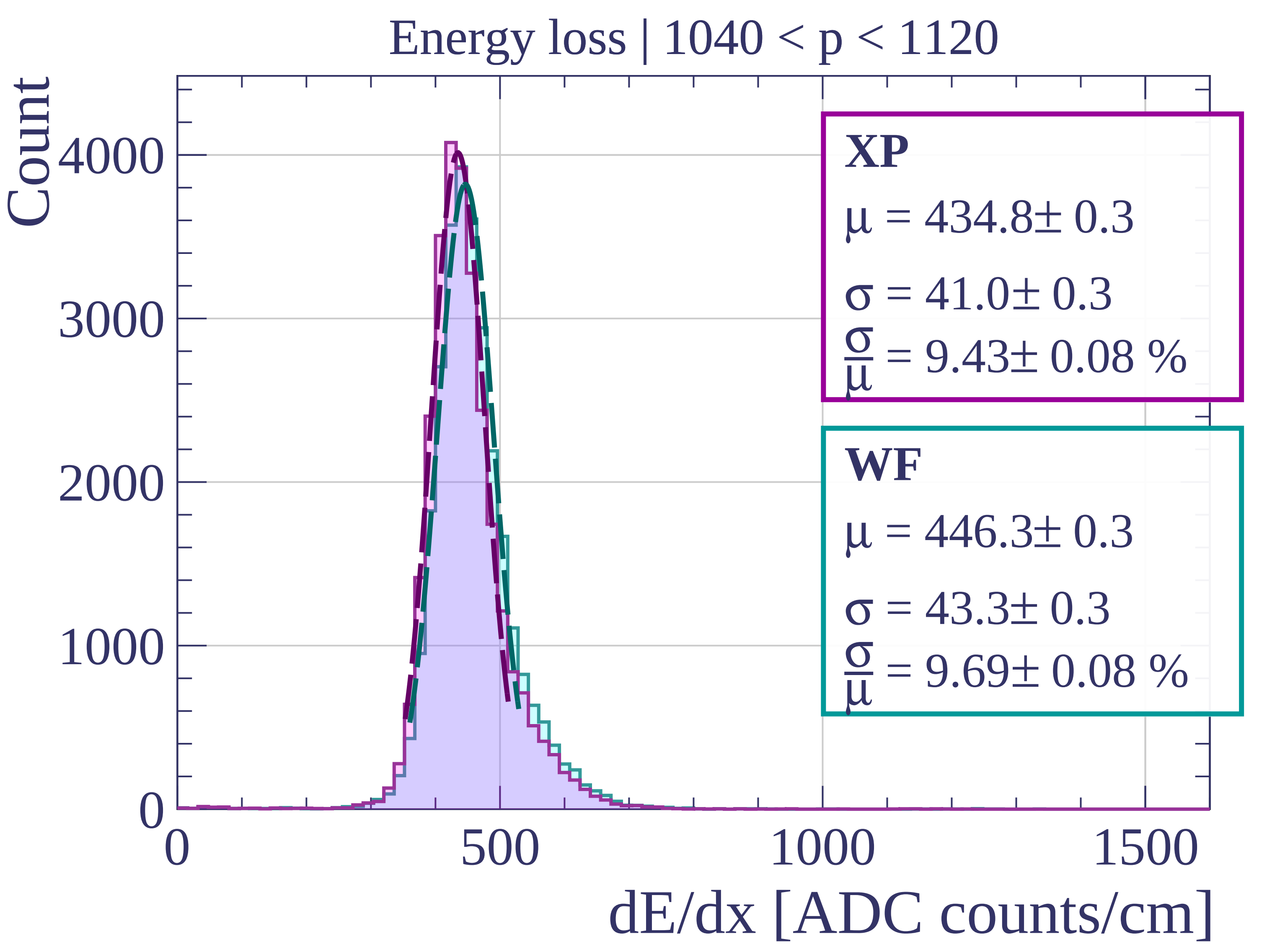

Energy loss | 800 < p < 880
Count
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 420.3± 0.3
σ = 39.1± 0.3
σ/μ = 9.31± 0.07 %
WF
μ = 430.9± 0.3
σ = 42.4± 0.3
σ/μ = 9.83± 0.08 %



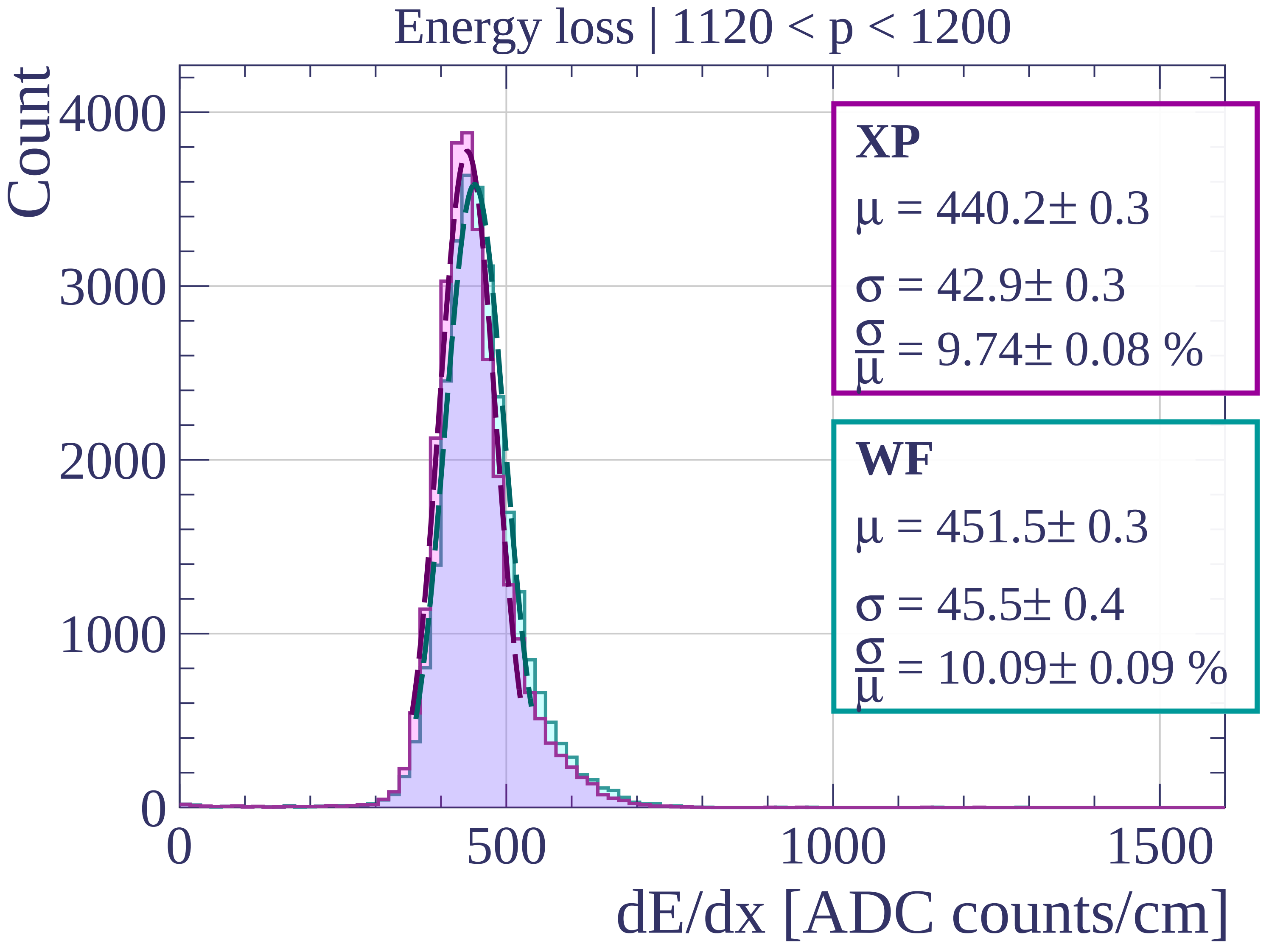
Energy loss | 880 < p < 960
Count
dE/dx [ADC counts/cm]
0
2000
4000
0
500
1000
1500
XP
μ = 426.1± 0.3
σ = 40.7± 0.3
σ/μ = 9.54± 0.07 %
WF
μ = 436.7± 0.3
σ = 43.8± 0.3
σ/μ = 10.03± 0.08 %



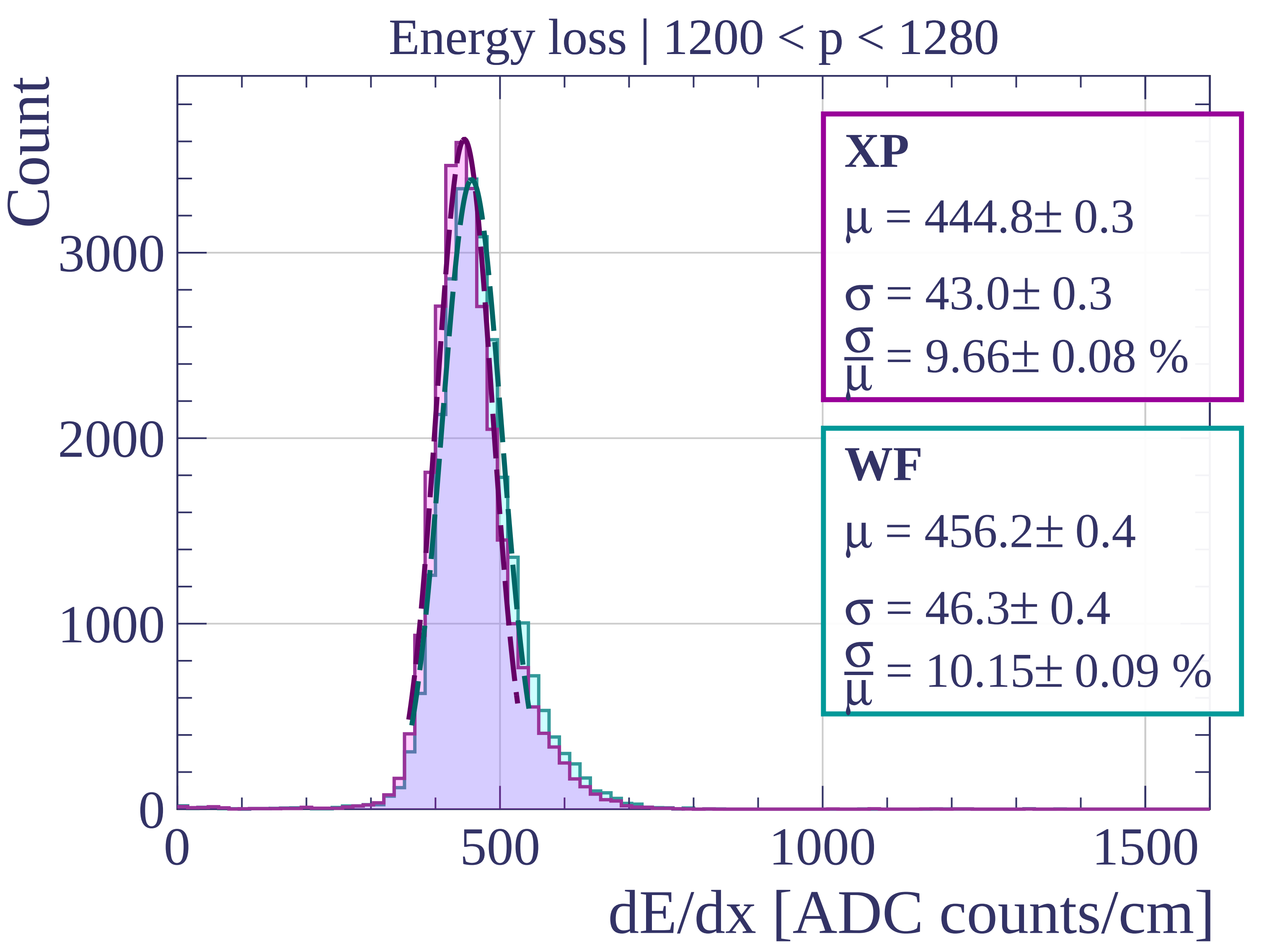
Energy loss | 960 < p < 1040
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 430.2± 0.3
σ = 41.0± 0.3
σ/μ = 9.54± 0.08 %
WF
μ = 441.8± 0.3
σ = 43.9± 0.3
σ/μ = 9.94± 0.08 %



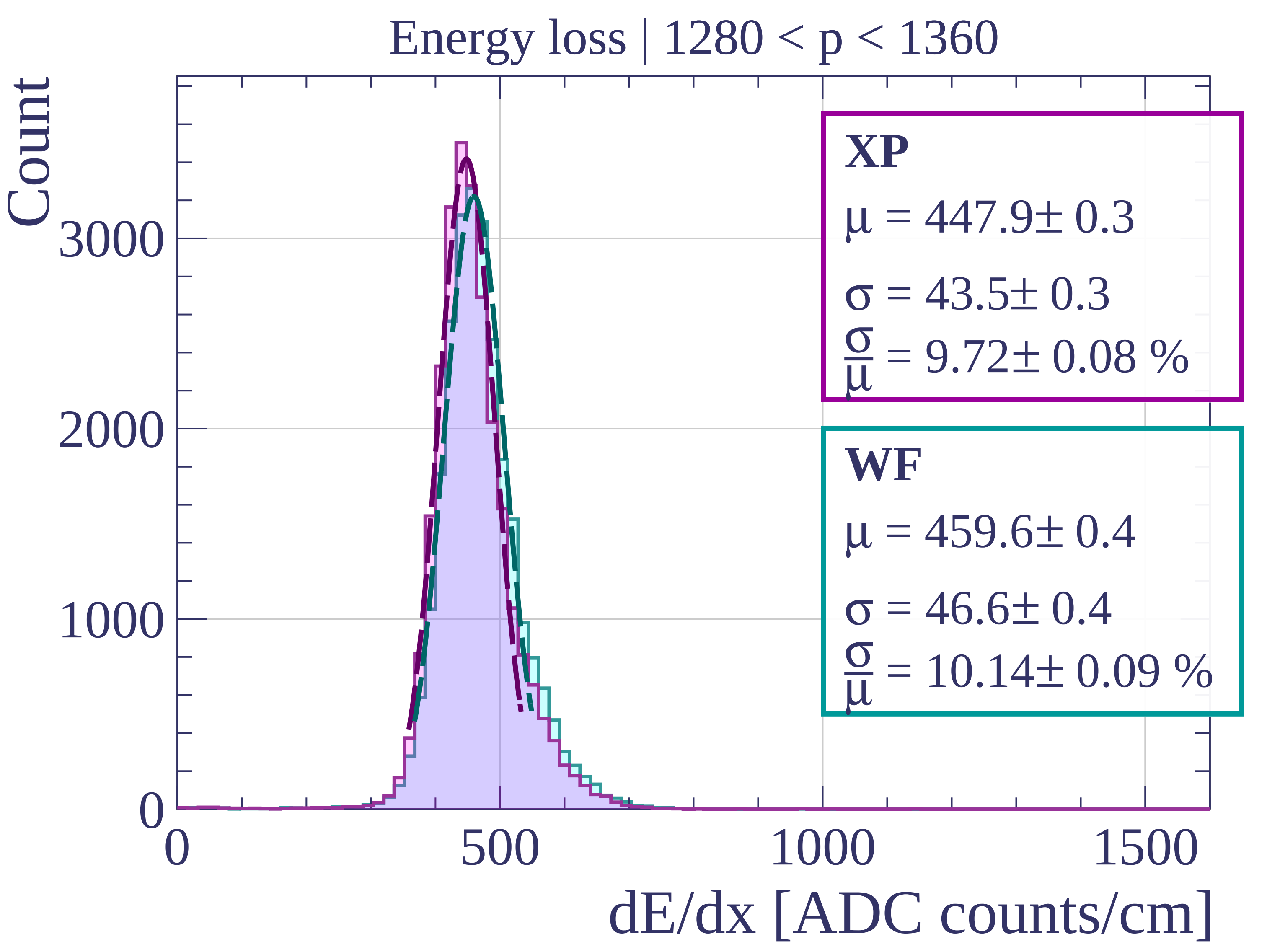
Energy loss | 1040 < p < 1120
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 434.8± 0.3
σ = 41.0± 0.3
σ/μ = 9.43± 0.08 %
WF
μ = 446.3± 0.3
σ = 43.3± 0.3
σ/μ = 9.69± 0.08 %

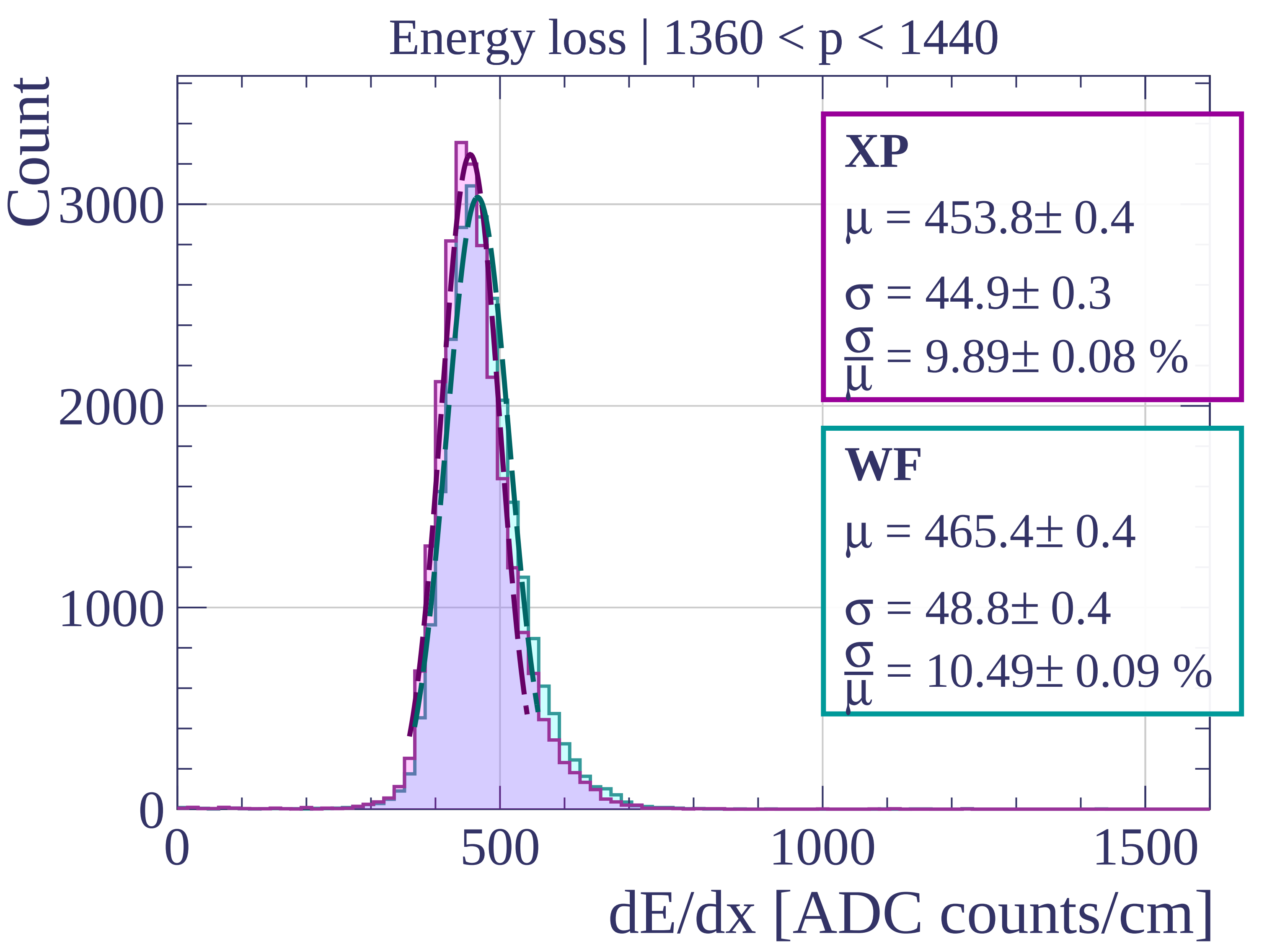


Energy loss | 1120 < p < 1200
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 440.2± 0.3
σ = 42.9± 0.3
σ/μ = 9.74± 0.08 %
WF
μ = 451.5± 0.3
σ = 45.5± 0.4
σ/μ = 10.09± 0.09 %

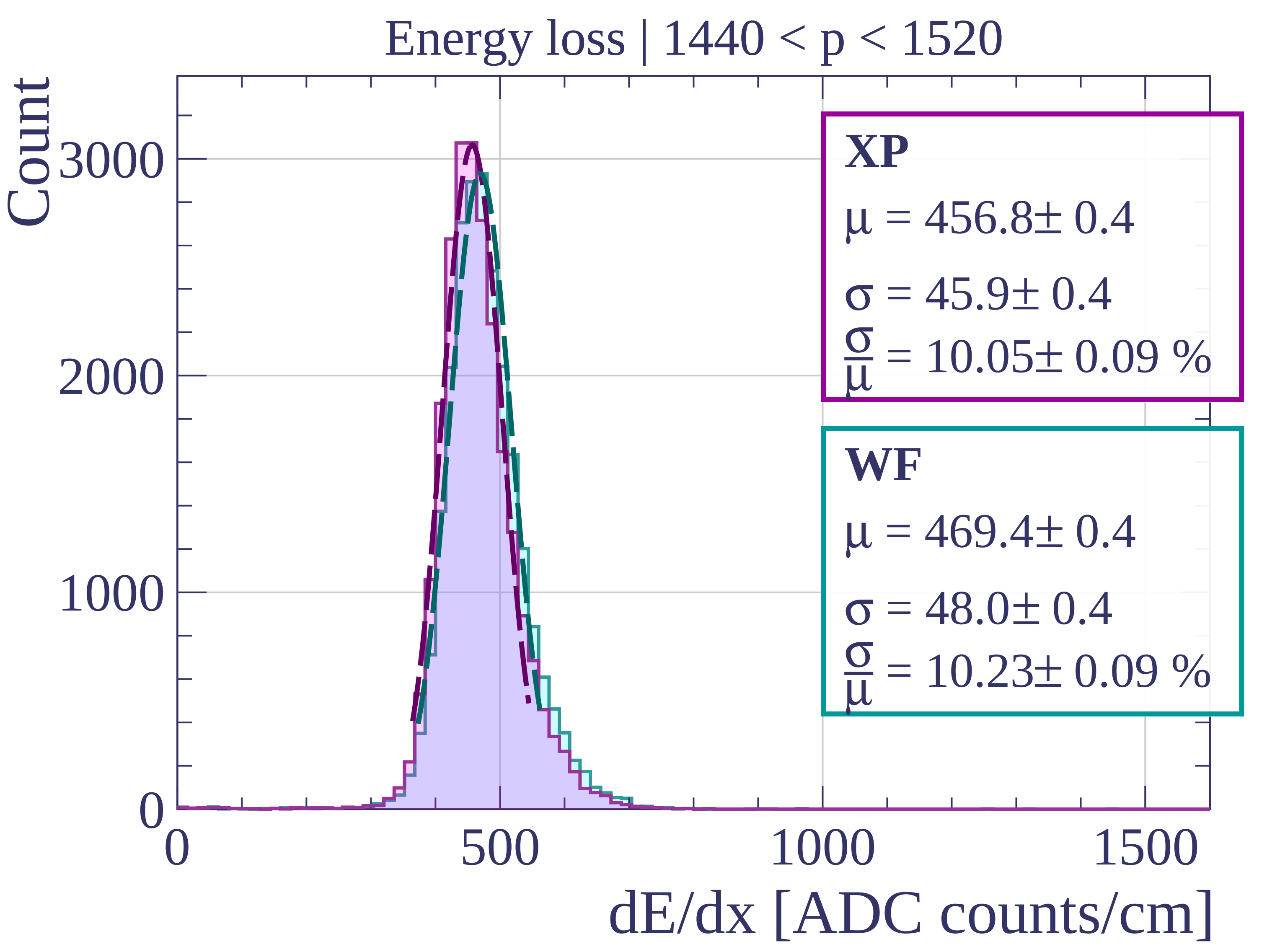


Energy loss | 1200 < p < 1280
Count
0
1000
2000
3000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 444.8± 0.3
σ = 43.0± 0.3
σ/μ = 9.66± 0.08 %
WF
μ = 456.2± 0.4
σ = 46.3± 0.4
σ/μ = 10.15± 0.09 %

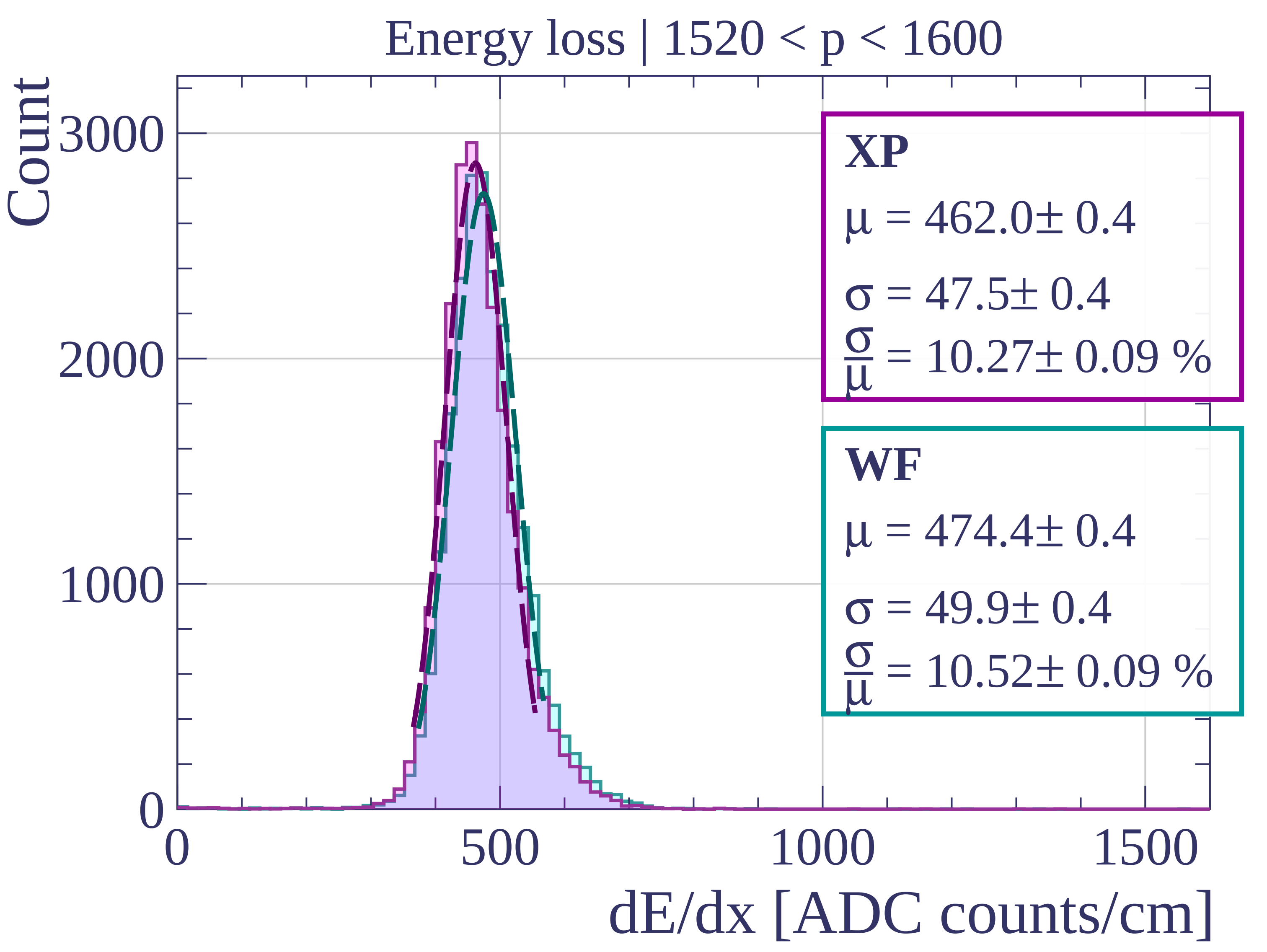

Energy loss | 1280 < p < 1360
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 447.9± 0.3
σ = 43.5± 0.3
σ/μ = 9.72± 0.08 %
WF
μ = 459.6± 0.4
σ = 46.6± 0.4
σ/μ = 10.14± 0.09 %



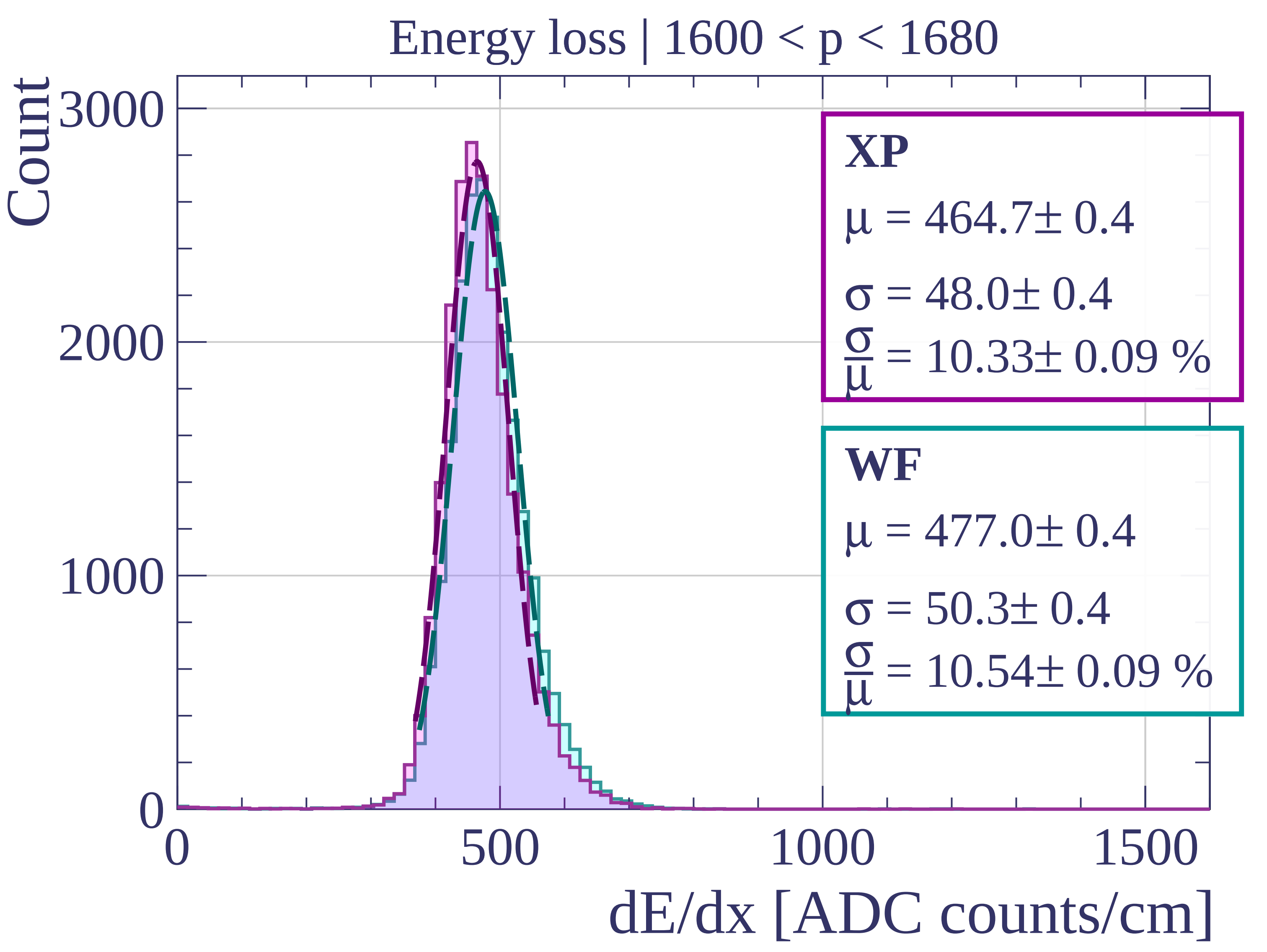
Energy loss | 1360 < p < 1440
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 453.8± 0.4
σ = 44.9± 0.3
σ/μ = 9.89± 0.08 %
WF
μ = 465.4± 0.4
σ = 48.8± 0.4
σ/μ = 10.49± 0.09 %



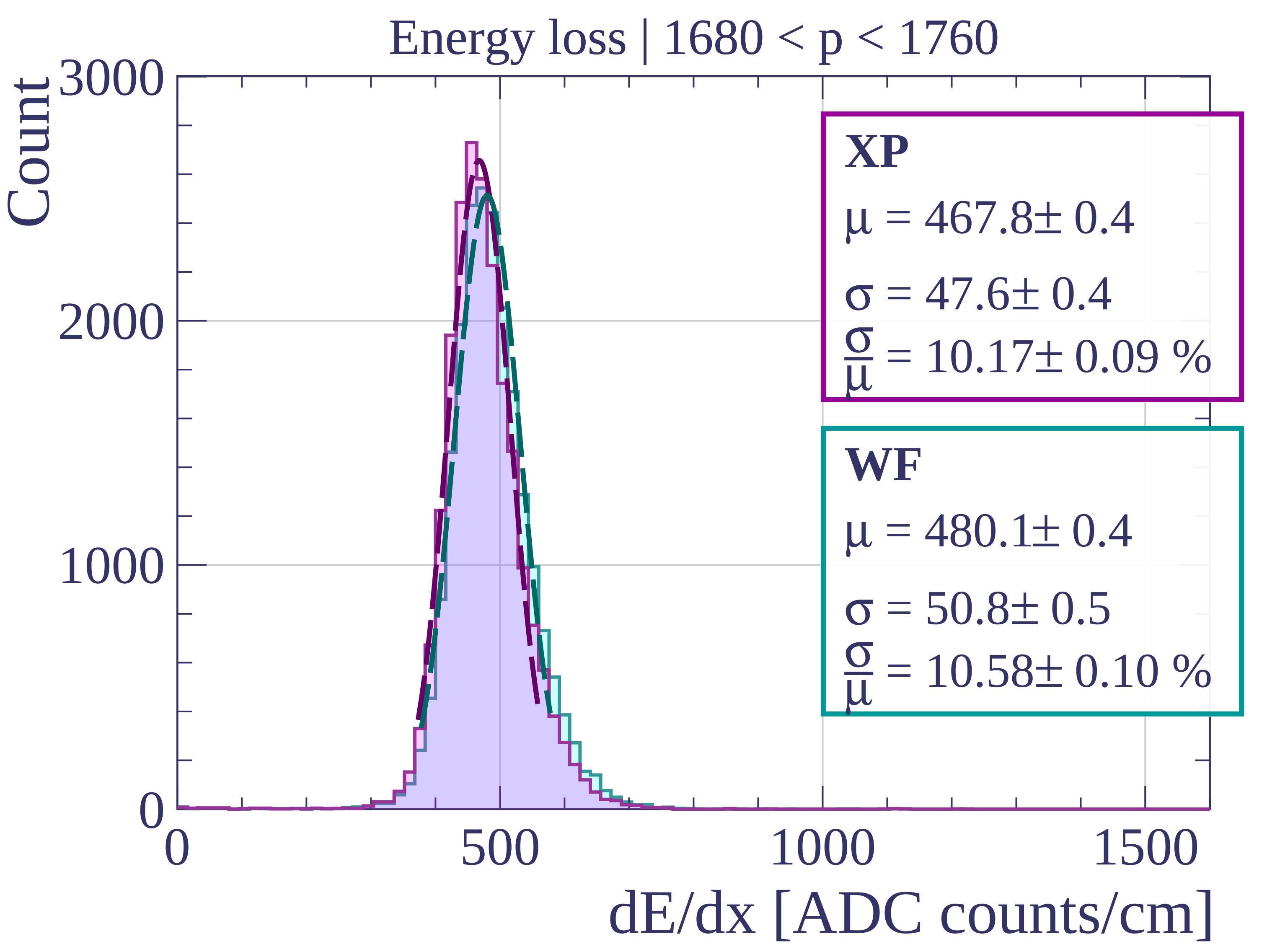
Energy loss | 1440 < p < 1520
Count
0
1000
2000
3000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 456.8± 0.4
σ = 45.9± 0.4
σ/μ = 10.05± 0.09 %
WF
μ = 469.4± 0.4
σ = 48.0± 0.4
σ/μ = 10.23± 0.09 %

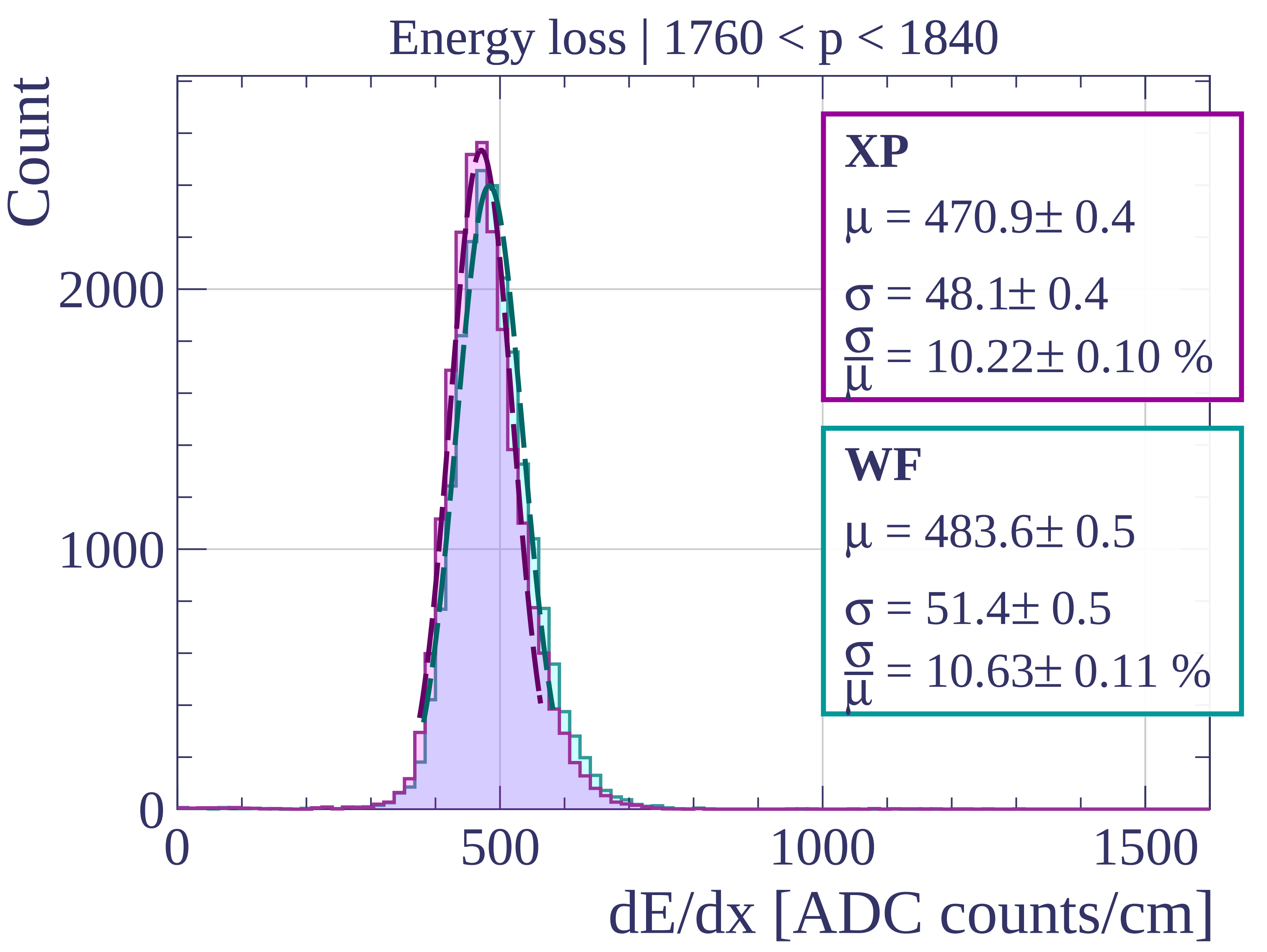


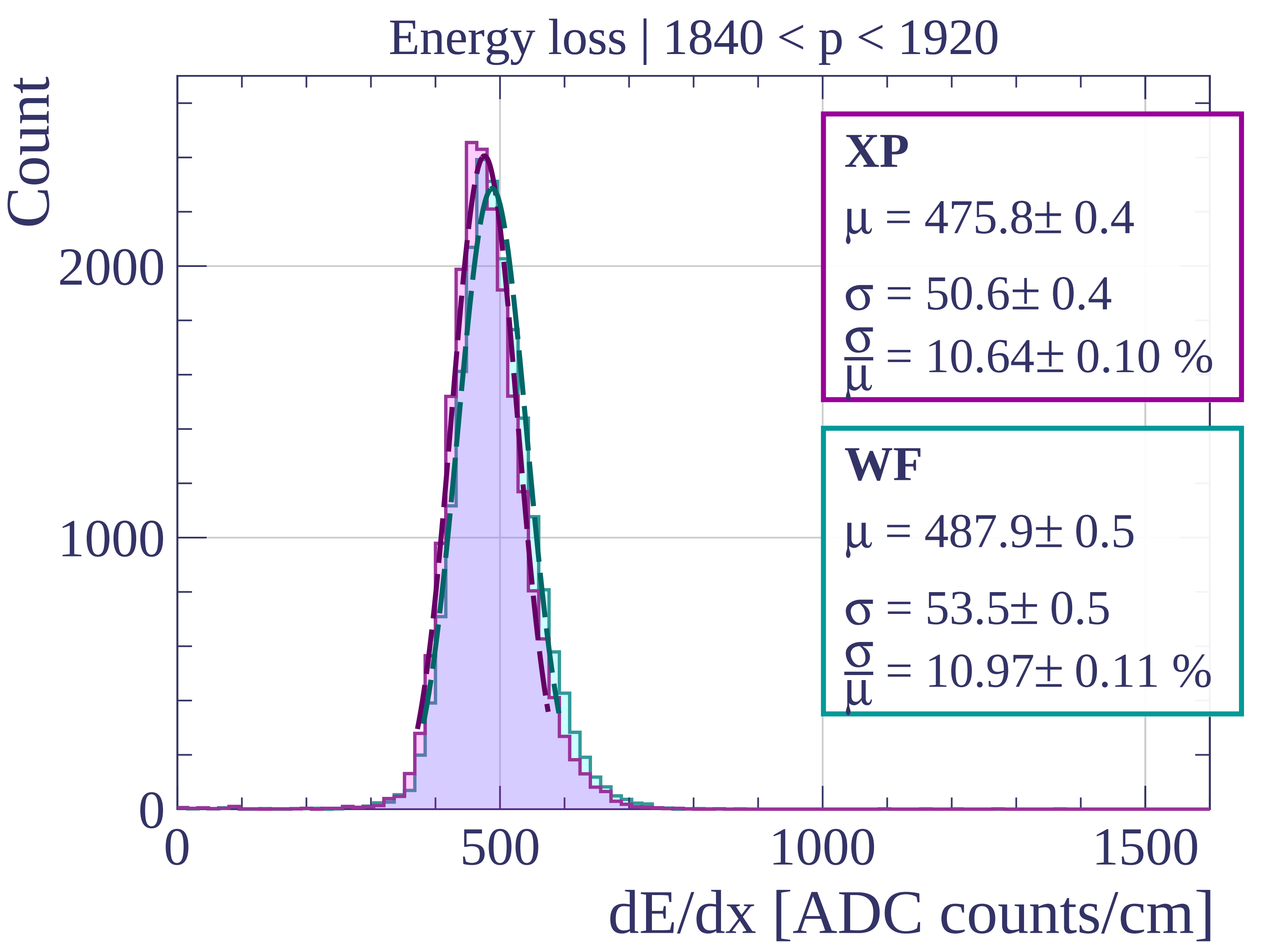
Energy loss | 1520 < p < 1600
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 462.0± 0.4
σ = 47.5± 0.4
σ/μ = 10.27± 0.09 %
WF
μ = 474.4± 0.4
σ = 49.9± 0.4
σ/μ = 10.52± 0.09 %

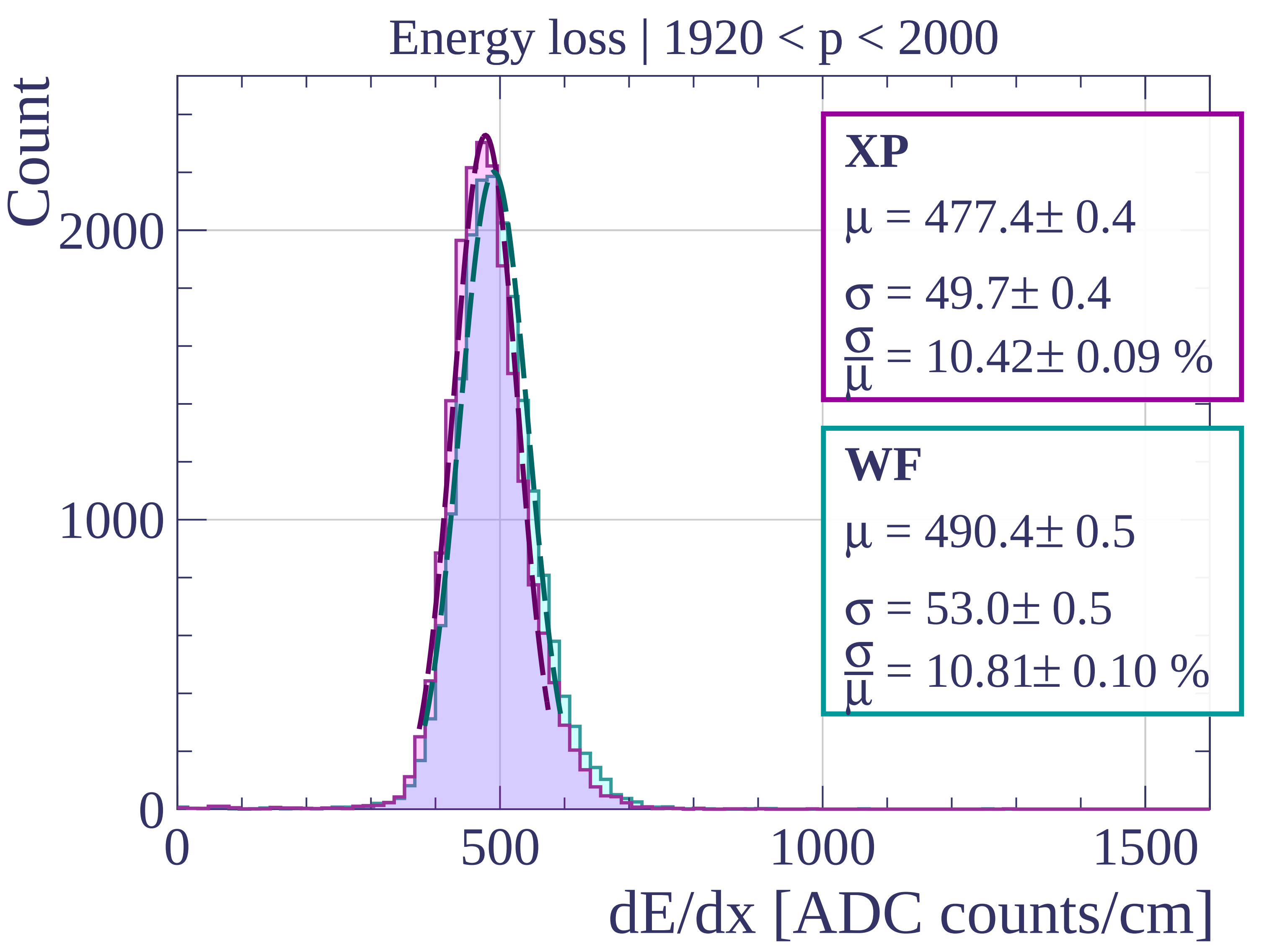


Energy loss | 1600 < p < 1680
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 464.7± 0.4
σ = 48.0± 0.4
σ/μ = 10.33± 0.09 %
WF
μ = 477.0± 0.4
σ = 50.3± 0.4
σ/μ = 10.54± 0.09 %

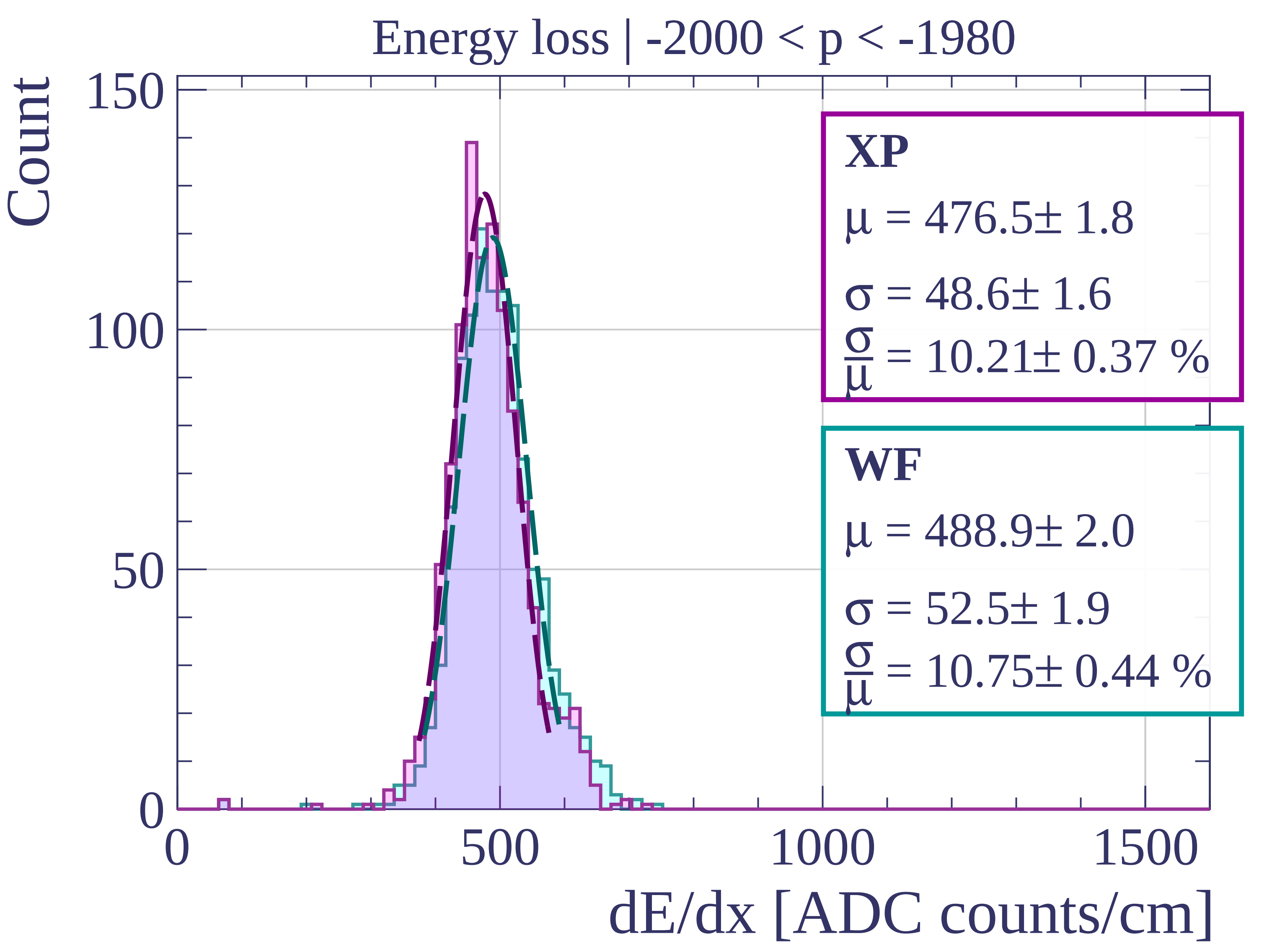

Energy loss | 1680 < p < 1760
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 467.8± 0.4
σ = 47.6± 0.4
σ/μ = 10.17± 0.09 %
WF
μ = 480.1± 0.4
σ = 50.8± 0.5
σ/μ = 10.58± 0.10 %


Energy loss | 1760 < p < 1840
Count
dE/dx [ADC counts/cm]
0
1000
2000
0
500
1000
1500
XP
μ = 470.9± 0.4
σ = 48.1± 0.4
σ/μ = 10.22± 0.10 %
WF
μ = 483.6± 0.5
σ = 51.4± 0.5
σ/μ = 10.63± 0.11 %

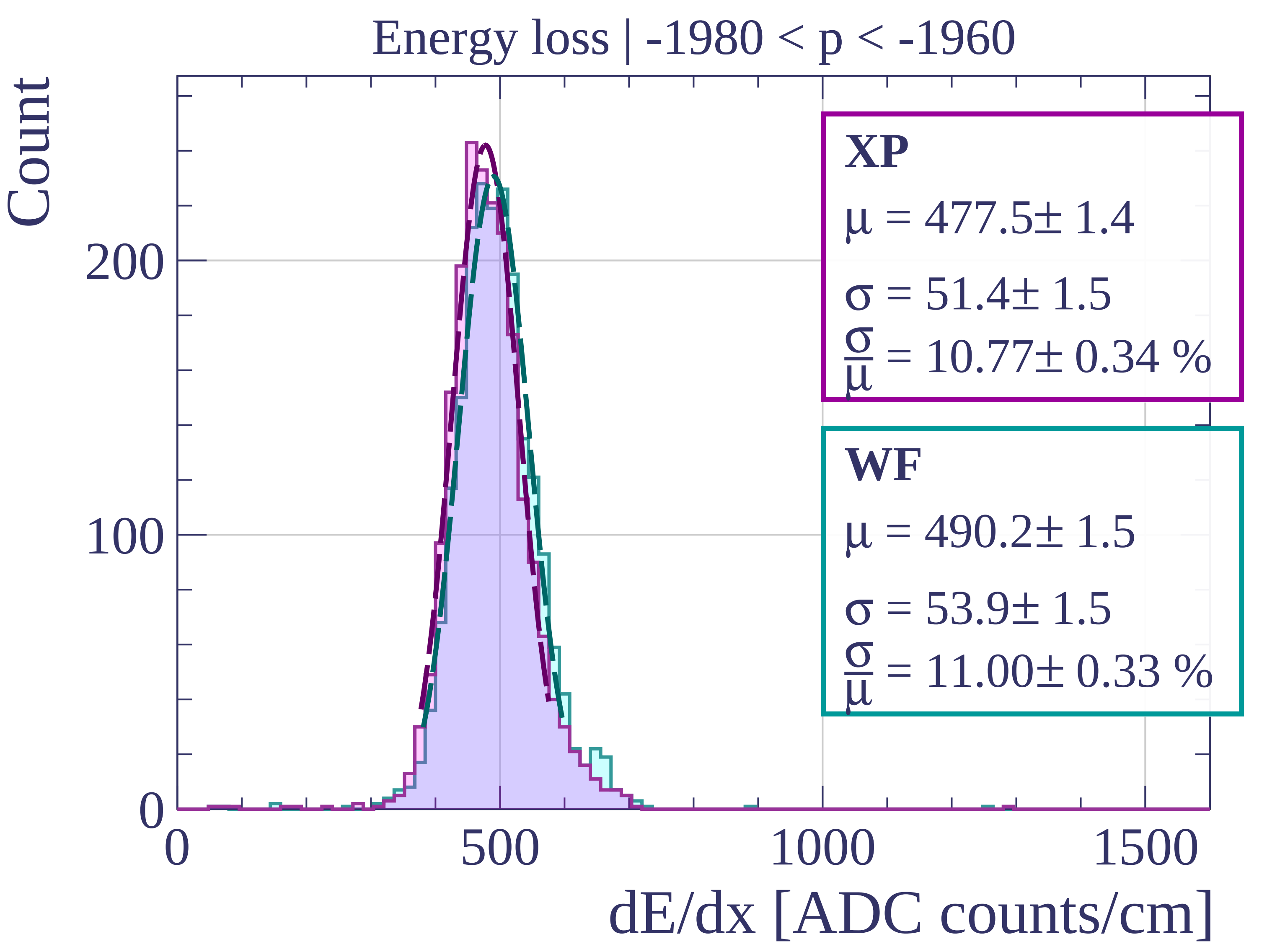

Energy loss | 1840 < p < 1920
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
1000
2000
XP
μ = 475.8± 0.4
σ = 50.6± 0.4
σ/μ = 10.64± 0.10 %
WF
μ = 487.9± 0.5
σ = 53.5± 0.5
σ/μ = 10.97± 0.11 %

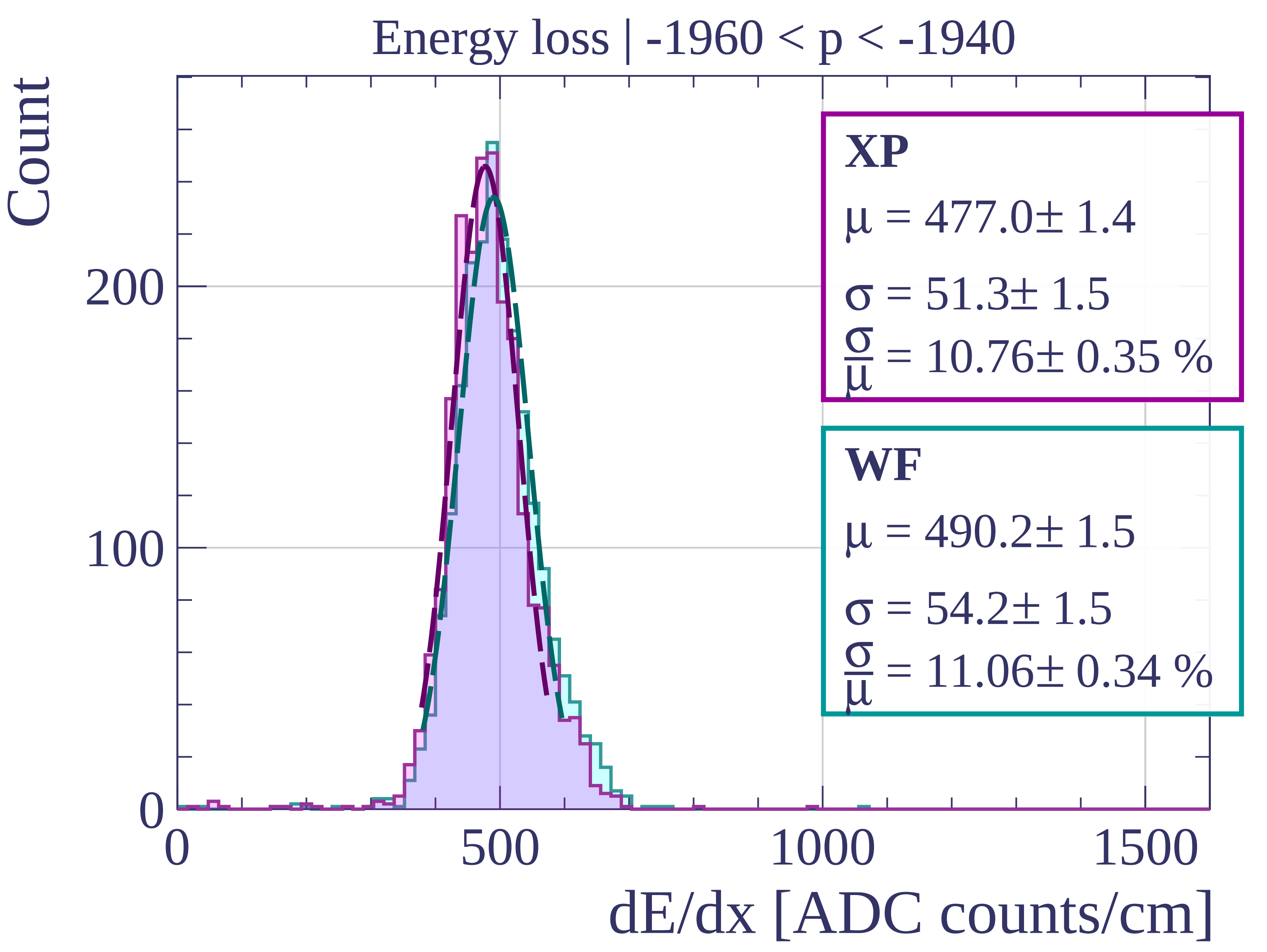


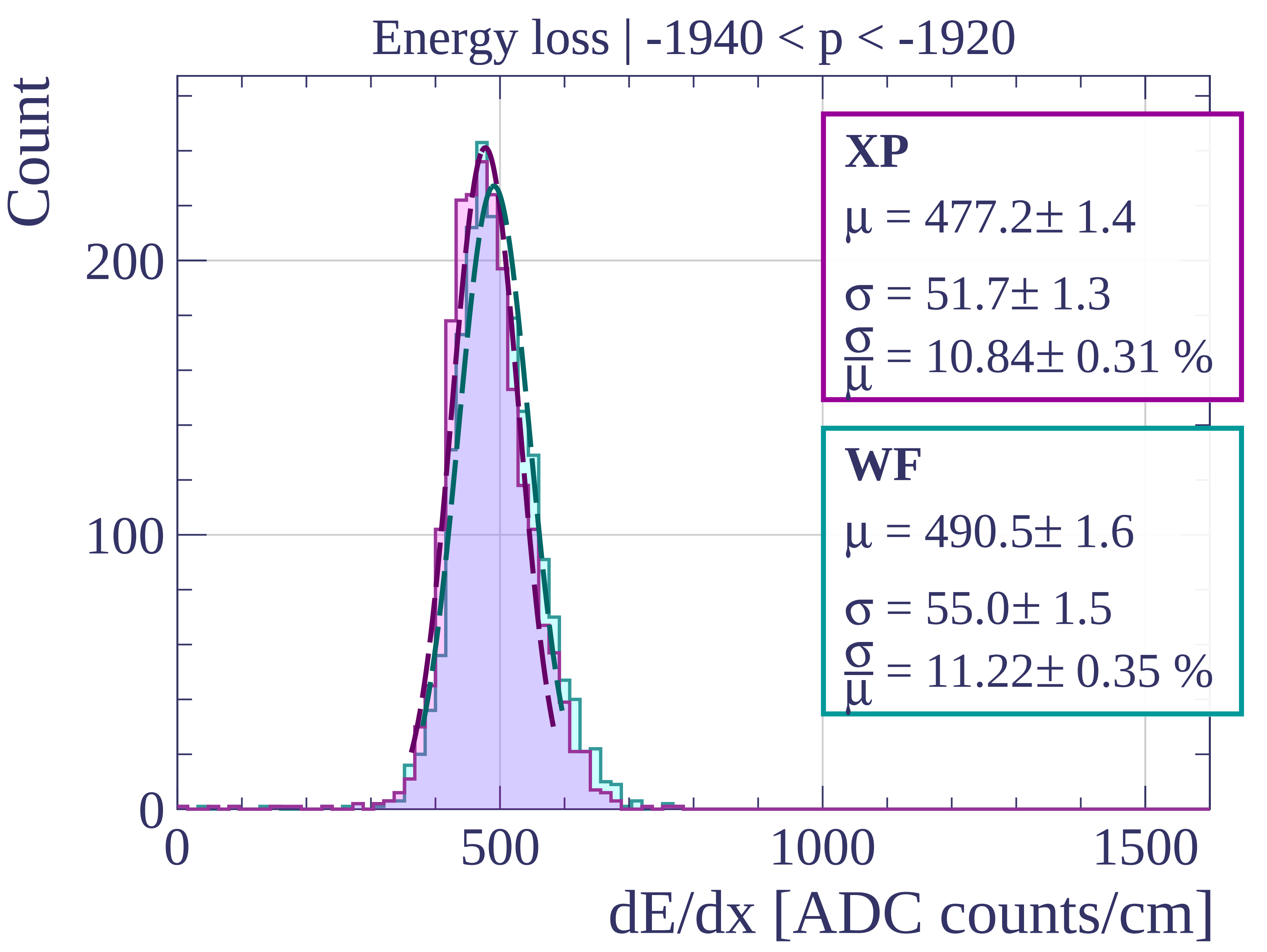
Energy loss | 1920 < p < 2000
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
1000
2000
XP
μ = 477.4± 0.4
σ = 49.7± 0.4
σ/μ = 10.42± 0.09 %
WF
μ = 490.4± 0.5
σ = 53.0± 0.5
σ/μ = 10.81± 0.10 %

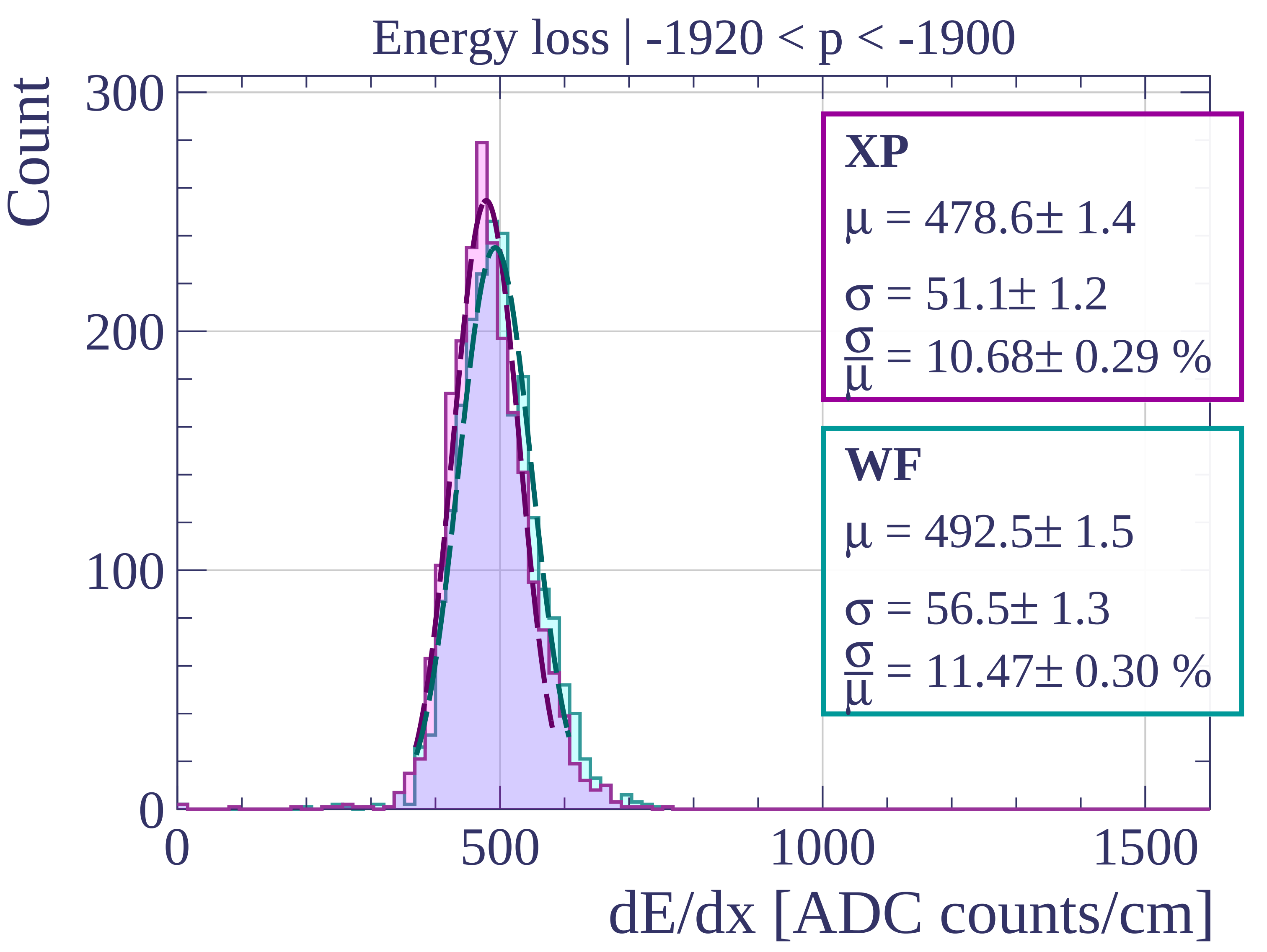


Energy loss | -2000 < p < -1980
Count
0
50
100
150
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 476.5± 1.8
σ = 48.6± 1.6
σ/μ = 10.21± 0.37 %
WF
μ = 488.9± 2.0
σ = 52.5± 1.9
σ/μ = 10.75± 0.44 %

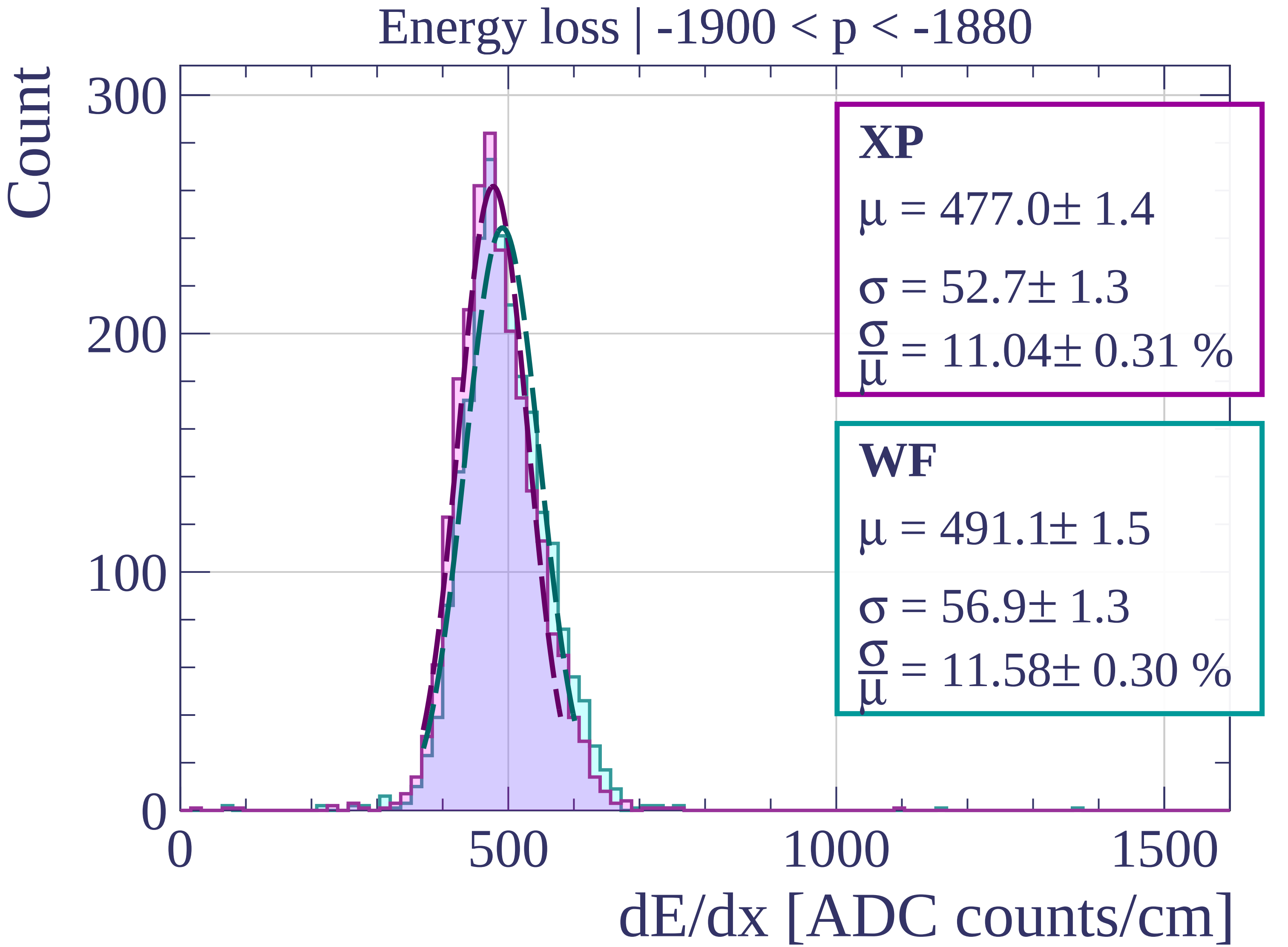

Energy loss | -1980 < p < -1960
Count
dE/dx [ADC counts/cm]
0
100
200
0
500
1000
1500
XP
μ = 477.5± 1.4
σ = 51.4± 1.5
σ/μ = 10.77± 0.34 %
WF
μ = 490.2± 1.5
σ = 53.9± 1.5
σ/μ = 11.00± 0.33 %


Energy loss | -1960 < p < -1940
Count
dE/dx [ADC counts/cm]
0
100
200
0
500
1000
1500
XP
μ = 477.0± 1.4
σ = 51.3± 1.5
σ/μ = 10.76± 0.35 %
WF
μ = 490.2± 1.5
σ = 54.2± 1.5
σ/μ = 11.06± 0.34 %

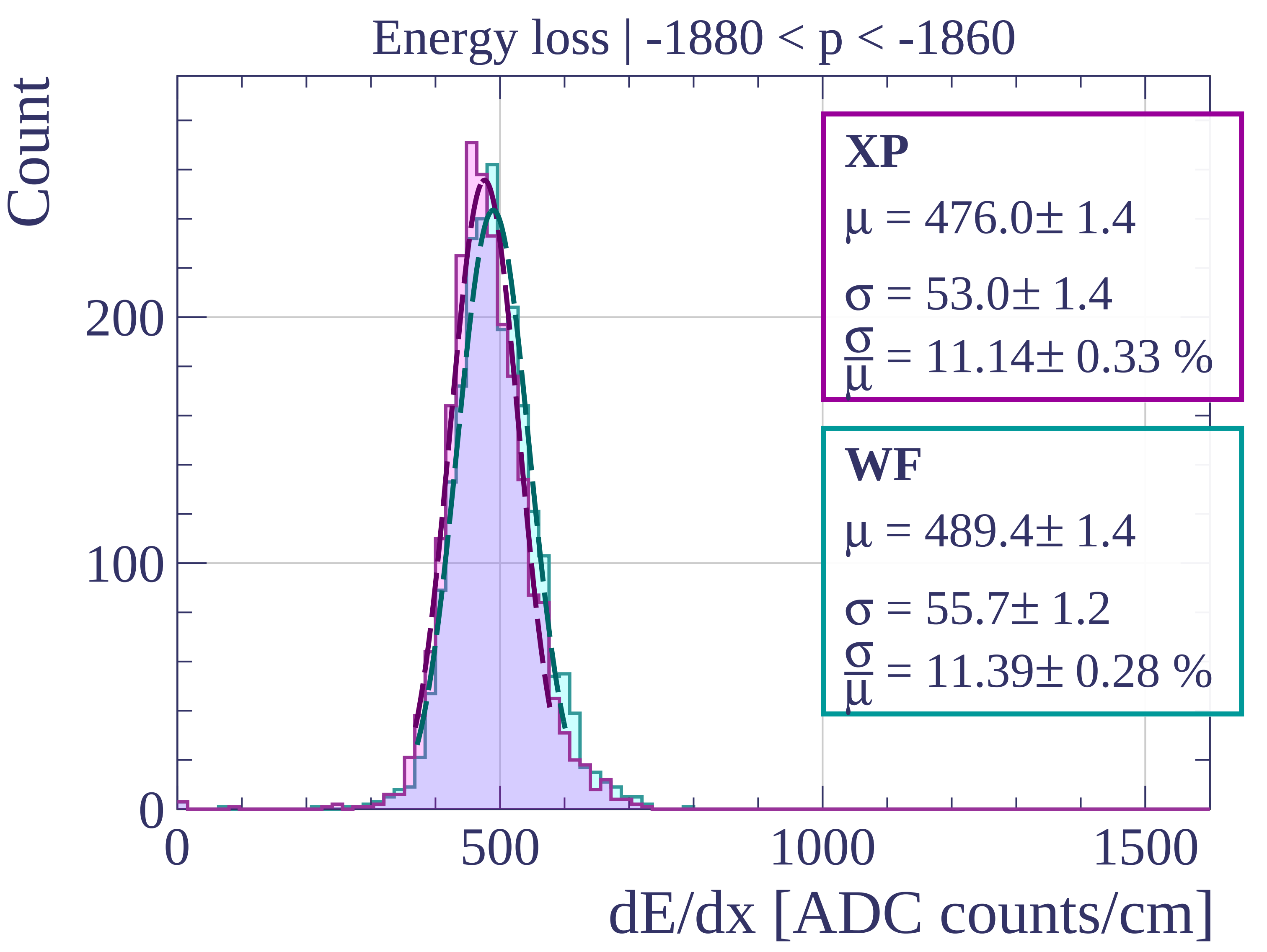

Energy loss | -1940 < p < -1920
Count
0
100
200
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 477.2± 1.4
σ = 51.7± 1.3
σ/μ = 10.84± 0.31 %
WF
μ = 490.5± 1.6
σ = 55.0± 1.5
σ/μ = 11.22± 0.35 %

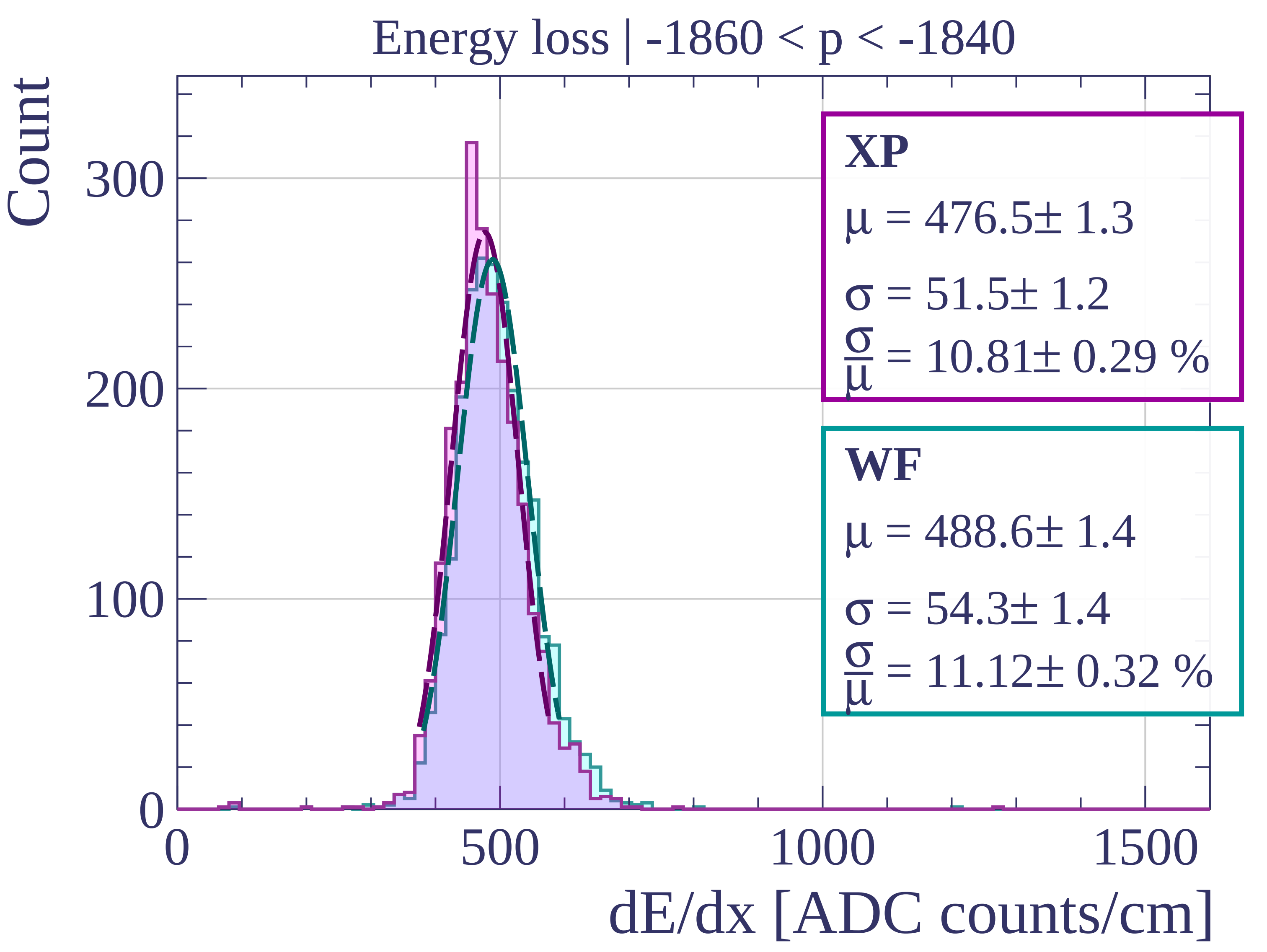


Energy loss | -1920 < p < -1900
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 478.6± 1.4
σ = 51.1± 1.2
σ/μ = 10.68± 0.29 %
WF
μ = 492.5± 1.5
σ = 56.5± 1.3
σ/μ = 11.47± 0.30 %

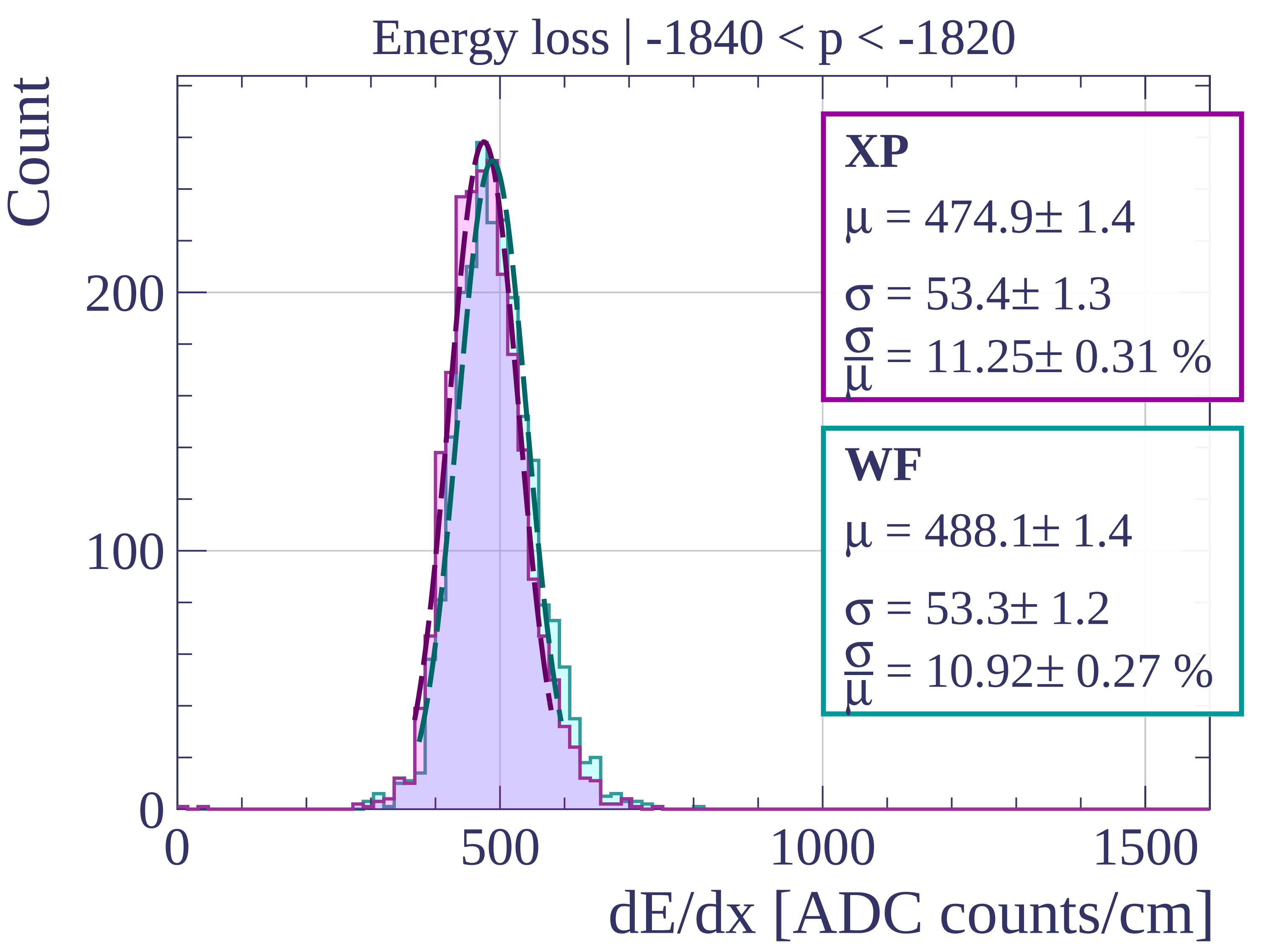


Energy loss | -1900 < p < -1880
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 477.0± 1.4
σ = 52.7± 1.3
σ/μ = 11.04± 0.31 %
WF
μ = 491.1± 1.5
σ = 56.9± 1.3
σ/μ = 11.58± 0.30 %

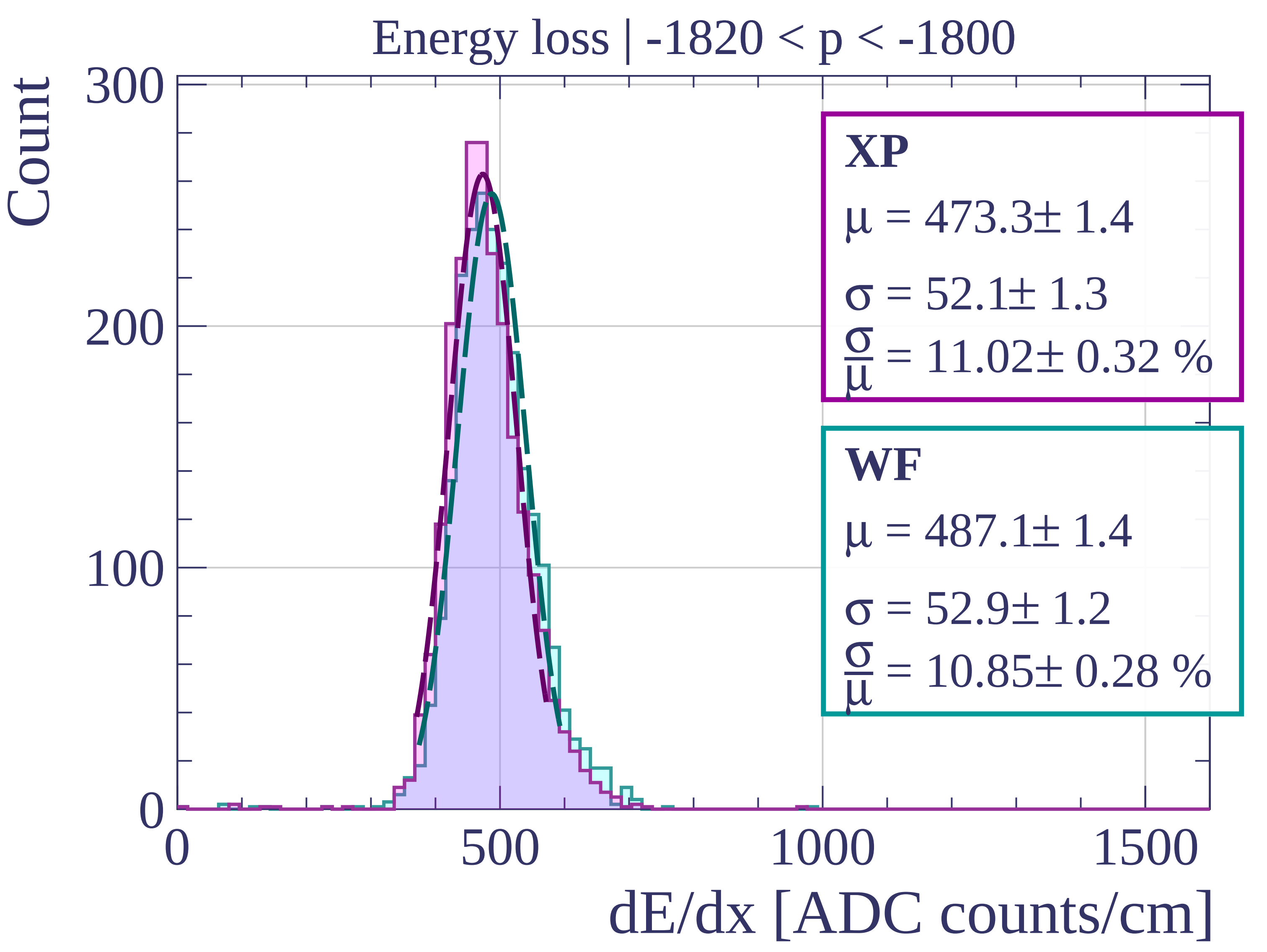

Energy loss | -1880 < p < -1860
Count
dE/dx [ADC counts/cm]
0
100
200
0
500
1000
1500
XP
μ = 476.0± 1.4
σ = 53.0± 1.4
σ/μ = 11.14± 0.33 %
WF
μ = 489.4± 1.4
σ = 55.7± 1.2
σ/μ = 11.39± 0.28 %

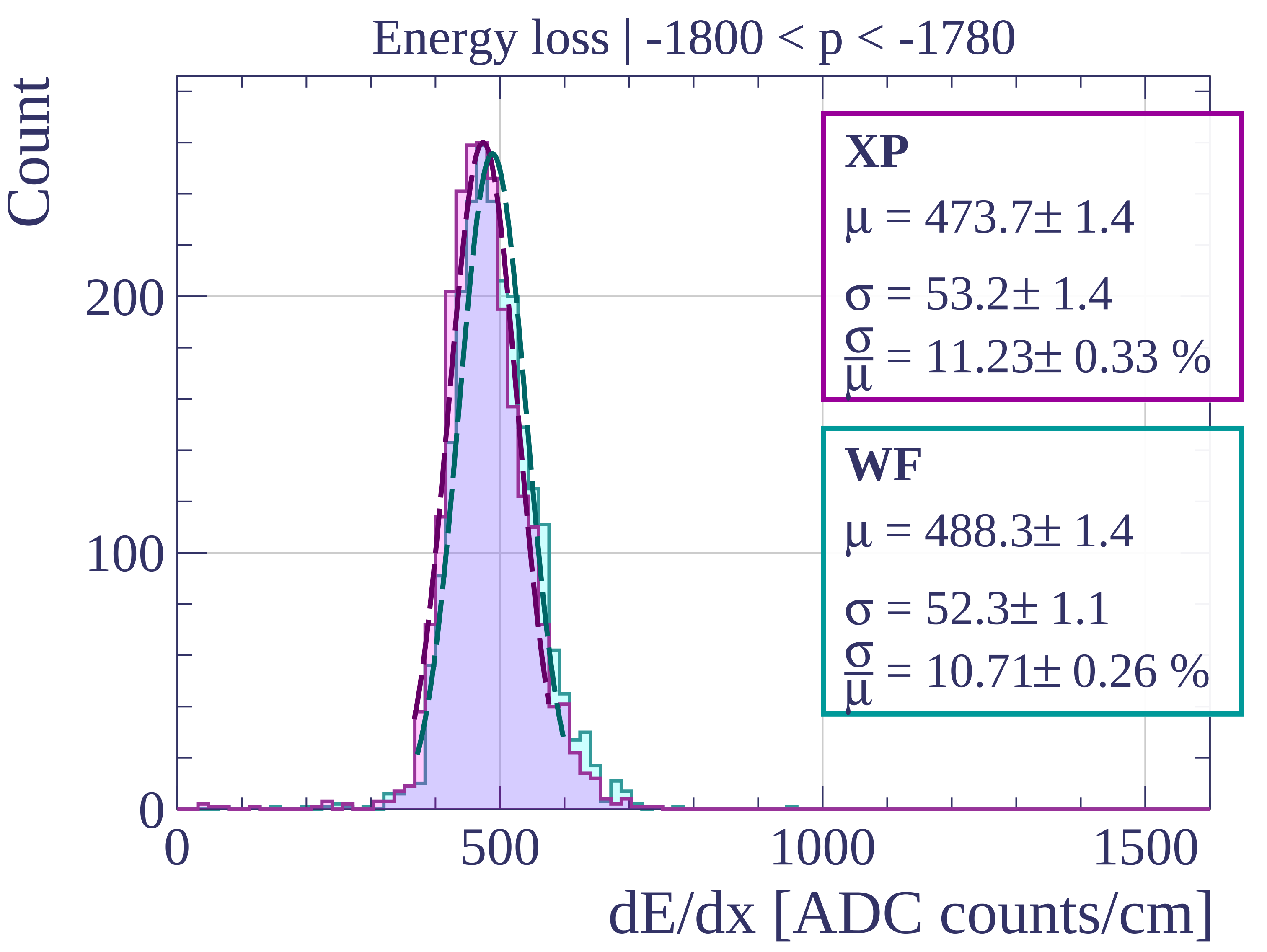


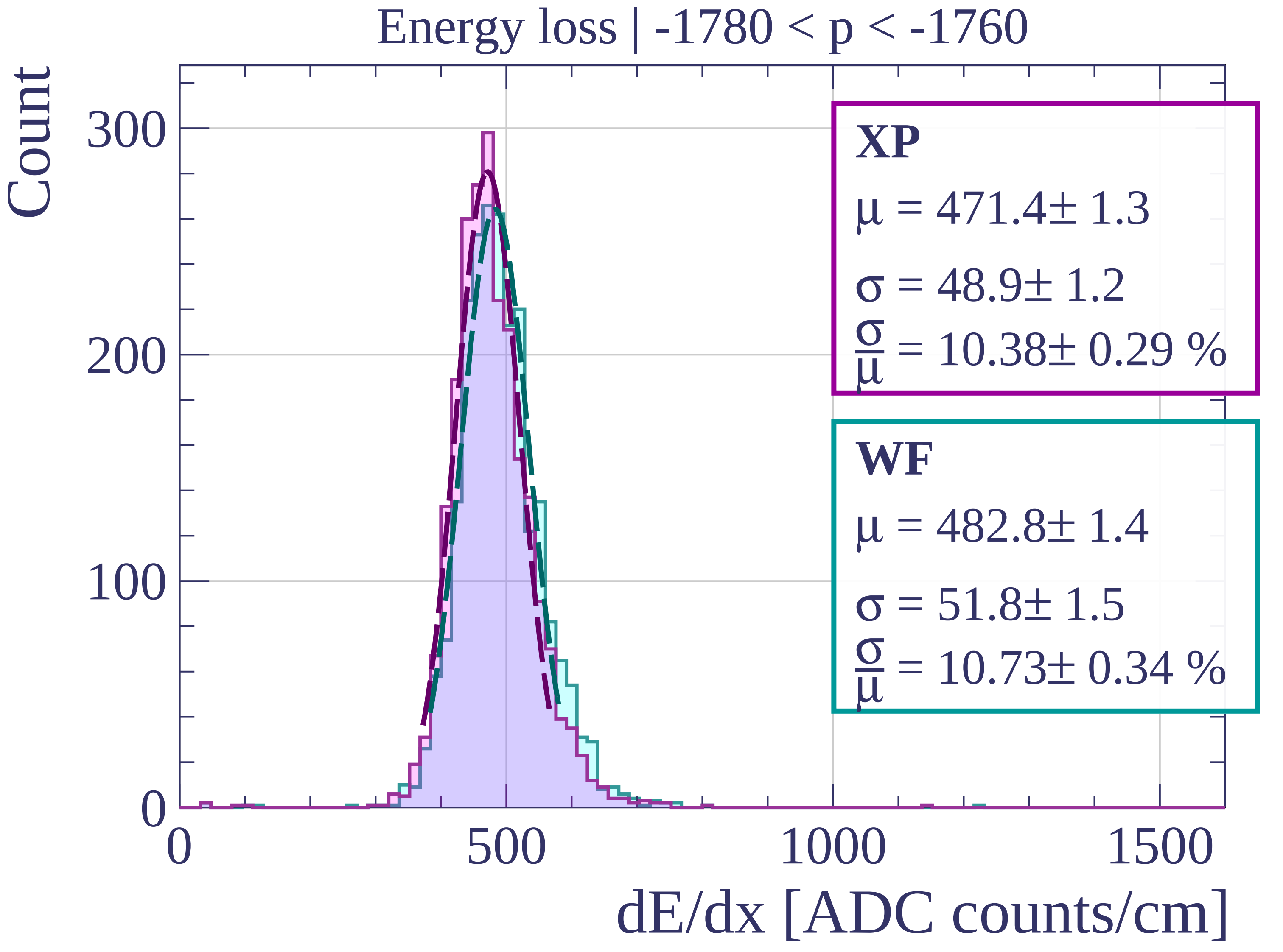
Energy loss | -1860 < p < -1840
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 476.5± 1.3
σ = 51.5± 1.2
σ/μ = 10.81± 0.29 %
WF
μ = 488.6± 1.4
σ = 54.3± 1.4
σ/μ = 11.12± 0.32 %



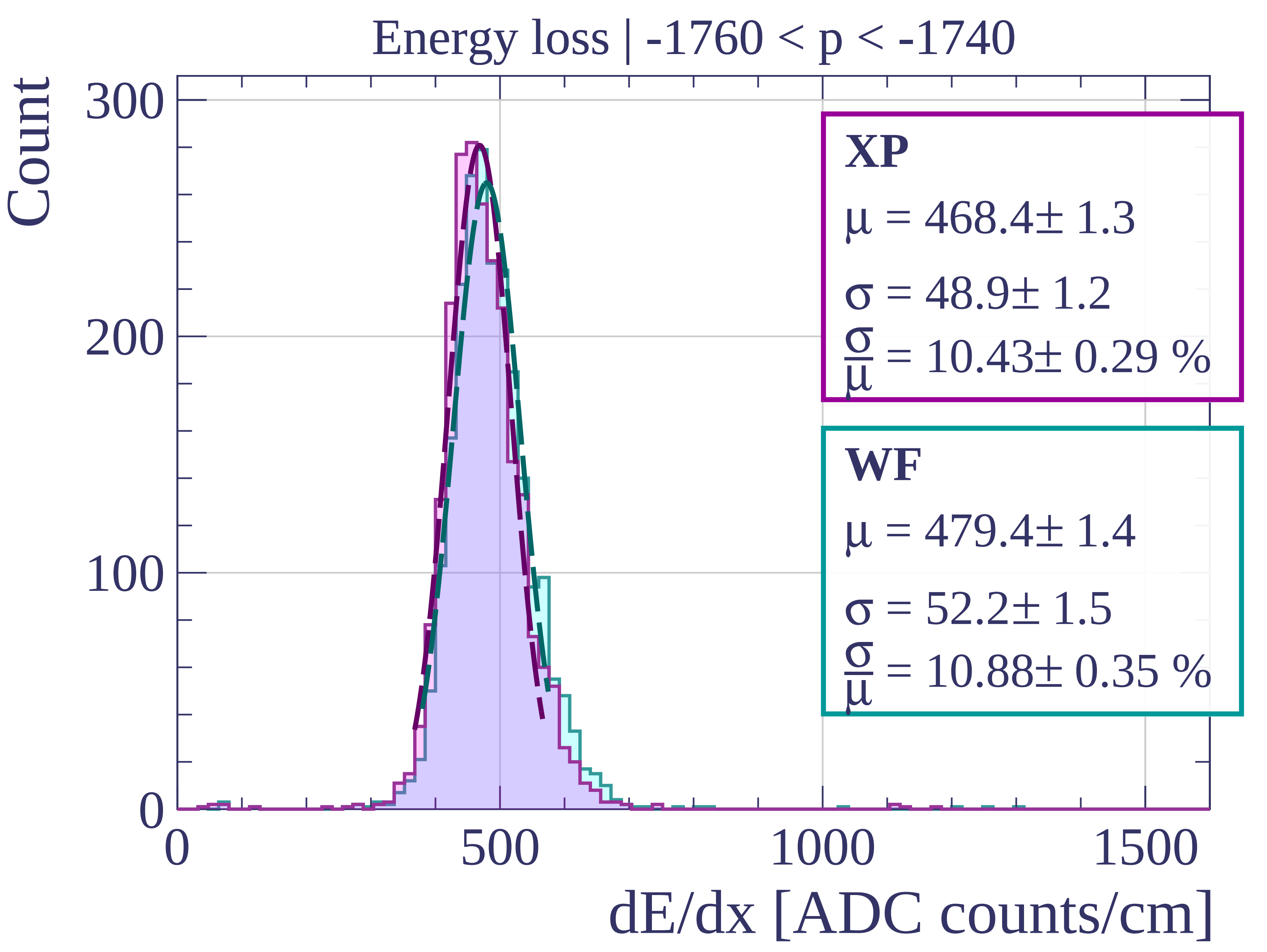
Energy loss | -1840 < p < -1820
Count
dE/dx [ADC counts/cm]
0
100
200
0
500
1000
1500
XP
μ = 474.9± 1.4
σ = 53.4± 1.3
σ/μ = 11.25± 0.31 %
WF
μ = 488.1± 1.4
σ = 53.3± 1.2
σ/μ = 10.92± 0.27 %

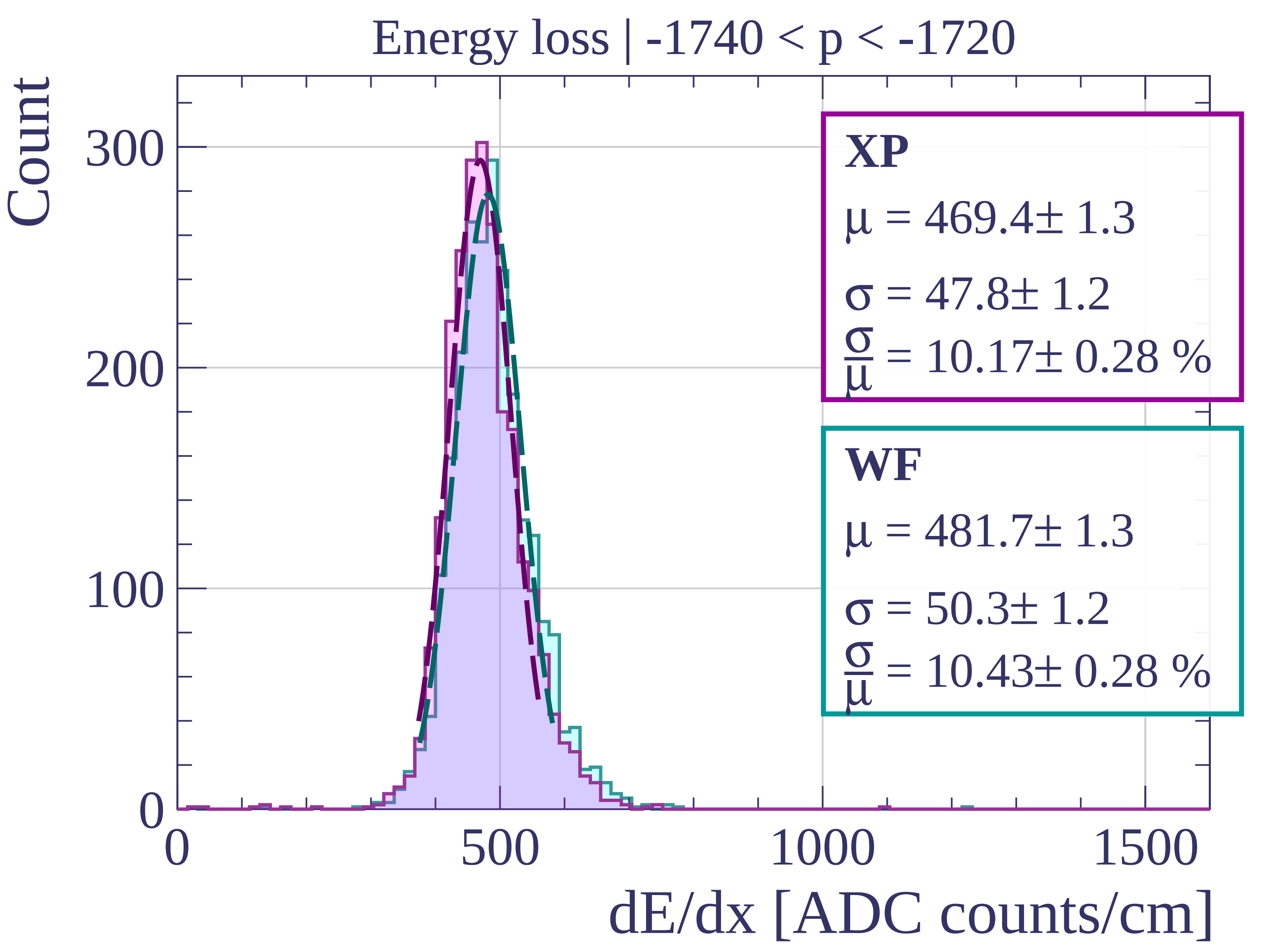

Energy loss | -1820 < p < -1800
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 473.3± 1.4
σ = 52.1± 1.3
σ/μ = 11.02± 0.32 %
WF
μ = 487.1± 1.4
σ = 52.9± 1.2
σ/μ = 10.85± 0.28 %


Energy loss | -1800 < p < -1780
Count
dE/dx [ADC counts/cm]
0
100
200
0
500
1000
1500
XP
μ = 473.7± 1.4
σ = 53.2± 1.4
σ/μ = 11.23± 0.33 %
WF
μ = 488.3± 1.4
σ = 52.3± 1.1
σ/μ = 10.71± 0.26 %

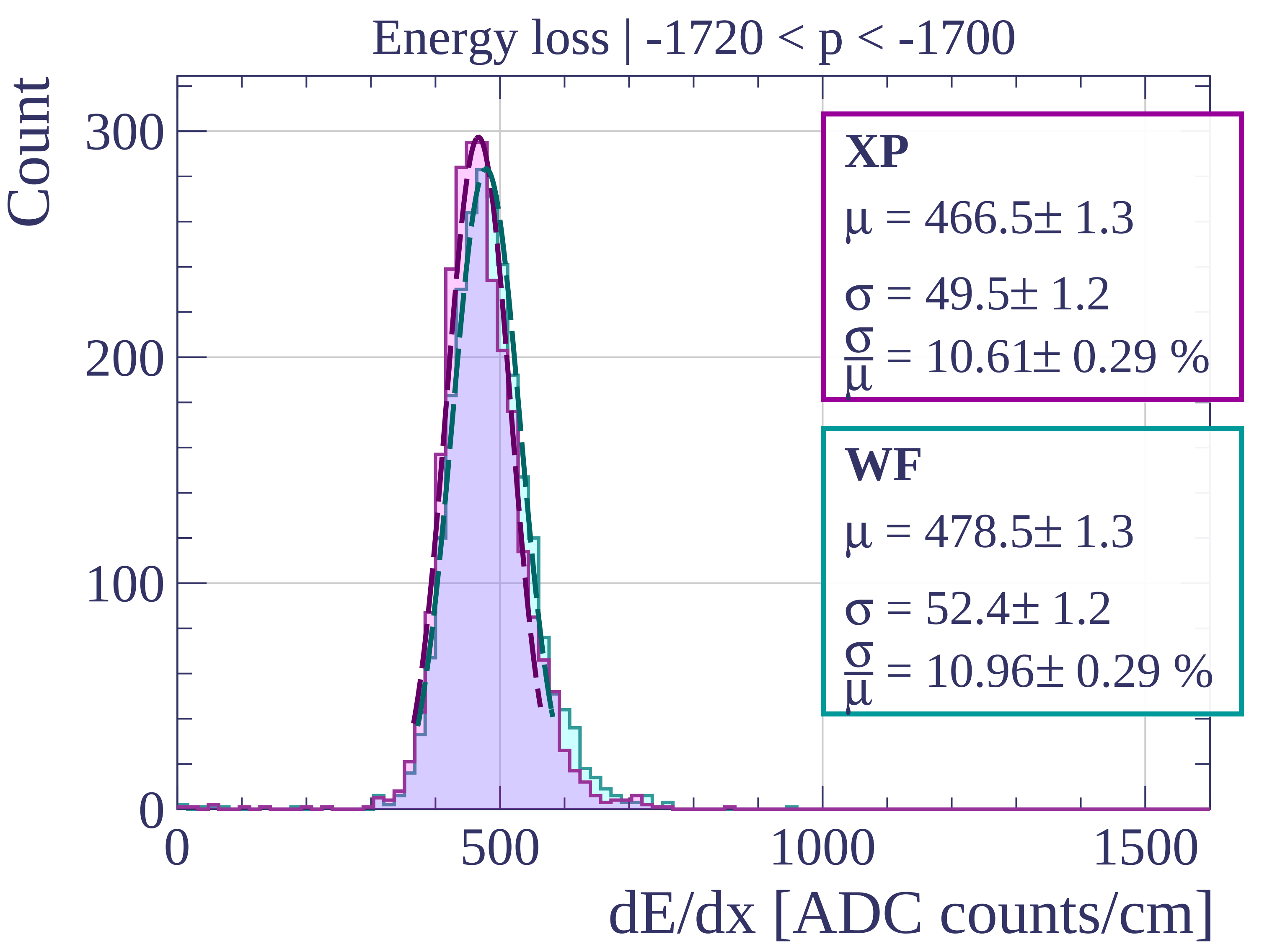


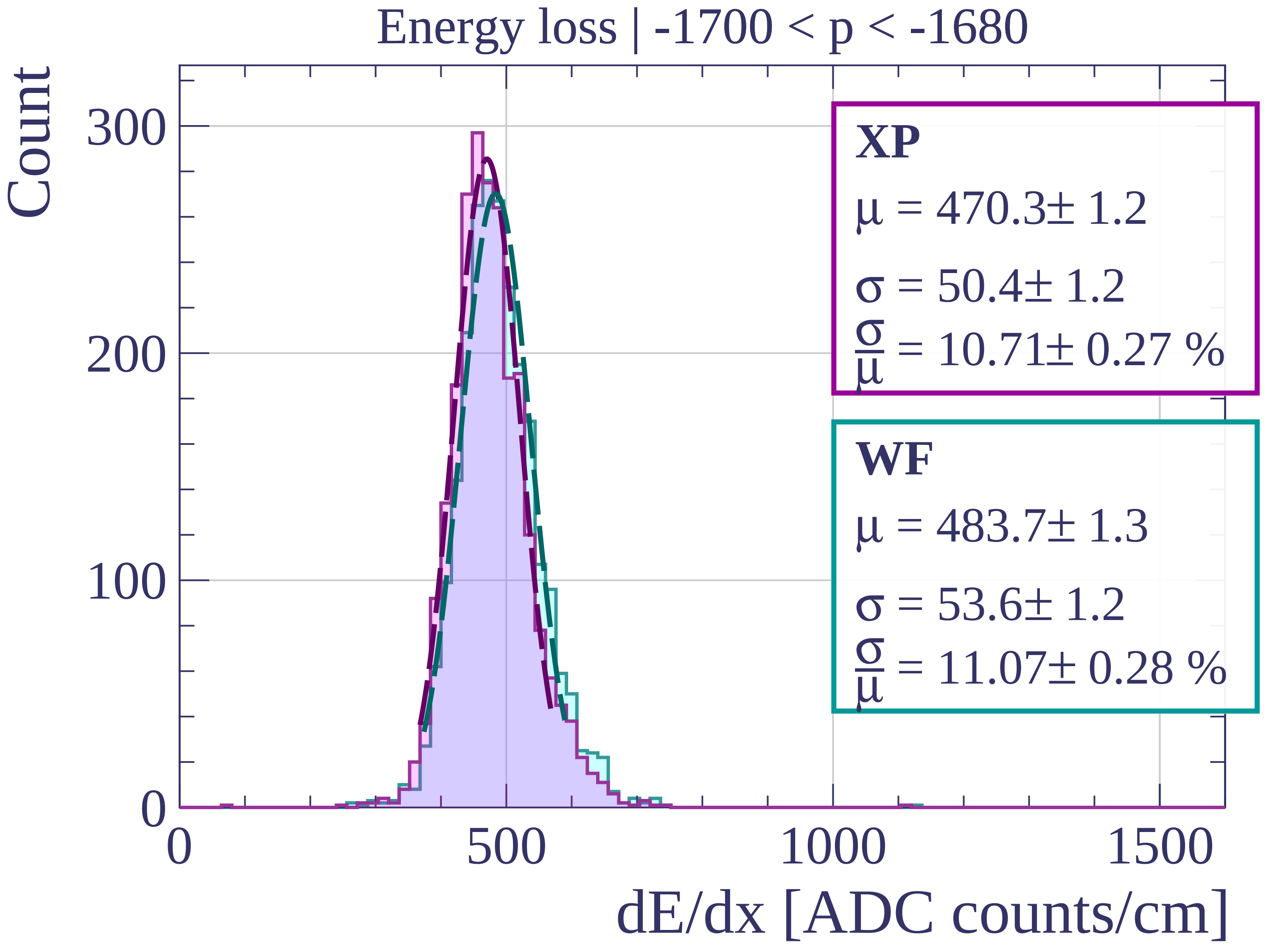
Energy loss | -1780 < p < -1760
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 471.4± 1.3
σ = 48.9± 1.2
σ/μ = 10.38± 0.29 %
WF
μ = 482.8± 1.4
σ = 51.8± 1.5
σ/μ = 10.73± 0.34 %

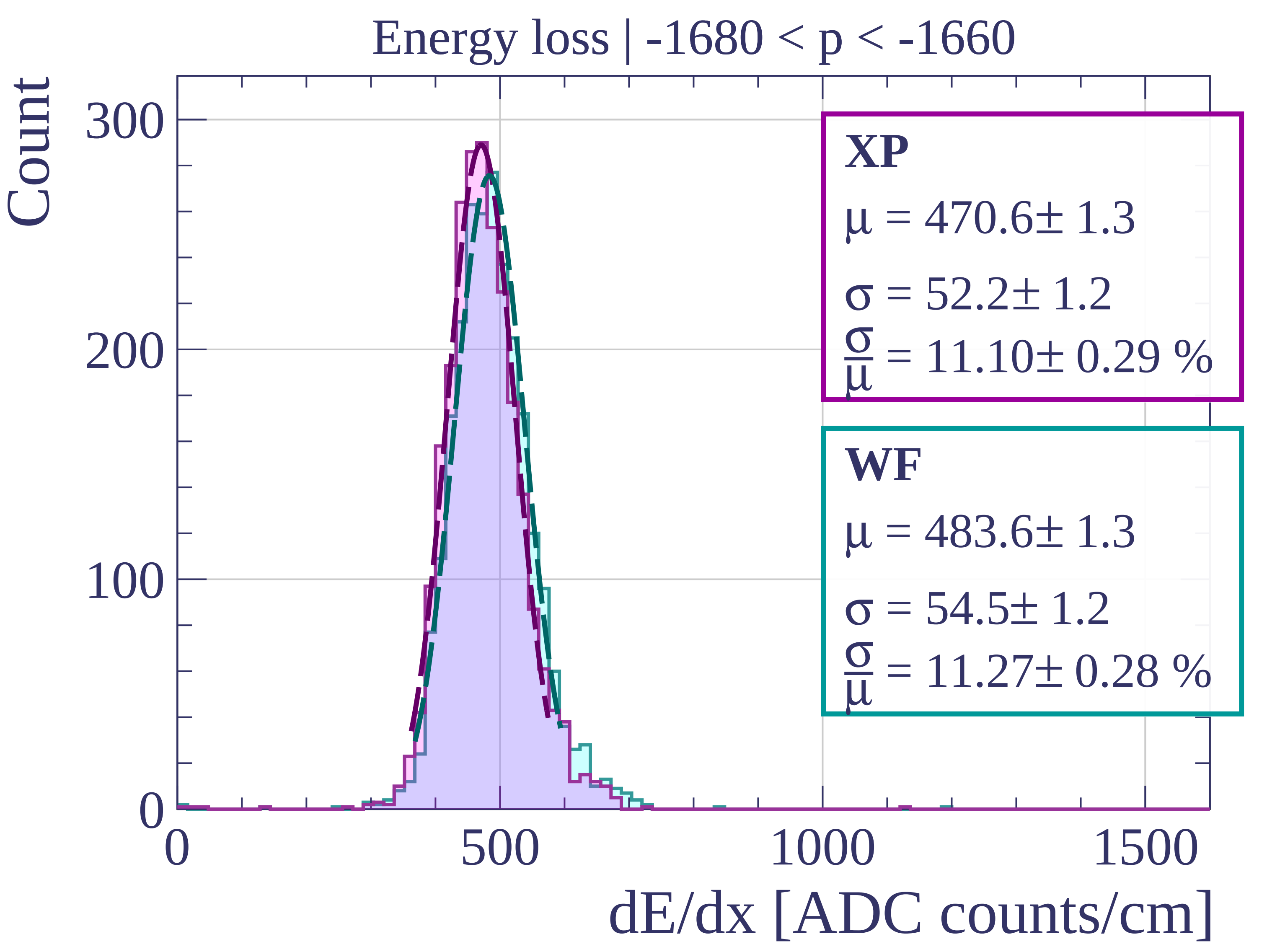

Energy loss | -1760 < p < -1740
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 468.4± 1.3
σ = 48.9± 1.2
σ/μ = 10.43± 0.29 %
WF
μ = 479.4± 1.4
σ = 52.2± 1.5
σ/μ = 10.88± 0.35 %

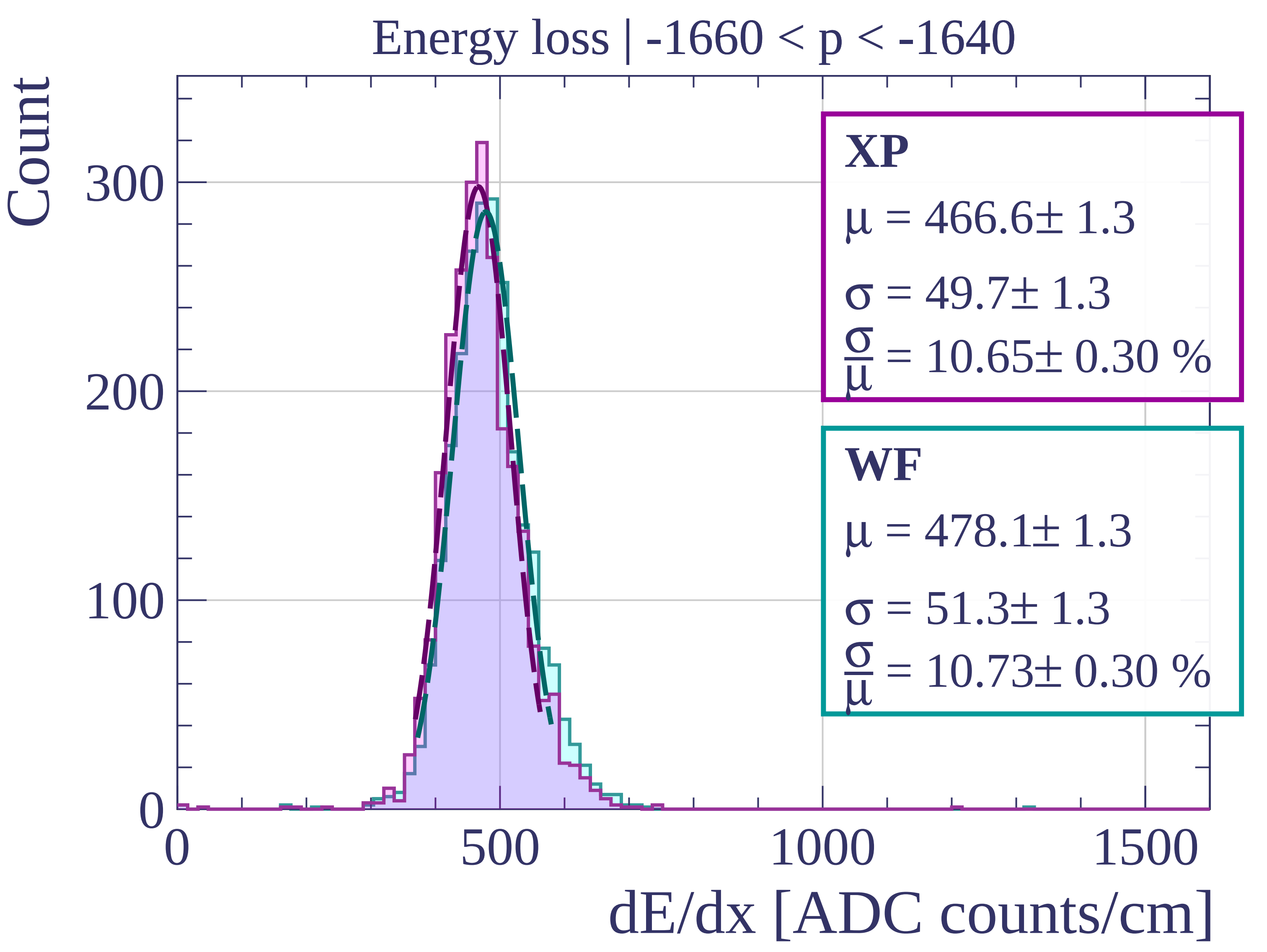


Energy loss | -1740 < p < -1720
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 469.4± 1.3
σ = 47.8± 1.2
σ/μ = 10.17± 0.28 %
WF
μ = 481.7± 1.3
σ = 50.3± 1.2
σ/μ = 10.43± 0.28 %

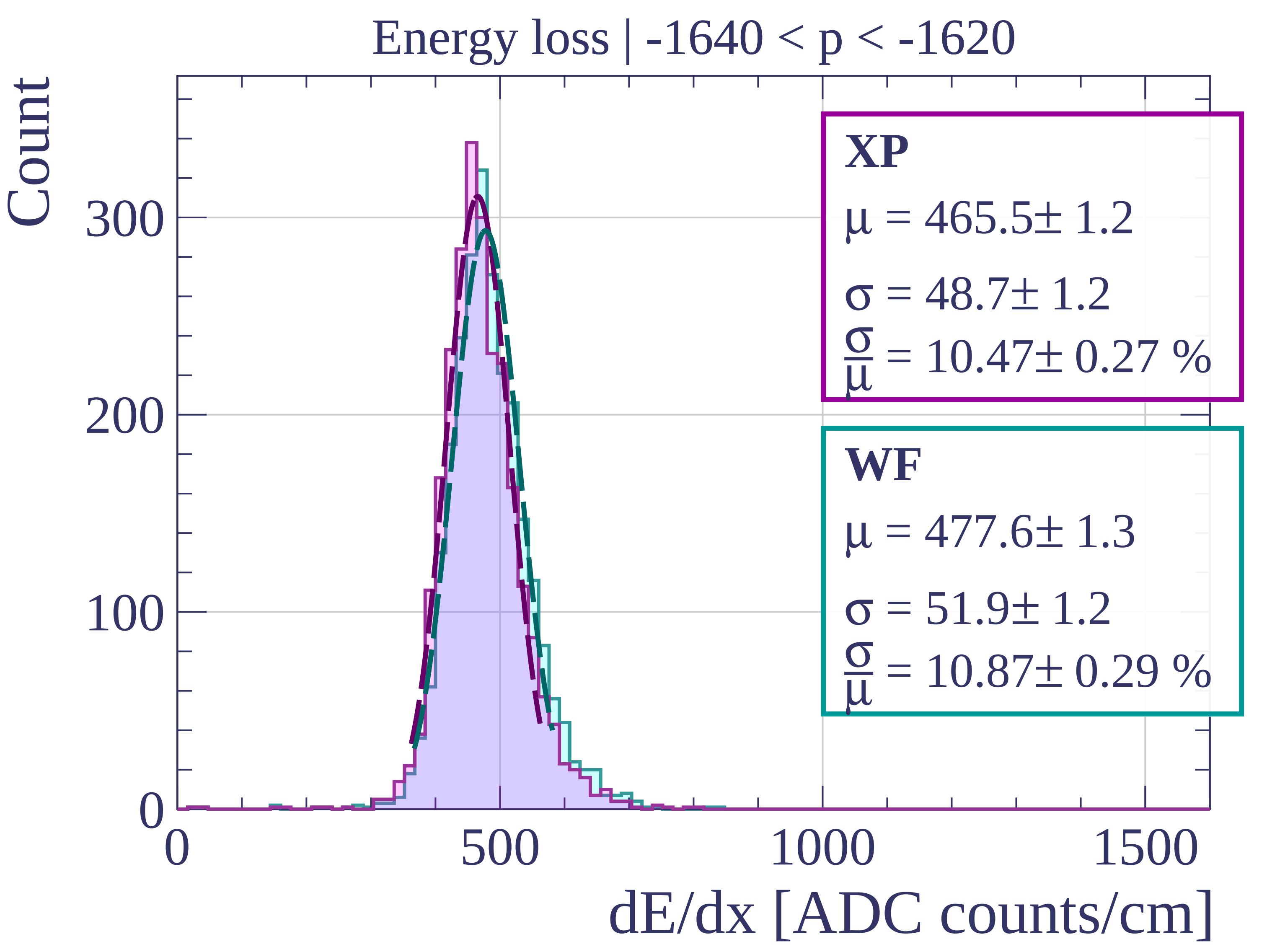

Energy loss | -1720 < p < -1700
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 466.5± 1.3
σ = 49.5± 1.2
σ/μ = 10.61± 0.29 %
WF
μ = 478.5± 1.3
σ = 52.4± 1.2
σ/μ = 10.96± 0.29 %


Energy loss | -1700 < p < -1680
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 470.3± 1.2
σ = 50.4± 1.2
σ/μ = 10.71± 0.27 %
WF
μ = 483.7± 1.3
σ = 53.6± 1.2
σ/μ = 11.07± 0.28 %

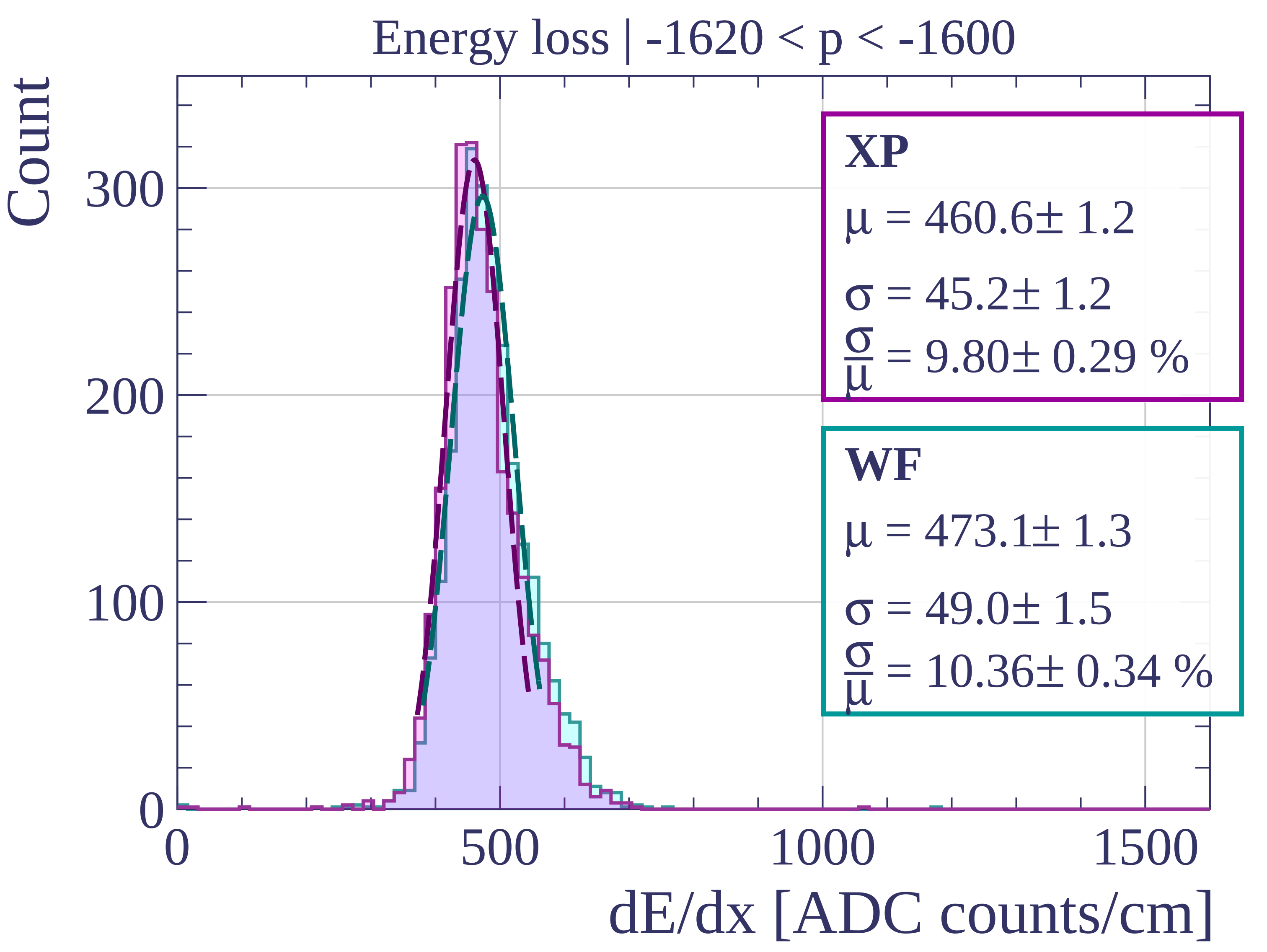

Energy loss | -1680 < p < -1660
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 470.6± 1.3
σ = 52.2± 1.2
σ/μ = 11.10± 0.29 %
WF
μ = 483.6± 1.3
σ = 54.5± 1.2
σ/μ = 11.27± 0.28 %

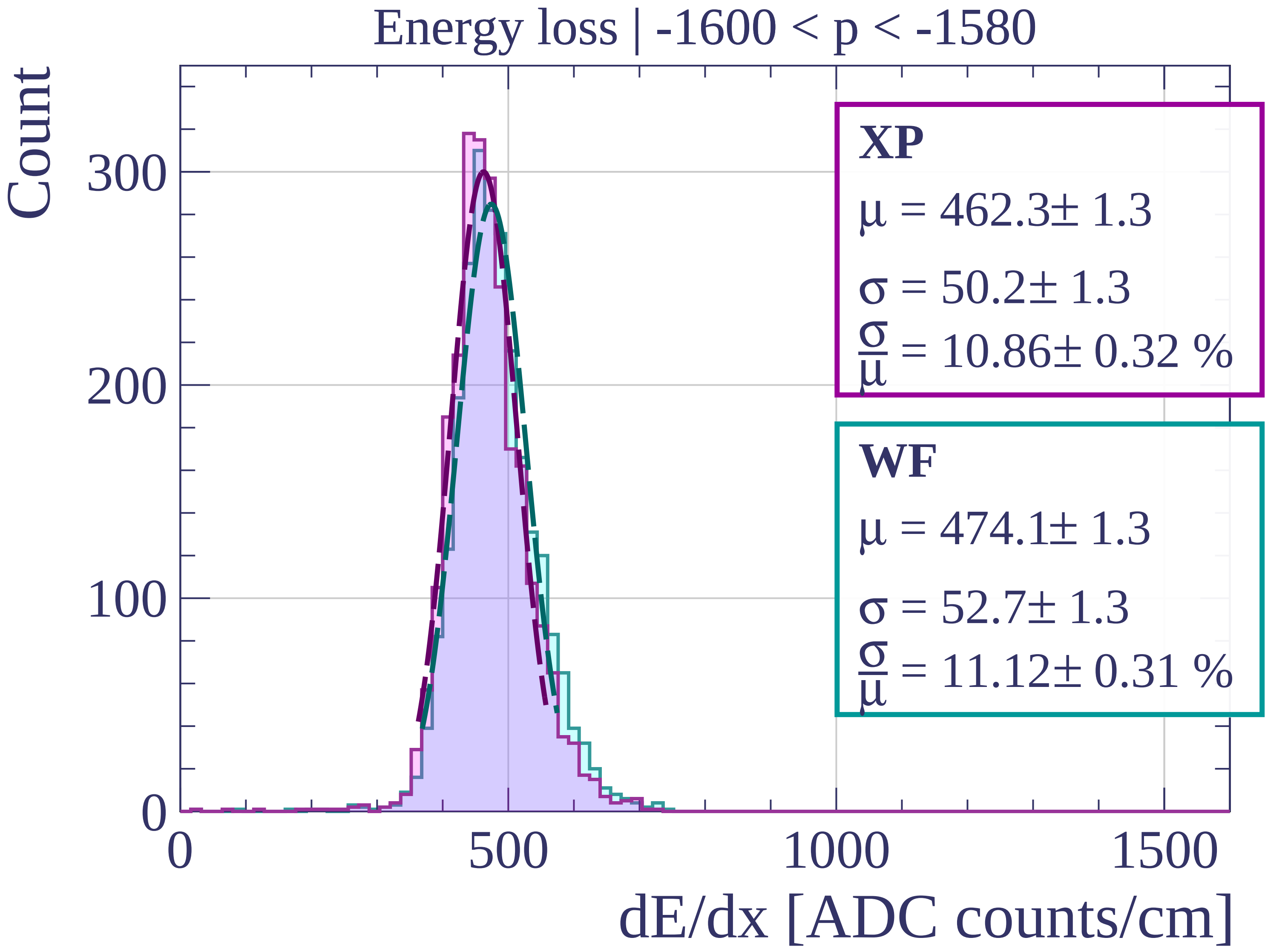


Energy loss | -1660 < p < -1640
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 466.6± 1.3
σ = 49.7± 1.3
σ/μ = 10.65± 0.30 %
WF
μ = 478.1± 1.3
σ = 51.3± 1.3
σ/μ = 10.73± 0.30 %

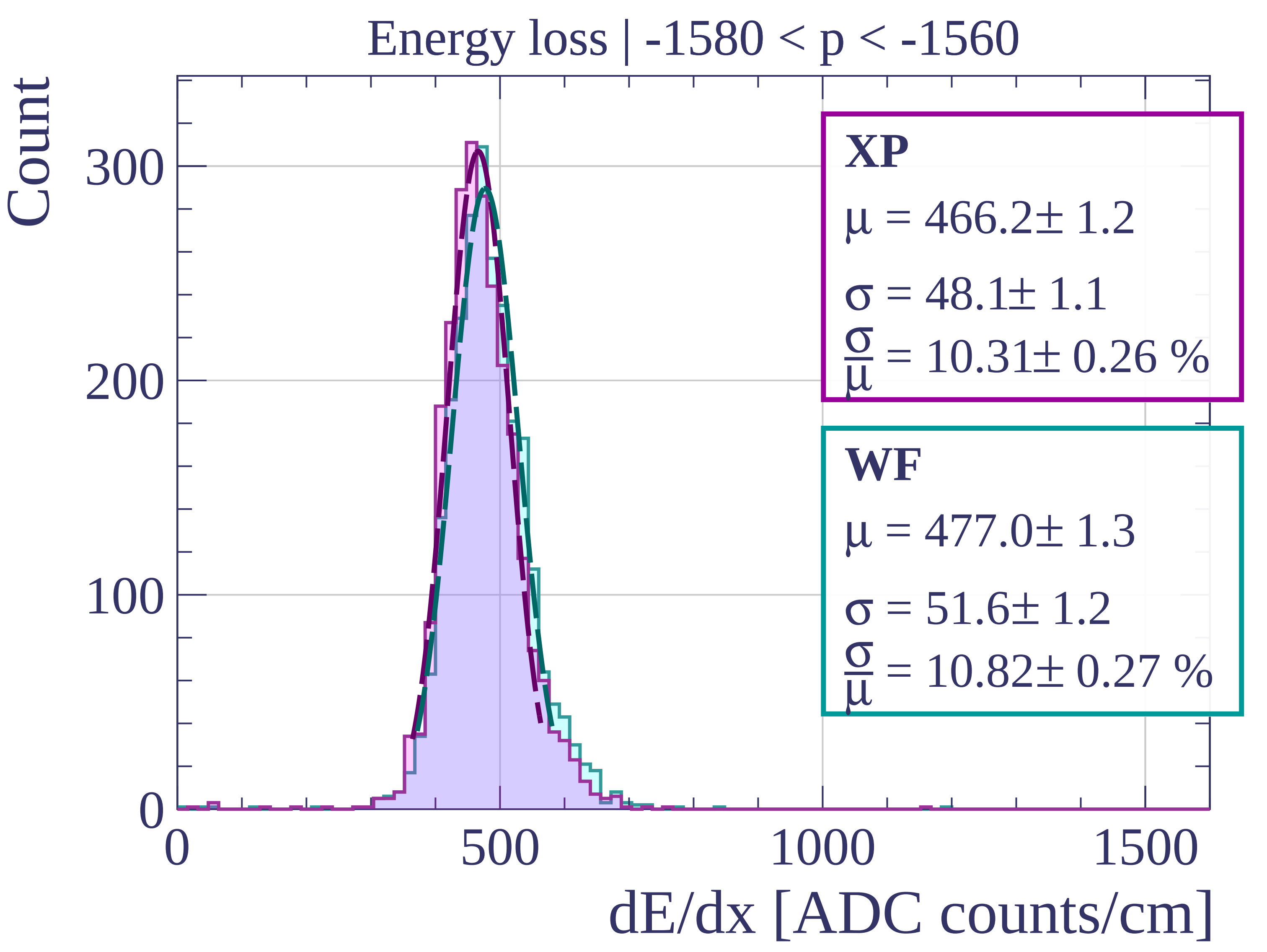


Energy loss | -1640 < p < -1620
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 465.5± 1.2
σ = 48.7± 1.2
σ/μ = 10.47± 0.27 %
WF
μ = 477.6± 1.3
σ = 51.9± 1.2
σ/μ = 10.87± 0.29 %

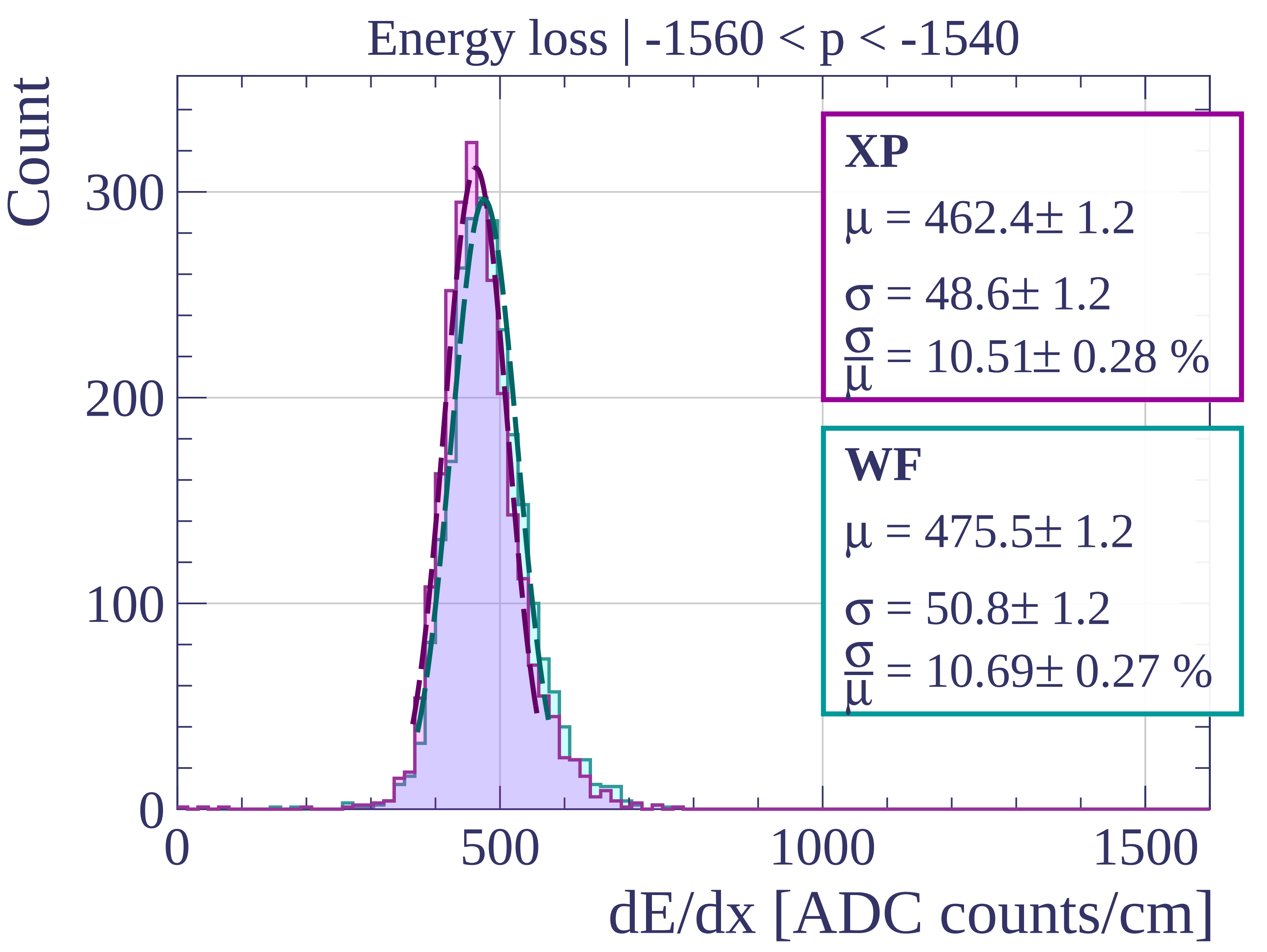

Energy loss | -1620 < p < -1600
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 460.6± 1.2
σ = 45.2± 1.2
σ/μ = 9.80± 0.29 %
WF
μ = 473.1± 1.3
σ = 49.0± 1.5
σ/μ = 10.36± 0.34 %



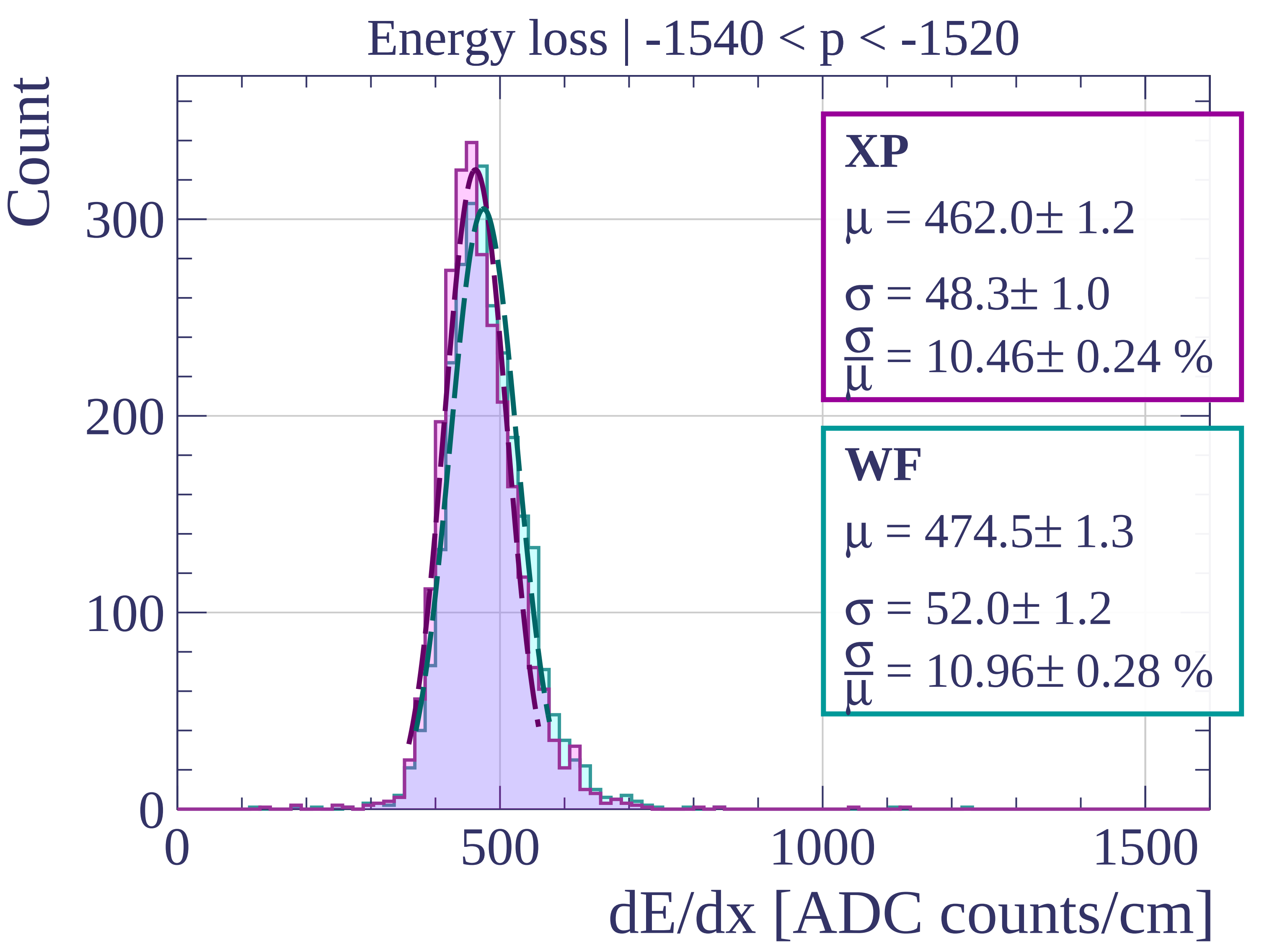
Energy loss | -1600 < p < -1580
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 462.3± 1.3
σ = 50.2± 1.3
σ/μ = 10.86± 0.32 %
WF
μ = 474.1± 1.3
σ = 52.7± 1.3
σ/μ = 11.12± 0.31 %



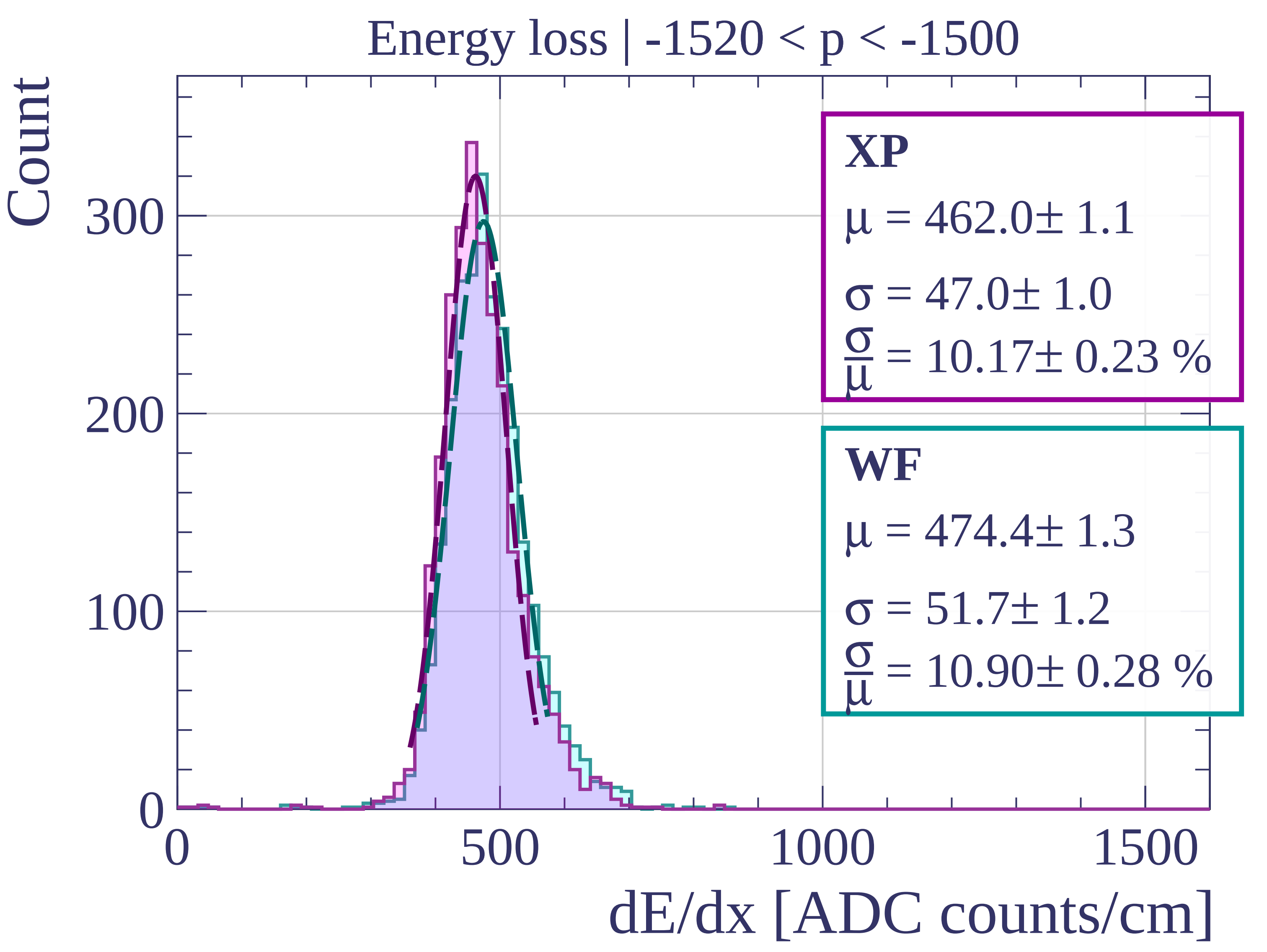
Energy loss | -1580 < p < -1560
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 466.2± 1.2
σ = 48.1± 1.1
σ/μ = 10.31± 0.26 %
WF
μ = 477.0± 1.3
σ = 51.6± 1.2
σ/μ = 10.82± 0.27 %

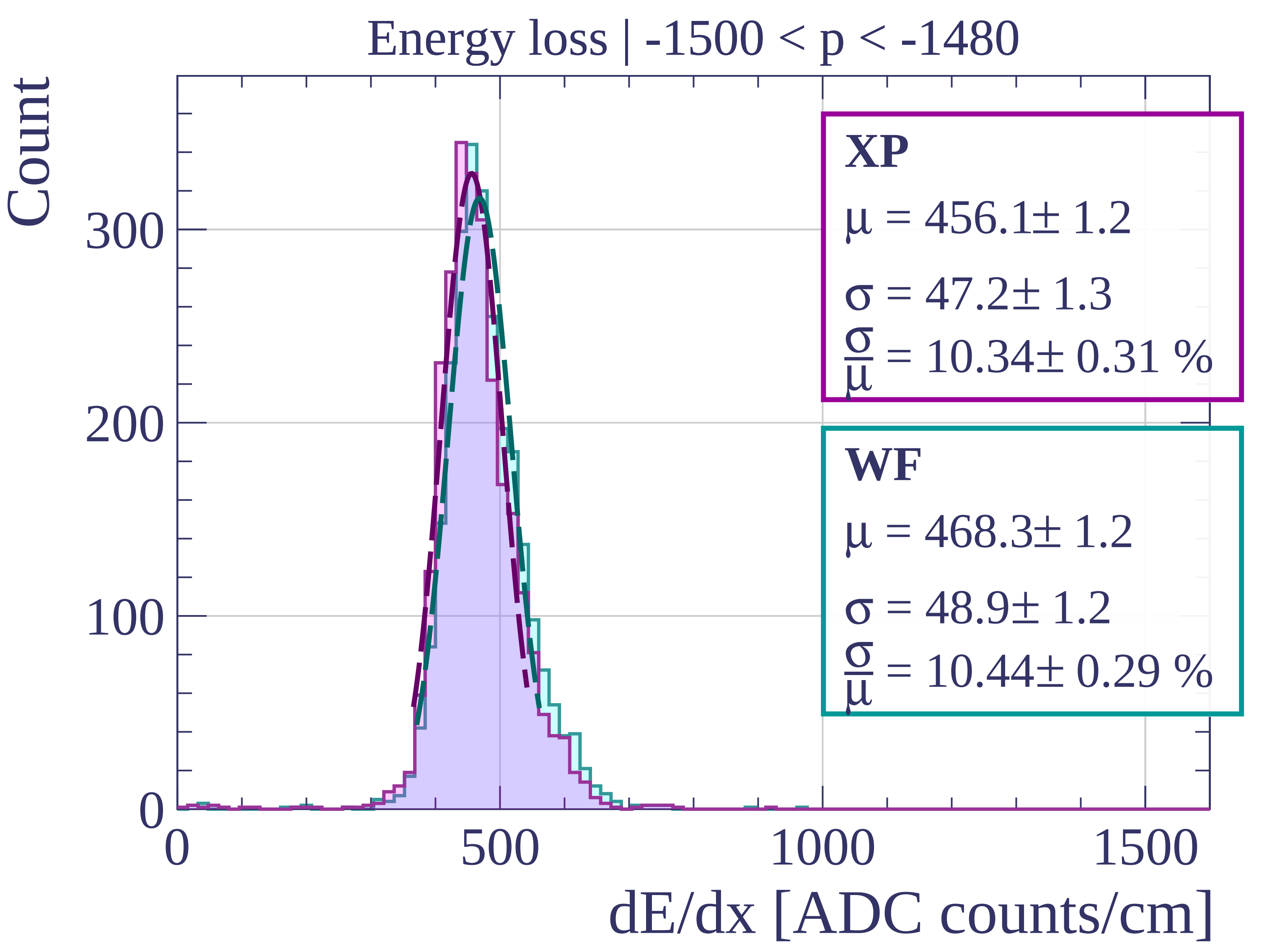


Energy loss | -1560 < p < -1540
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 462.4± 1.2
σ = 48.6± 1.2
σ/μ = 10.51± 0.28 %
WF
μ = 475.5± 1.2
σ = 50.8± 1.2
σ/μ = 10.69± 0.27 %

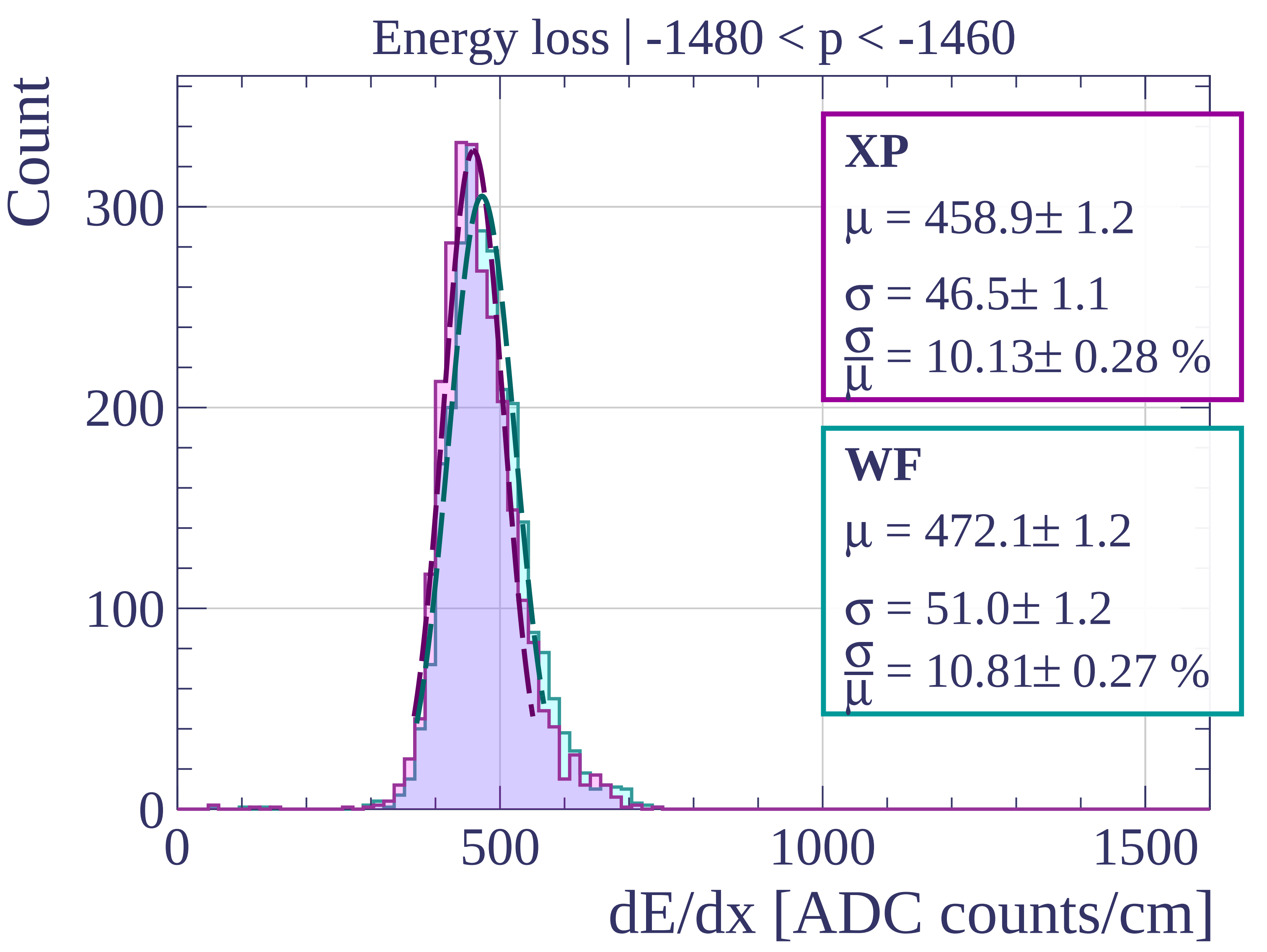


Energy loss | -1540 < p < -1520
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 462.0± 1.2
σ = 48.3± 1.0
σ/μ = 10.46± 0.24 %
WF
μ = 474.5± 1.3
σ = 52.0± 1.2
σ/μ = 10.96± 0.28 %

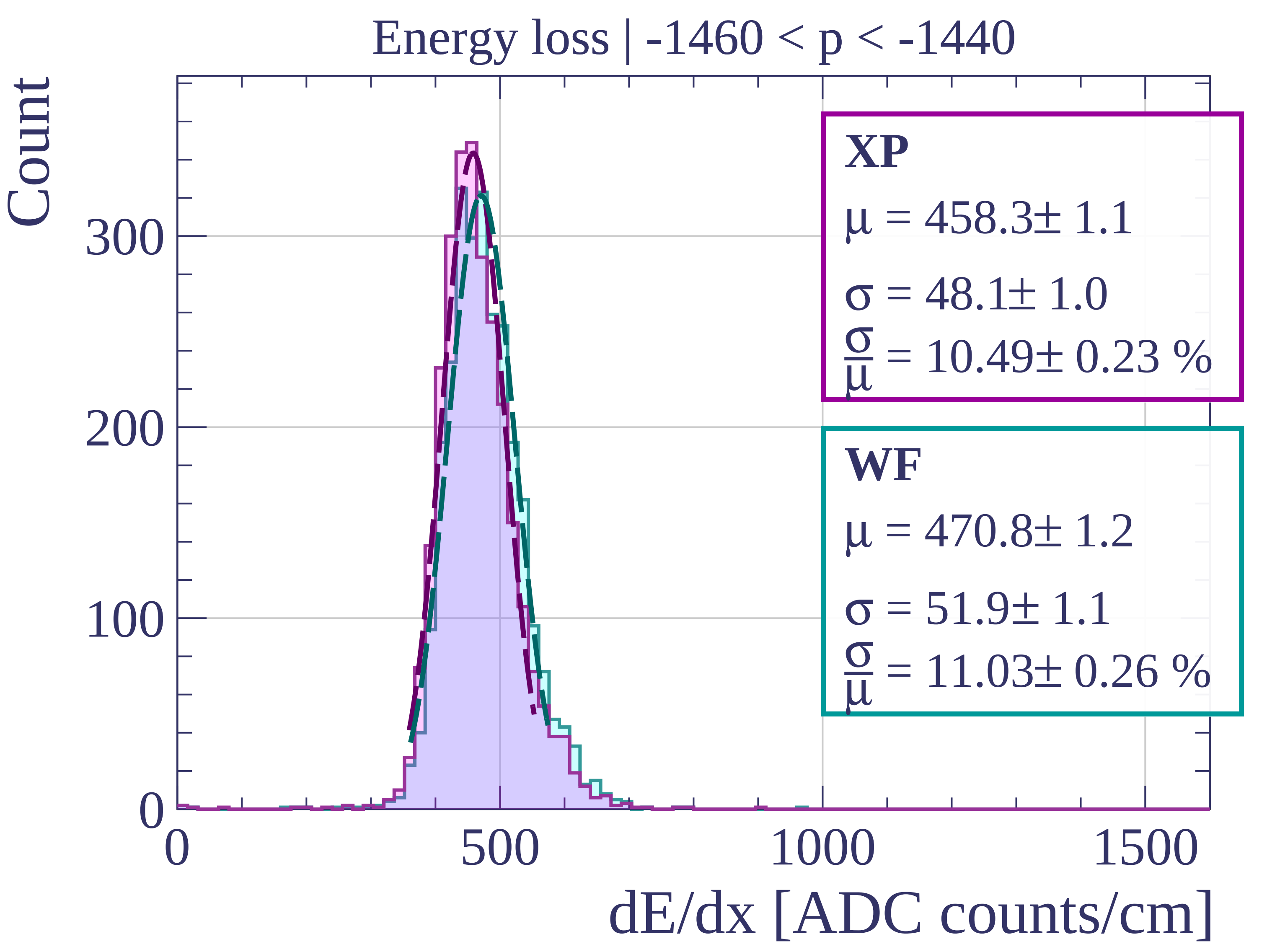

Energy loss | -1520 < p < -1500
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 462.0± 1.1
σ = 47.0± 1.0
σ/μ = 10.17± 0.23 %
WF
μ = 474.4± 1.3
σ = 51.7± 1.2
σ/μ = 10.90± 0.28 %

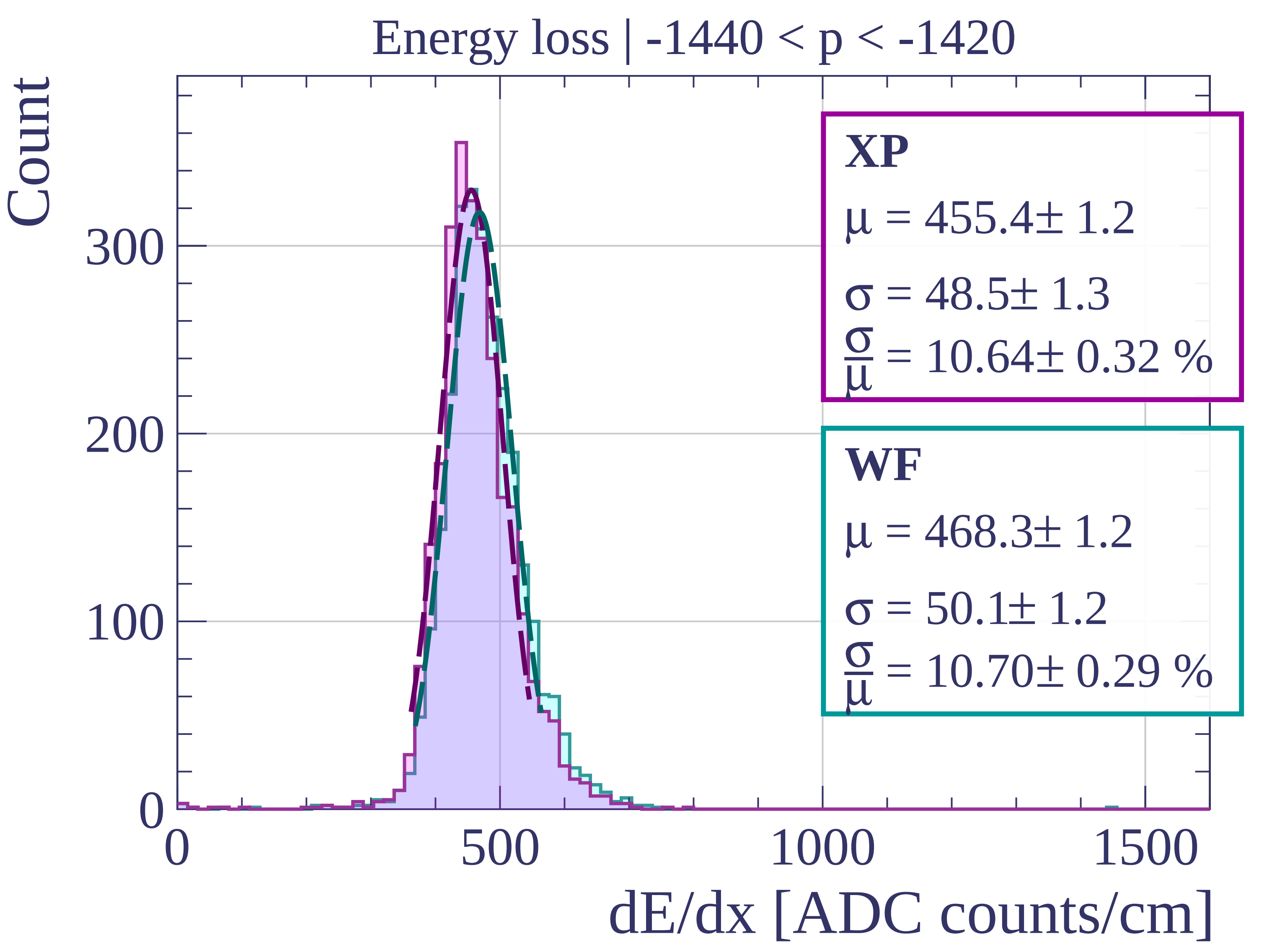


Energy loss | -1500 < p < -1480
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 456.1± 1.2
σ = 47.2± 1.3
σ/μ = 10.34± 0.31 %
WF
μ = 468.3± 1.2
σ = 48.9± 1.2
σ/μ = 10.44± 0.29 %

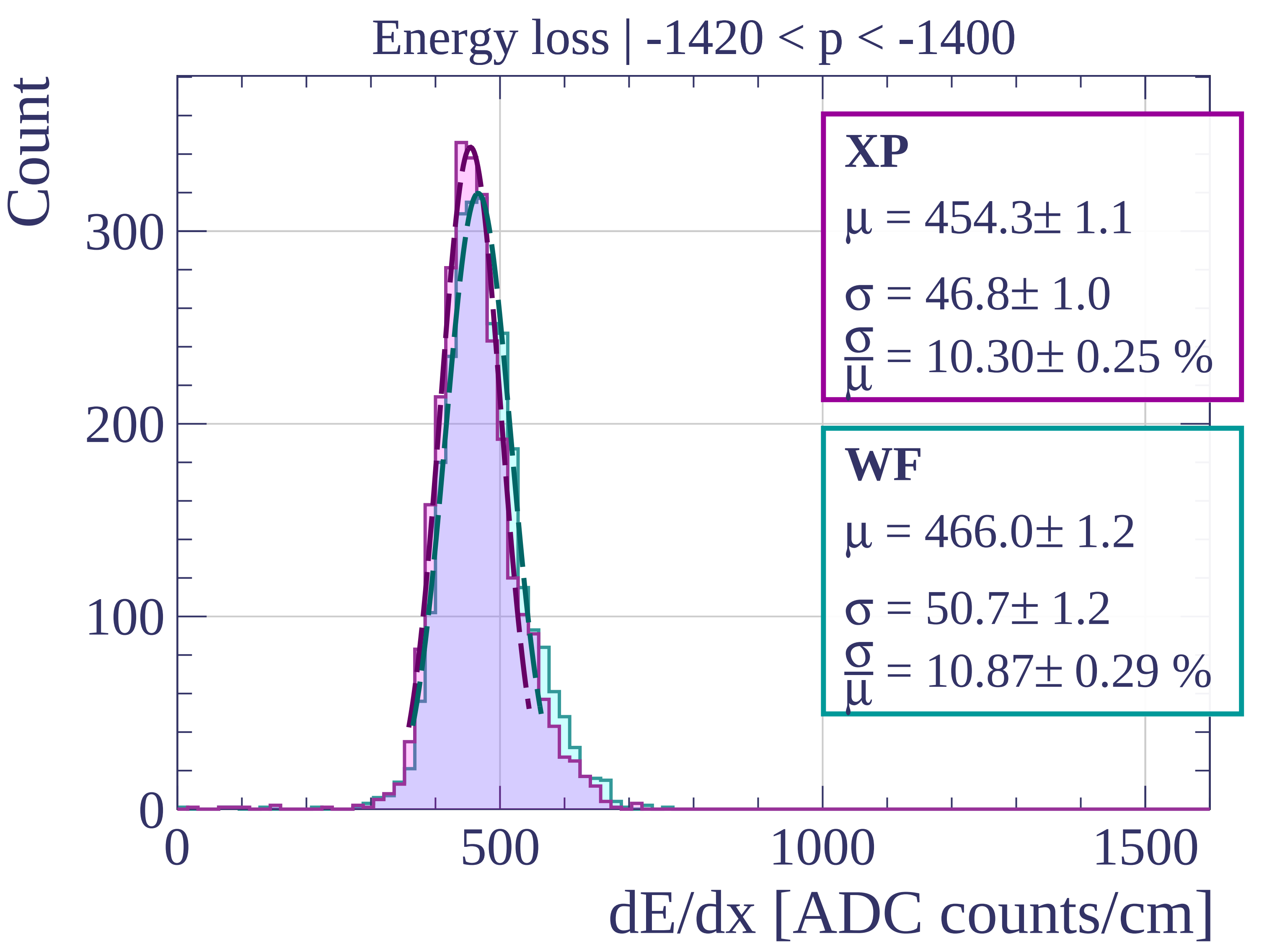


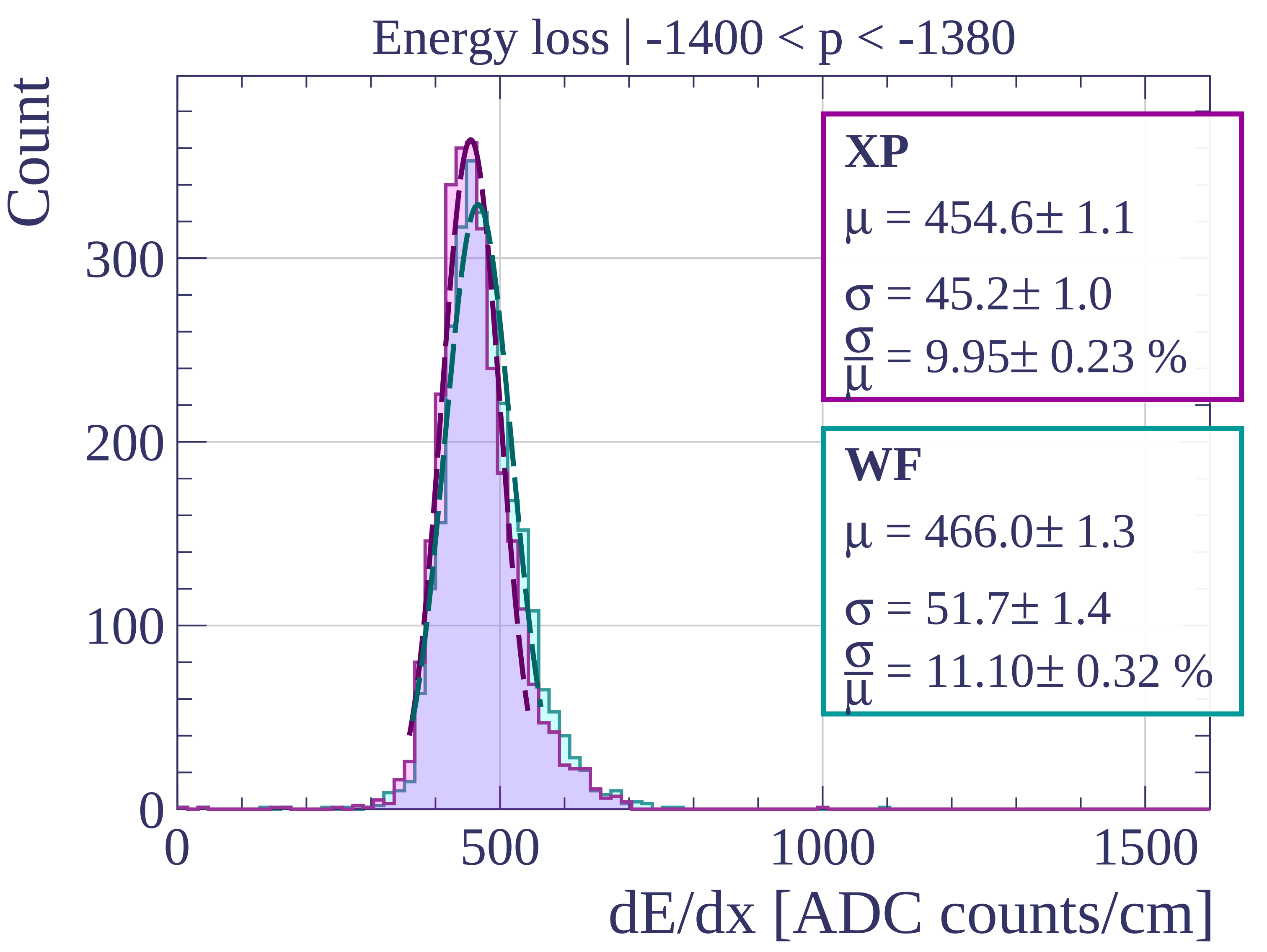
Energy loss | -1480 < p < -1460
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 458.9± 1.2
σ = 46.5± 1.1
σ/μ = 10.13± 0.28 %
WF
μ = 472.1± 1.2
σ = 51.0± 1.2
σ/μ = 10.81± 0.27 %


Energy loss | -1460 < p < -1440
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 458.3± 1.1
σ = 48.1± 1.0
σ/μ = 10.49± 0.23 %
WF
μ = 470.8± 1.2
σ = 51.9± 1.1
σ/μ = 11.03± 0.26 %

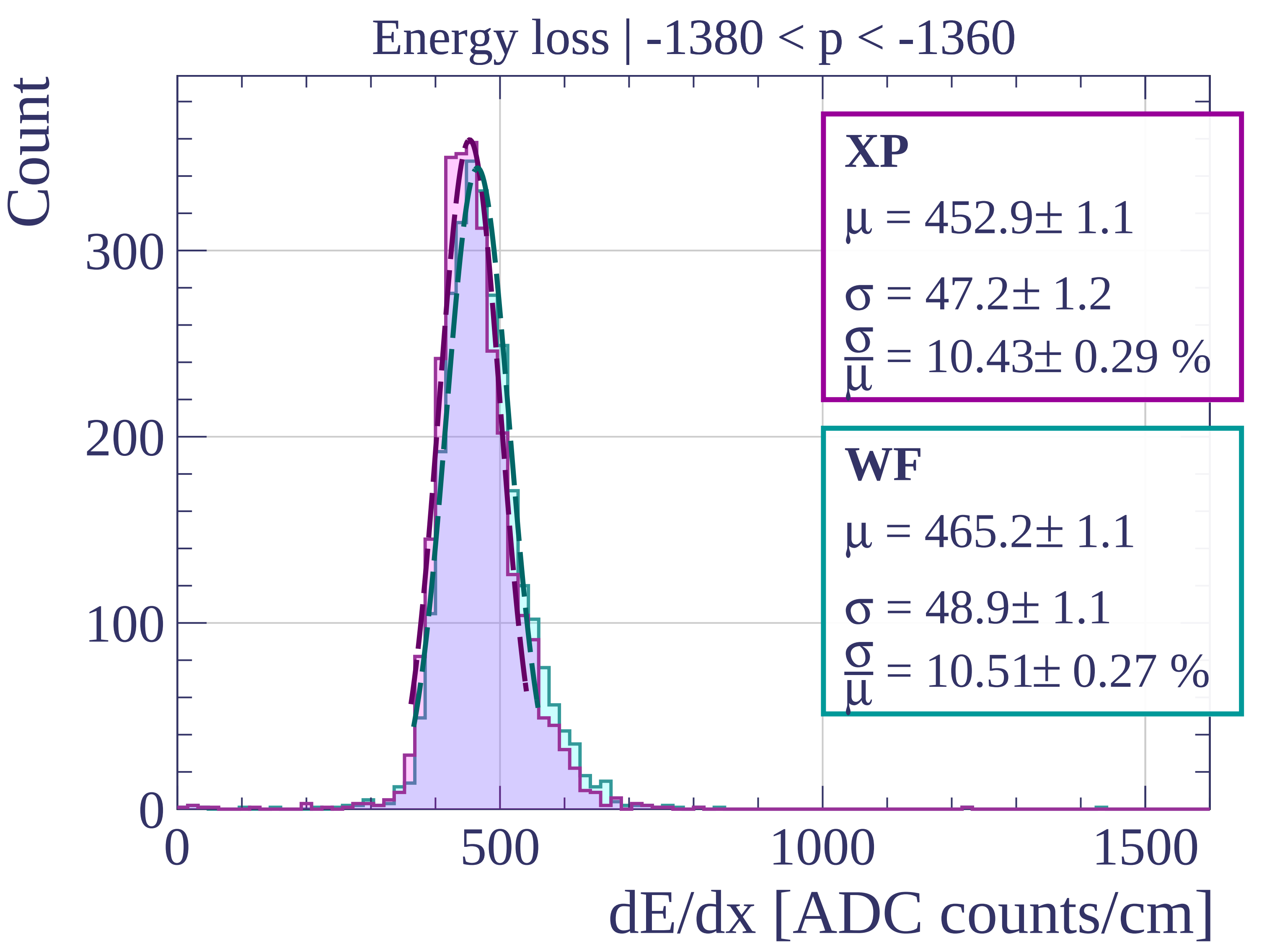


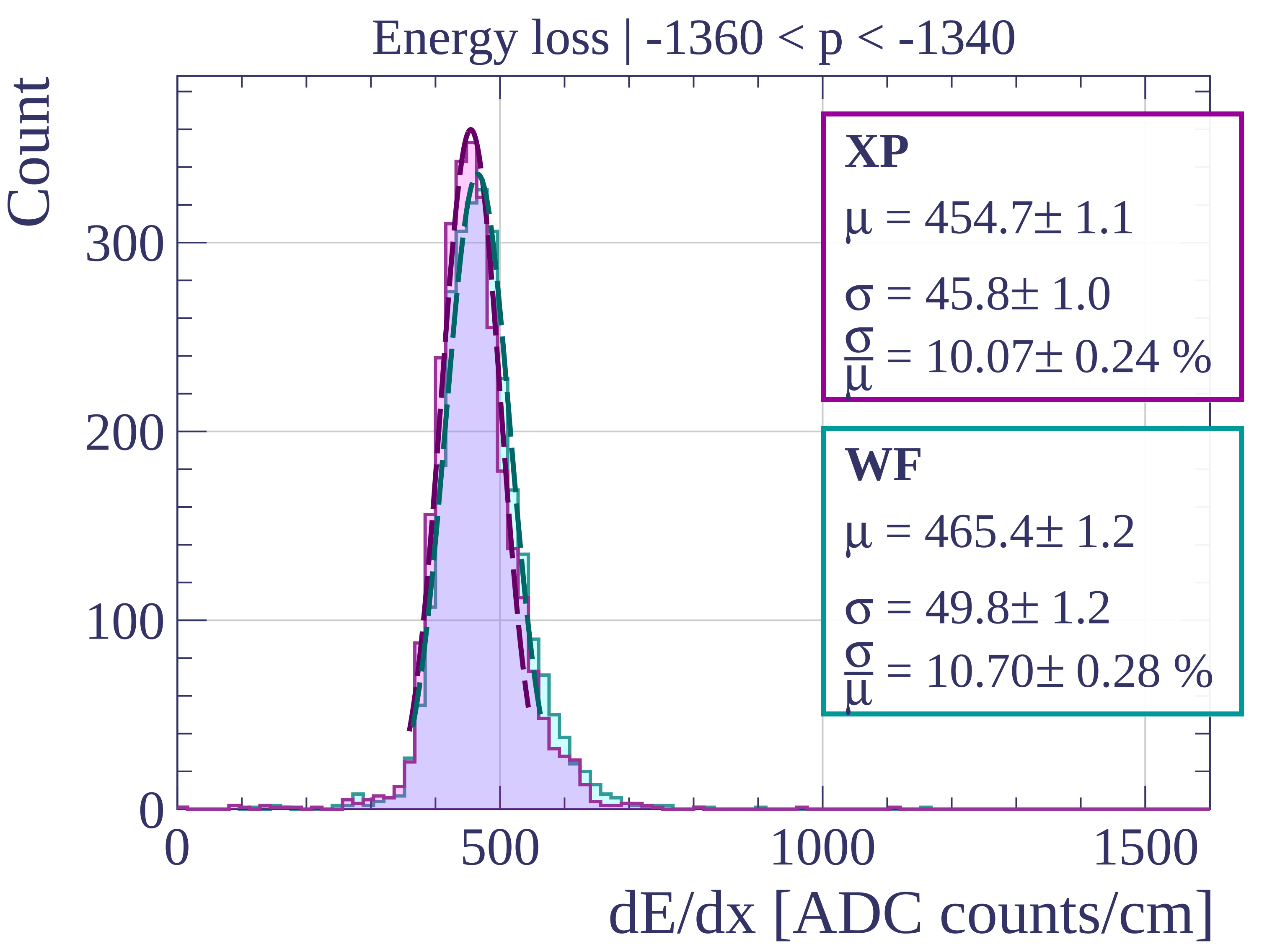
Energy loss | -1440 < p < -1420
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 455.4± 1.2
σ = 48.5± 1.3
σ/μ = 10.64± 0.32 %
WF
μ = 468.3± 1.2
σ = 50.1± 1.2
σ/μ = 10.70± 0.29 %


Energy loss | -1420 < p < -1400
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 454.3± 1.1
σ = 46.8± 1.0
σ/μ = 10.30± 0.25 %
WF
μ = 466.0± 1.2
σ = 50.7± 1.2
σ/μ = 10.87± 0.29 %

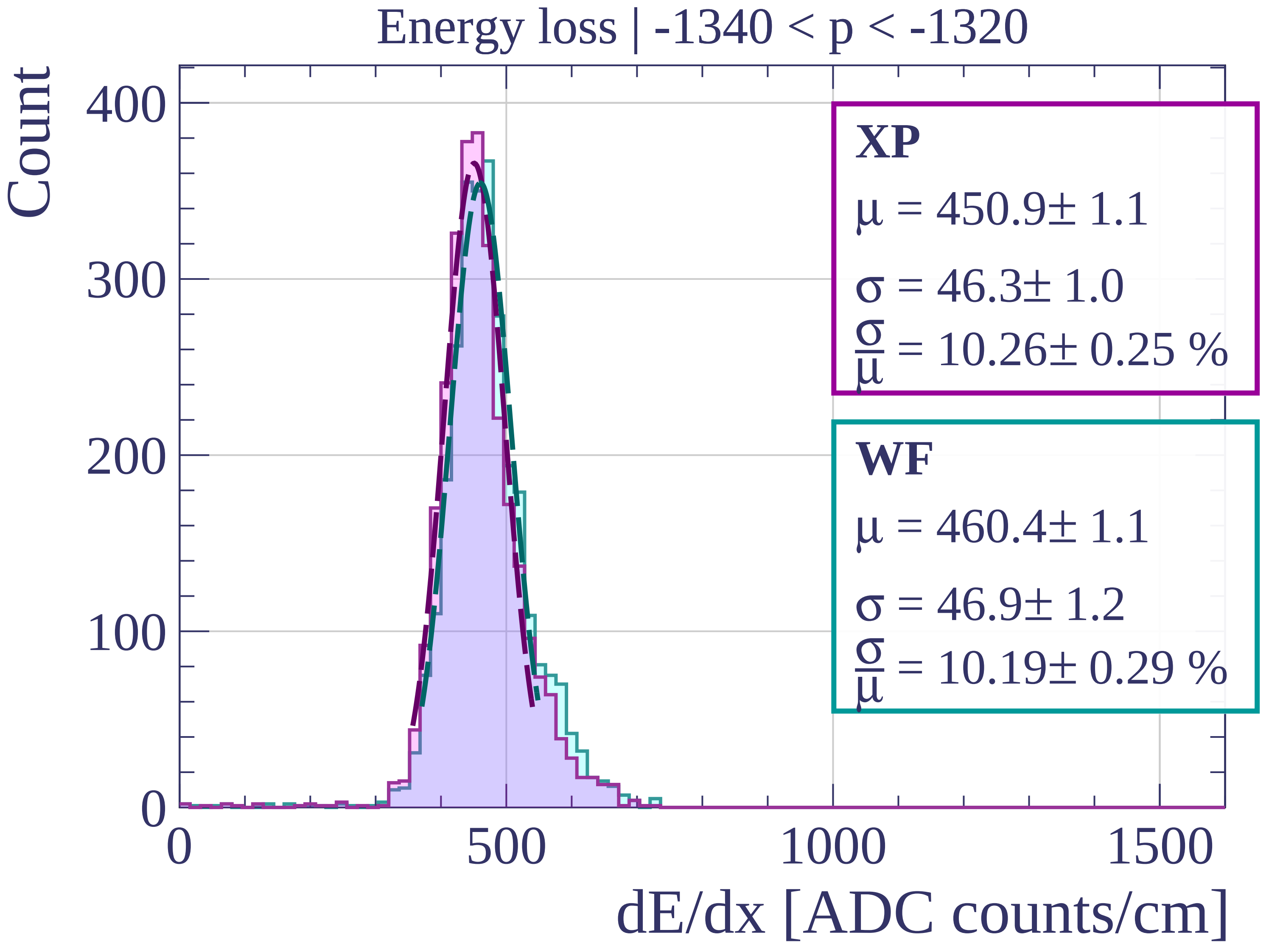


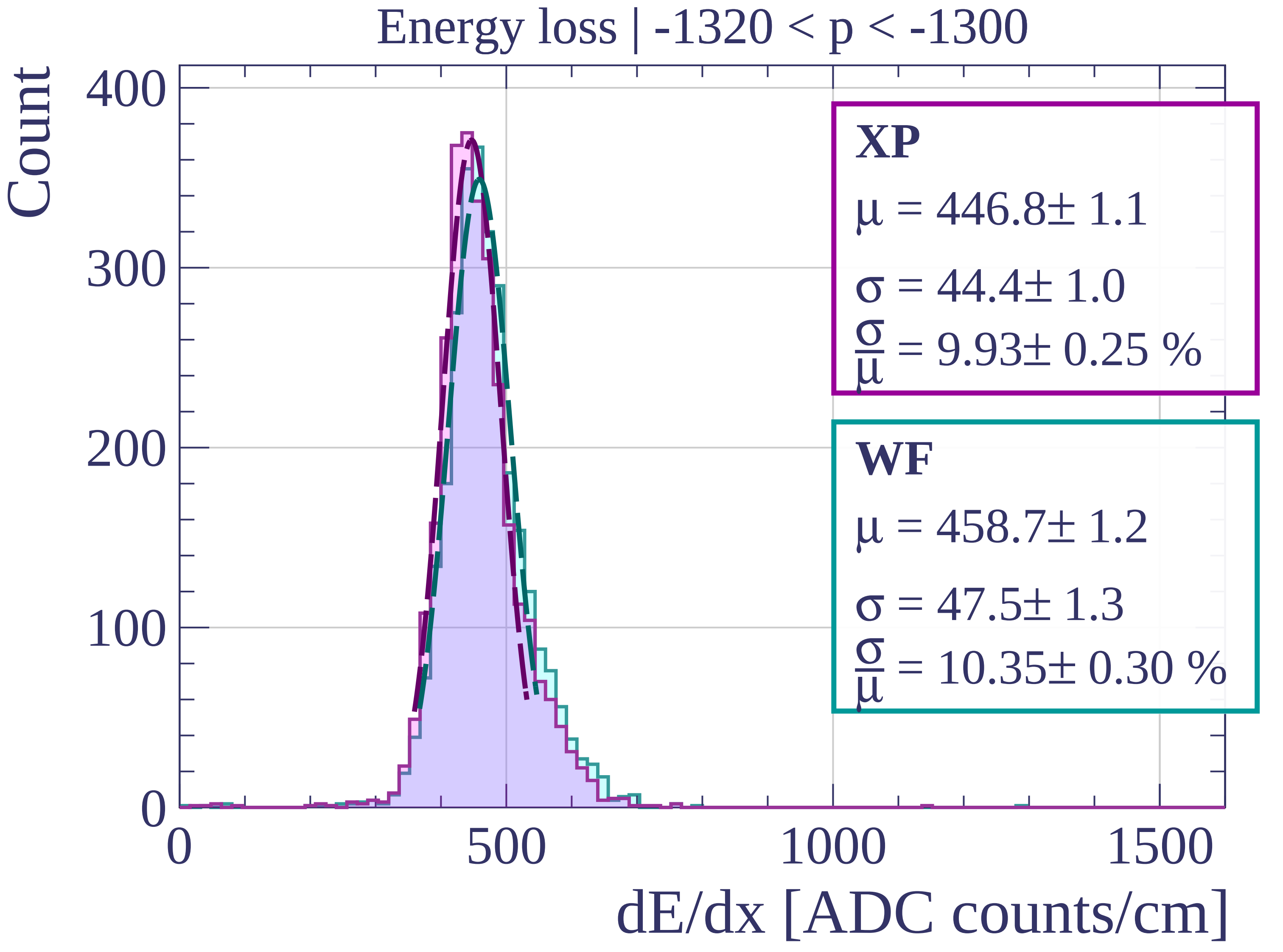
Energy loss | -1400 < p < -1380
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 454.6± 1.1
σ = 45.2± 1.0
σ/μ = 9.95± 0.23 %
WF
μ = 466.0± 1.3
σ = 51.7± 1.4
σ/μ = 11.10± 0.32 %


Energy loss | -1380 < p < -1360
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 452.9± 1.1
σ = 47.2± 1.2
σ/μ = 10.43± 0.29 %
WF
μ = 465.2± 1.1
σ = 48.9± 1.1
σ/μ = 10.51± 0.27 %

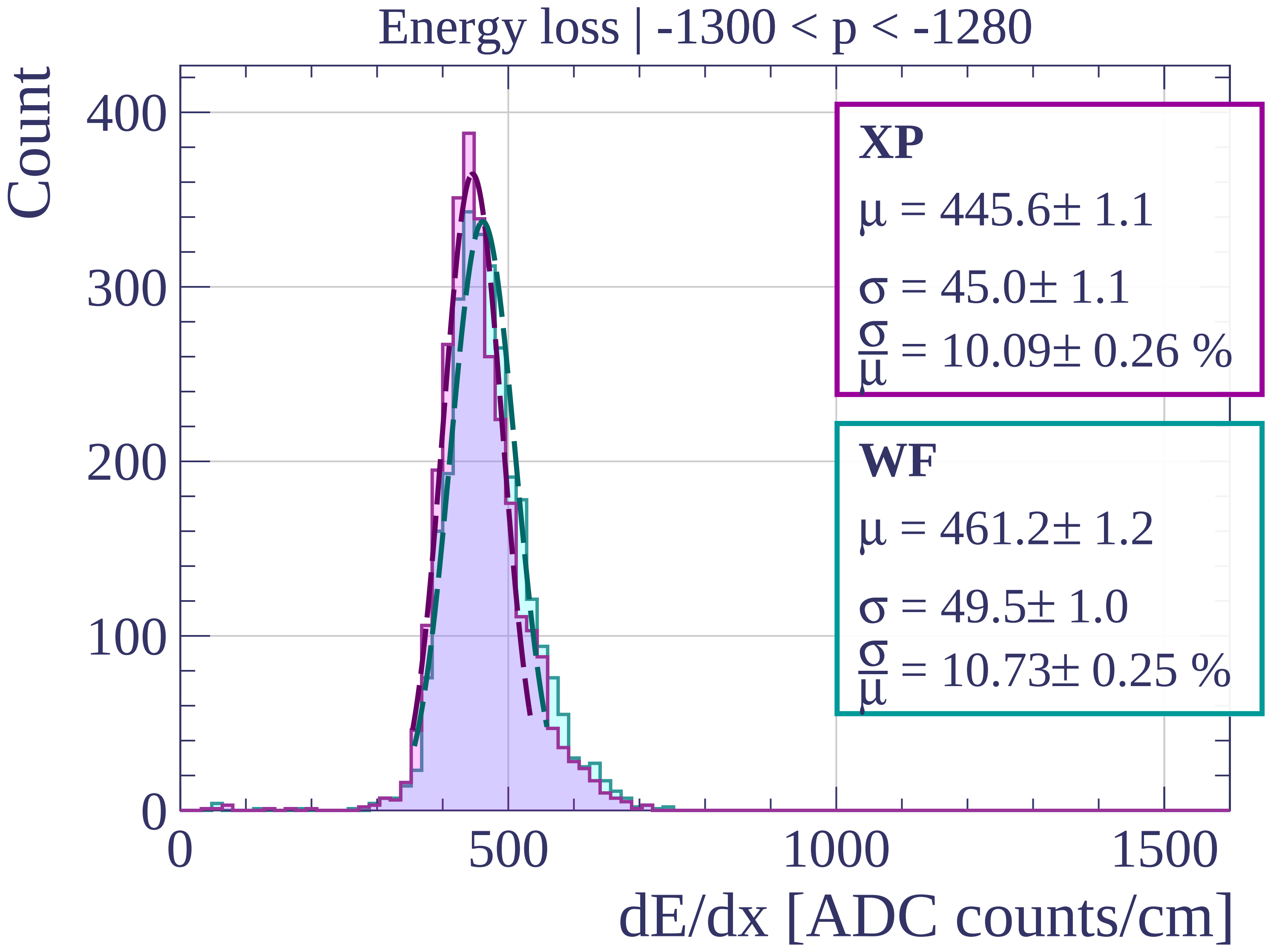


Energy loss | -1360 < p < -1340
Count
0
100
200
300
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 454.7± 1.1
σ = 45.8± 1.0
σ/μ = 10.07± 0.24 %
WF
μ = 465.4± 1.2
σ = 49.8± 1.2
σ/μ = 10.70± 0.28 %

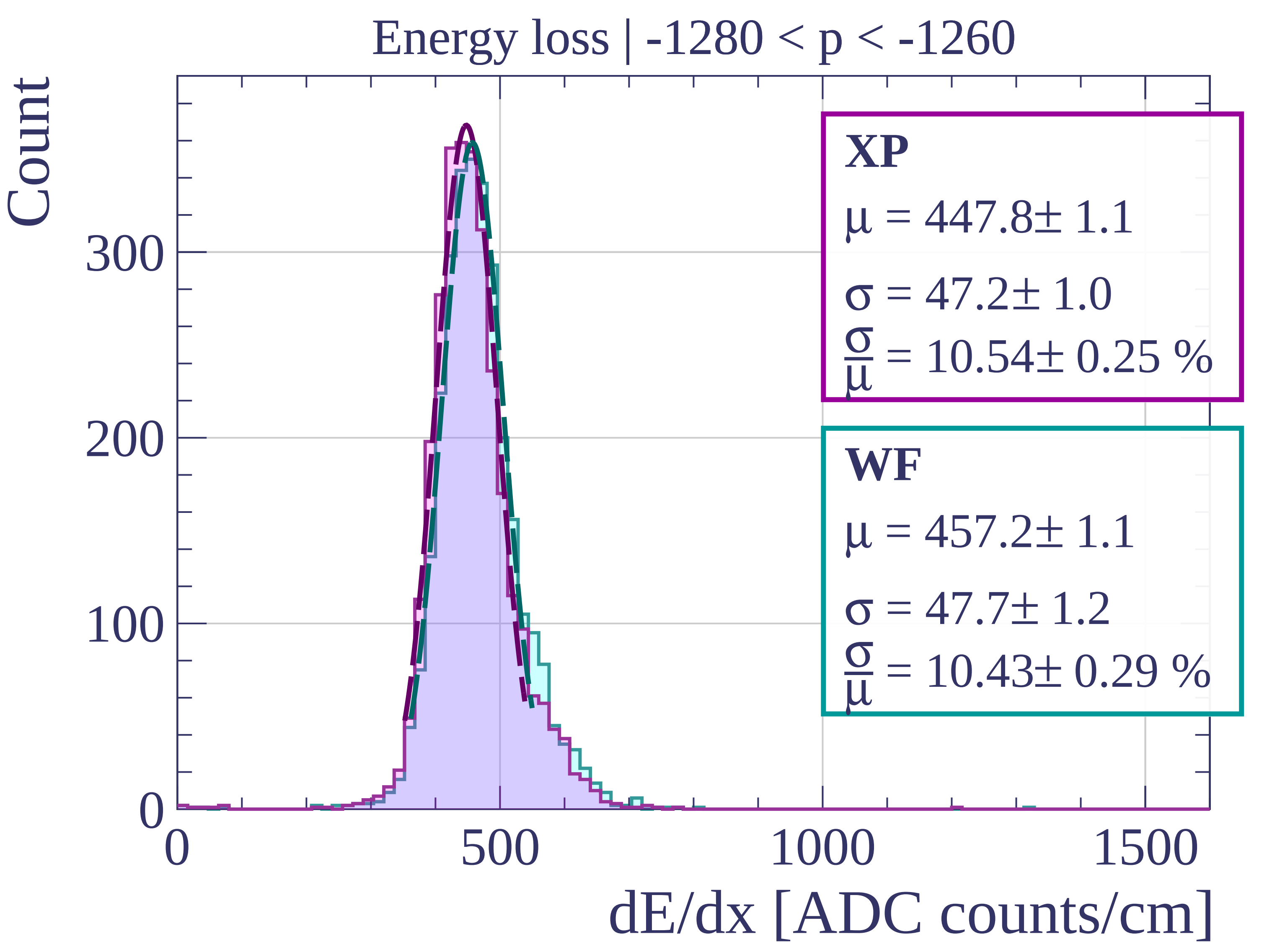

Energy loss | -1340 < p < -1320
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 450.9± 1.1
σ = 46.3± 1.0
σ/μ = 10.26± 0.25 %
WF
μ = 460.4± 1.1
σ = 46.9± 1.2
σ/μ = 10.19± 0.29 %



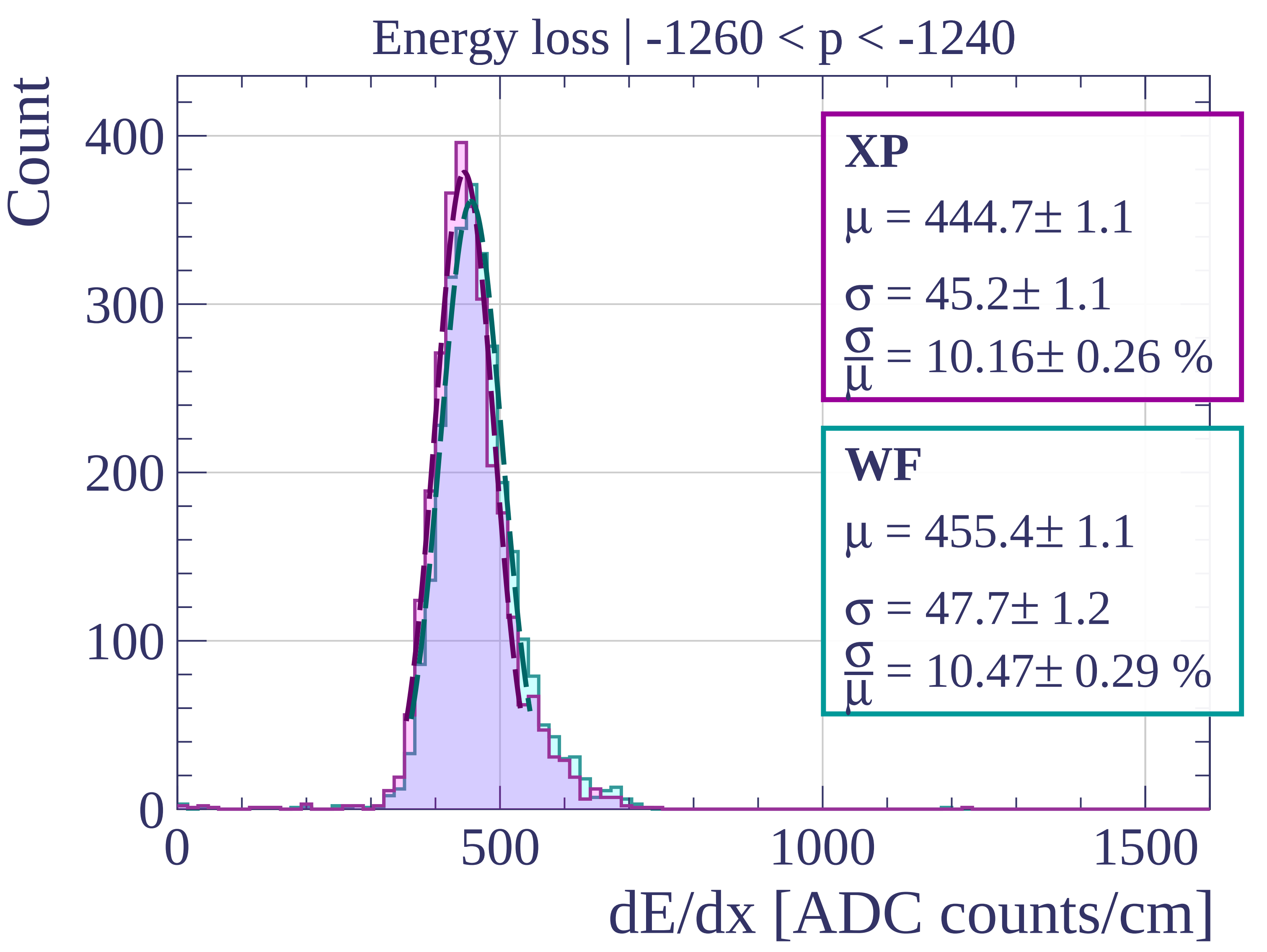
Energy loss | -1320 < p < -1300
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 446.8± 1.1
σ = 44.4± 1.0
σ/μ = 9.93± 0.25 %
WF
μ = 458.7± 1.2
σ = 47.5± 1.3
σ/μ = 10.35± 0.30 %



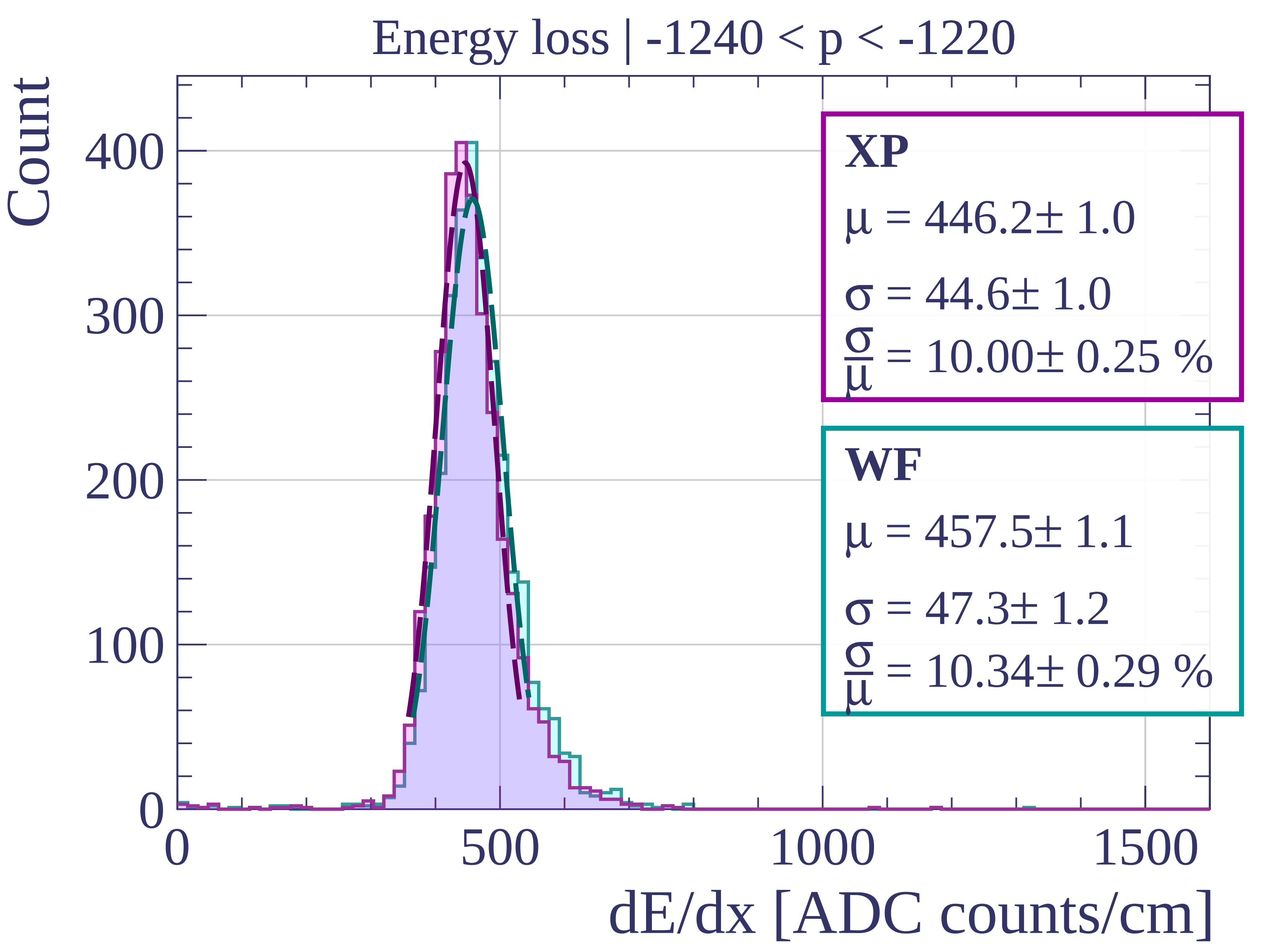
Energy loss | -1300 < p < -1280
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 445.6± 1.1
σ = 45.0± 1.1
σ/μ = 10.09± 0.26 %
WF
μ = 461.2± 1.2
σ = 49.5± 1.0
σ/μ = 10.73± 0.25 %



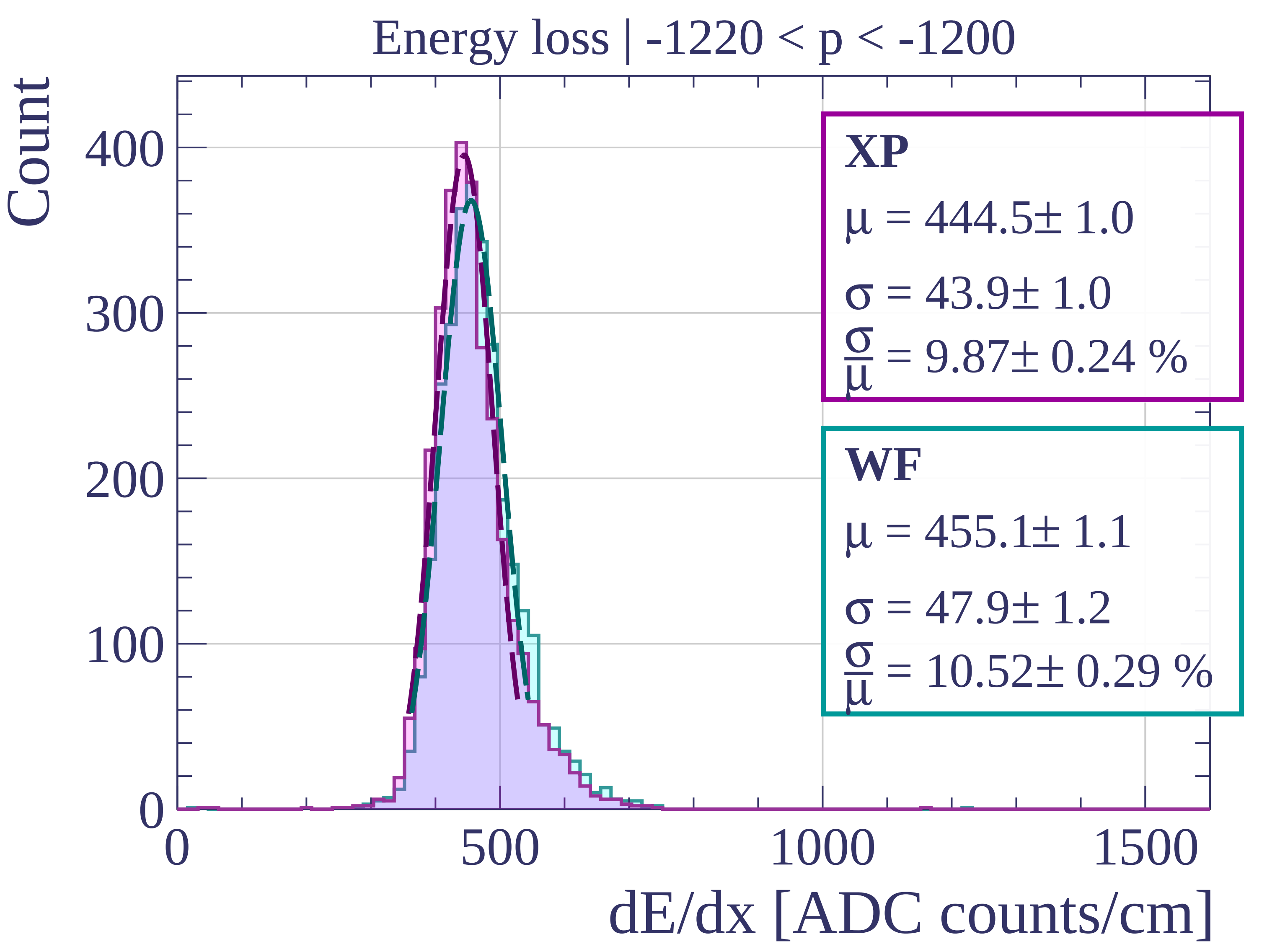
Energy loss | -1280 < p < -1260
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 447.8± 1.1
σ = 47.2± 1.0
σ/μ = 10.54± 0.25 %
WF
μ = 457.2± 1.1
σ = 47.7± 1.2
σ/μ = 10.43± 0.29 %



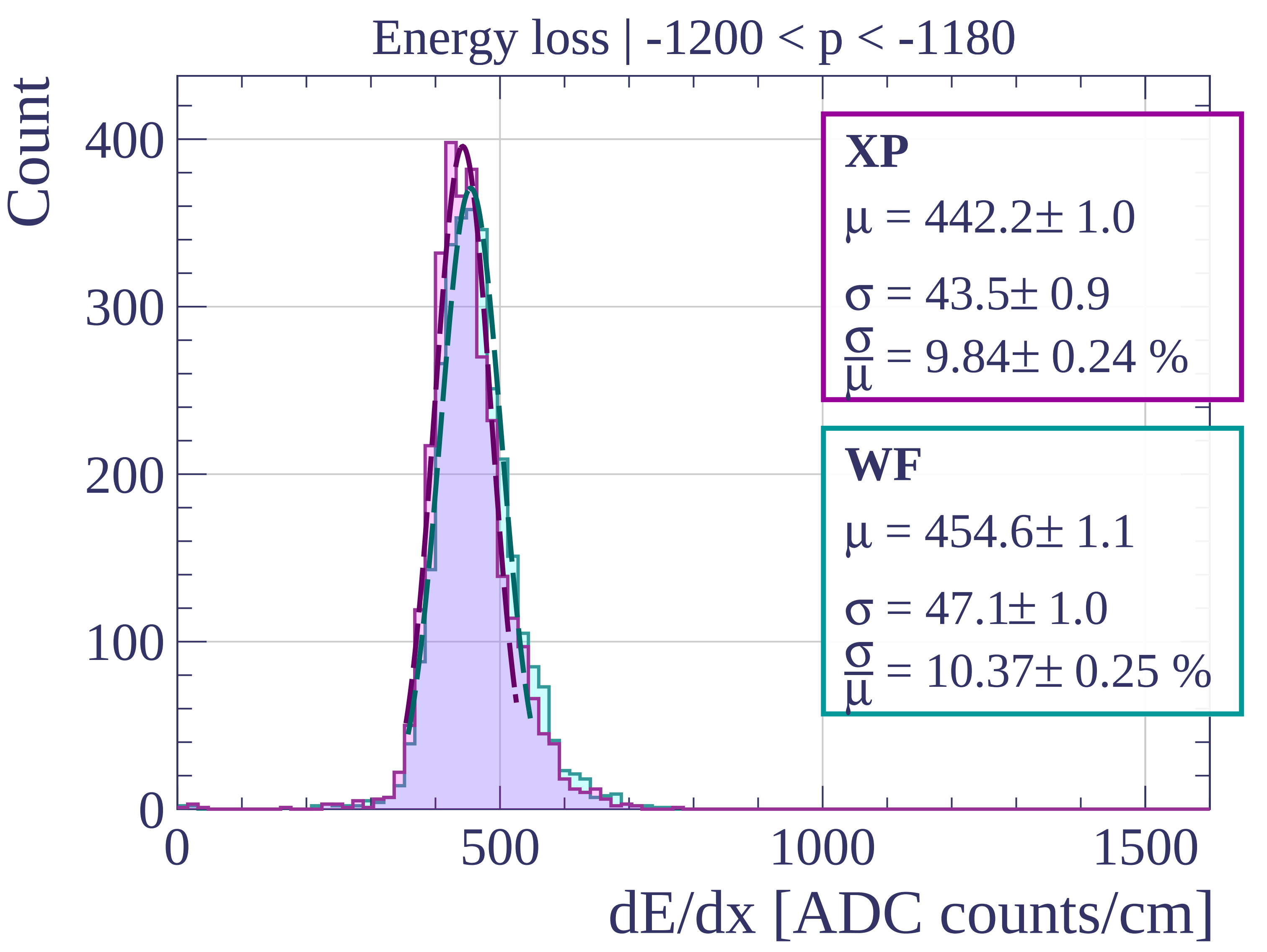
Energy loss | -1260 < p < -1240
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
XP
μ = 444.7± 1.1
σ = 45.2± 1.1
σ/μ = 10.16± 0.26 %
WF
μ = 455.4± 1.1
σ = 47.7± 1.2
σ/μ = 10.47± 0.29 %



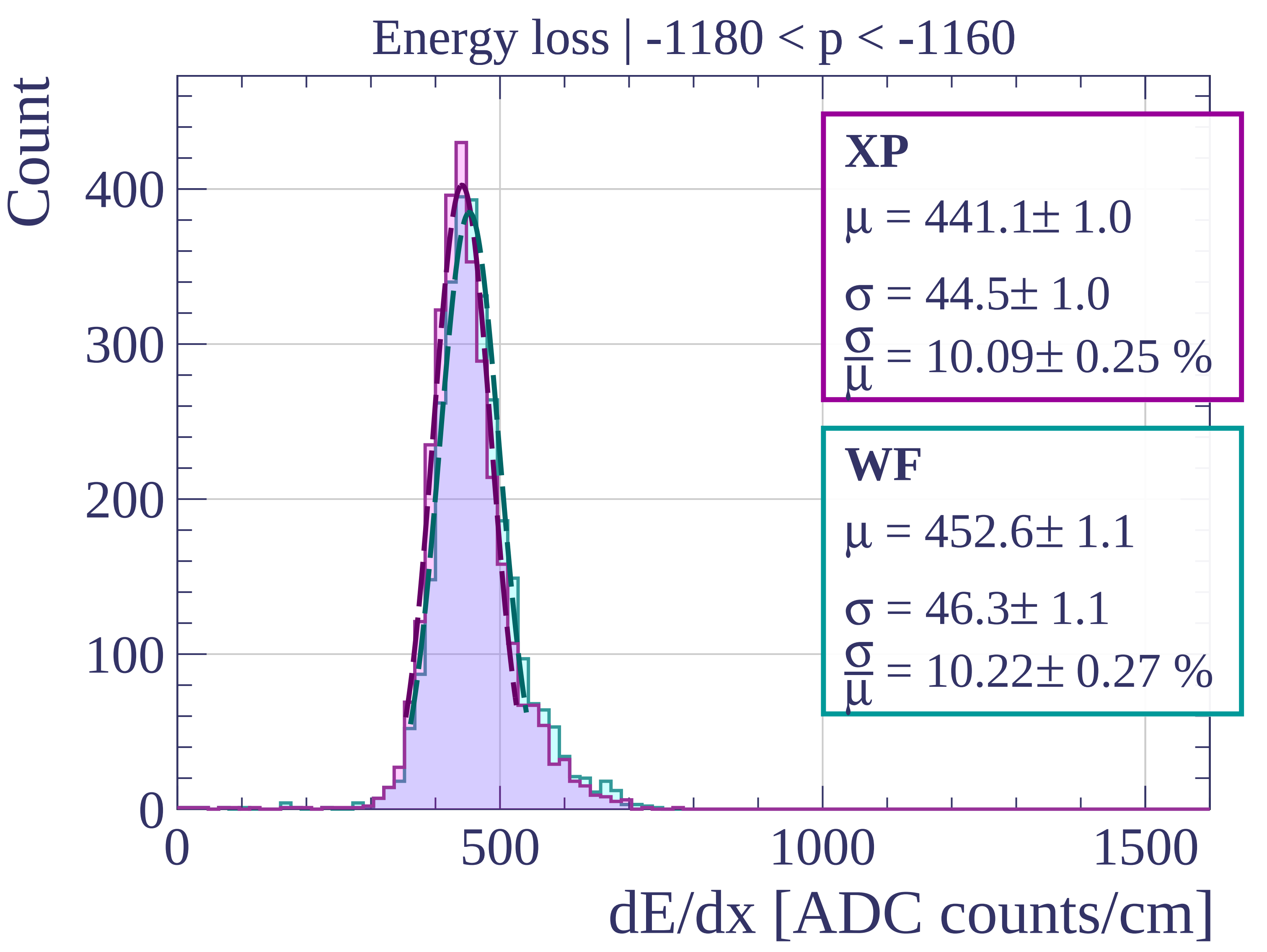
Energy loss | -1240 < p < -1220
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 446.2± 1.0
σ = 44.6± 1.0
σ/μ = 10.00± 0.25 %
WF
μ = 457.5± 1.1
σ = 47.3± 1.2
σ/μ = 10.34± 0.29 %



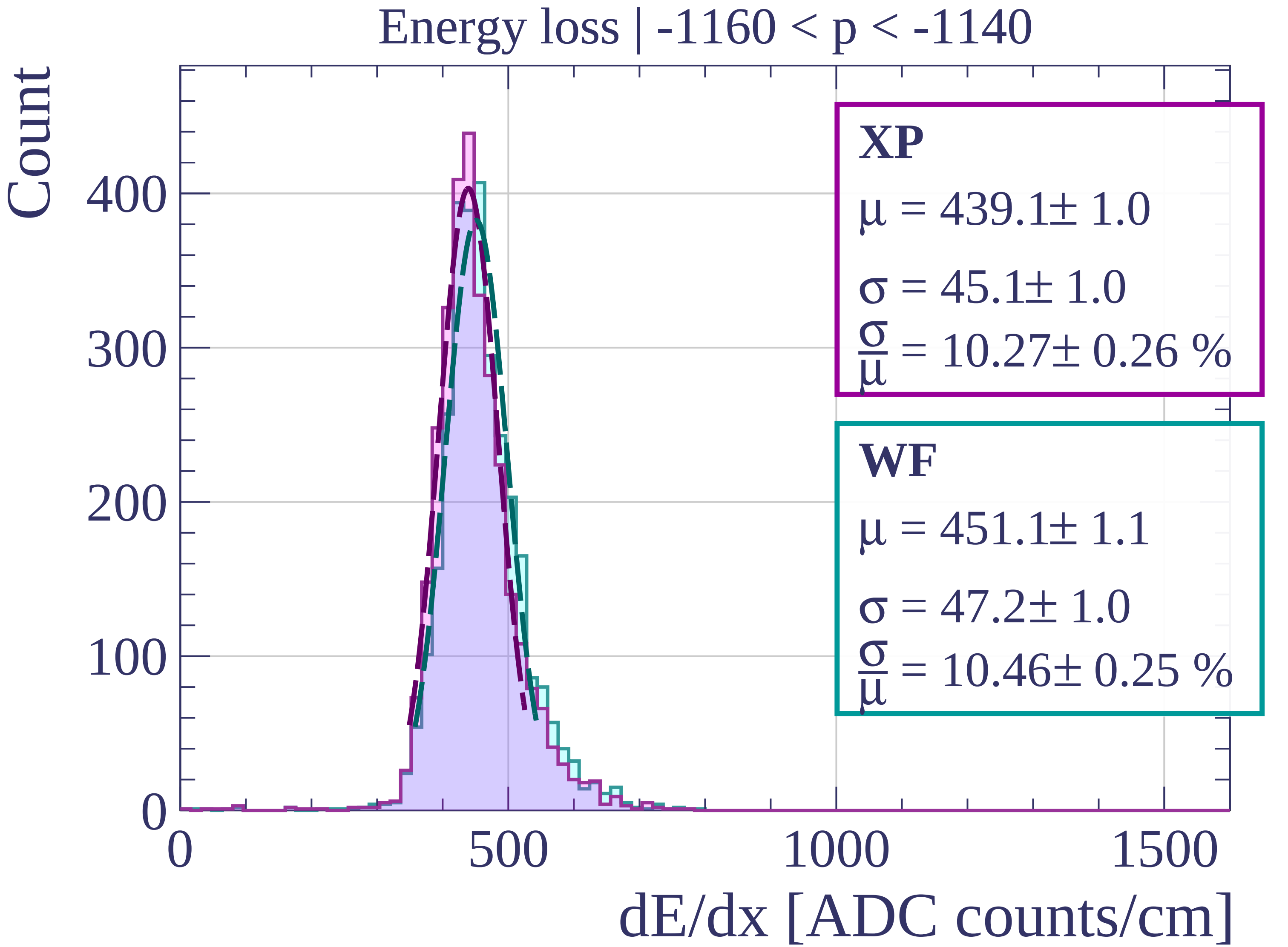
Energy loss | -1220 < p < -1200
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
XP
μ = 444.5± 1.0
σ = 43.9± 1.0
σ/μ = 9.87± 0.24 %
WF
μ = 455.1± 1.1
σ = 47.9± 1.2
σ/μ = 10.52± 0.29 %

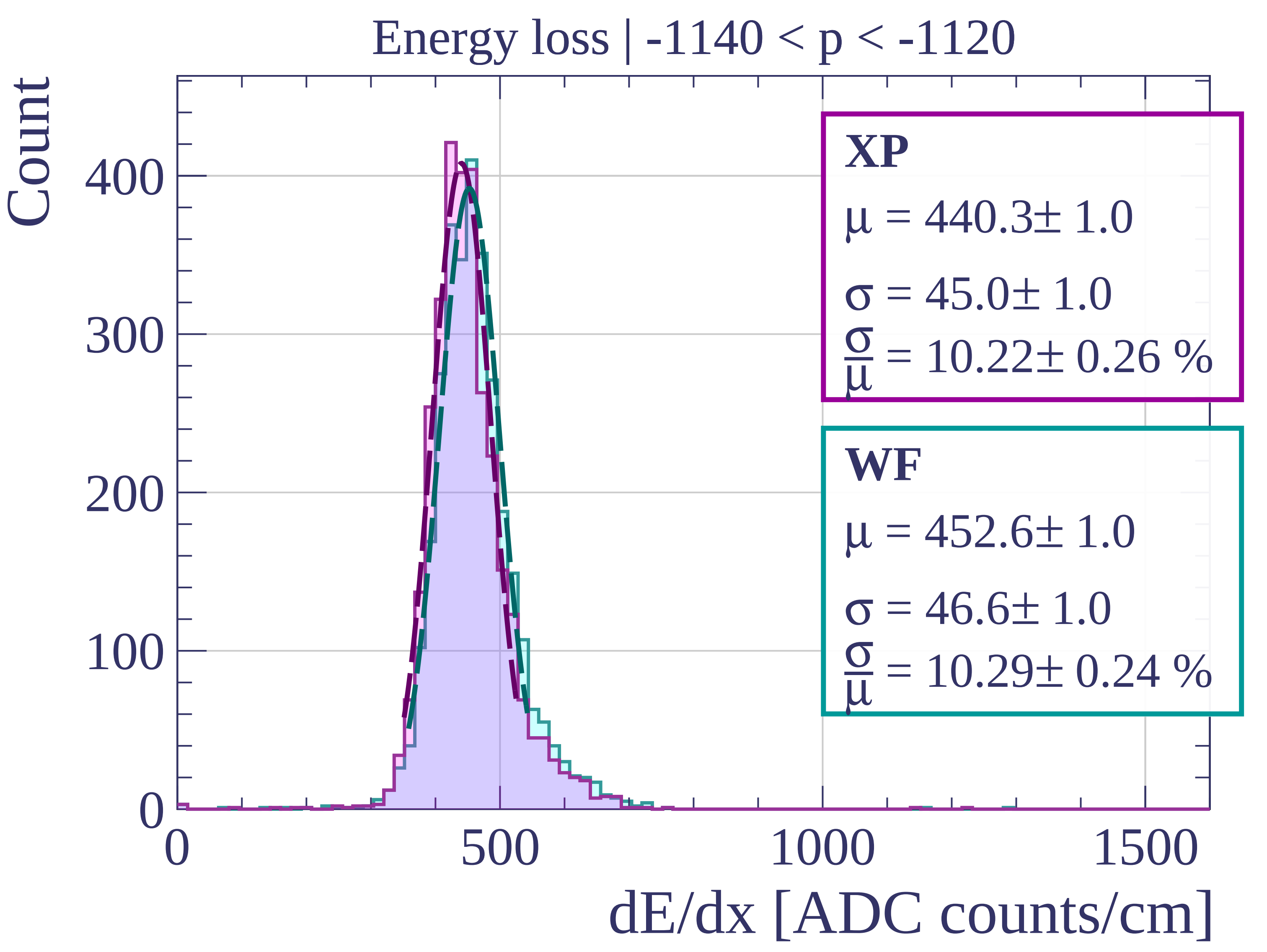


Energy loss | -1200 < p < -1180
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 442.2± 1.0
σ = 43.5± 0.9
σ/μ = 9.84± 0.24 %
WF
μ = 454.6± 1.1
σ = 47.1± 1.0
σ/μ = 10.37± 0.25 %

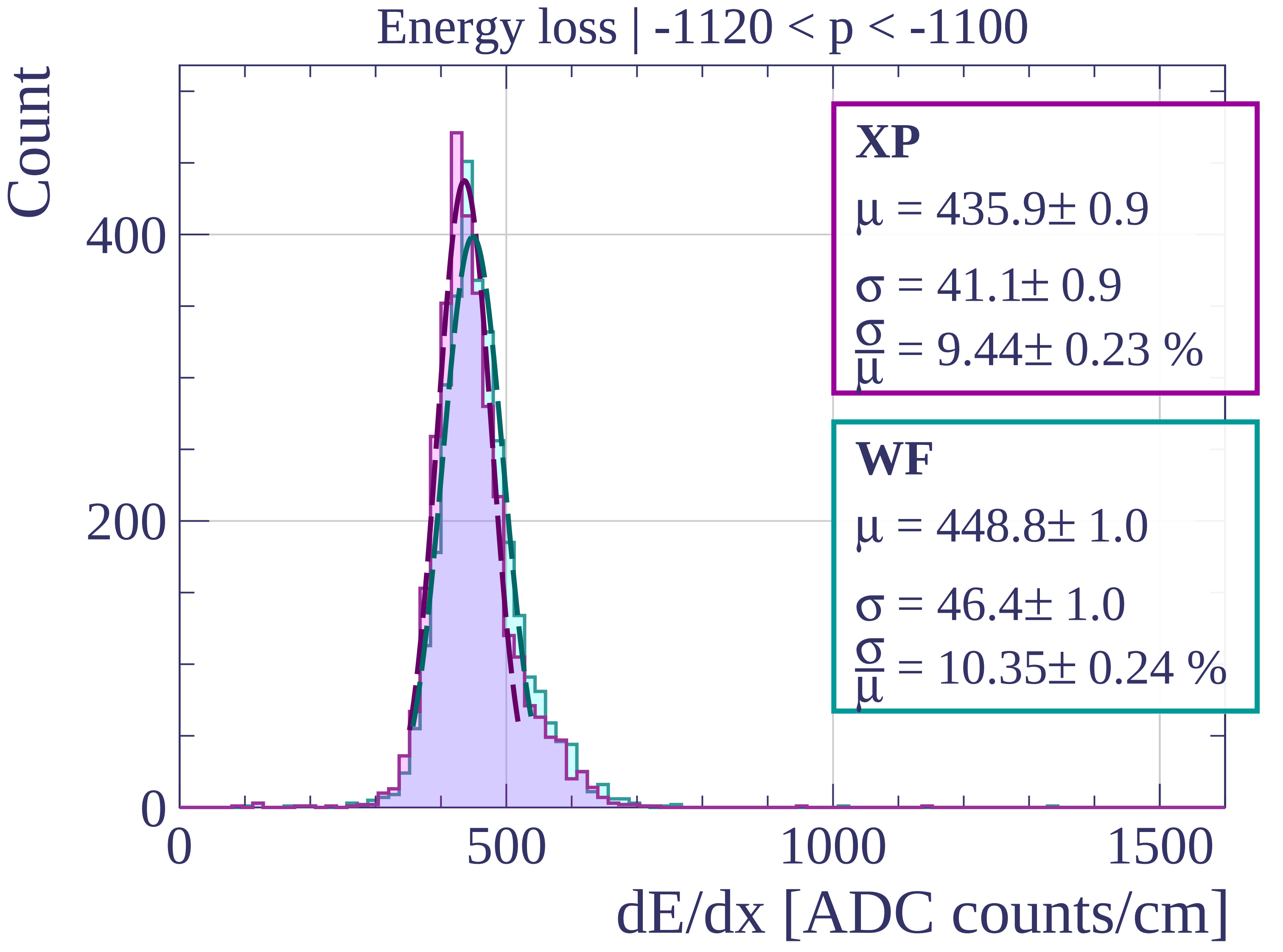


Energy loss | -1180 < p < -1160
Count
0
100
200
300
400
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 441.1± 1.0
σ = 44.5± 1.0
σ/μ = 10.09± 0.25 %
WF
μ = 452.6± 1.1
σ = 46.3± 1.1
σ/μ = 10.22± 0.27 %

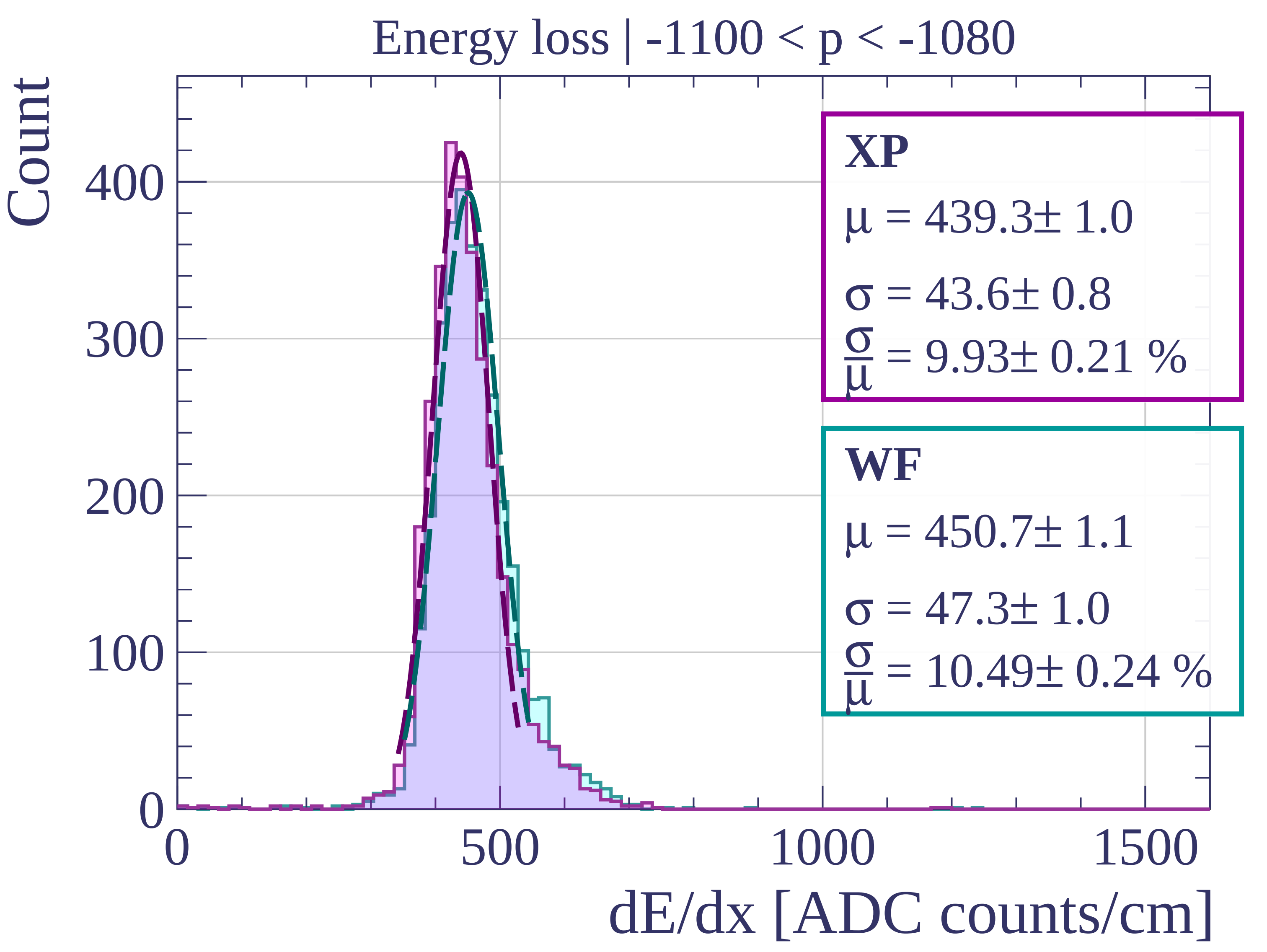

Energy loss | -1160 < p < -1140
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 439.1± 1.0
σ = 45.1± 1.0
σ/μ = 10.27± 0.26 %
WF
μ = 451.1± 1.1
σ = 47.2± 1.0
σ/μ = 10.46± 0.25 %



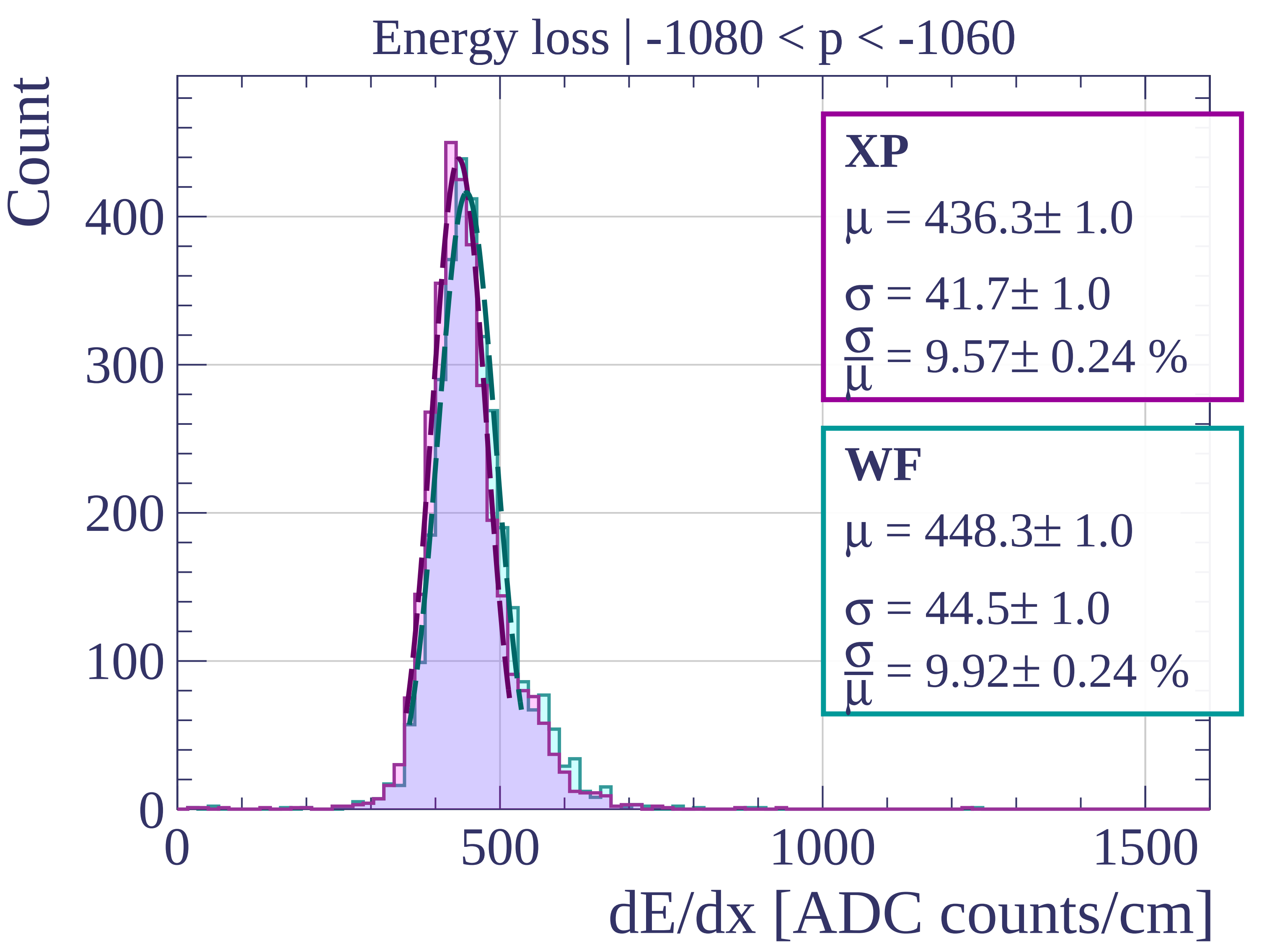
Energy loss | -1140 < p < -1120
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 440.3± 1.0
σ = 45.0± 1.0
σ/μ = 10.22± 0.26 %
WF
μ = 452.6± 1.0
σ = 46.6± 1.0
σ/μ = 10.29± 0.24 %

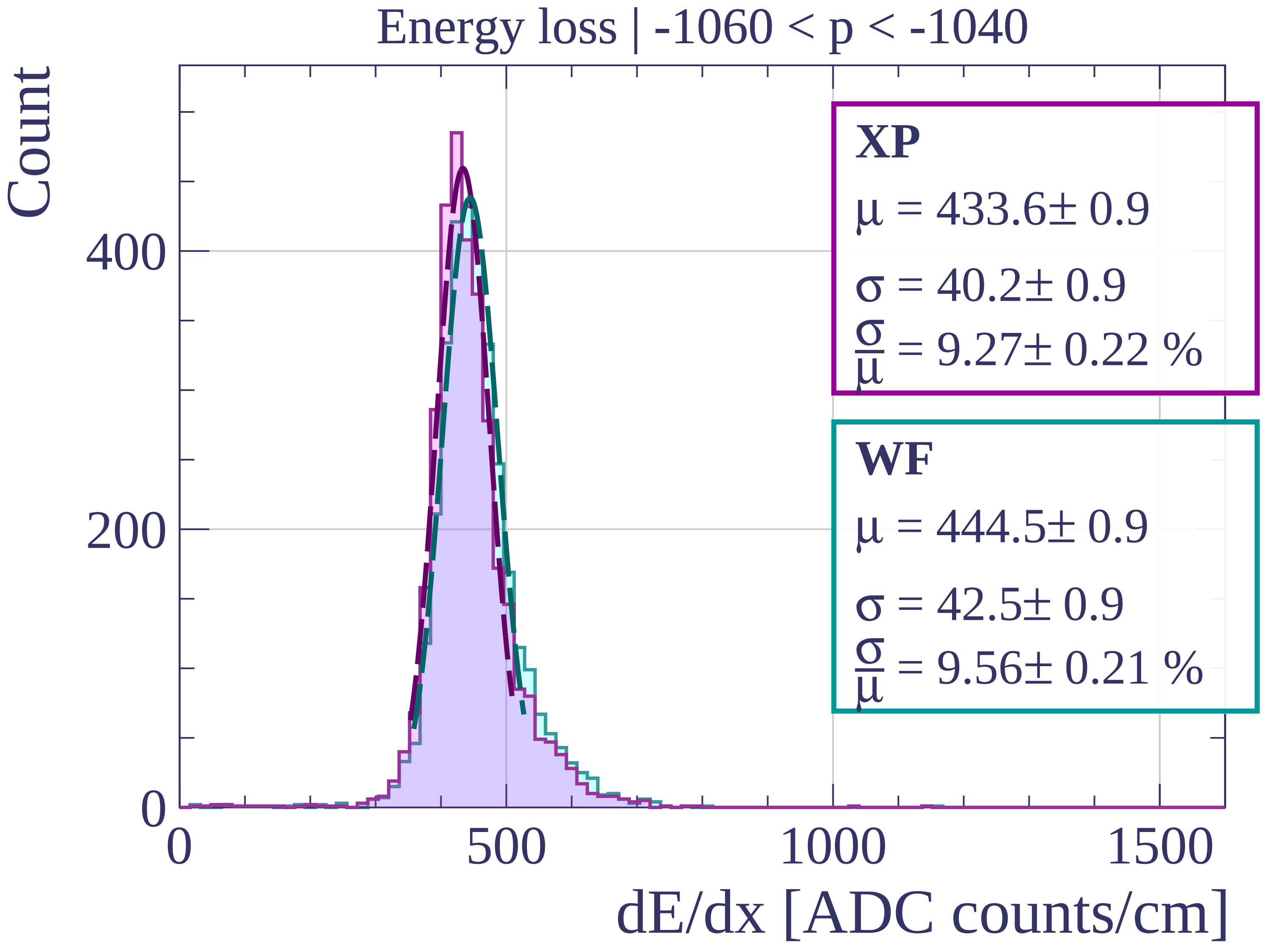


Energy loss | -1120 < p < -1100
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 435.9± 0.9
σ = 41.1± 0.9
σ/μ = 9.44± 0.23 %
WF
μ = 448.8± 1.0
σ = 46.4± 1.0
σ/μ = 10.35± 0.24 %

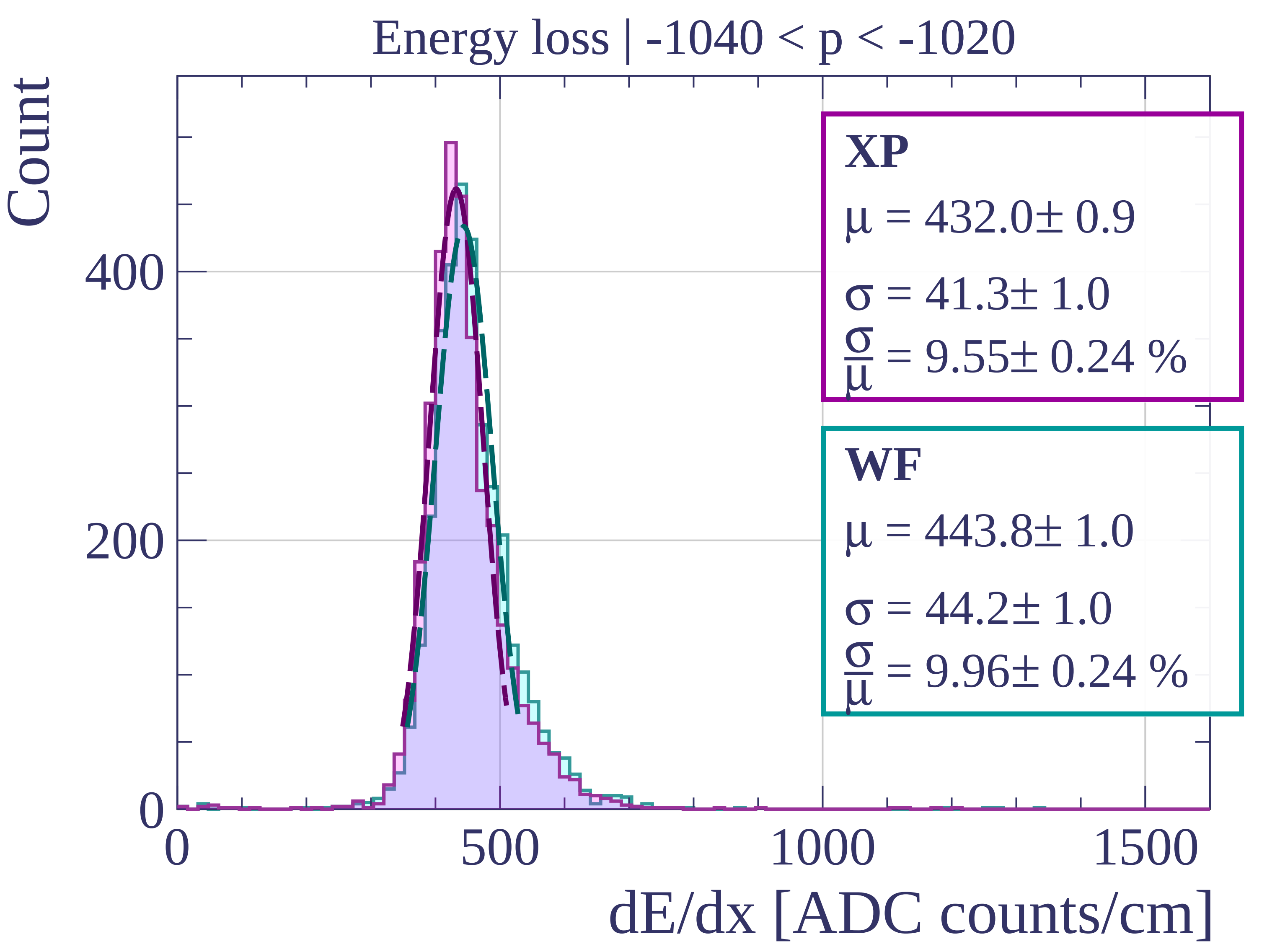


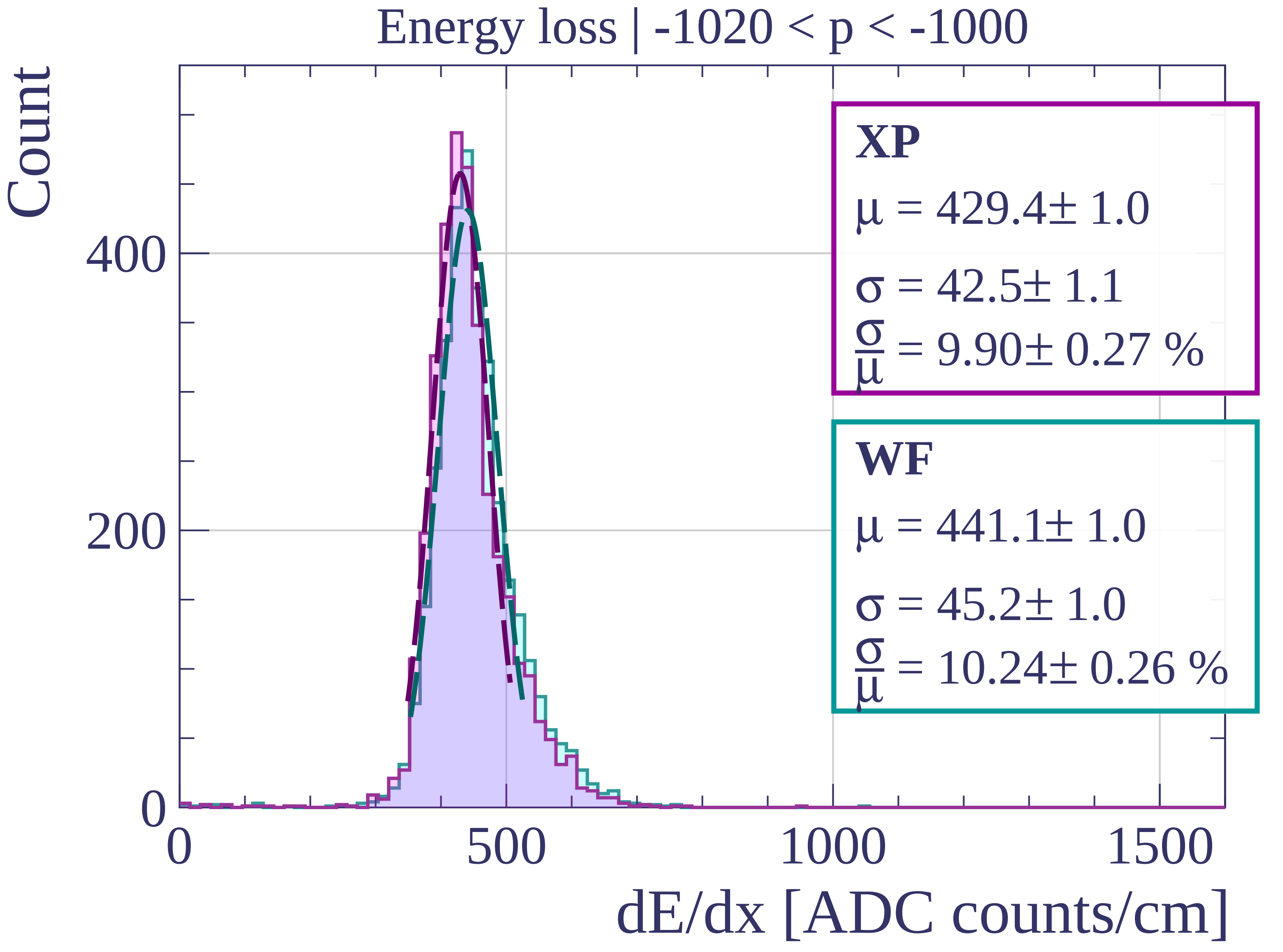
Energy loss | -1100 < p < -1080
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 439.3± 1.0
σ = 43.6± 0.8
σ/μ = 9.93± 0.21 %
WF
μ = 450.7± 1.1
σ = 47.3± 1.0
σ/μ = 10.49± 0.24 %

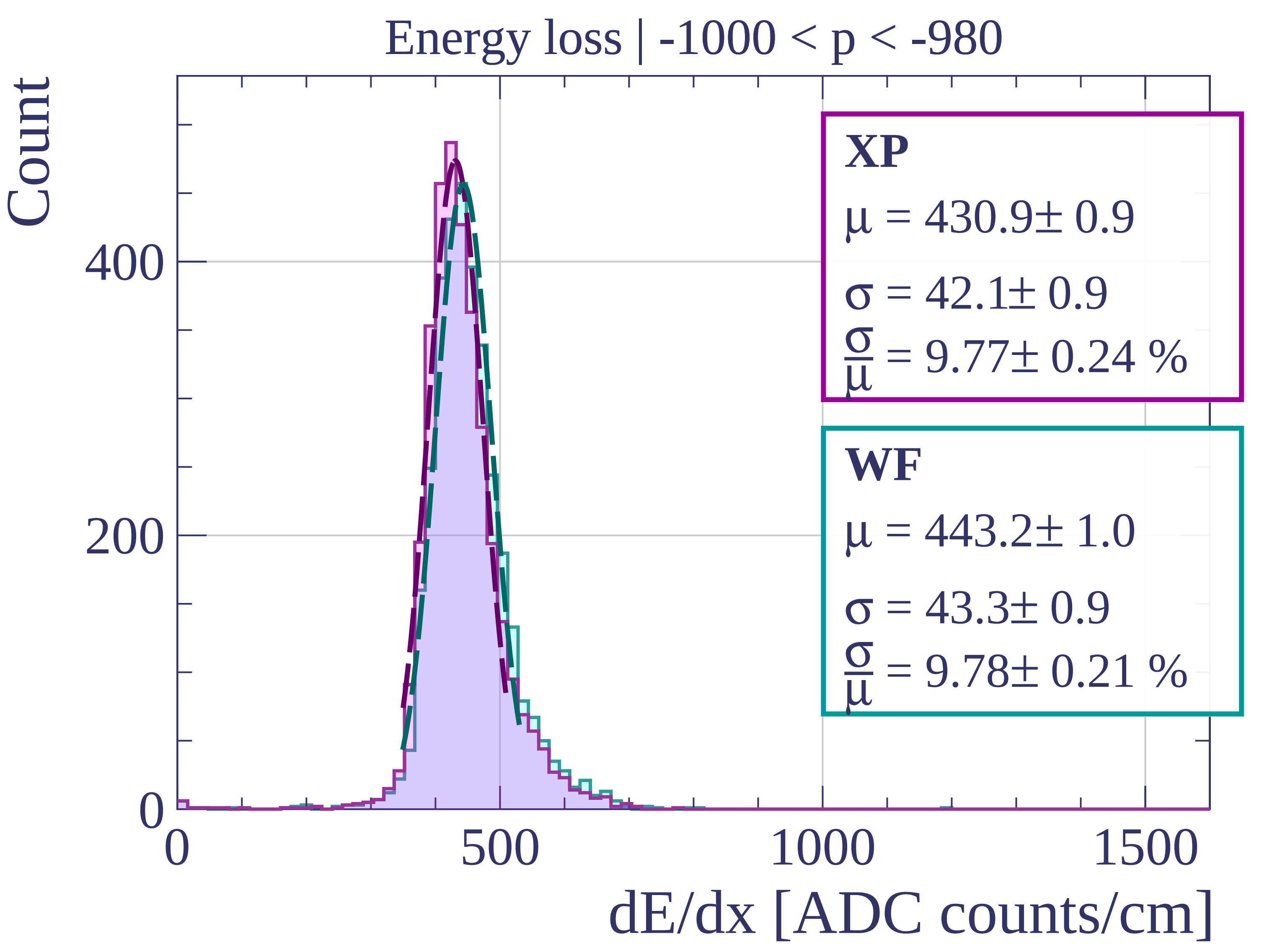

Energy loss | -1080 < p < -1060
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
XP
μ = 436.3± 1.0
σ = 41.7± 1.0
σ/μ = 9.57± 0.24 %
WF
μ = 448.3± 1.0
σ = 44.5± 1.0
σ/μ = 9.92± 0.24 %



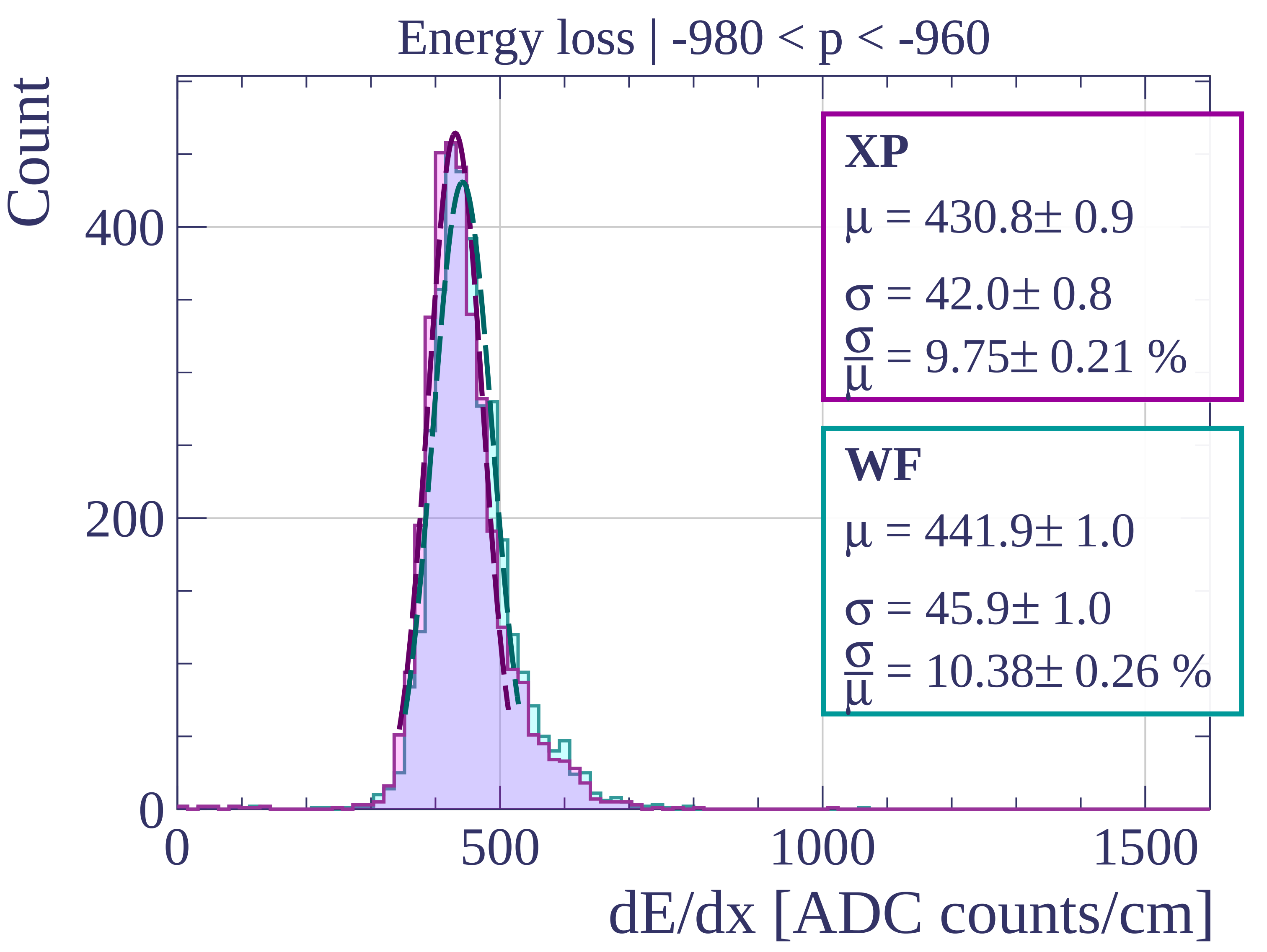
Energy loss | -1060 < p < -1040
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 433.6± 0.9
σ = 40.2± 0.9
σ/μ = 9.27± 0.22 %
WF
μ = 444.5± 0.9
σ = 42.5± 0.9
σ/μ = 9.56± 0.21 %


Energy loss | -1040 < p < -1020
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 432.0± 0.9
σ = 41.3± 1.0
σ/μ = 9.55± 0.24 %
WF
μ = 443.8± 1.0
σ = 44.2± 1.0
σ/μ = 9.96± 0.24 %

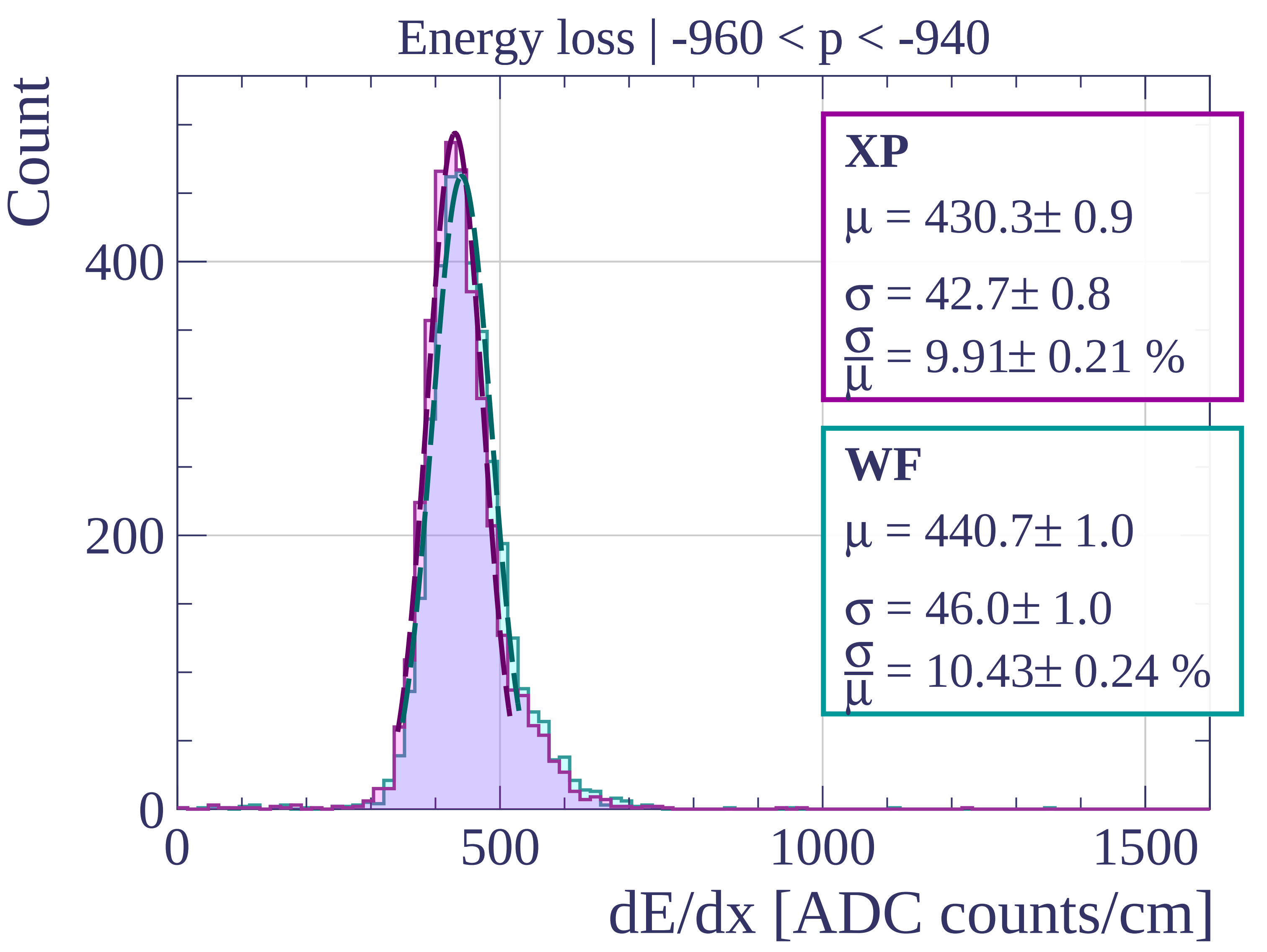

Energy loss | -1020 < p < -1000
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 429.4± 1.0
σ = 42.5± 1.1
σ/μ = 9.90± 0.27 %
WF
μ = 441.1± 1.0
σ = 45.2± 1.0
σ/μ = 10.24± 0.26 %



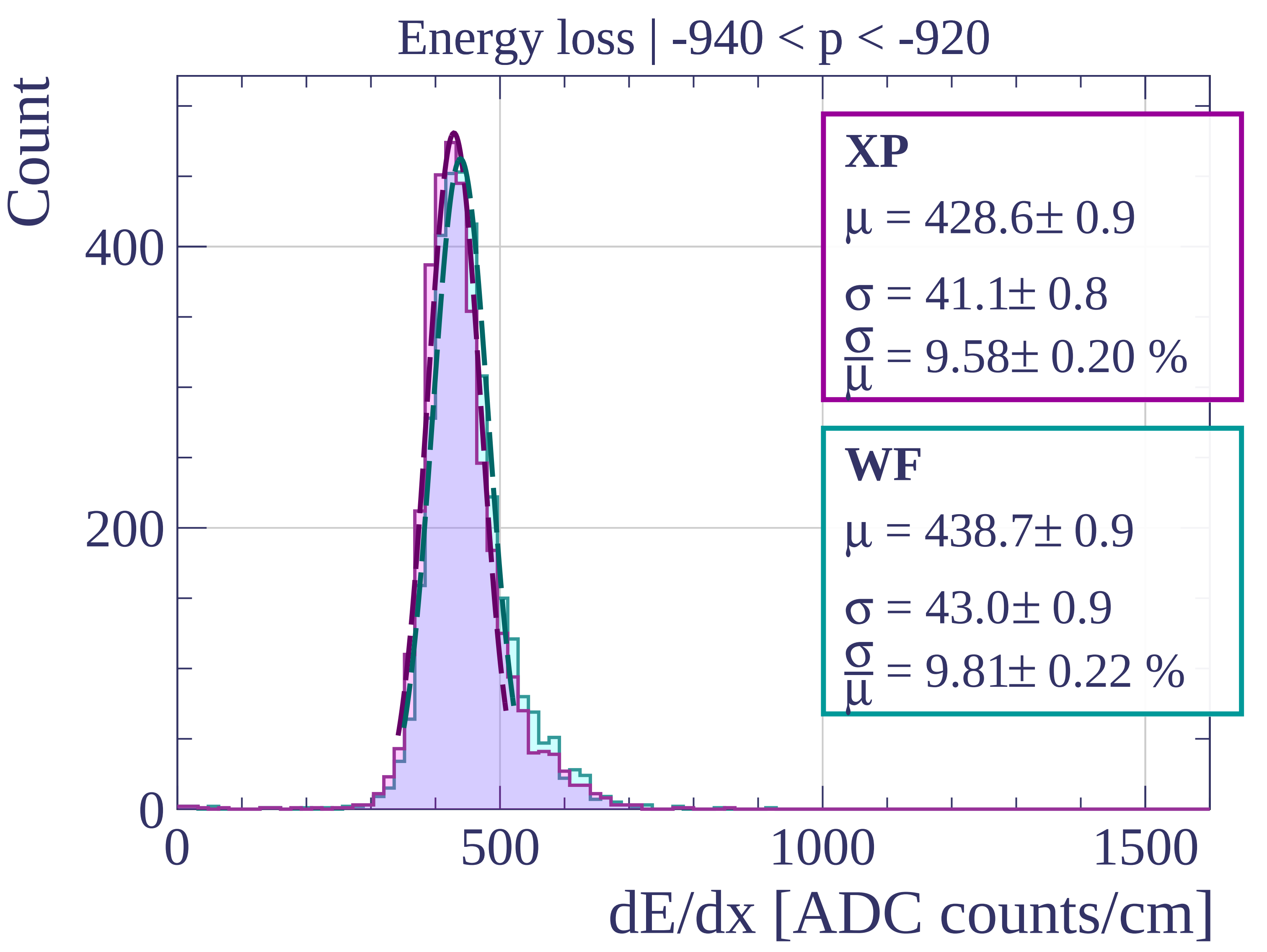
Energy loss | -1000 < p < -980
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 430.9± 0.9
σ = 42.1± 0.9
σ/μ = 9.77± 0.24 %
WF
μ = 443.2± 1.0
σ = 43.3± 0.9
σ/μ = 9.78± 0.21 %



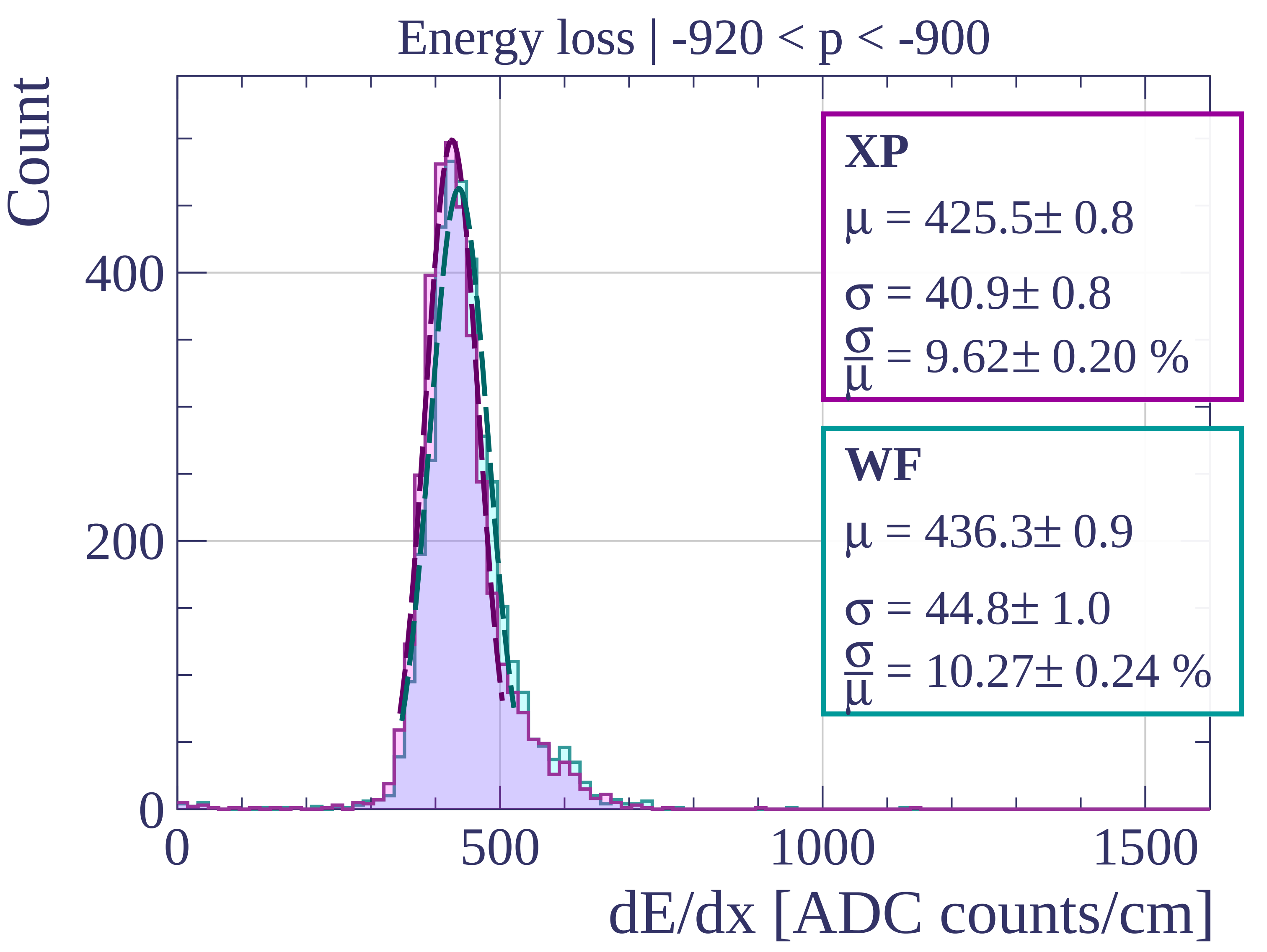
Energy loss | -980 < p < -960
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 430.8± 0.9
σ = 42.0± 0.8
σ/μ = 9.75± 0.21 %
WF
μ = 441.9± 1.0
σ = 45.9± 1.0
σ/μ = 10.38± 0.26 %



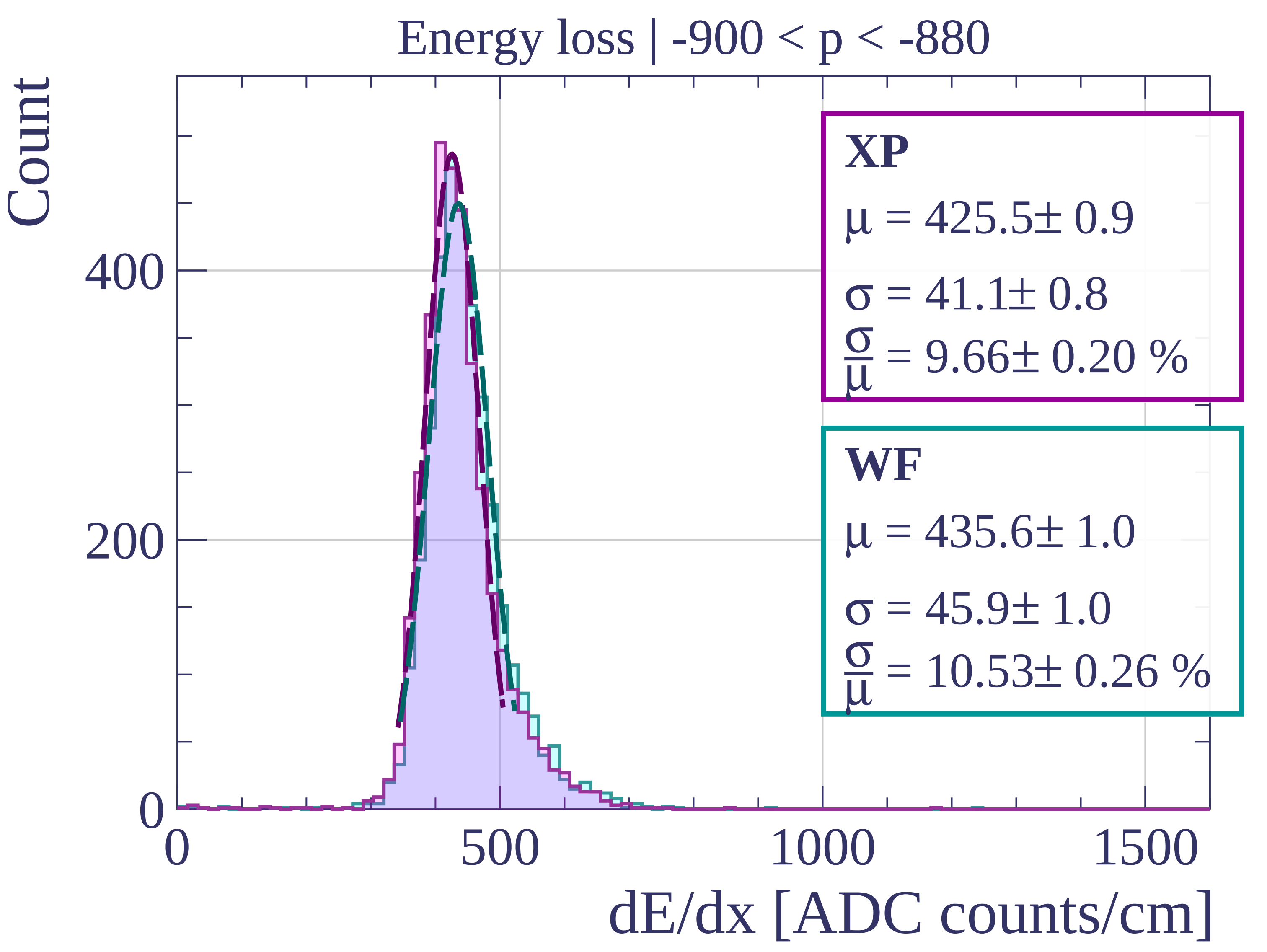
Energy loss | -960 < p < -940
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 430.3± 0.9
σ = 42.7± 0.8
σ/μ = 9.91± 0.21 %
WF
μ = 440.7± 1.0
σ = 46.0± 1.0
σ/μ = 10.43± 0.24 %

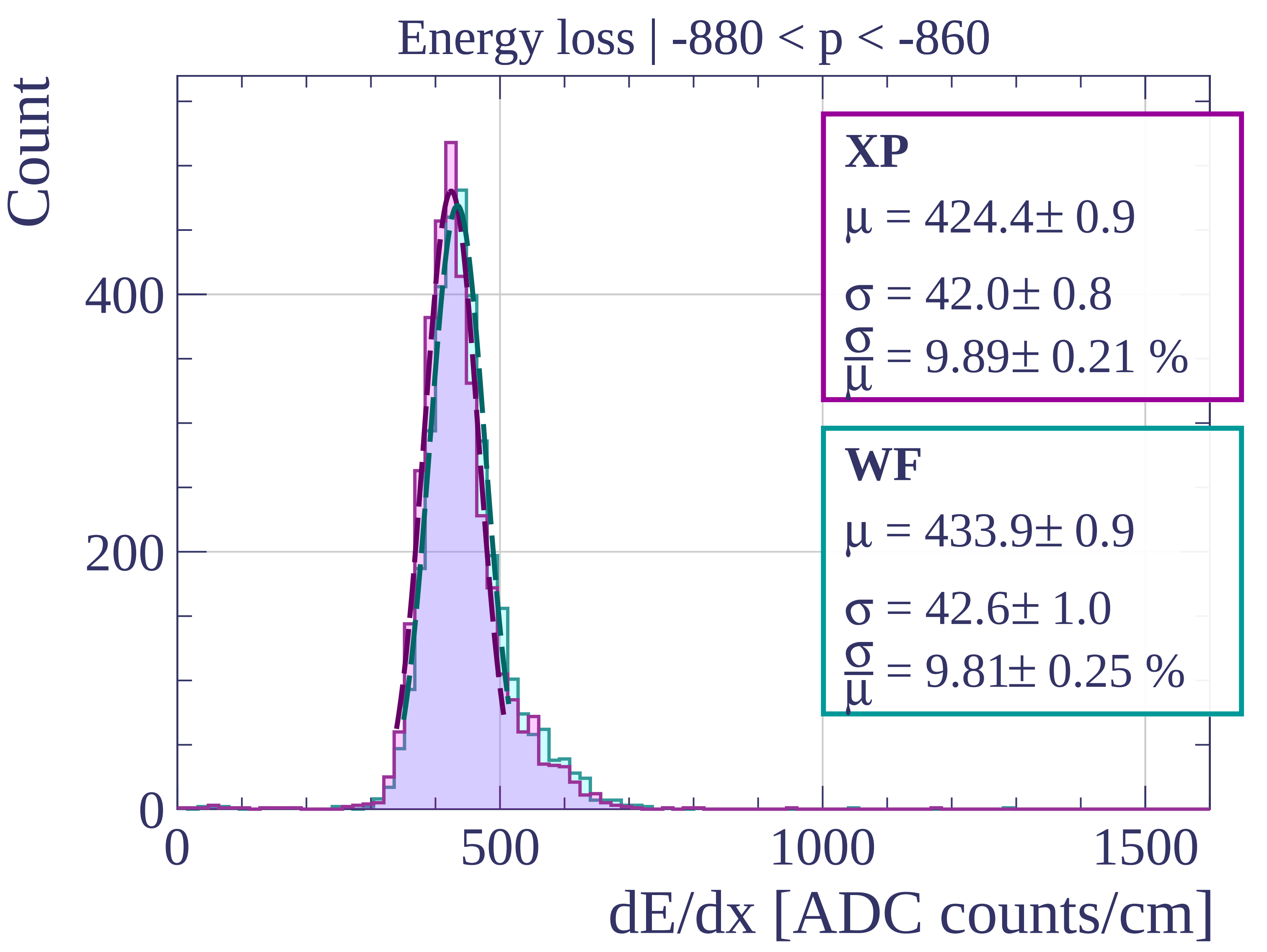


Energy loss | -940 < p < -920
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 428.6± 0.9
σ = 41.1± 0.8
σ/μ = 9.58± 0.20 %
WF
μ = 438.7± 0.9
σ = 43.0± 0.9
σ/μ = 9.81± 0.22 %

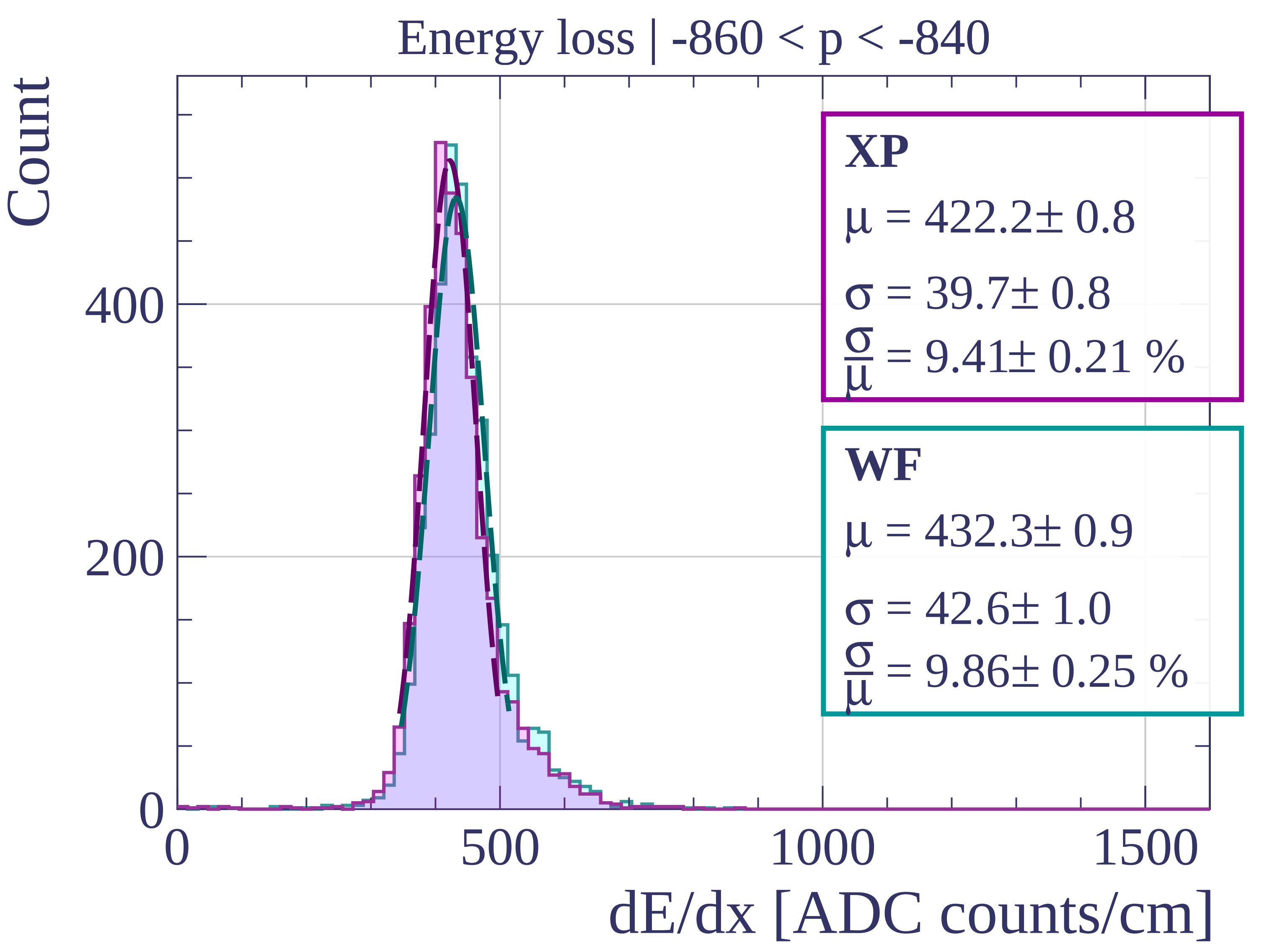


Energy loss | -920 < p < -900
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 425.5± 0.8
σ = 40.9± 0.8
σ/μ = 9.62± 0.20 %
WF
μ = 436.3± 0.9
σ = 44.8± 1.0
σ/μ = 10.27± 0.24 %

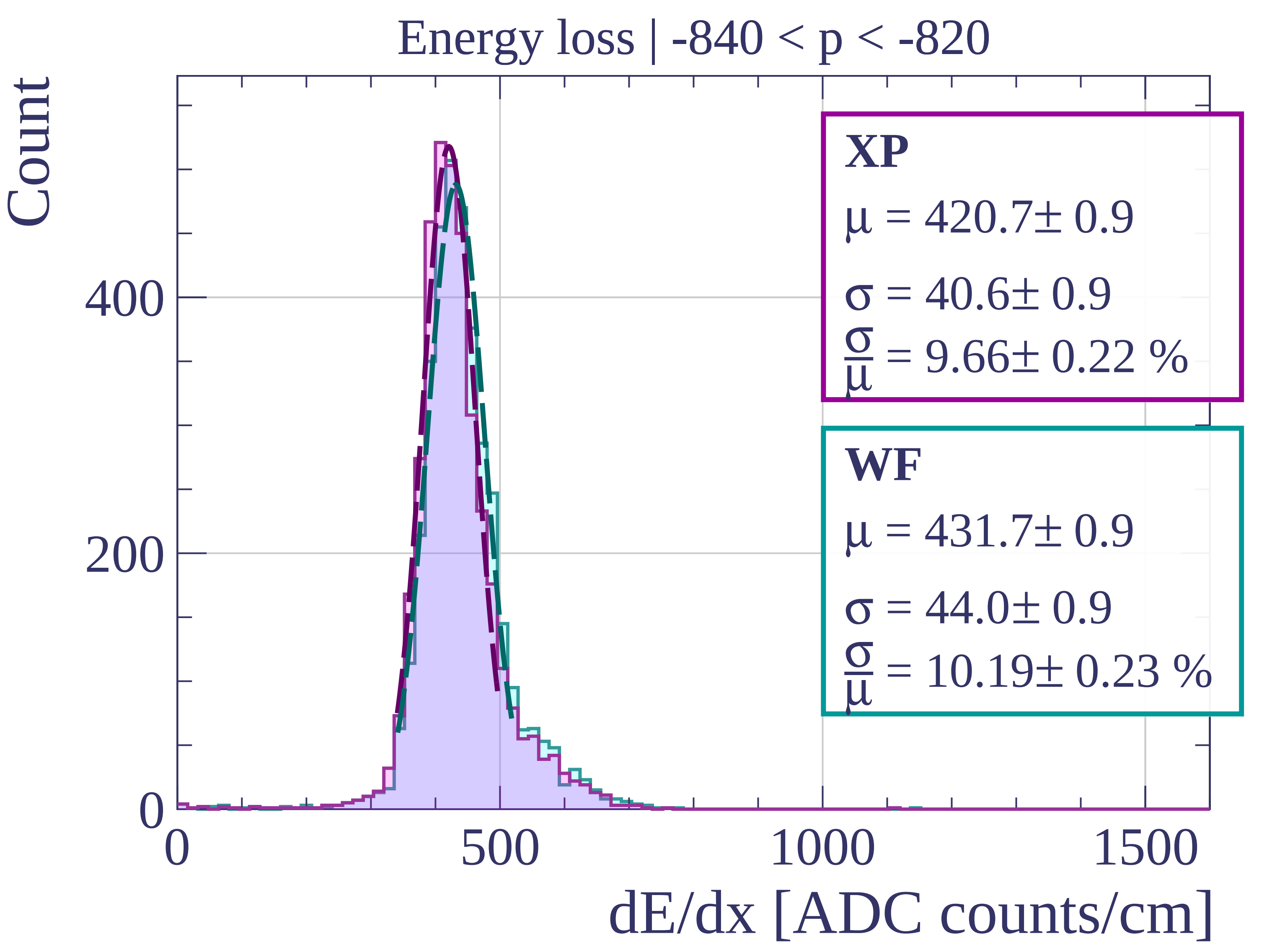

Energy loss | -900 < p < -880
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 425.5± 0.9
σ = 41.1± 0.8
σ/μ = 9.66± 0.20 %
WF
μ = 435.6± 1.0
σ = 45.9± 1.0
σ/μ = 10.53± 0.26 %



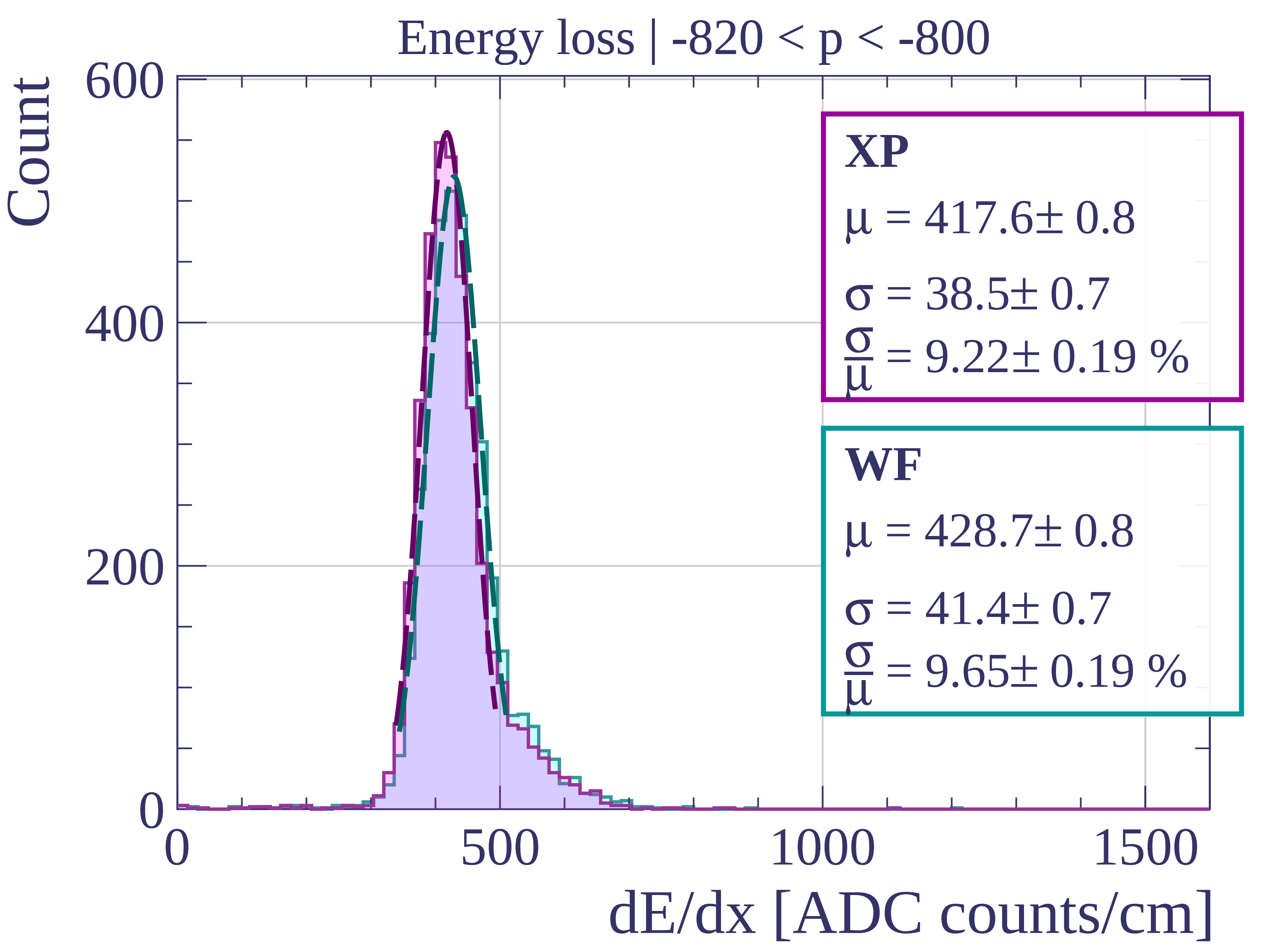
Energy loss | -880 < p < -860
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 424.4± 0.9
σ = 42.0± 0.8
σ/μ = 9.89± 0.21 %
WF
μ = 433.9± 0.9
σ = 42.6± 1.0
σ/μ = 9.81± 0.25 %

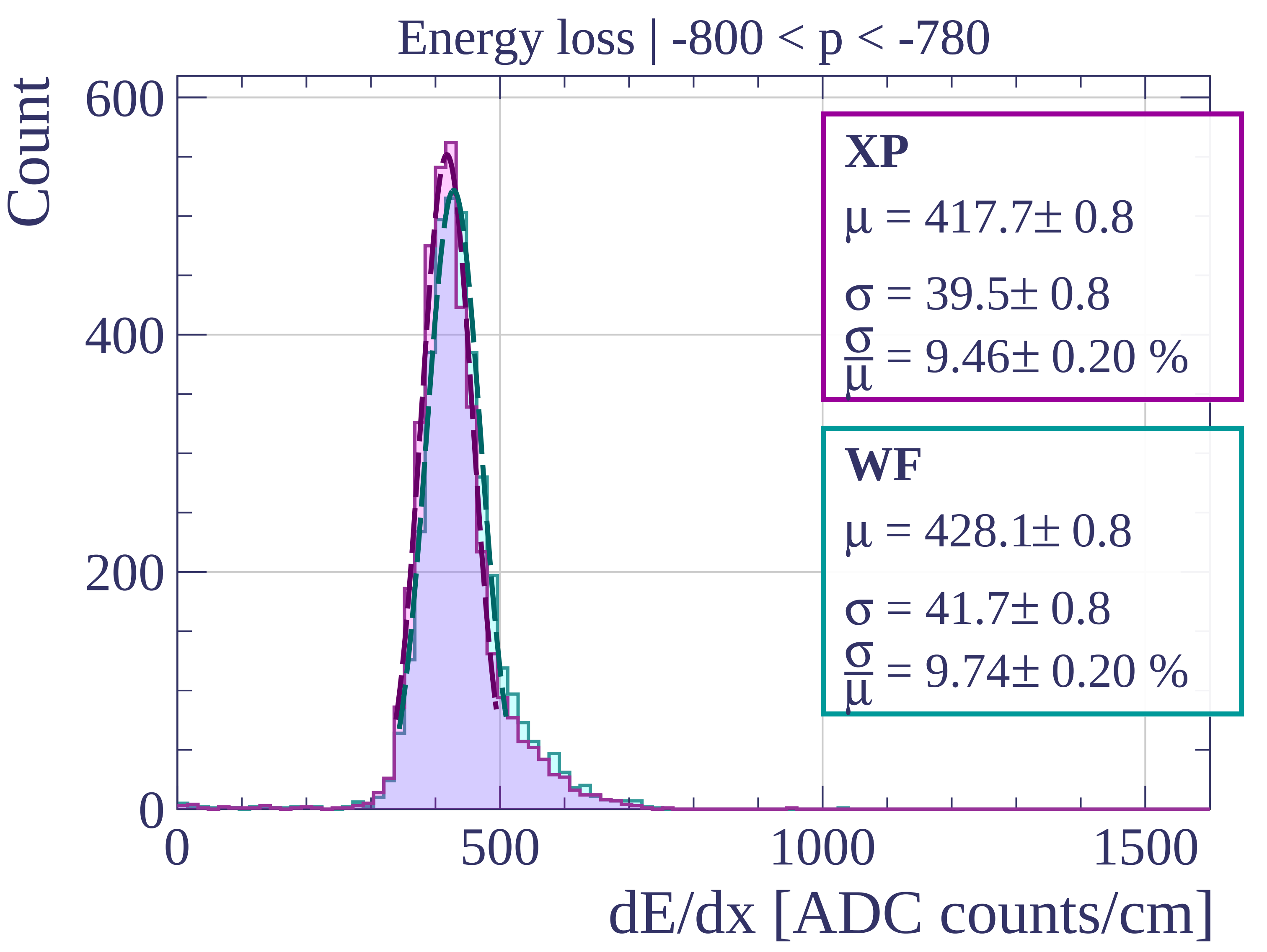


Energy loss | -860 < p < -840
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 422.2± 0.8
σ = 39.7± 0.8
σ/μ = 9.41± 0.21 %
WF
μ = 432.3± 0.9
σ = 42.6± 1.0
σ/μ = 9.86± 0.25 %

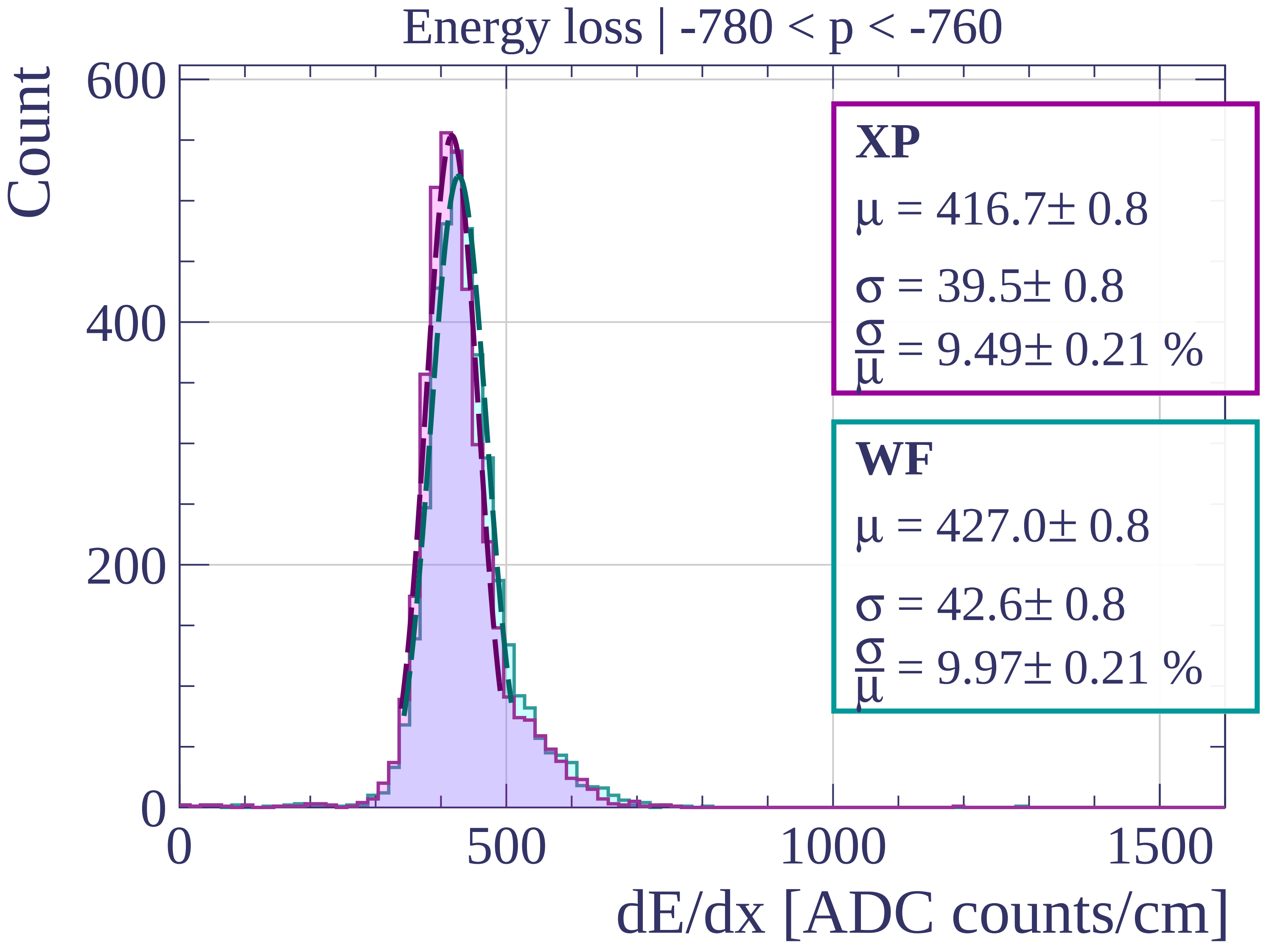


Energy loss | -840 < p < -820
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 420.7± 0.9
σ = 40.6± 0.9
σ/μ = 9.66± 0.22 %
WF
μ = 431.7± 0.9
σ = 44.0± 0.9
σ/μ = 10.19± 0.23 %

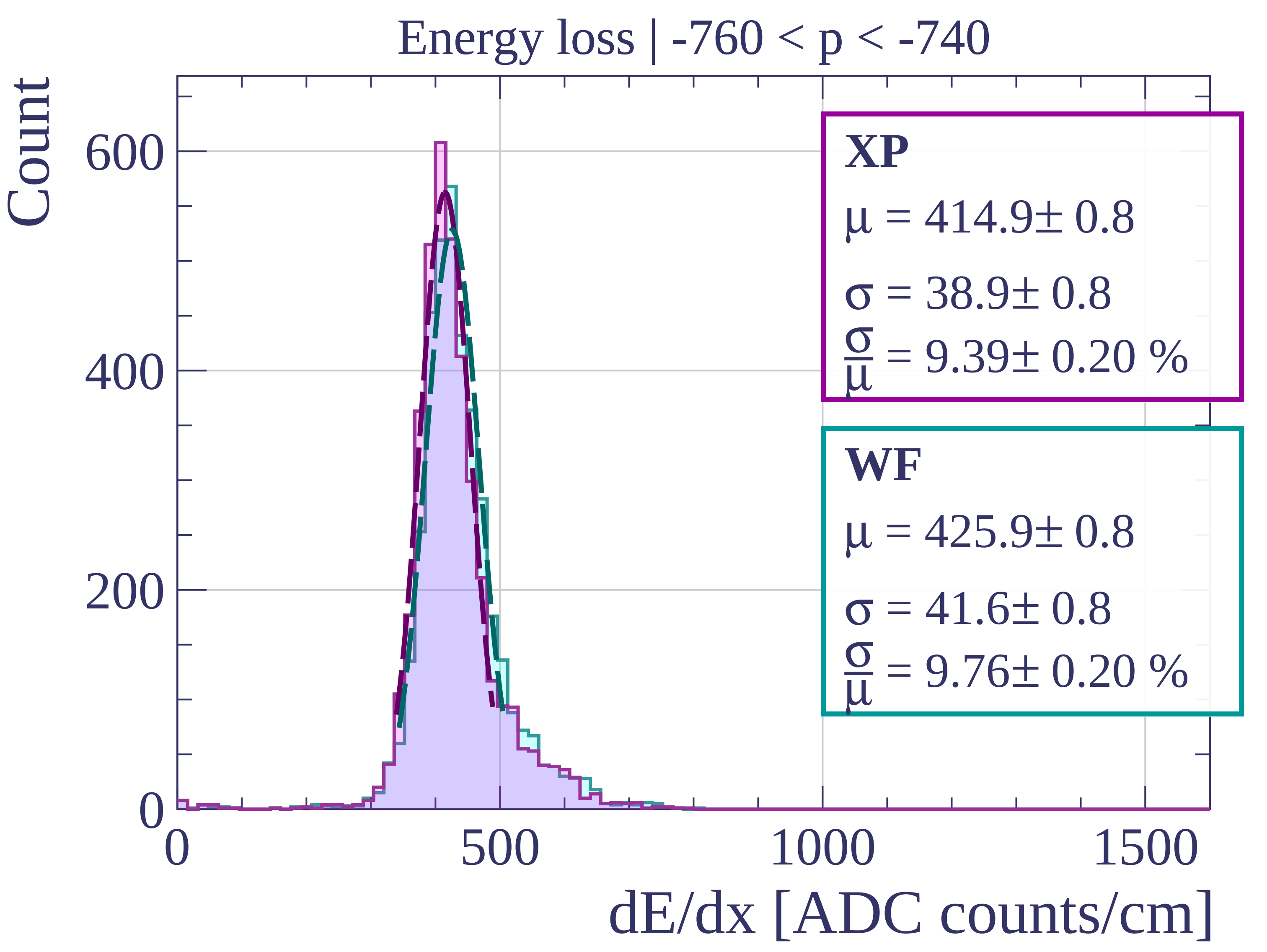

Energy loss | -820 < p < -800
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 417.6± 0.8
σ = 38.5± 0.7
σ/μ = 9.22± 0.19 %
WF
μ = 428.7± 0.8
σ = 41.4± 0.7
σ/μ = 9.65± 0.19 %

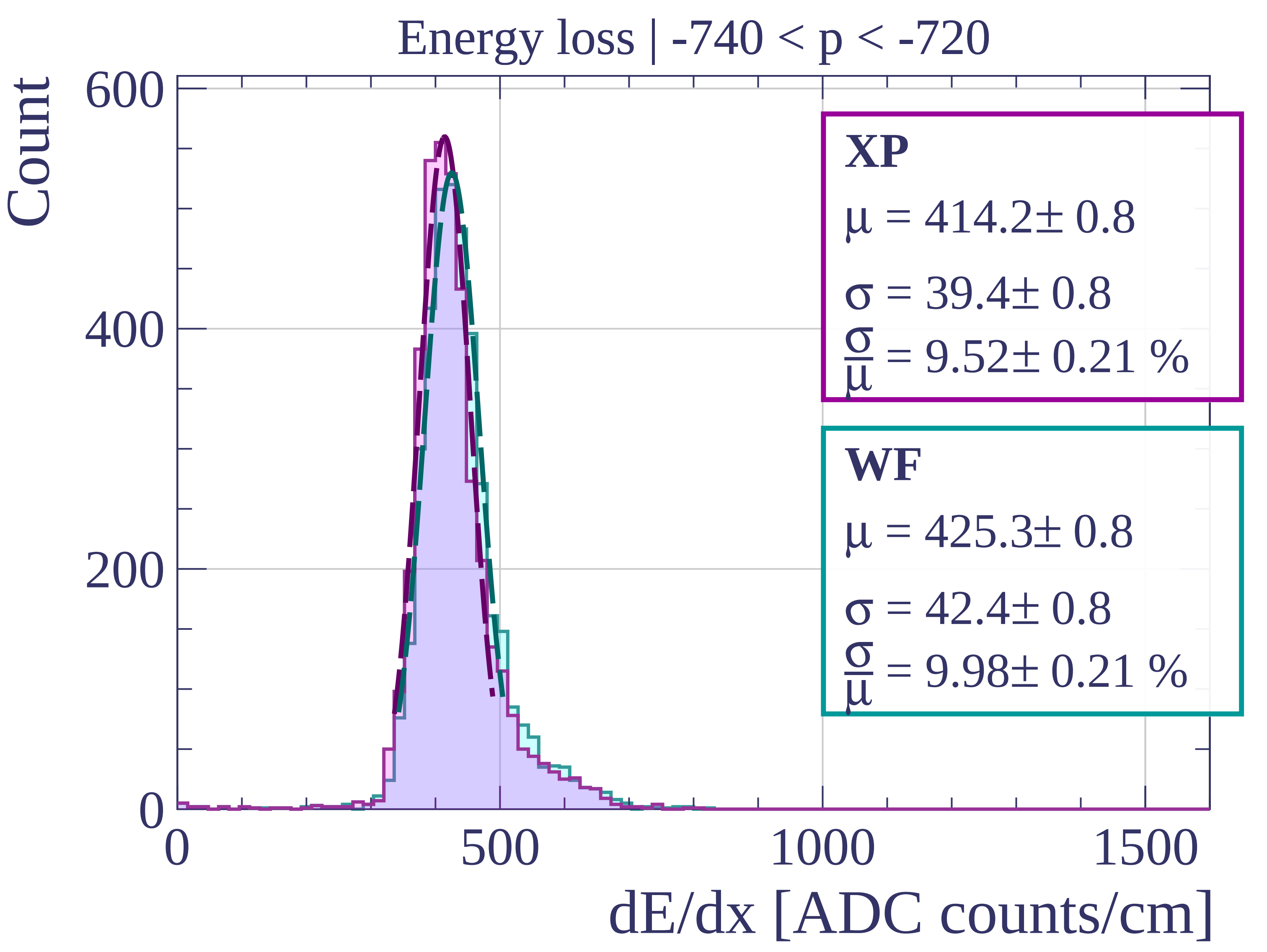


Energy loss | -800 < p < -780
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 417.7± 0.8
σ = 39.5± 0.8
σ/μ = 9.46± 0.20 %
WF
μ = 428.1± 0.8
σ = 41.7± 0.8
σ/μ = 9.74± 0.20 %

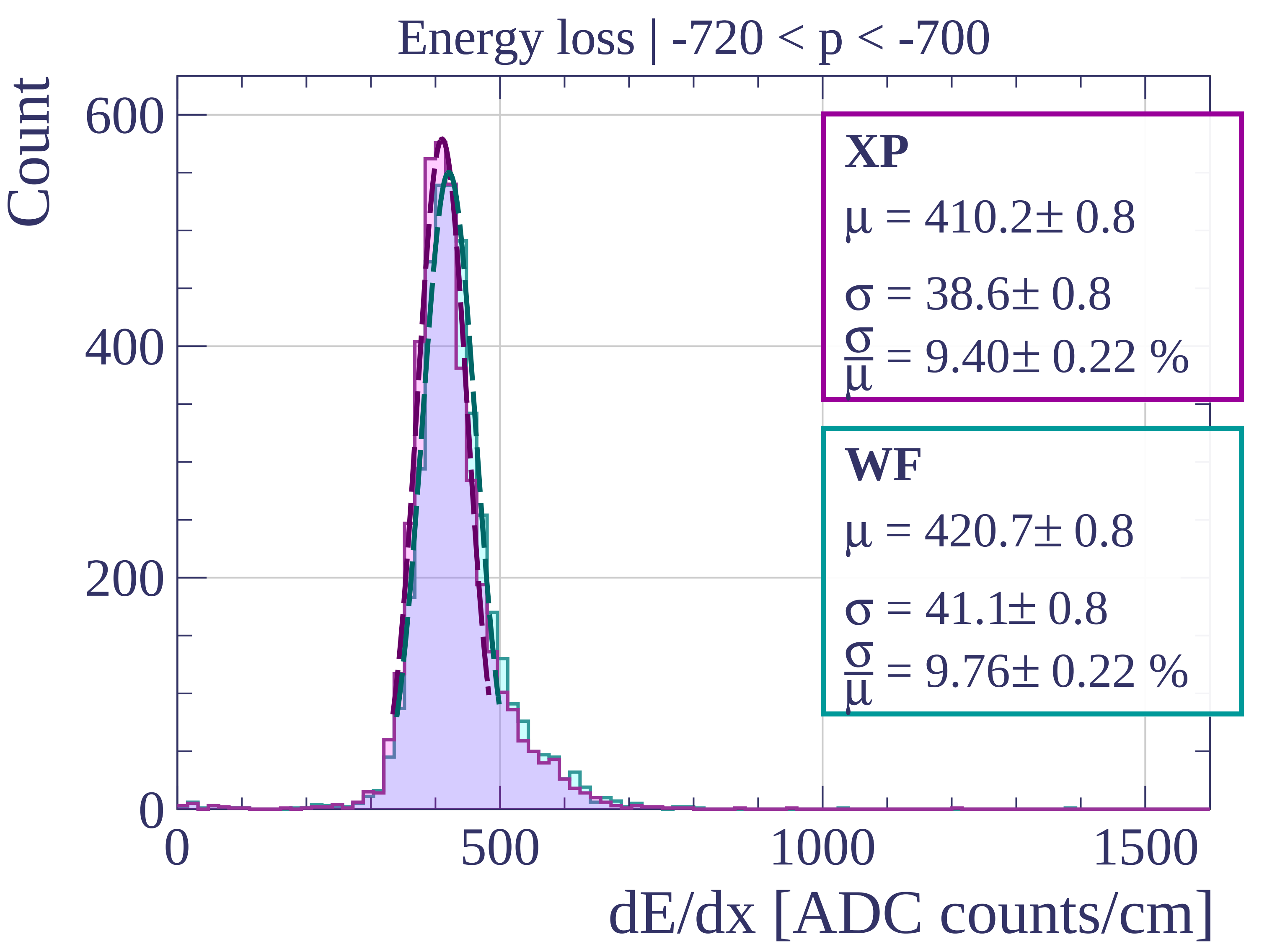


Energy loss | -780 < p < -760
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 416.7± 0.8
σ = 39.5± 0.8
σ/μ = 9.49± 0.21 %
WF
μ = 427.0± 0.8
σ = 42.6± 0.8
σ/μ = 9.97± 0.21 %

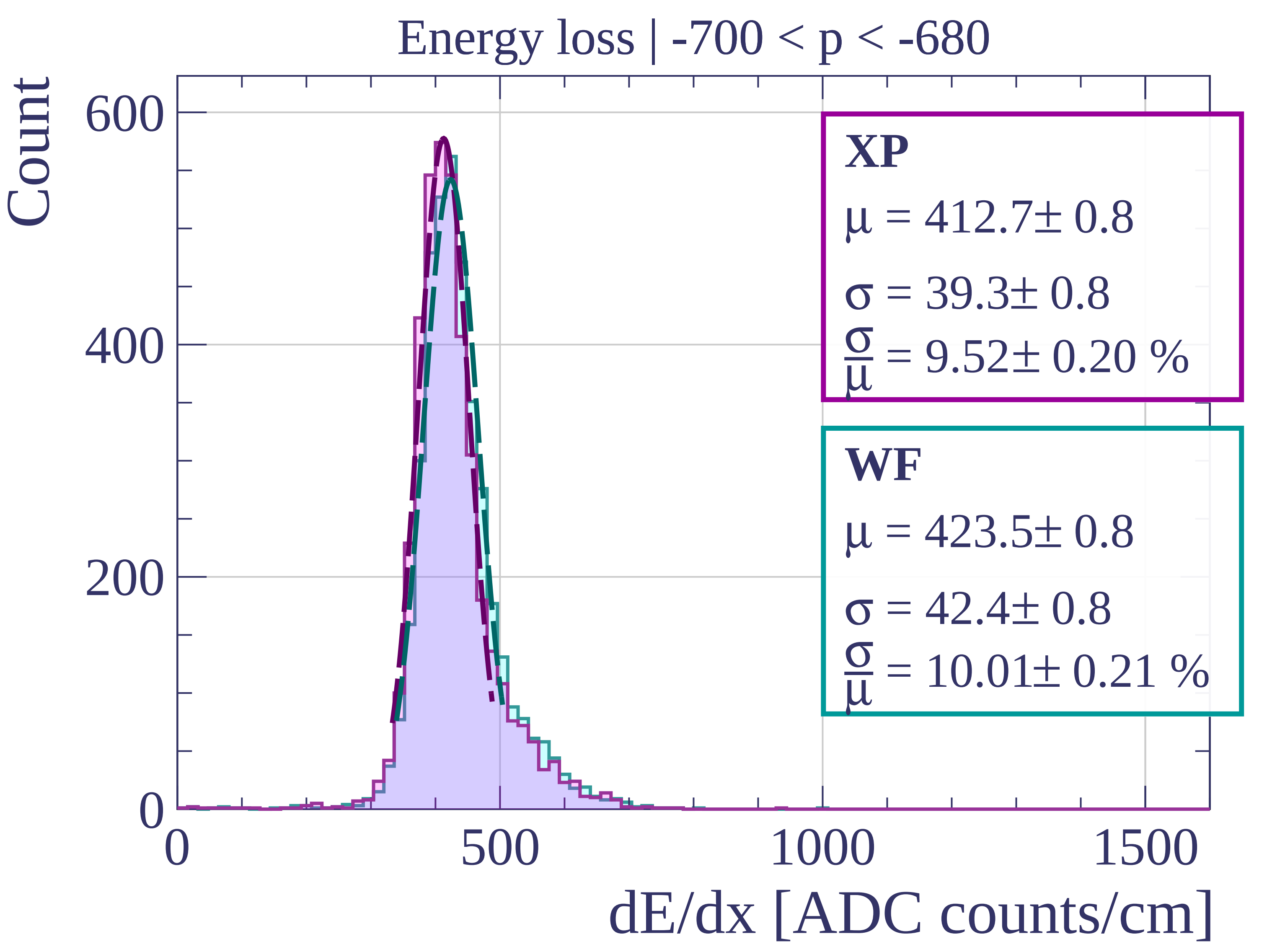

Energy loss | -760 < p < -740
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 414.9± 0.8
σ = 38.9± 0.8
σ/μ = 9.39± 0.20 %
WF
μ = 425.9± 0.8
σ = 41.6± 0.8
σ/μ = 9.76± 0.20 %

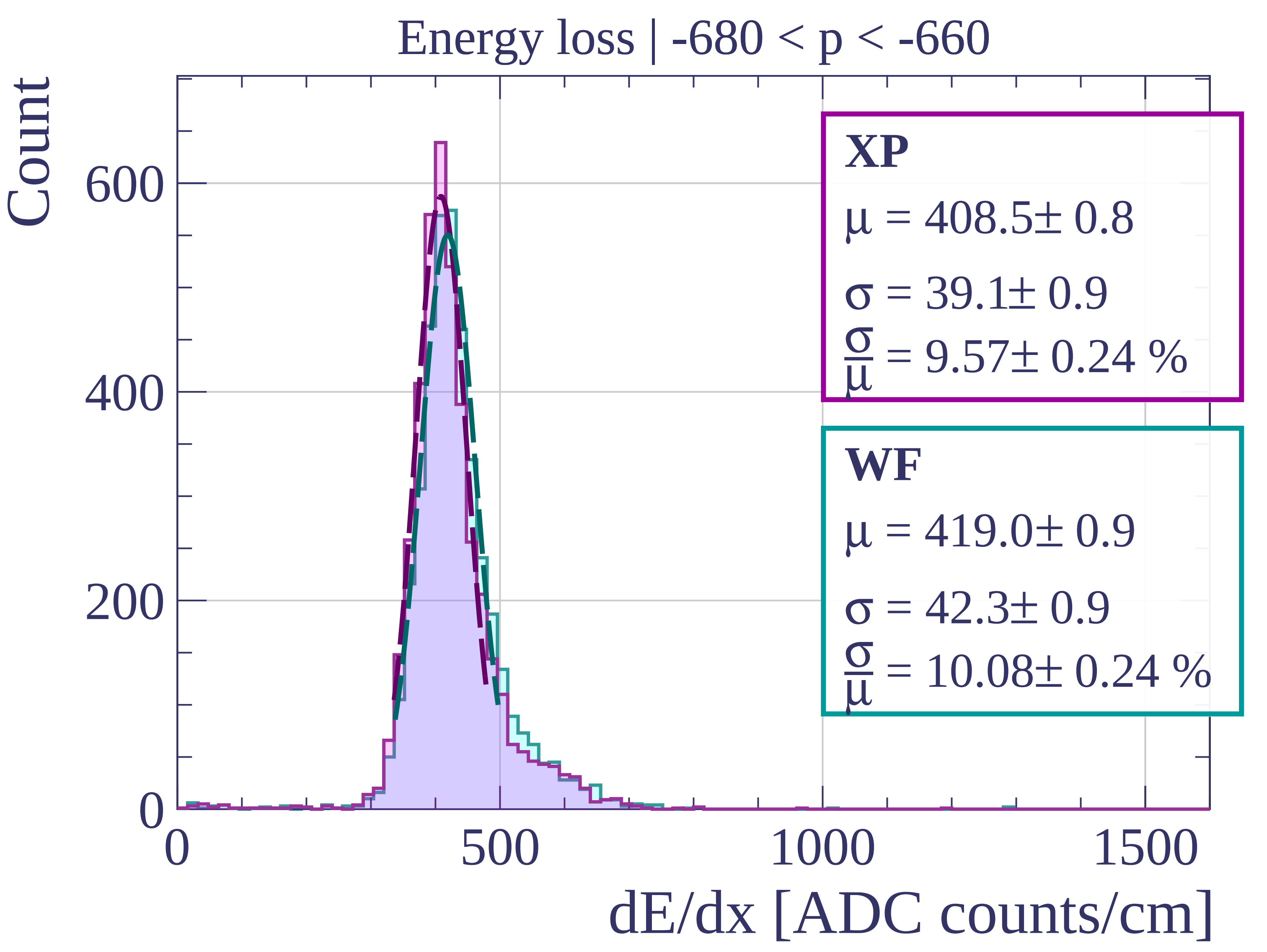


Energy loss | -740 < p < -720
Count
600
400
200
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 414.2± 0.8
σ = 39.4± 0.8
σ/μ = 9.52± 0.21 %
WF
μ = 425.3± 0.8
σ = 42.4± 0.8
σ/μ = 9.98± 0.21 %

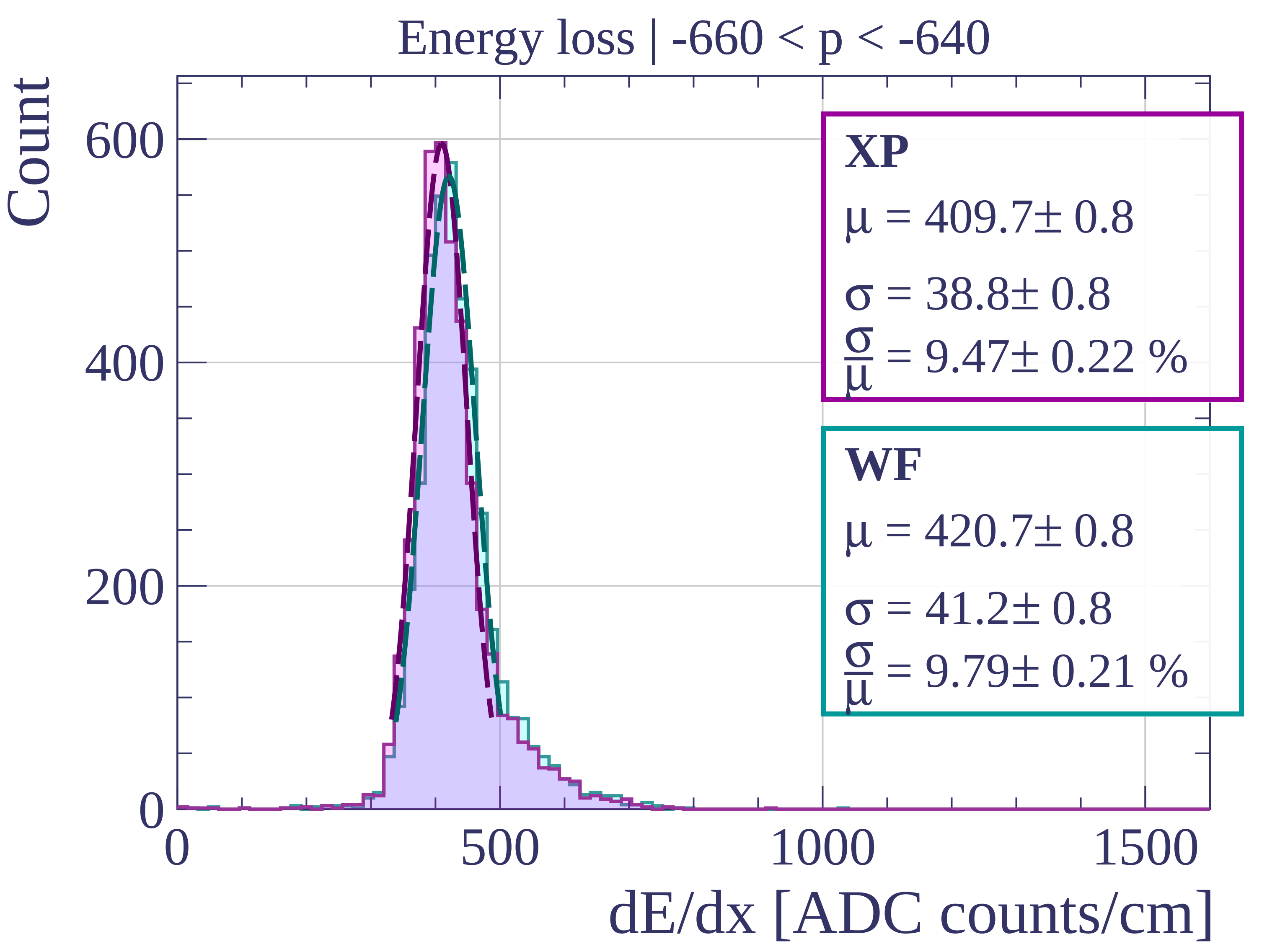


Energy loss | -720 < p < -700
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 410.2± 0.8
σ = 38.6± 0.8
σ/μ = 9.40± 0.22 %
WF
μ = 420.7± 0.8
σ = 41.1± 0.8
σ/μ = 9.76± 0.22 %

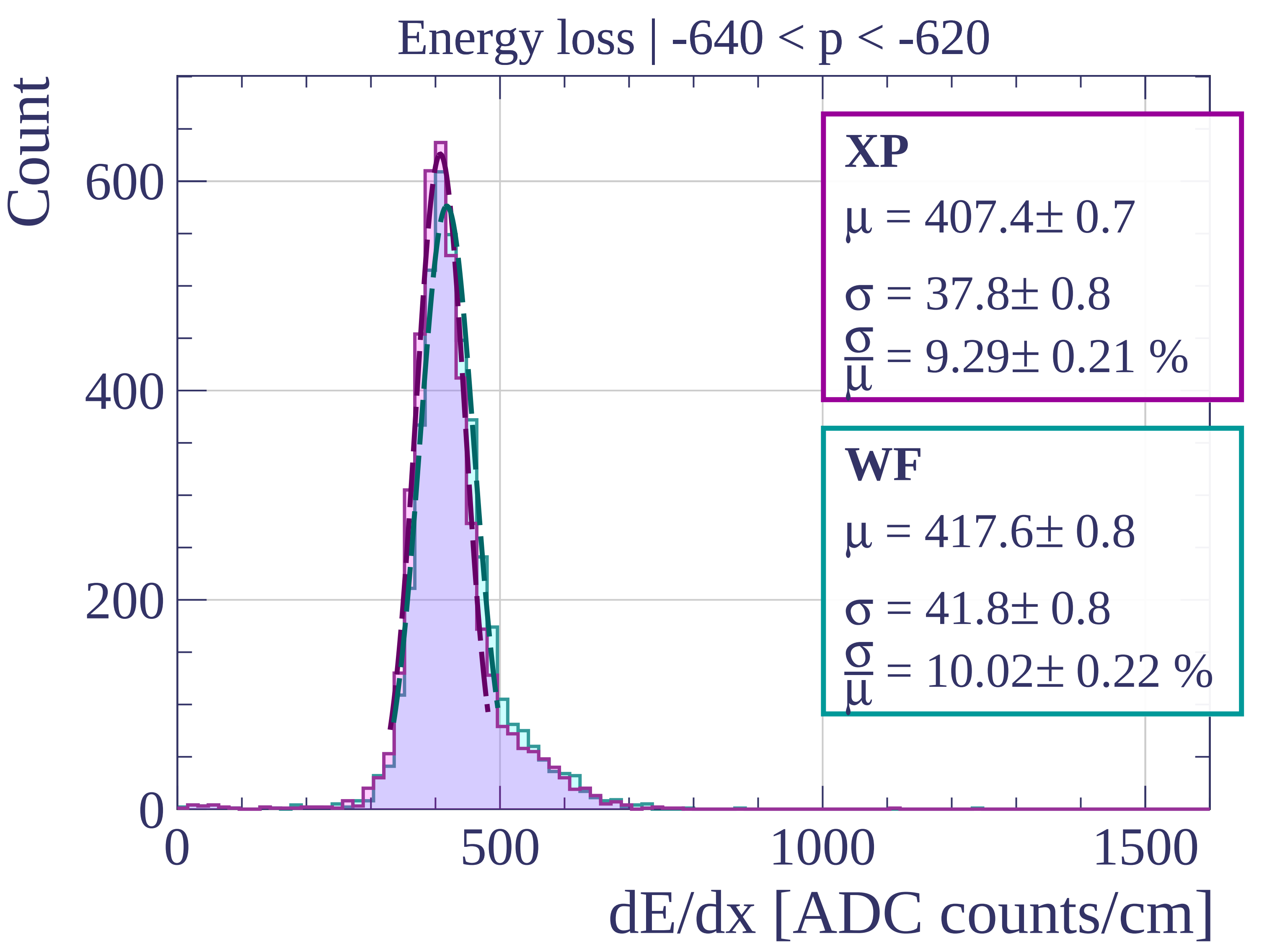

Energy loss | -700 < p < -680
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 412.7± 0.8
σ = 39.3± 0.8
σ/μ = 9.52± 0.20 %
WF
μ = 423.5± 0.8
σ = 42.4± 0.8
σ/μ = 10.01± 0.21 %

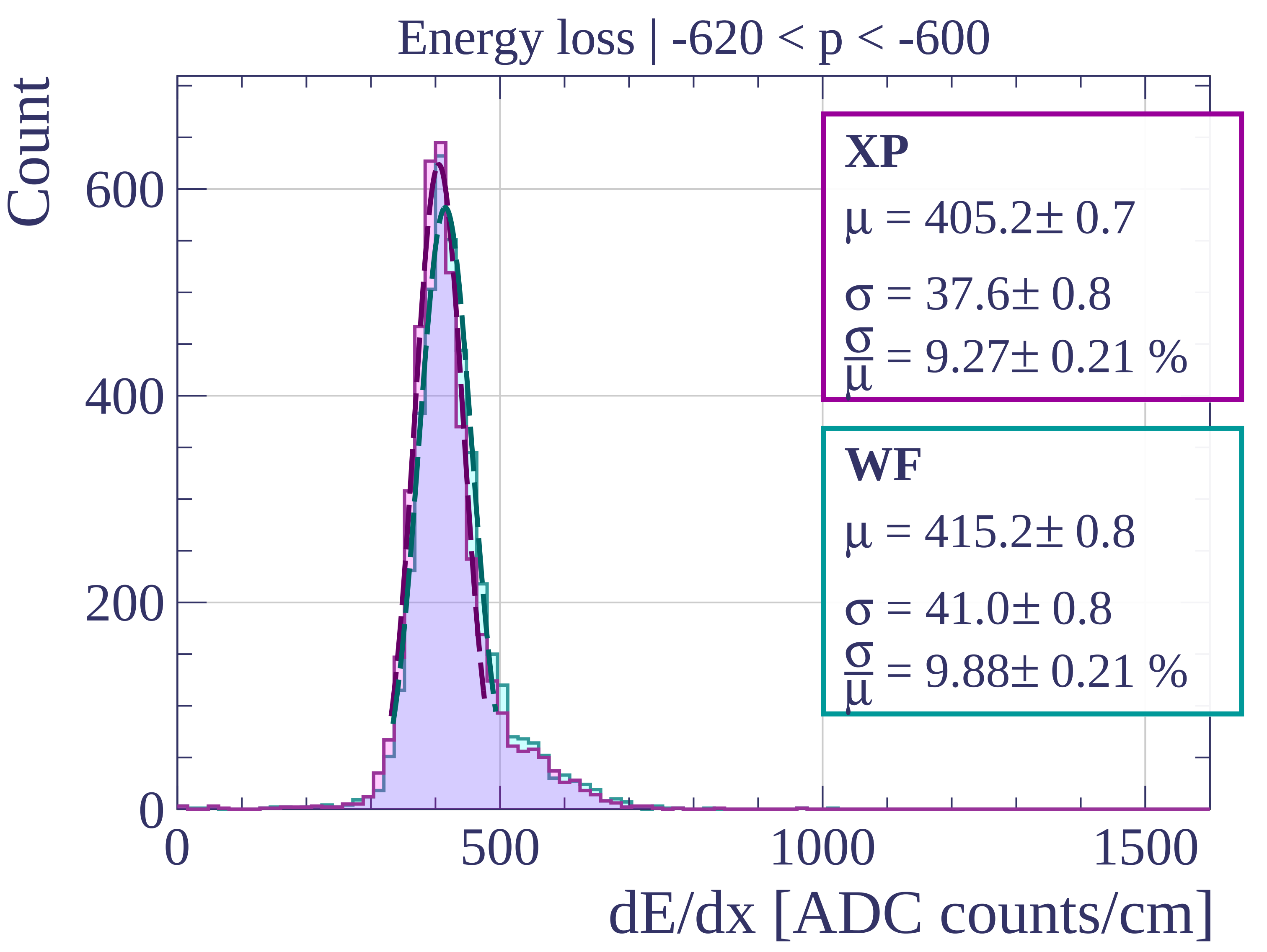


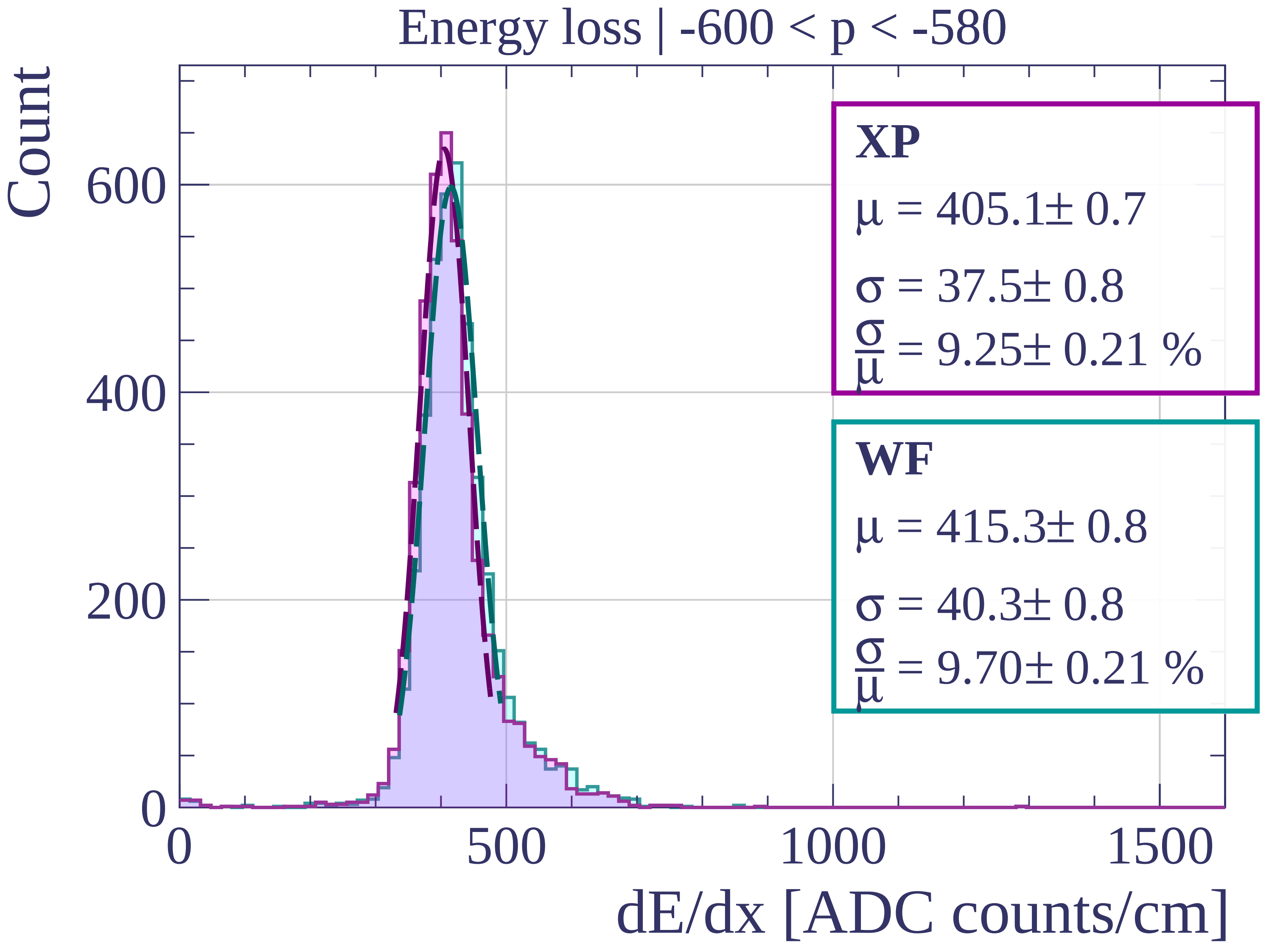
Energy loss | -680 < p < -660
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 408.5± 0.8
σ = 39.1± 0.9
σ/μ = 9.57± 0.24 %
WF
μ = 419.0± 0.9
σ = 42.3± 0.9
σ/μ = 10.08± 0.24 %



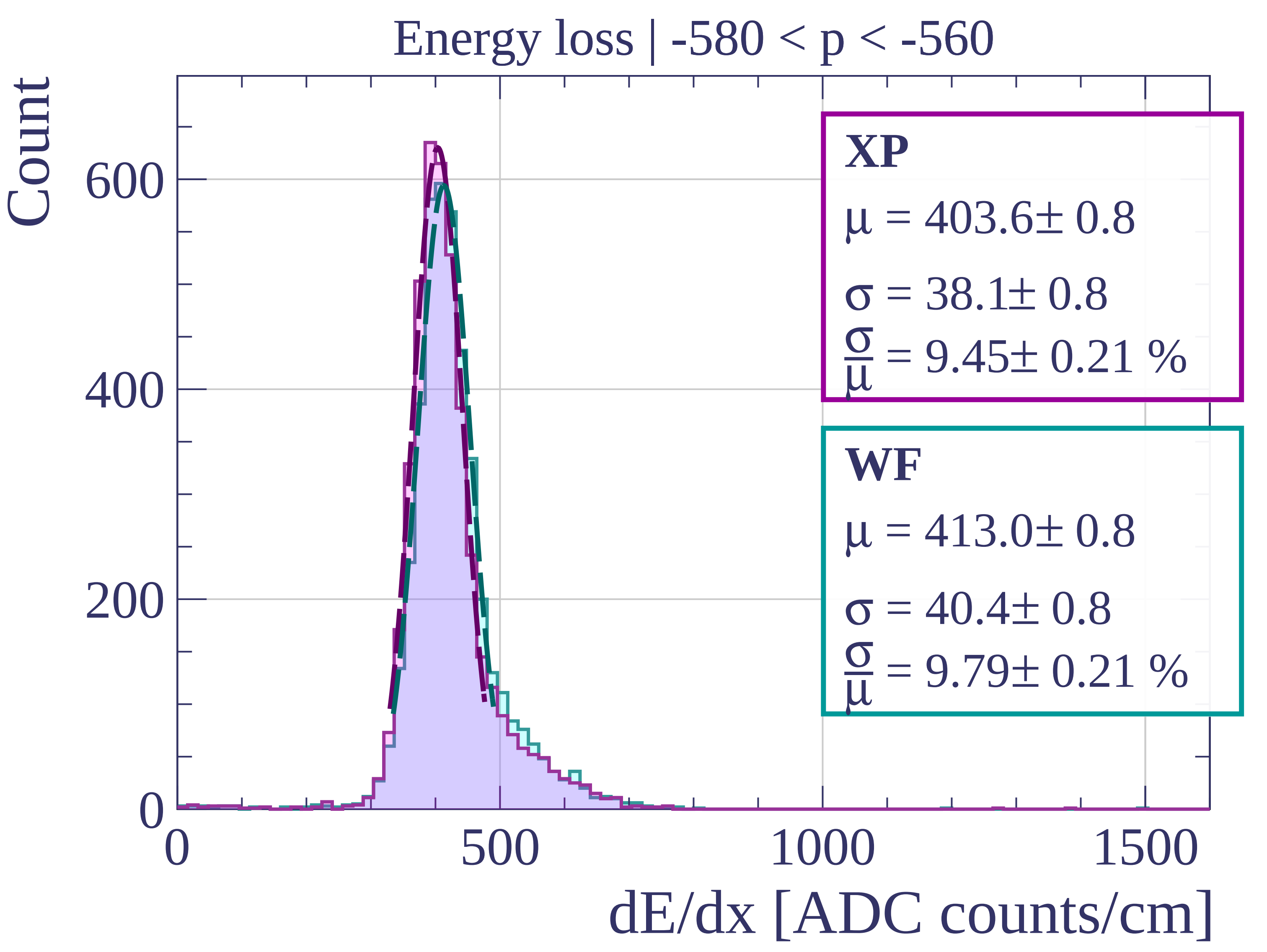
Energy loss | -660 < p < -640
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 409.7± 0.8
σ = 38.8± 0.8
σ/μ = 9.47± 0.22 %
WF
μ = 420.7± 0.8
σ = 41.2± 0.8
σ/μ = 9.79± 0.21 %


Energy loss | -640 < p < -620
Count
dE/dx [ADC counts/cm]
0
200
400
600
500
1000
1500
XP
μ = 407.4± 0.7
σ = 37.8± 0.8
σ/μ = 9.29± 0.21 %
WF
μ = 417.6± 0.8
σ = 41.8± 0.8
σ/μ = 10.02± 0.22 %

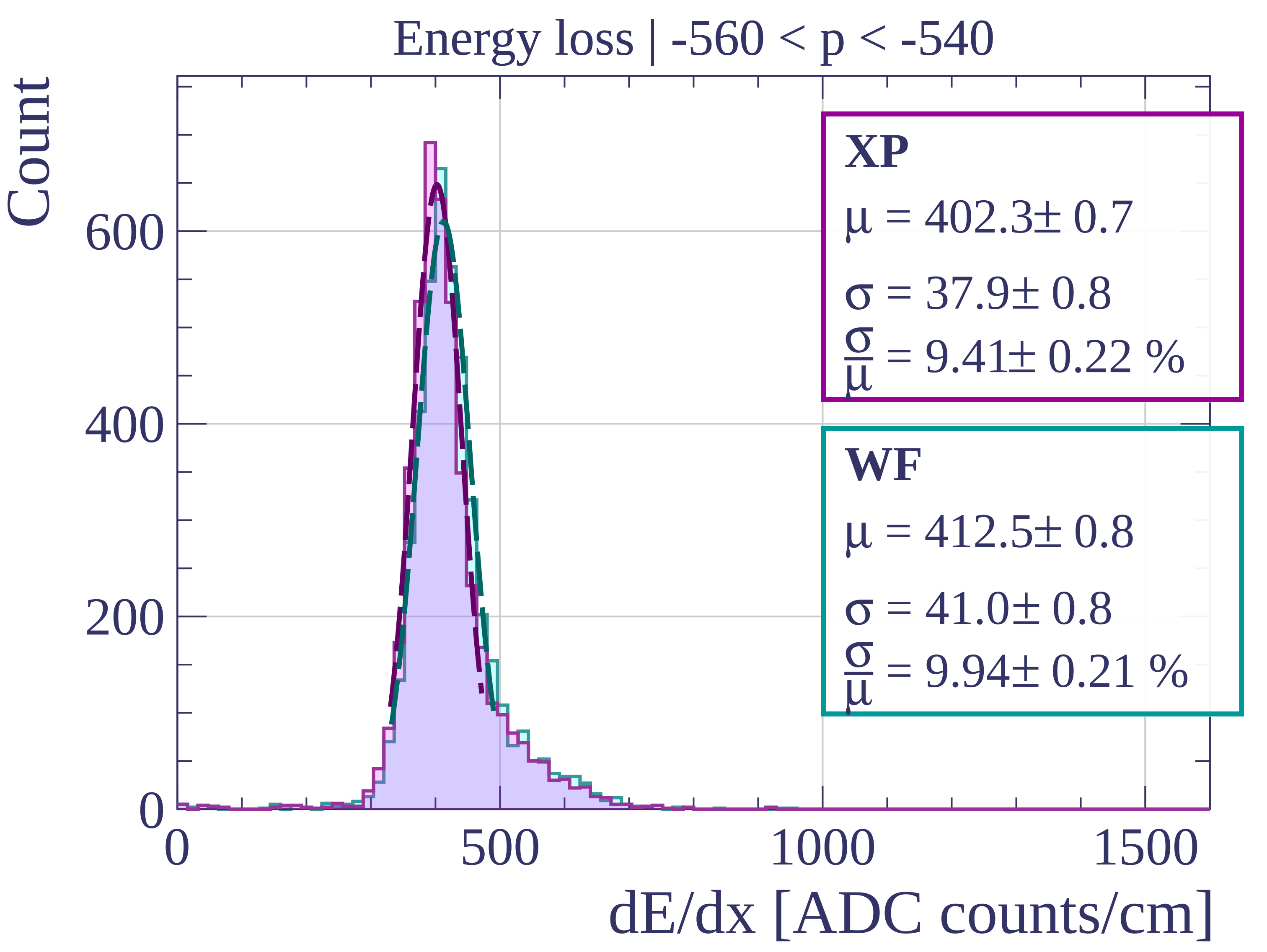

Energy loss | -620 < p < -600
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 405.2± 0.7
σ = 37.6± 0.8
σ/μ = 9.27± 0.21 %
WF
μ = 415.2± 0.8
σ = 41.0± 0.8
σ/μ = 9.88± 0.21 %



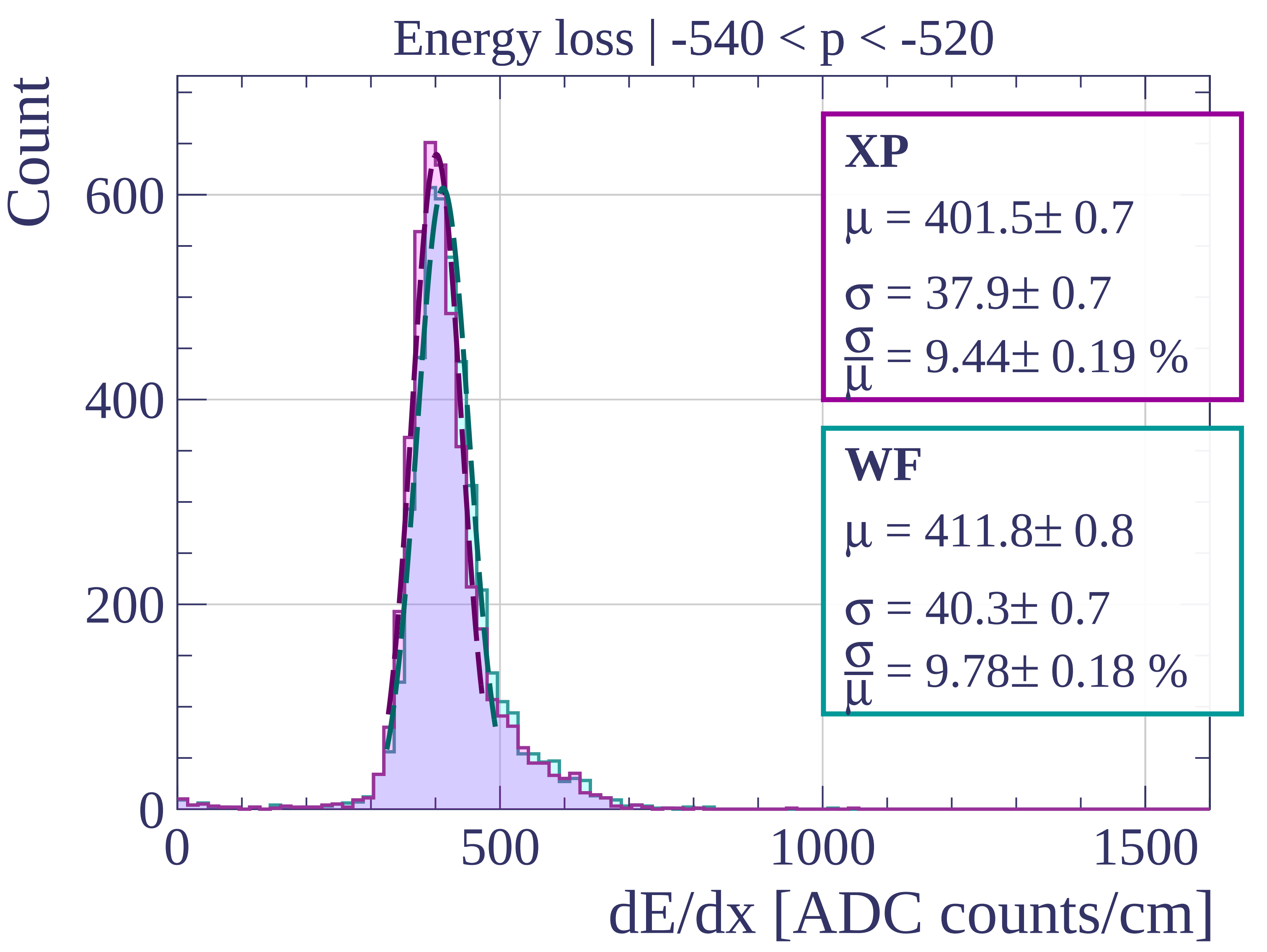
Energy loss | -600 < p < -580
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 405.1± 0.7
σ = 37.5± 0.8
σ/μ = 9.25± 0.21 %
WF
μ = 415.3± 0.8
σ = 40.3± 0.8
σ/μ = 9.70± 0.21 %



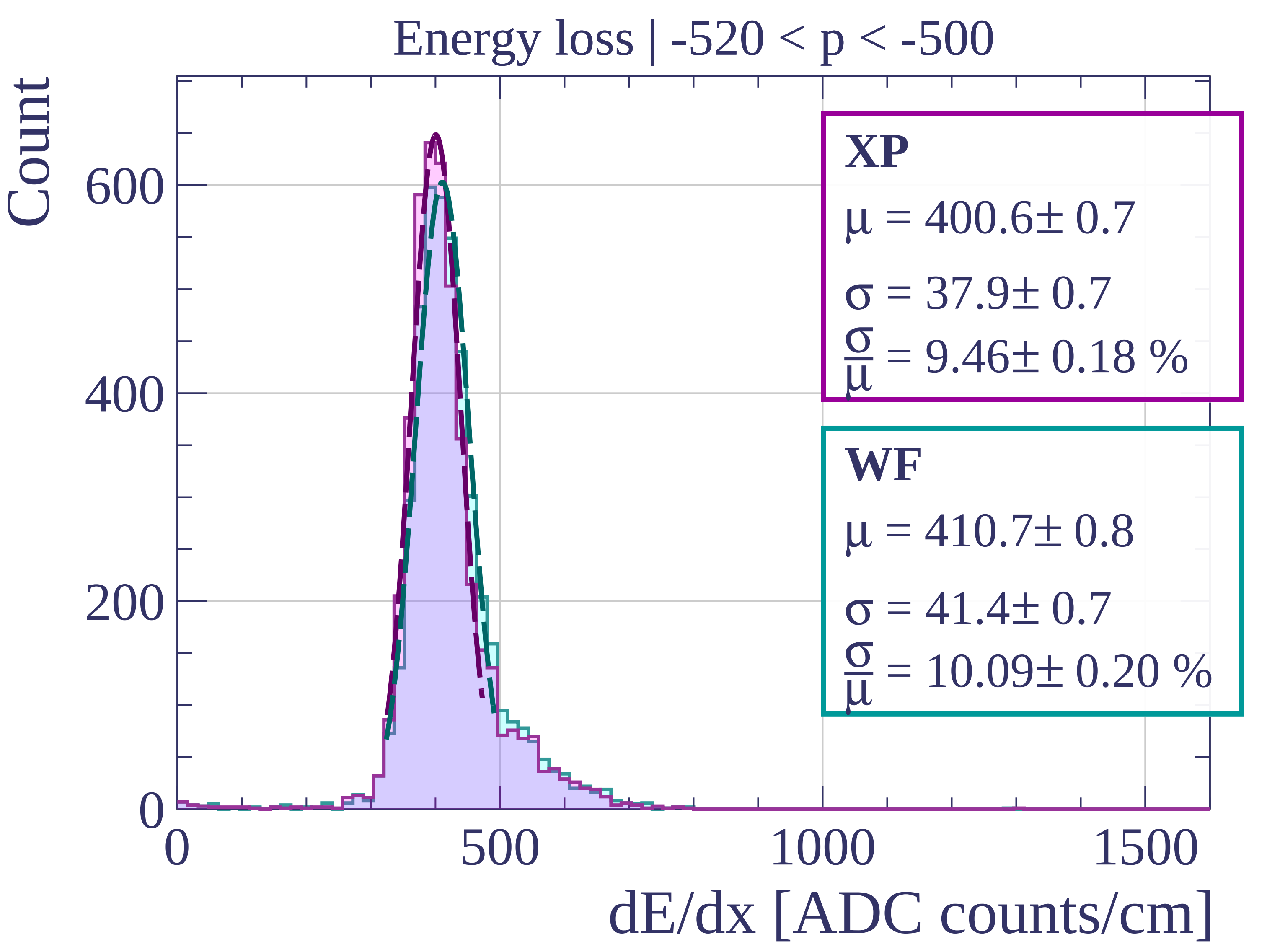
Energy loss | -580 < p < -560
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 403.6± 0.8
σ = 38.1± 0.8
σ/μ = 9.45± 0.21 %
WF
μ = 413.0± 0.8
σ = 40.4± 0.8
σ/μ = 9.79± 0.21 %



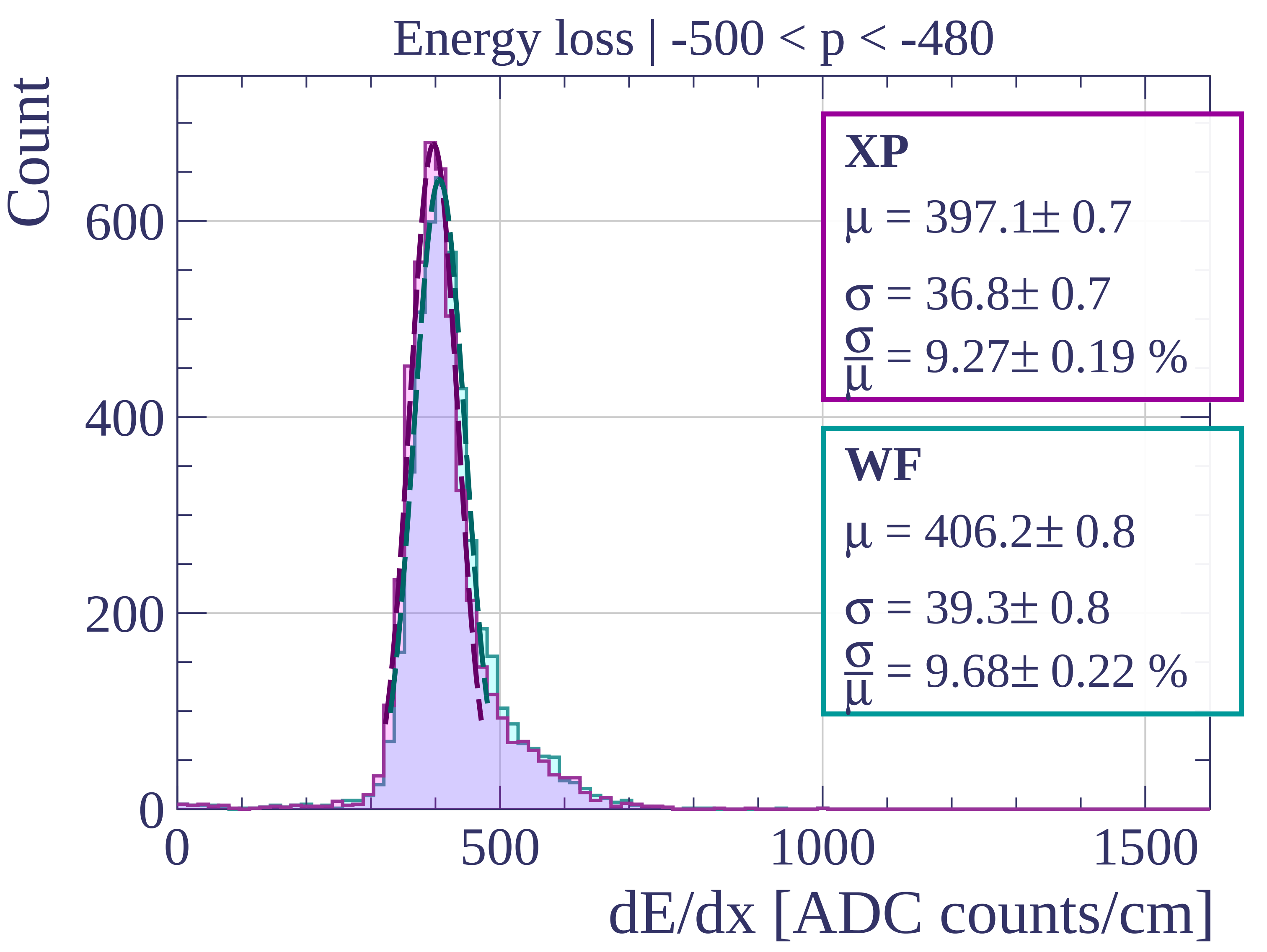
Energy loss | -560 < p < -540
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 402.3± 0.7
σ = 37.9± 0.8
σ/μ = 9.41± 0.22 %
WF
μ = 412.5± 0.8
σ = 41.0± 0.8
σ/μ = 9.94± 0.21 %



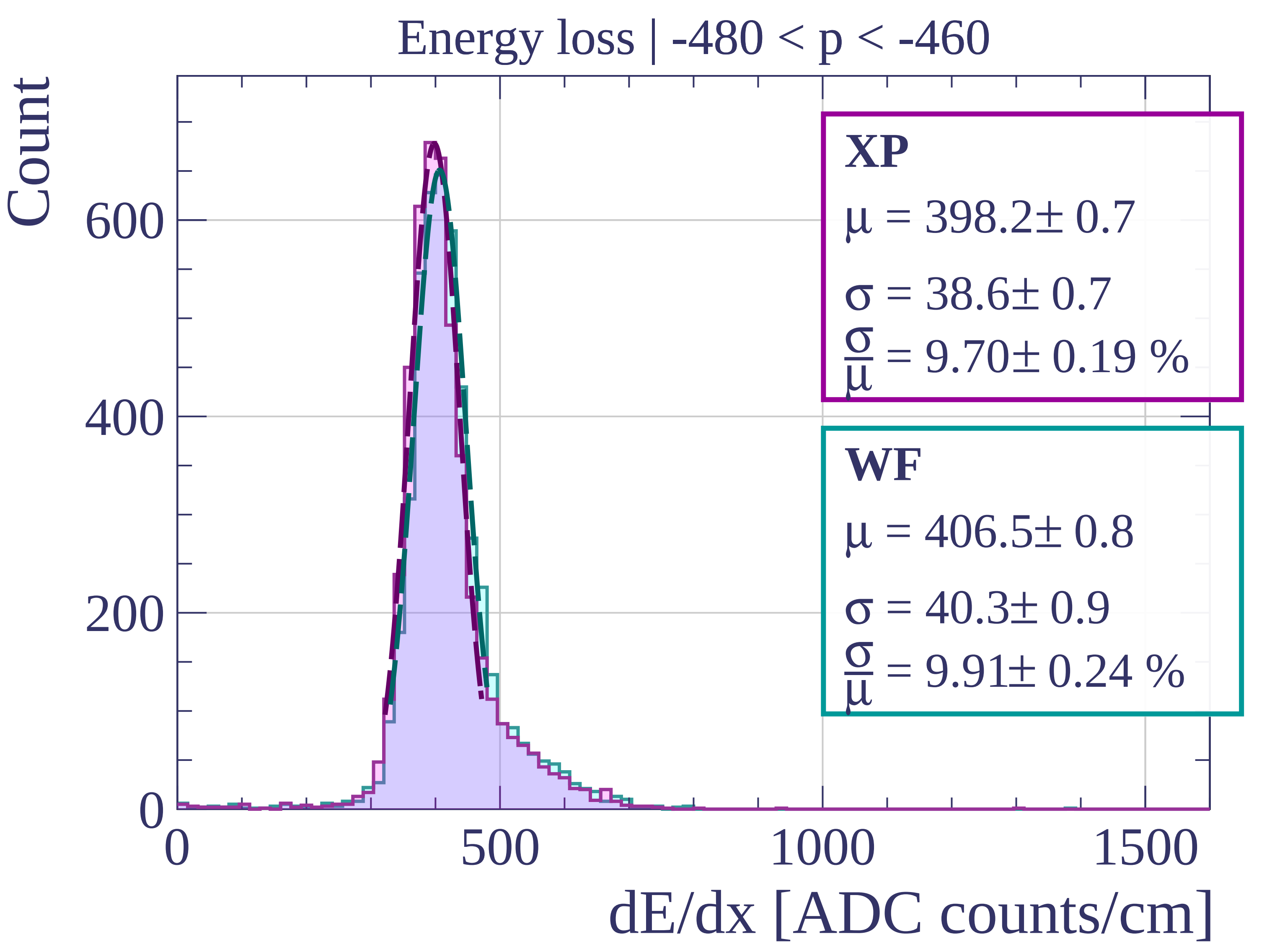
Energy loss | -540 < p < -520
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 401.5± 0.7
σ = 37.9± 0.7
σ/μ = 9.44± 0.19 %
WF
μ = 411.8± 0.8
σ = 40.3± 0.7
σ/μ = 9.78± 0.18 %



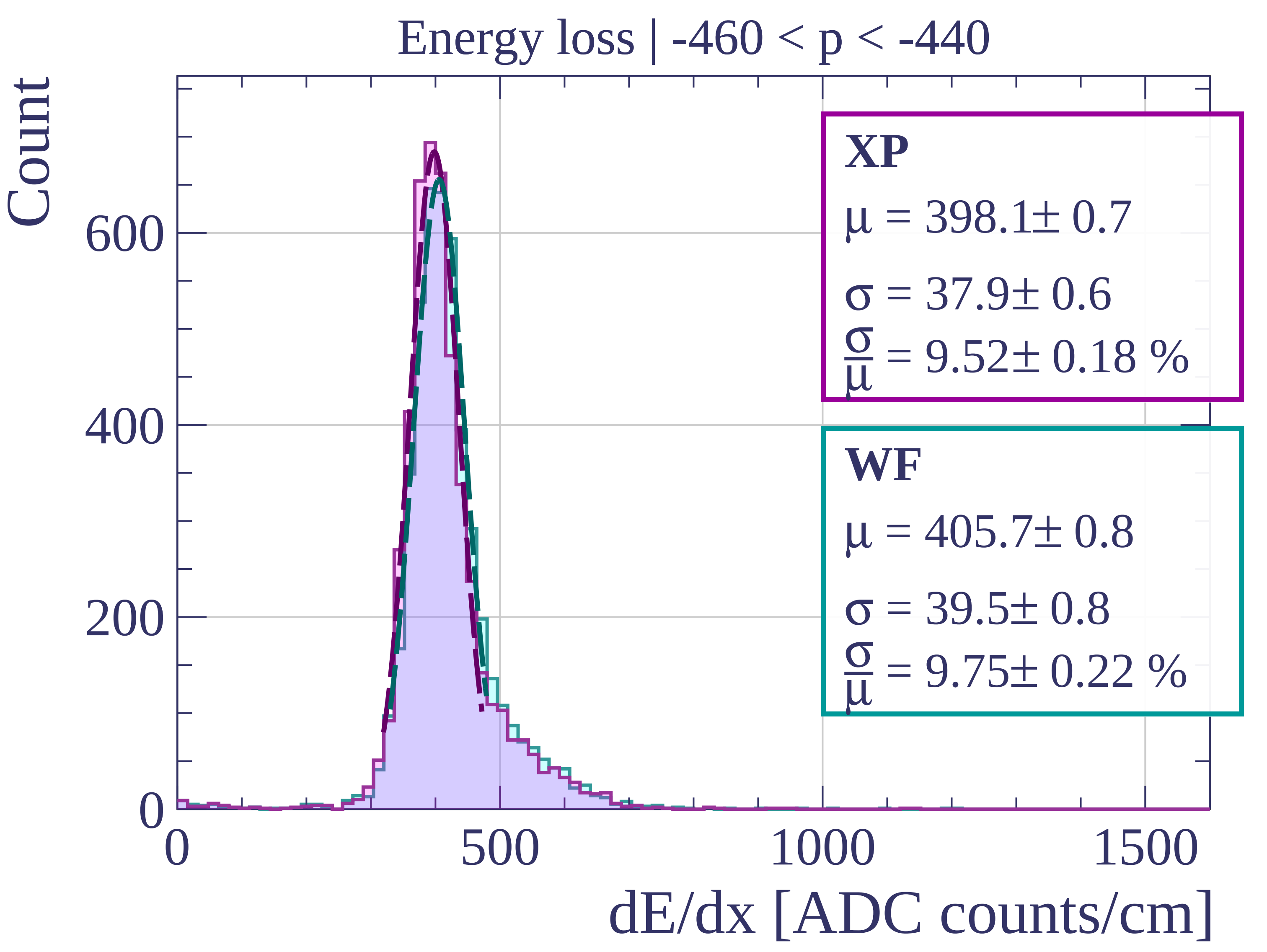
Energy loss | -520 < p < -500
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 400.6± 0.7
σ = 37.9± 0.7
σ/μ = 9.46± 0.18 %
WF
μ = 410.7± 0.8
σ = 41.4± 0.7
σ/μ = 10.09± 0.20 %

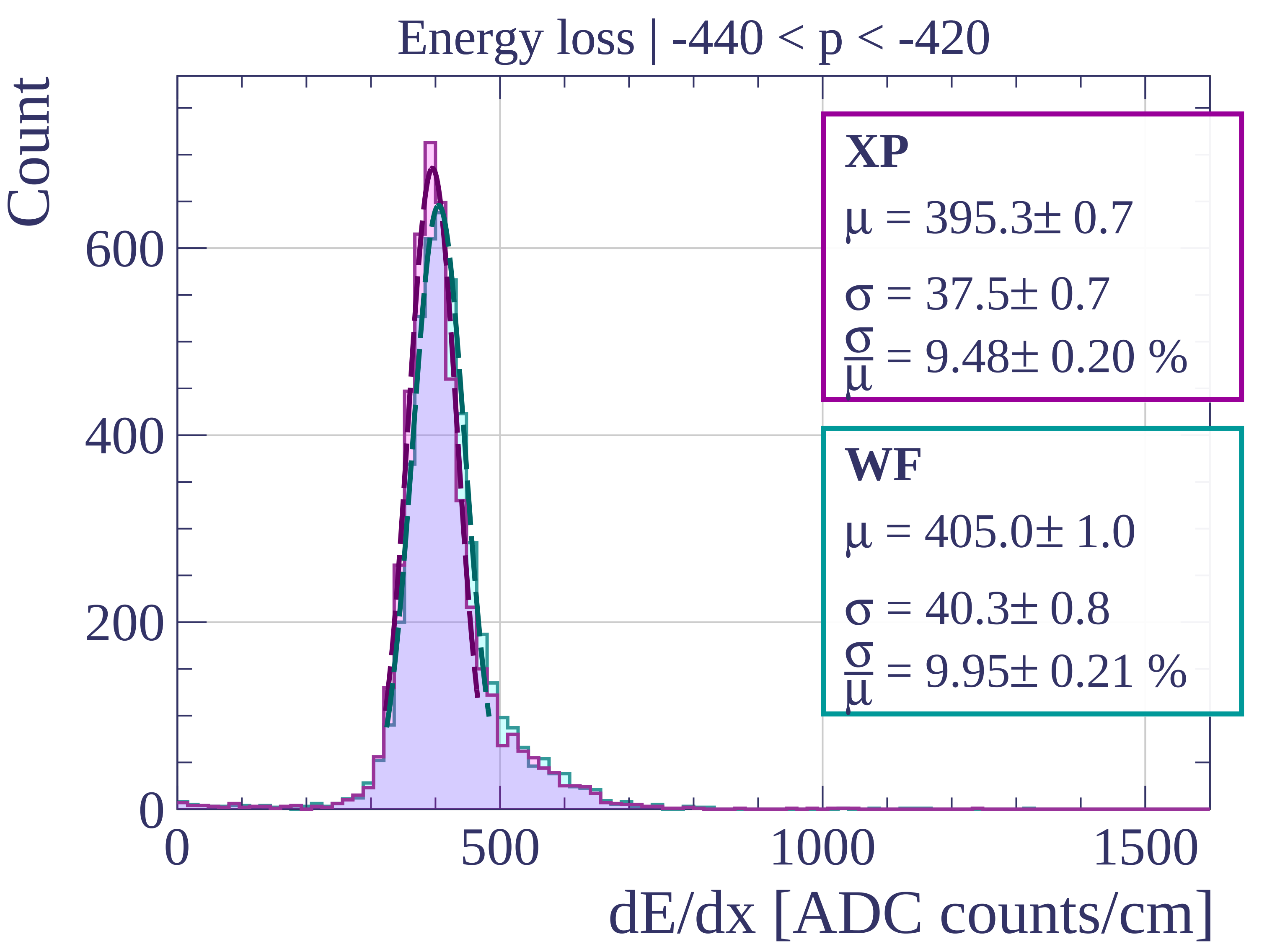


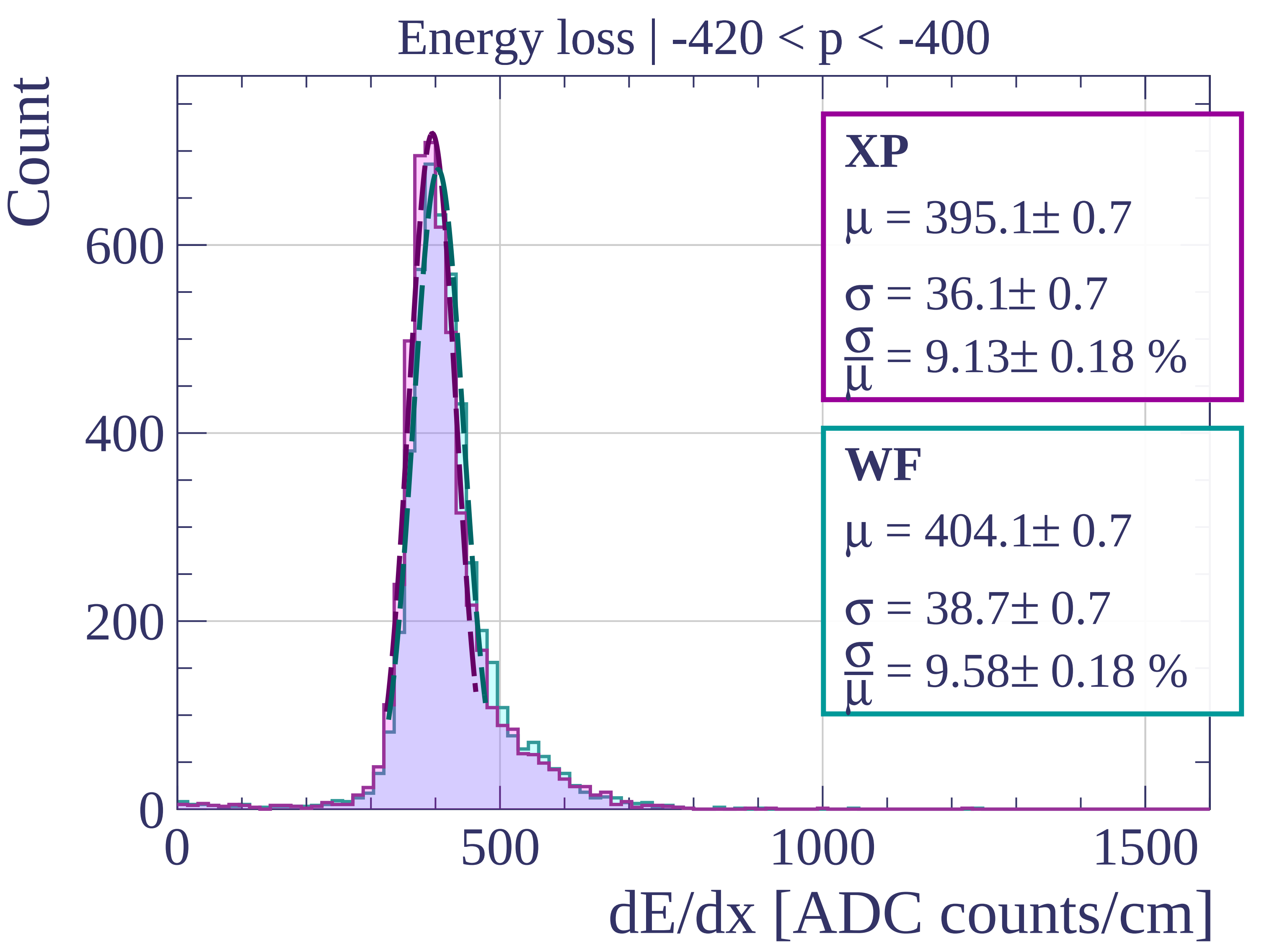
Energy loss | -500 < p < -480
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 397.1± 0.7
σ = 36.8± 0.7
σ/μ = 9.27± 0.19 %
WF
μ = 406.2± 0.8
σ = 39.3± 0.8
σ/μ = 9.68± 0.22 %



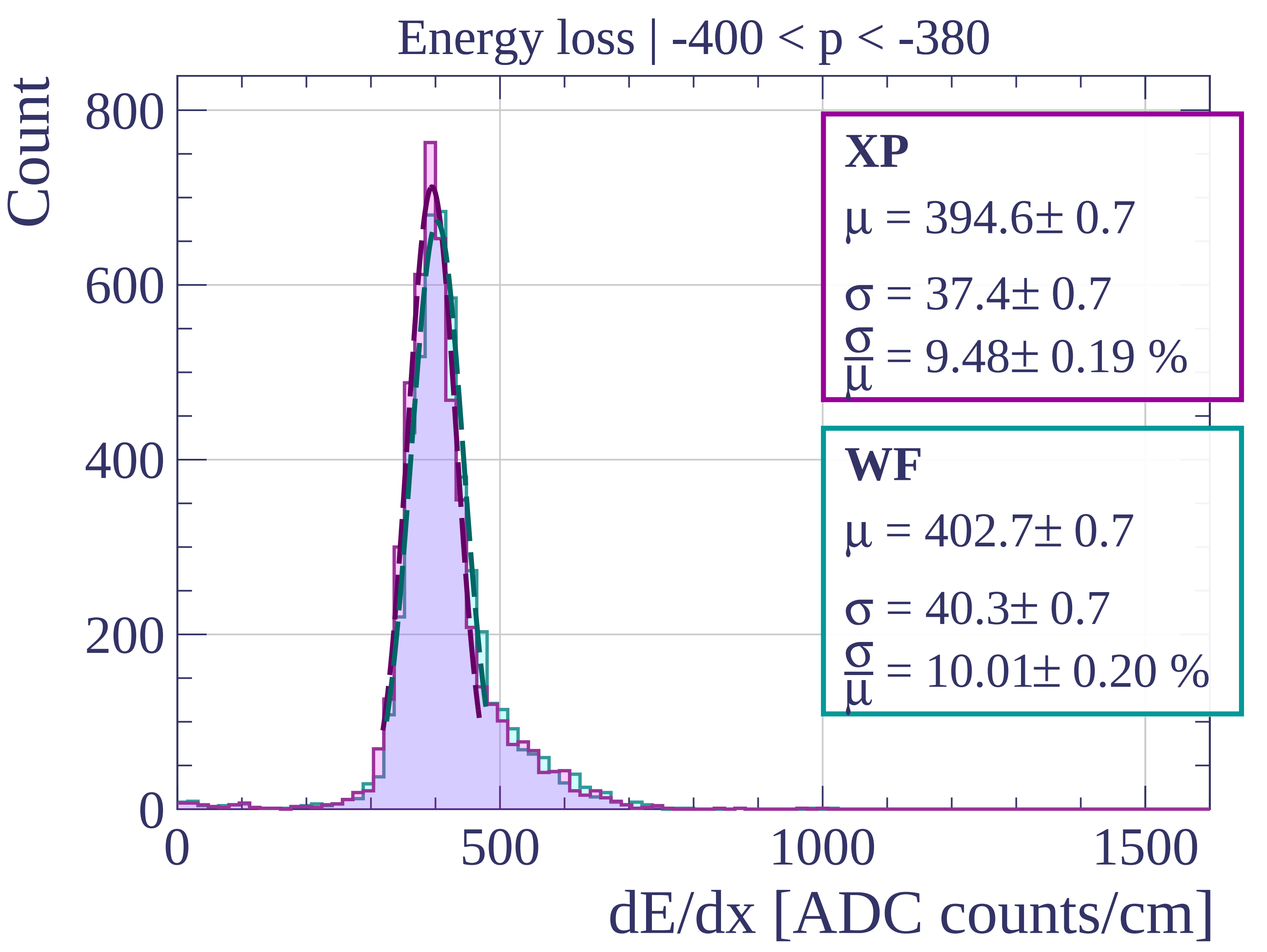
Energy loss | -480 < p < -460
Count
0
200
400
600
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 398.2± 0.7
σ = 38.6± 0.7
σ/μ = 9.70± 0.19 %
WF
μ = 406.5± 0.8
σ = 40.3± 0.9
σ/μ = 9.91± 0.24 %


Energy loss | -460 < p < -440
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 398.1± 0.7
σ = 37.9± 0.6
σ/μ = 9.52± 0.18 %
WF
μ = 405.7± 0.8
σ = 39.5± 0.8
σ/μ = 9.75± 0.22 %

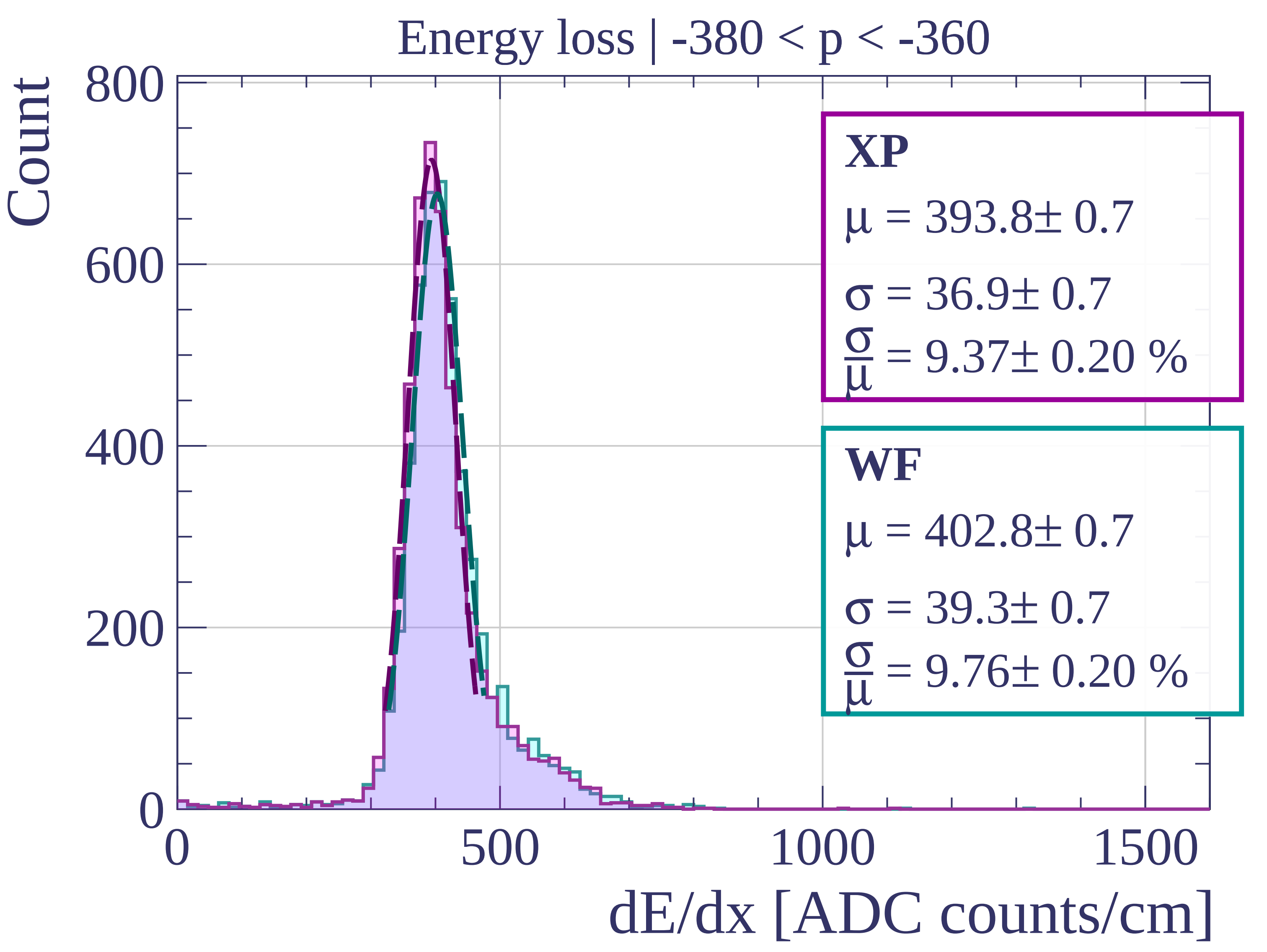

Energy loss | -440 < p < -420
Count
0
200
400
600
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 395.3± 0.7
σ = 37.5± 0.7
σ/μ = 9.48± 0.20 %
WF
μ = 405.0± 1.0
σ = 40.3± 0.8
σ/μ = 9.95± 0.21 %

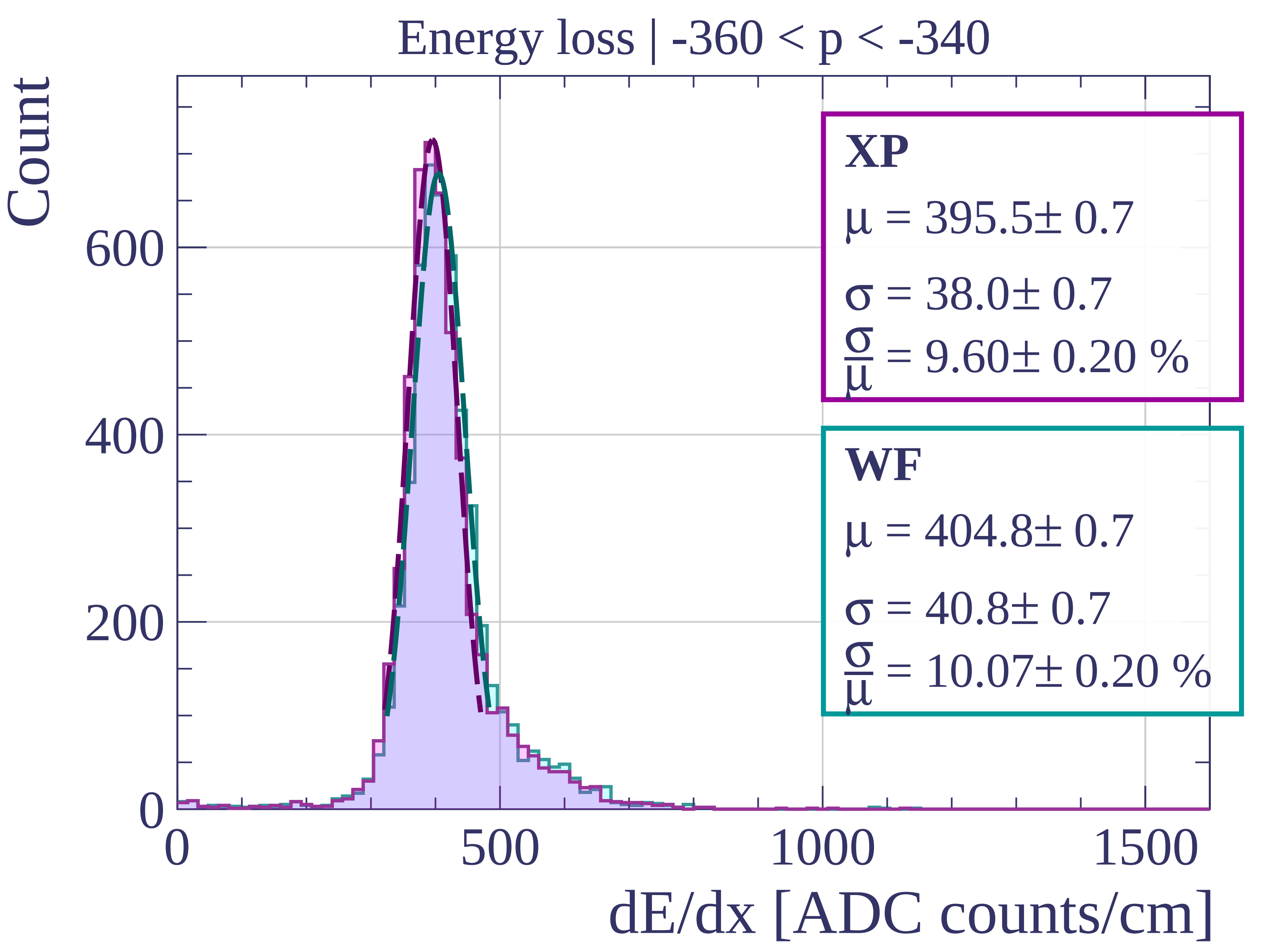


Energy loss | -420 < p < -400
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 395.1± 0.7
σ = 36.1± 0.7
σ/μ = 9.13± 0.18 %
WF
μ = 404.1± 0.7
σ = 38.7± 0.7
σ/μ = 9.58± 0.18 %

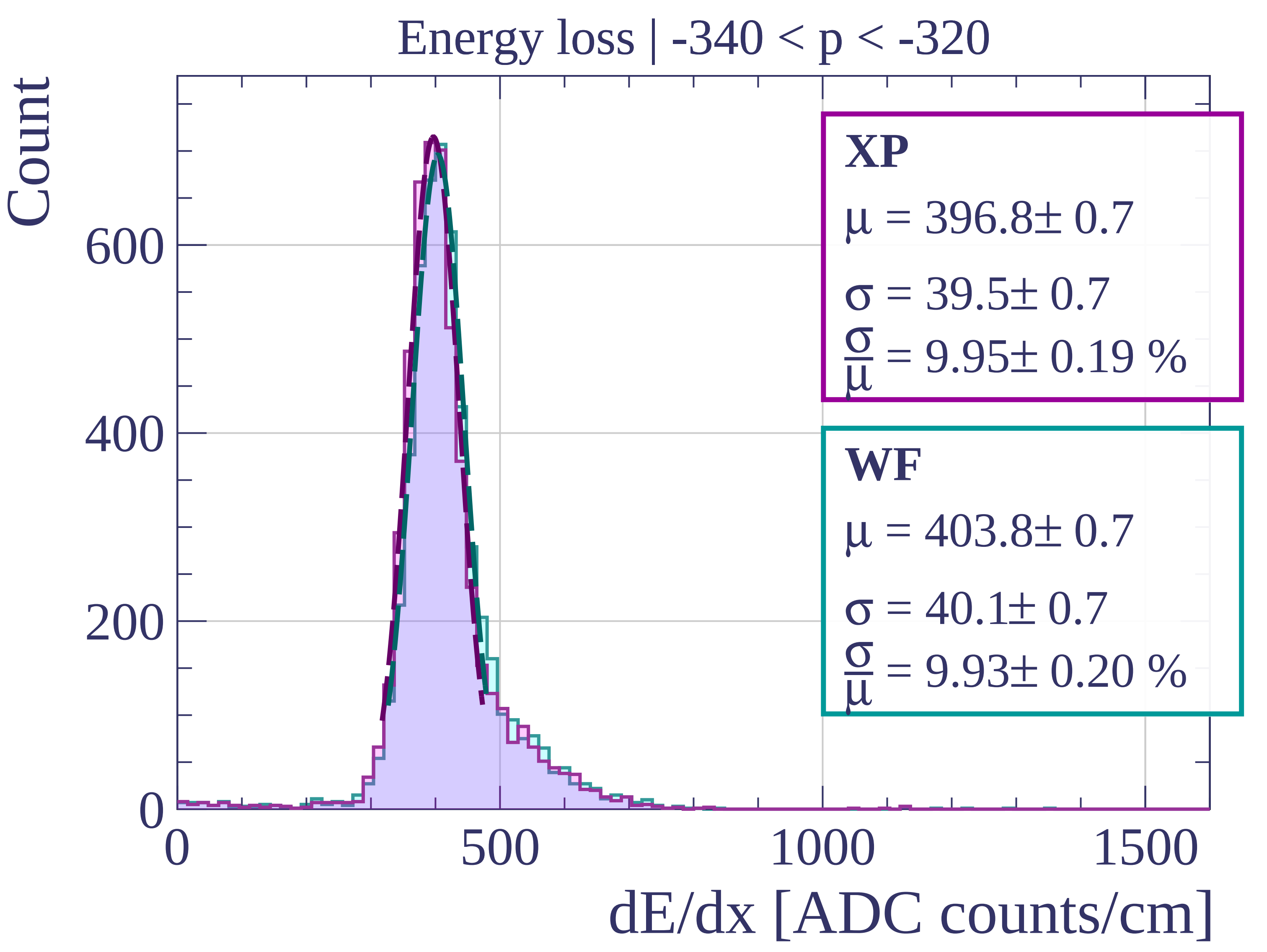


Energy loss | -400 < p < -380
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 394.6± 0.7
σ = 37.4± 0.7
σ/μ = 9.48± 0.19 %
WF
μ = 402.7± 0.7
σ = 40.3± 0.7
σ/μ = 10.01± 0.20 %

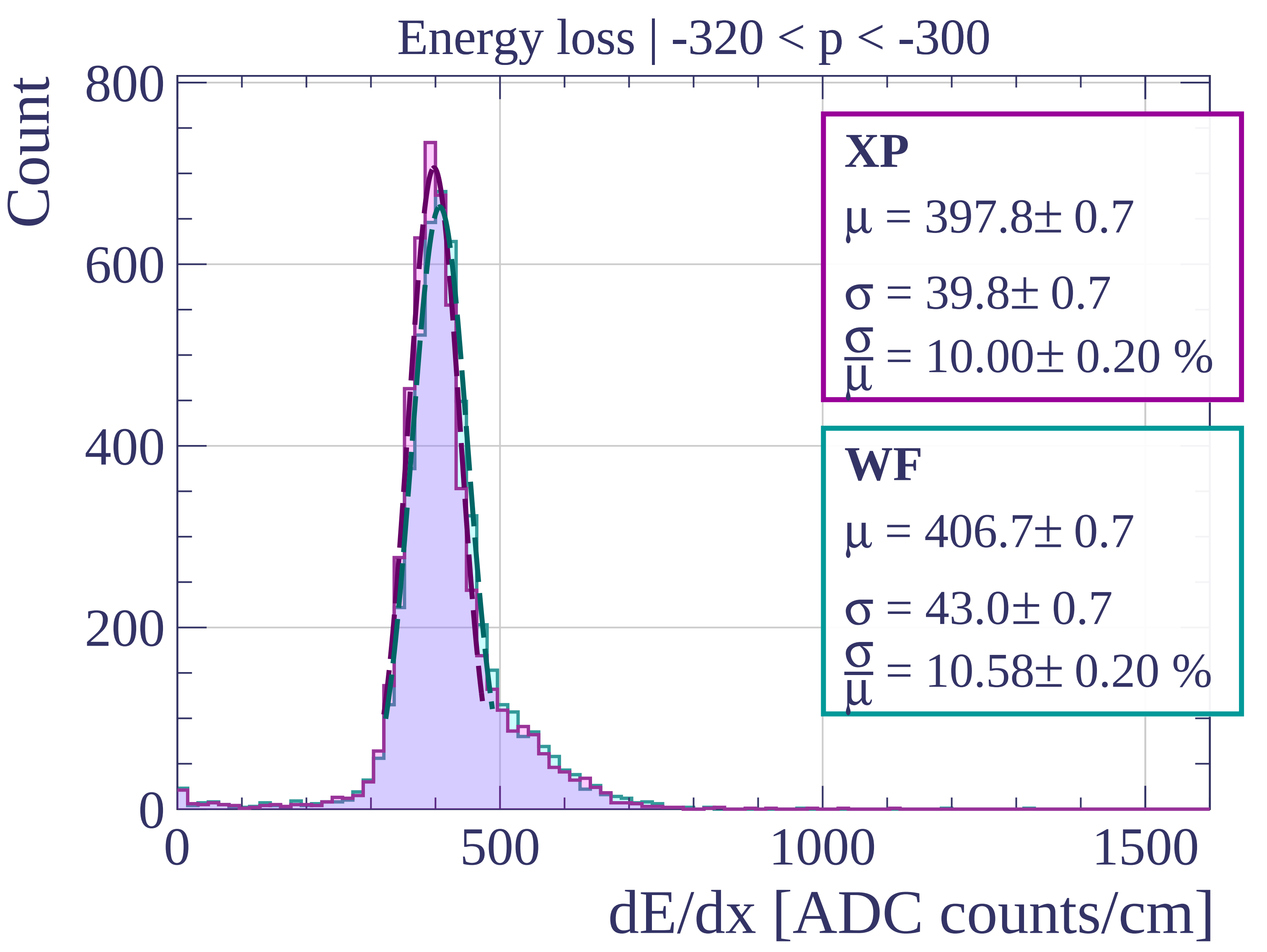

Energy loss | -380 < p < -360
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 393.8± 0.7
σ = 36.9± 0.7
σ/μ = 9.37± 0.20 %
WF
μ = 402.8± 0.7
σ = 39.3± 0.7
σ/μ = 9.76± 0.20 %



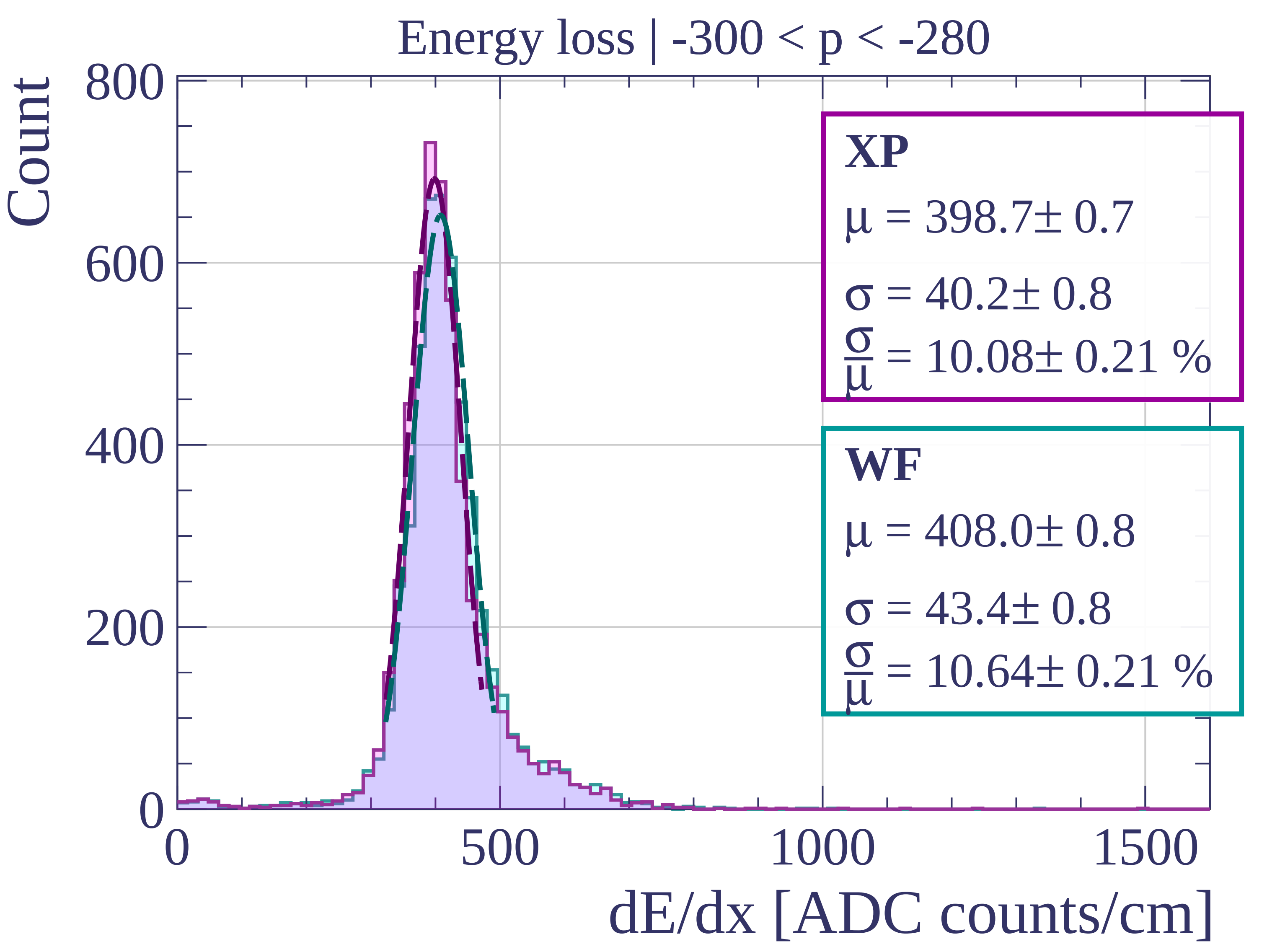
Energy loss | -360 < p < -340
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 395.5± 0.7
σ = 38.0± 0.7
σ/μ = 9.60± 0.20 %
WF
μ = 404.8± 0.7
σ = 40.8± 0.7
σ/μ = 10.07± 0.20 %



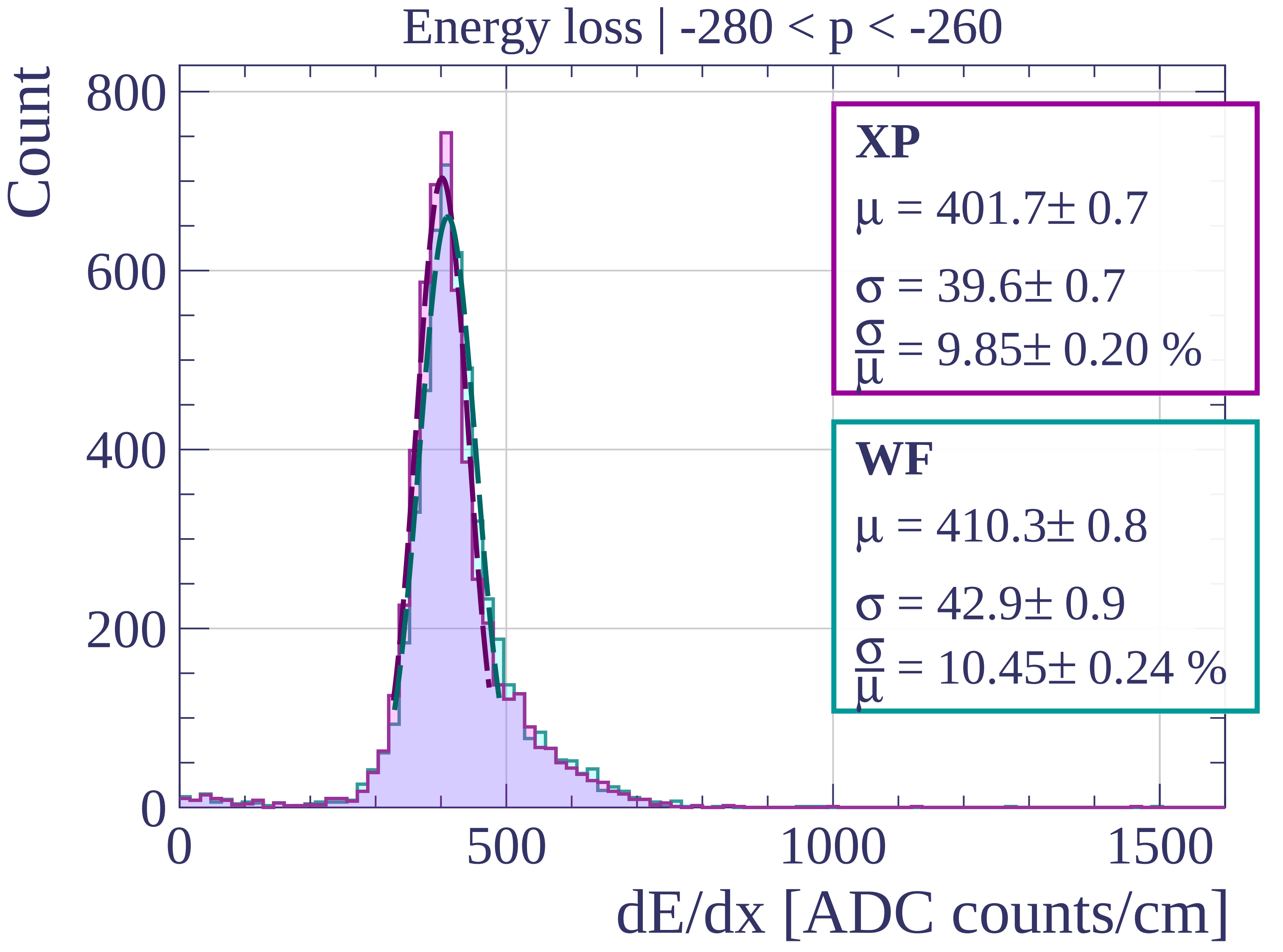
Energy loss | -340 < p < -320
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 396.8± 0.7
σ = 39.5± 0.7
σ/μ = 9.95± 0.19 %
WF
μ = 403.8± 0.7
σ = 40.1± 0.7
σ/μ = 9.93± 0.20 %



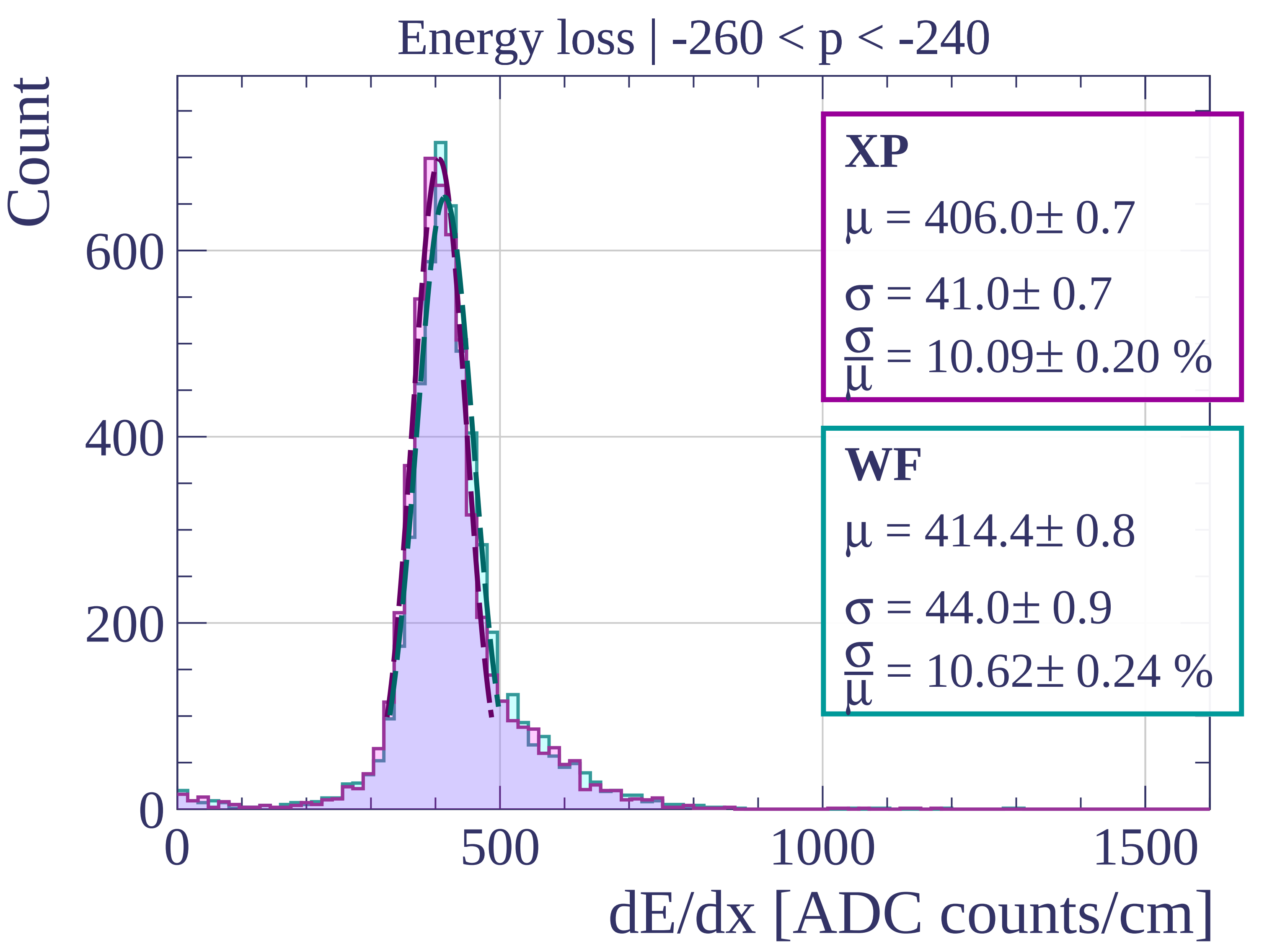
Energy loss | -320 < p < -300
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.8± 0.7
σ = 39.8± 0.7
σ/μ = 10.00± 0.20 %
WF
μ = 406.7± 0.7
σ = 43.0± 0.7
σ/μ = 10.58± 0.20 %



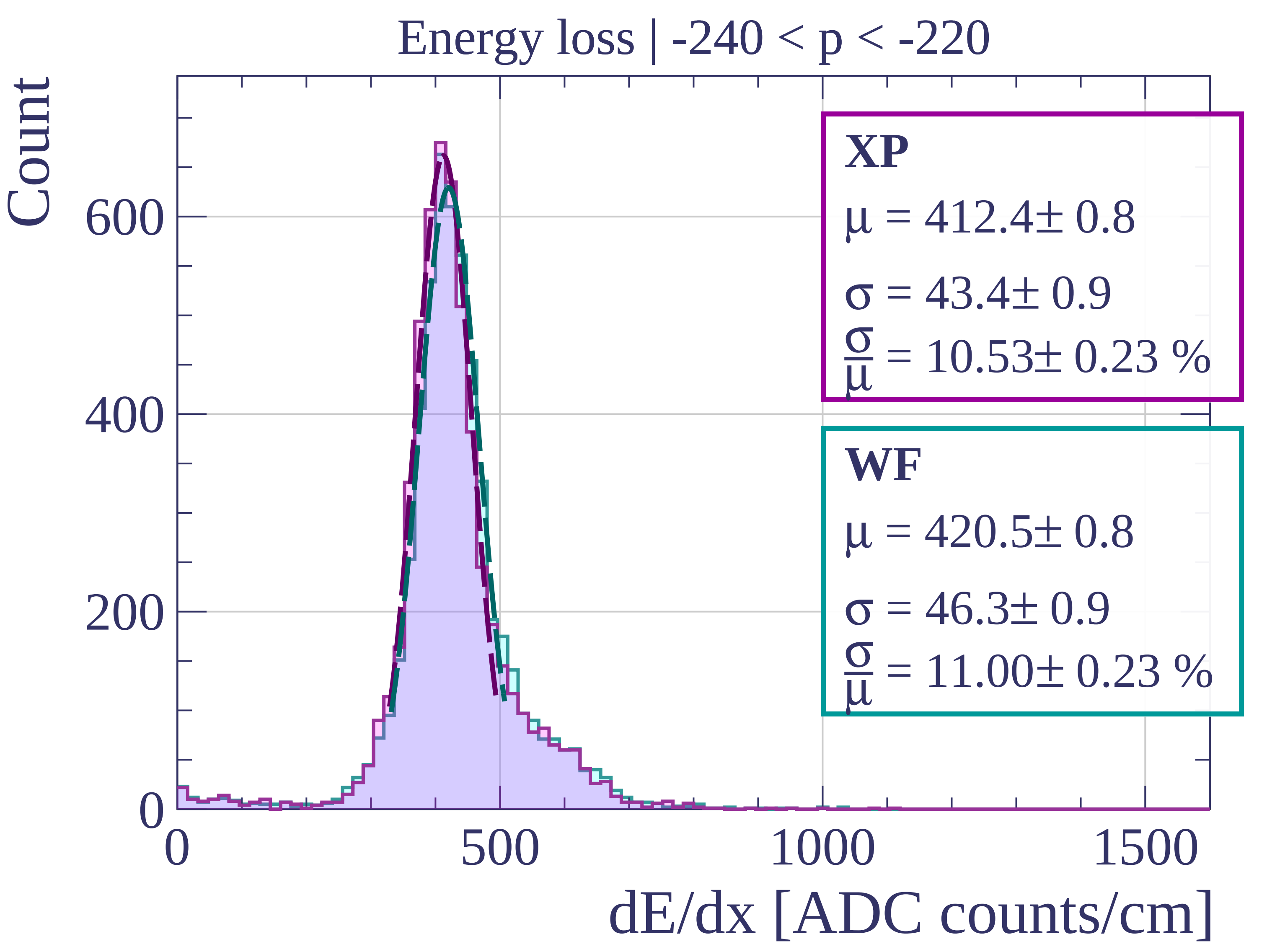
Energy loss | -300 < p < -280
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 398.7± 0.7
σ = 40.2± 0.8
σ/μ = 10.08± 0.21 %
WF
μ = 408.0± 0.8
σ = 43.4± 0.8
σ/μ = 10.64± 0.21 %



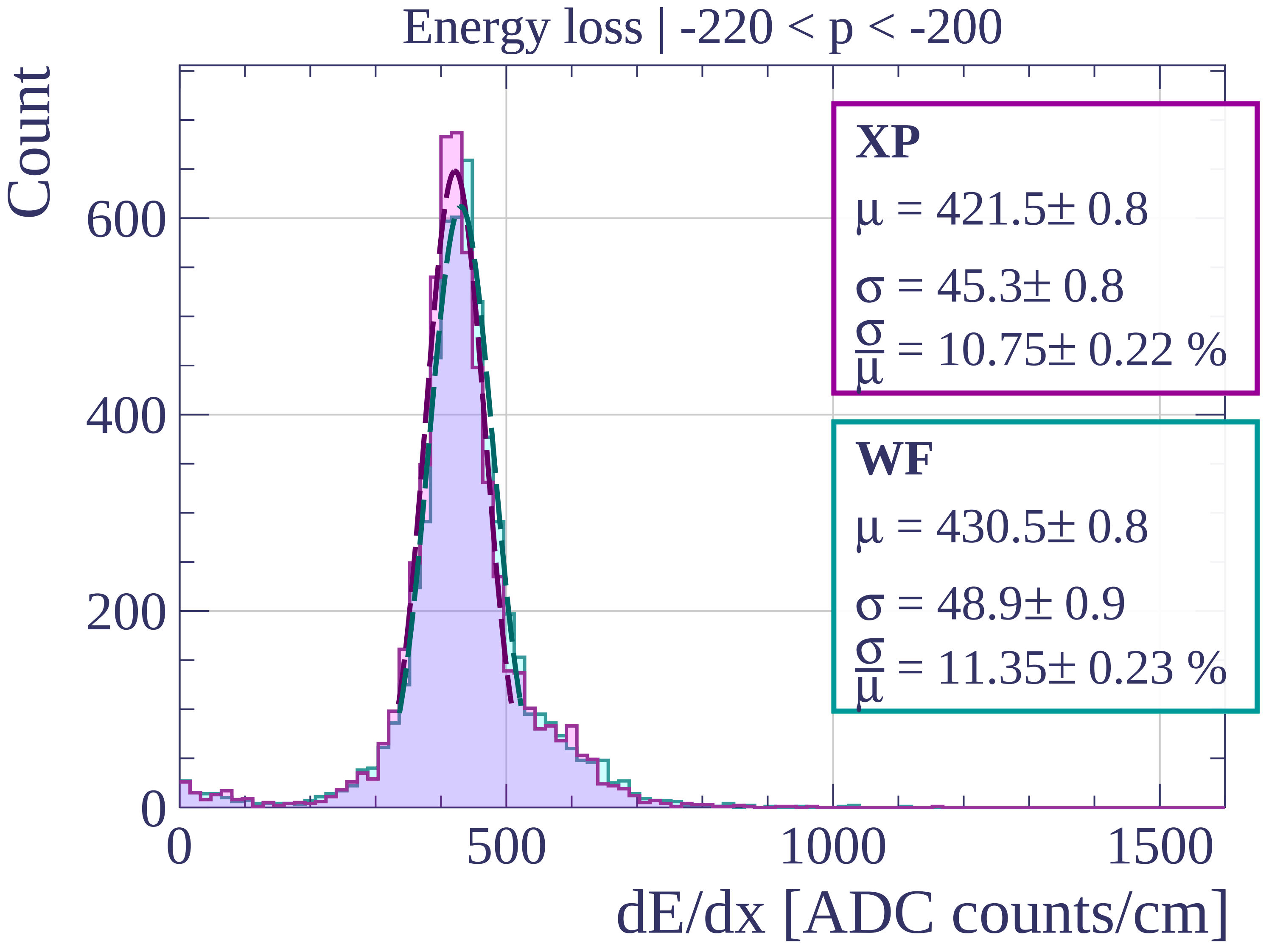
Energy loss | -280 < p < -260
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 401.7± 0.7
σ = 39.6± 0.7
σ/μ = 9.85± 0.20 %
WF
μ = 410.3± 0.8
σ = 42.9± 0.9
σ/μ = 10.45± 0.24 %



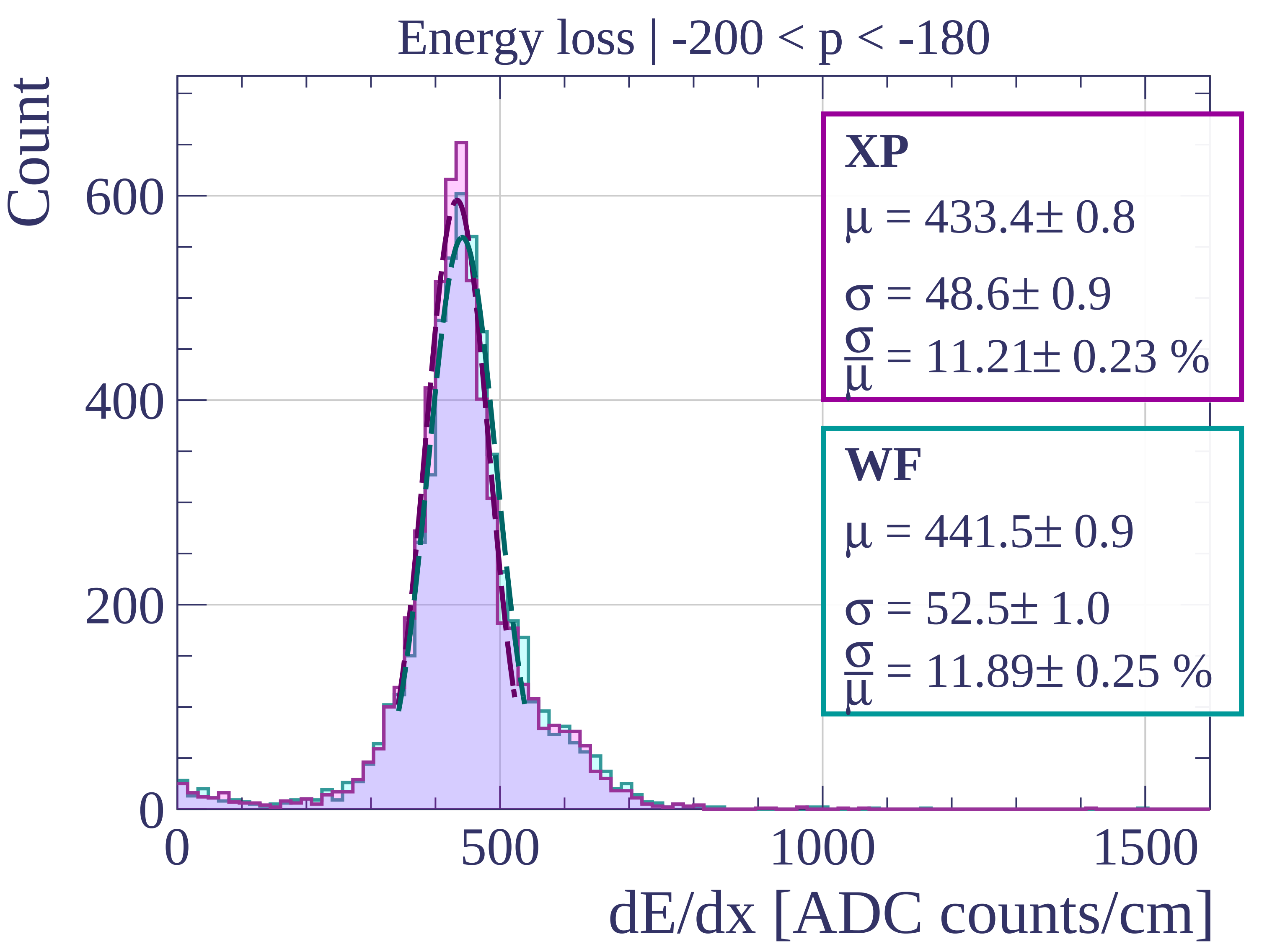
Energy loss | -260 < p < -240
Count
0
200
400
600
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 406.0± 0.7
σ = 41.0± 0.7
σ/μ = 10.09± 0.20 %
WF
μ = 414.4± 0.8
σ = 44.0± 0.9
σ/μ = 10.62± 0.24 %



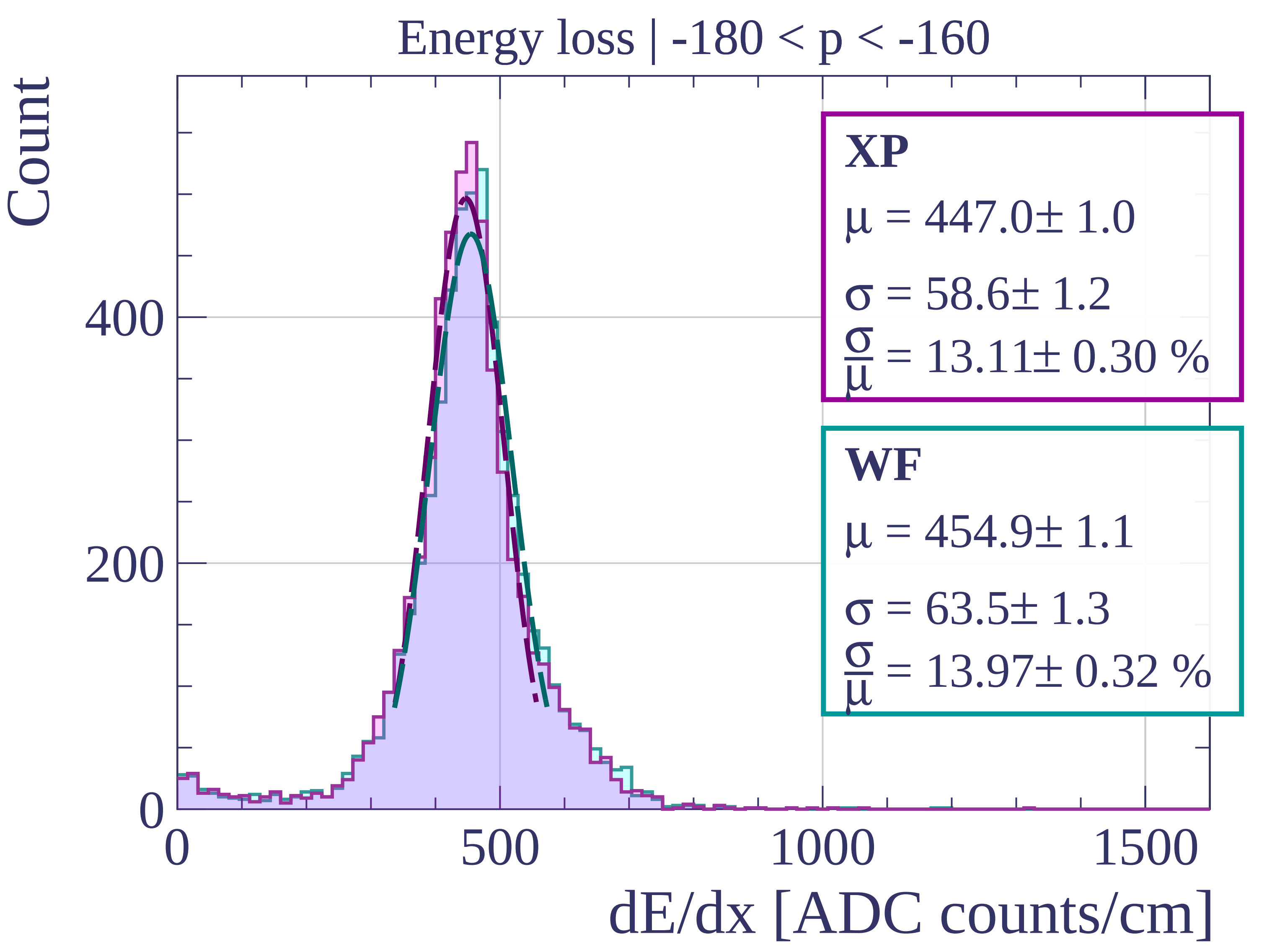
Energy loss | -240 < p < -220
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 412.4± 0.8
σ = 43.4± 0.9
σ/μ = 10.53± 0.23 %
WF
μ = 420.5± 0.8
σ = 46.3± 0.9
σ/μ = 11.00± 0.23 %



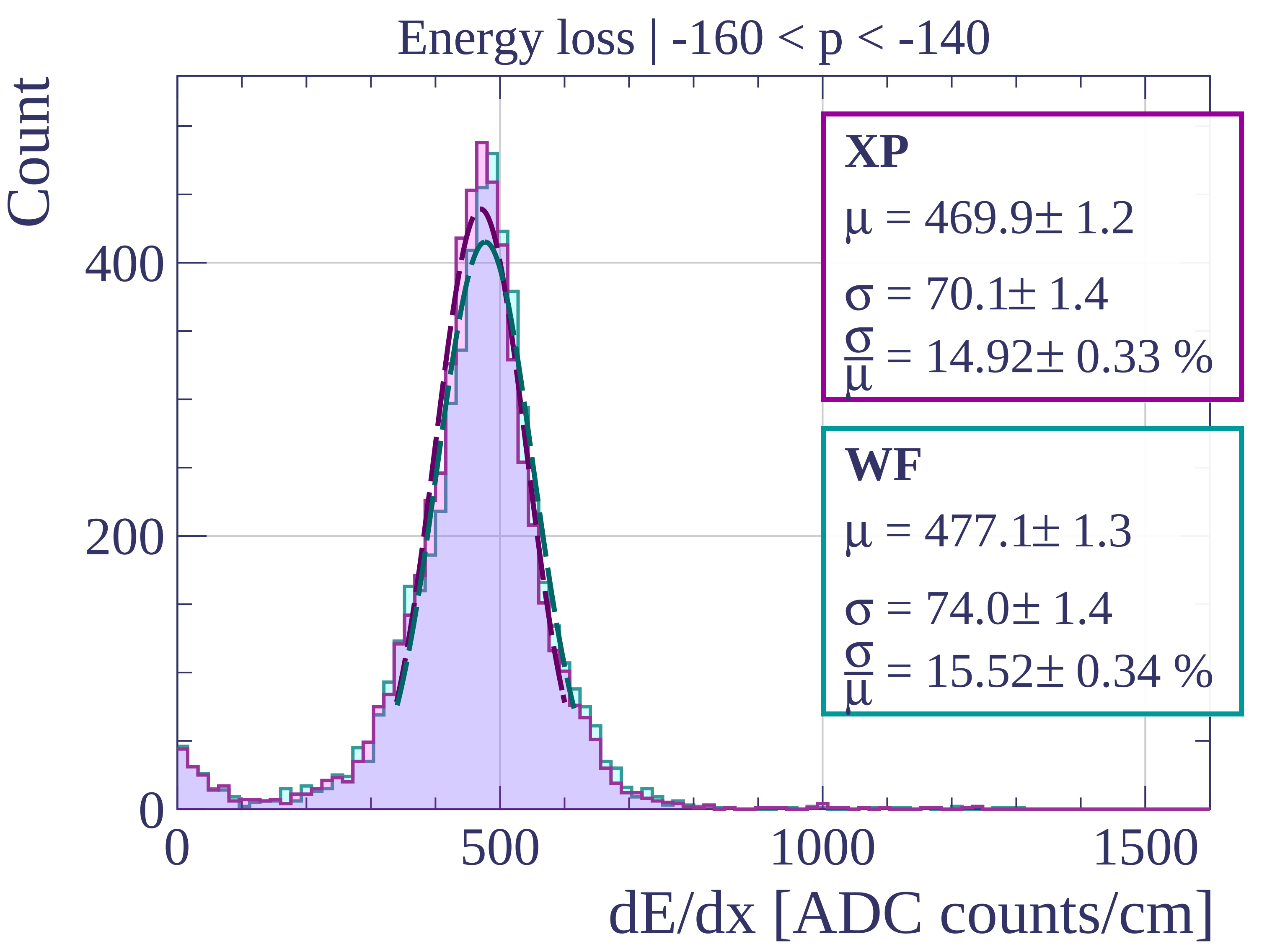
Energy loss | -220 < p < -200
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 421.5± 0.8
σ = 45.3± 0.8
σ/μ = 10.75± 0.22 %
WF
μ = 430.5± 0.8
σ = 48.9± 0.9
σ/μ = 11.35± 0.23 %

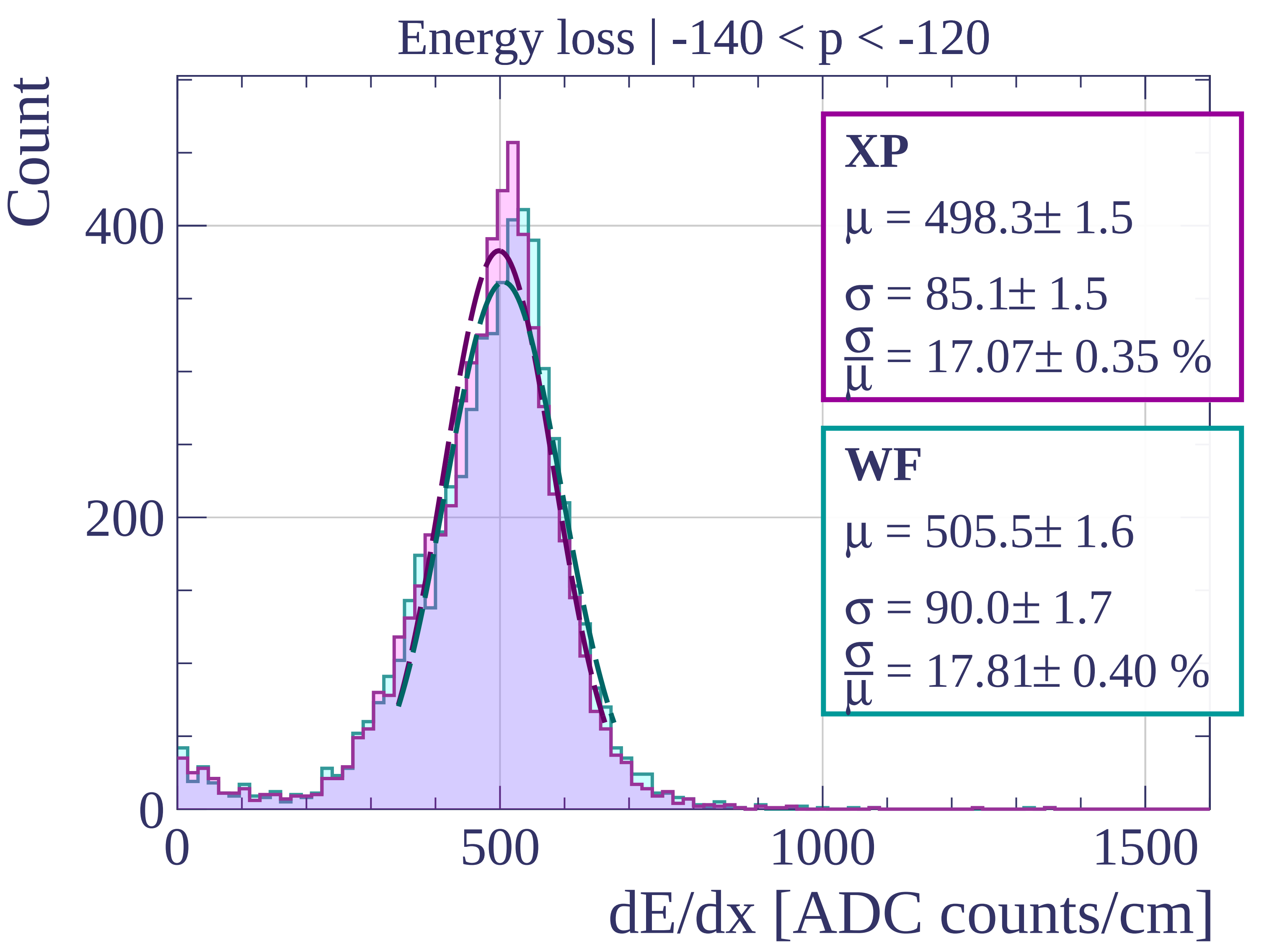


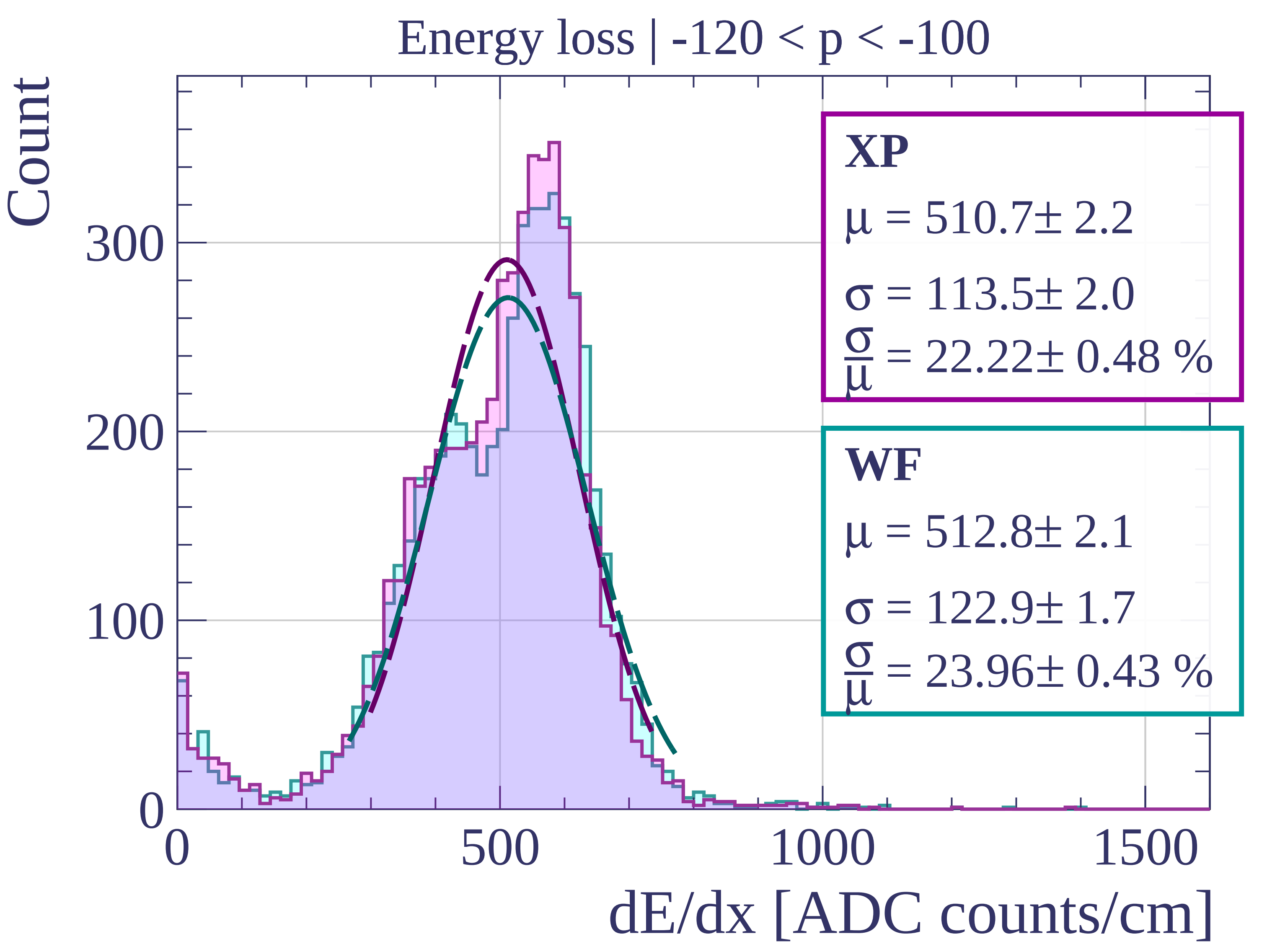
Energy loss | -200 < p < -180
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 433.4± 0.8
σ = 48.6± 0.9
σ/μ = 11.21± 0.23 %
WF
μ = 441.5± 0.9
σ = 52.5± 1.0
σ/μ = 11.89± 0.25 %

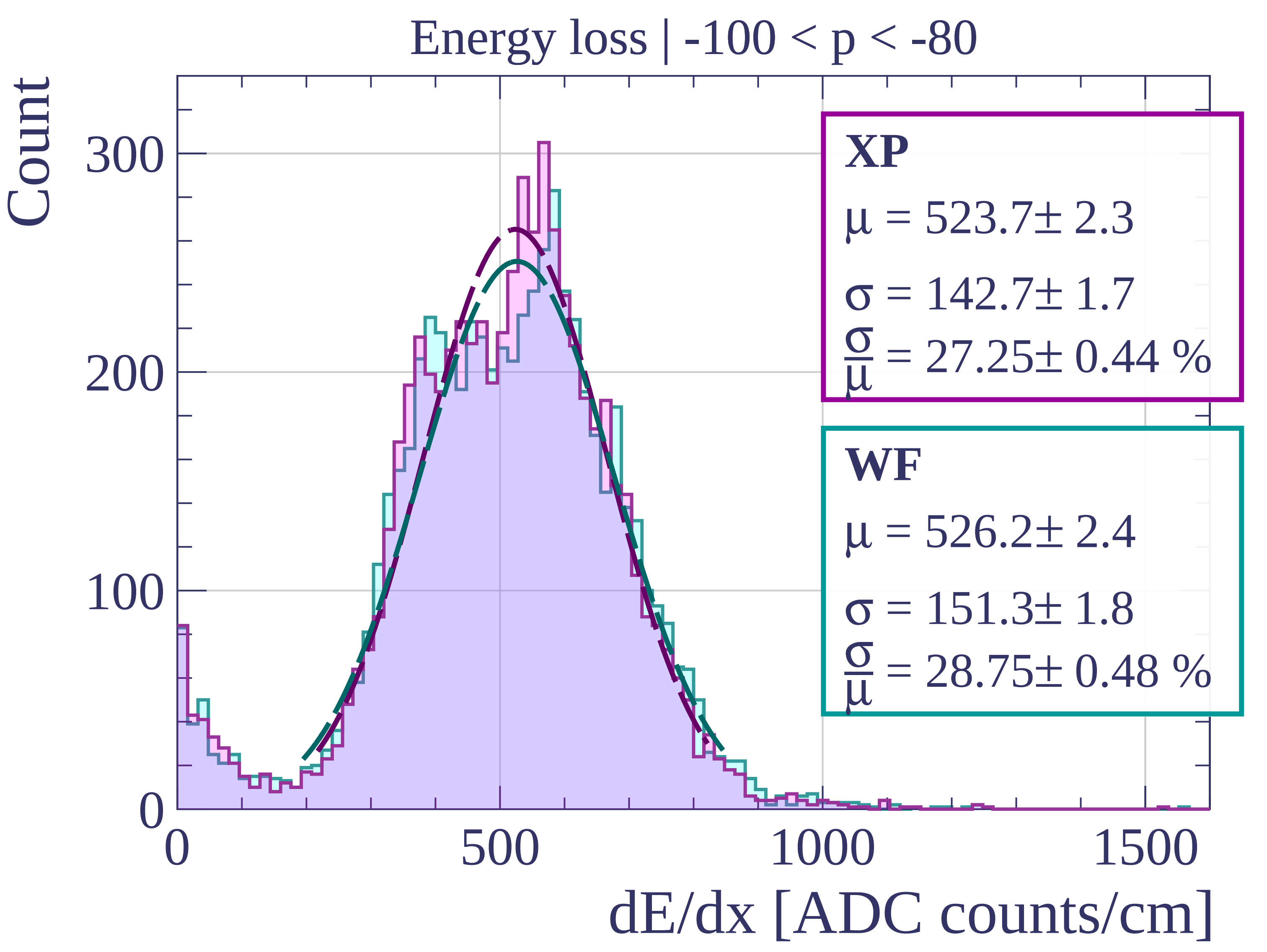


Energy loss | -180 < p < -160
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 447.0± 1.0
σ = 58.6± 1.2
σ/μ = 13.11± 0.30 %
WF
μ = 454.9± 1.1
σ = 63.5± 1.3
σ/μ = 13.97± 0.32 %

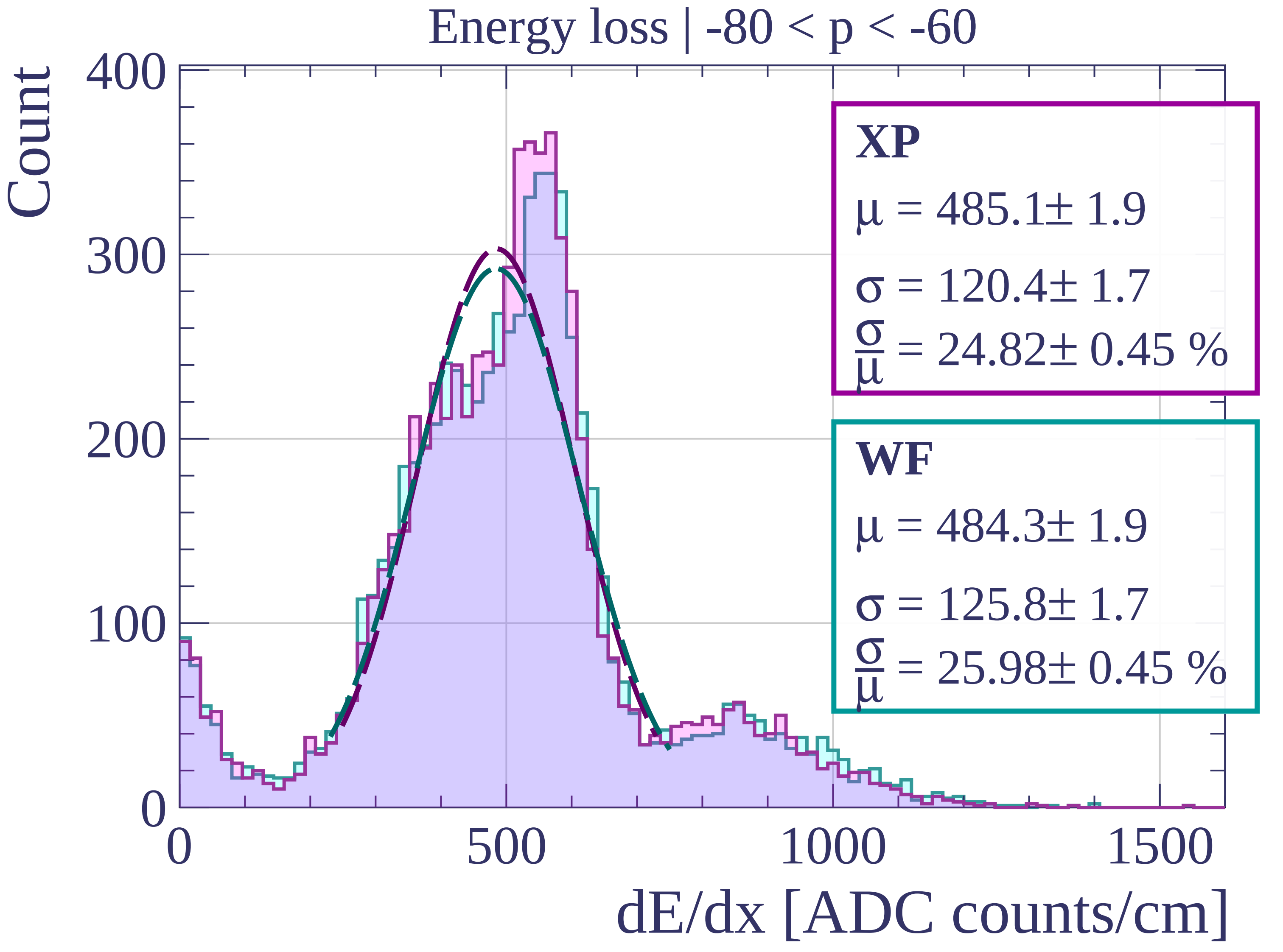

Energy loss | -160 < p < -140
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 469.9± 1.2
σ = 70.1± 1.4
σ/μ = 14.92± 0.33 %
WF
μ = 477.1± 1.3
σ = 74.0± 1.4
σ/μ = 15.52± 0.34 %


Energy loss | -140 < p < -120
Count
400
200
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 498.3± 1.5
σ = 85.1± 1.5
σ/μ = 17.07± 0.35 %
WF
μ = 505.5± 1.6
σ = 90.0± 1.7
σ/μ = 17.81± 0.40 %

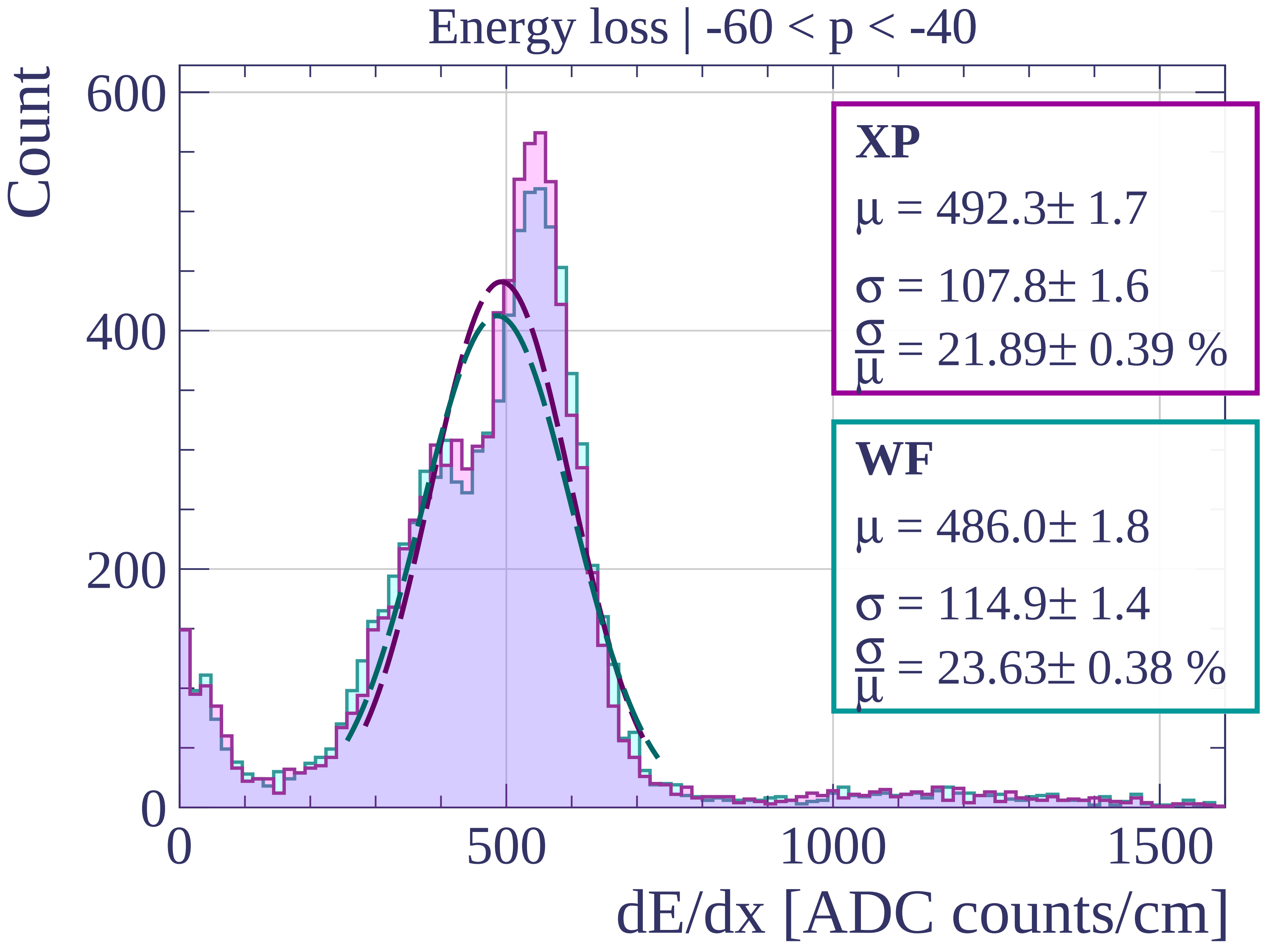

Energy loss | -120 < p < -100
Count
0
100
200
300
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 510.7± 2.2
σ = 113.5± 2.0
σ/μ = 22.22± 0.48 %
WF
μ = 512.8± 2.1
σ = 122.9± 1.7
σ/μ = 23.96± 0.43 %



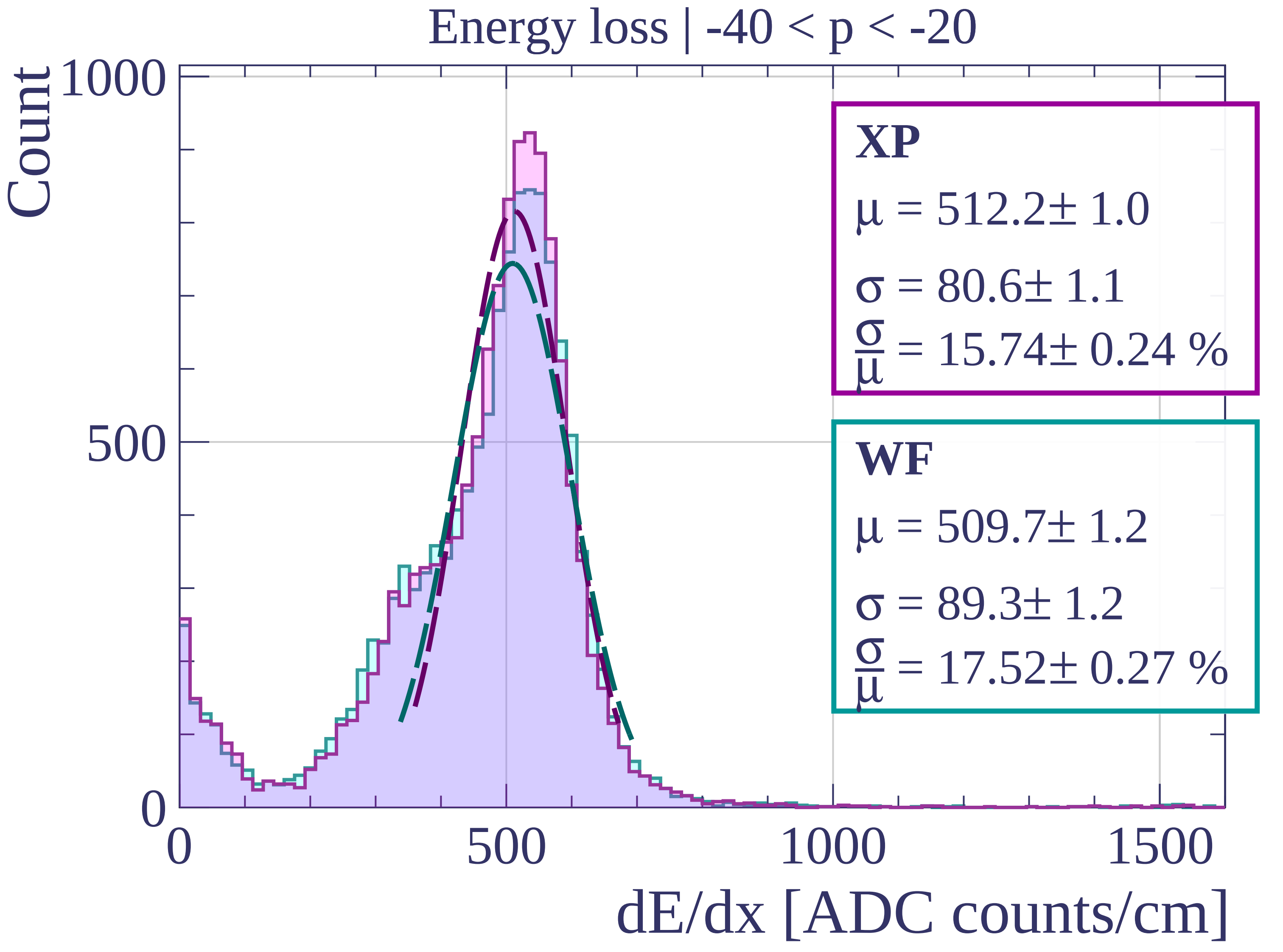
Energy loss | -100 < p < -80
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 523.7± 2.3
σ = 142.7± 1.7
σ/μ = 27.25± 0.44 %
WF
μ = 526.2± 2.4
σ = 151.3± 1.8
σ/μ = 28.75± 0.48 %



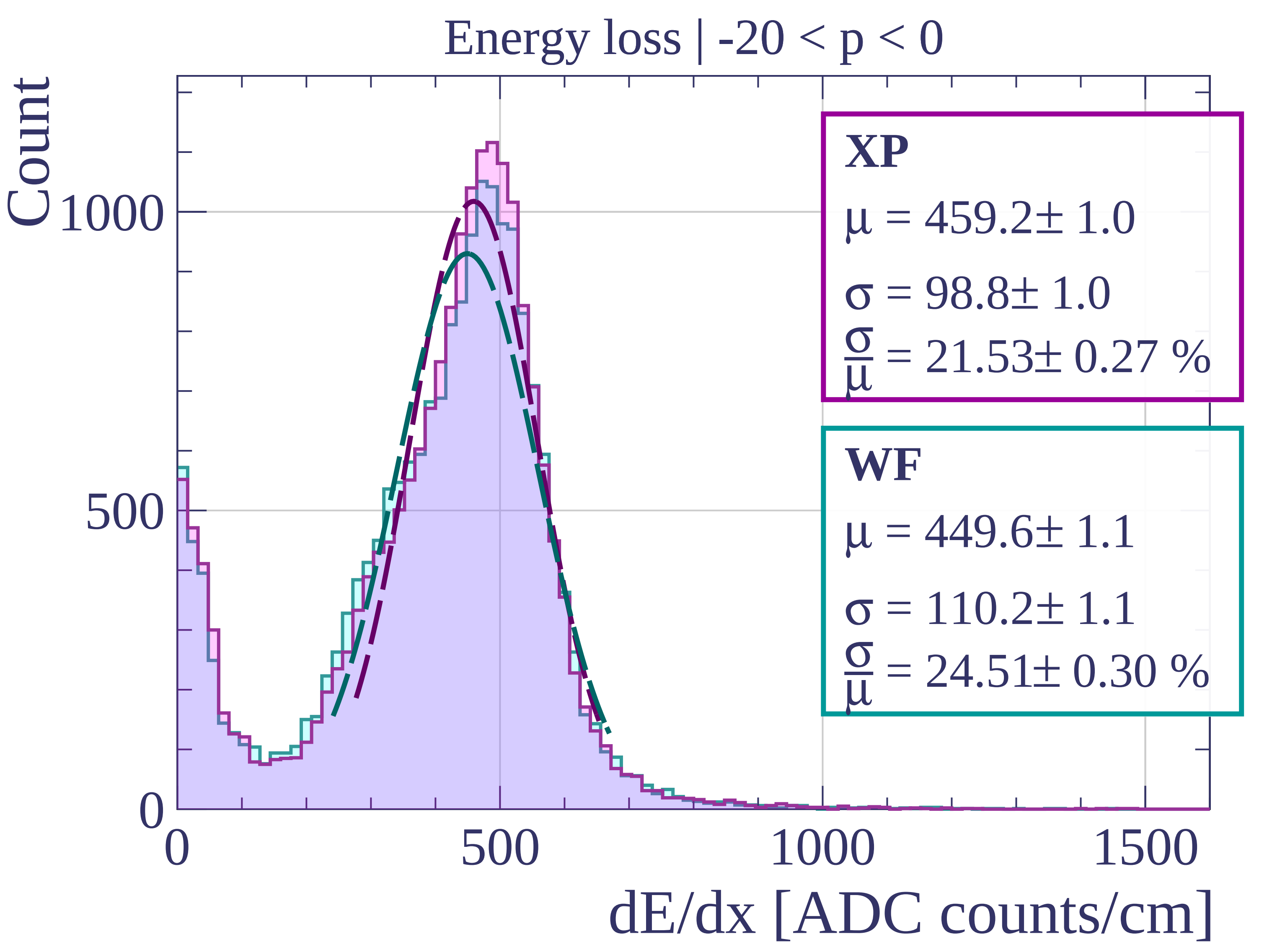
Energy loss | -80 < p < -60
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 485.1± 1.9
σ = 120.4± 1.7
σ/μ = 24.82± 0.45 %
WF
μ = 484.3± 1.9
σ = 125.8± 1.7
σ/μ = 25.98± 0.45 %

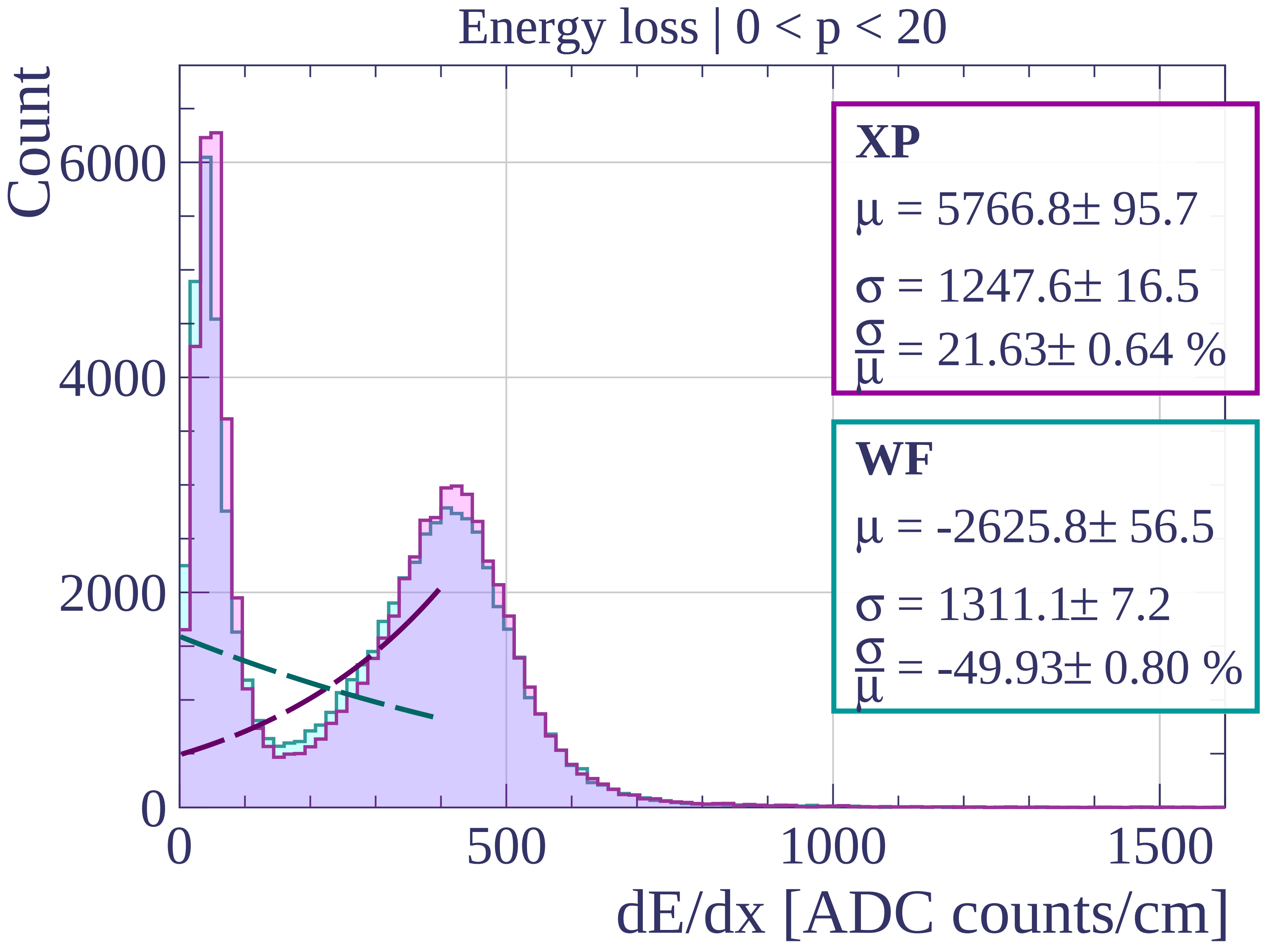


Energy loss | -60 < p < -40
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 492.3± 1.7
σ = 107.8± 1.6
σ/μ = 21.89± 0.39 %
WF
μ = 486.0± 1.8
σ = 114.9± 1.4
σ/μ = 23.63± 0.38 %

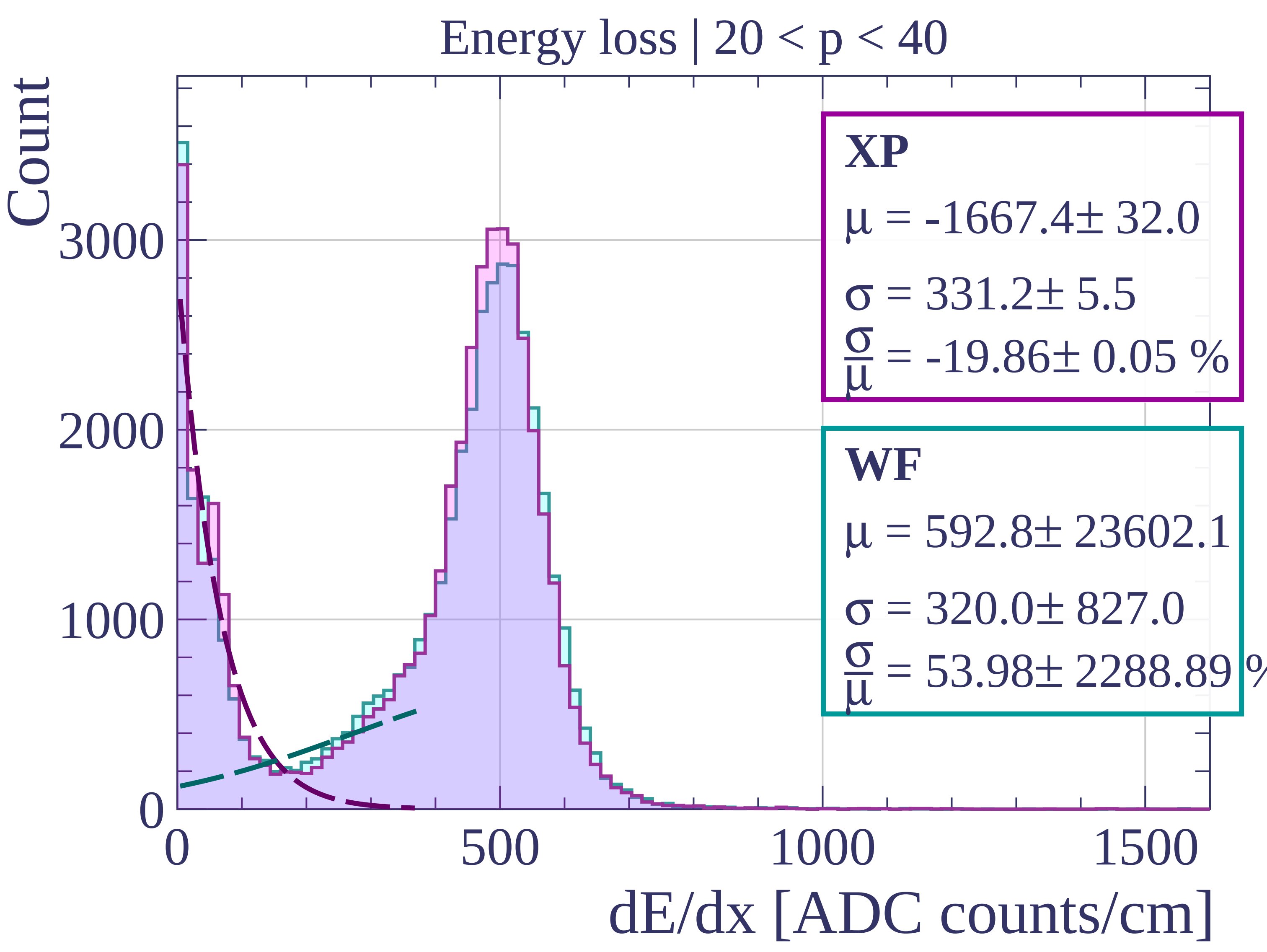


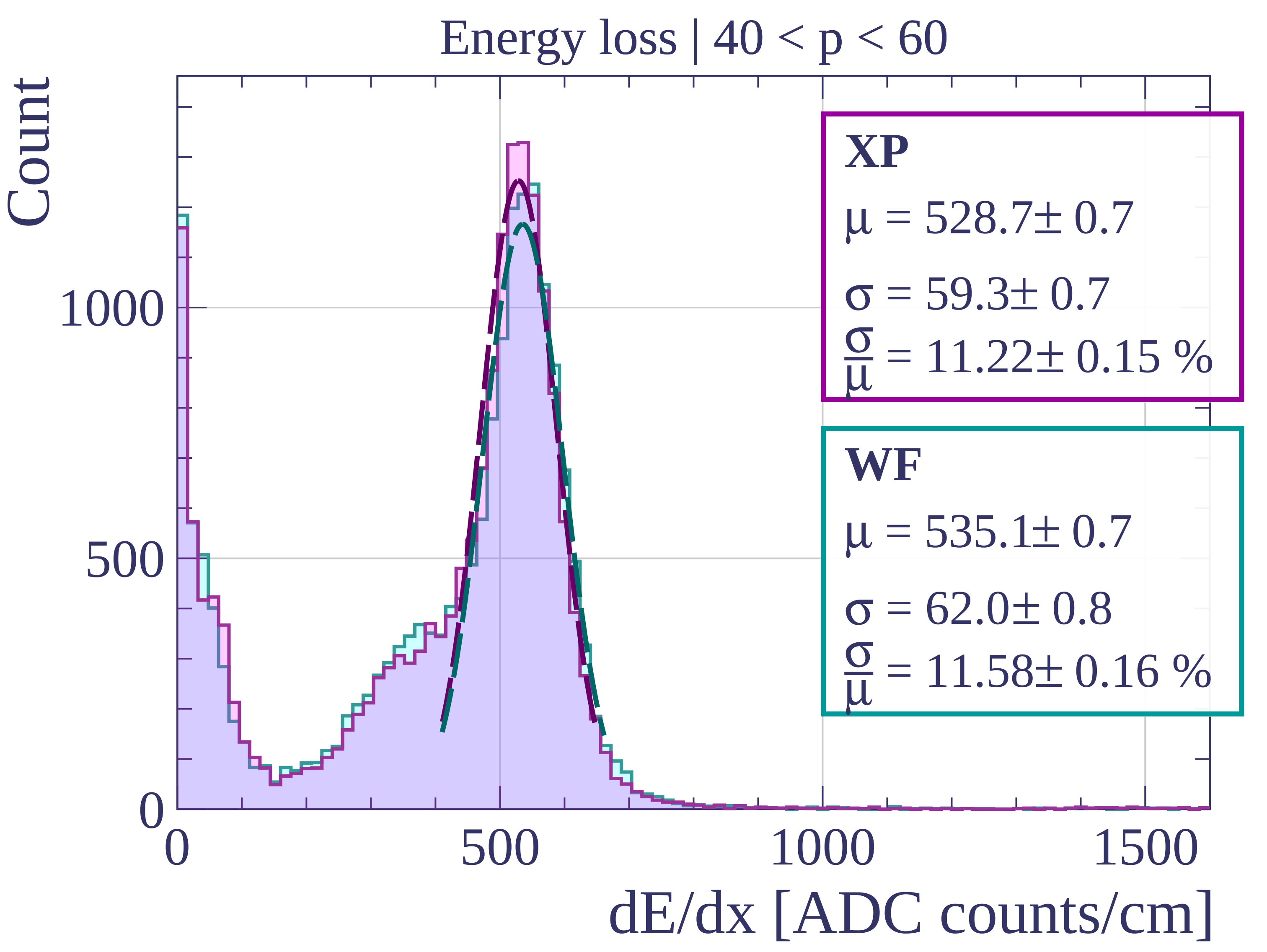
Energy loss | -40 < p < -20
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = 512.2± 1.0
σ = 80.6± 1.1
σ/μ = 15.74± 0.24 %
WF
μ = 509.7± 1.2
σ = 89.3± 1.2
σ/μ = 17.52± 0.27 %


Energy loss | -20 < p < 0
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 459.2± 1.0
σ = 98.8± 1.0
σ/μ = 21.53± 0.27 %
WF
μ = 449.6± 1.1
σ = 110.2± 1.1
σ/μ = 24.51± 0.30 %

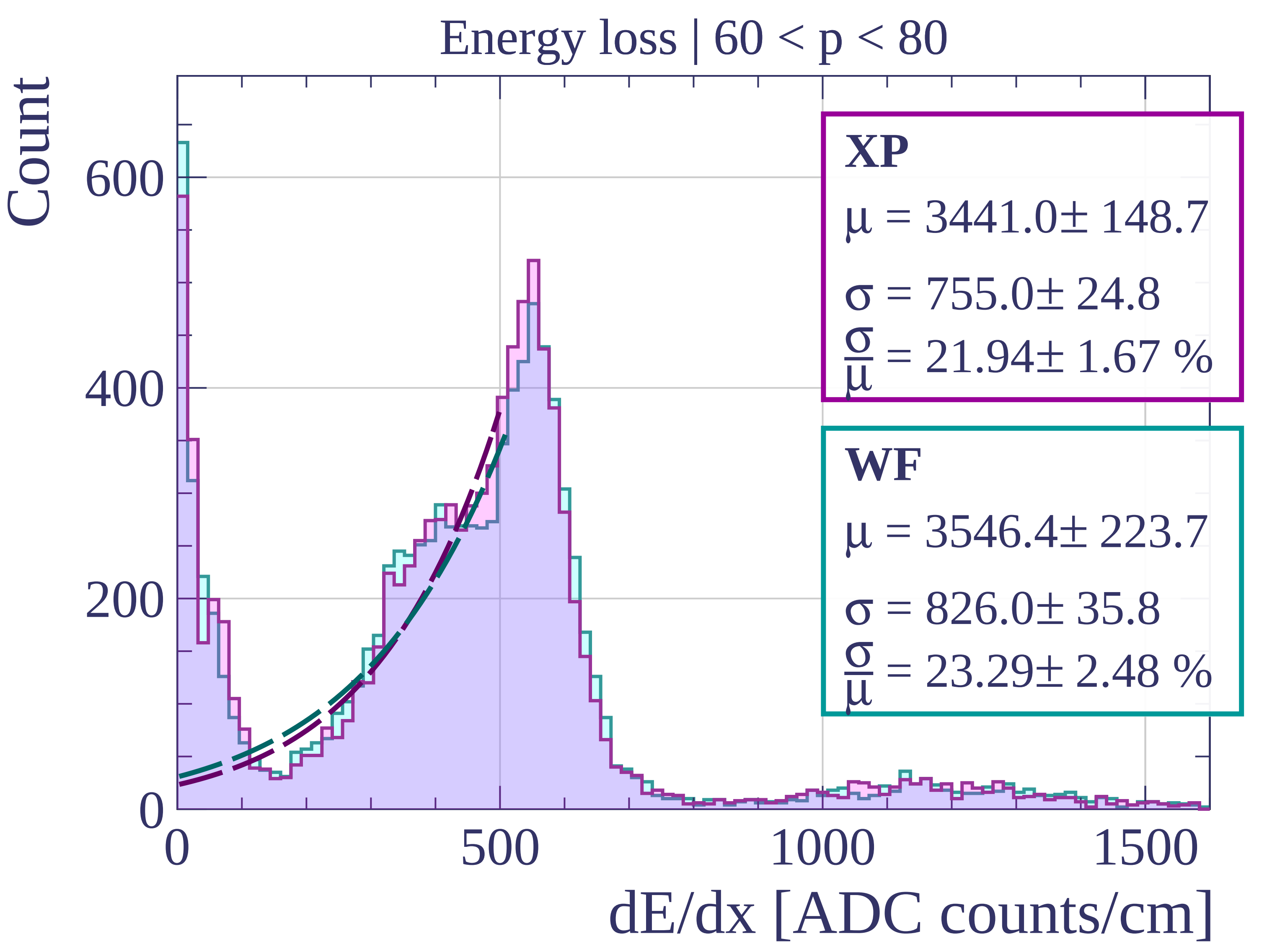


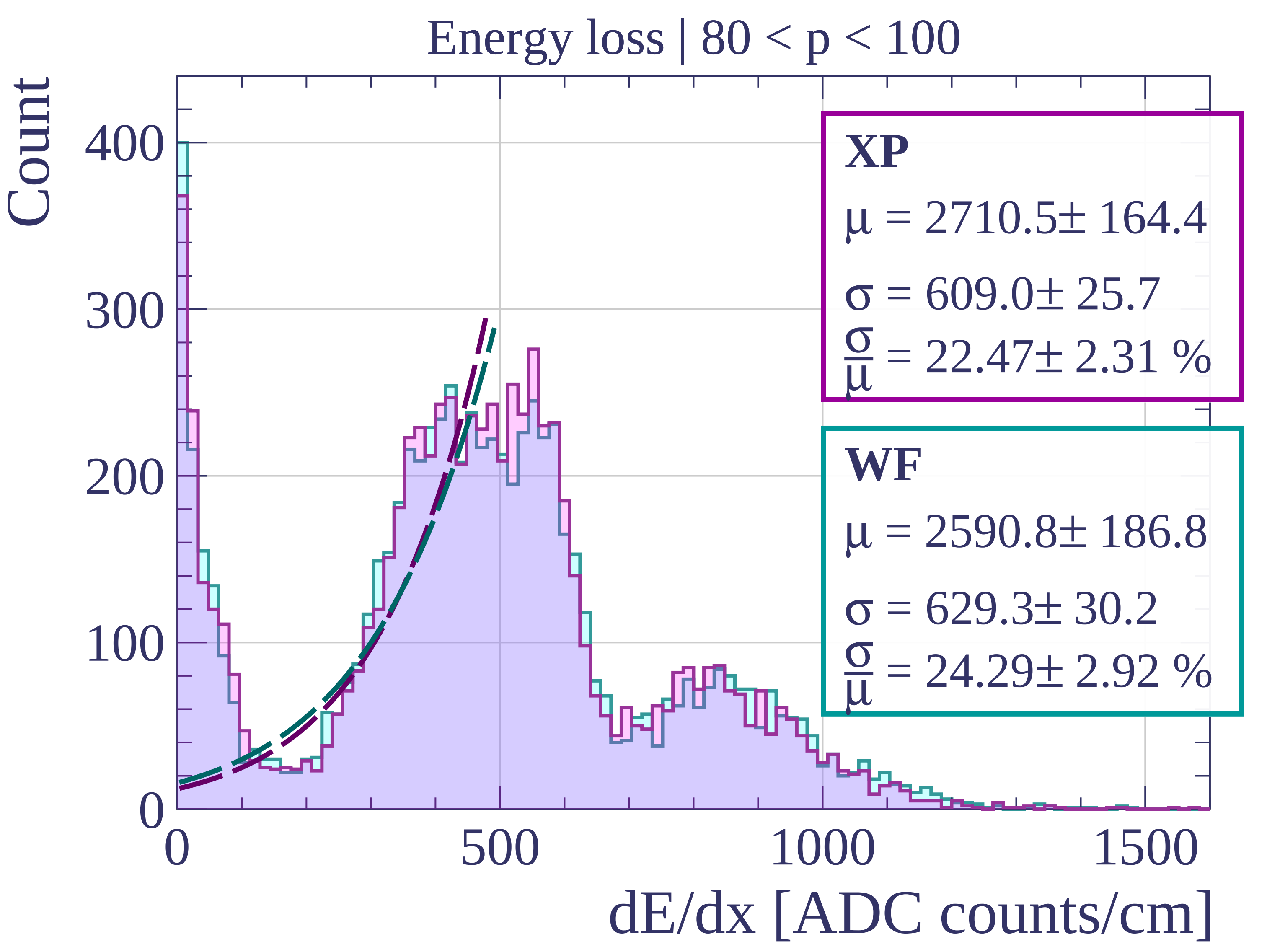
Energy loss | 0 < p < 20
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 5766.8± 95.7
σ = 1247.6± 16.5
σ/μ = 21.63± 0.64 %
WF
μ = -2625.8± 56.5
σ = 1311.1± 7.2
σ/μ = -49.93± 0.80 %

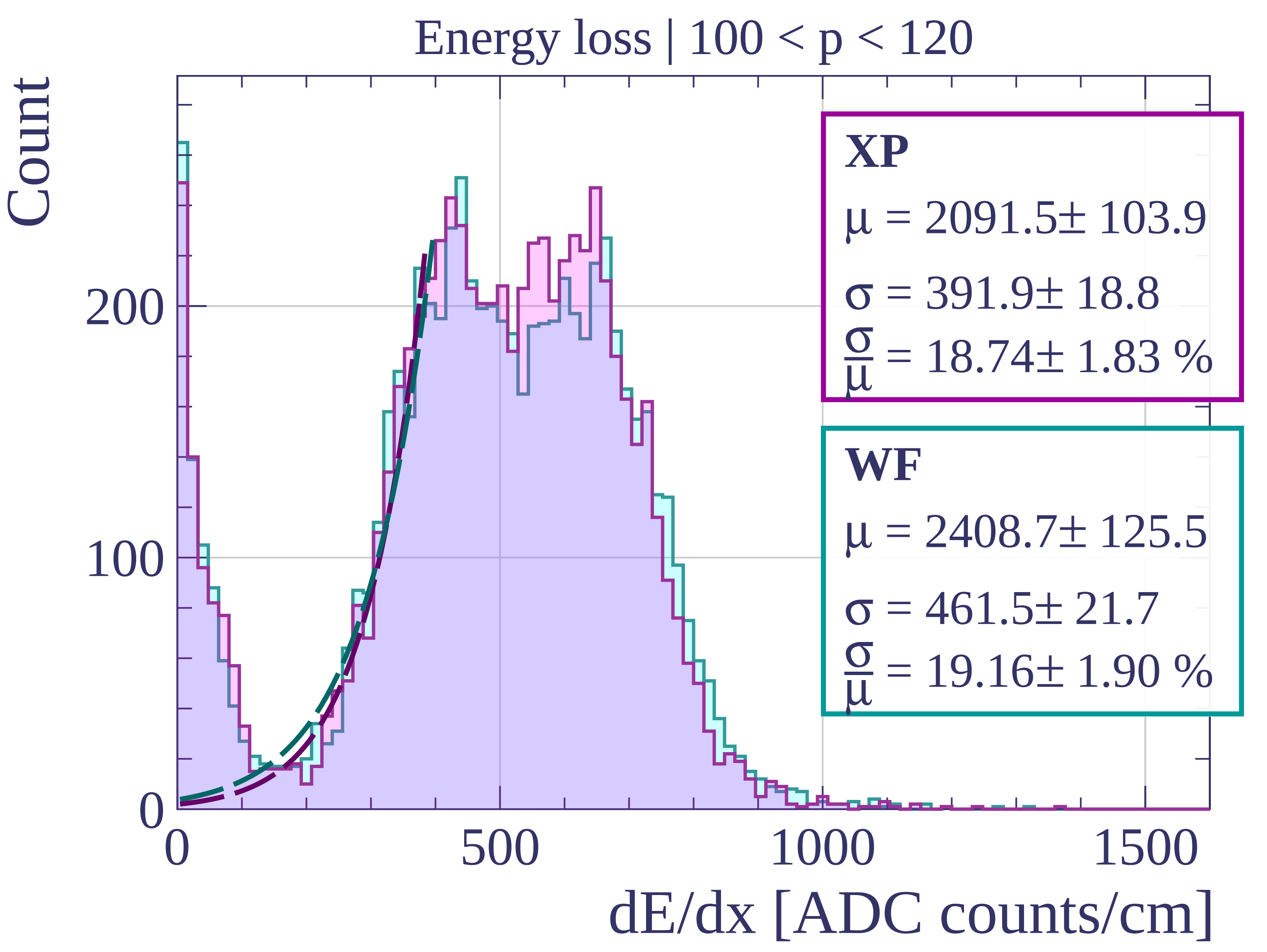

Energy loss | 20 < p < 40
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = -1667.4± 32.0
σ = 331.2± 5.5
σ/μ = -19.86± 0.05 %
WF
μ = 592.8± 23602.1
σ = 320.0± 827.0
σ/μ = 53.98± 2288.89 %



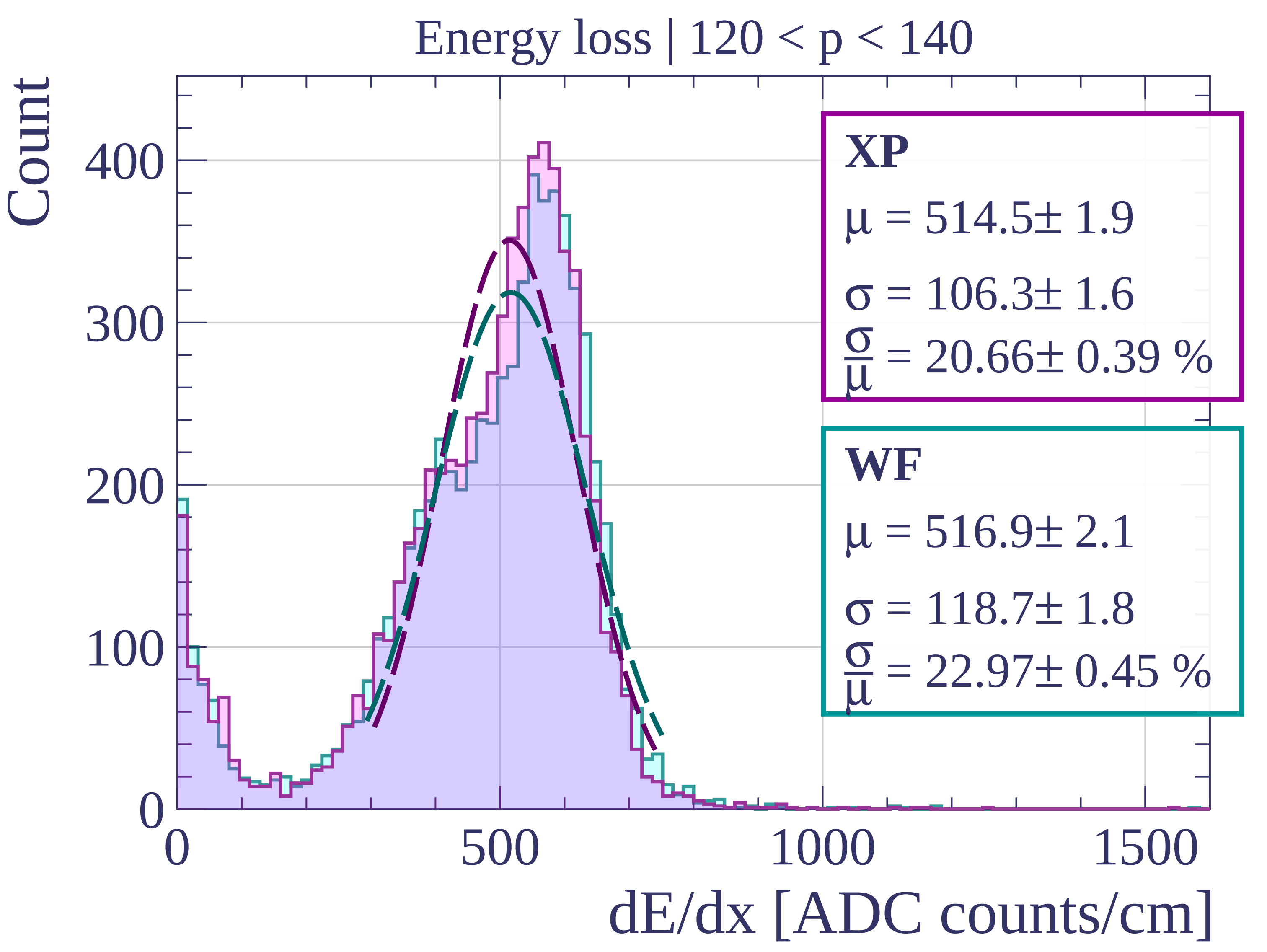
Energy loss | 40 < p < 60
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 528.7± 0.7
σ = 59.3± 0.7
σ/μ = 11.22± 0.15 %
WF
μ = 535.1± 0.7
σ = 62.0± 0.8
σ/μ = 11.58± 0.16 %


Energy loss | 60 < p < 80
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 3441.0± 148.7
σ = 755.0± 24.8
σ/μ = 21.94± 1.67 %
WF
μ = 3546.4± 223.7
σ = 826.0± 35.8
σ/μ = 23.29± 2.48 %

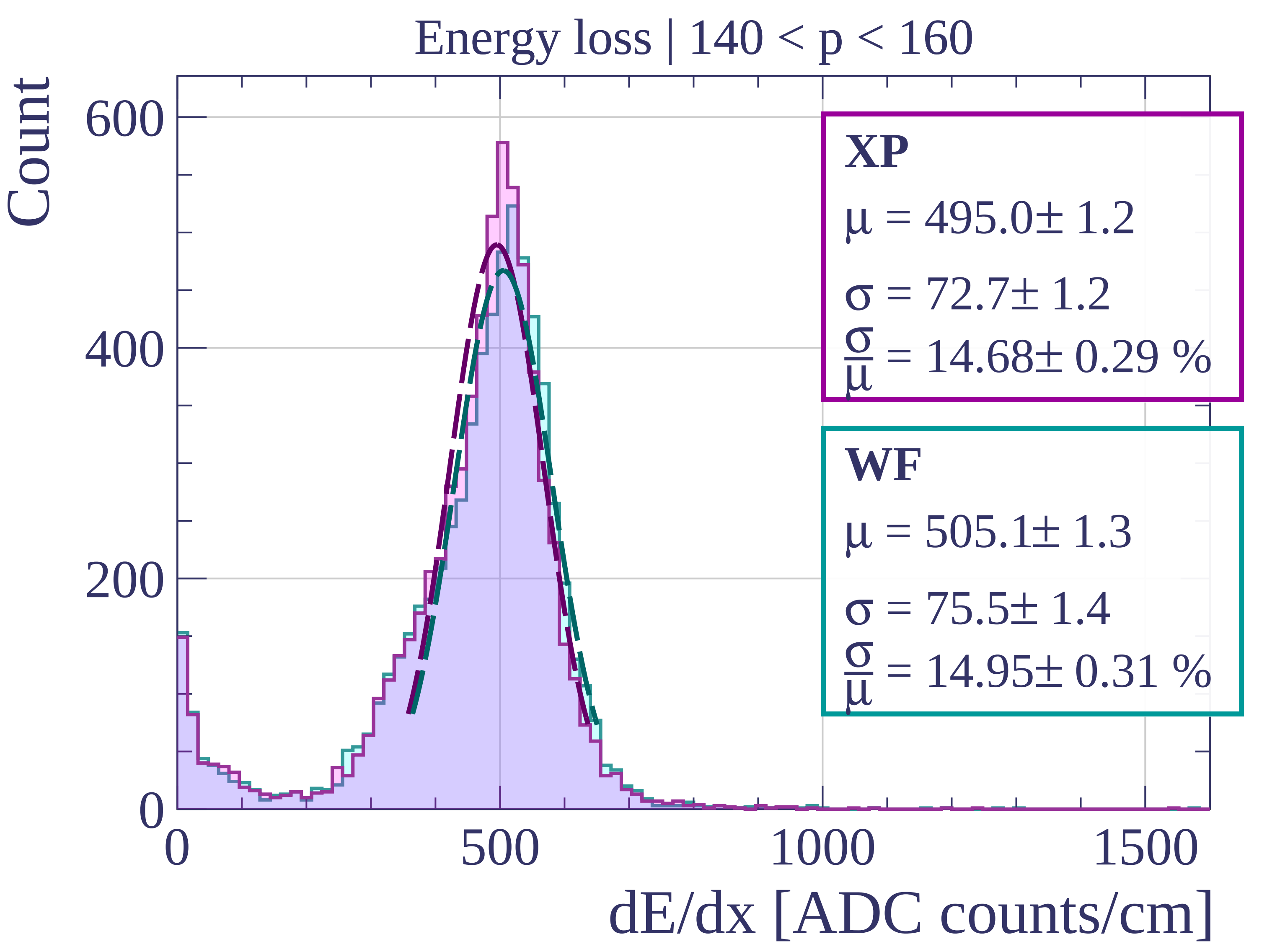

Energy loss | 80 < p < 100
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
XP
μ = 2710.5± 164.4
σ = 609.0± 25.7
σ/μ = 22.47± 2.31 %
WF
μ = 2590.8± 186.8
σ = 629.3± 30.2
σ/μ = 24.29± 2.92 %



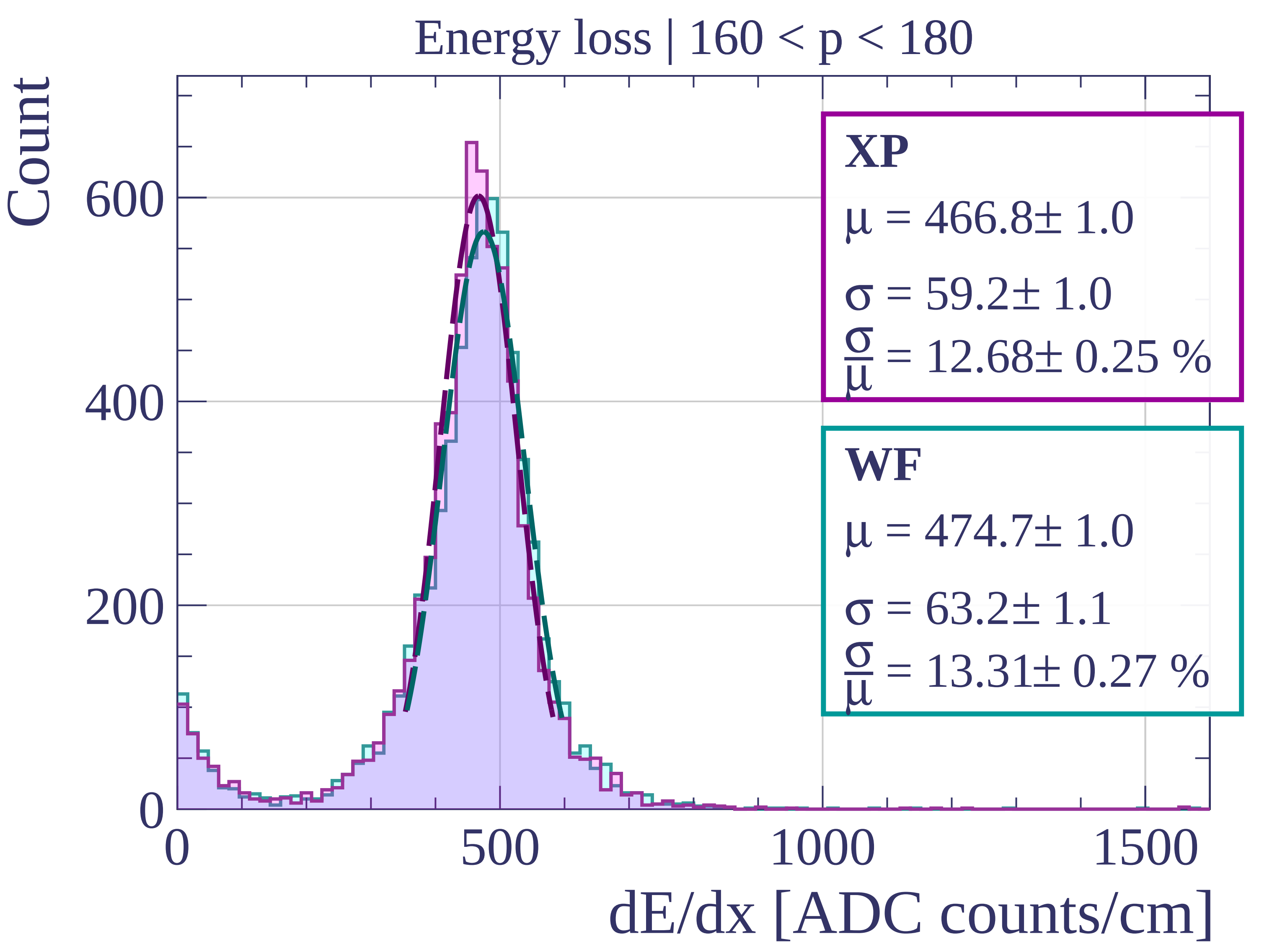
Energy loss | 100 < p < 120
Count
0
100
200
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 2091.5± 103.9
σ = 391.9± 18.8
σ/μ = 18.74± 1.83 %
WF
μ = 2408.7± 125.5
σ = 461.5± 21.7
σ/μ = 19.16± 1.90 %

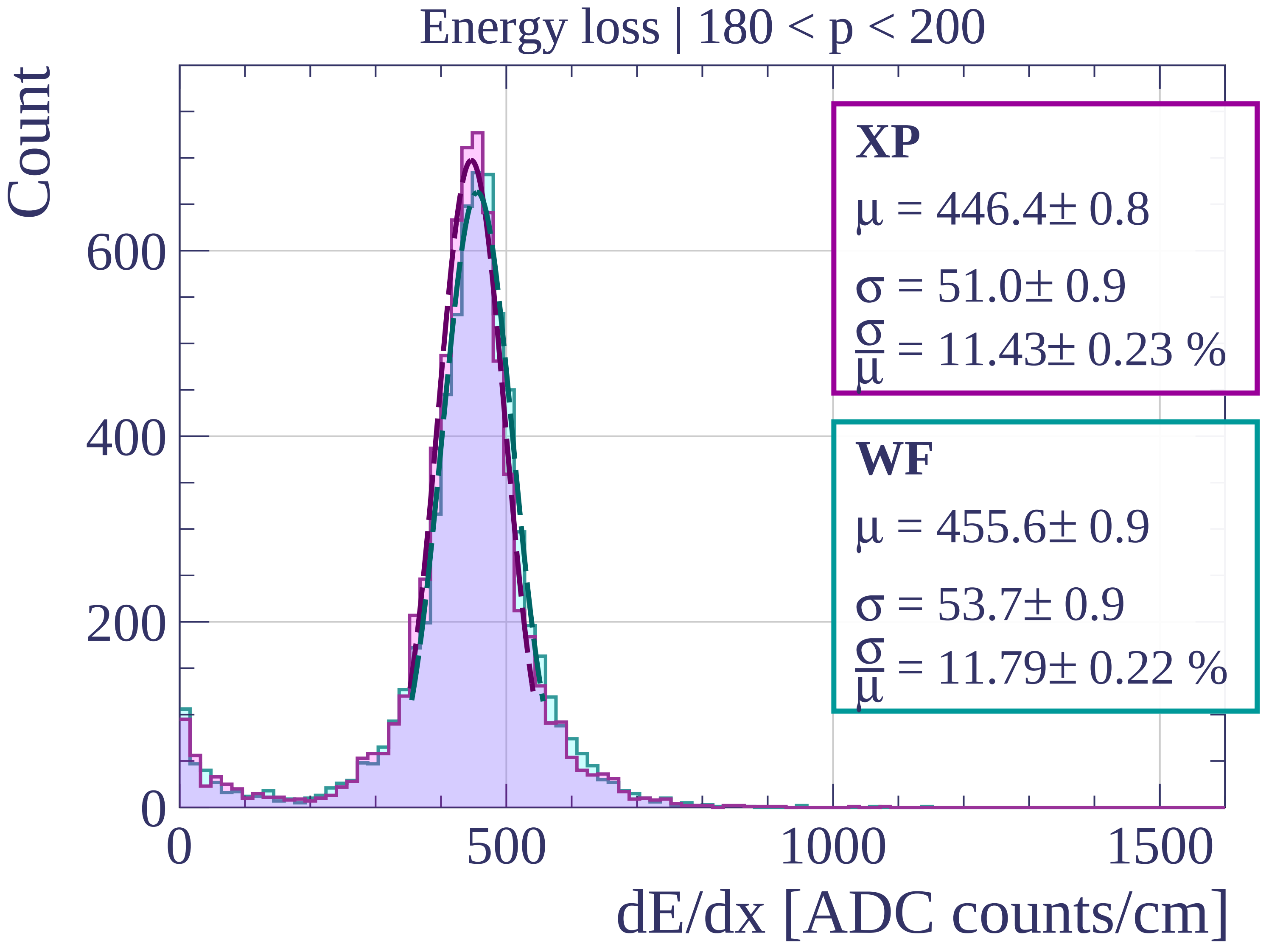


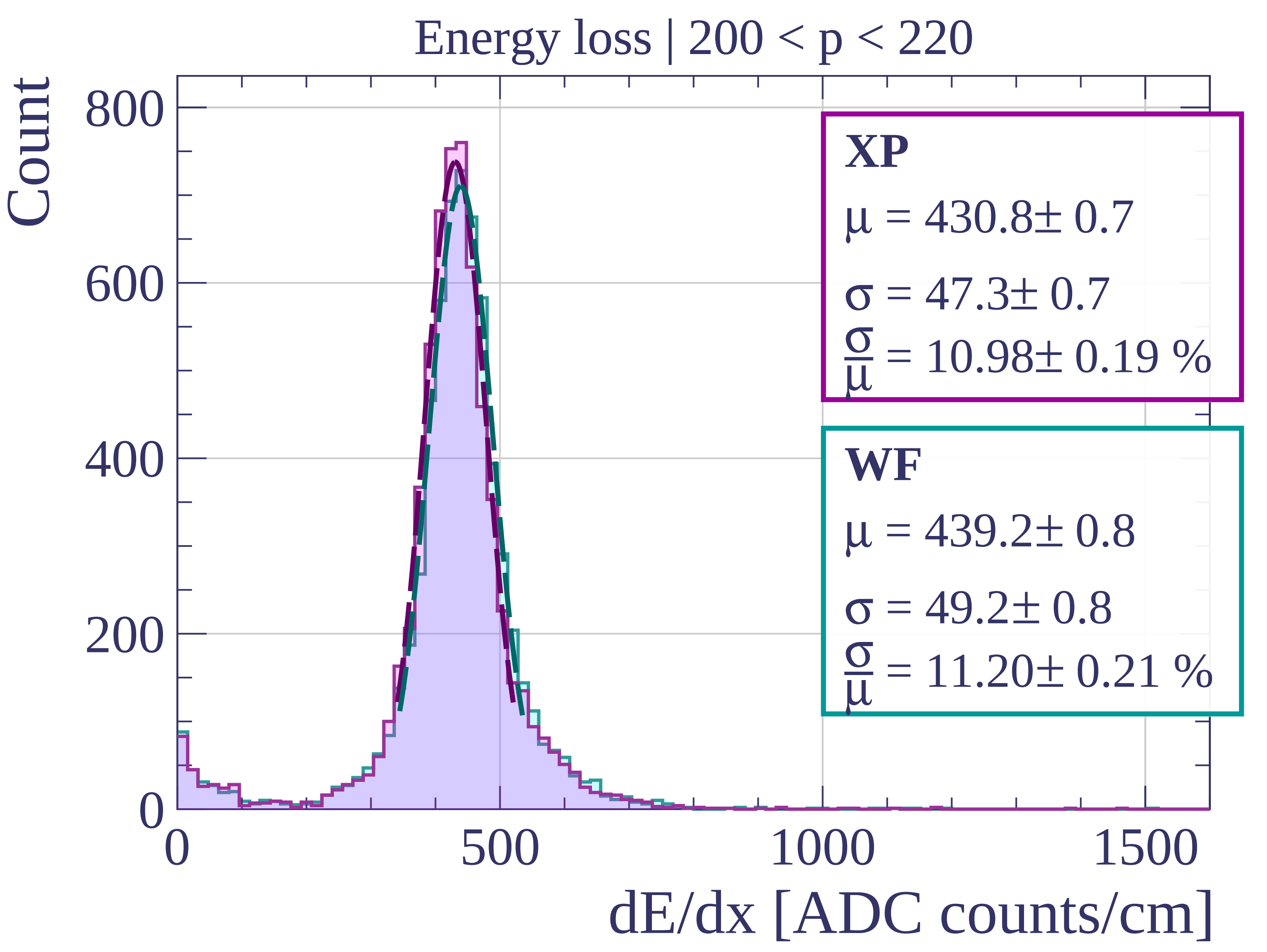
Energy loss | 120 < p < 140
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
500
1000
1500
XP
μ = 514.5± 1.9
σ = 106.3± 1.6
σ/μ = 20.66± 0.39 %
WF
μ = 516.9± 2.1
σ = 118.7± 1.8
σ/μ = 22.97± 0.45 %

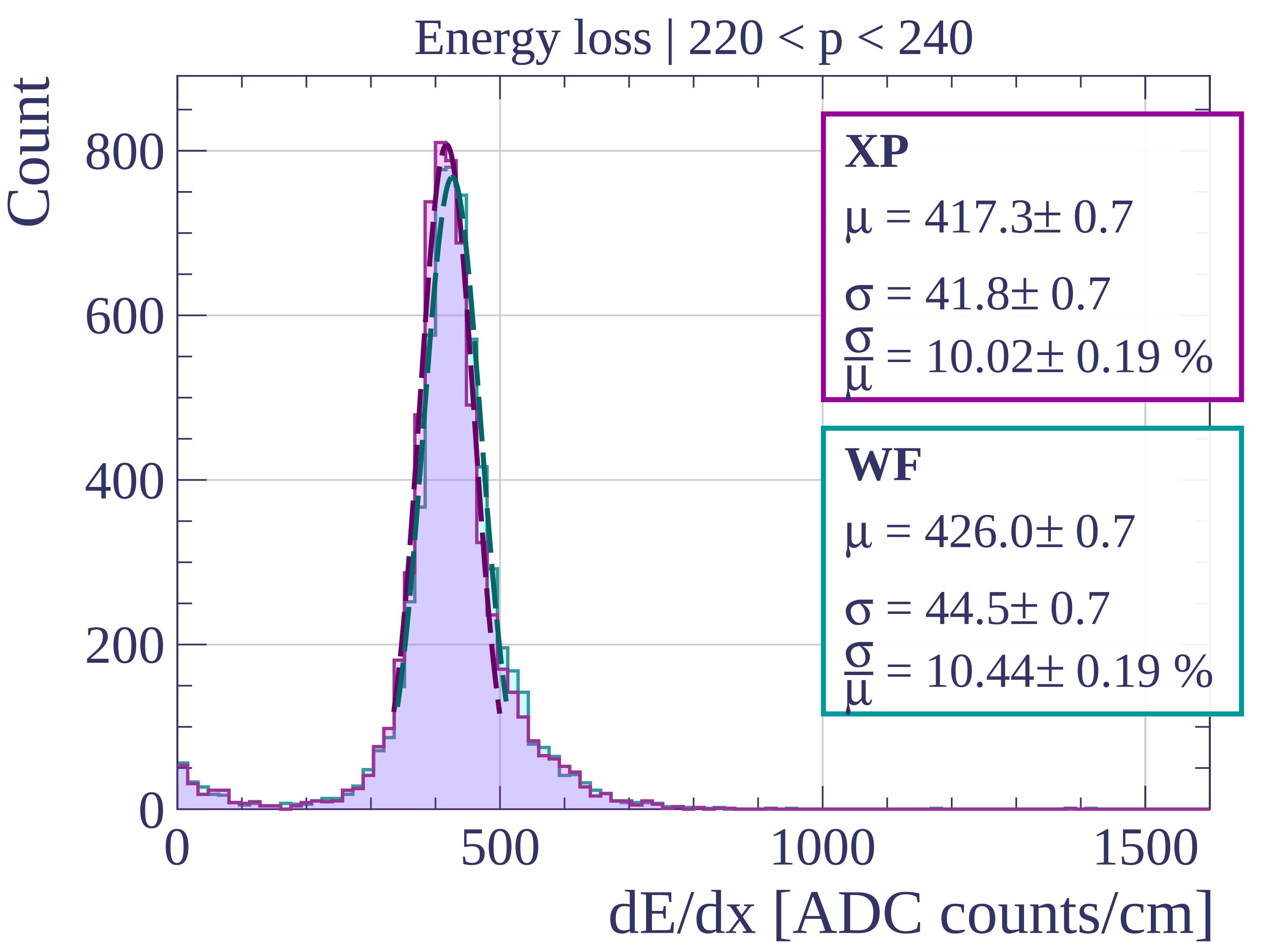


Energy loss | 140 < p < 160
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 495.0± 1.2
σ = 72.7± 1.2
σ/μ = 14.68± 0.29 %
WF
μ = 505.1± 1.3
σ = 75.5± 1.4
σ/μ = 14.95± 0.31 %

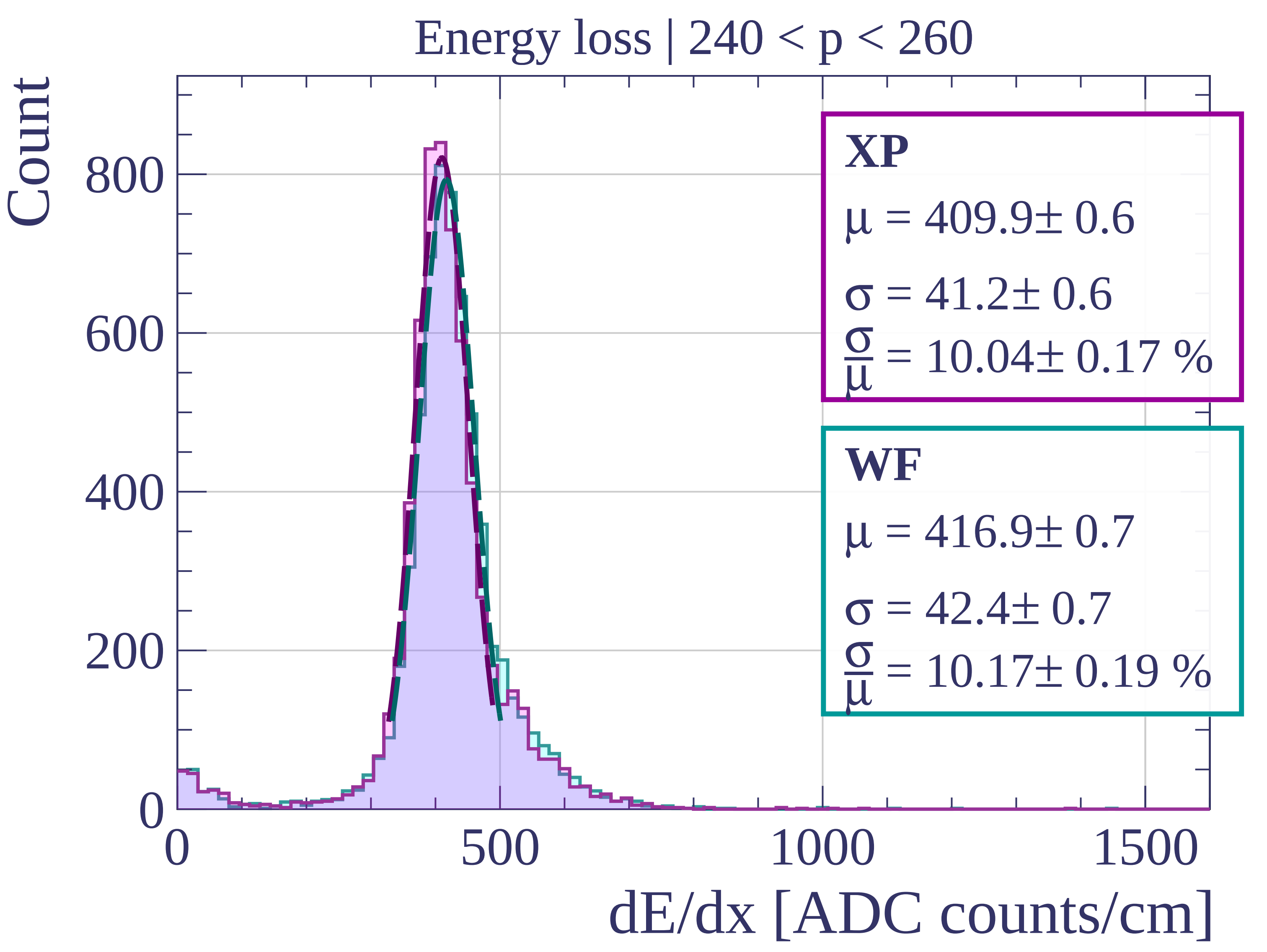

Energy loss | 160 < p < 180
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 466.8± 1.0
σ = 59.2± 1.0
σ/μ = 12.68± 0.25 %
WF
μ = 474.7± 1.0
σ = 63.2± 1.1
σ/μ = 13.31± 0.27 %


Energy loss | 180 < p < 200
Count
XP
μ = 446.4± 0.8
σ = 51.0± 0.9
σ/μ = 11.43± 0.23 %
WF
μ = 455.6± 0.9
σ = 53.7± 0.9
σ/μ = 11.79± 0.22 %
0
200
400
600
0
500
1000
1500
dE/dx [ADC counts/cm]

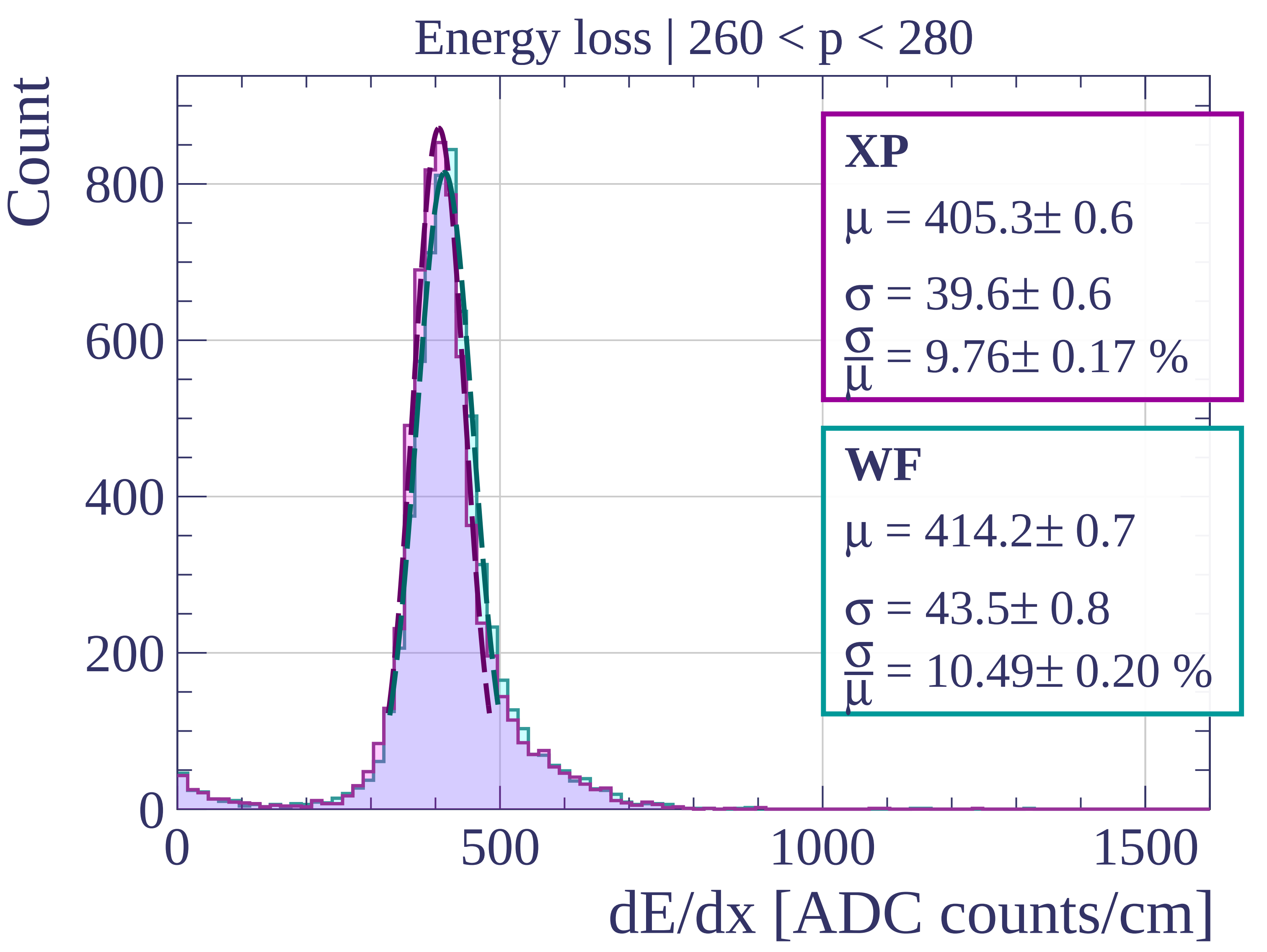

Energy loss | 200 < p < 220
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 430.8± 0.7
σ = 47.3± 0.7
σ/μ = 10.98± 0.19 %
WF
μ = 439.2± 0.8
σ = 49.2± 0.8
σ/μ = 11.20± 0.21 %



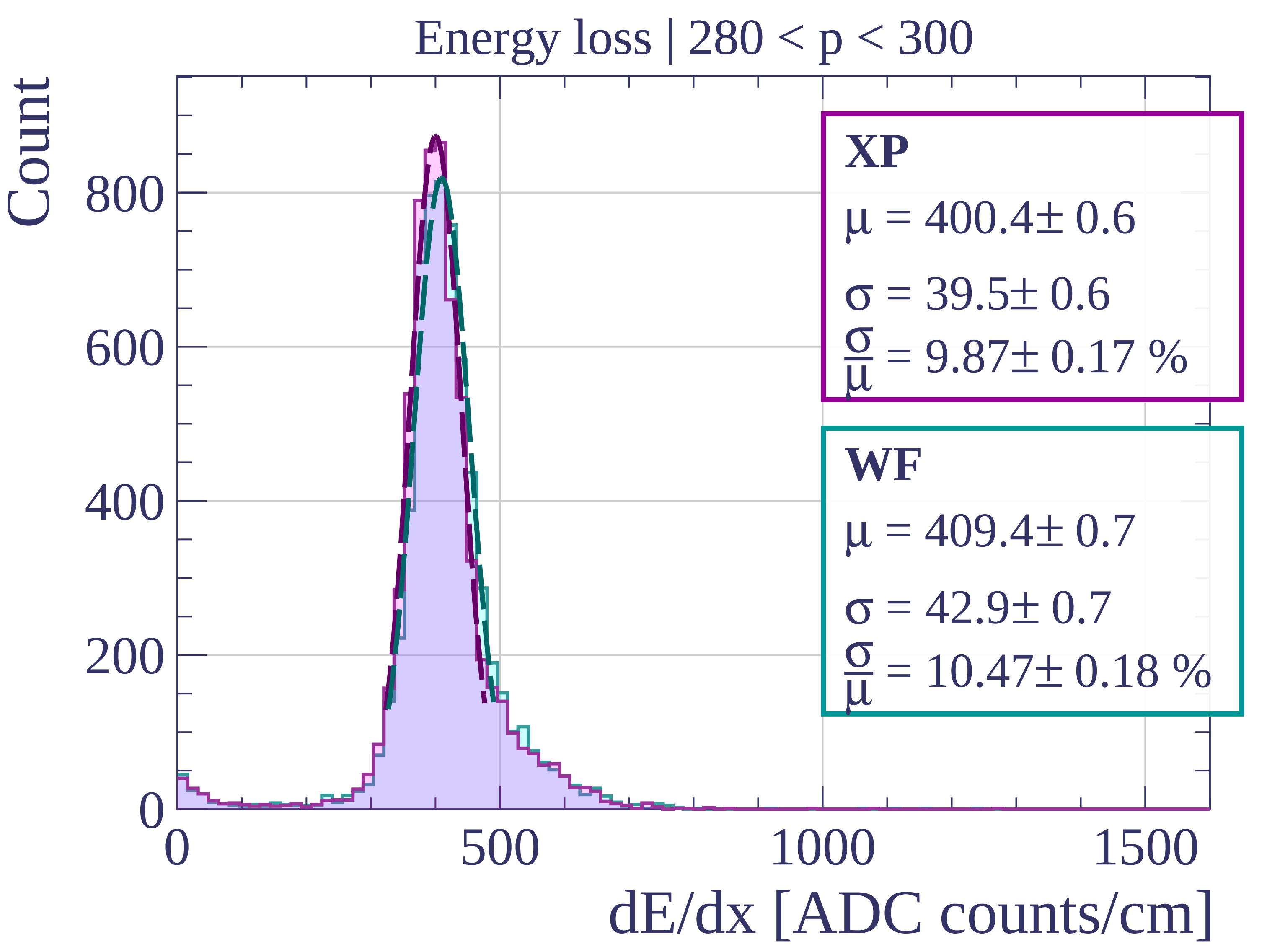
Energy loss | 220 < p < 240
Count
0
200
400
600
800
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 417.3± 0.7
σ = 41.8± 0.7
σ/μ = 10.02± 0.19 %
WF
μ = 426.0± 0.7
σ = 44.5± 0.7
σ/μ = 10.44± 0.19 %



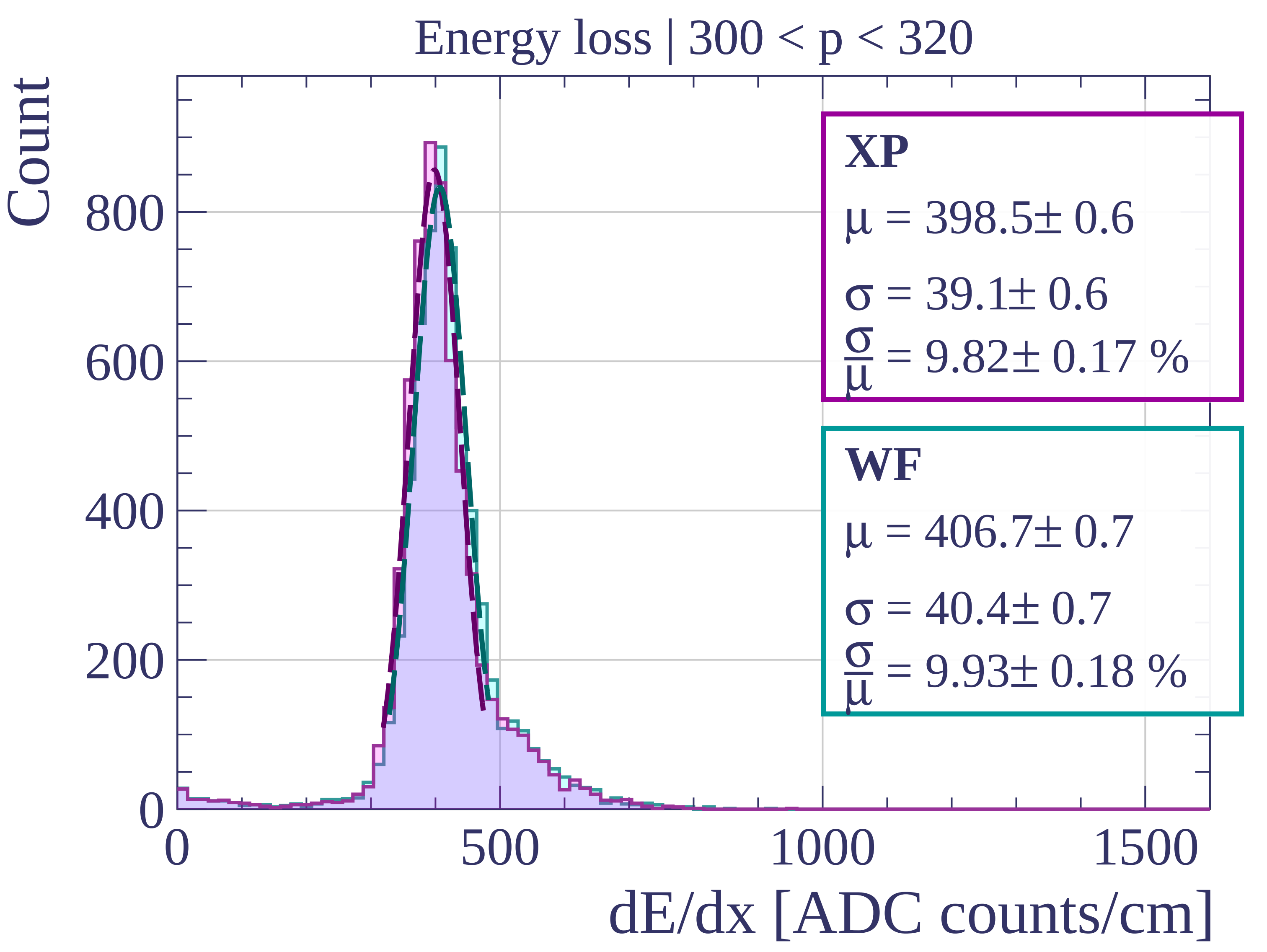
Energy loss | 240 < p < 260
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 409.9± 0.6
σ = 41.2± 0.6
σ/μ = 10.04± 0.17 %
WF
μ = 416.9± 0.7
σ = 42.4± 0.7
σ/μ = 10.17± 0.19 %



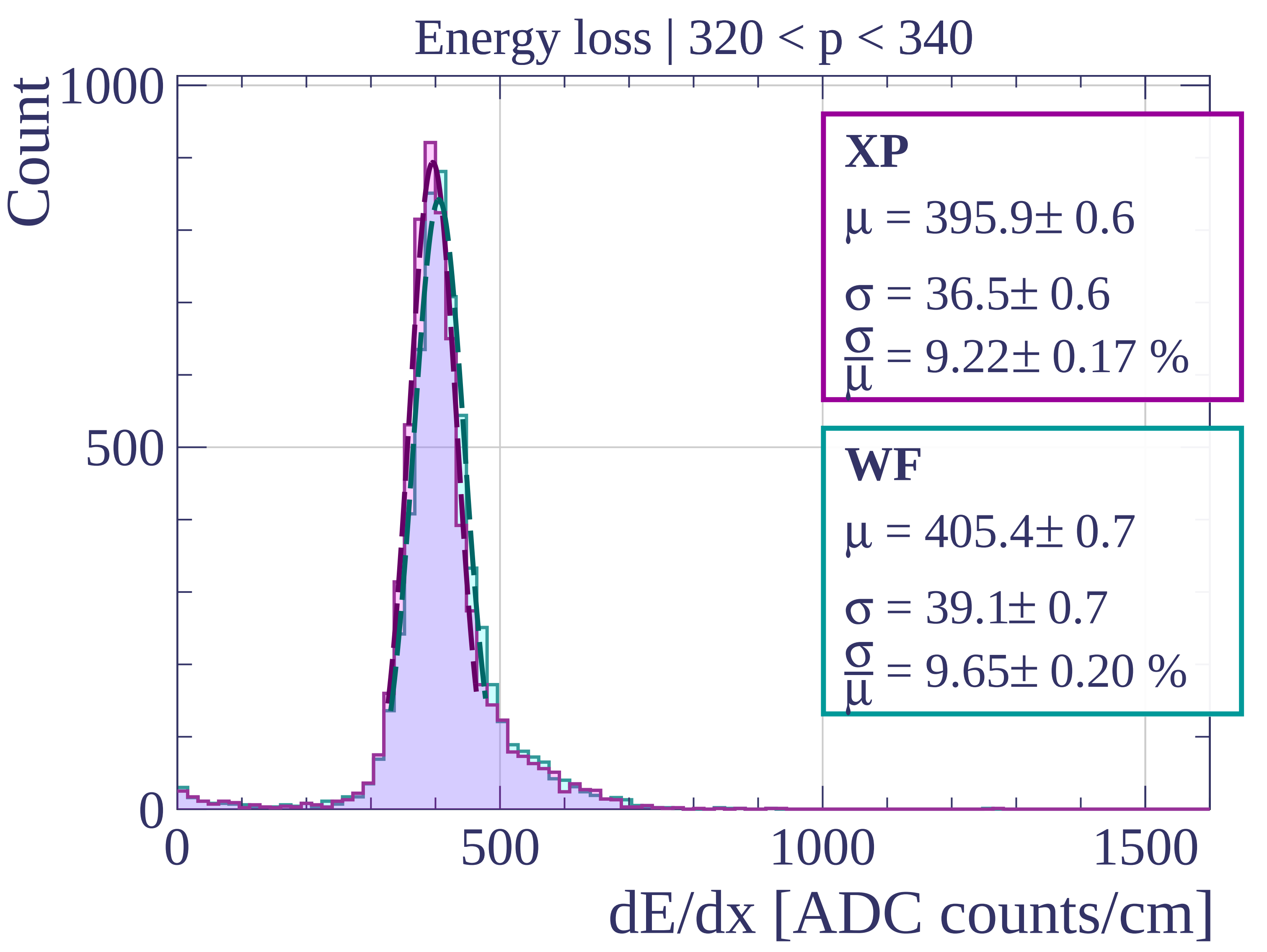
Energy loss | 260 < p < 280
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
500
1000
1500
XP
μ = 405.3± 0.6
σ = 39.6± 0.6
σ/μ = 9.76± 0.17 %
WF
μ = 414.2± 0.7
σ = 43.5± 0.8
σ/μ = 10.49± 0.20 %

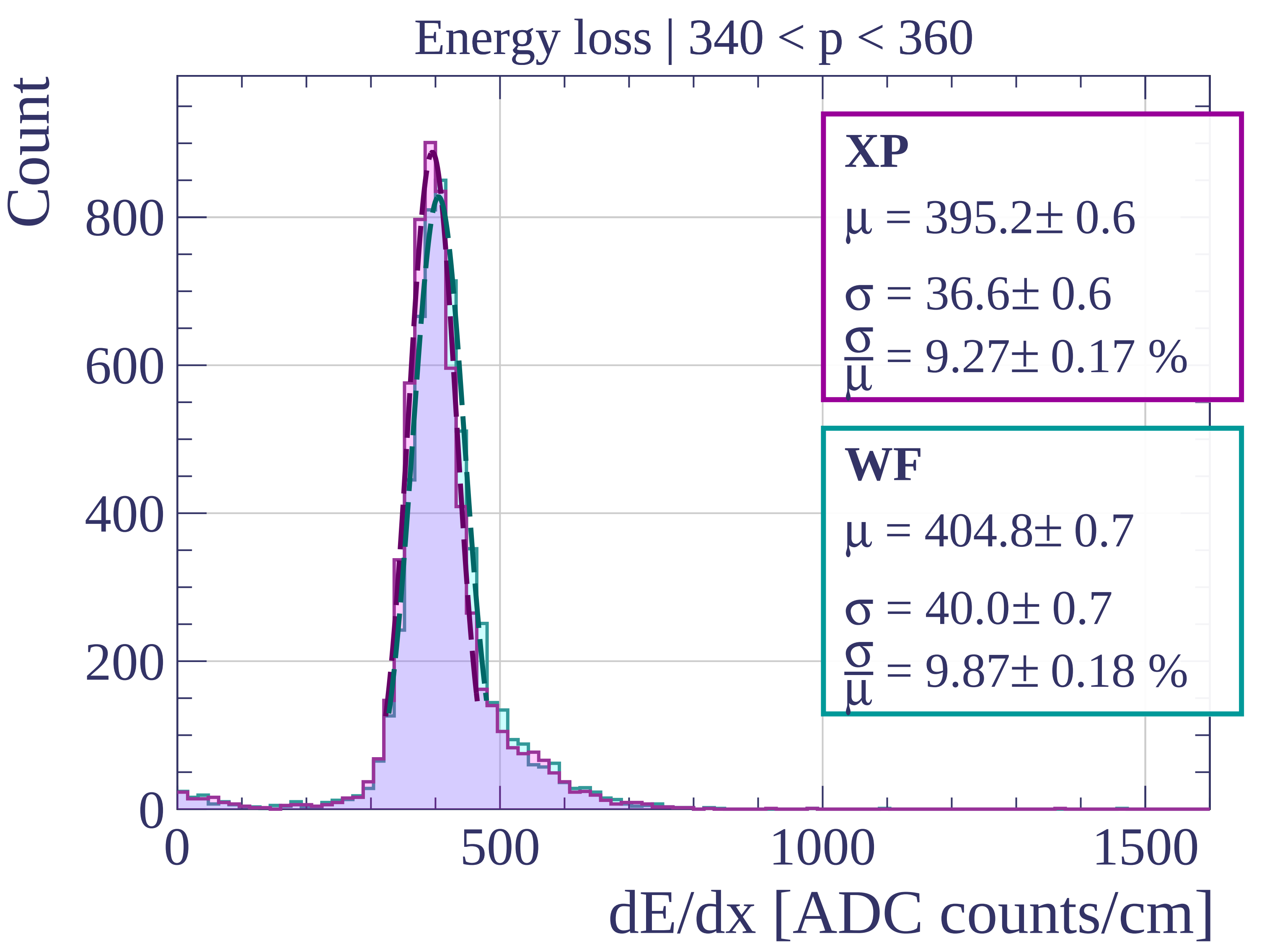


Energy loss | 280 < p < 300
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 400.4± 0.6
σ = 39.5± 0.6
σ/μ = 9.87± 0.17 %
WF
μ = 409.4± 0.7
σ = 42.9± 0.7
σ/μ = 10.47± 0.18 %

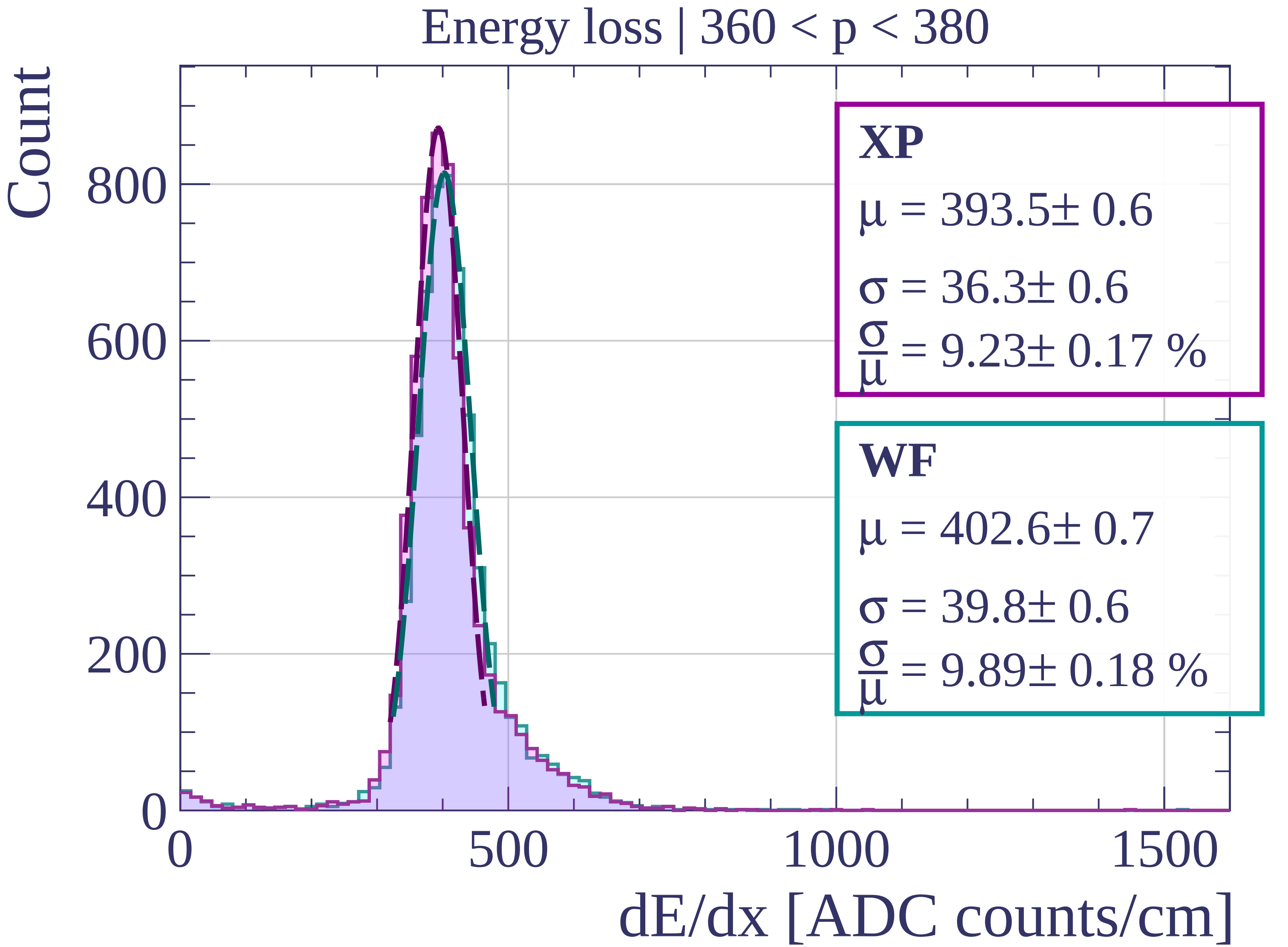


Energy loss | 300 < p < 320
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 398.5± 0.6
σ = 39.1± 0.6
σ/μ = 9.82± 0.17 %
WF
μ = 406.7± 0.7
σ = 40.4± 0.7
σ/μ = 9.93± 0.18 %

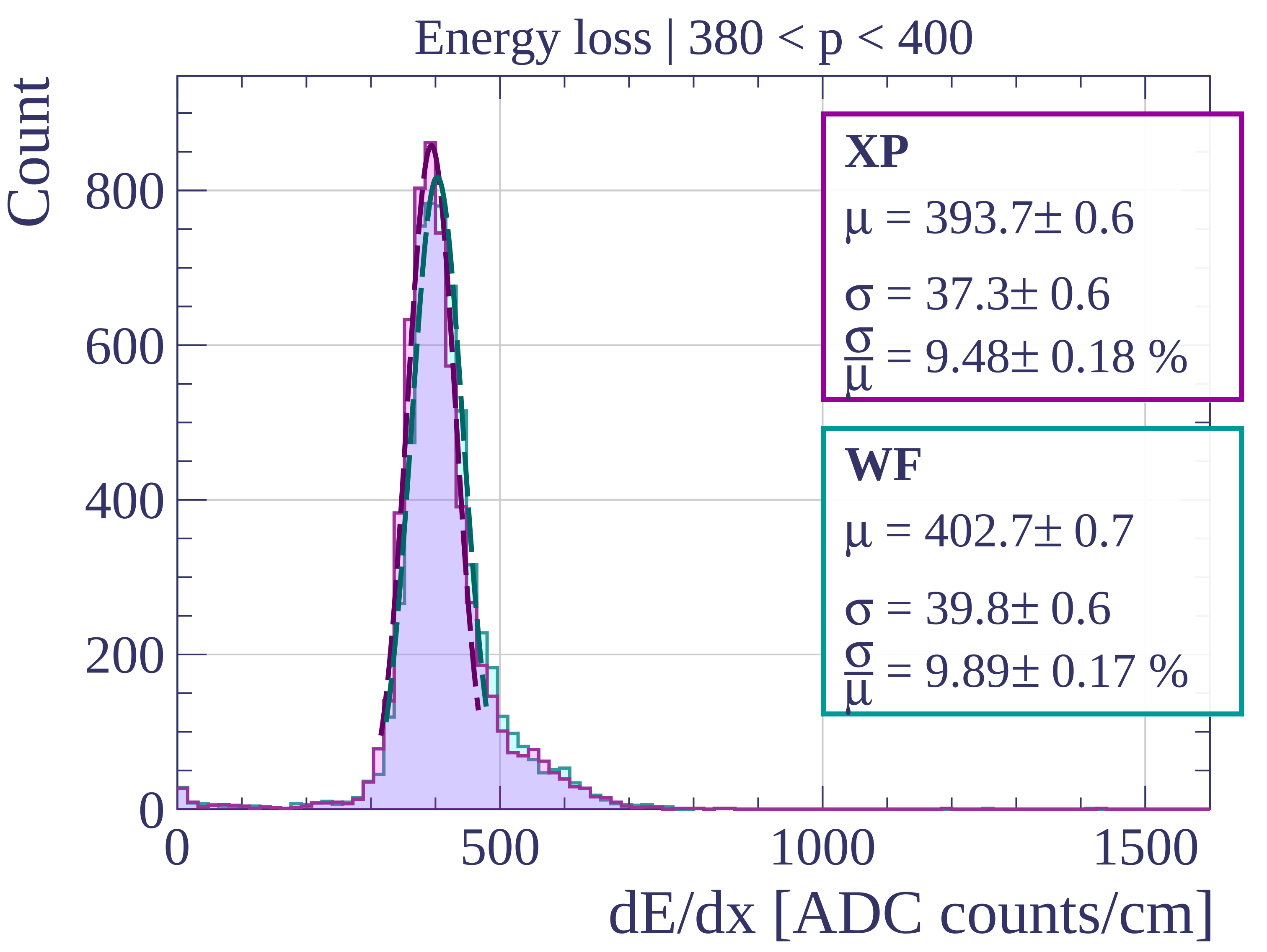

Energy loss | 320 < p < 340
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 395.9± 0.6
σ = 36.5± 0.6
σ/μ = 9.22± 0.17 %
WF
μ = 405.4± 0.7
σ = 39.1± 0.7
σ/μ = 9.65± 0.20 %



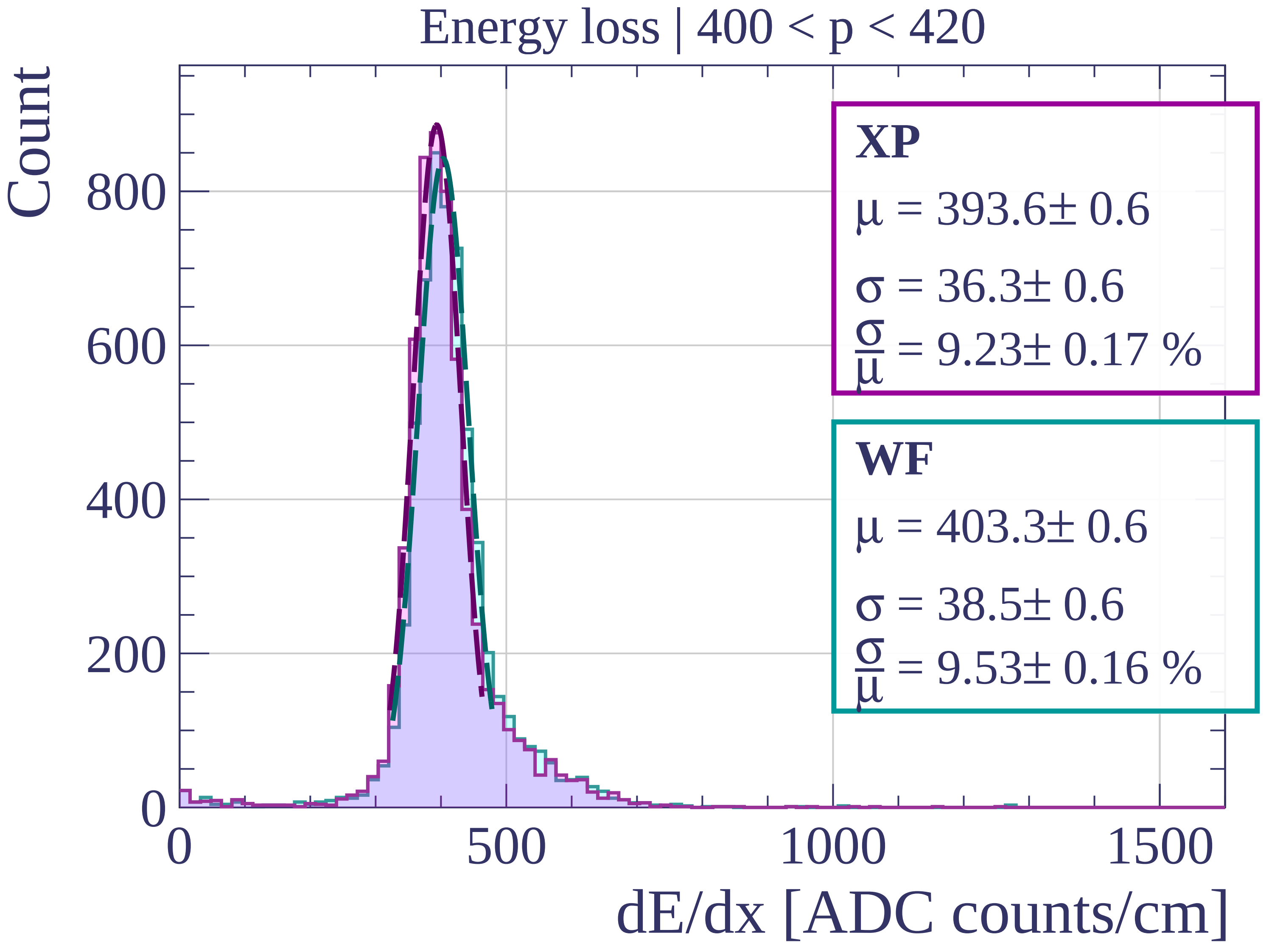
Energy loss | 340 < p < 360
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 395.2± 0.6
σ = 36.6± 0.6
σ/μ = 9.27± 0.17 %
WF
μ = 404.8± 0.7
σ = 40.0± 0.7
σ/μ = 9.87± 0.18 %

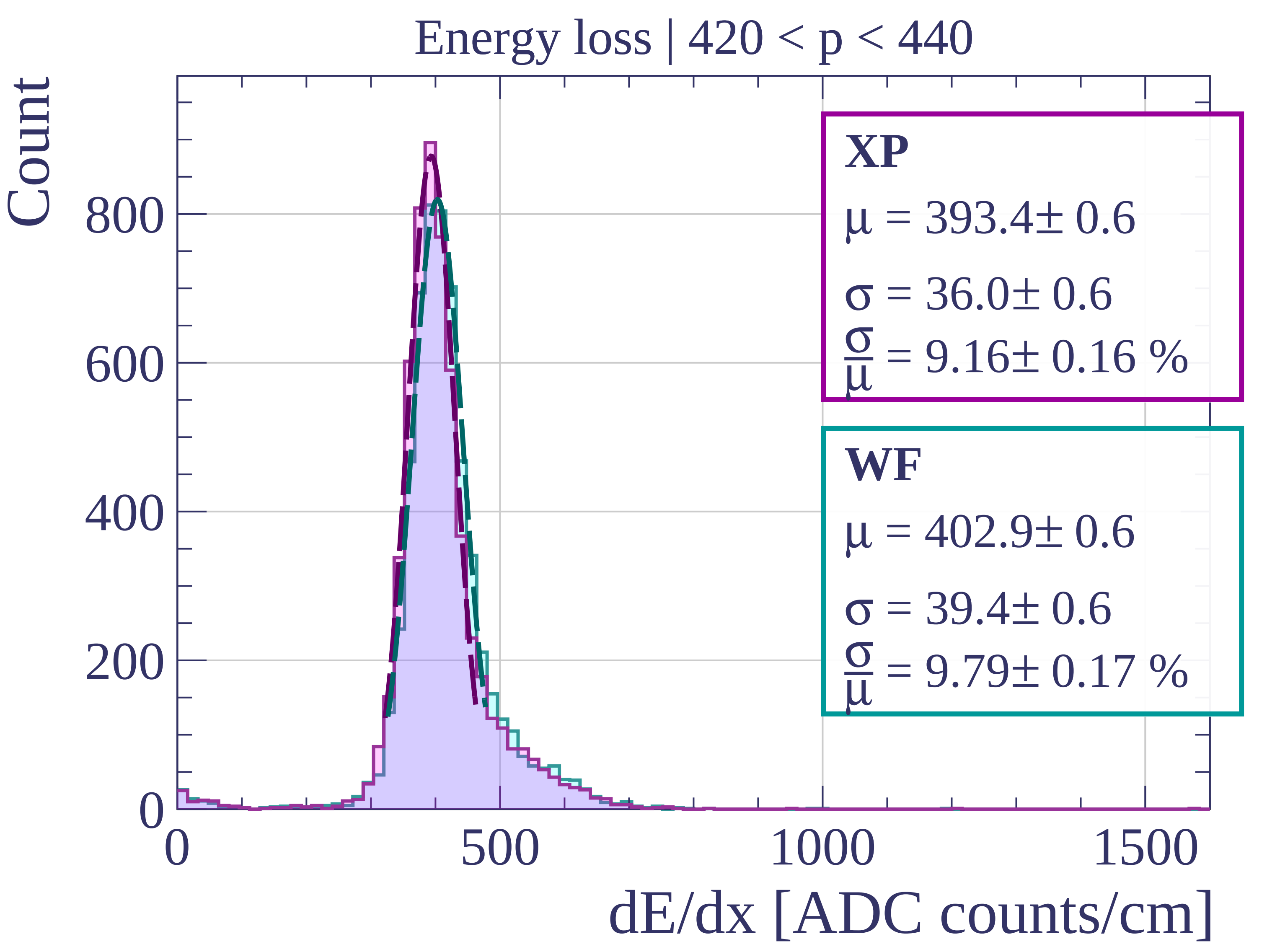


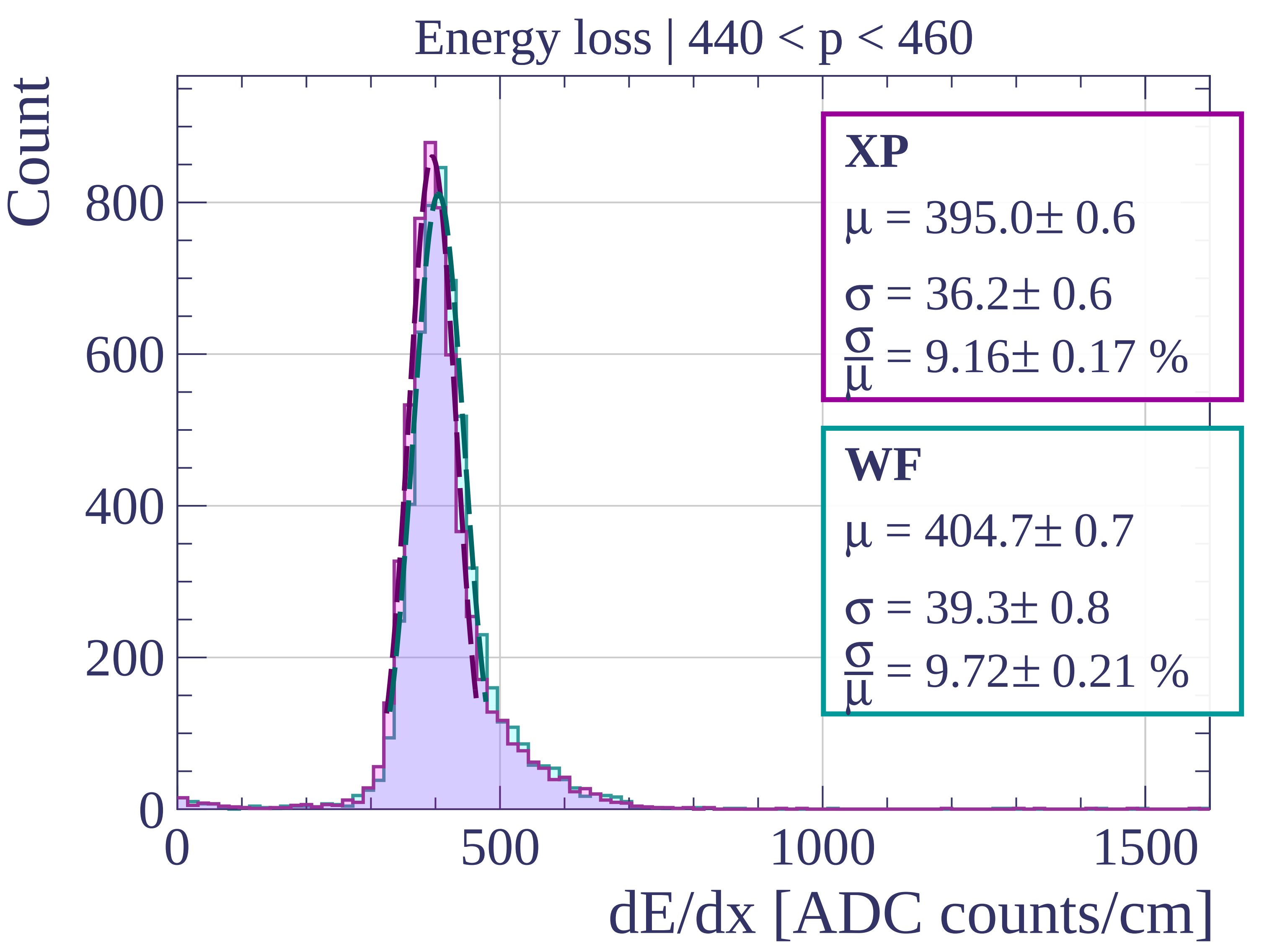
Energy loss | 360 < p < 380
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 393.5± 0.6
σ = 36.3± 0.6
σ/μ = 9.23± 0.17 %
WF
μ = 402.6± 0.7
σ = 39.8± 0.6
σ/μ = 9.89± 0.18 %



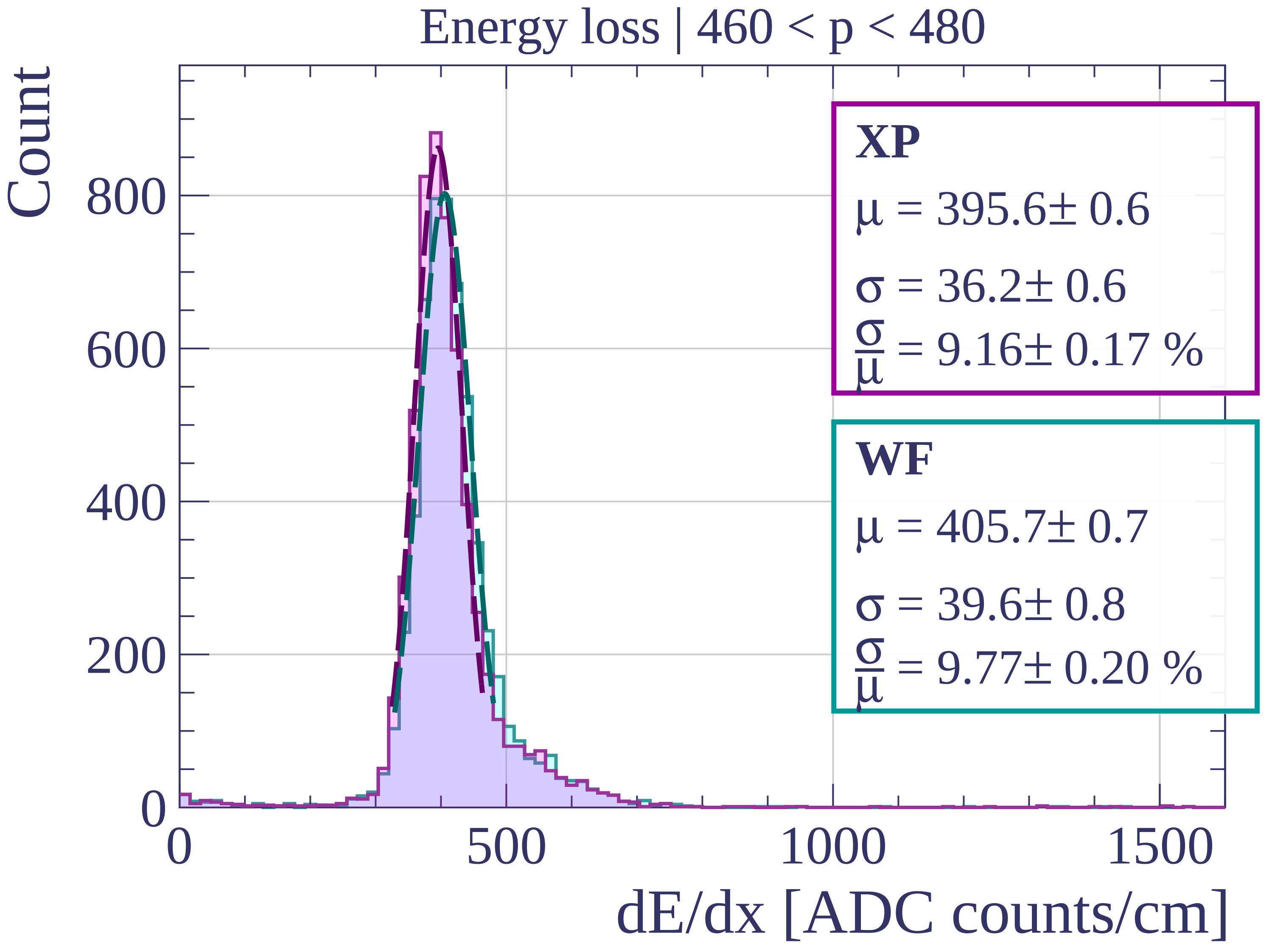
Energy loss | 380 < p < 400
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 393.7± 0.6
σ = 37.3± 0.6
σ/μ = 9.48± 0.18 %
WF
μ = 402.7± 0.7
σ = 39.8± 0.6
σ/μ = 9.89± 0.17 %


Energy loss | 400 < p < 420
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 393.6± 0.6
σ = 36.3± 0.6
σ/μ = 9.23± 0.17 %
WF
μ = 403.3± 0.6
σ = 38.5± 0.6
σ/μ = 9.53± 0.16 %

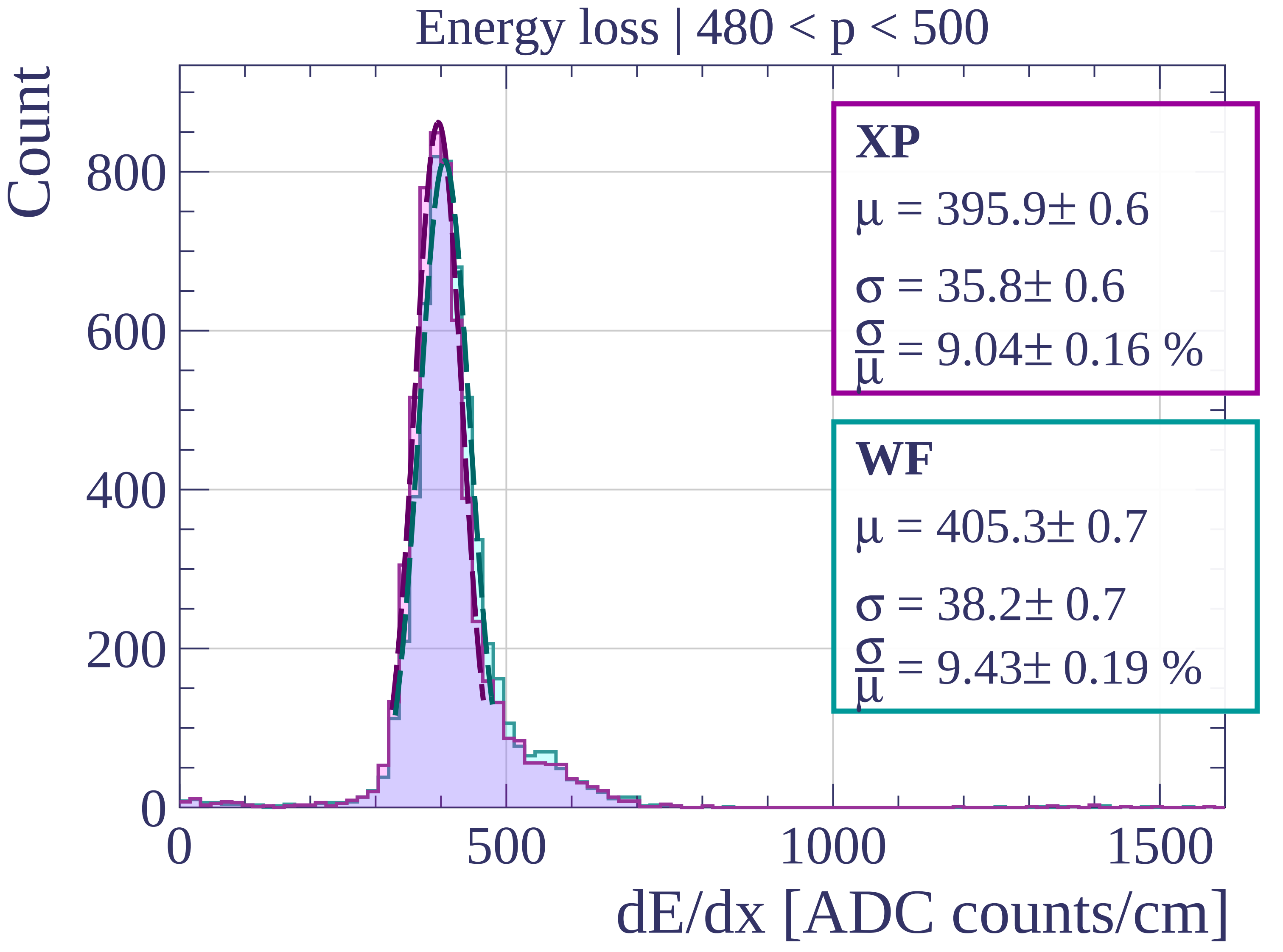

Energy loss | 420 < p < 440
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 393.4± 0.6
σ = 36.0± 0.6
σ/μ = 9.16± 0.16 %
WF
μ = 402.9± 0.6
σ = 39.4± 0.6
σ/μ = 9.79± 0.17 %



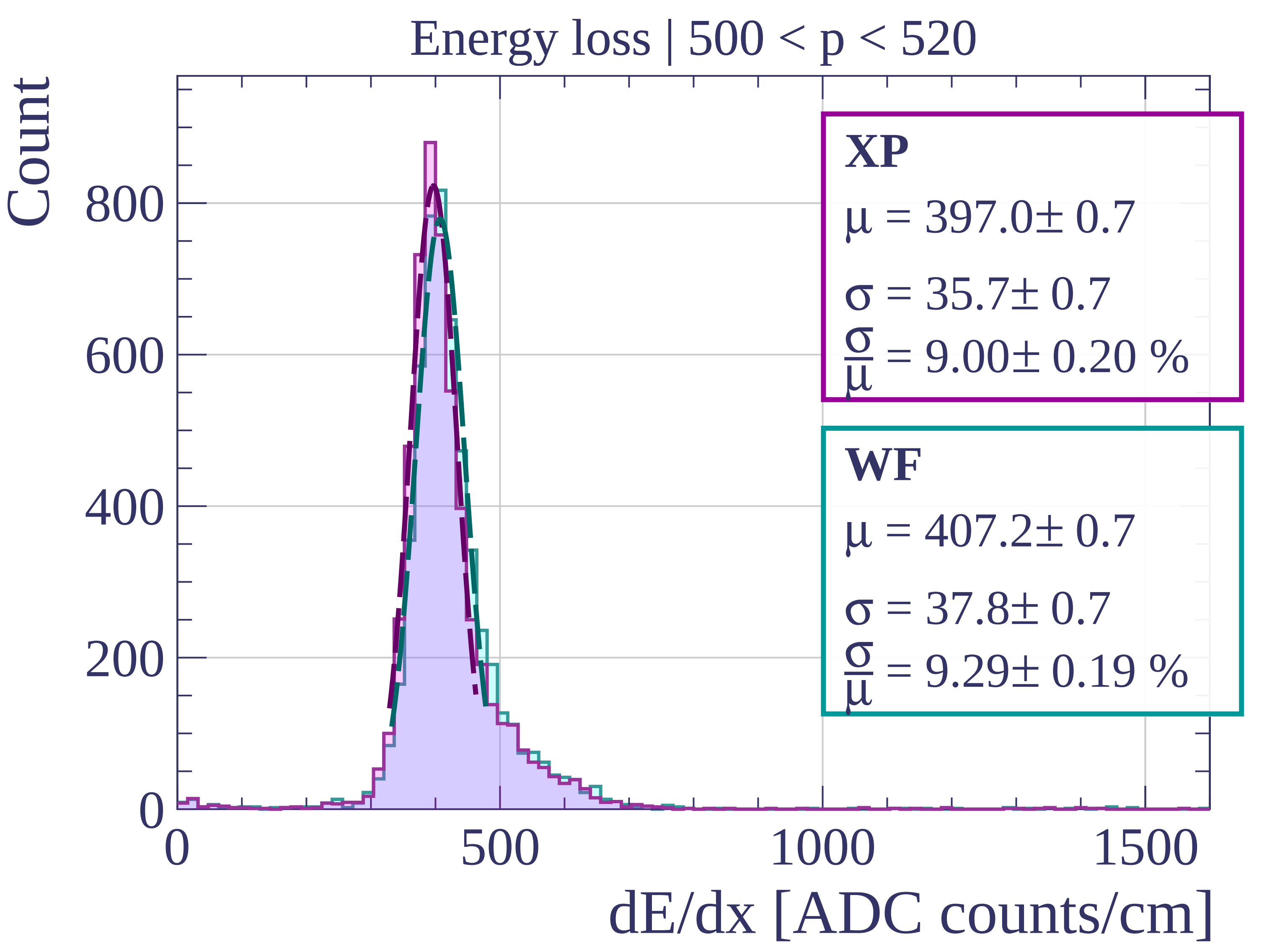
Energy loss | 440 < p < 460
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 395.0± 0.6
σ = 36.2± 0.6
σ/μ = 9.16± 0.17 %
WF
μ = 404.7± 0.7
σ = 39.3± 0.8
σ/μ = 9.72± 0.21 %



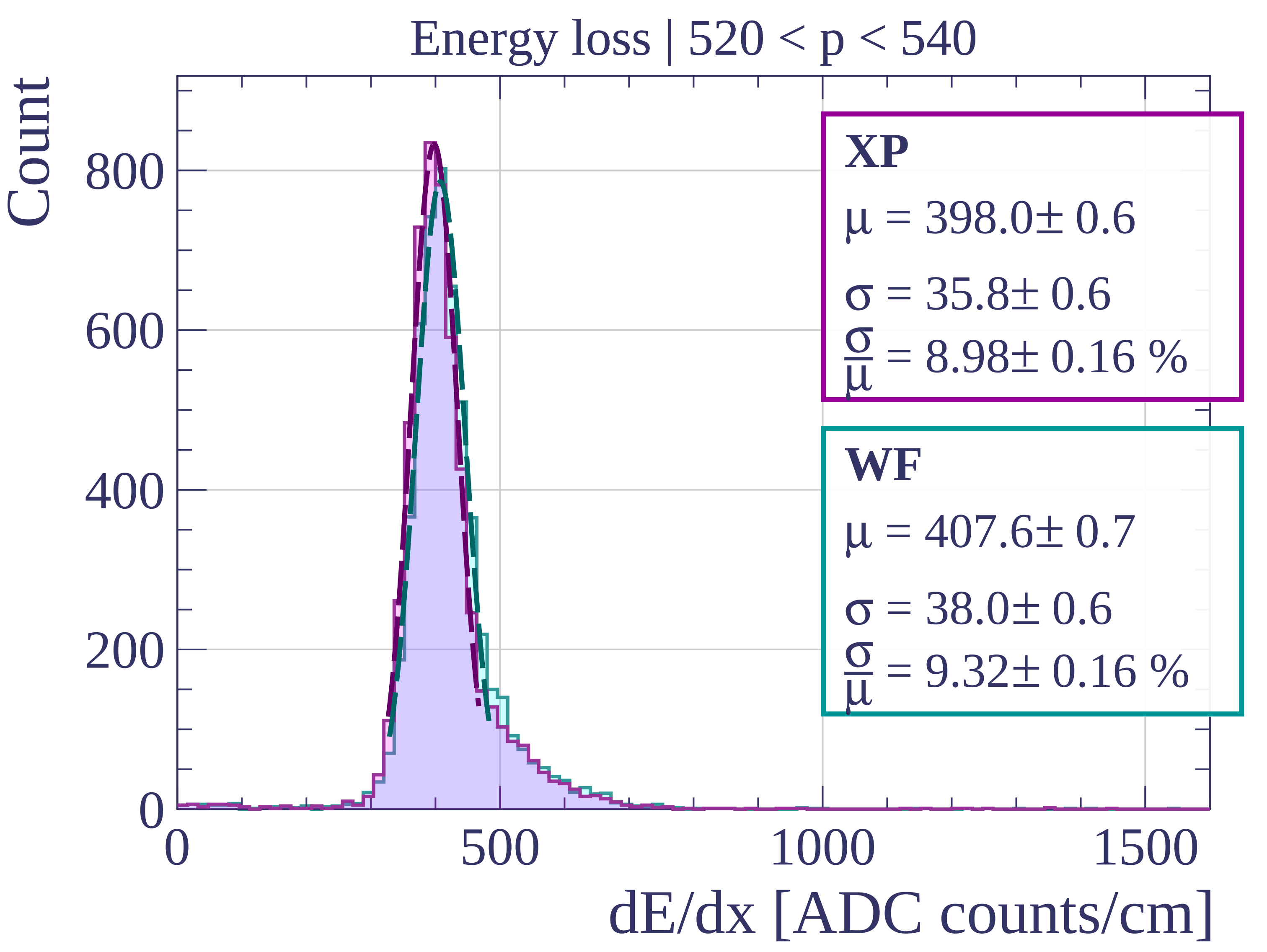
Energy loss | 460 < p < 480
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 395.6± 0.6
σ = 36.2± 0.6
σ/μ = 9.16± 0.17 %
WF
μ = 405.7± 0.7
σ = 39.6± 0.8
σ/μ = 9.77± 0.20 %



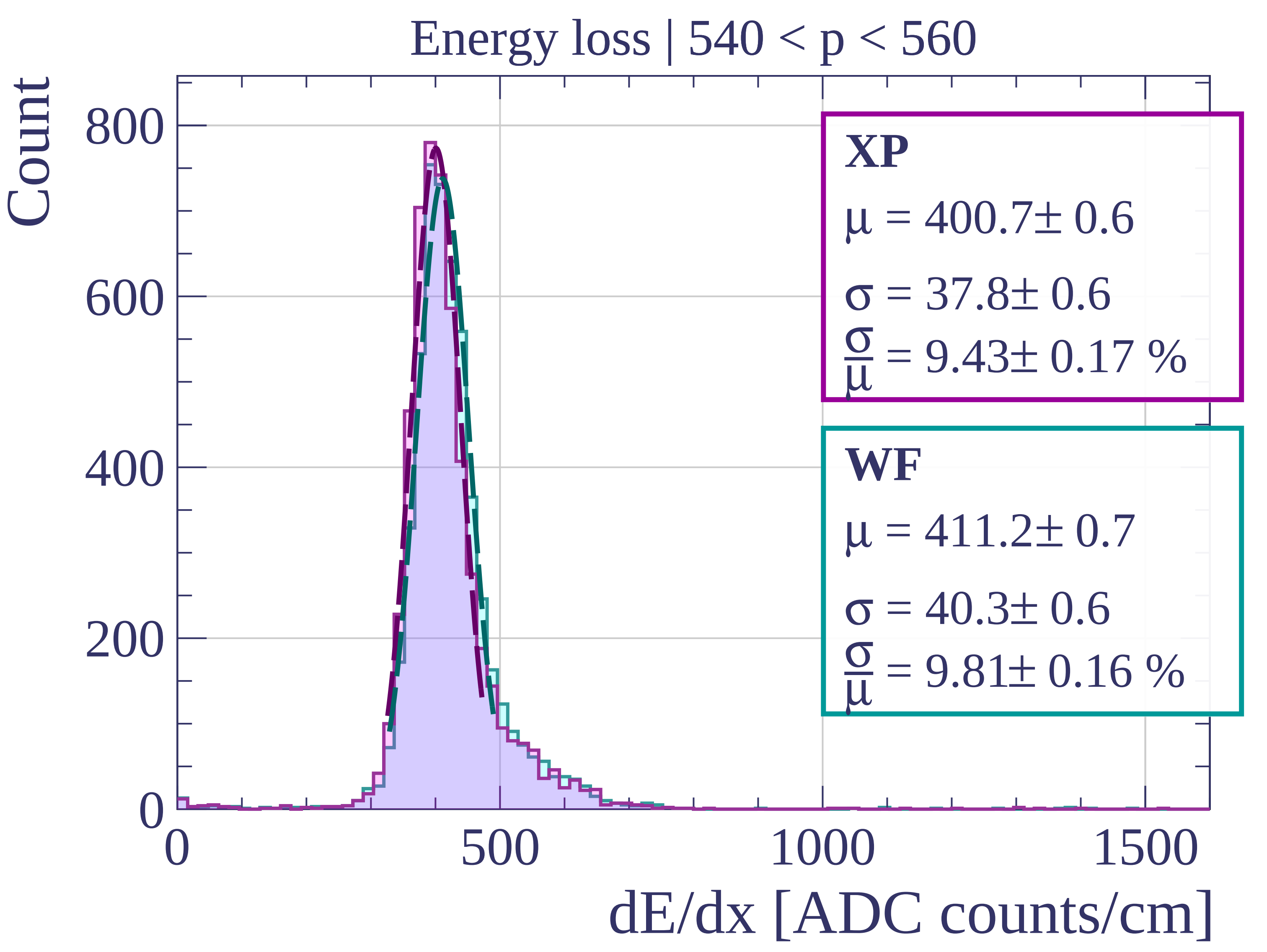
Energy loss | 480 < p < 500
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 395.9± 0.6
σ = 35.8± 0.6
σ/μ = 9.04± 0.16 %
WF
μ = 405.3± 0.7
σ = 38.2± 0.7
σ/μ = 9.43± 0.19 %



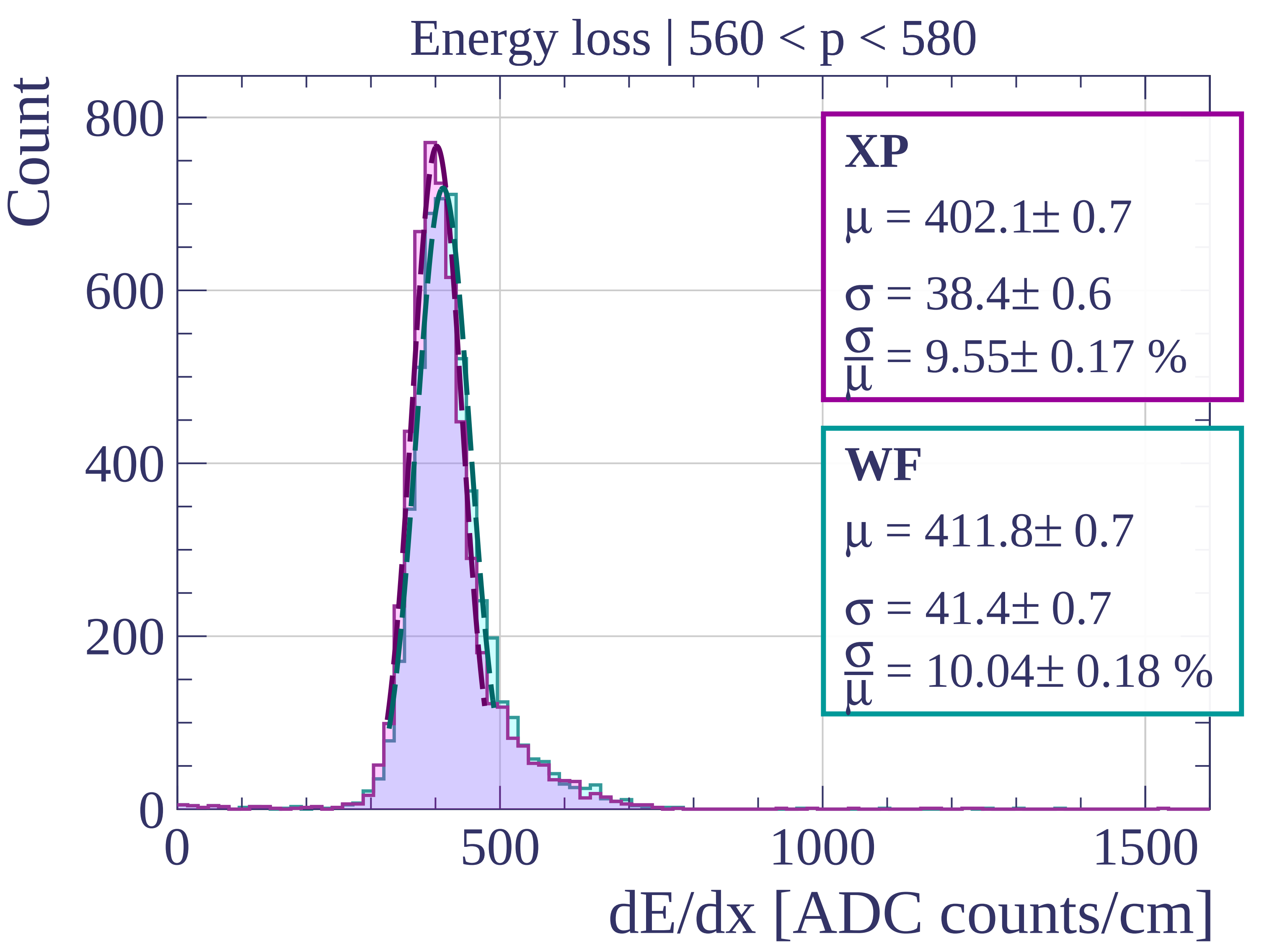
Energy loss | 500 < p < 520
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 397.0± 0.7
σ = 35.7± 0.7
σ/μ = 9.00± 0.20 %
WF
μ = 407.2± 0.7
σ = 37.8± 0.7
σ/μ = 9.29± 0.19 %



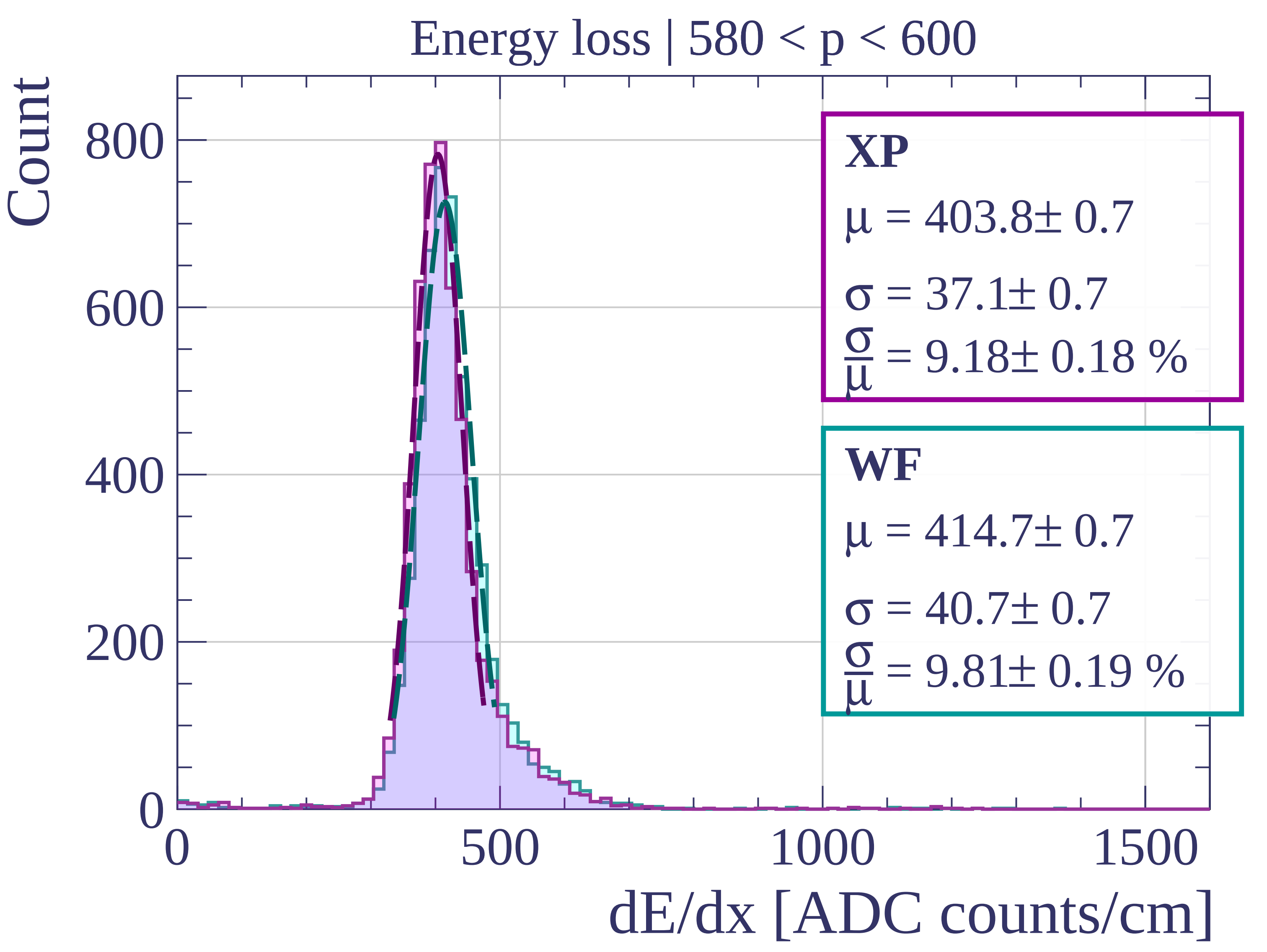
Energy loss | 520 < p < 540
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 398.0± 0.6
σ = 35.8± 0.6
σ/μ = 8.98± 0.16 %
WF
μ = 407.6± 0.7
σ = 38.0± 0.6
σ/μ = 9.32± 0.16 %



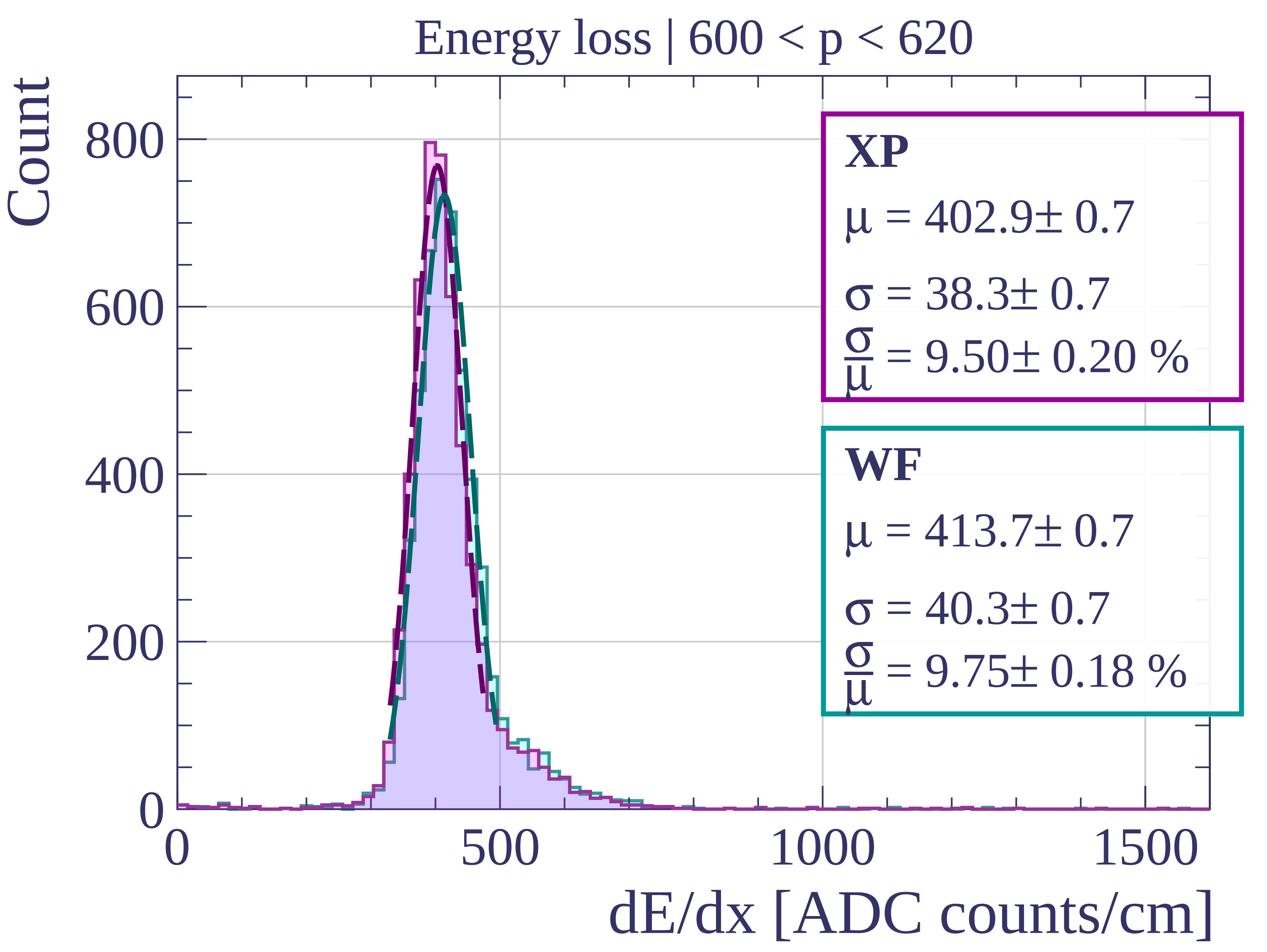
Energy loss | 540 < p < 560
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 400.7± 0.6
σ = 37.8± 0.6
σ/μ = 9.43± 0.17 %
WF
μ = 411.2± 0.7
σ = 40.3± 0.6
σ/μ = 9.81± 0.16 %



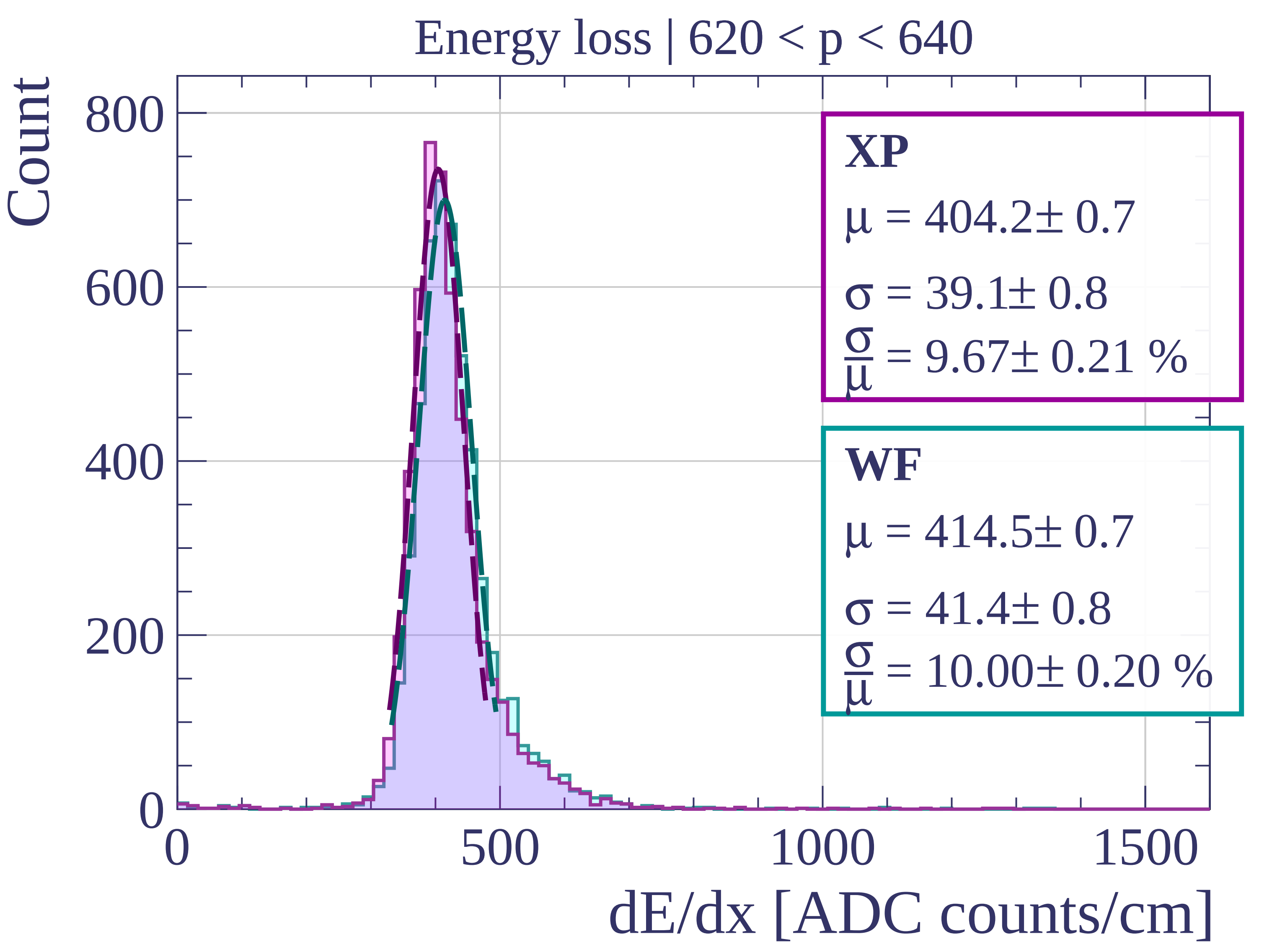
Energy loss | 560 < p < 580
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 402.1± 0.7
σ = 38.4± 0.6
σ/μ = 9.55± 0.17 %
WF
μ = 411.8± 0.7
σ = 41.4± 0.7
σ/μ = 10.04± 0.18 %

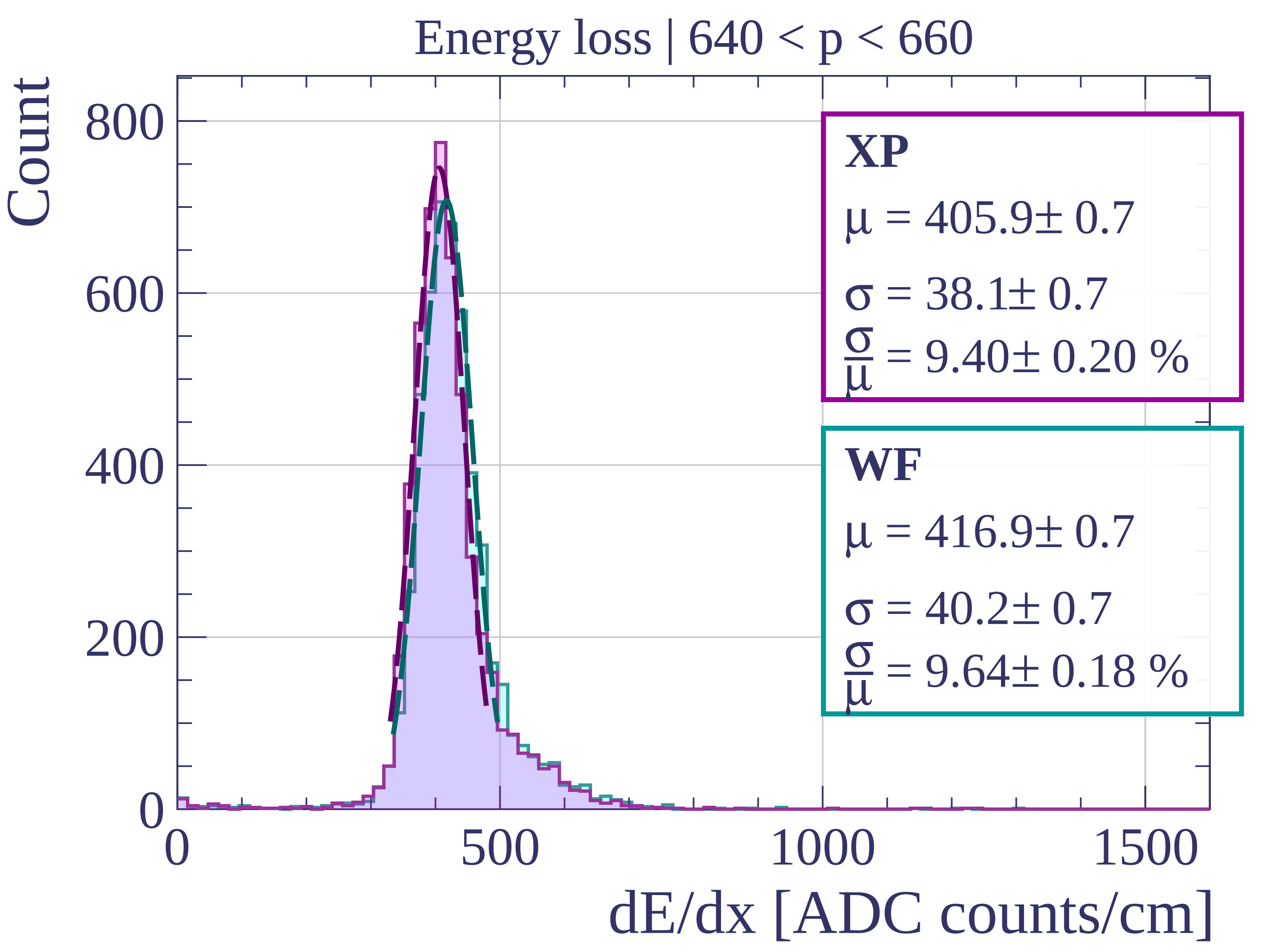


Energy loss | 580 < p < 600
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 403.8± 0.7
σ = 37.1± 0.7
σ/μ = 9.18± 0.18 %
WF
μ = 414.7± 0.7
σ = 40.7± 0.7
σ/μ = 9.81± 0.19 %

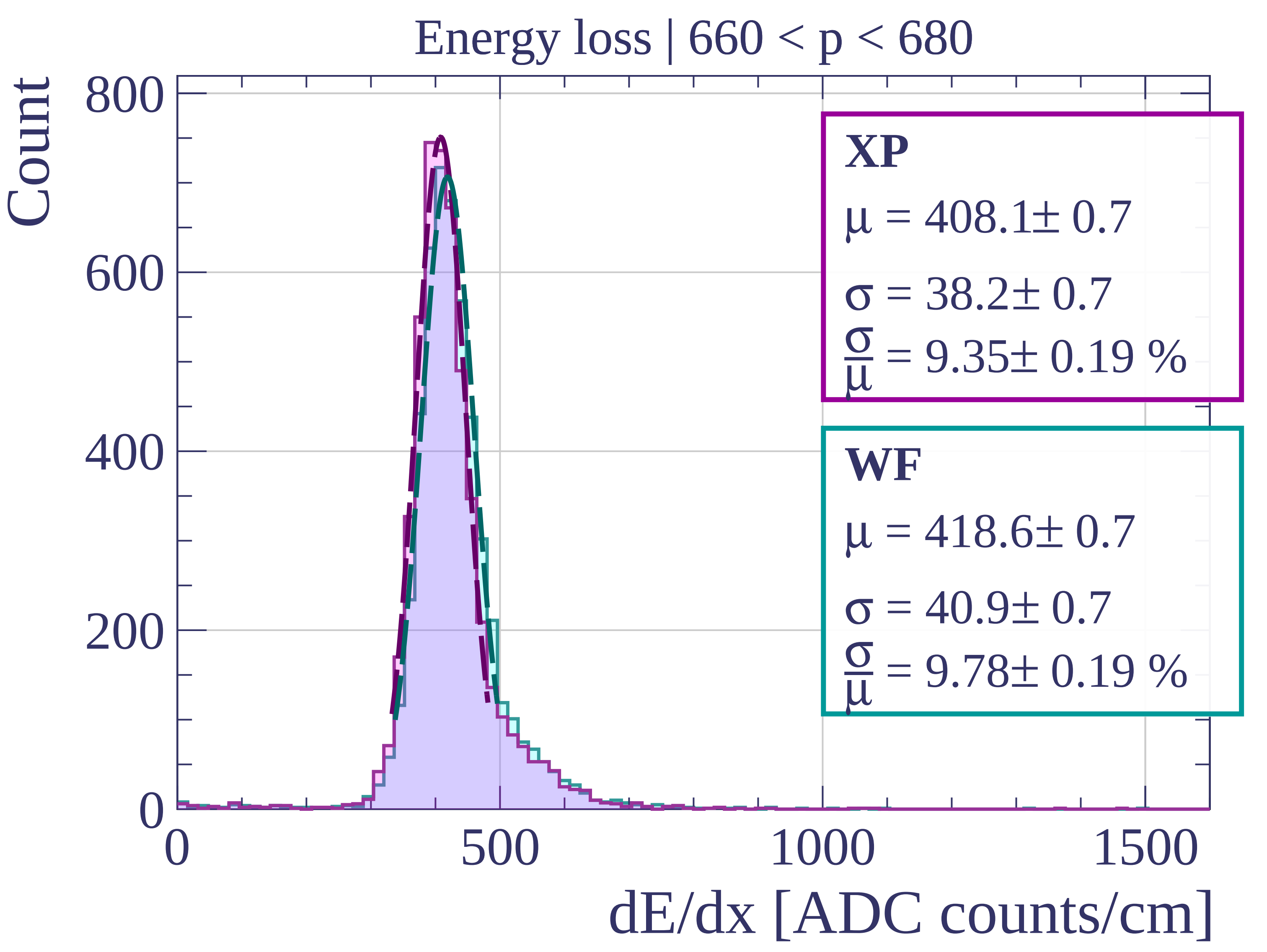


Energy loss | 600 < p < 620
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 402.9± 0.7
σ = 38.3± 0.7
σ/μ = 9.50± 0.20 %
WF
μ = 413.7± 0.7
σ = 40.3± 0.7
σ/μ = 9.75± 0.18 %

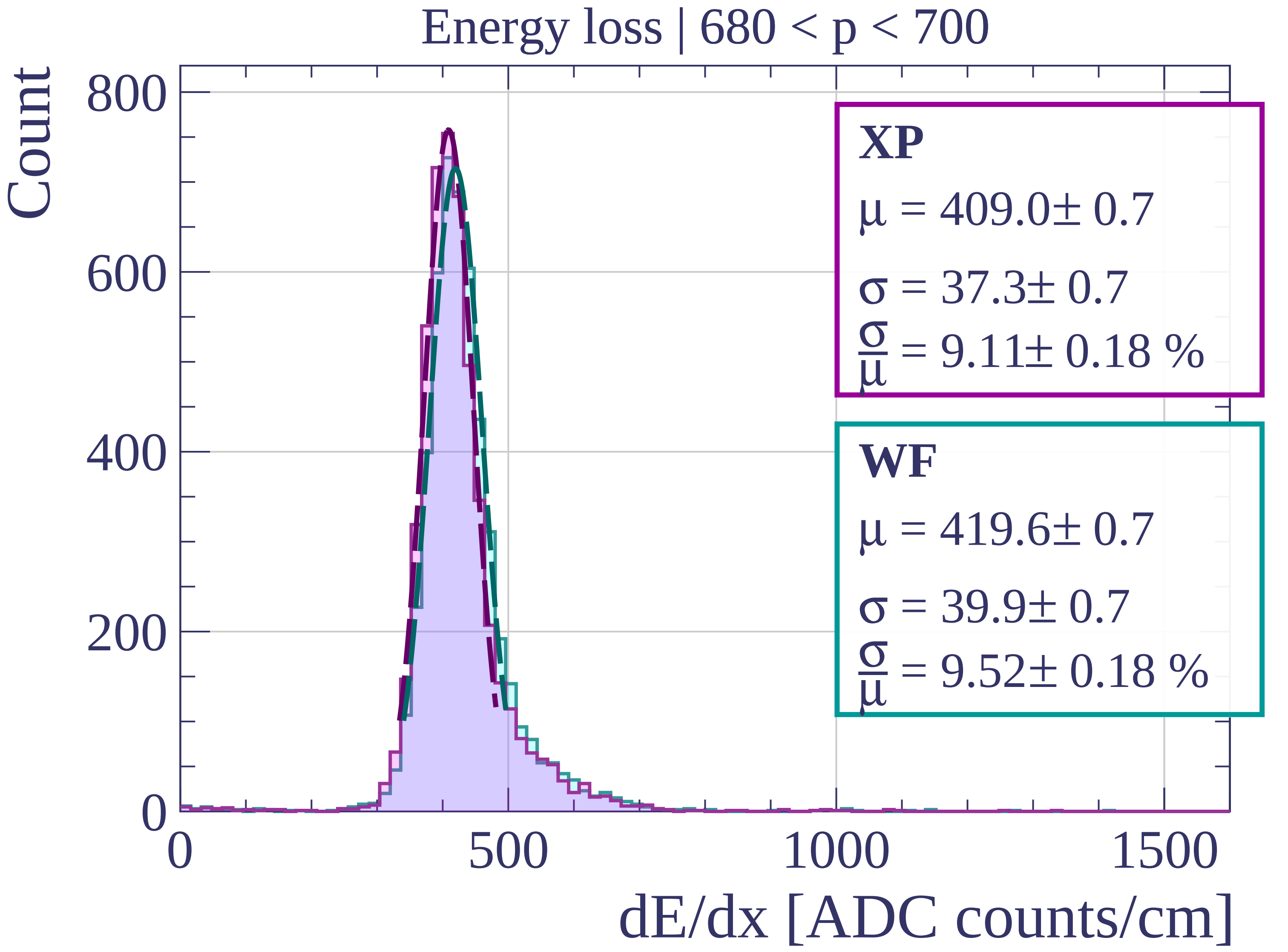

Energy loss | 620 < p < 640
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
μ = 404.2± 0.7
σ = 39.1± 0.8
σ/μ = 9.67± 0.21 %
WF
μ = 414.5± 0.7
σ = 41.4± 0.8
σ/μ = 10.00± 0.20 %

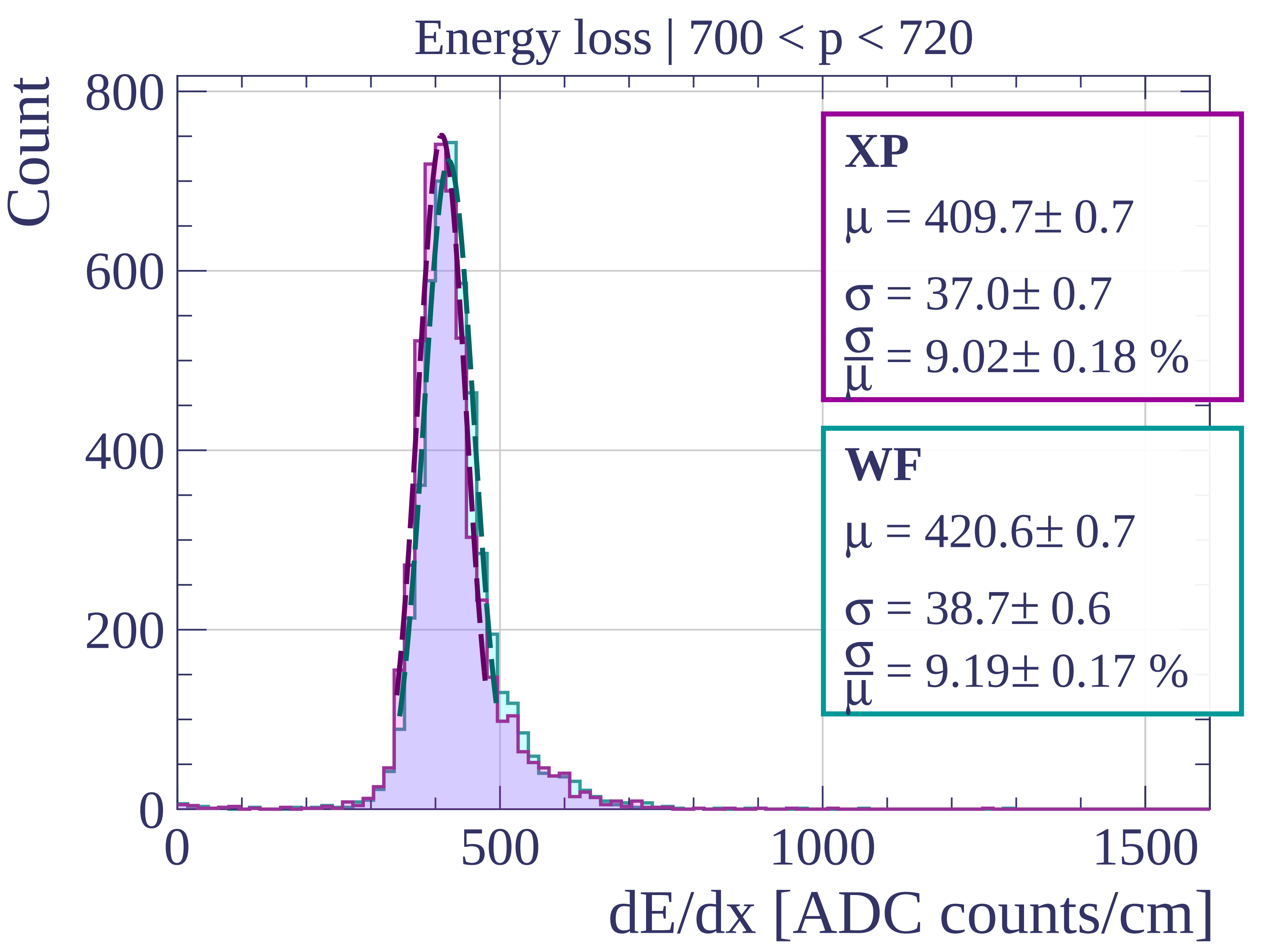


Energy loss | 640 < p < 660
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 405.9± 0.7
σ = 38.1± 0.7
σ/μ = 9.40± 0.20 %
WF
μ = 416.9± 0.7
σ = 40.2± 0.7
σ/μ = 9.64± 0.18 %

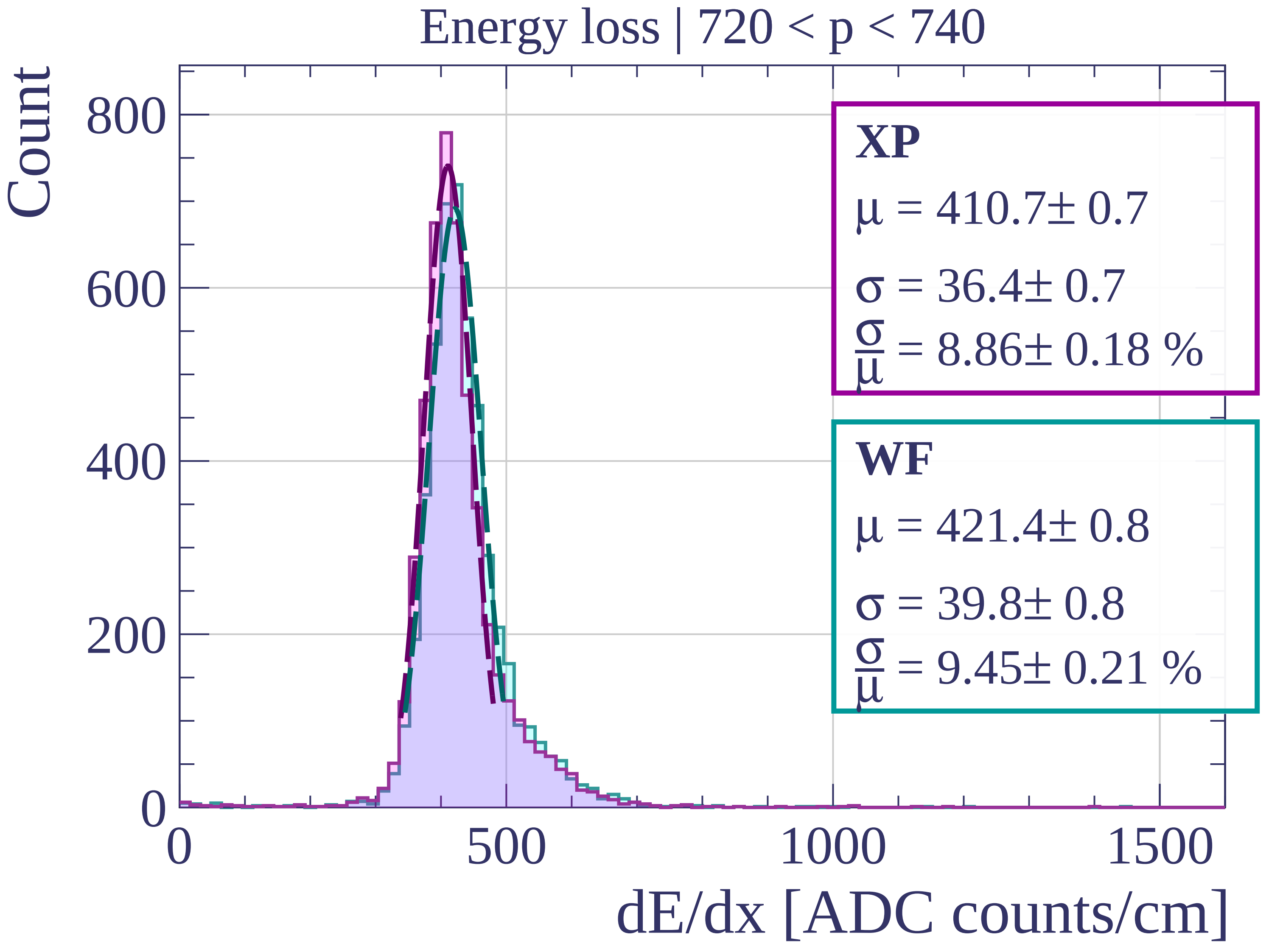


Energy loss | 660 < p < 680
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 408.1± 0.7
σ = 38.2± 0.7
σ/μ = 9.35± 0.19 %
WF
μ = 418.6± 0.7
σ = 40.9± 0.7
σ/μ = 9.78± 0.19 %

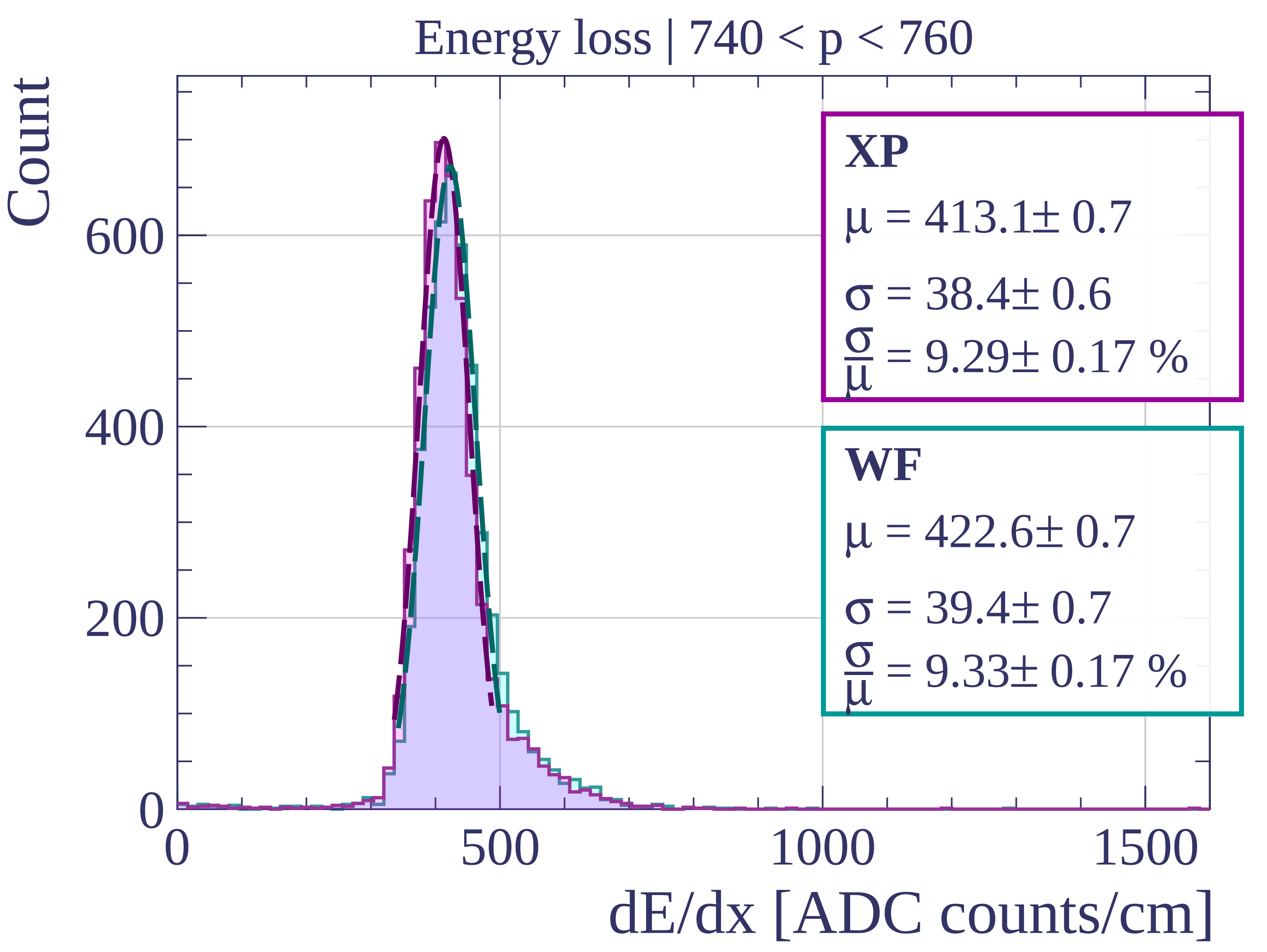

Energy loss | 680 < p < 700
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 409.0± 0.7
σ = 37.3± 0.7
σ/μ = 9.11± 0.18 %
WF
μ = 419.6± 0.7
σ = 39.9± 0.7
σ/μ = 9.52± 0.18 %

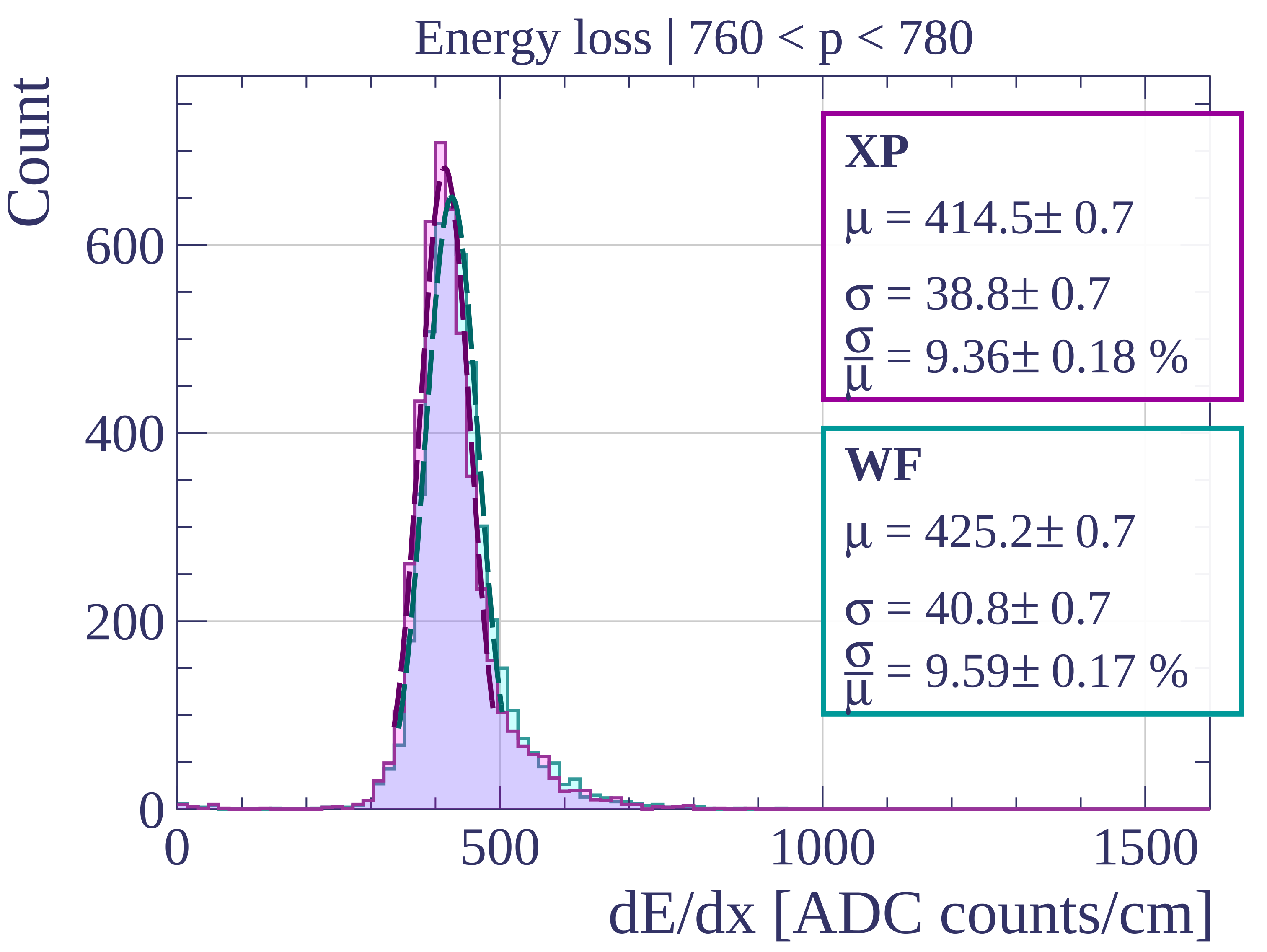


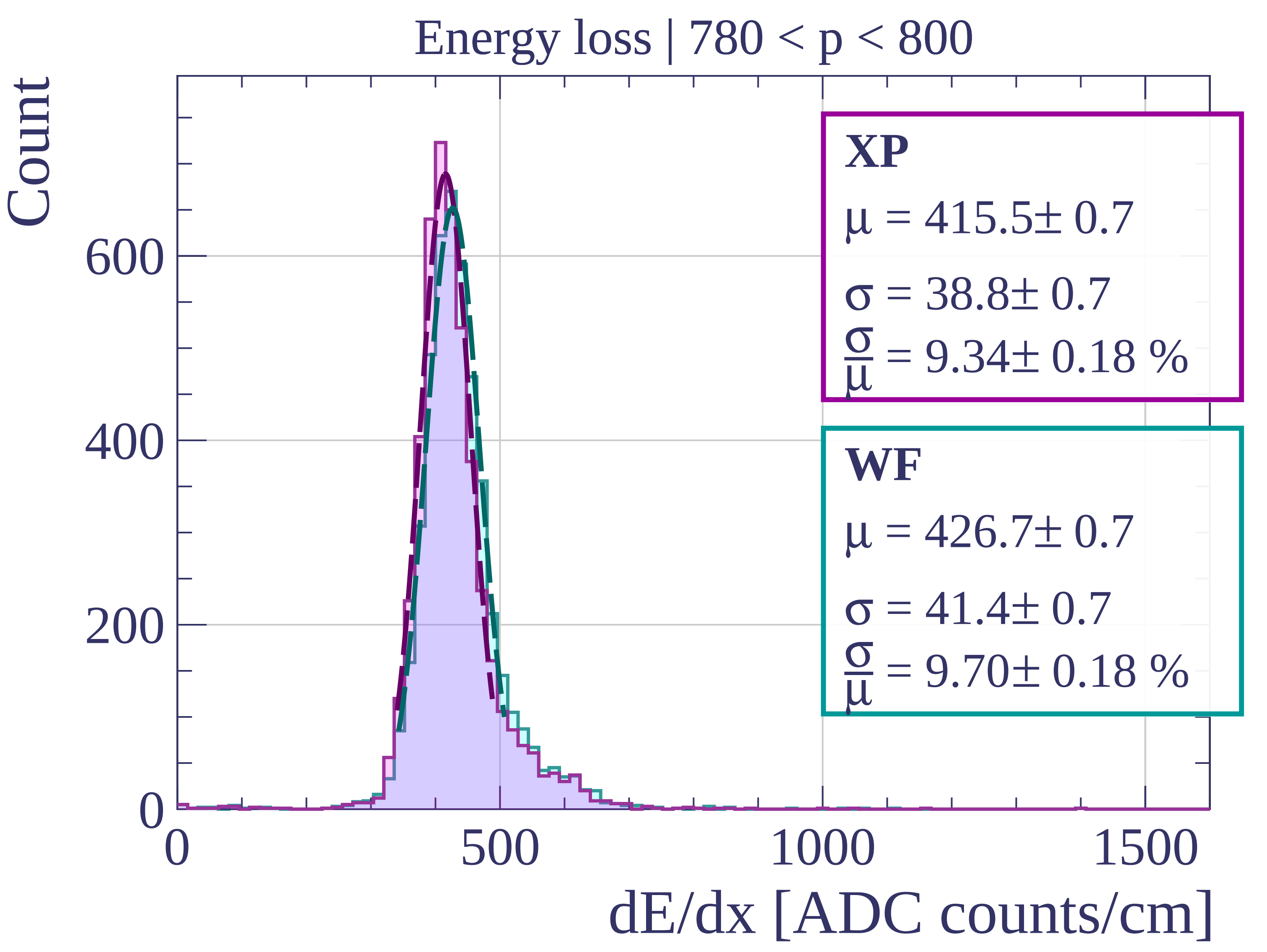
Energy loss | 700 < p < 720
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 409.7± 0.7
σ = 37.0± 0.7
σ/μ = 9.02± 0.18 %
WF
μ = 420.6± 0.7
σ = 38.7± 0.6
σ/μ = 9.19± 0.17 %



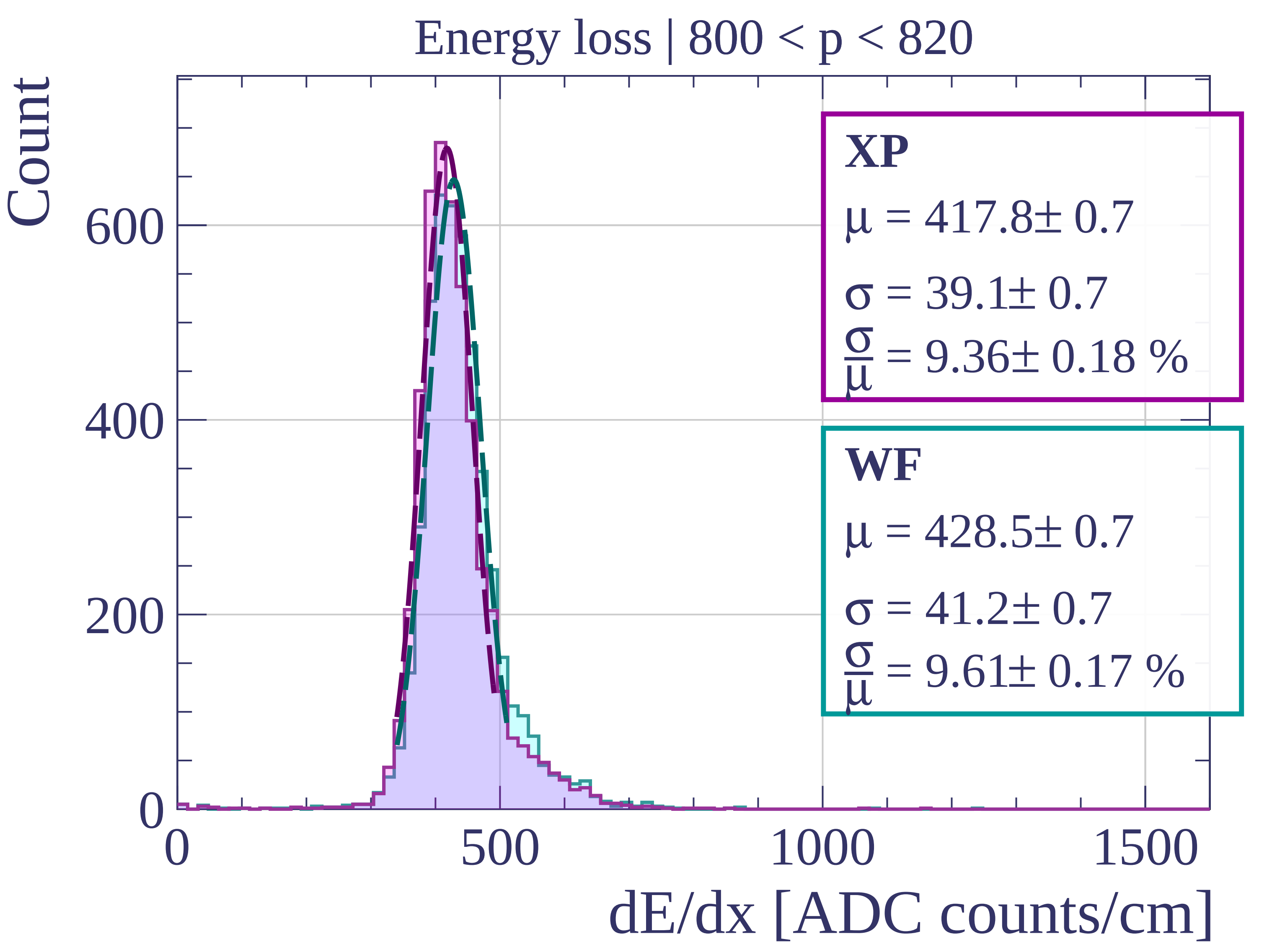
Energy loss | 720 < p < 740
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 410.7± 0.7
σ = 36.4± 0.7
σ/μ = 8.86± 0.18 %
WF
μ = 421.4± 0.8
σ = 39.8± 0.8
σ/μ = 9.45± 0.21 %


Energy loss | 740 < p < 760
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 413.1± 0.7
σ = 38.4± 0.6
σ/μ = 9.29± 0.17 %
WF
μ = 422.6± 0.7
σ = 39.4± 0.7
σ/μ = 9.33± 0.17 %

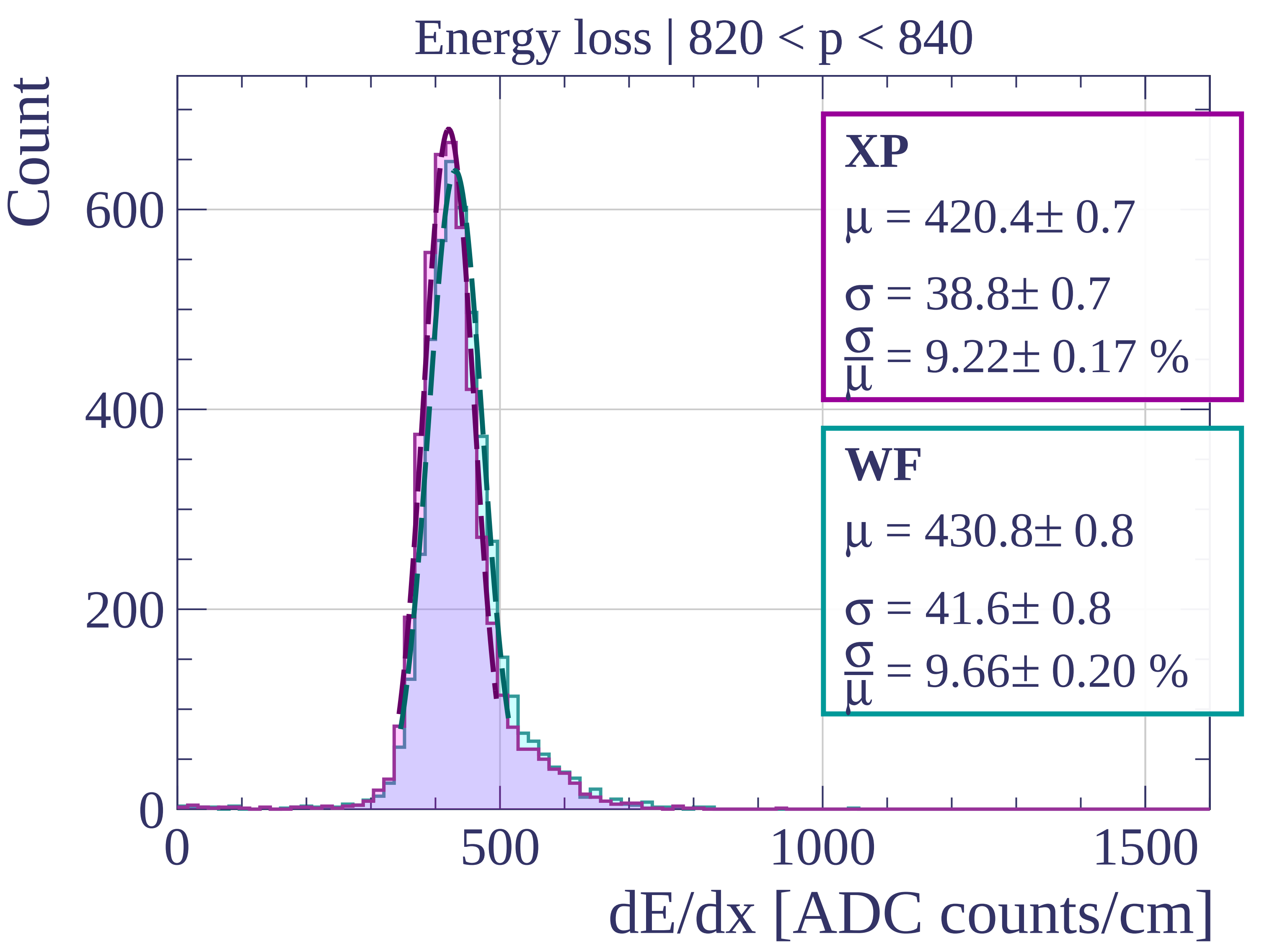

Energy loss | 760 < p < 780
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 414.5± 0.7
σ = 38.8± 0.7
σ/μ = 9.36± 0.18 %
WF
μ = 425.2± 0.7
σ = 40.8± 0.7
σ/μ = 9.59± 0.17 %

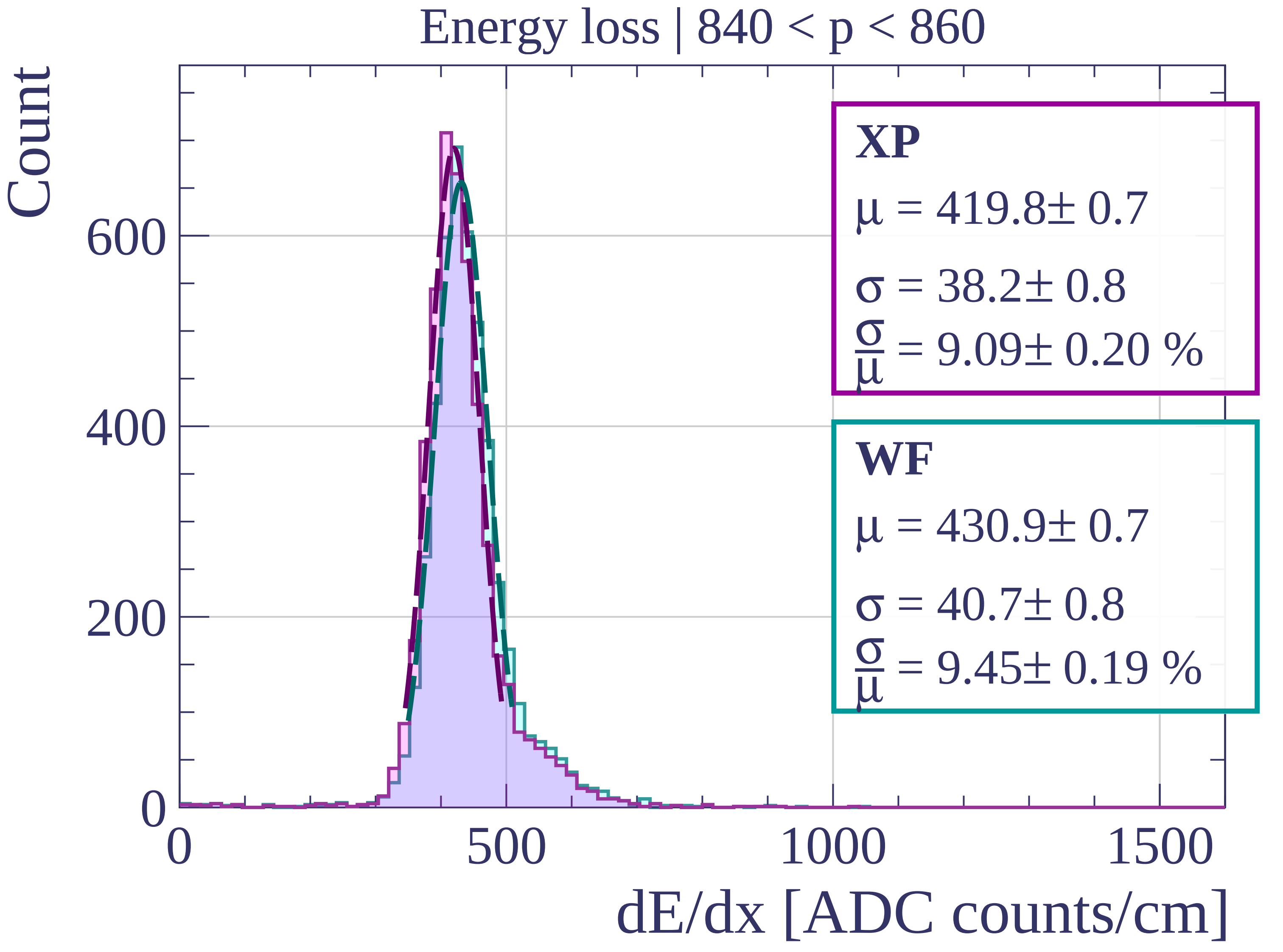


Energy loss | 780 < p < 800
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 415.5± 0.7
σ = 38.8± 0.7
σ/μ = 9.34± 0.18 %
WF
μ = 426.7± 0.7
σ = 41.4± 0.7
σ/μ = 9.70± 0.18 %

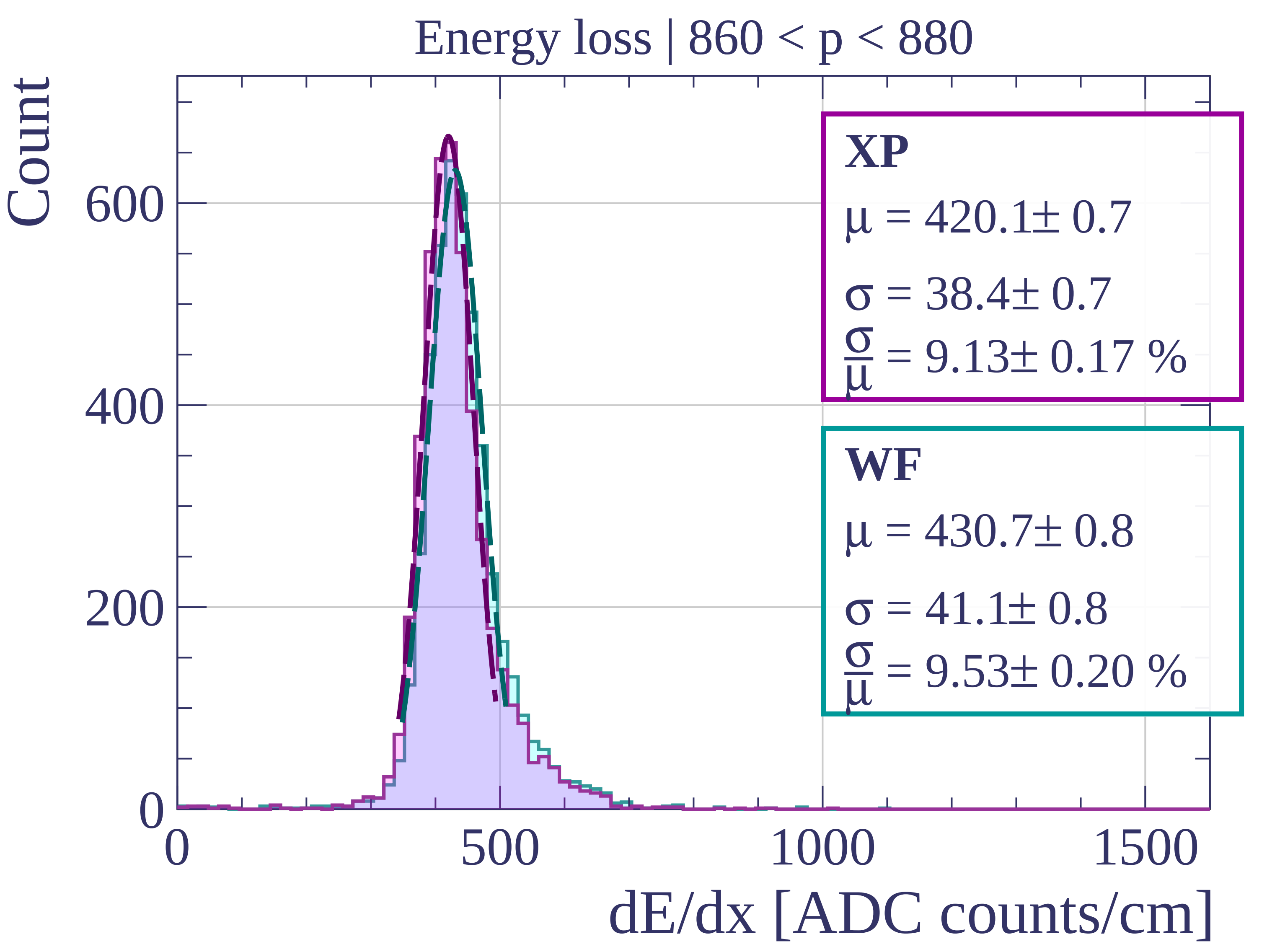


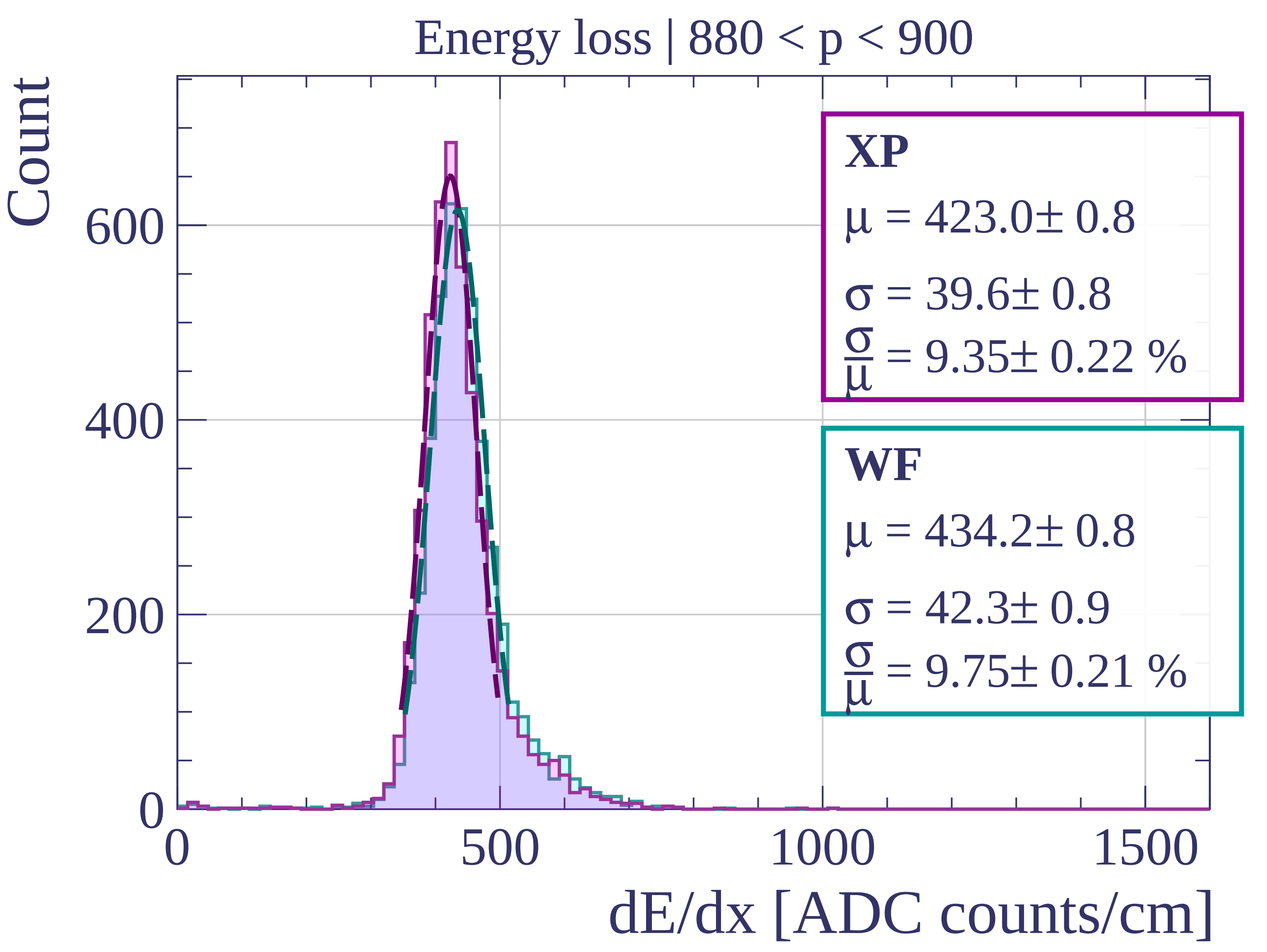
Energy loss | 800 < p < 820
Count
0
200
400
600
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 417.8± 0.7
σ = 39.1± 0.7
σ/μ = 9.36± 0.18 %
WF
μ = 428.5± 0.7
σ = 41.2± 0.7
σ/μ = 9.61± 0.17 %


Energy loss | 820 < p < 840
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 420.4± 0.7
σ = 38.8± 0.7
σ/μ = 9.22± 0.17 %
WF
μ = 430.8± 0.8
σ = 41.6± 0.8
σ/μ = 9.66± 0.20 %

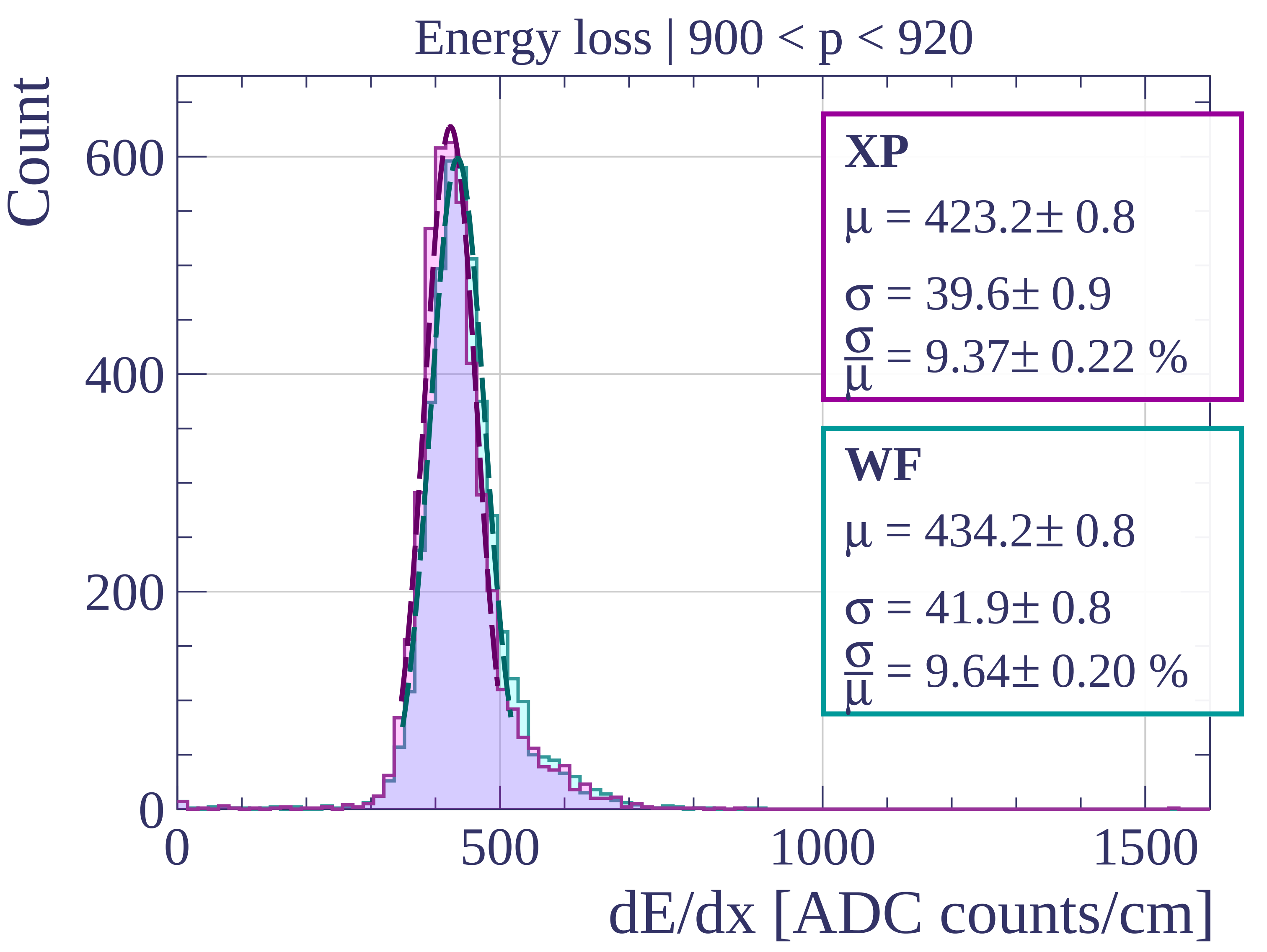


Energy loss | 840 < p < 860
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 419.8± 0.7
σ = 38.2± 0.8
σ/μ = 9.09± 0.20 %
WF
μ = 430.9± 0.7
σ = 40.7± 0.8
σ/μ = 9.45± 0.19 %

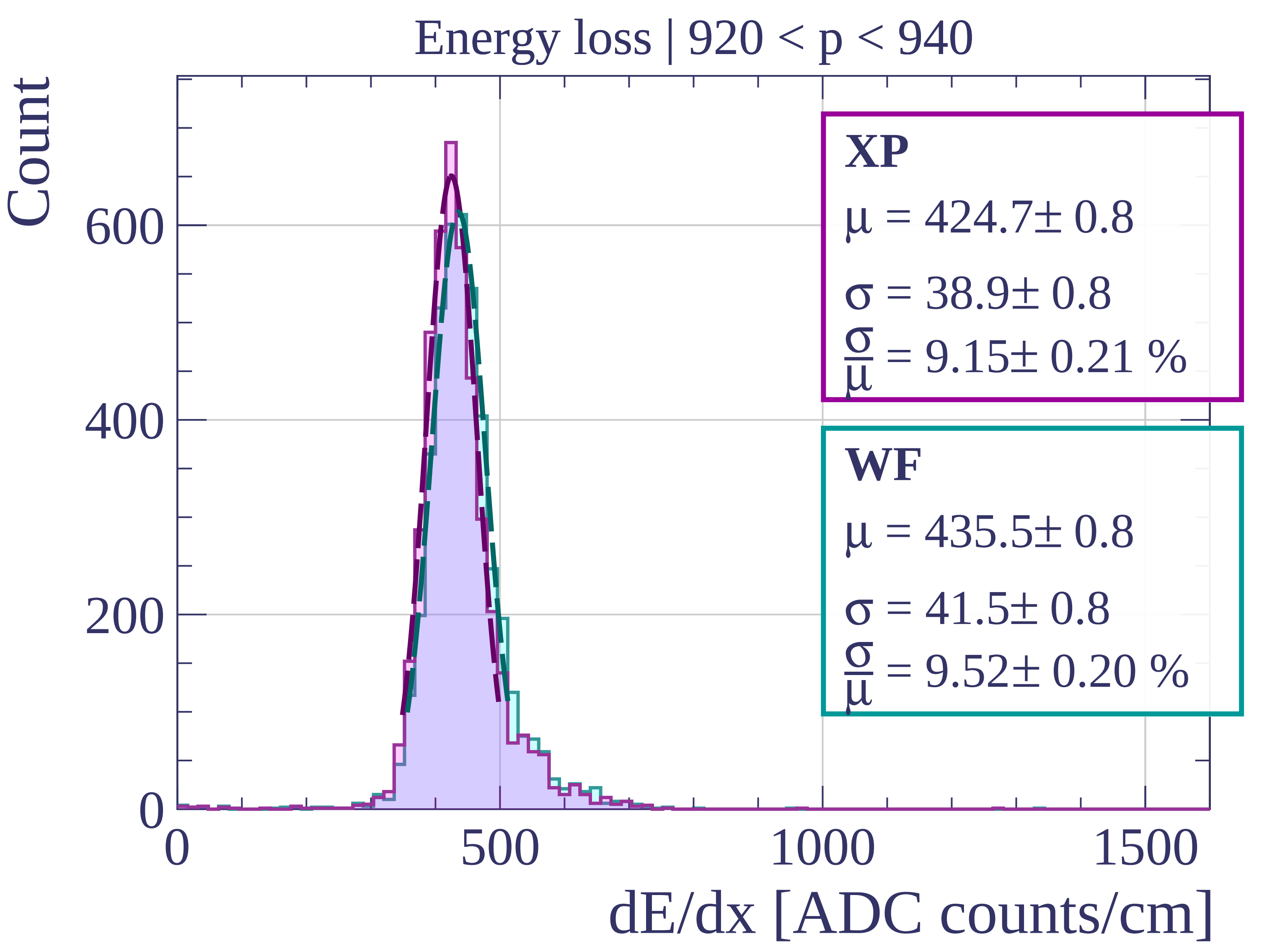

Energy loss | 860 < p < 880
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 420.1± 0.7
σ = 38.4± 0.7
σ/μ = 9.13± 0.17 %
WF
μ = 430.7± 0.8
σ = 41.1± 0.8
σ/μ = 9.53± 0.20 %



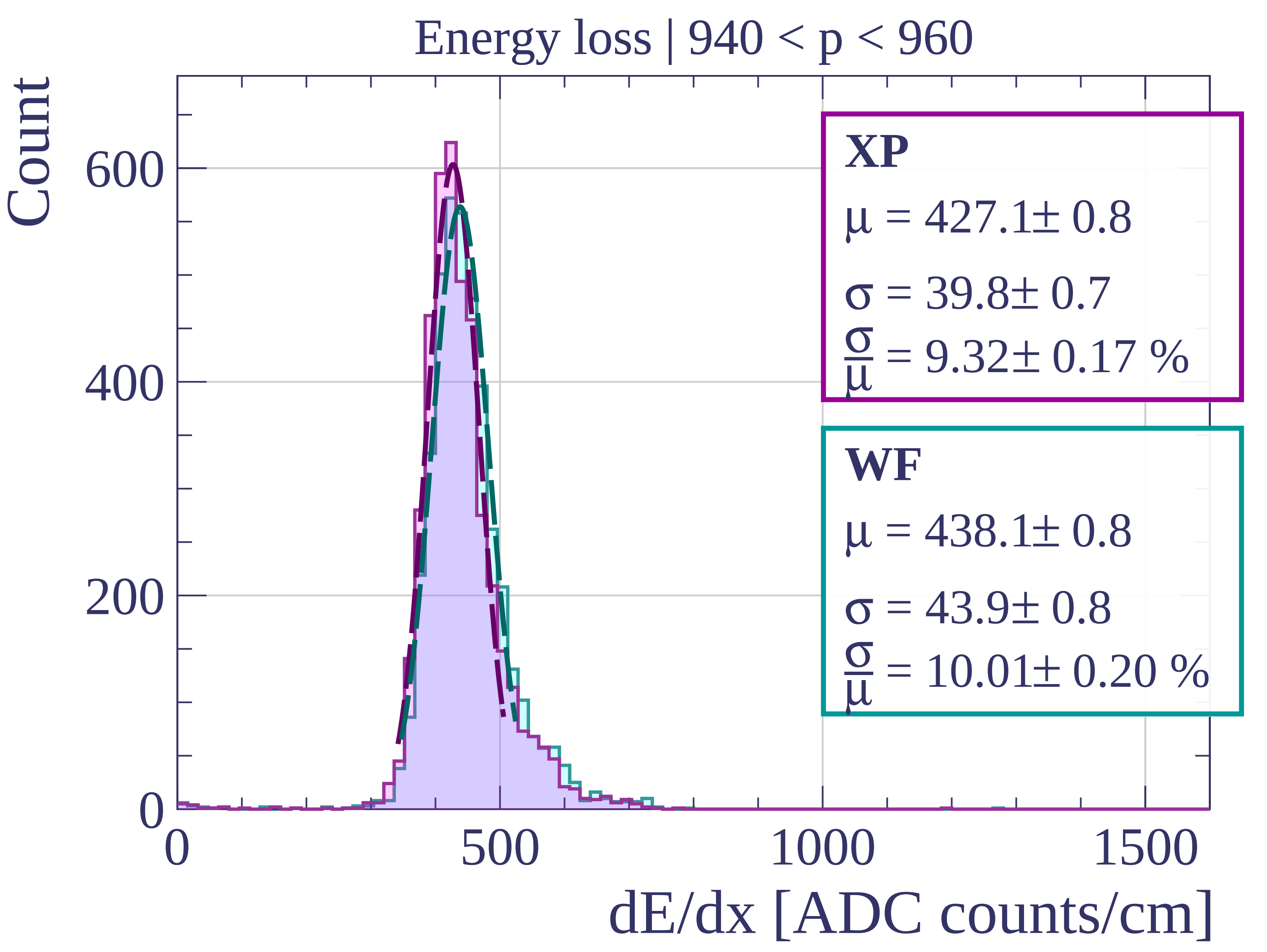
Energy loss | 880 < p < 900
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 423.0± 0.8
σ = 39.6± 0.8
σ/μ = 9.35± 0.22 %
WF
μ = 434.2± 0.8
σ = 42.3± 0.9
σ/μ = 9.75± 0.21 %



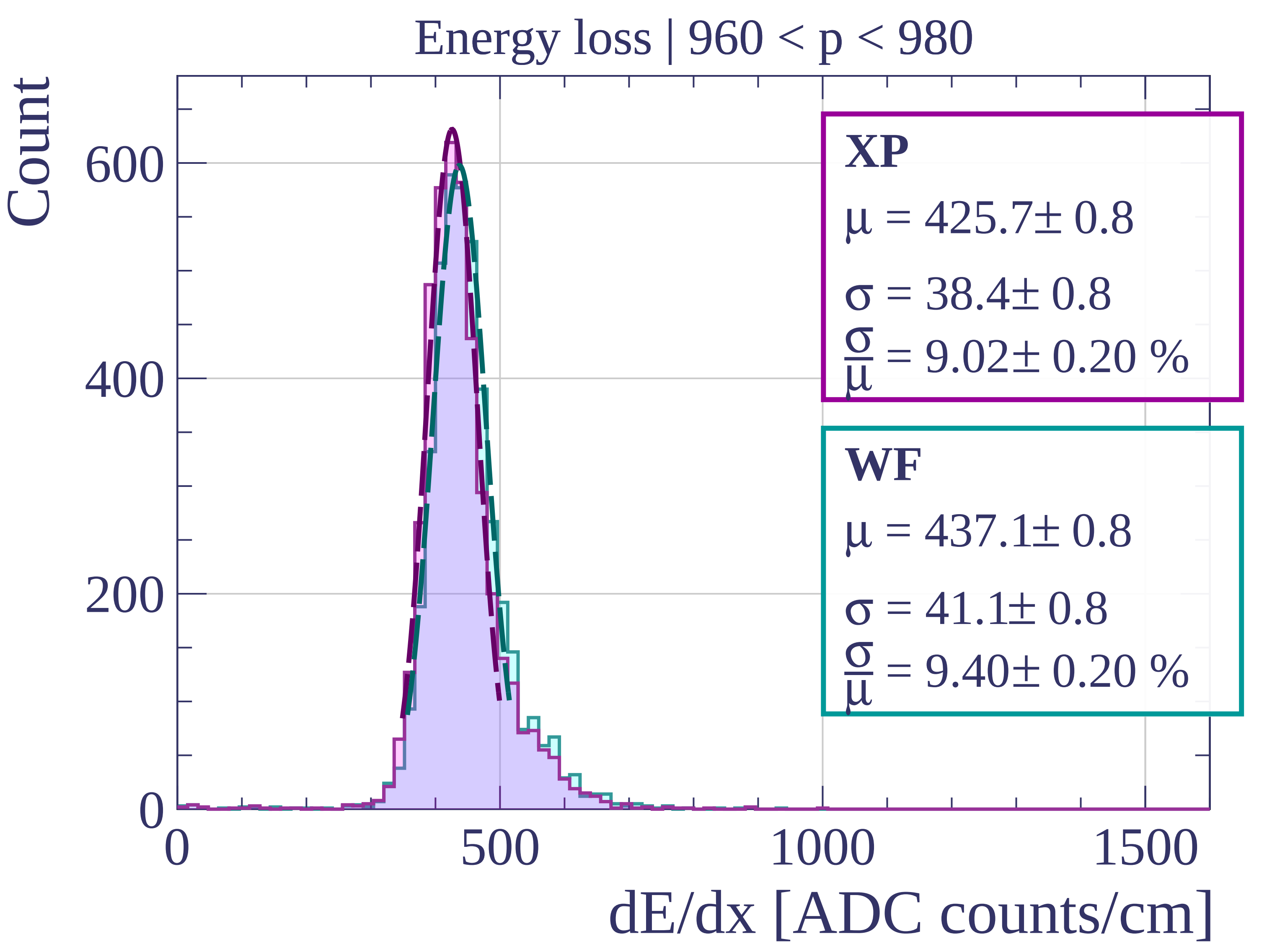
Energy loss | 900 < p < 920
Count
0
200
400
600
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 423.2± 0.8
σ = 39.6± 0.9
σ/μ = 9.37± 0.22 %
WF
μ = 434.2± 0.8
σ = 41.9± 0.8
σ/μ = 9.64± 0.20 %



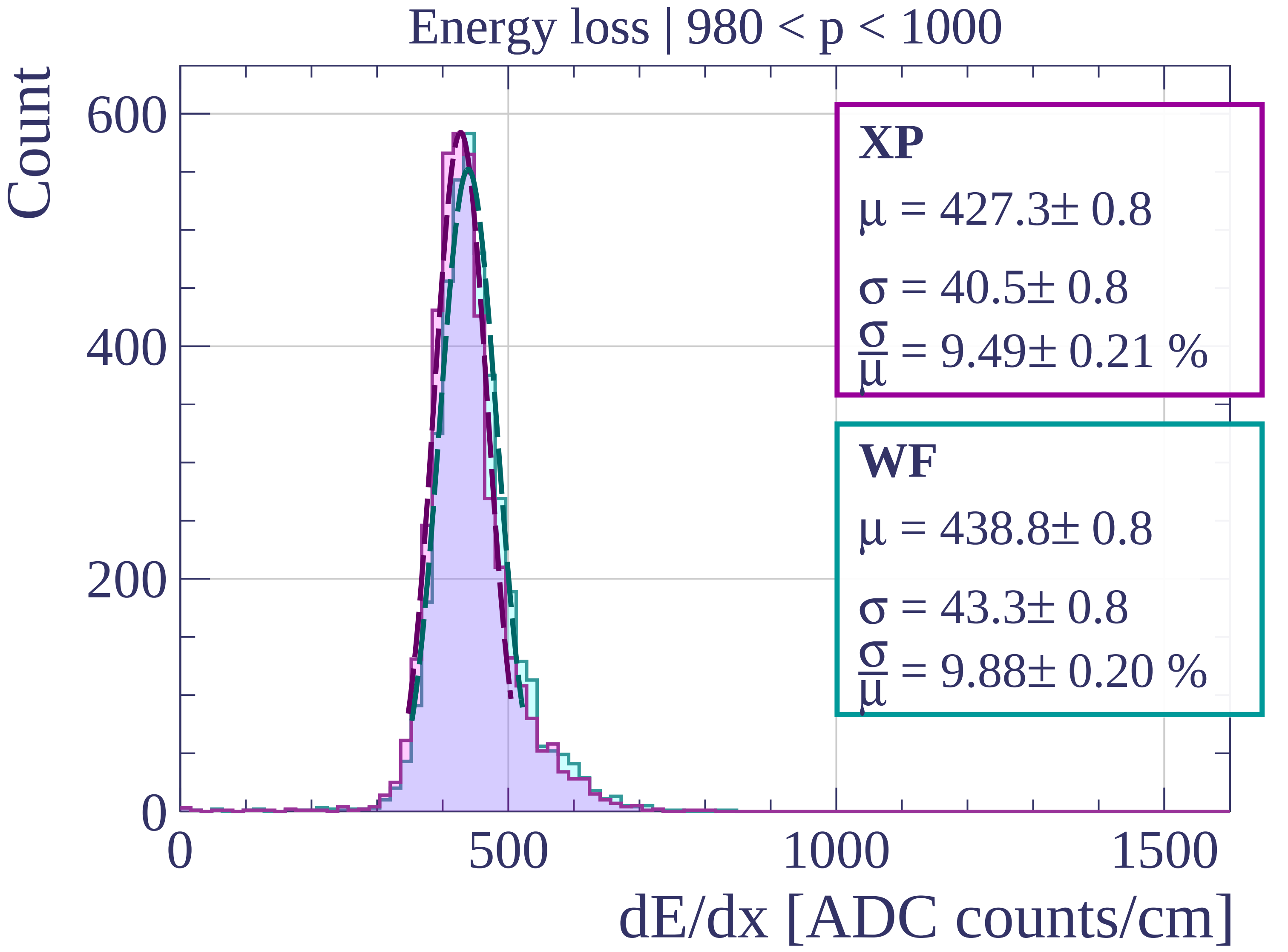
Energy loss | 920 < p < 940
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 424.7± 0.8
σ = 38.9± 0.8
σ/μ = 9.15± 0.21 %
WF
μ = 435.5± 0.8
σ = 41.5± 0.8
σ/μ = 9.52± 0.20 %



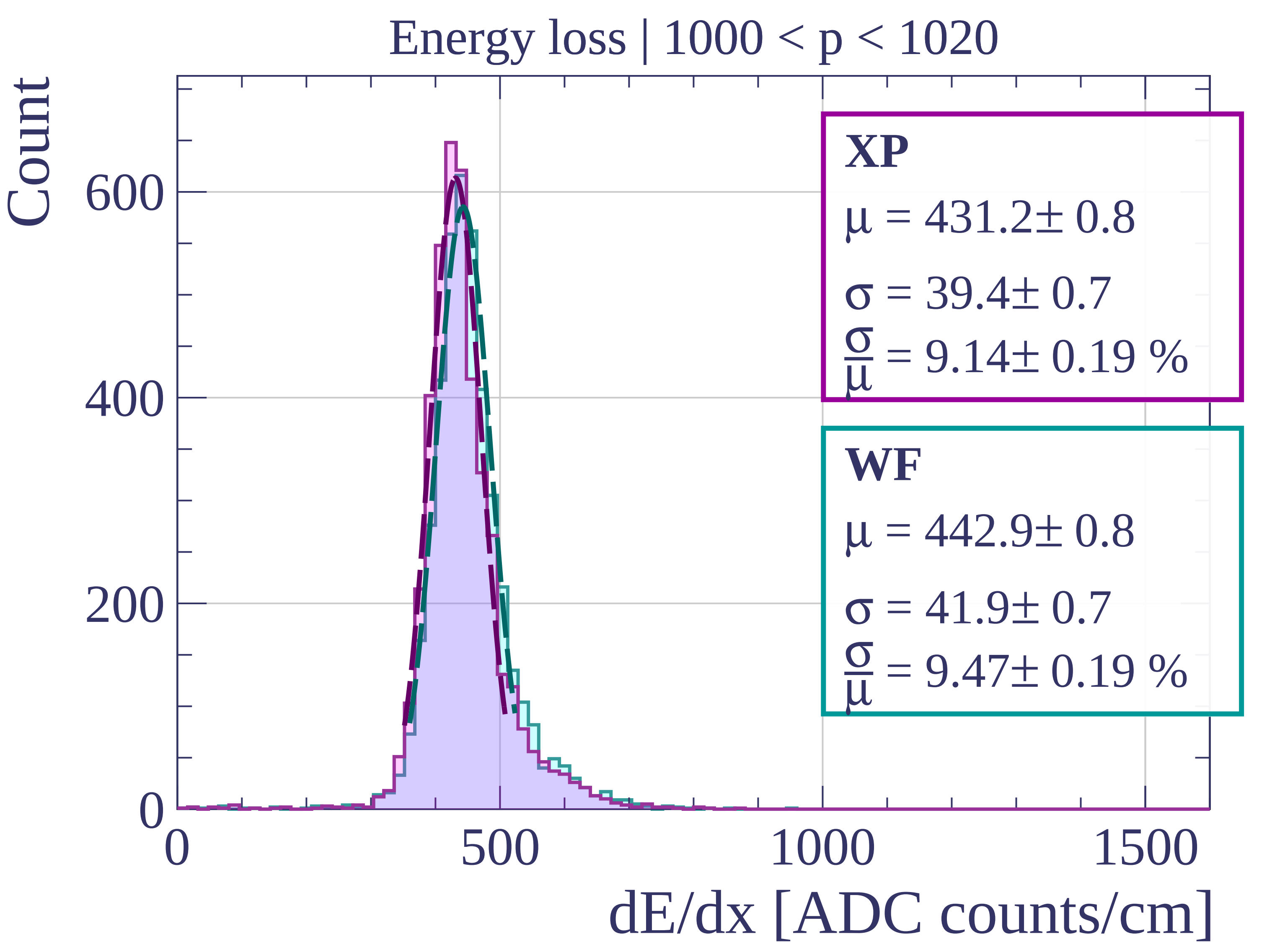
Energy loss | 940 < p < 960
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 427.1± 0.8
σ = 39.8± 0.7
σ/μ = 9.32± 0.17 %
WF
μ = 438.1± 0.8
σ = 43.9± 0.8
σ/μ = 10.01± 0.20 %



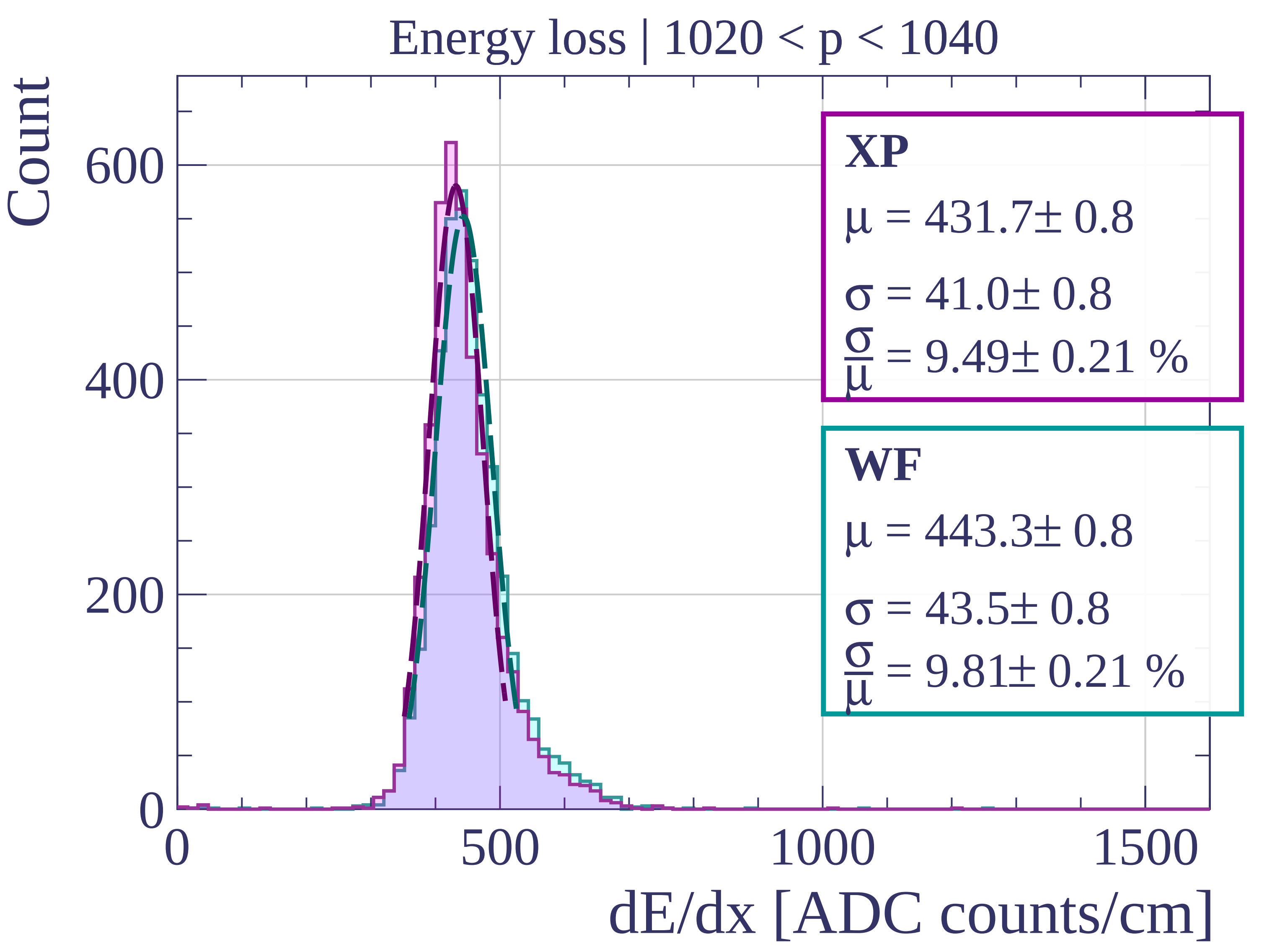
Energy loss | 960 < p < 980
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 425.7± 0.8
σ = 38.4± 0.8
σ/μ = 9.02± 0.20 %
WF
μ = 437.1± 0.8
σ = 41.1± 0.8
σ/μ = 9.40± 0.20 %



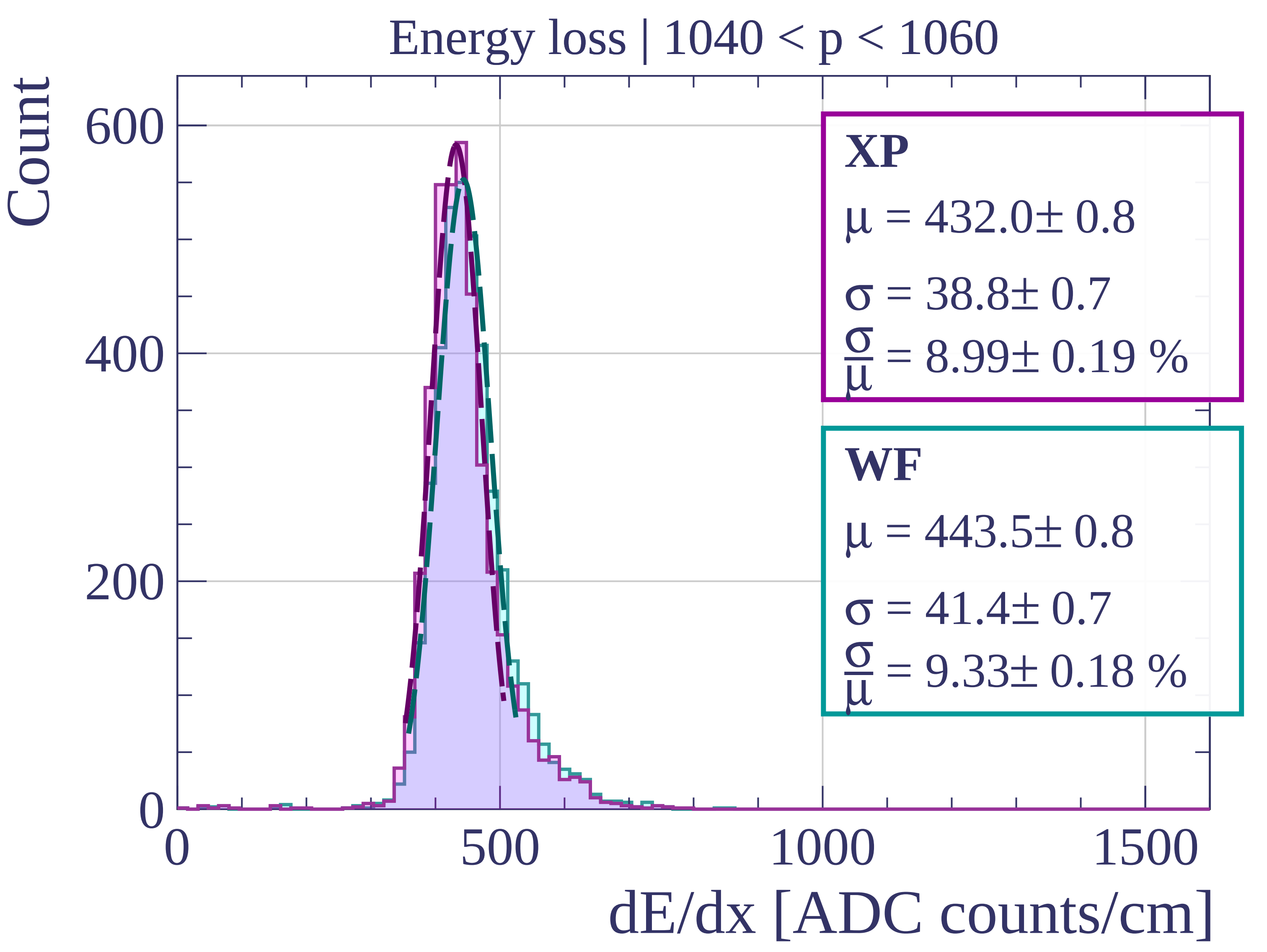
Energy loss | 980 < p < 1000
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 427.3± 0.8
σ = 40.5± 0.8
σ/μ = 9.49± 0.21 %
WF
μ = 438.8± 0.8
σ = 43.3± 0.8
σ/μ = 9.88± 0.20 %



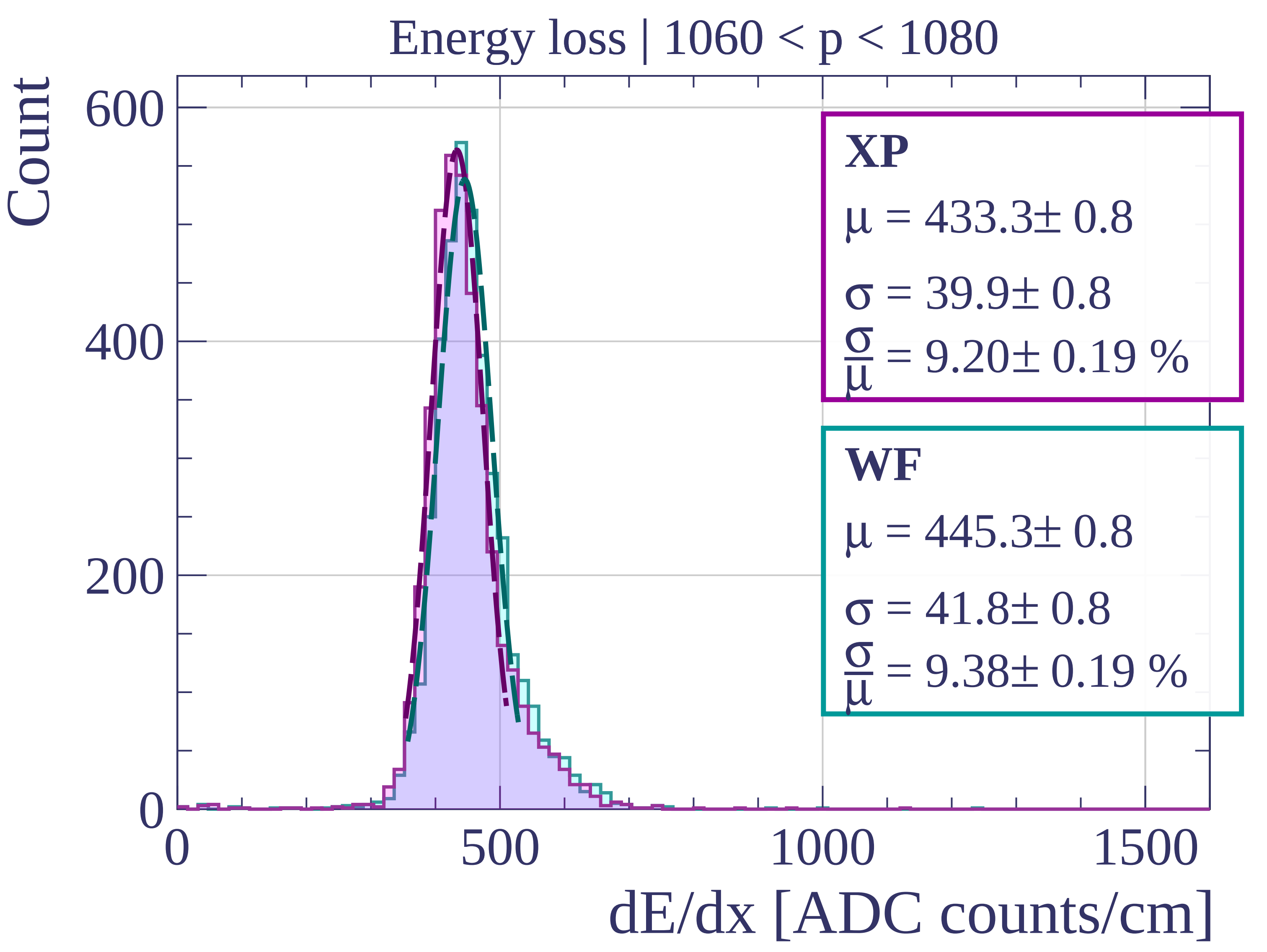
Energy loss | 1000 < p < 1020
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 431.2± 0.8
σ = 39.4± 0.7
σ/μ = 9.14± 0.19 %
WF
μ = 442.9± 0.8
σ = 41.9± 0.7
σ/μ = 9.47± 0.19 %



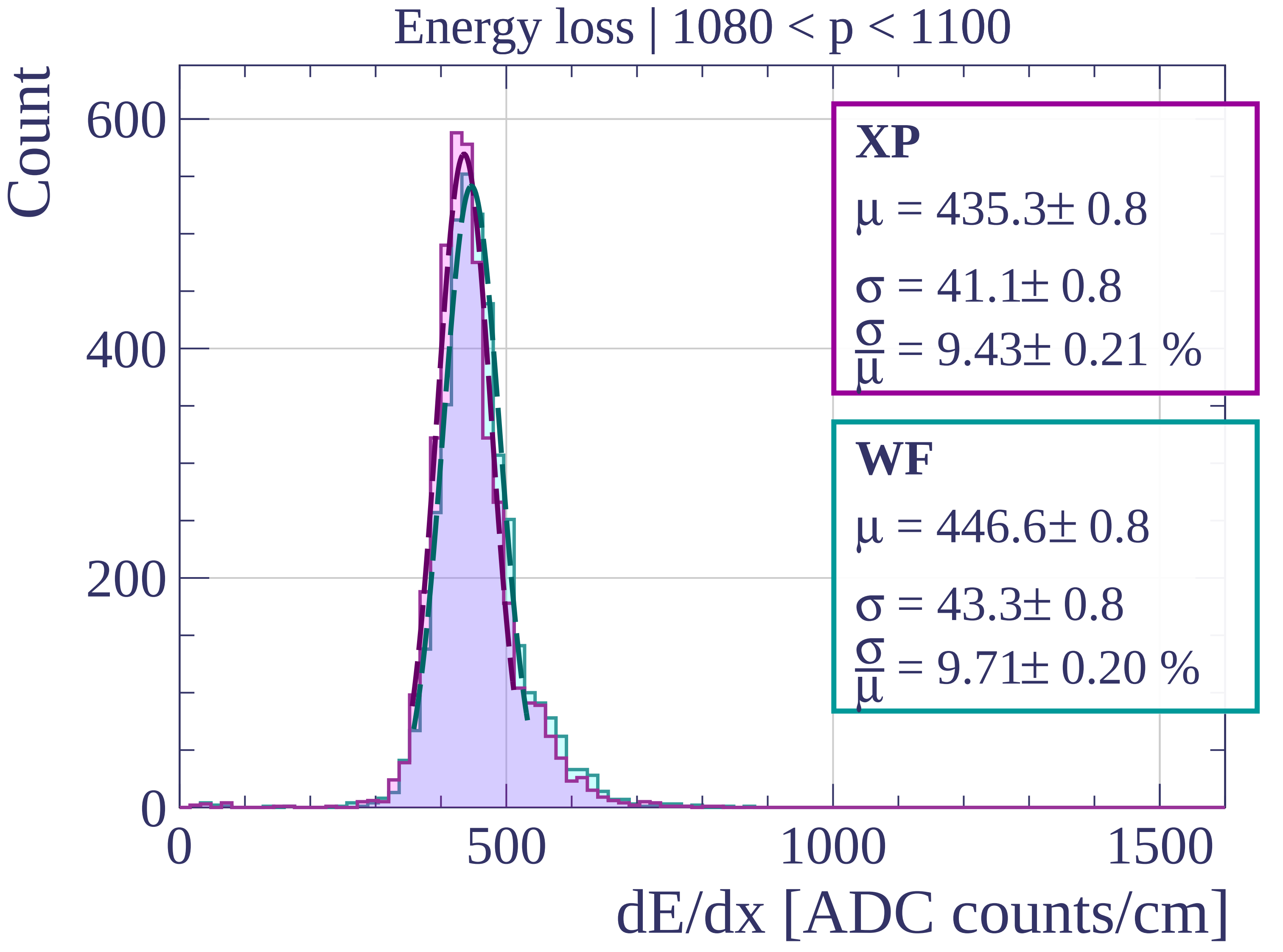
Energy loss | 1020 < p < 1040
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 431.7± 0.8
σ = 41.0± 0.8
σ/μ = 9.49± 0.21 %
WF
μ = 443.3± 0.8
σ = 43.5± 0.8
σ/μ = 9.81± 0.21 %



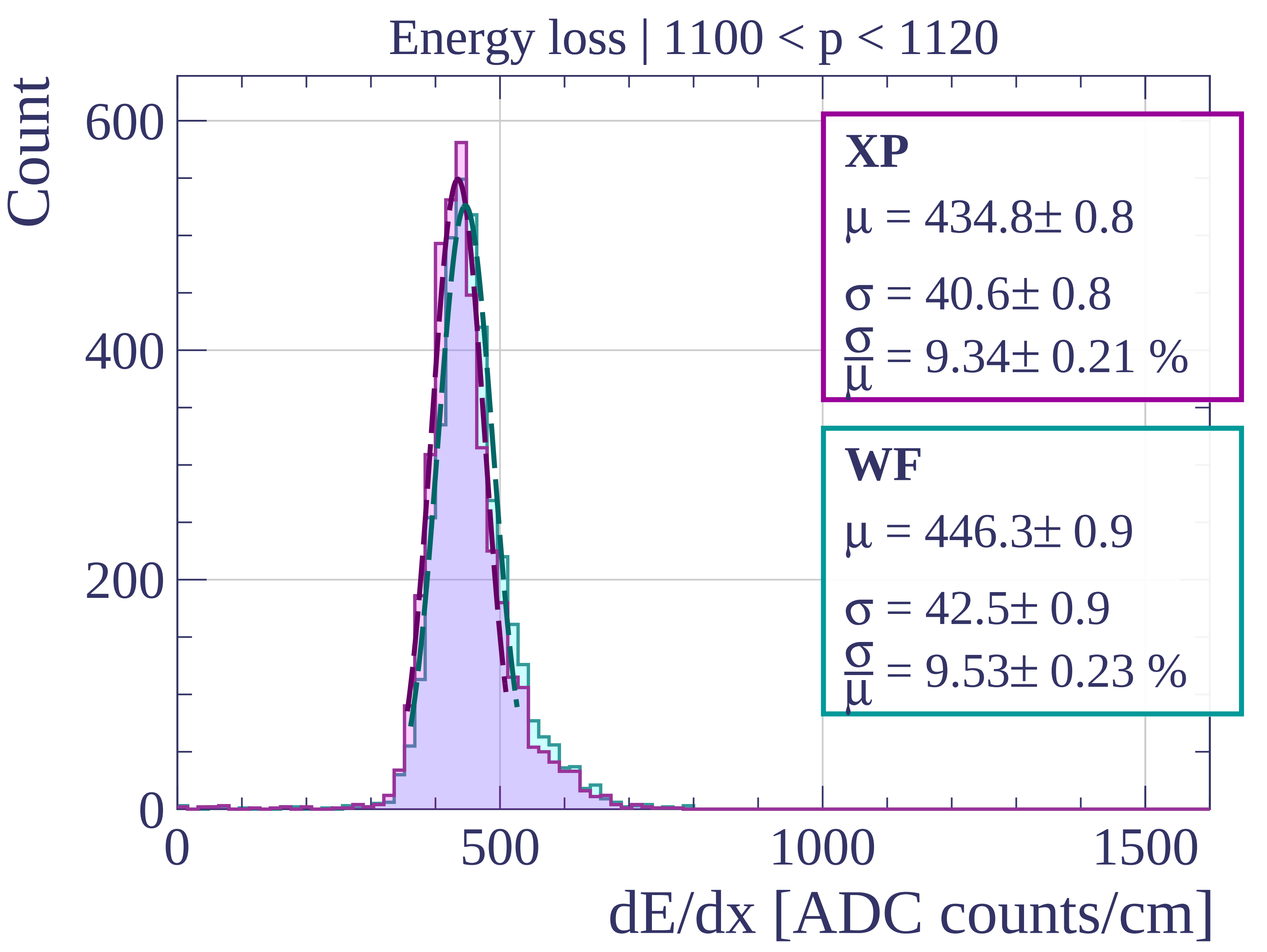
Energy loss | 1040 < p < 1060
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 432.0± 0.8
σ = 38.8± 0.7
σ/μ = 8.99± 0.19 %
WF
μ = 443.5± 0.8
σ = 41.4± 0.7
σ/μ = 9.33± 0.18 %



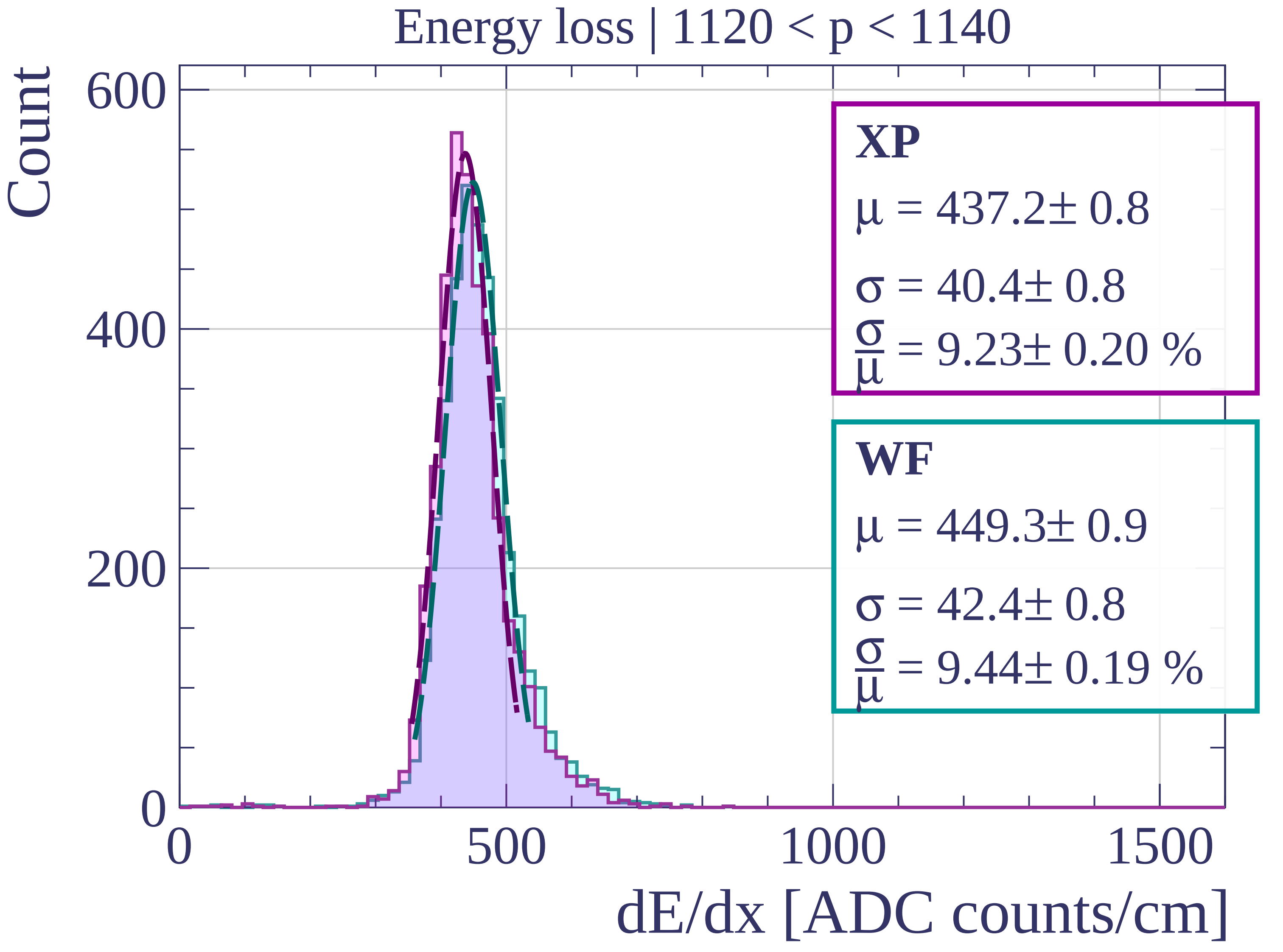
Energy loss | 1060 < p < 1080
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 433.3± 0.8
σ = 39.9± 0.8
σ/μ = 9.20± 0.19 %
WF
μ = 445.3± 0.8
σ = 41.8± 0.8
σ/μ = 9.38± 0.19 %



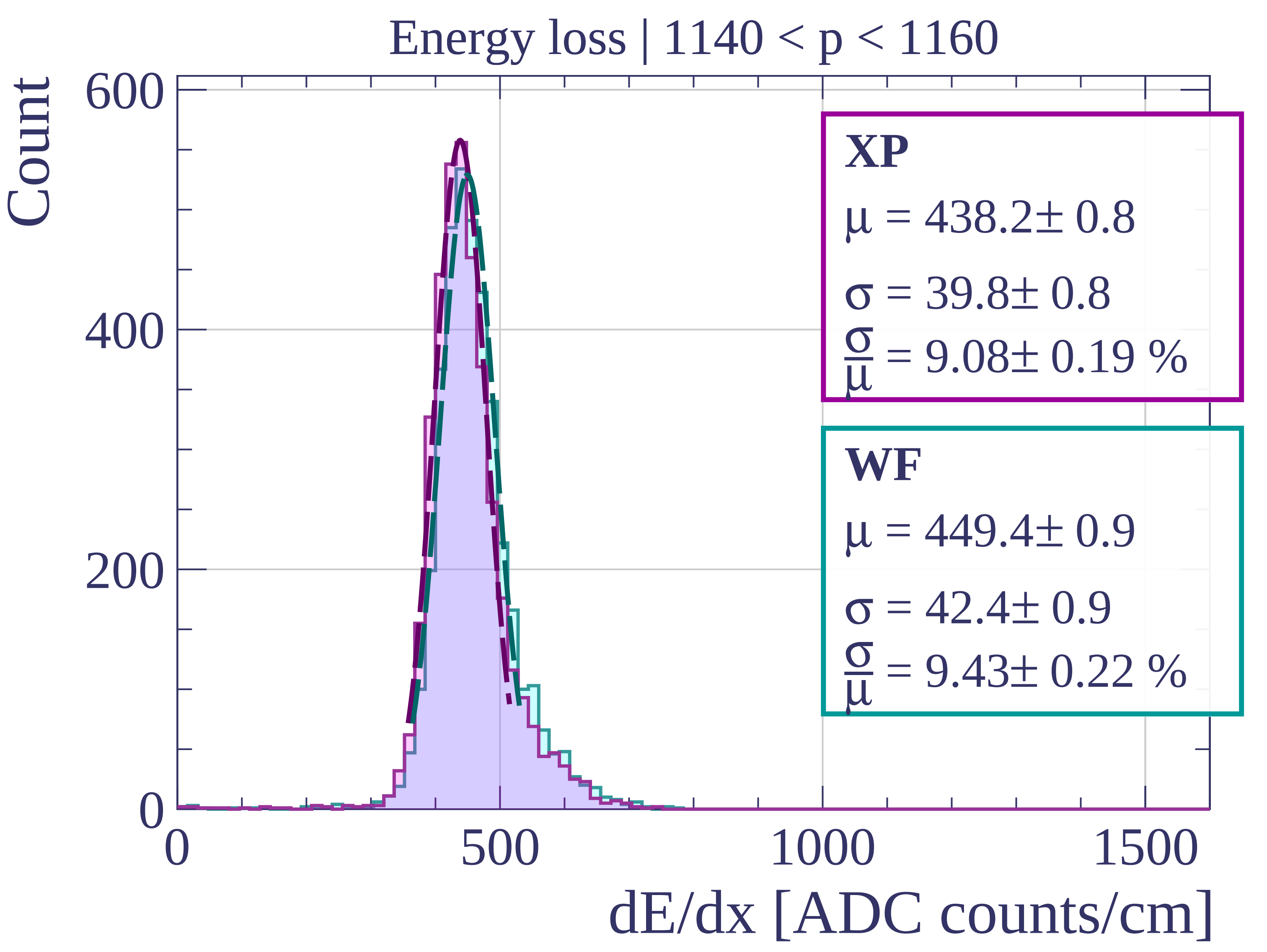
Energy loss | 1080 < p < 1100
Count
0
200
400
600
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 435.3± 0.8
σ = 41.1± 0.8
σ/μ = 9.43± 0.21 %
WF
μ = 446.6± 0.8
σ = 43.3± 0.8
σ/μ = 9.71± 0.20 %



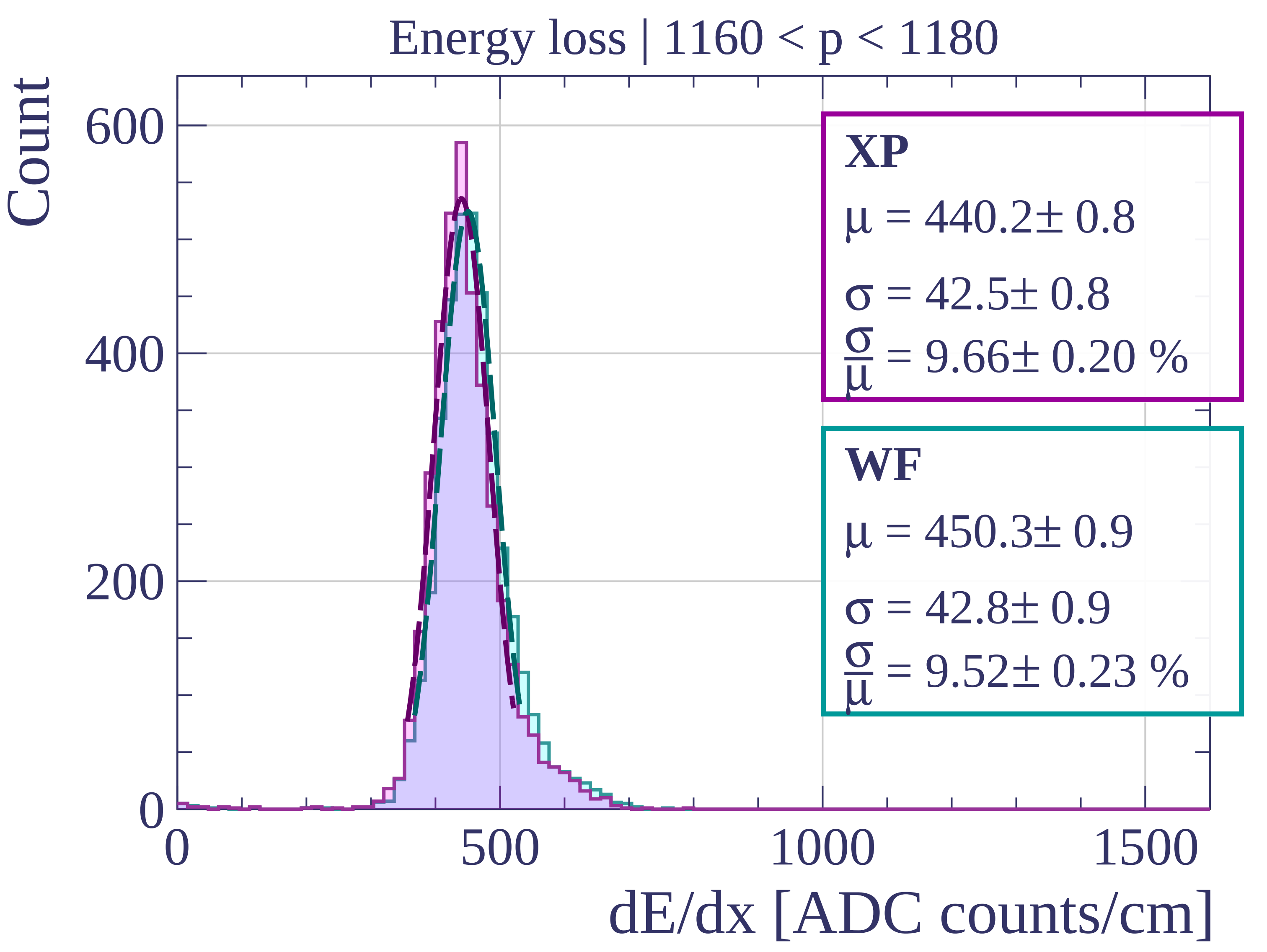
Energy loss | 1100 < p < 1120
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 434.8± 0.8
σ = 40.6± 0.8
σ/μ = 9.34± 0.21 %
WF
μ = 446.3± 0.9
σ = 42.5± 0.9
σ/μ = 9.53± 0.23 %

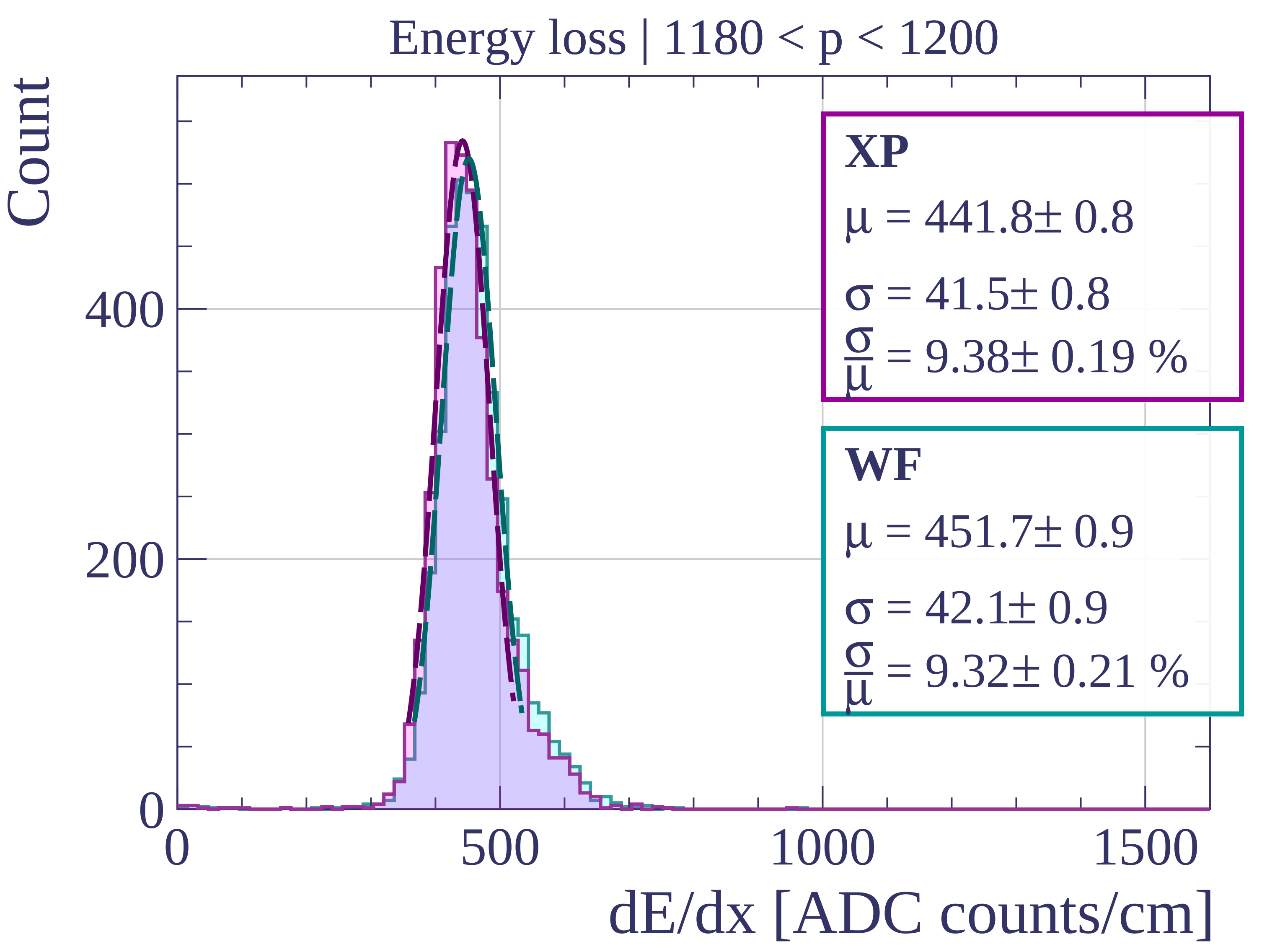


Energy loss | 1120 < p < 1140
Count
0
200
400
600
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 437.2± 0.8
σ = 40.4± 0.8
σ/μ = 9.23± 0.20 %
WF
μ = 449.3± 0.9
σ = 42.4± 0.8
σ/μ = 9.44± 0.19 %

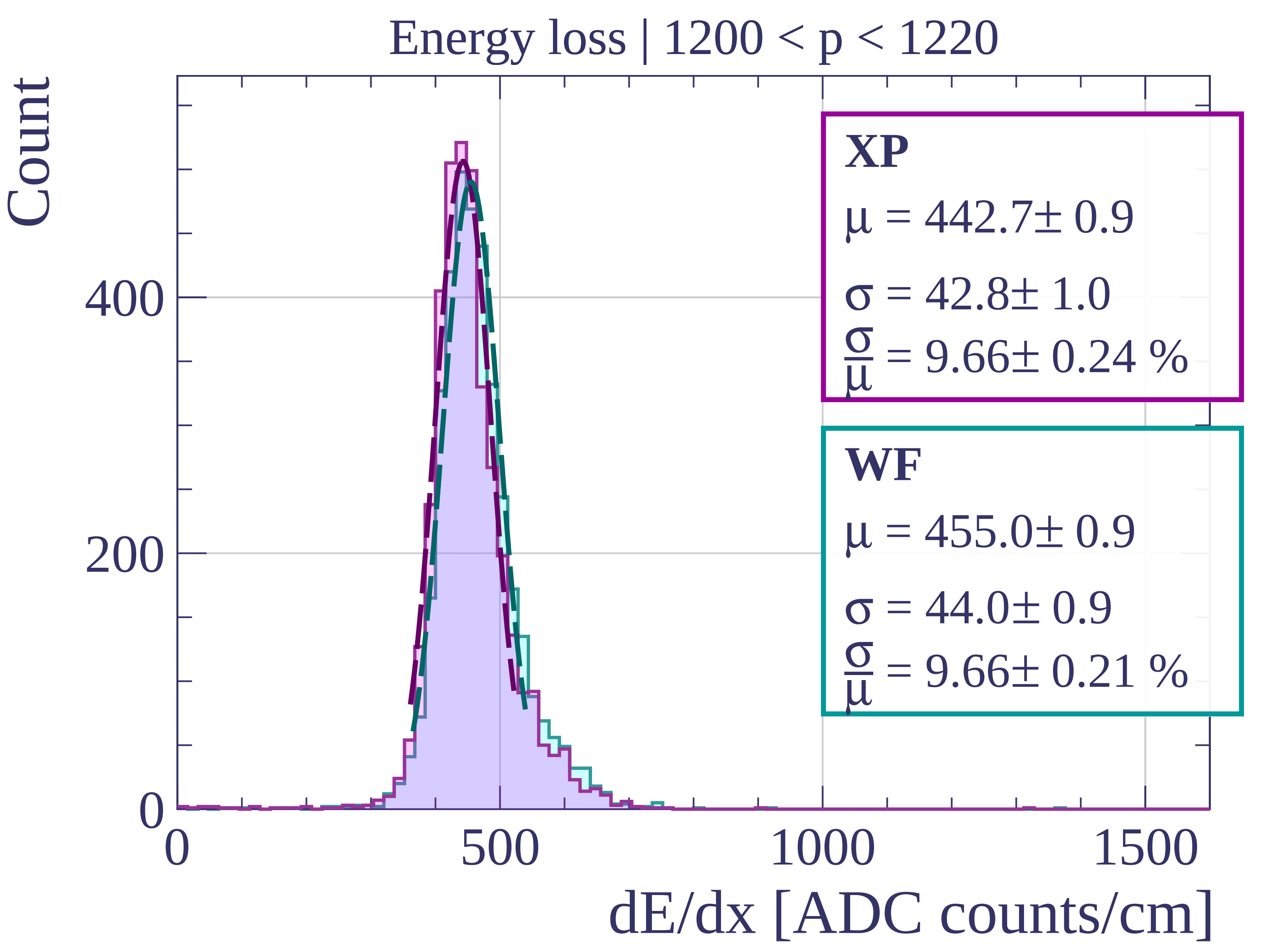


Energy loss | 1140 < p < 1160
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 438.2± 0.8
σ = 39.8± 0.8
σ/μ = 9.08± 0.19 %
WF
μ = 449.4± 0.9
σ = 42.4± 0.9
σ/μ = 9.43± 0.22 %

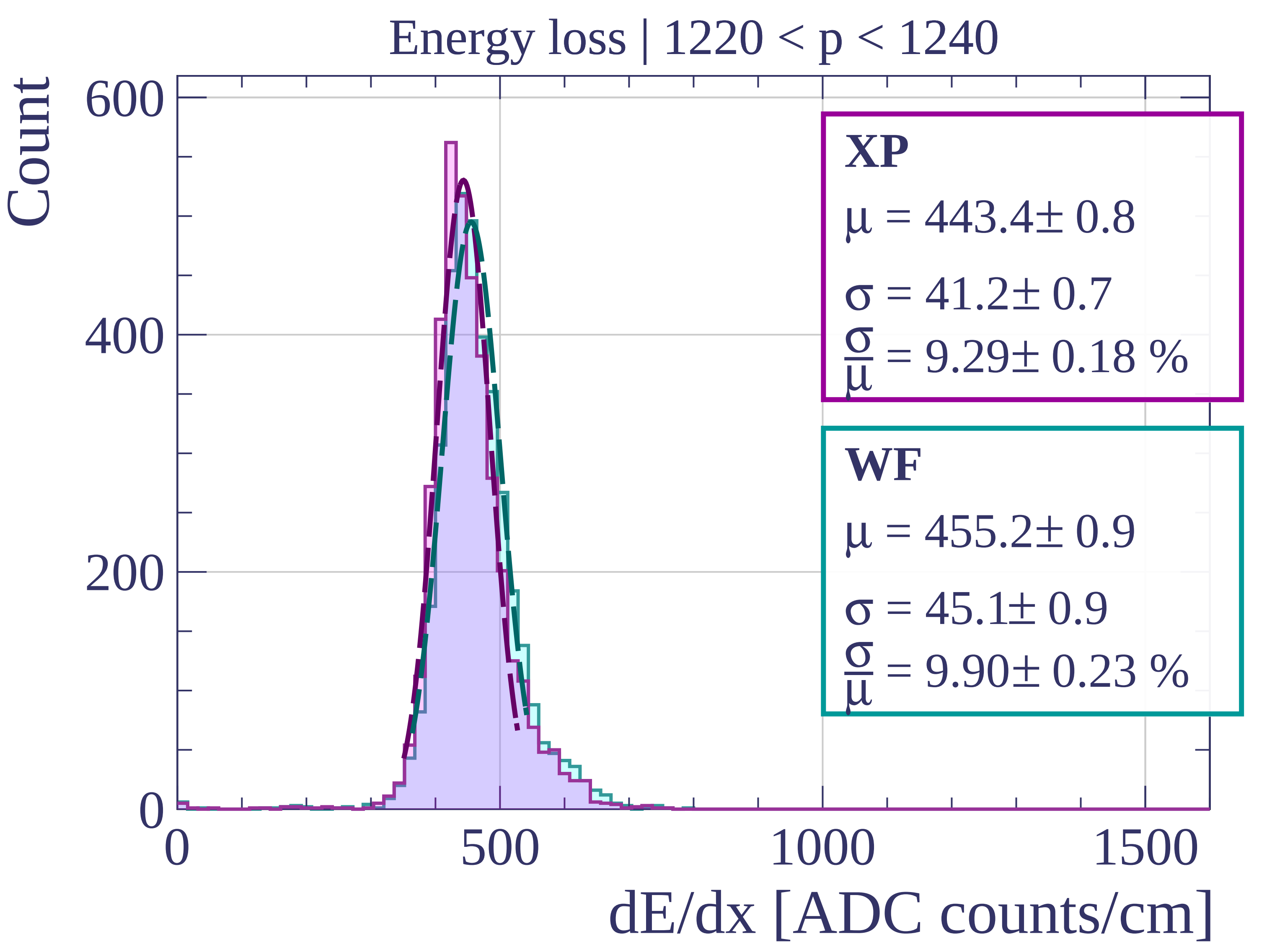

Energy loss | 1160 < p < 1180
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 440.2± 0.8
σ = 42.5± 0.8
σ/μ = 9.66± 0.20 %
WF
μ = 450.3± 0.9
σ = 42.8± 0.9
σ/μ = 9.52± 0.23 %



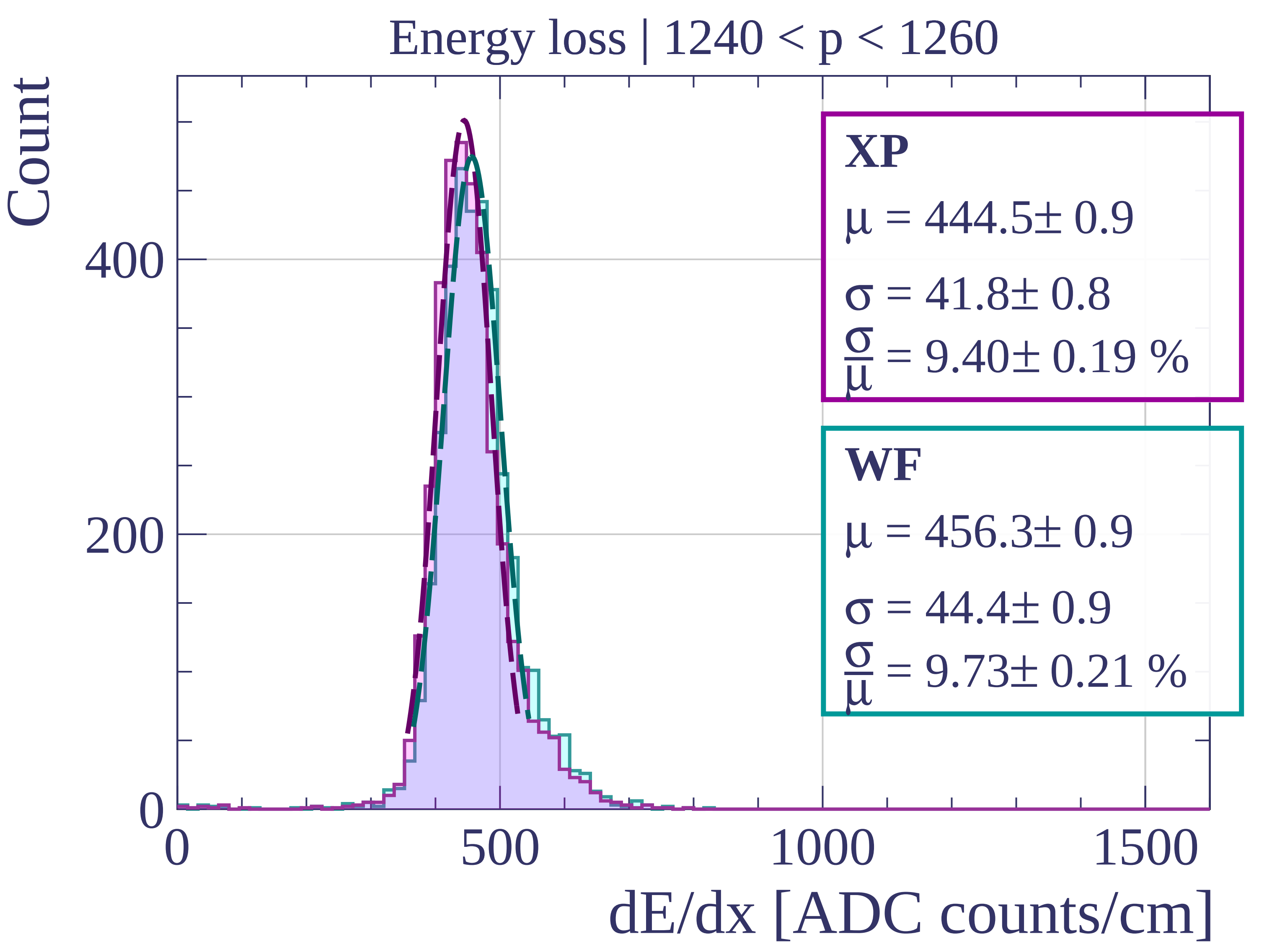
Energy loss | 1180 < p < 1200
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 441.8± 0.8
σ = 41.5± 0.8
σ/μ = 9.38± 0.19 %
WF
μ = 451.7± 0.9
σ = 42.1± 0.9
σ/μ = 9.32± 0.21 %



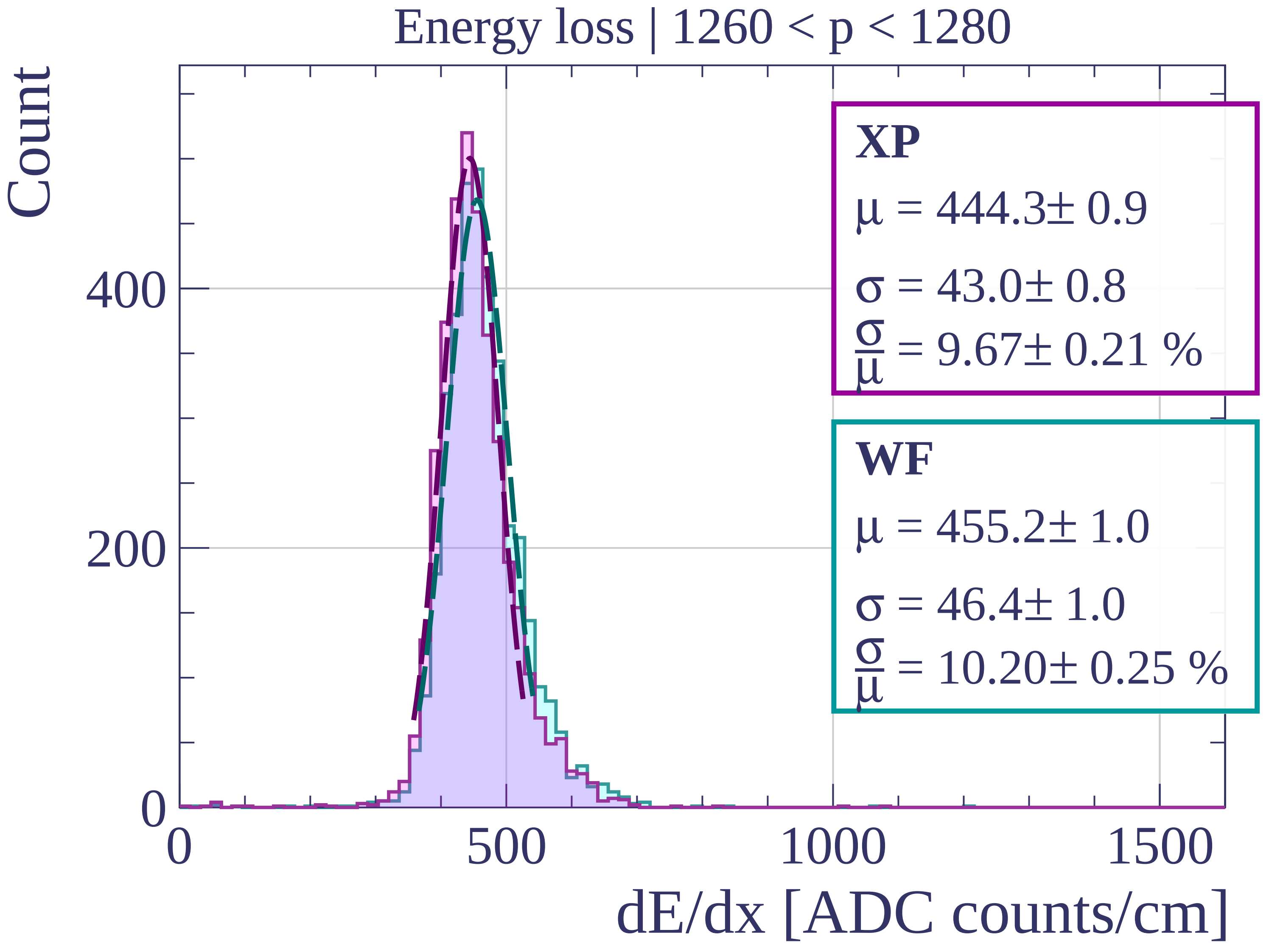
Energy loss | 1200 < p < 1220
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 442.7± 0.9
σ = 42.8± 1.0
σ/μ = 9.66± 0.24 %
WF
μ = 455.0± 0.9
σ = 44.0± 0.9
σ/μ = 9.66± 0.21 %

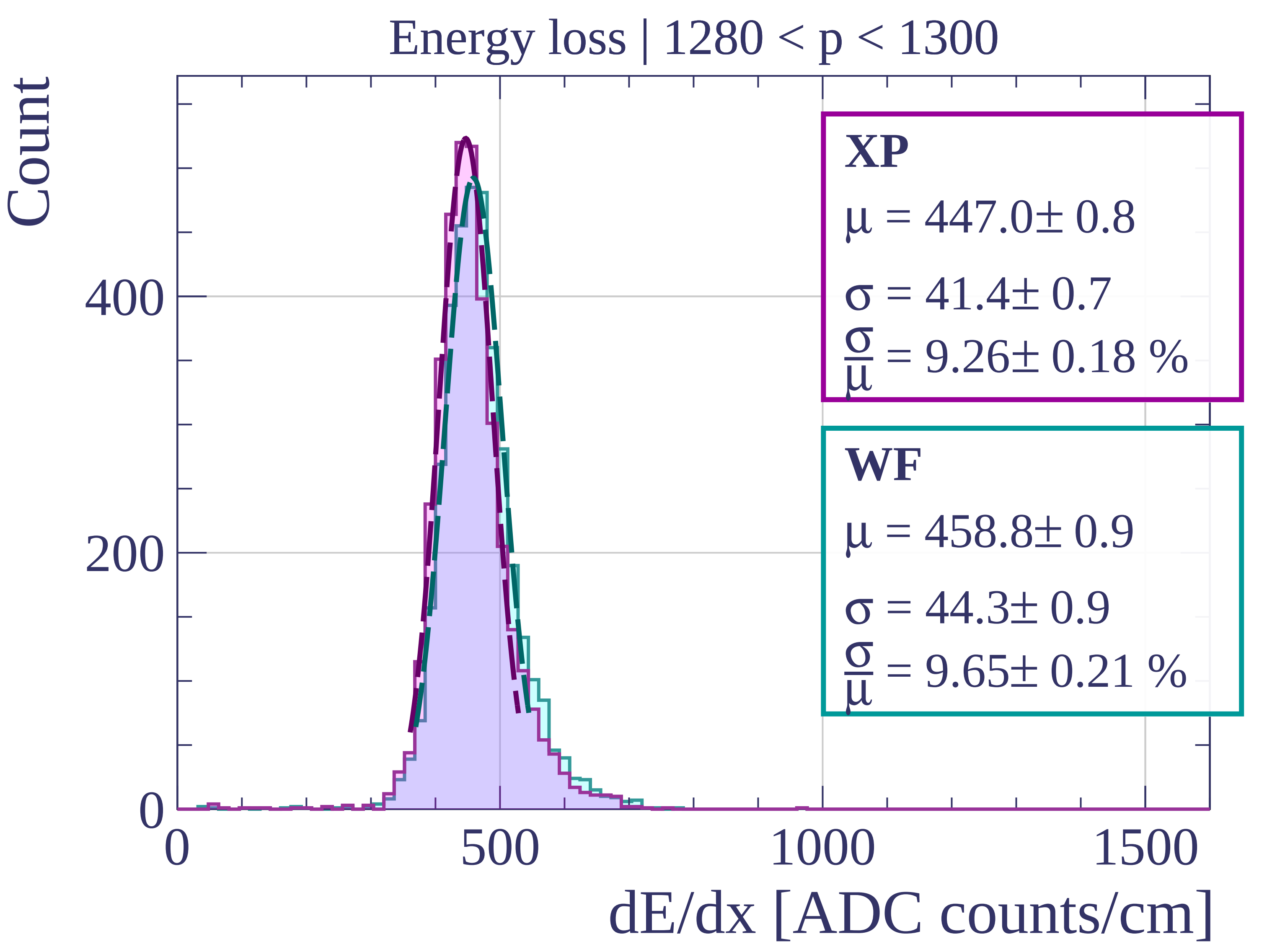


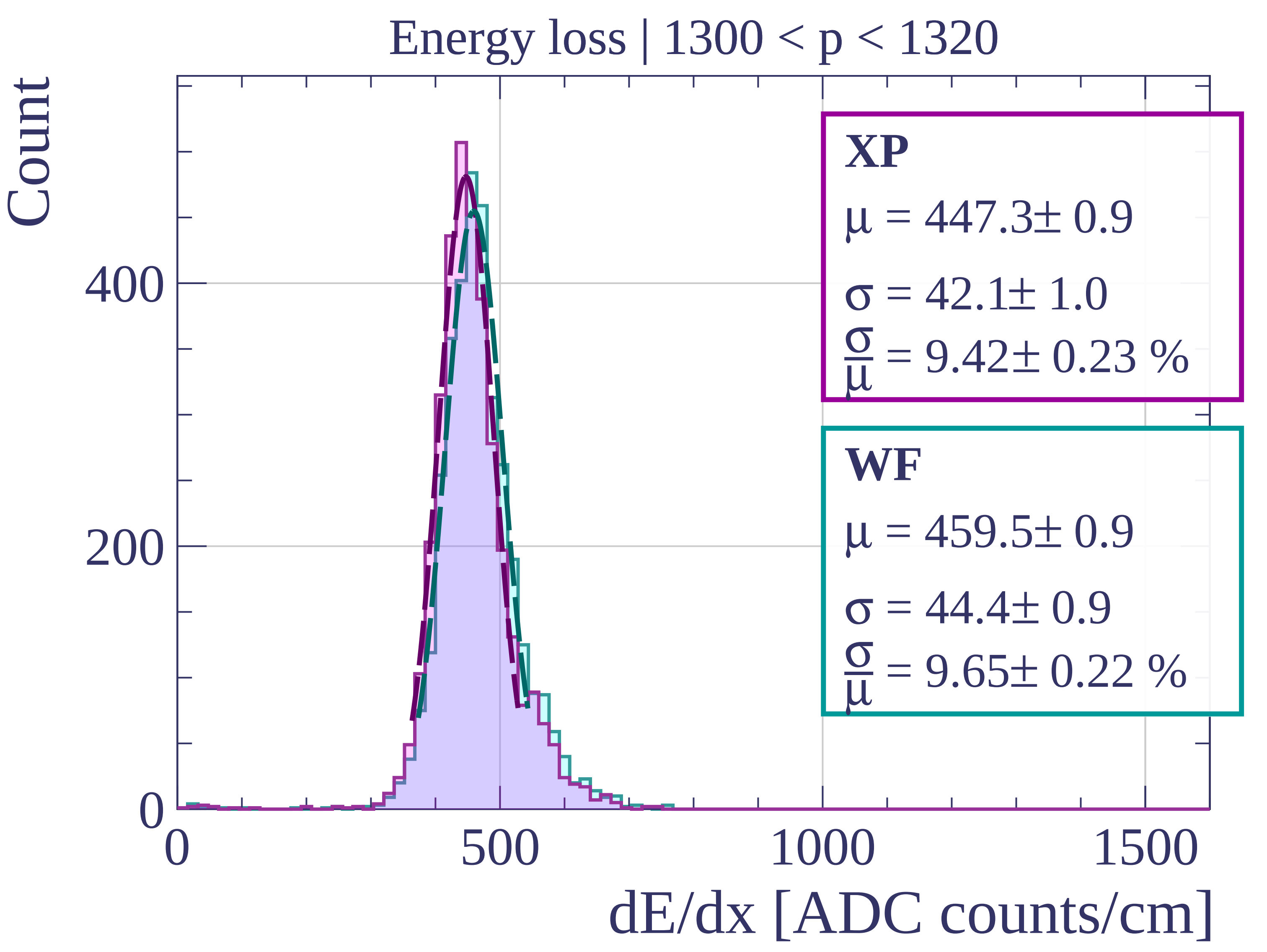
Energy loss | 1220 < p < 1240
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 443.4± 0.8
σ = 41.2± 0.7
σ/μ = 9.29± 0.18 %
WF
μ = 455.2± 0.9
σ = 45.1± 0.9
σ/μ = 9.90± 0.23 %



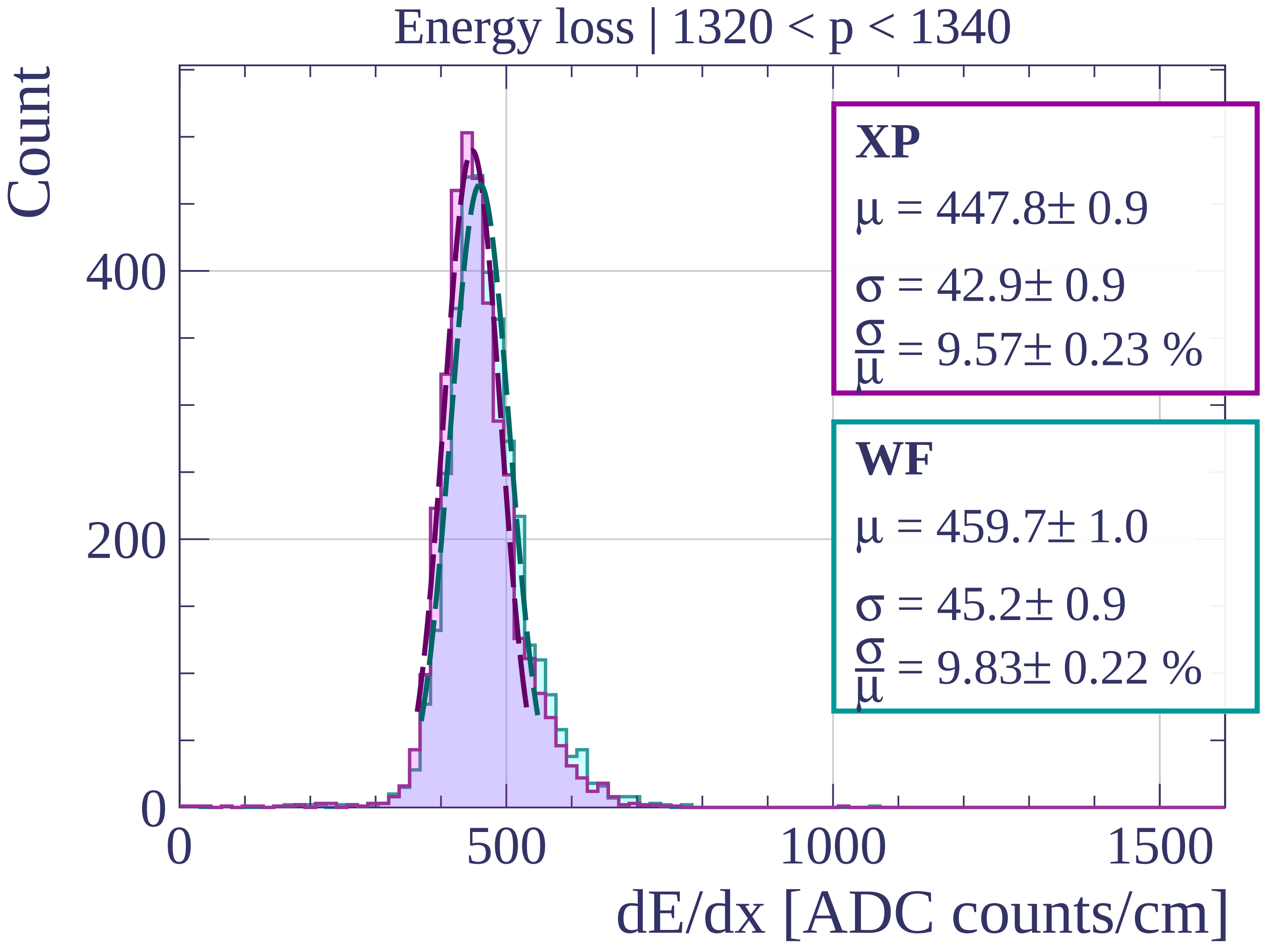
Energy loss | 1240 < p < 1260
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 444.5± 0.9
σ = 41.8± 0.8
σ/μ = 9.40± 0.19 %
WF
μ = 456.3± 0.9
σ = 44.4± 0.9
σ/μ = 9.73± 0.21 %


Energy loss | 1260 < p < 1280
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 444.3± 0.9
σ = 43.0± 0.8
σ/μ = 9.67± 0.21 %
WF
μ = 455.2± 1.0
σ = 46.4± 1.0
σ/μ = 10.20± 0.25 %

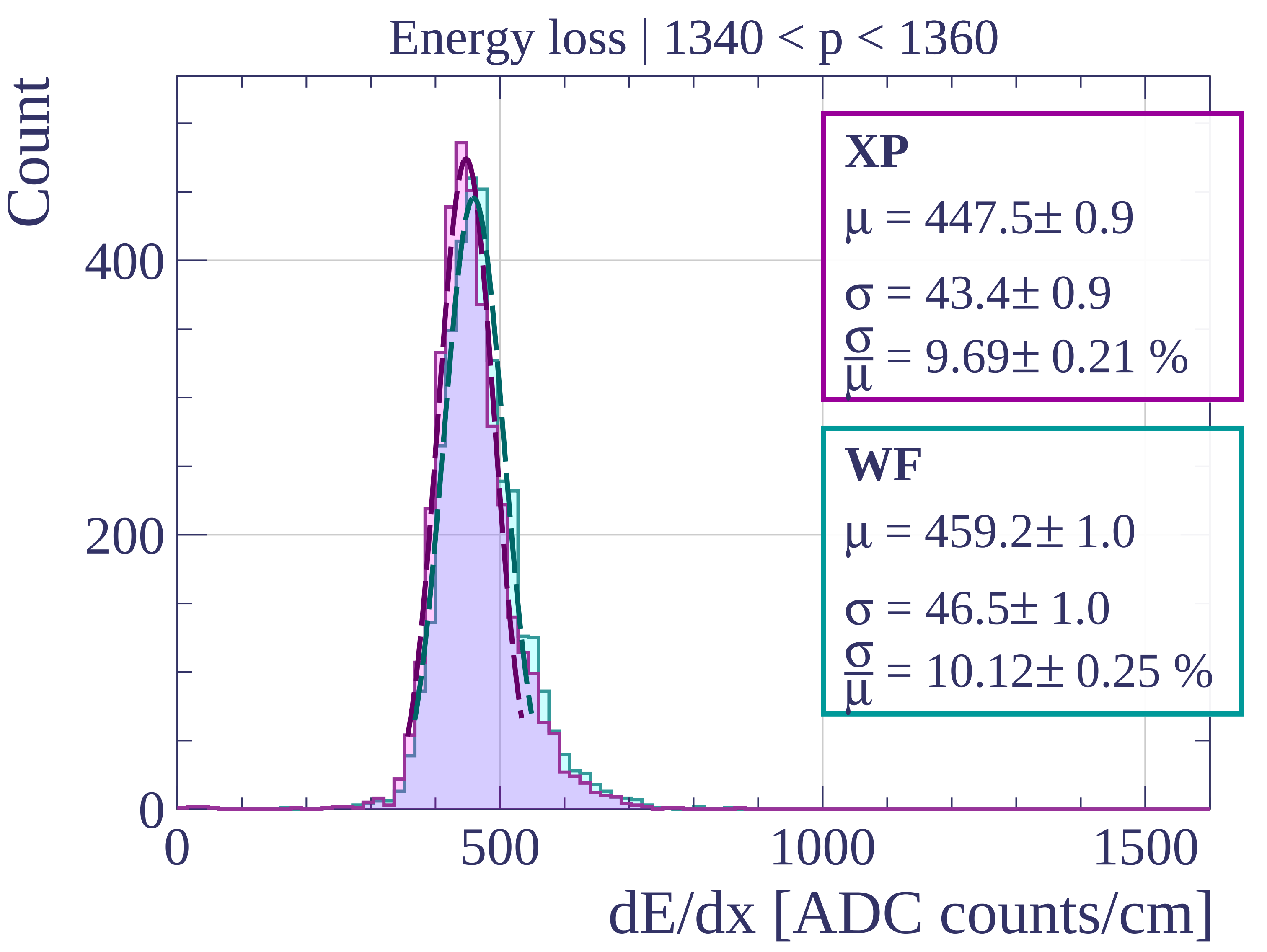

Energy loss | 1280 < p < 1300
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 447.0± 0.8
σ = 41.4± 0.7
σ/μ = 9.26± 0.18 %
WF
μ = 458.8± 0.9
σ = 44.3± 0.9
σ/μ = 9.65± 0.21 %

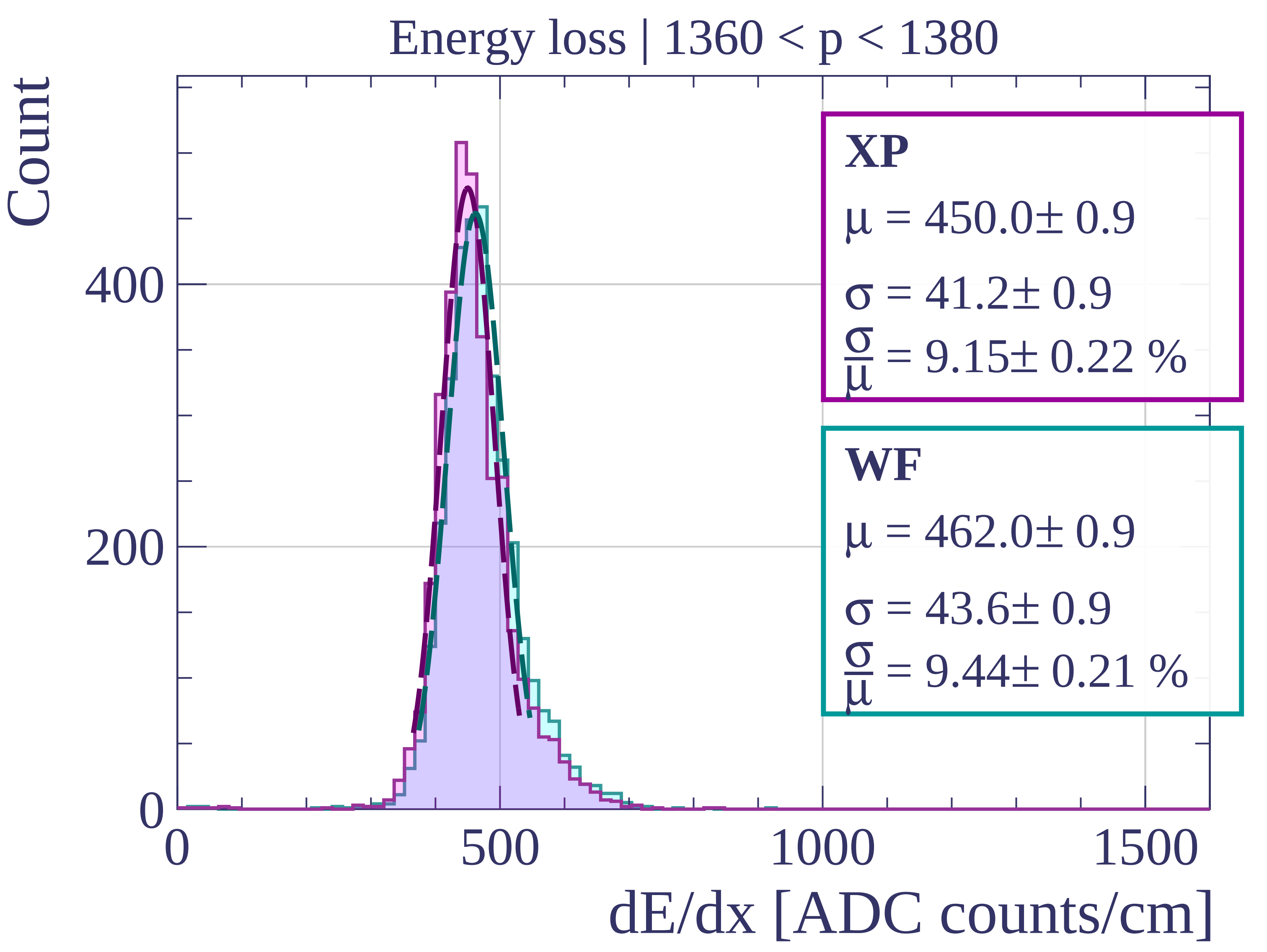


Energy loss | 1300 < p < 1320
Count
0
200
400
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 447.3± 0.9
σ = 42.1± 1.0
σ/μ = 9.42± 0.23 %
WF
μ = 459.5± 0.9
σ = 44.4± 0.9
σ/μ = 9.65± 0.22 %

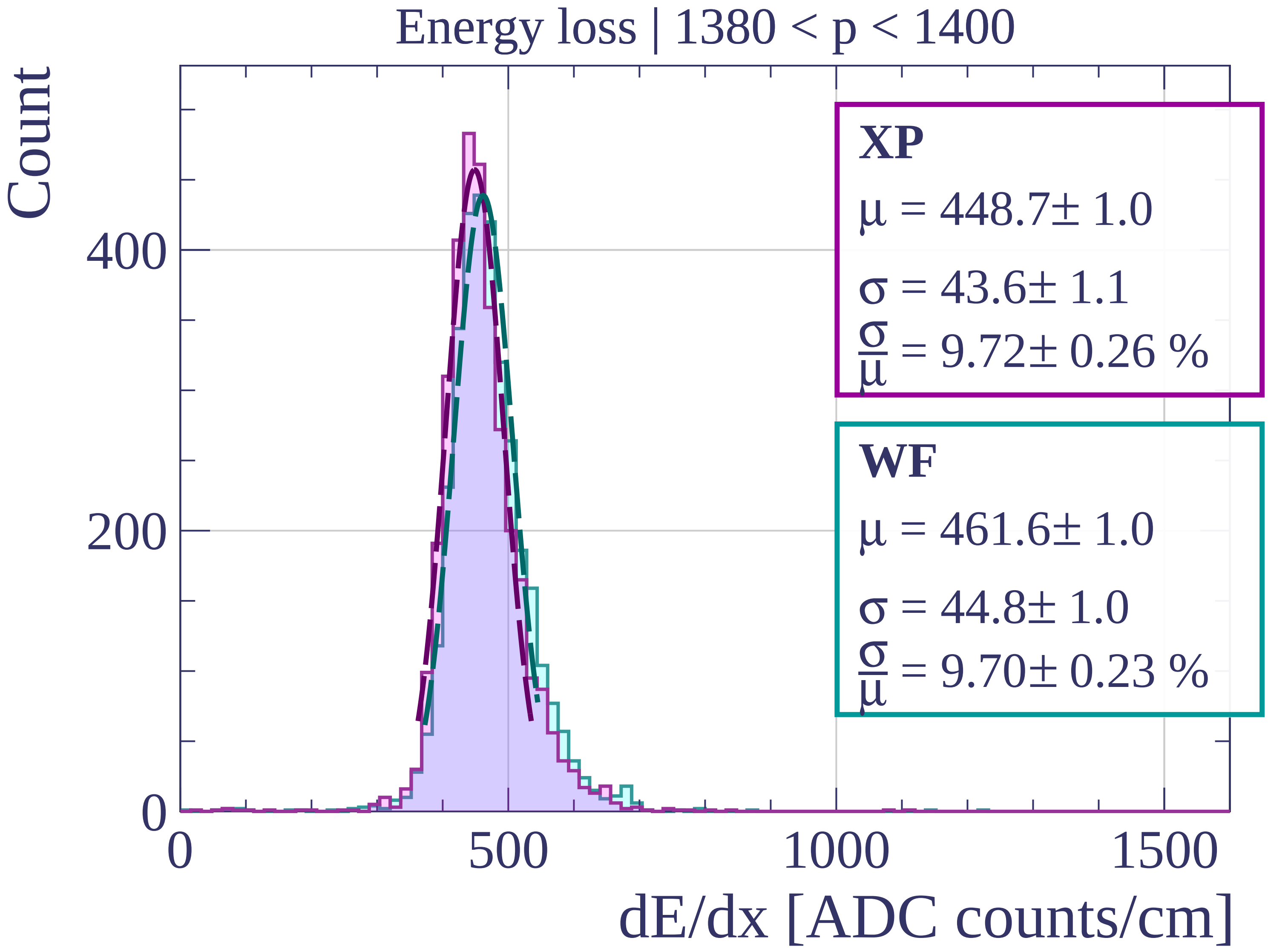


Energy loss | 1320 < p < 1340
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 447.8± 0.9
σ = 42.9± 0.9
σ/μ = 9.57± 0.23 %
WF
μ = 459.7± 1.0
σ = 45.2± 0.9
σ/μ = 9.83± 0.22 %

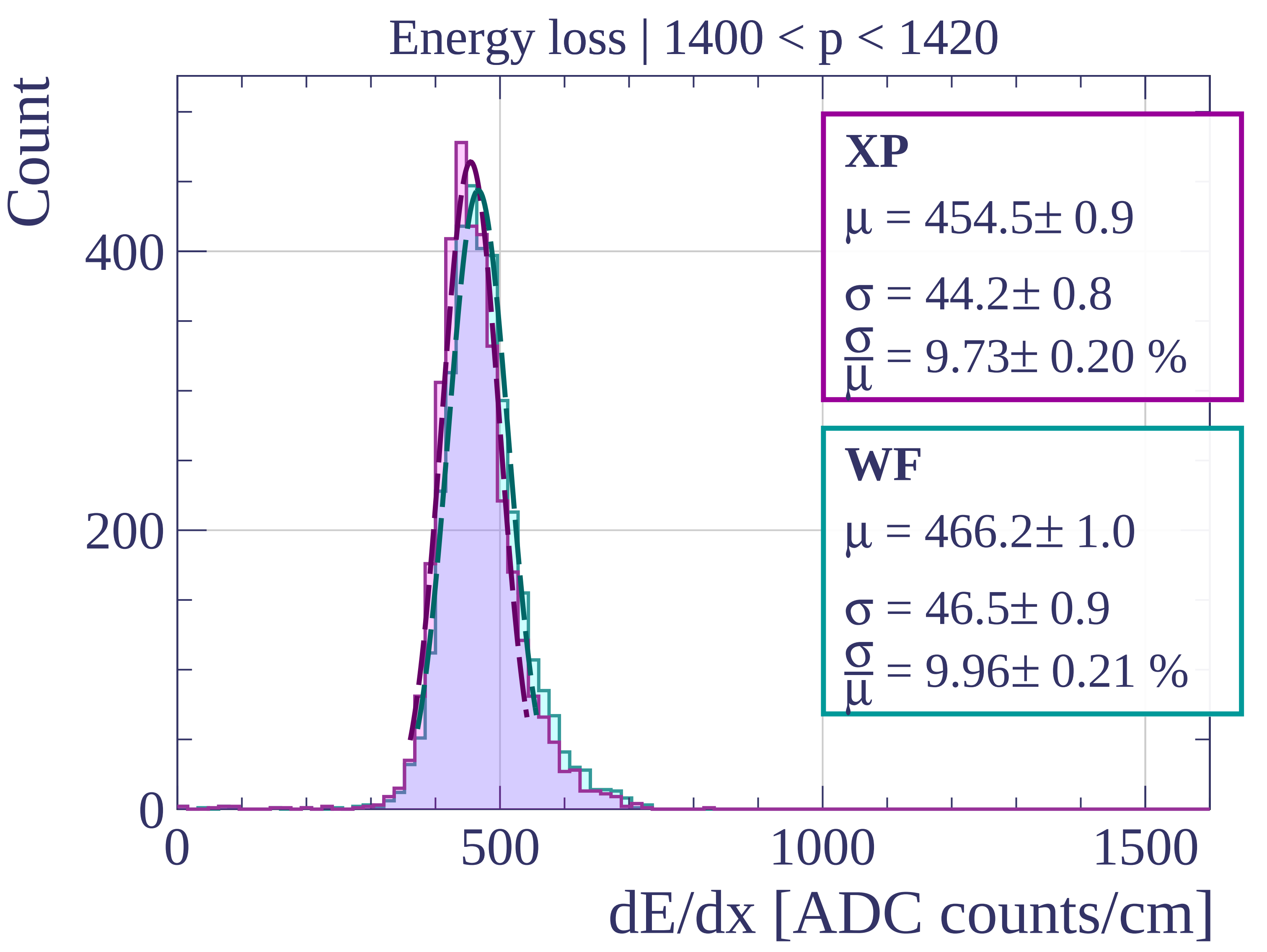

Energy loss | 1340 < p < 1360
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 447.5± 0.9
σ = 43.4± 0.9
σ/μ = 9.69± 0.21 %
WF
μ = 459.2± 1.0
σ = 46.5± 1.0
σ/μ = 10.12± 0.25 %



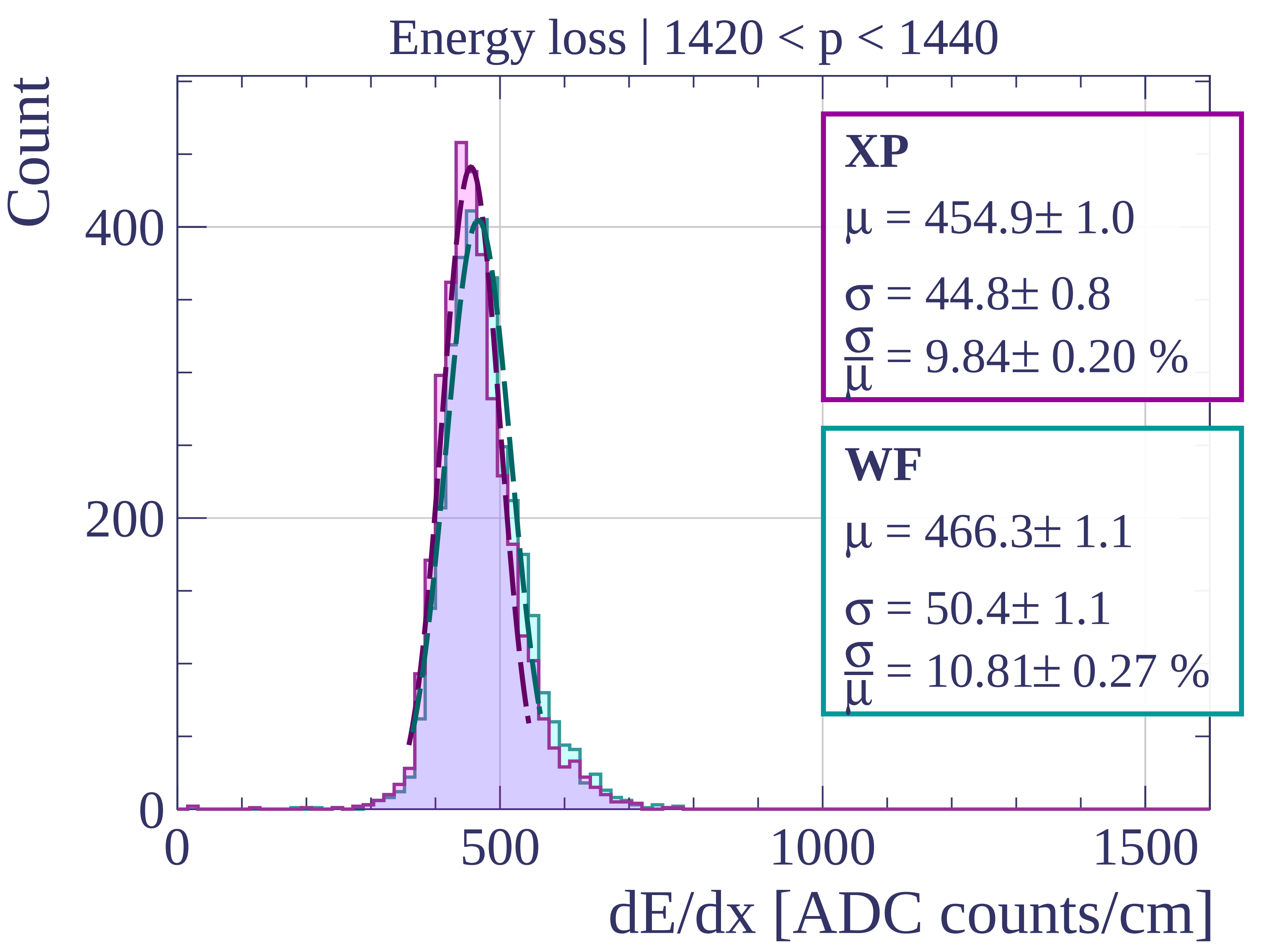
Energy loss | 1360 < p < 1380
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 450.0± 0.9
σ = 41.2± 0.9
σ/μ = 9.15± 0.22 %
WF
μ = 462.0± 0.9
σ = 43.6± 0.9
σ/μ = 9.44± 0.21 %

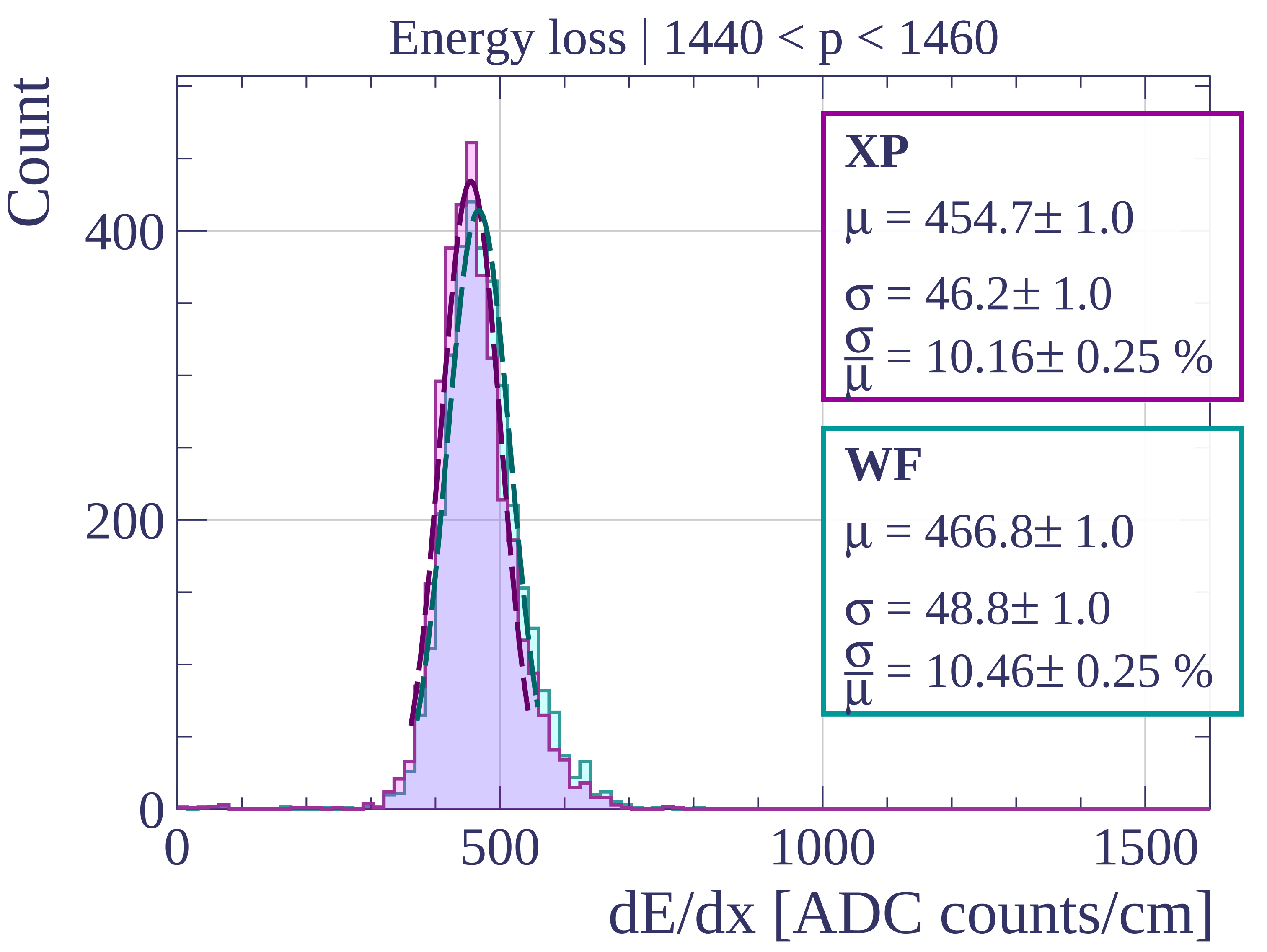


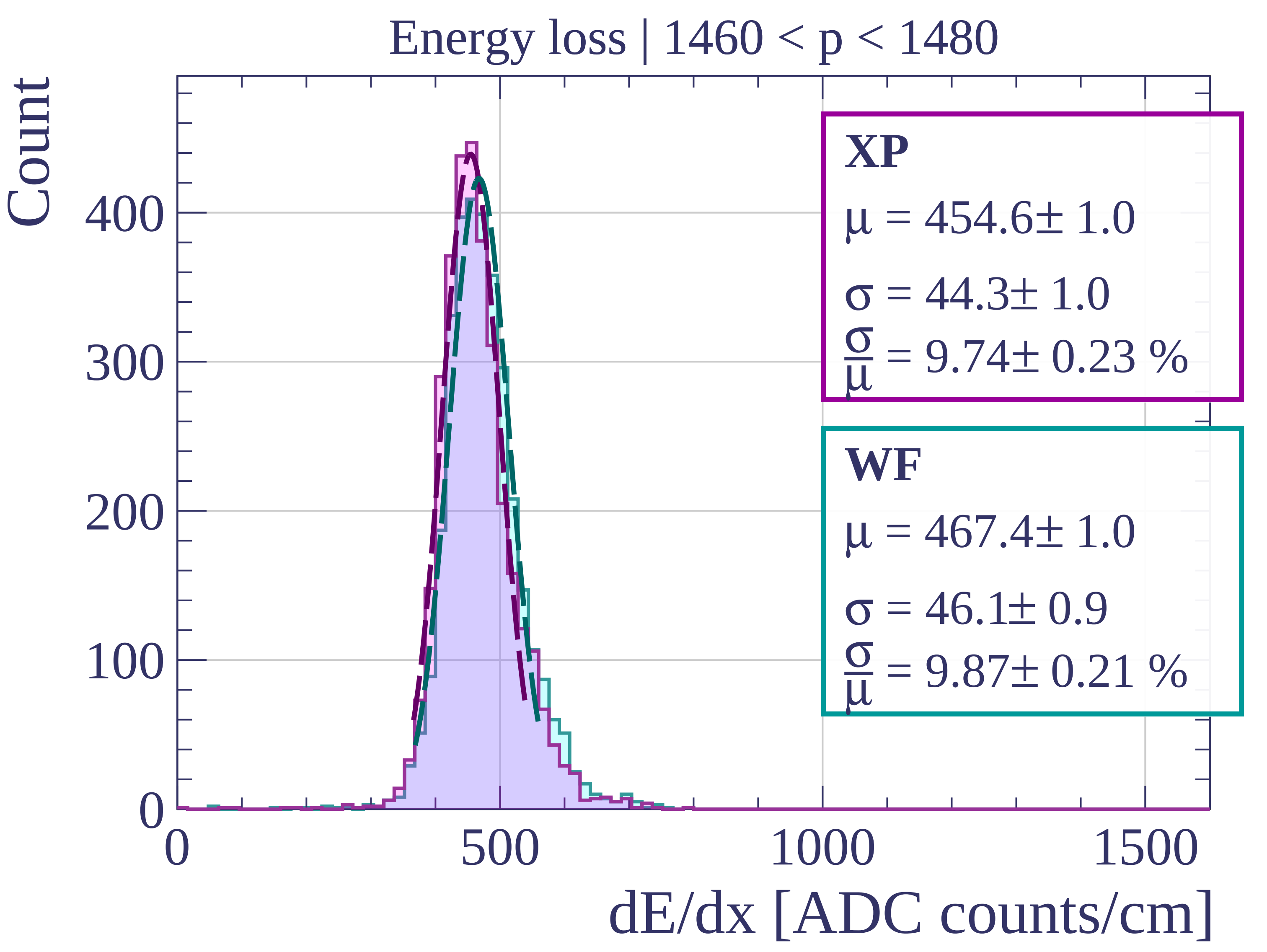
Energy loss | 1380 < p < 1400
Count
0
200
400
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 448.7± 1.0
σ = 43.6± 1.1
σ/μ = 9.72± 0.26 %
WF
μ = 461.6± 1.0
σ = 44.8± 1.0
σ/μ = 9.70± 0.23 %

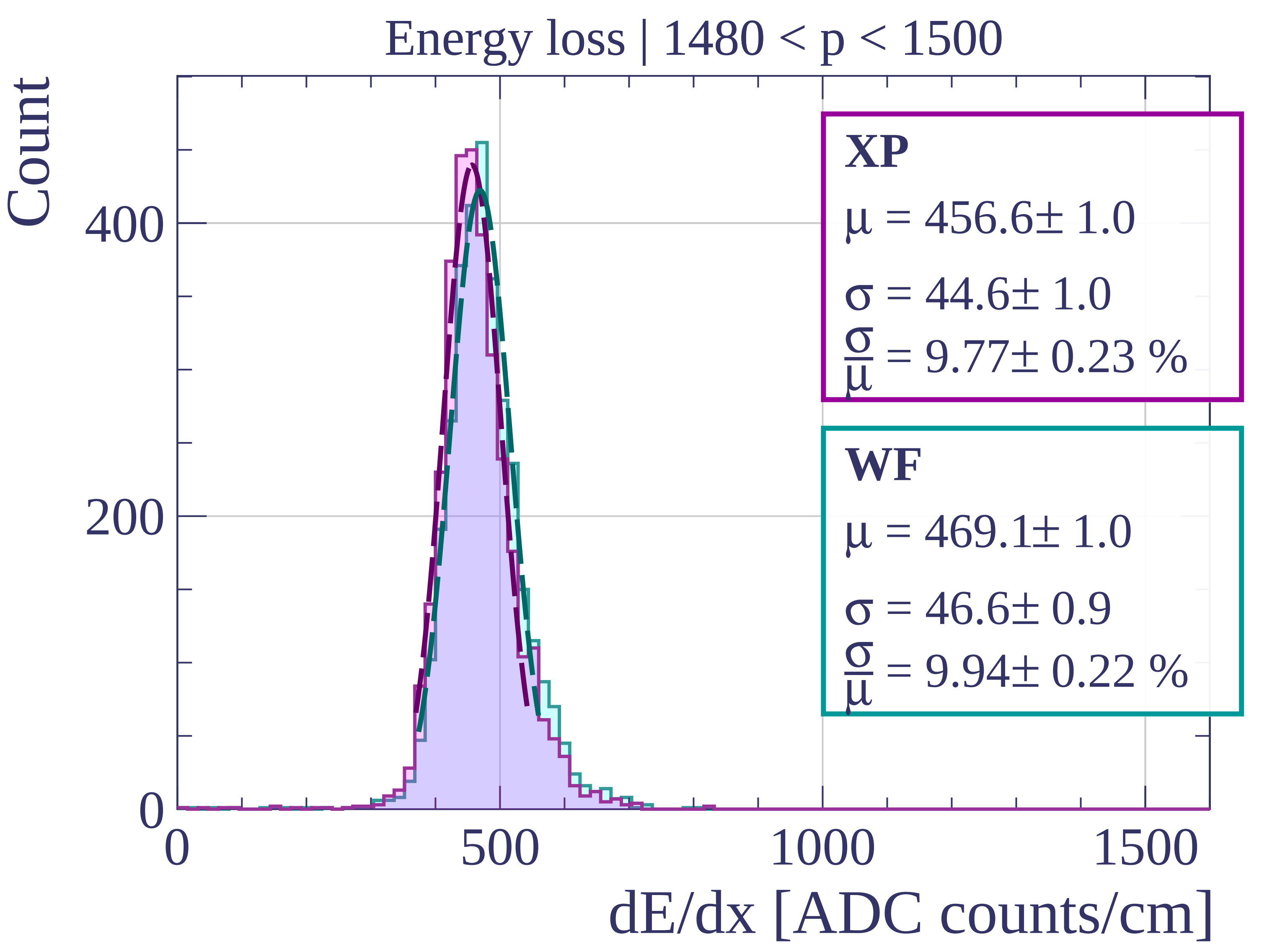


Energy loss | 1400 < p < 1420
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 454.5± 0.9
σ = 44.2± 0.8
σ/μ = 9.73± 0.20 %
WF
μ = 466.2± 1.0
σ = 46.5± 0.9
σ/μ = 9.96± 0.21 %

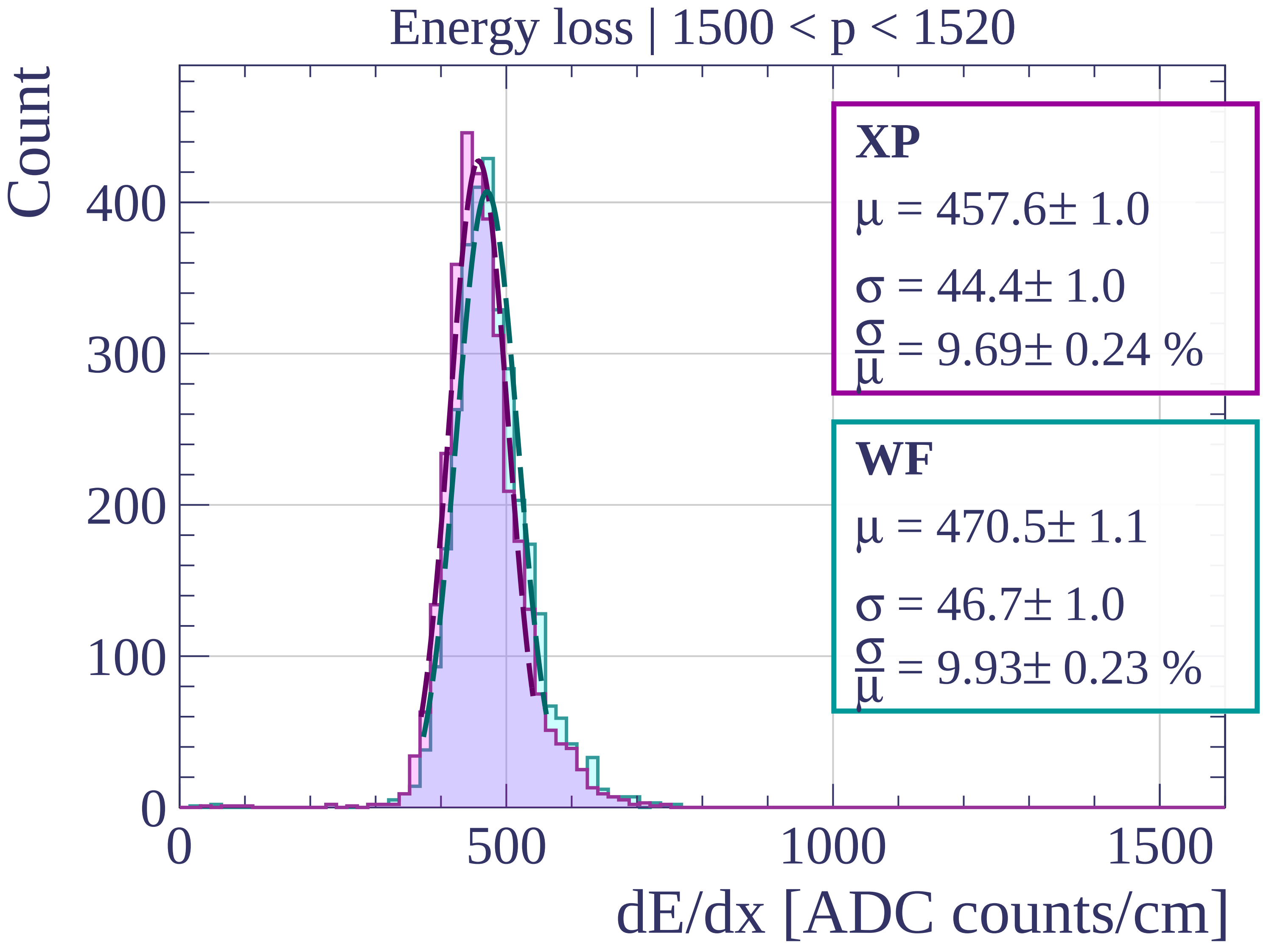

Energy loss | 1420 < p < 1440
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 454.9± 1.0
σ = 44.8± 0.8
σ/μ = 9.84± 0.20 %
WF
μ = 466.3± 1.1
σ = 50.4± 1.1
σ/μ = 10.81± 0.27 %


Energy loss | 1440 < p < 1460
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 454.7± 1.0
σ = 46.2± 1.0
σ/μ = 10.16± 0.25 %
WF
μ = 466.8± 1.0
σ = 48.8± 1.0
σ/μ = 10.46± 0.25 %

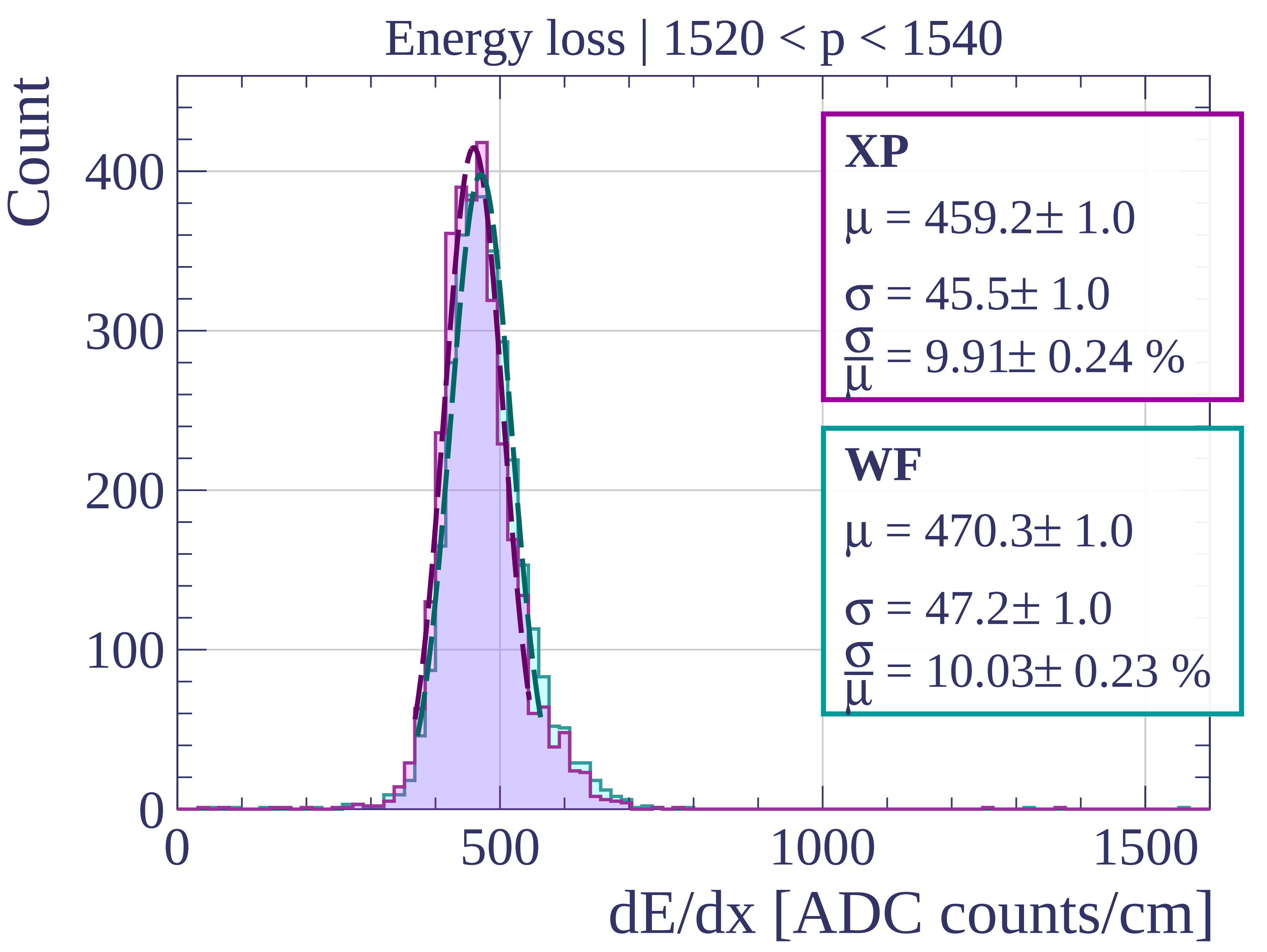

Energy loss | 1460 < p < 1480
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
XP
μ = 454.6± 1.0
σ = 44.3± 1.0
σ/μ = 9.74± 0.23 %
WF
μ = 467.4± 1.0
σ = 46.1± 0.9
σ/μ = 9.87± 0.21 %



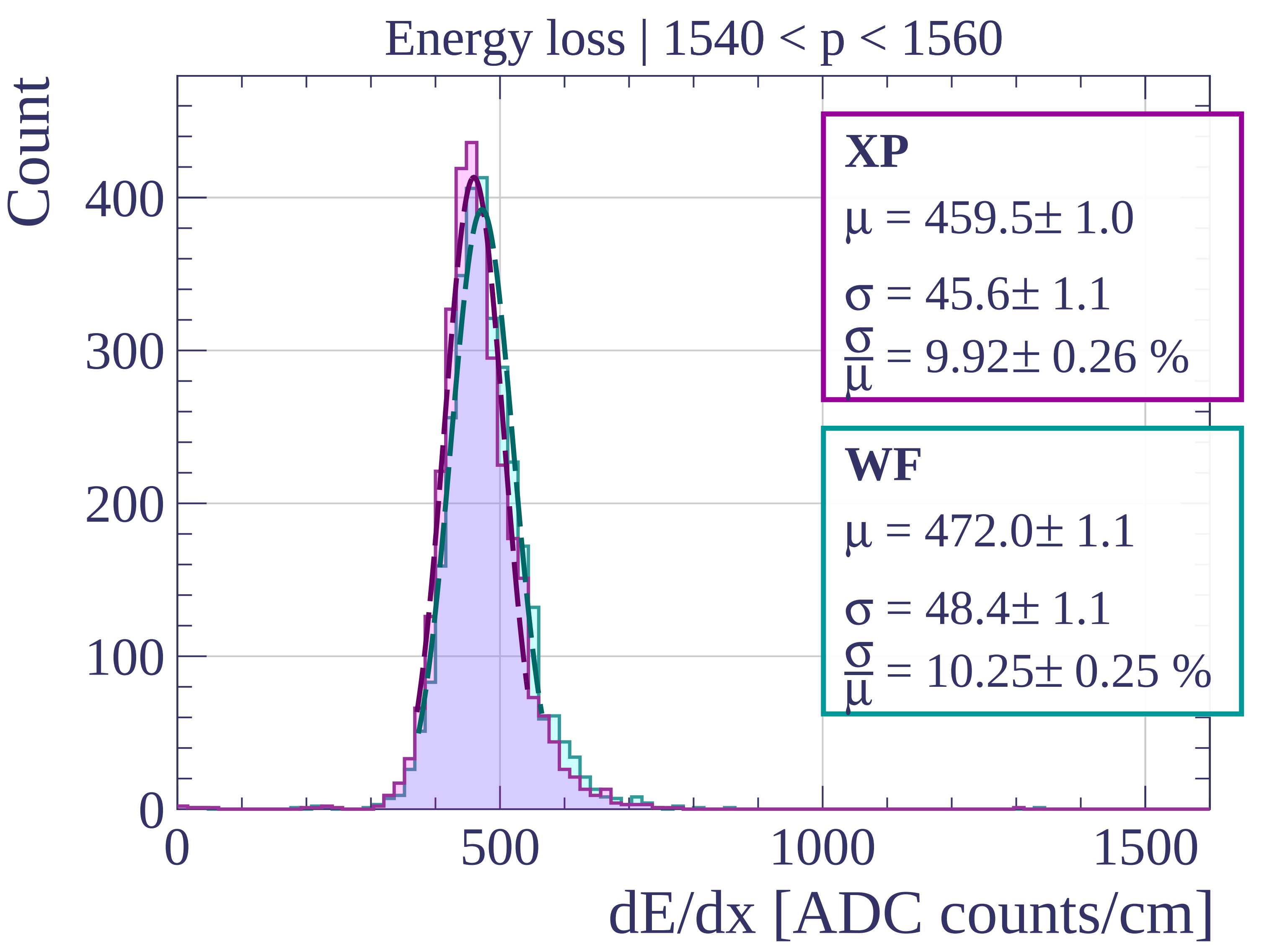
Energy loss | 1480 < p < 1500
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 456.6± 1.0
σ = 44.6± 1.0
σ/μ = 9.77± 0.23 %
WF
μ = 469.1± 1.0
σ = 46.6± 0.9
σ/μ = 9.94± 0.22 %



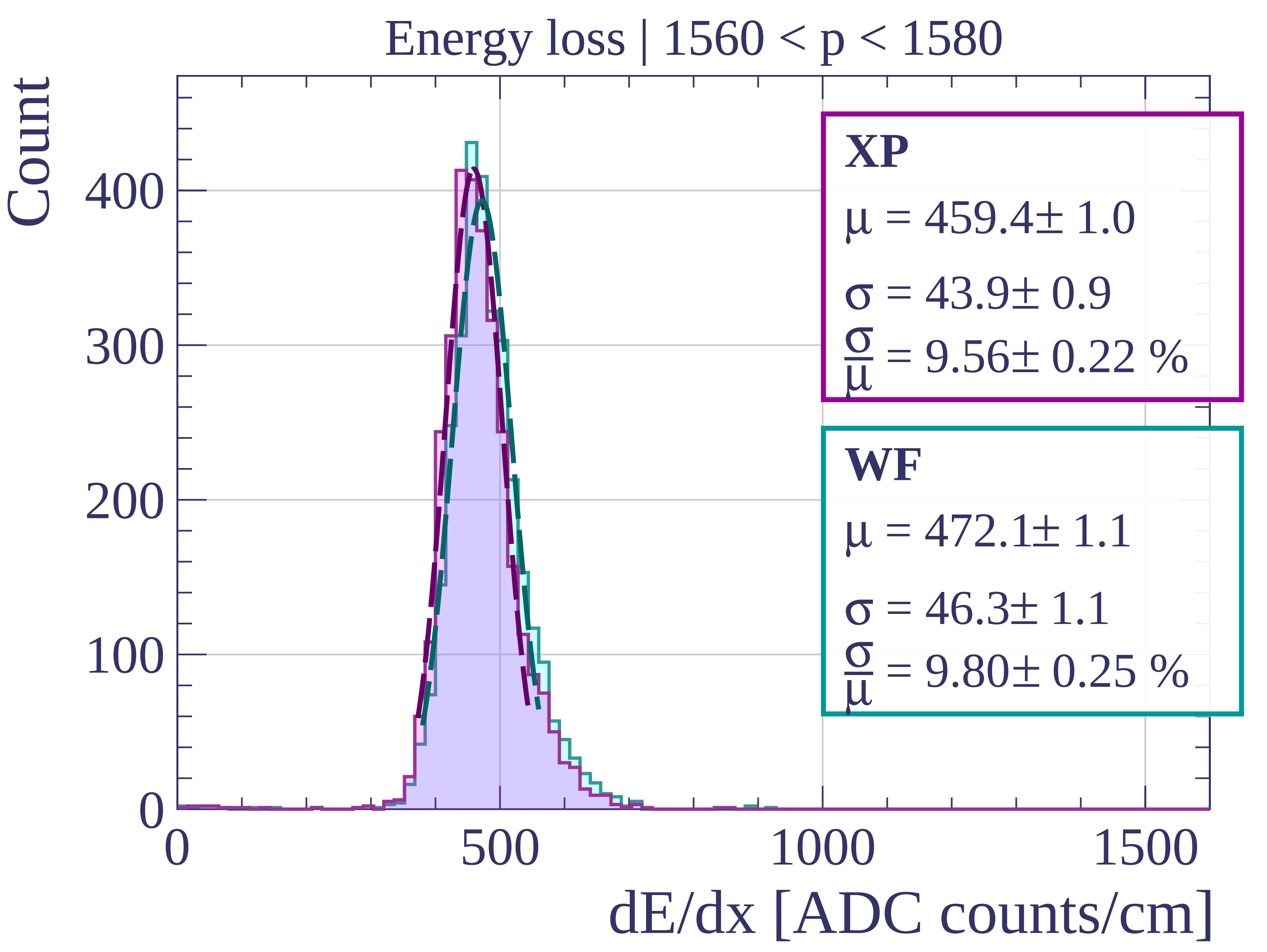
Energy loss | 1500 < p < 1520
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
XP
μ = 457.6± 1.0
σ = 44.4± 1.0
σ/μ = 9.69± 0.24 %
WF
μ = 470.5± 1.1
σ = 46.7± 1.0
σ/μ = 9.93± 0.23 %

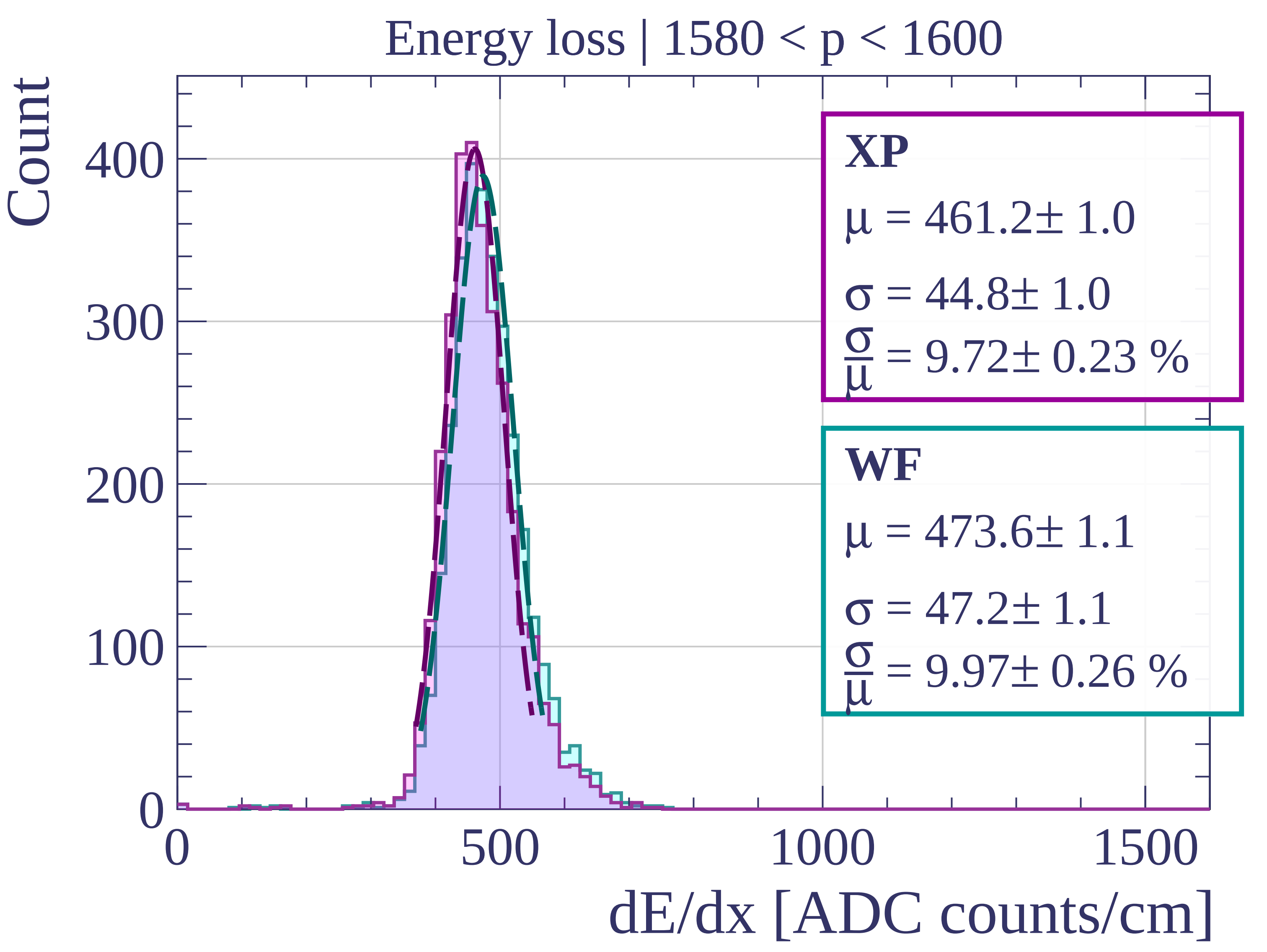


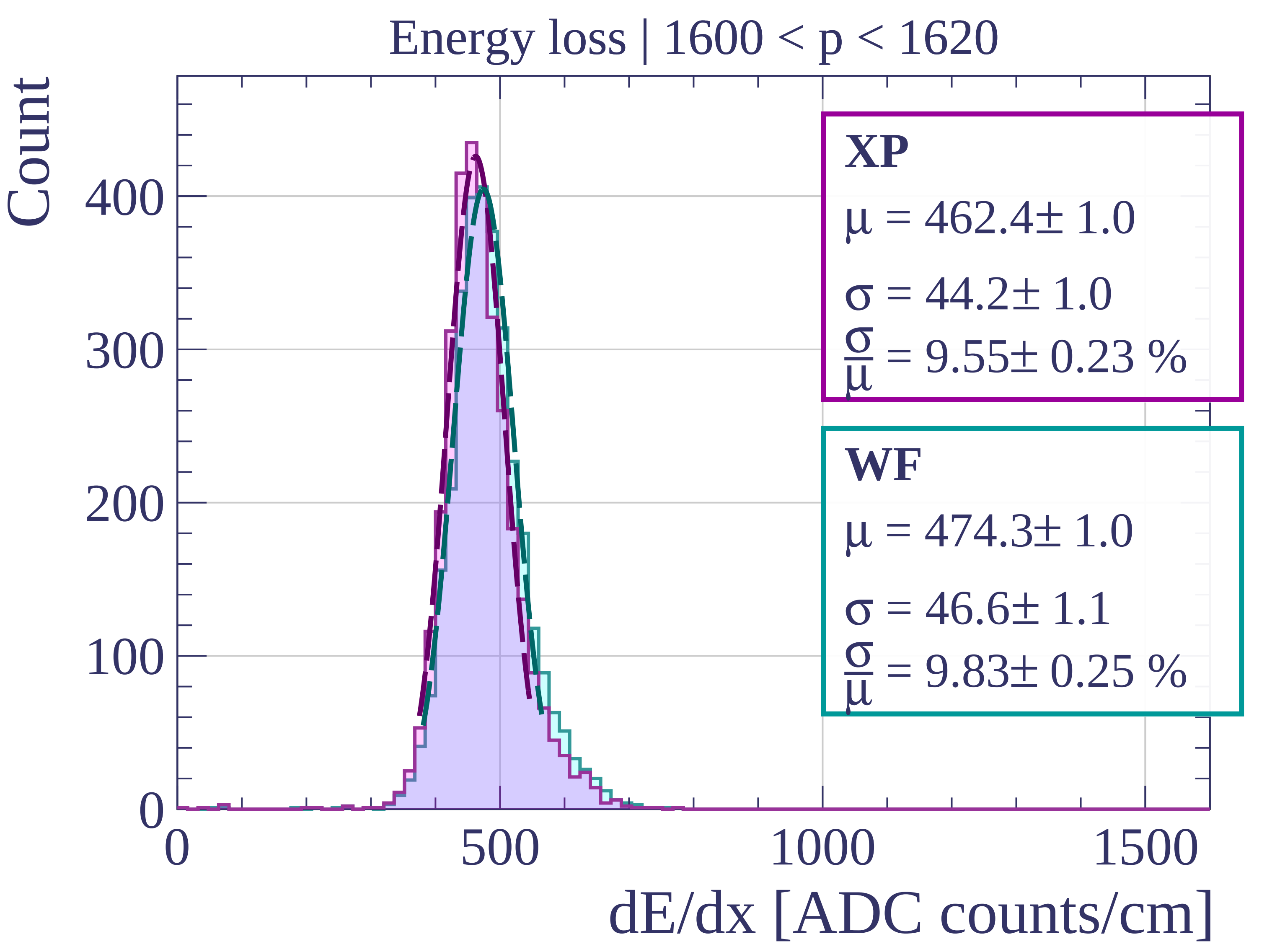
Energy loss | 1520 < p < 1540
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 459.2± 1.0
σ = 45.5± 1.0
σ/μ = 9.91± 0.24 %
WF
μ = 470.3± 1.0
σ = 47.2± 1.0
σ/μ = 10.03± 0.23 %



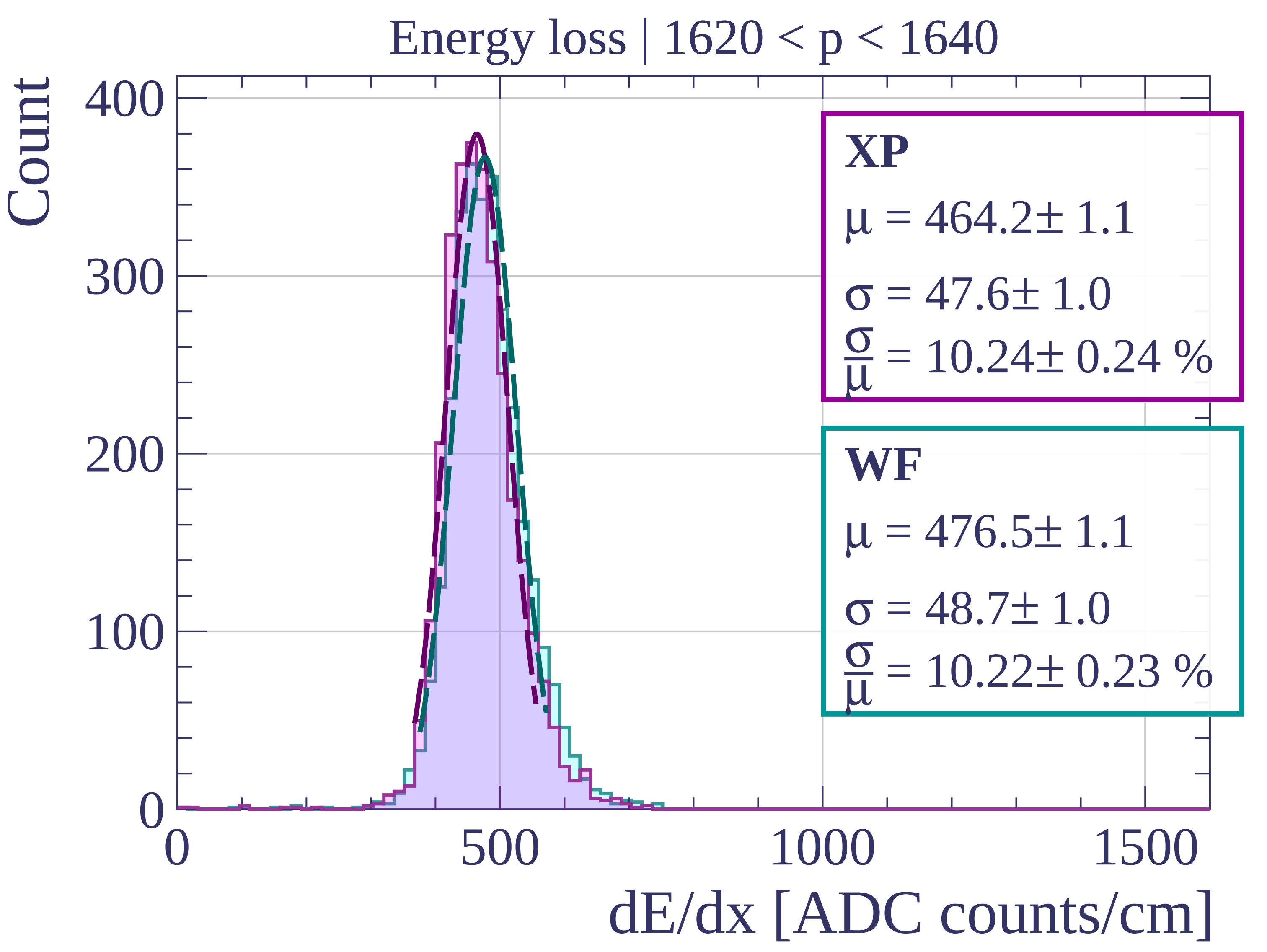
Energy loss | 1540 < p < 1560
Count
0
100
200
300
400
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 459.5± 1.0
σ = 45.6± 1.1
σ/μ = 9.92± 0.26 %
WF
μ = 472.0± 1.1
σ = 48.4± 1.1
σ/μ = 10.25± 0.25 %

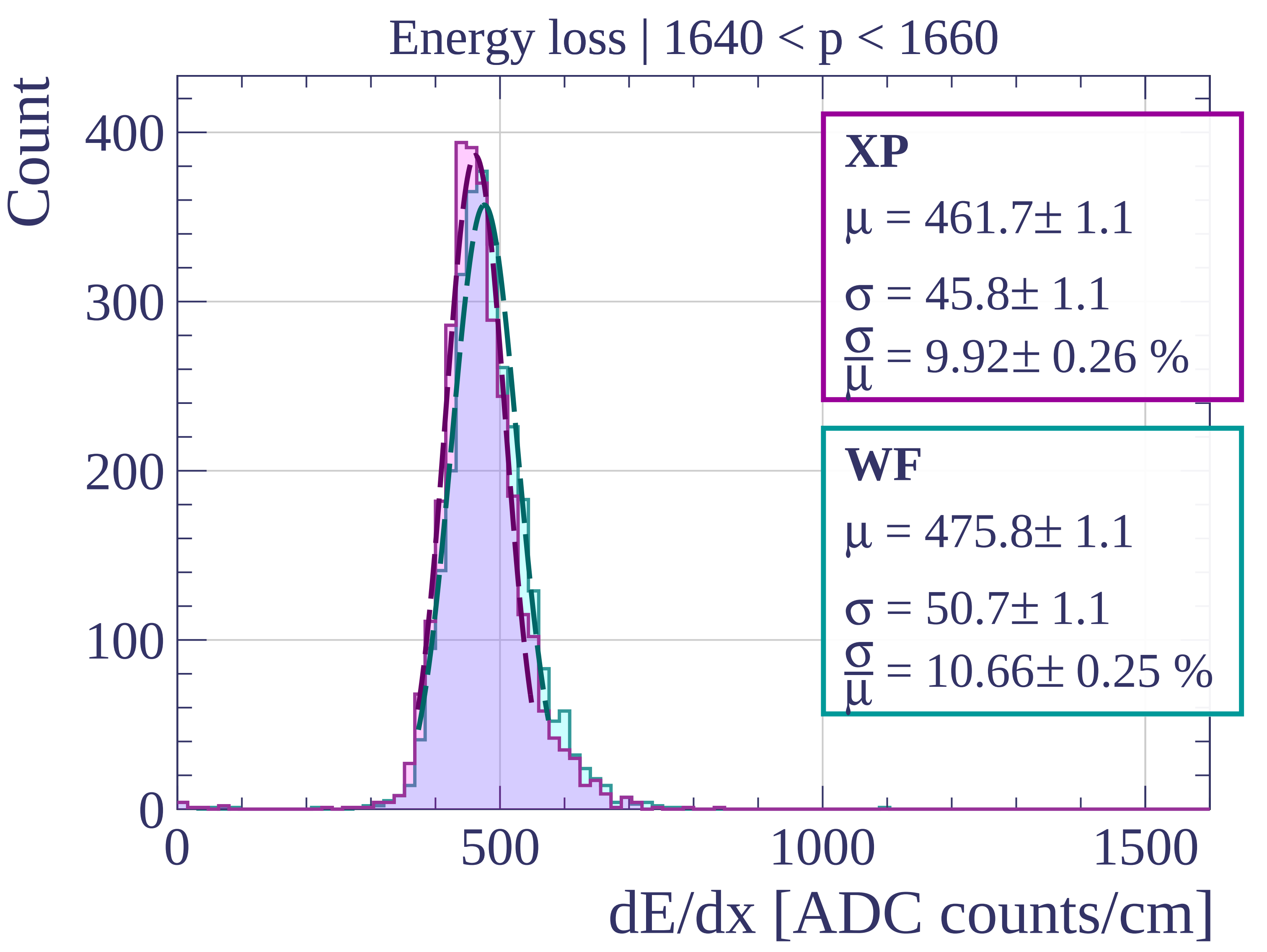

Energy loss | 1560 < p < 1580
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
XP
μ = 459.4± 1.0
σ = 43.9± 0.9
σ/μ = 9.56± 0.22 %
WF
μ = 472.1± 1.1
σ = 46.3± 1.1
σ/μ = 9.80± 0.25 %


Energy loss | 1580 < p < 1600
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 461.2± 1.0
σ = 44.8± 1.0
σ/μ = 9.72± 0.23 %
WF
μ = 473.6± 1.1
σ = 47.2± 1.1
σ/μ = 9.97± 0.26 %

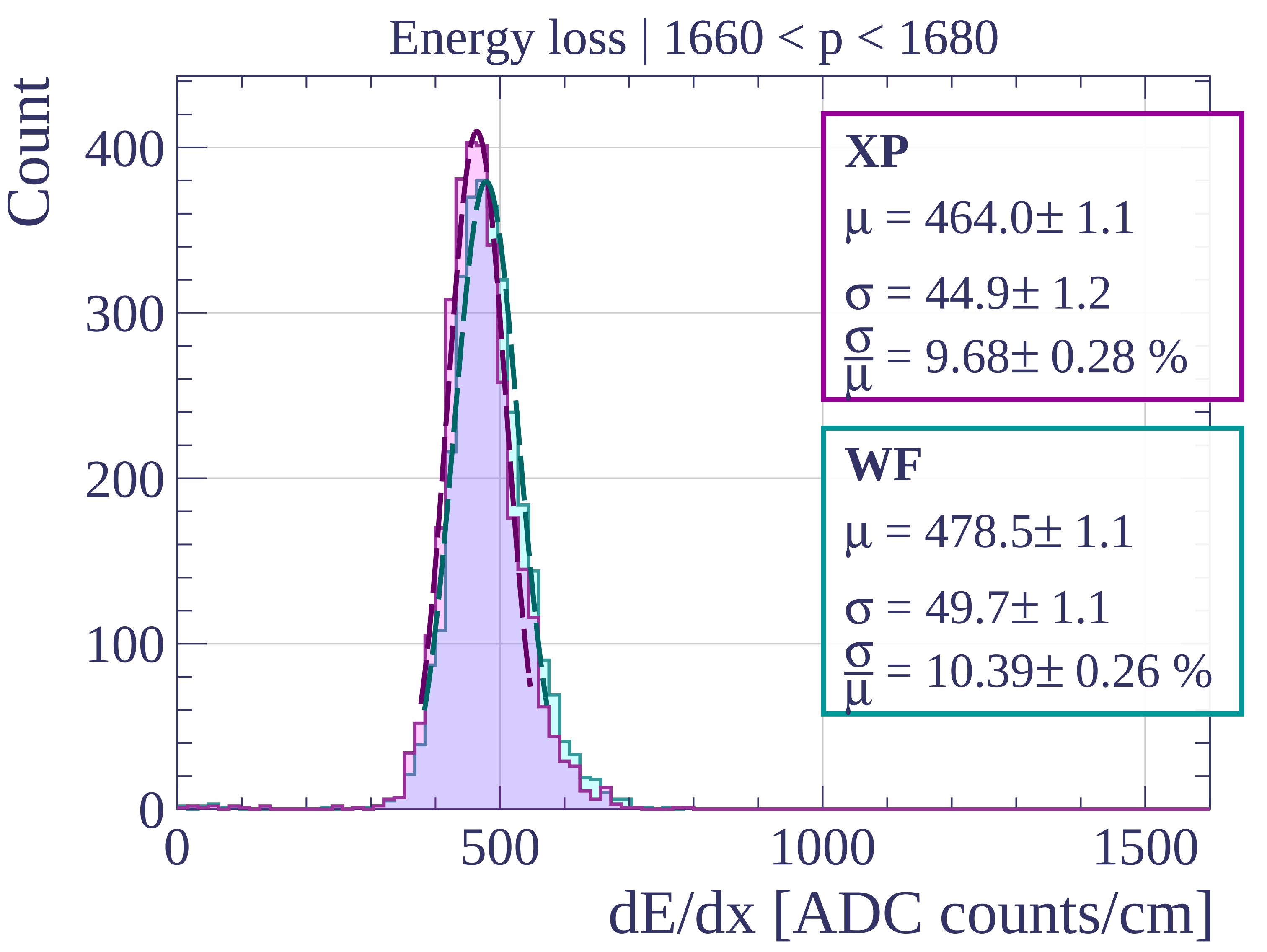


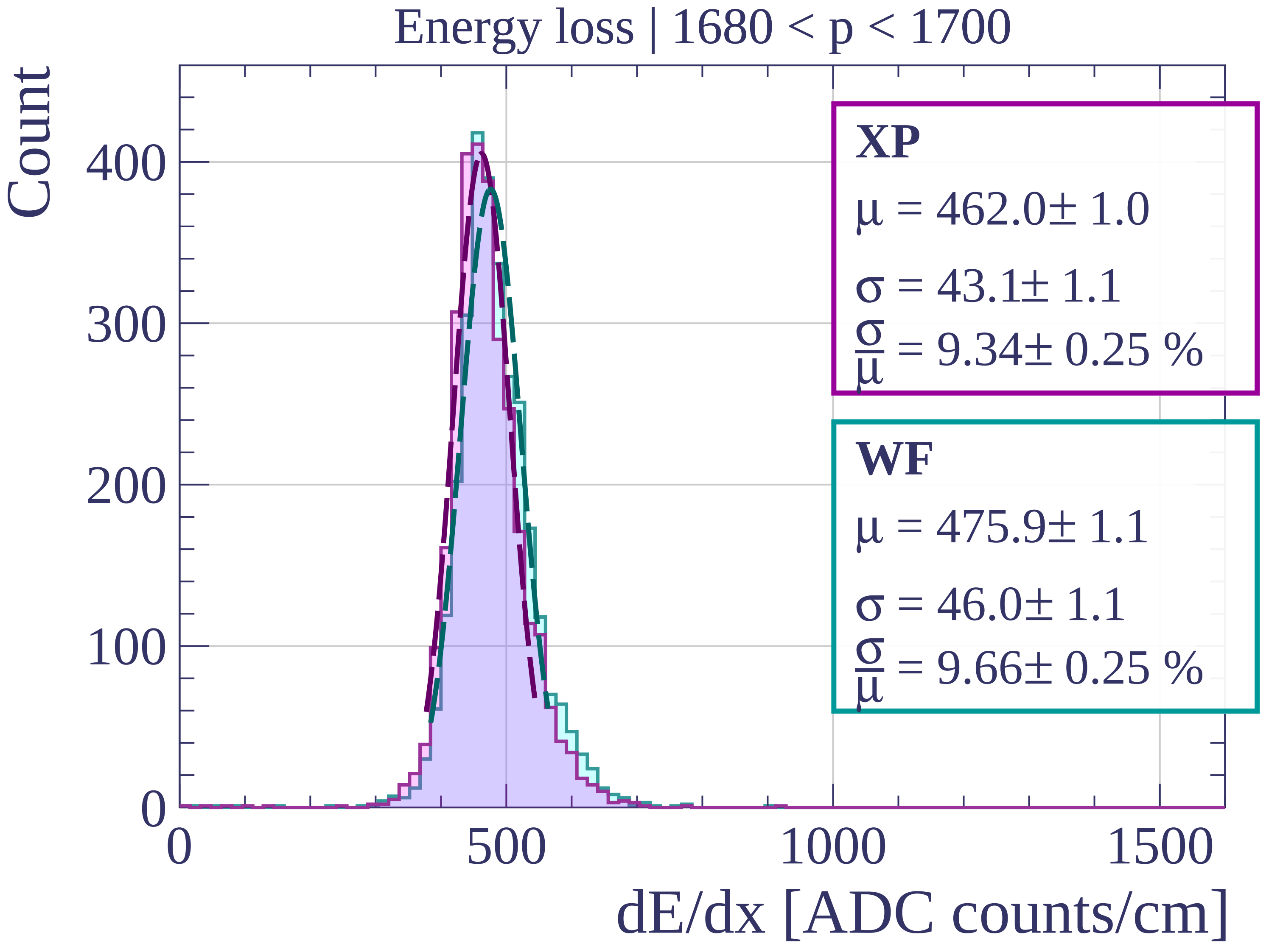
Energy loss | 1600 < p < 1620
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 462.4± 1.0
σ = 44.2± 1.0
σ/μ = 9.55± 0.23 %
WF
μ = 474.3± 1.0
σ = 46.6± 1.1
σ/μ = 9.83± 0.25 %

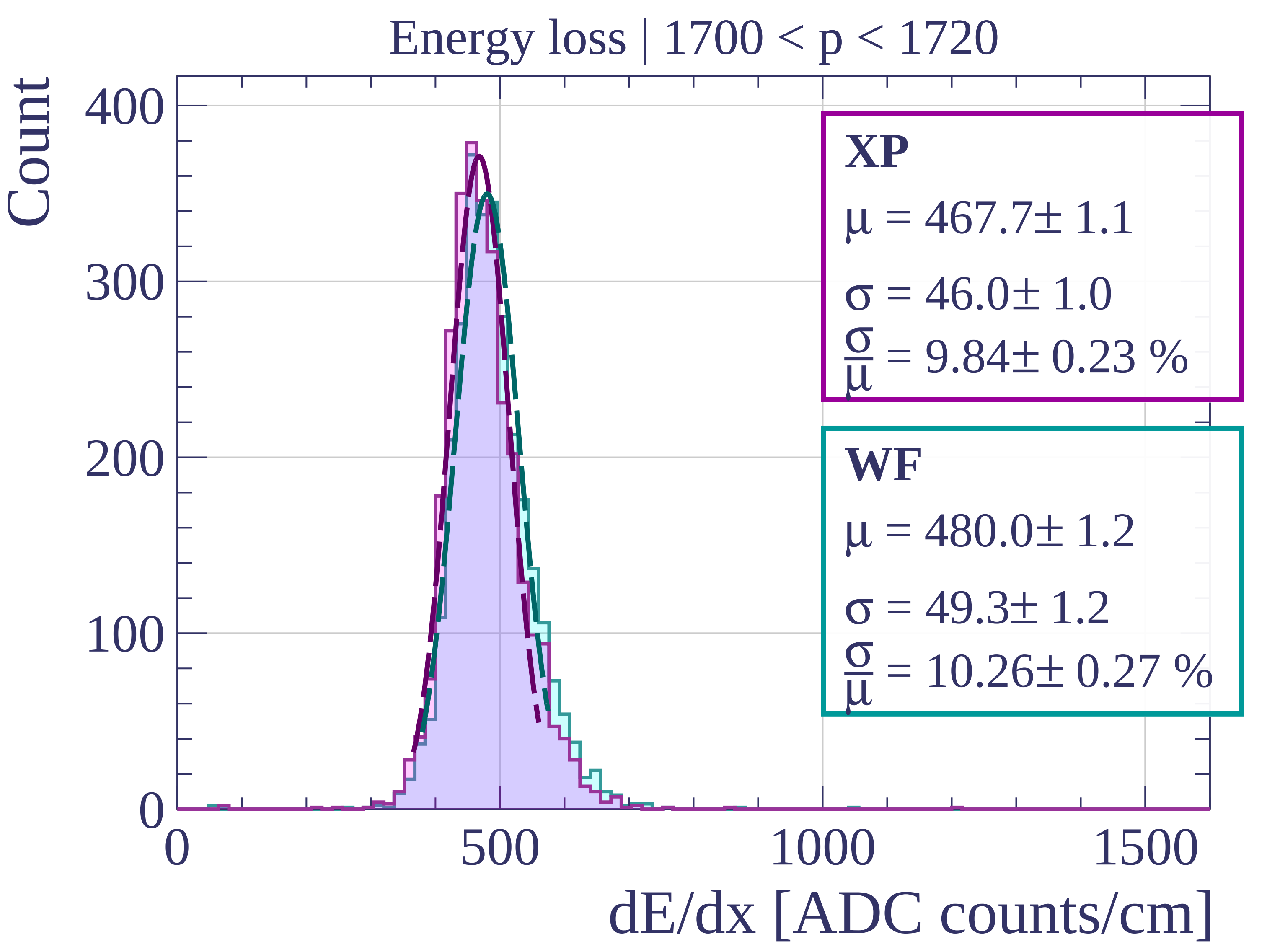

Energy loss | 1620 < p < 1640
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 464.2± 1.1
σ = 47.6± 1.0
σ/μ = 10.24± 0.24 %
WF
μ = 476.5± 1.1
σ = 48.7± 1.0
σ/μ = 10.22± 0.23 %



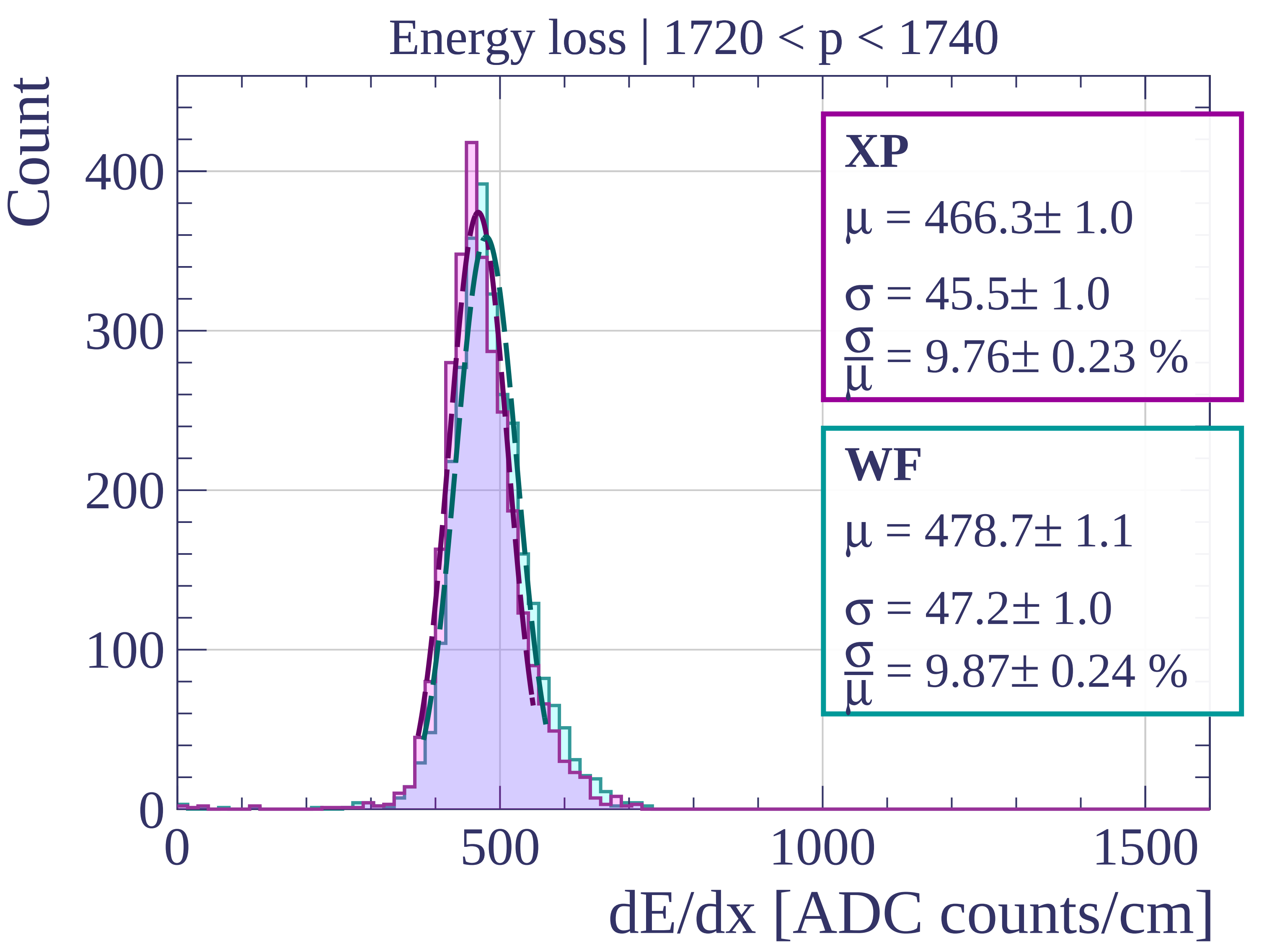
Energy loss | 1640 < p < 1660
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 461.7± 1.1
σ = 45.8± 1.1
σ/μ = 9.92± 0.26 %
WF
μ = 475.8± 1.1
σ = 50.7± 1.1
σ/μ = 10.66± 0.25 %

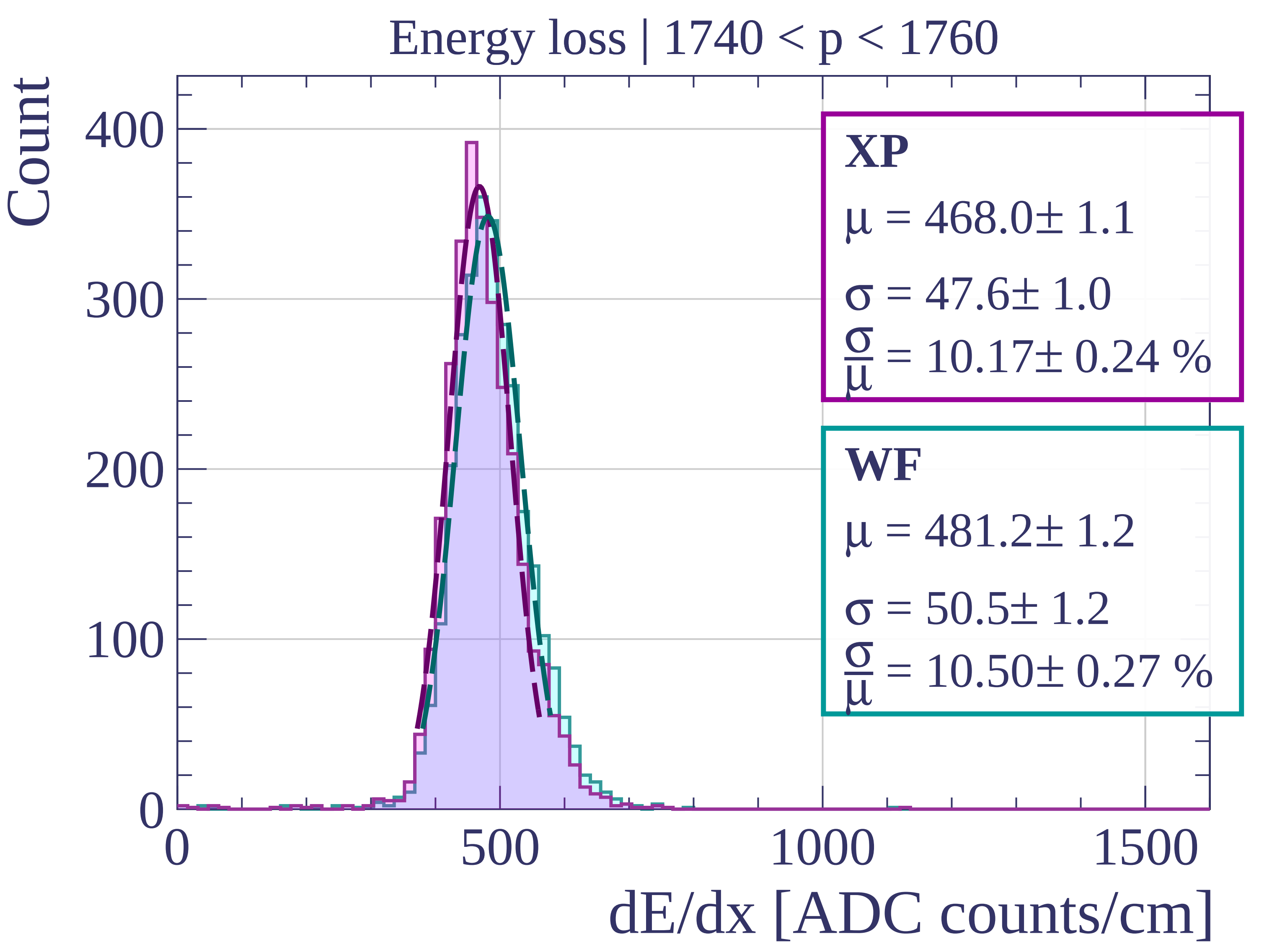

Energy loss | 1660 < p < 1680
Count
0
100
200
300
400
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 464.0± 1.1
σ = 44.9± 1.2
σ/μ = 9.68± 0.28 %
WF
μ = 478.5± 1.1
σ = 49.7± 1.1
σ/μ = 10.39± 0.26 %

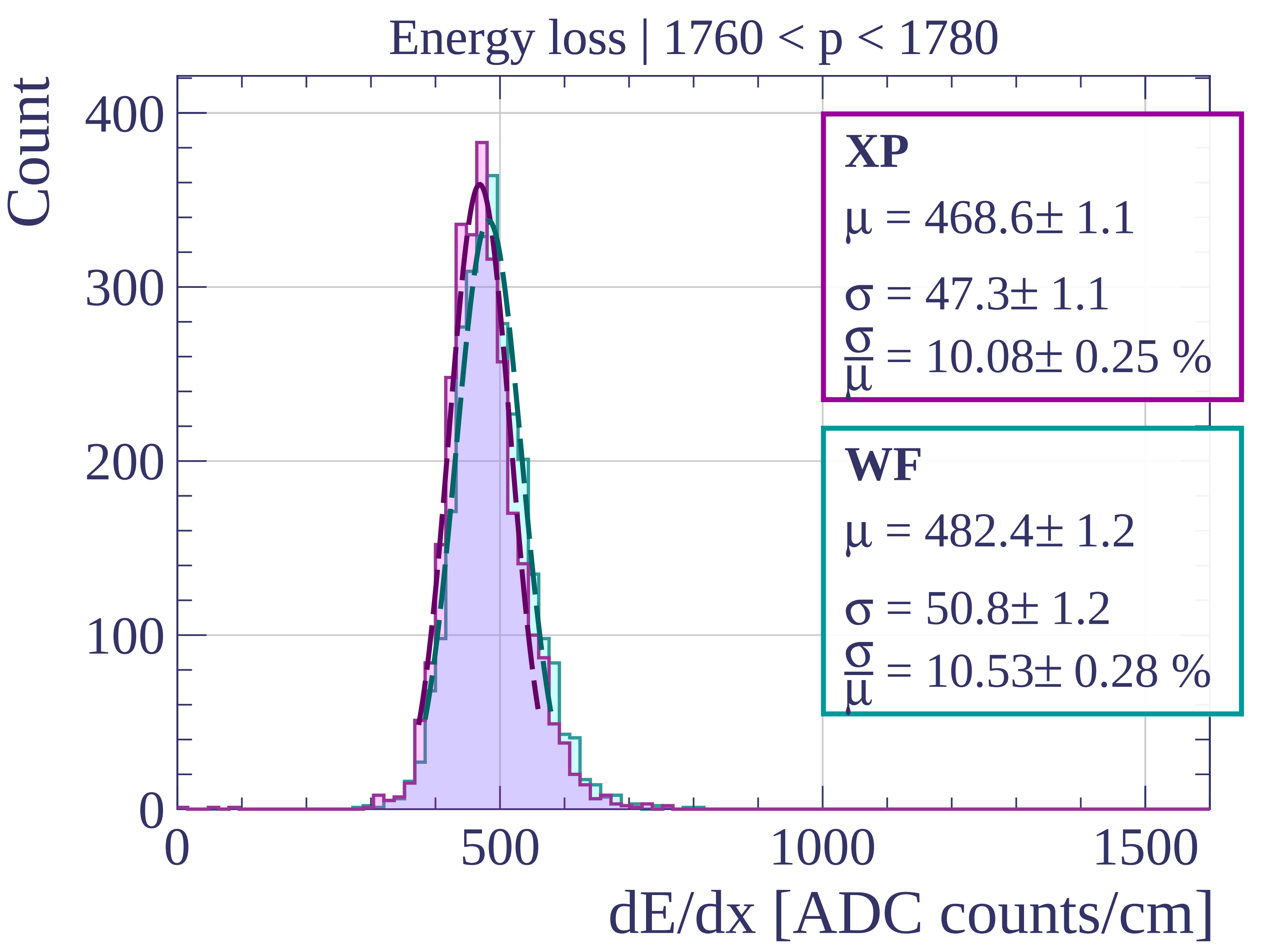

Energy loss | 1680 < p < 1700
Count
dE/dx [ADC counts/cm]
XP
μ = 462.0± 1.0
σ = 43.1± 1.1
σ/μ = 9.34± 0.25 %
WF
μ = 475.9± 1.1
σ = 46.0± 1.1
σ/μ = 9.66± 0.25 %



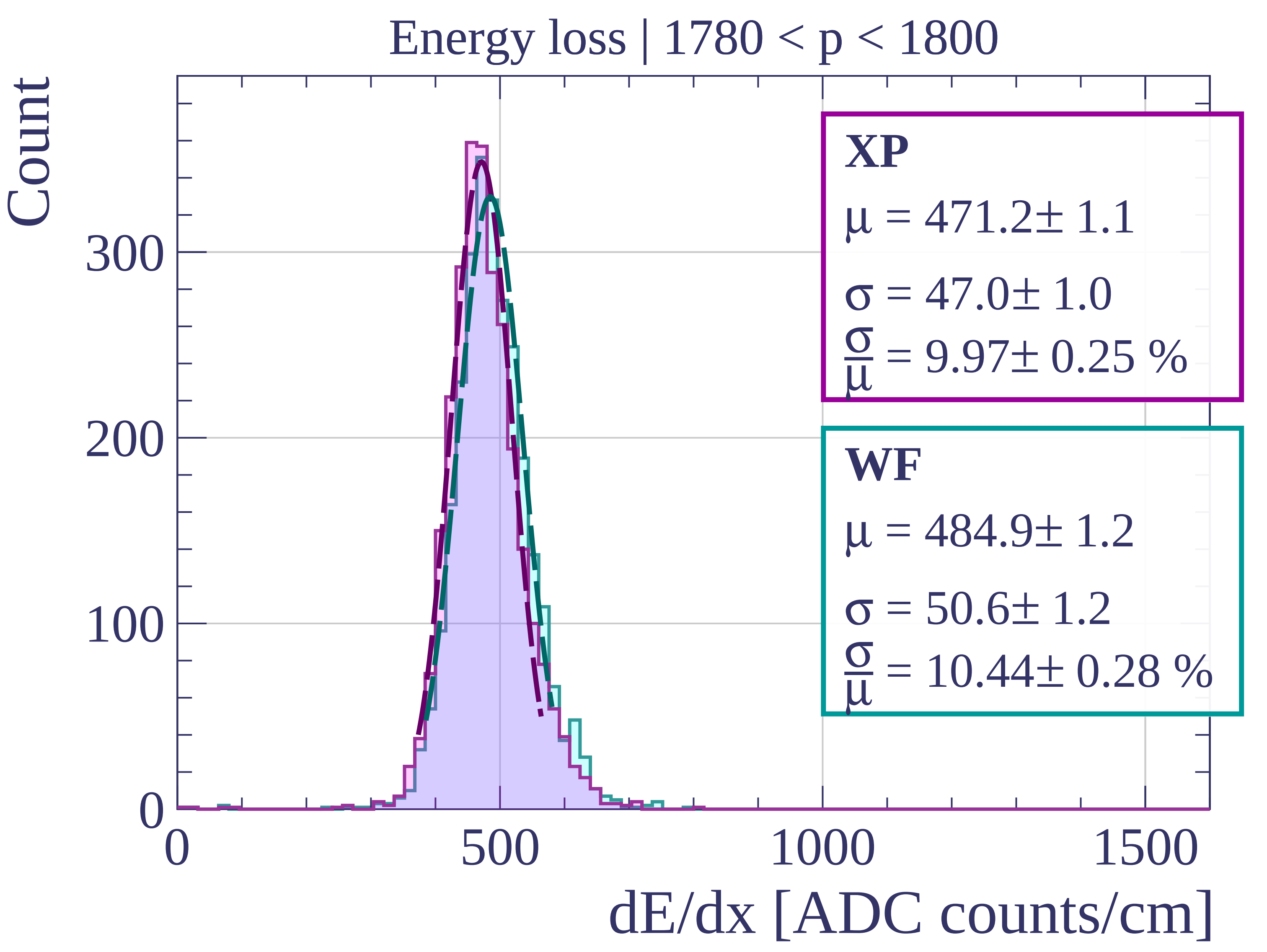
Energy loss | 1700 < p < 1720
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 467.7± 1.1
σ = 46.0± 1.0
σ/μ = 9.84± 0.23 %
WF
μ = 480.0± 1.2
σ = 49.3± 1.2
σ/μ = 10.26± 0.27 %


Energy loss | 1720 < p < 1740
Count
0
100
200
300
400
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 466.3± 1.0
σ = 45.5± 1.0
σ/μ = 9.76± 0.23 %
WF
μ = 478.7± 1.1
σ = 47.2± 1.0
σ/μ = 9.87± 0.24 %

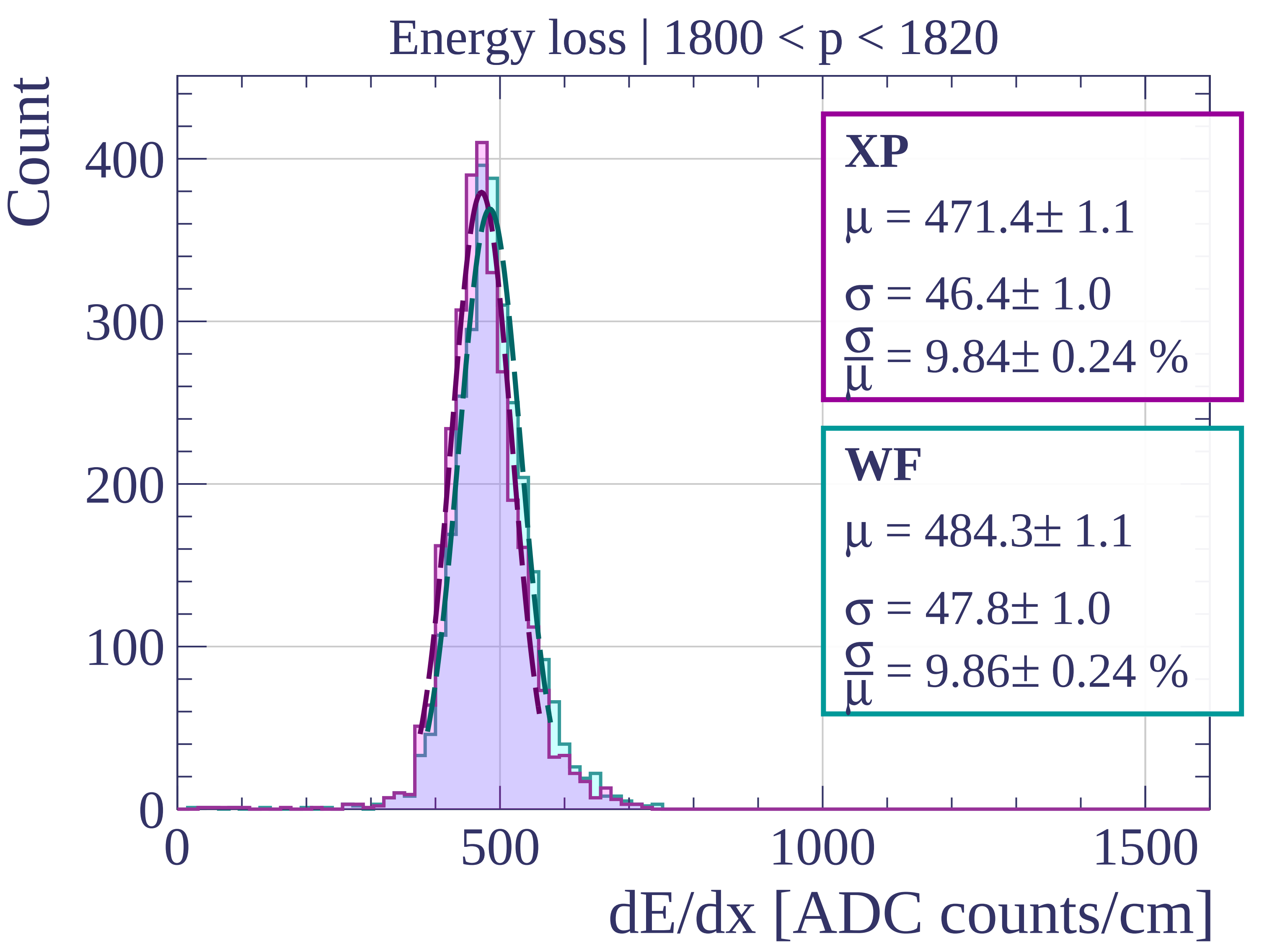

Energy loss | 1740 < p < 1760
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 468.0± 1.1
σ = 47.6± 1.0
σ/μ = 10.17± 0.24 %
WF
μ = 481.2± 1.2
σ = 50.5± 1.2
σ/μ = 10.50± 0.27 %



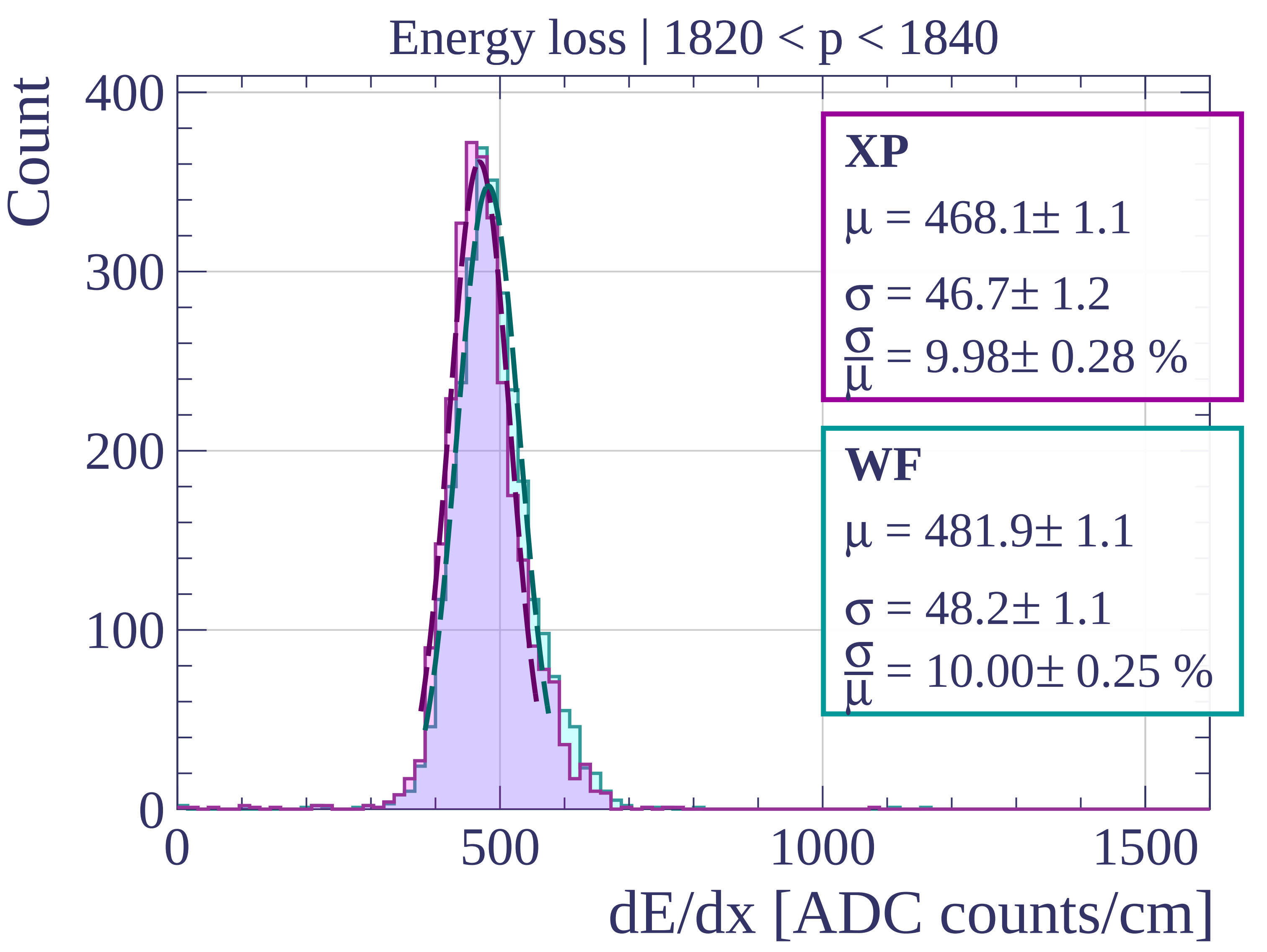
Energy loss | 1760 < p < 1780
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
XP
μ = 468.6± 1.1
σ = 47.3± 1.1
σ/μ = 10.08± 0.25 %
WF
μ = 482.4± 1.2
σ = 50.8± 1.2
σ/μ = 10.53± 0.28 %



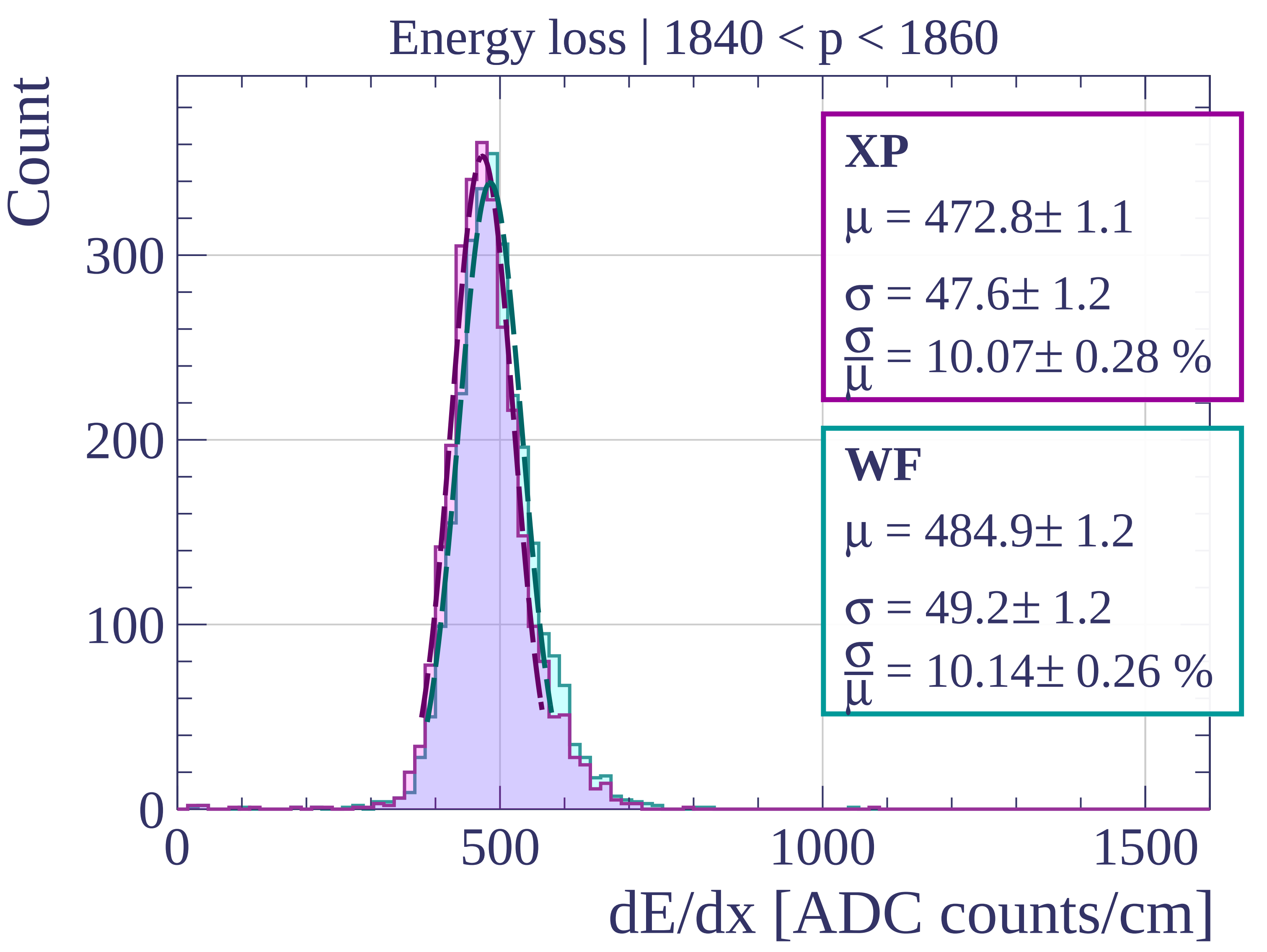
Energy loss | 1780 < p < 1800
Count
dE/dx [ADC counts/cm]
0
100
200
300
500
1000
1500
XP
μ = 471.2± 1.1
σ = 47.0± 1.0
σ/μ = 9.97± 0.25 %
WF
μ = 484.9± 1.2
σ = 50.6± 1.2
σ/μ = 10.44± 0.28 %



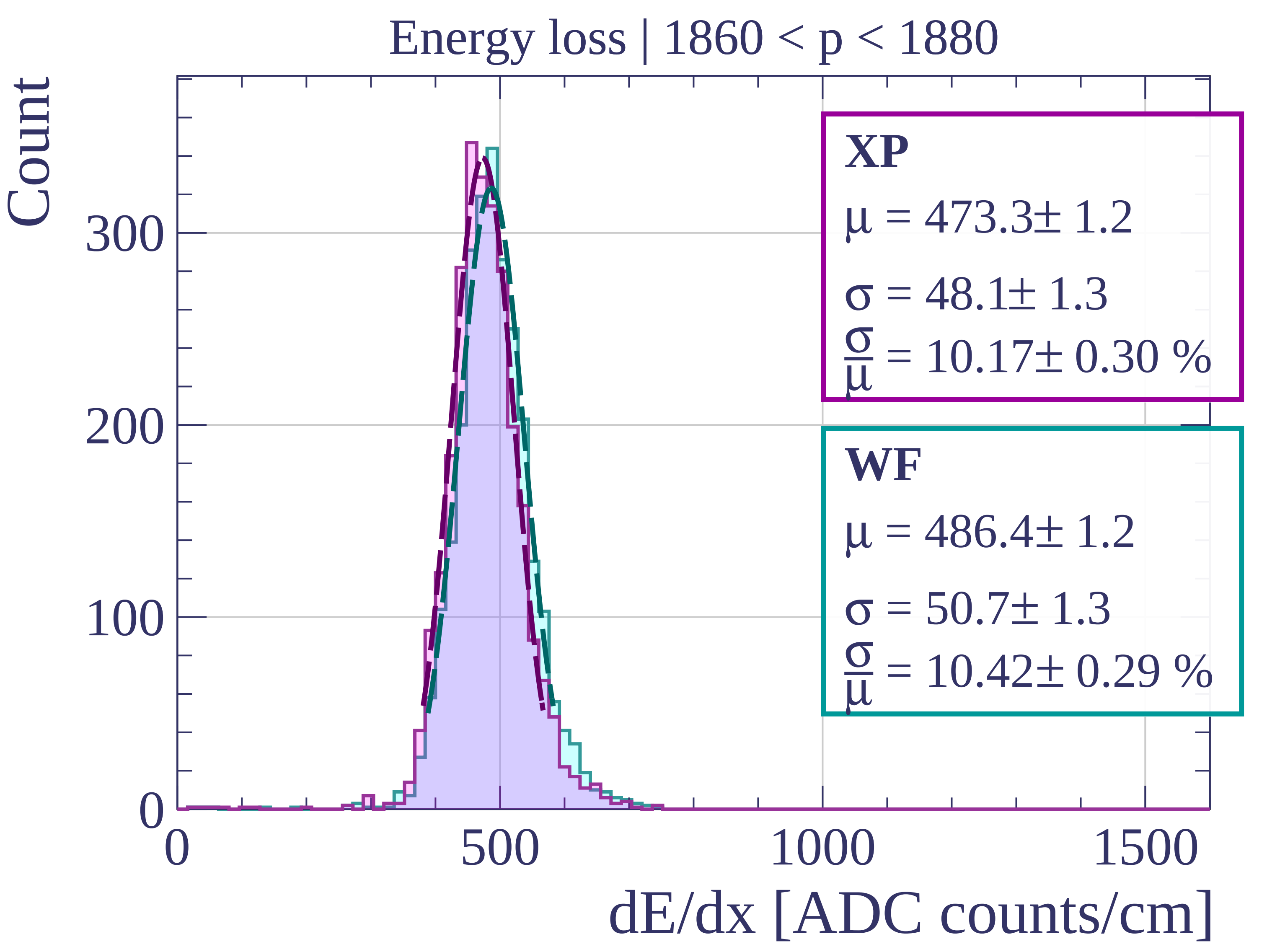
Energy loss | 1800 < p < 1820
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 471.4± 1.1
σ = 46.4± 1.0
σ/μ = 9.84± 0.24 %
WF
μ = 484.3± 1.1
σ = 47.8± 1.0
σ/μ = 9.86± 0.24 %



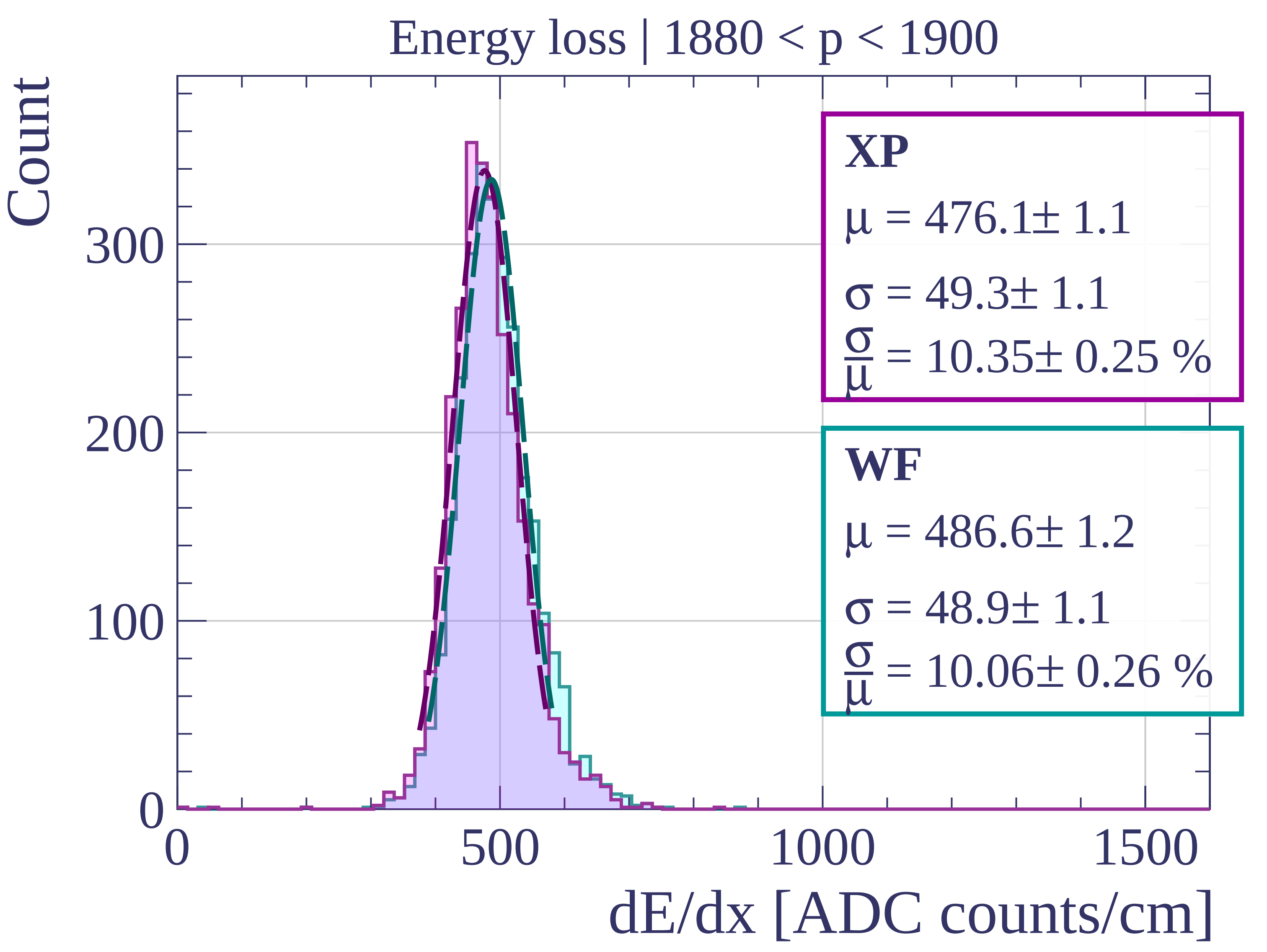
Energy loss | 1820 < p < 1840
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 468.1± 1.1
σ = 46.7± 1.2
σ/μ = 9.98± 0.28 %
WF
μ = 481.9± 1.1
σ = 48.2± 1.1
σ/μ = 10.00± 0.25 %



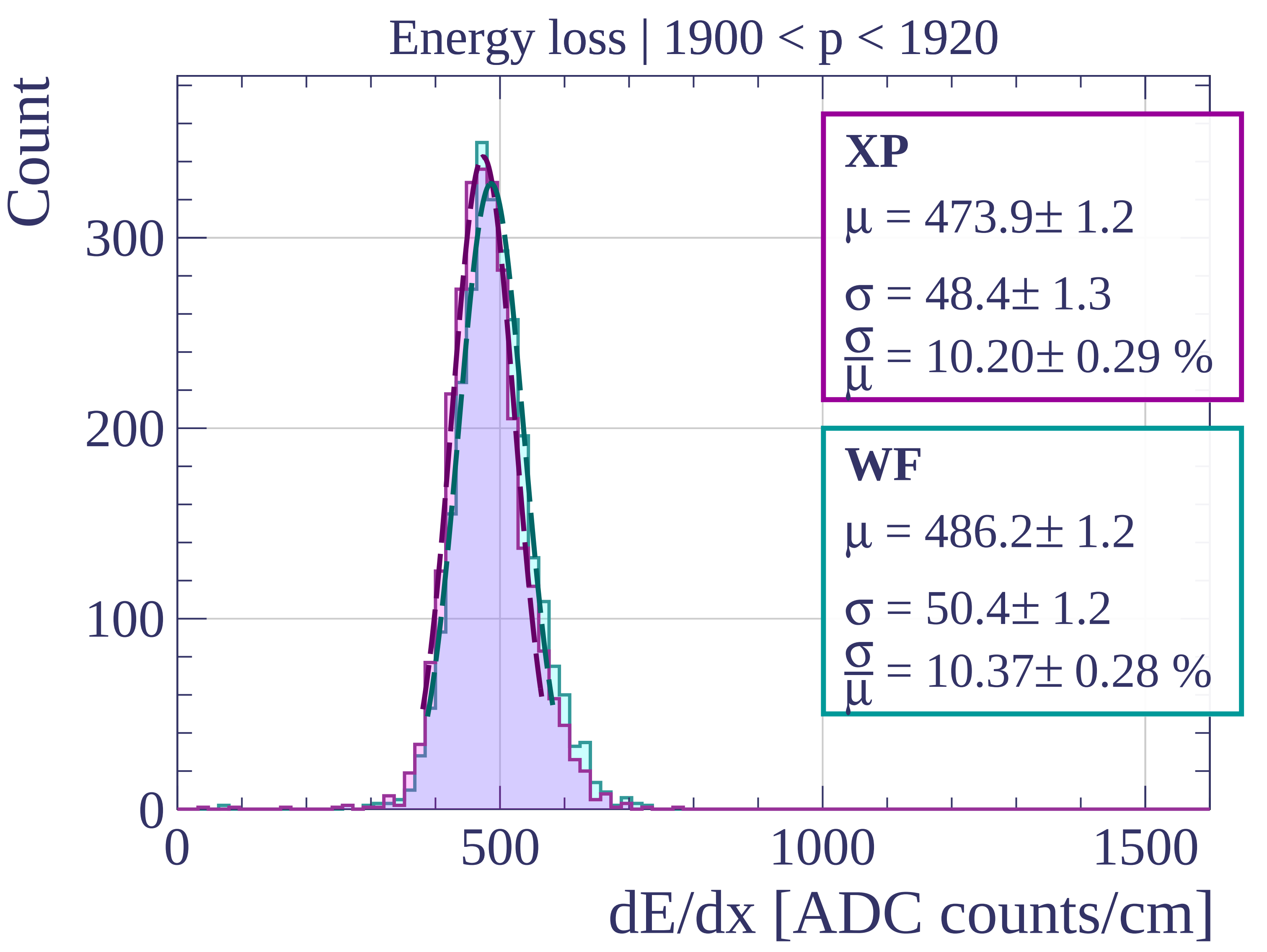
Energy loss | 1840 < p < 1860
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 472.8± 1.1
σ = 47.6± 1.2
σ/μ = 10.07± 0.28 %
WF
μ = 484.9± 1.2
σ = 49.2± 1.2
σ/μ = 10.14± 0.26 %



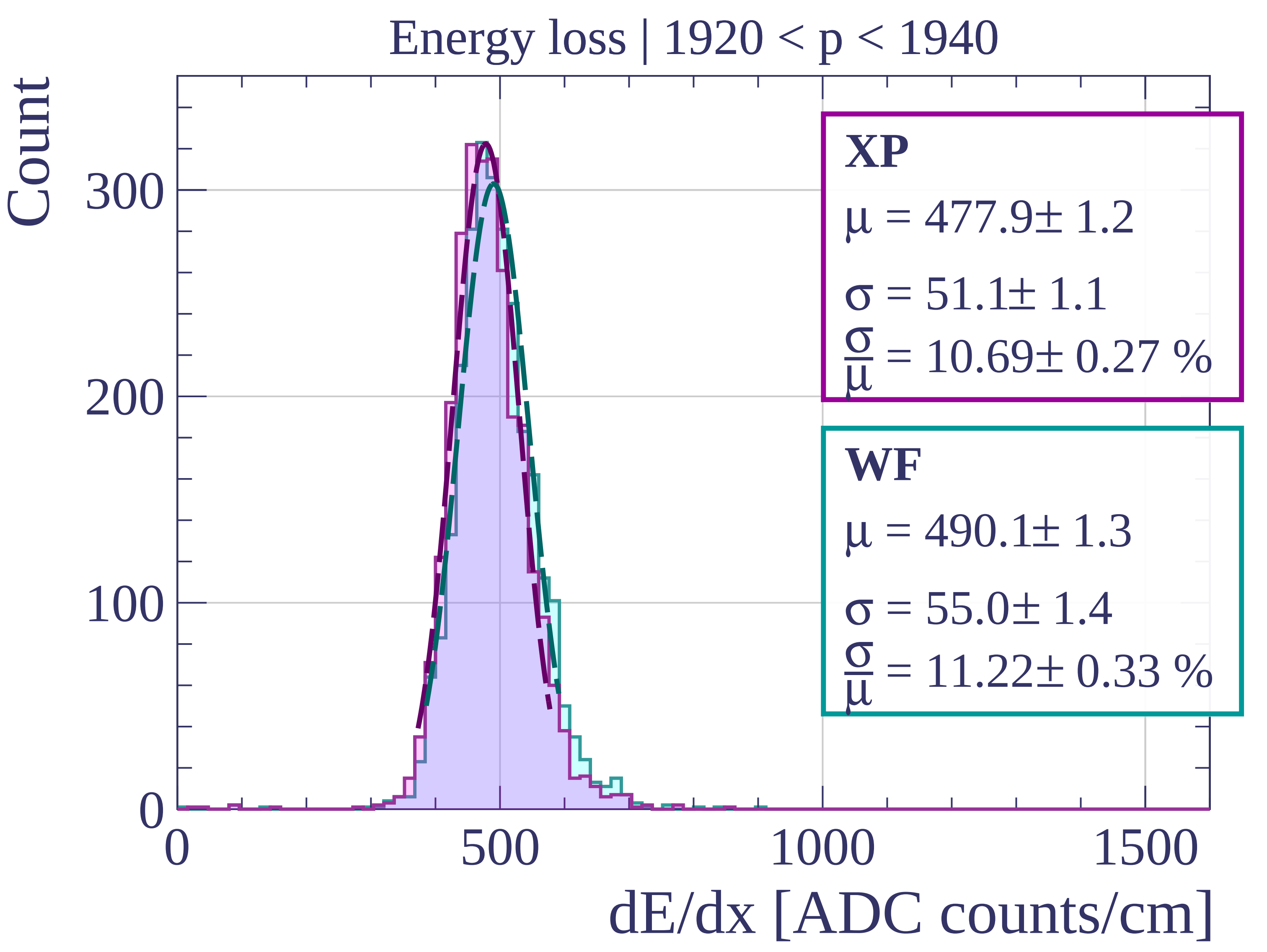
Energy loss | 1860 < p < 1880
Count
0
100
200
300
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 473.3± 1.2
σ = 48.1± 1.3
σ/μ = 10.17± 0.30 %
WF
μ = 486.4± 1.2
σ = 50.7± 1.3
σ/μ = 10.42± 0.29 %



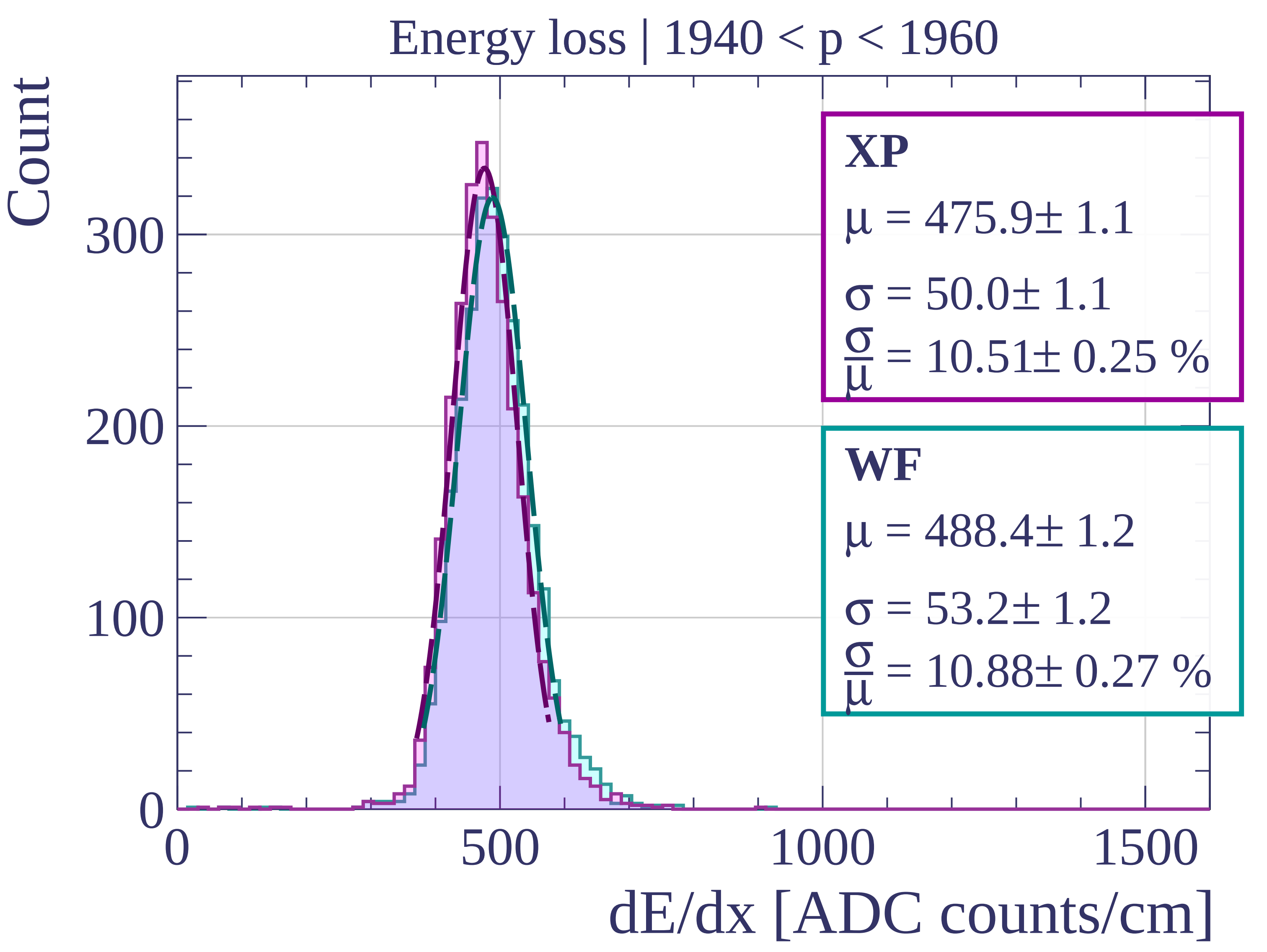
Energy loss | 1880 < p < 1900
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 476.1± 1.1
σ = 49.3± 1.1
σ/μ = 10.35± 0.25 %
WF
μ = 486.6± 1.2
σ = 48.9± 1.1
σ/μ = 10.06± 0.26 %



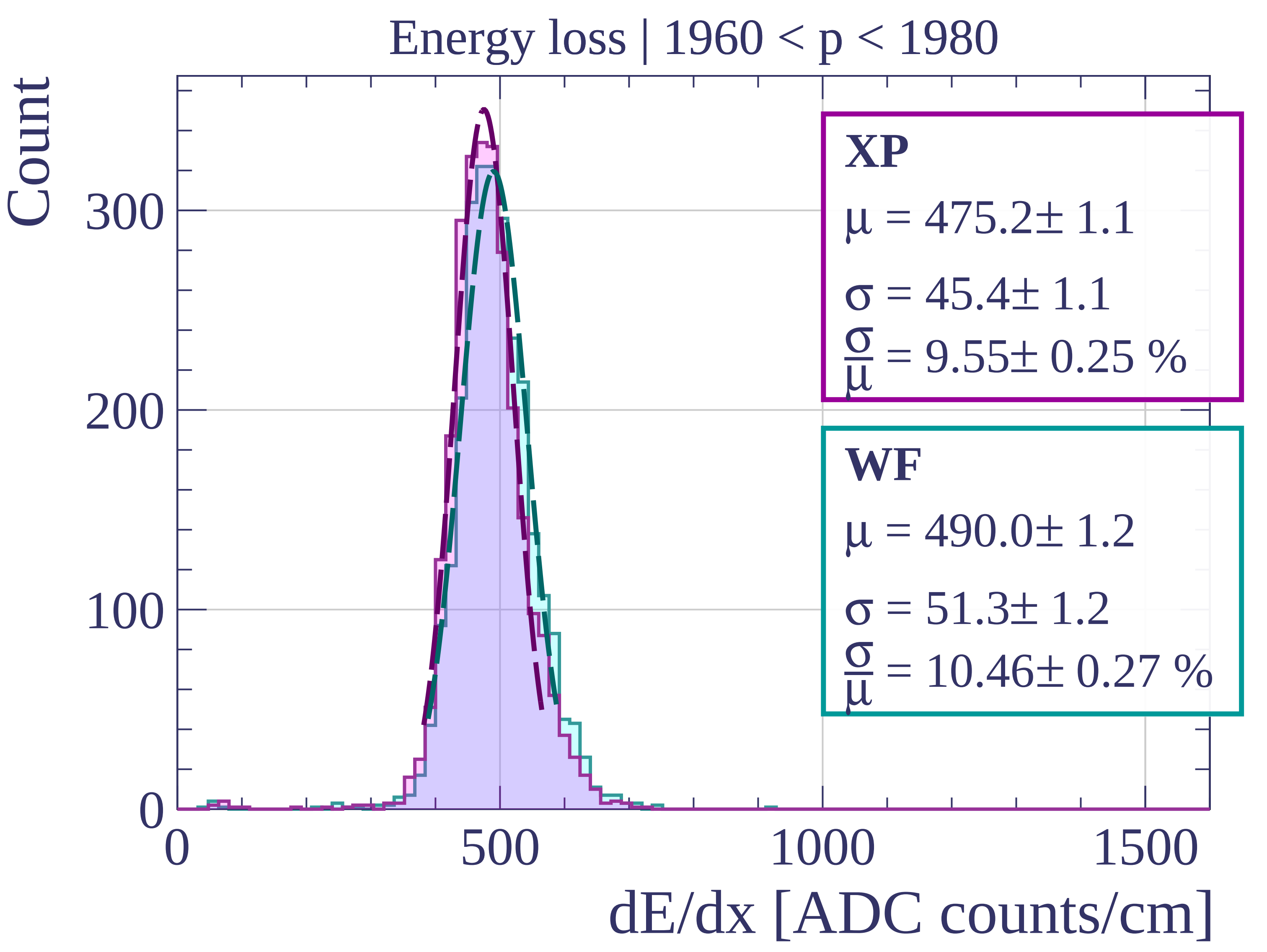
Energy loss | 1900 < p < 1920
Count
0
100
200
300
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 473.9± 1.2
σ = 48.4± 1.3
σ/μ = 10.20± 0.29 %
WF
μ = 486.2± 1.2
σ = 50.4± 1.2
σ/μ = 10.37± 0.28 %

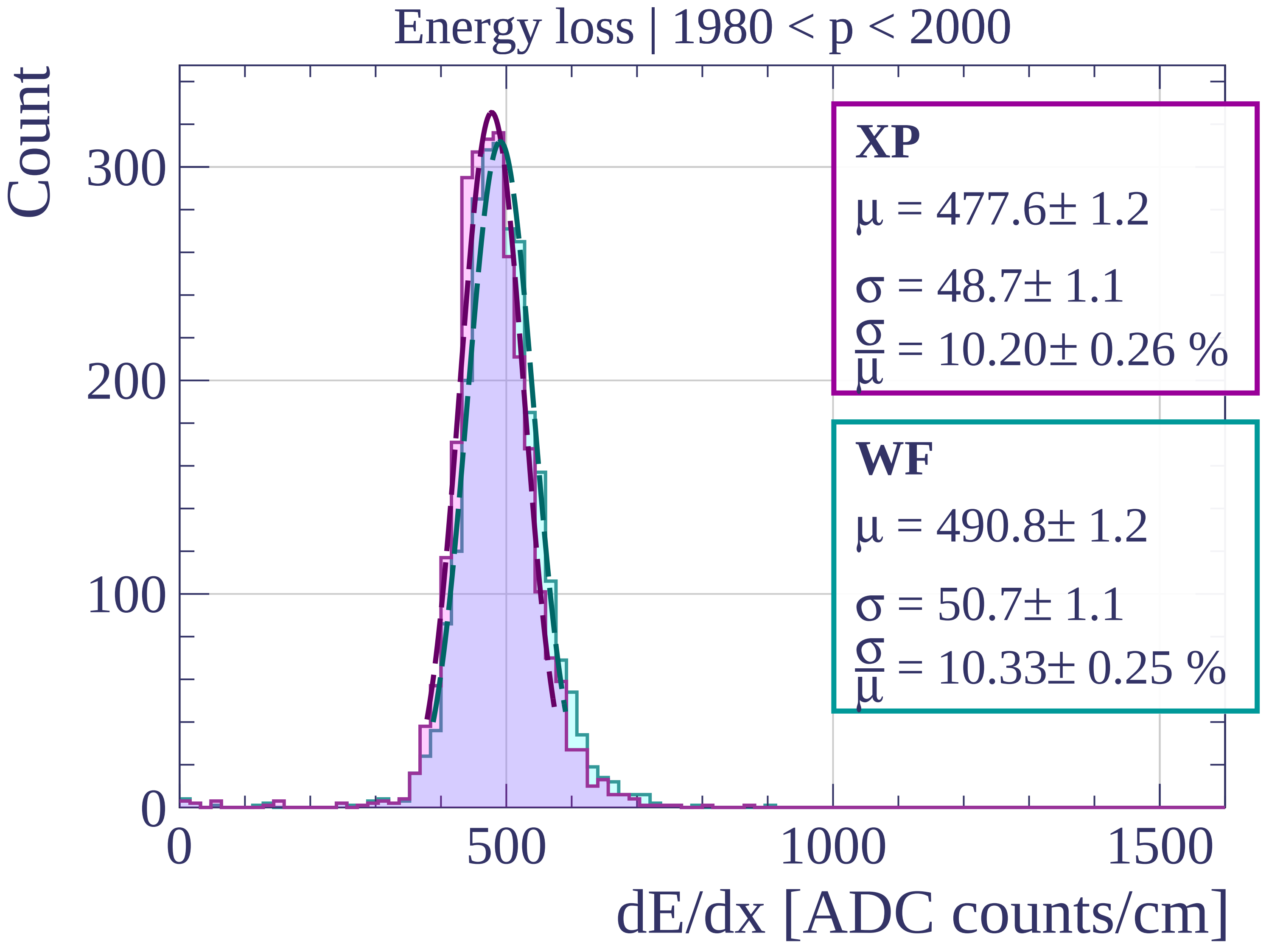


Energy loss | 1920 < p < 1940
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 477.9± 1.2
σ = 51.1± 1.1
σ/μ = 10.69± 0.27 %
WF
μ = 490.1± 1.3
σ = 55.0± 1.4
σ/μ = 11.22± 0.33 %

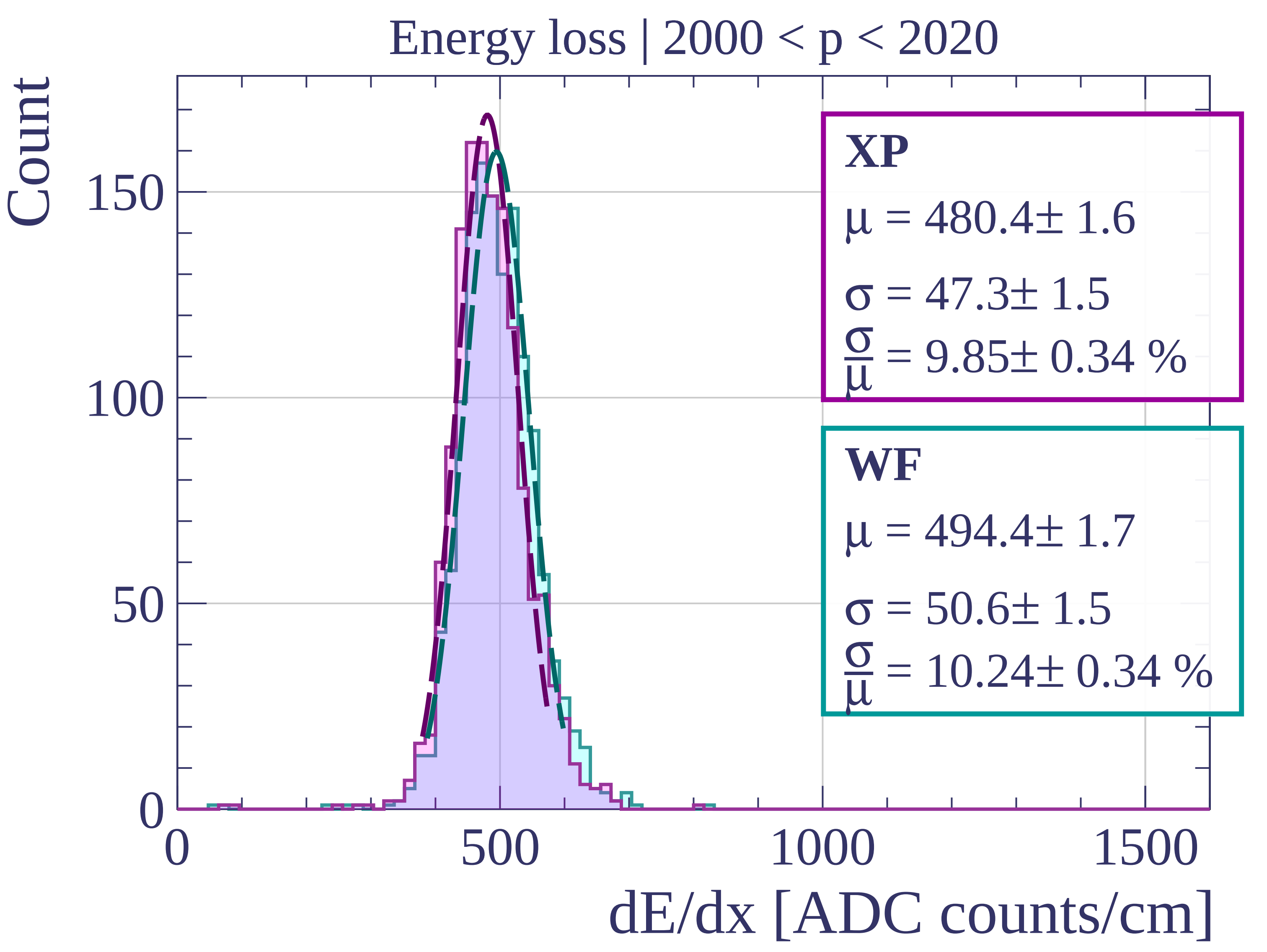


Energy loss | 1940 < p < 1960
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 475.9± 1.1
σ = 50.0± 1.1
σ/μ = 10.51± 0.25 %
WF
μ = 488.4± 1.2
σ = 53.2± 1.2
σ/μ = 10.88± 0.27 %

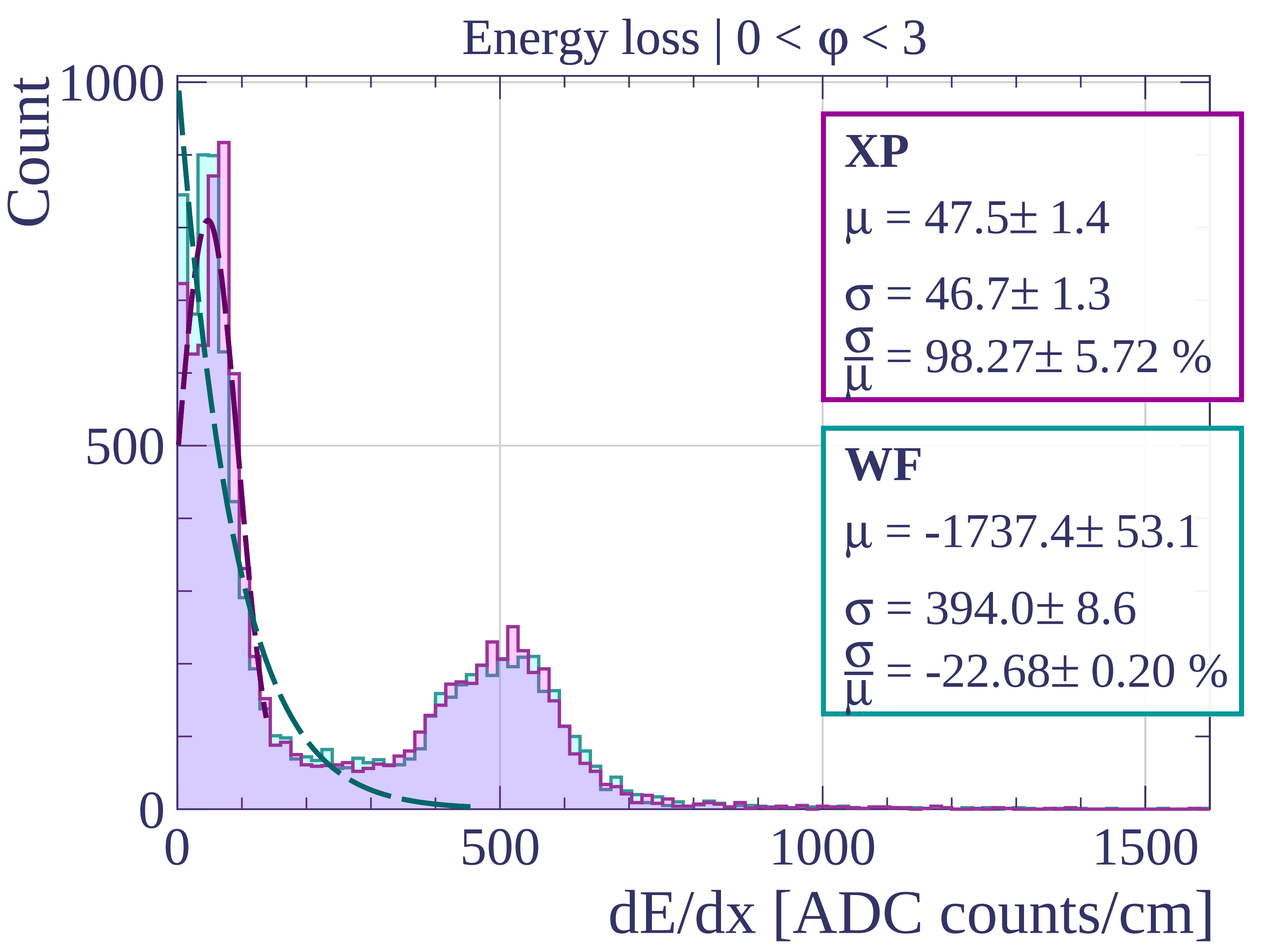

Energy loss | 1960 < p < 1980
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
μ = 475.2± 1.1
σ = 45.4± 1.1
σ/μ = 9.55± 0.25 %
WF
μ = 490.0± 1.2
σ = 51.3± 1.2
σ/μ = 10.46± 0.27 %



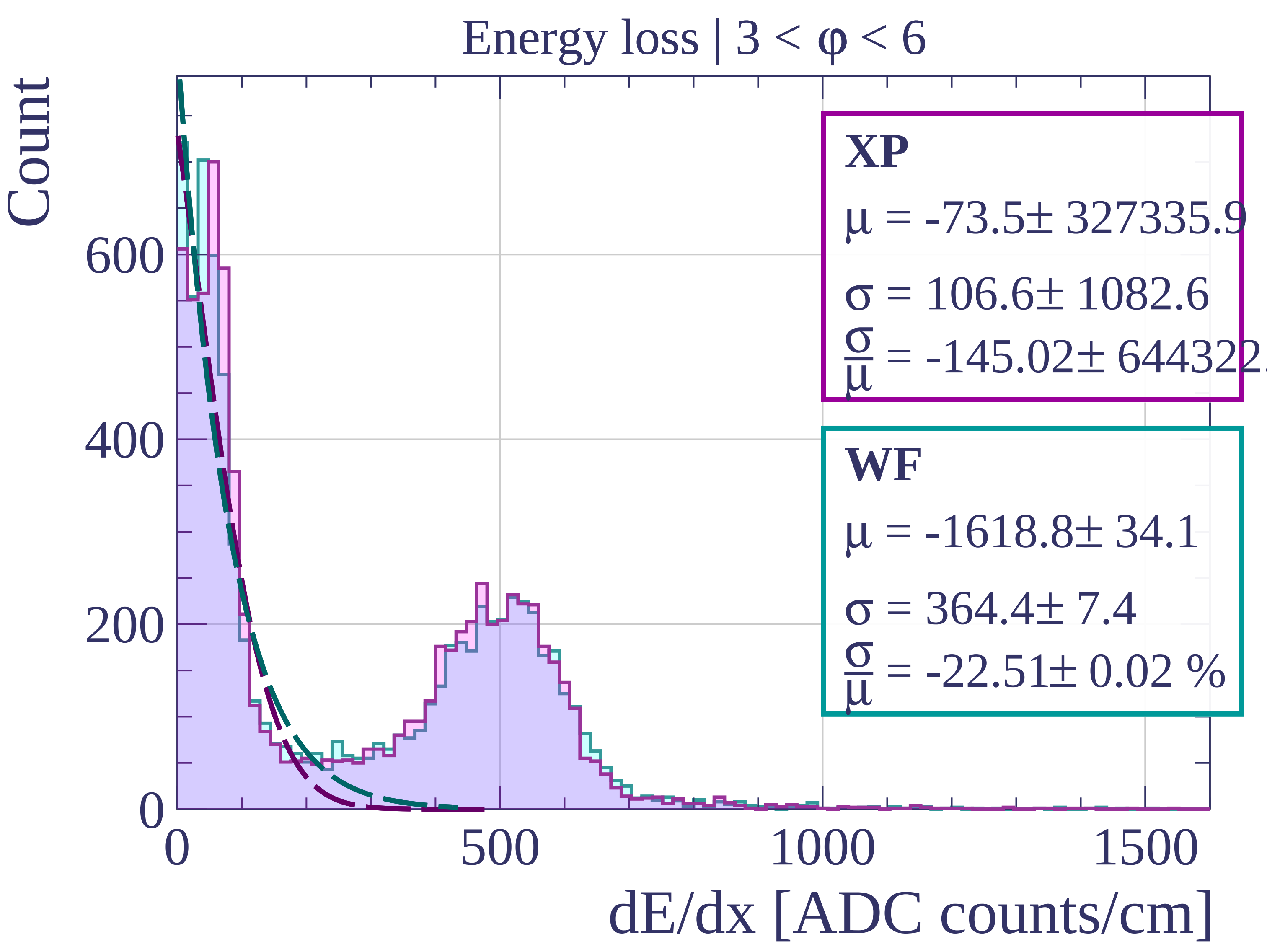
Energy loss | 1980 < p < 2000
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 477.6± 1.2
σ = 48.7± 1.1
σ/μ = 10.20± 0.26 %
WF
μ = 490.8± 1.2
σ = 50.7± 1.1
σ/μ = 10.33± 0.25 %



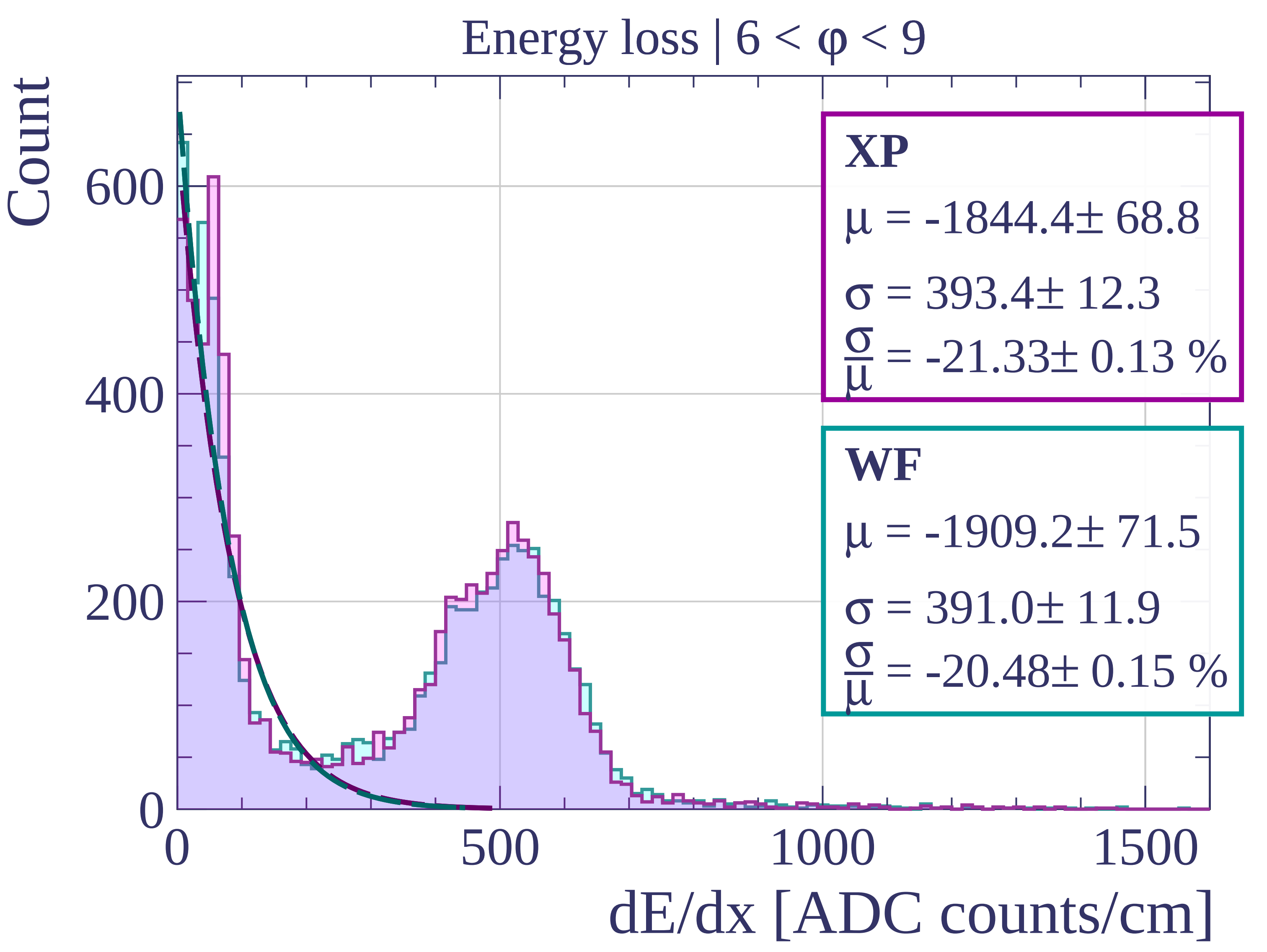
Energy loss | 2000 < p < 2020
Count
0
50
100
150
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 480.4± 1.6
σ = 47.3± 1.5
σ/μ = 9.85± 0.34 %
WF
μ = 494.4± 1.7
σ = 50.6± 1.5
σ/μ = 10.24± 0.34 %



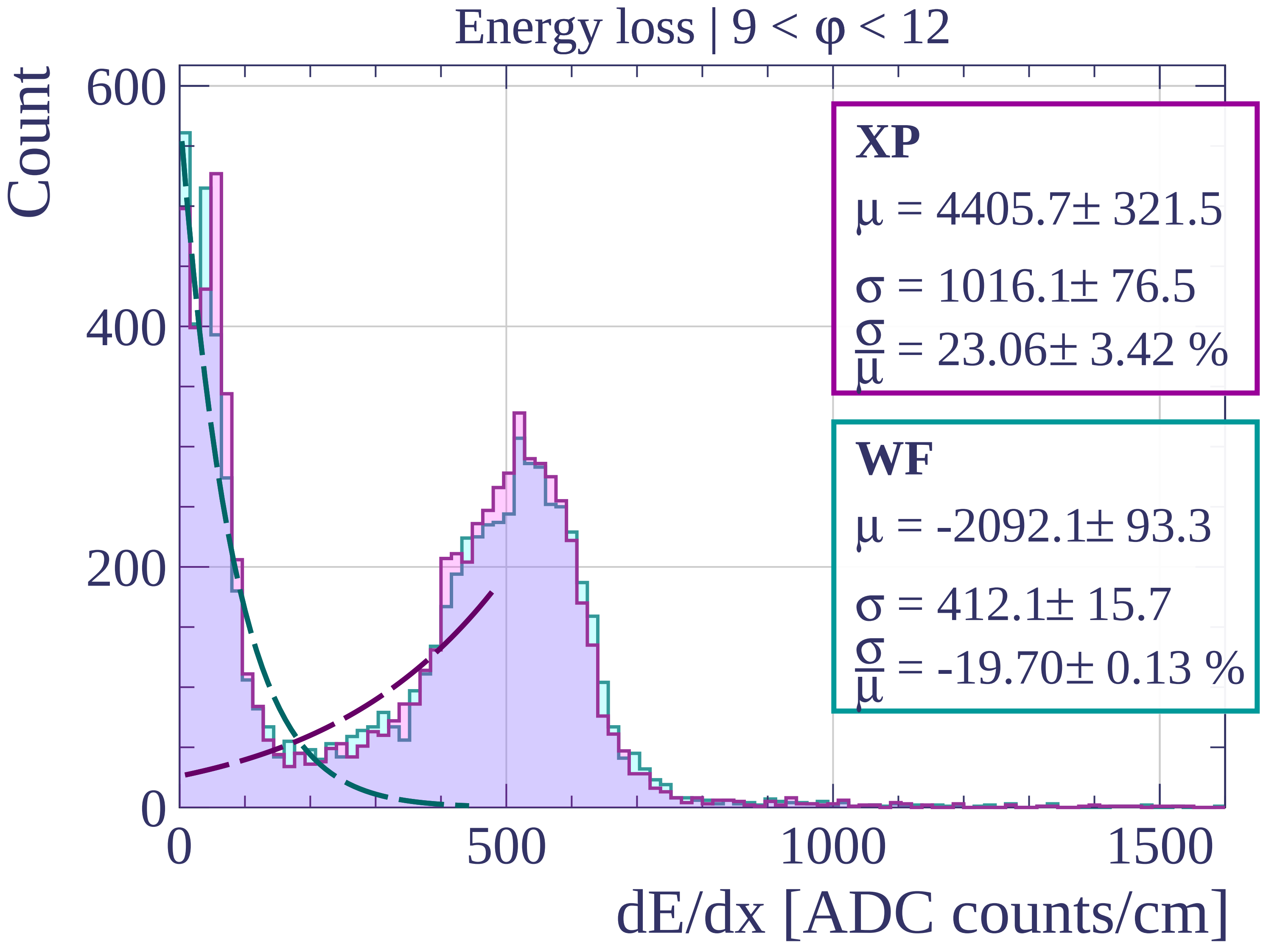
Energy loss | 0 < φ < 3
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 47.5± 1.4
σ = 46.7± 1.3
σ/μ = 98.27± 5.72 %
WF
μ = -1737.4± 53.1
σ = 394.0± 8.6
σ/μ = -22.68± 0.20 %



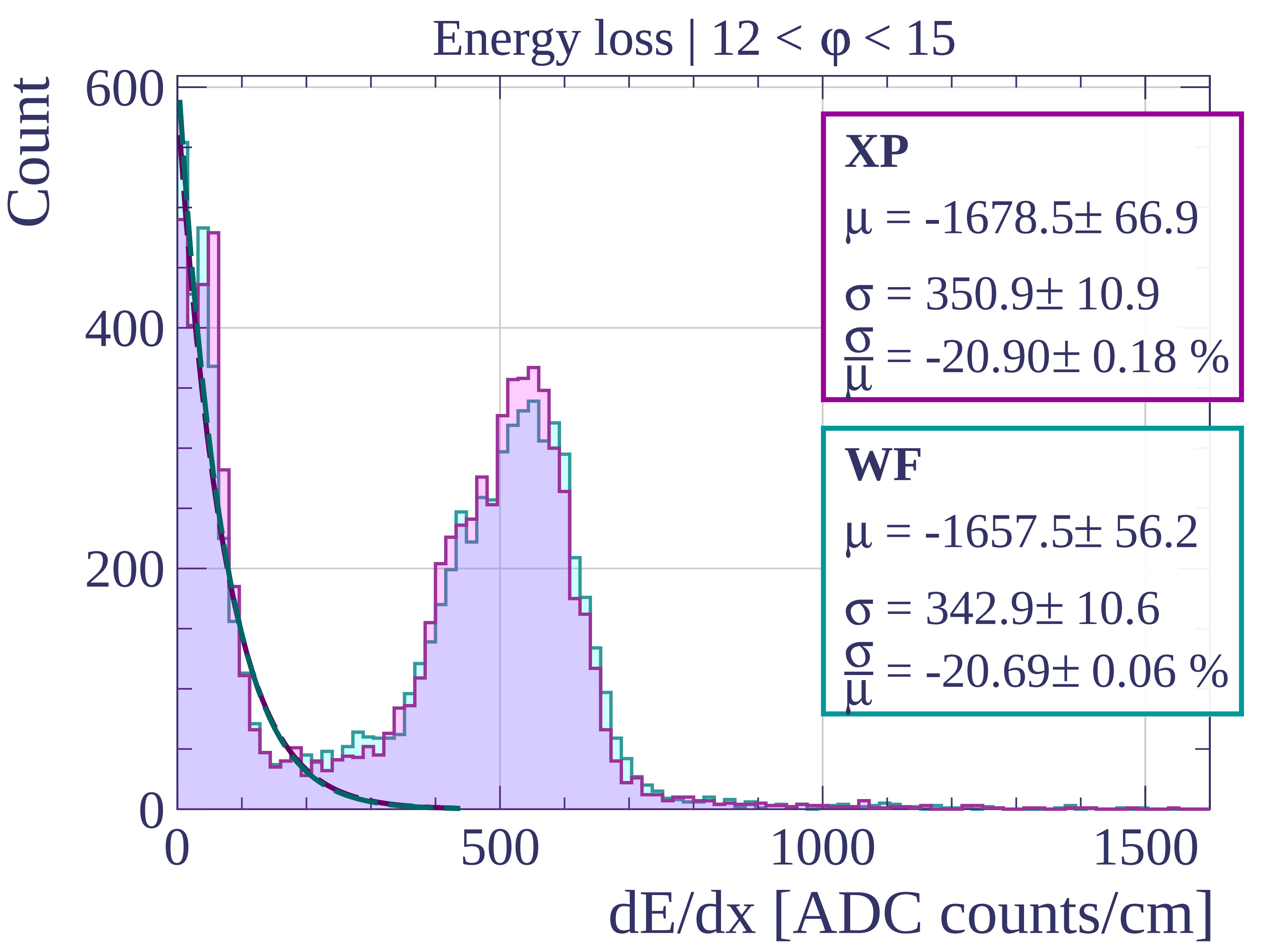
Energy loss | 3 < φ < 6
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = -73.5± 327335.9
σ = 106.6± 1082.6
σ/μ = -145.02± 644322.
WF
μ = -1618.8± 34.1
σ = 364.4± 7.4
σ/μ = -22.51± 0.02 %



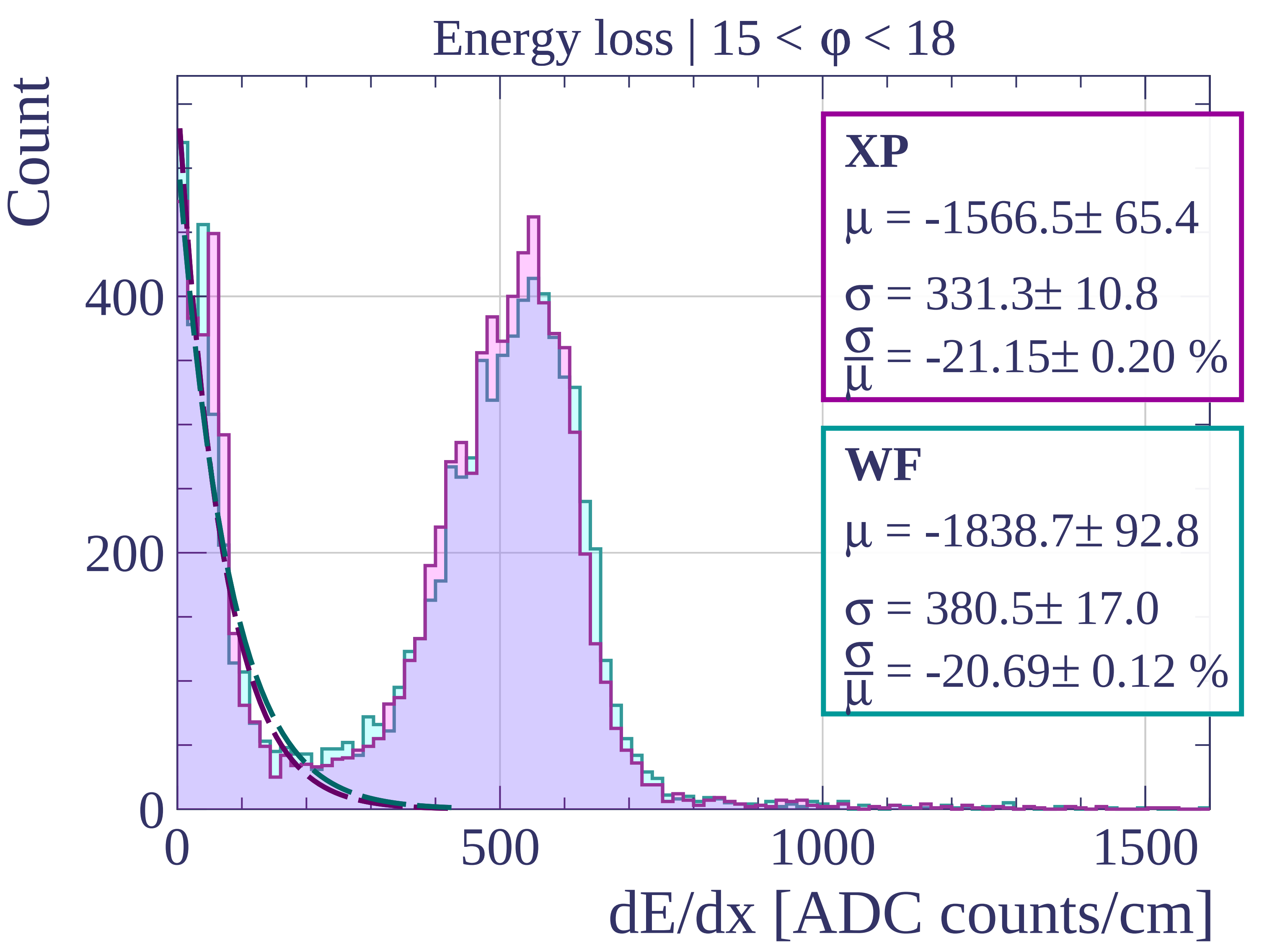
Energy loss | 6 < φ < 9
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = -1844.4± 68.8
σ = 393.4± 12.3
σ/μ = -21.33± 0.13 %
WF
μ = -1909.2± 71.5
σ = 391.0± 11.9
σ/μ = -20.48± 0.15 %



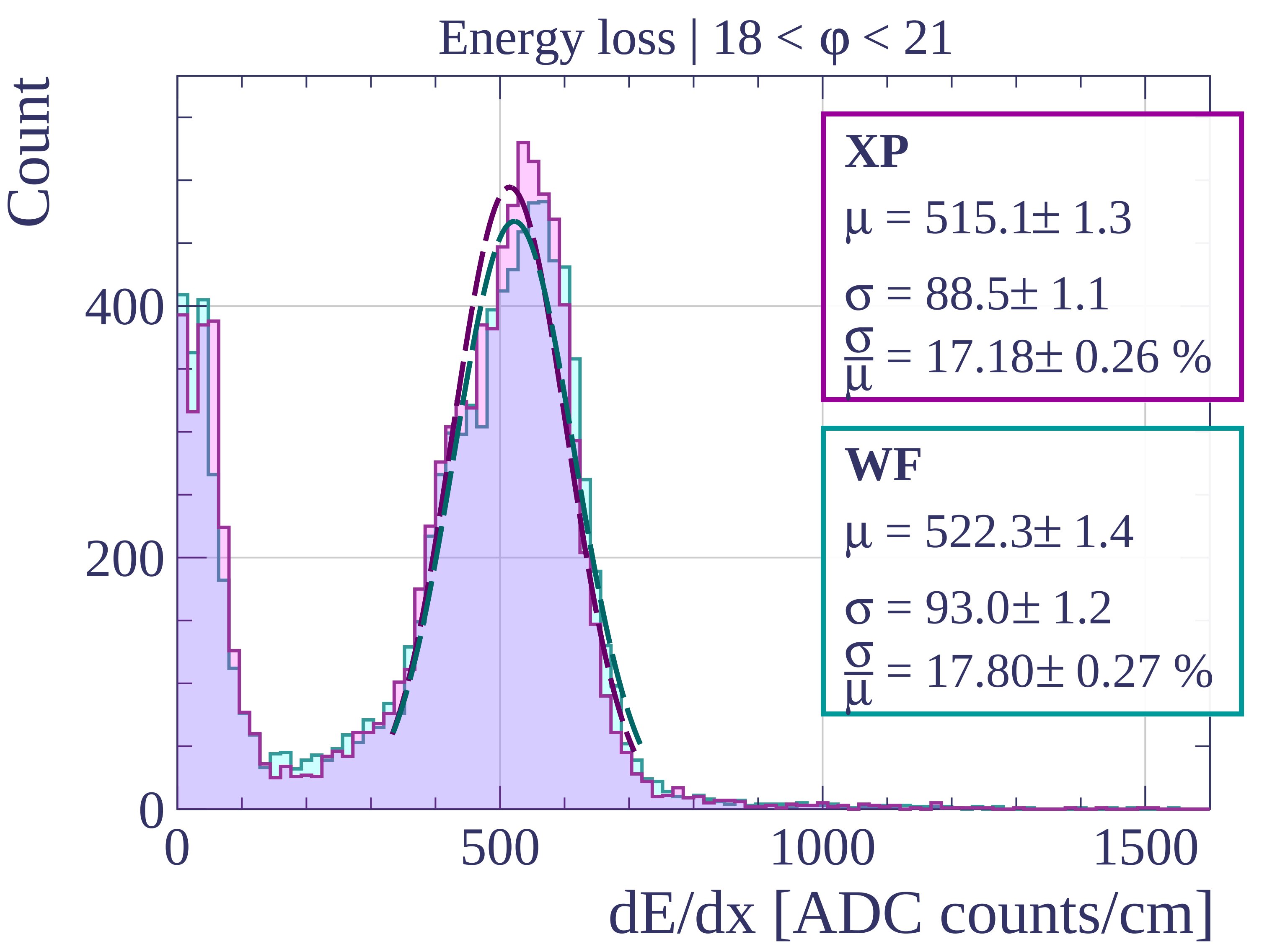
Energy loss | 9 < φ < 12
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 4405.7± 321.5
σ = 1016.1± 76.5
σ/μ = 23.06± 3.42 %
WF
μ = -2092.1± 93.3
σ = 412.1± 15.7
σ/μ = -19.70± 0.13 %



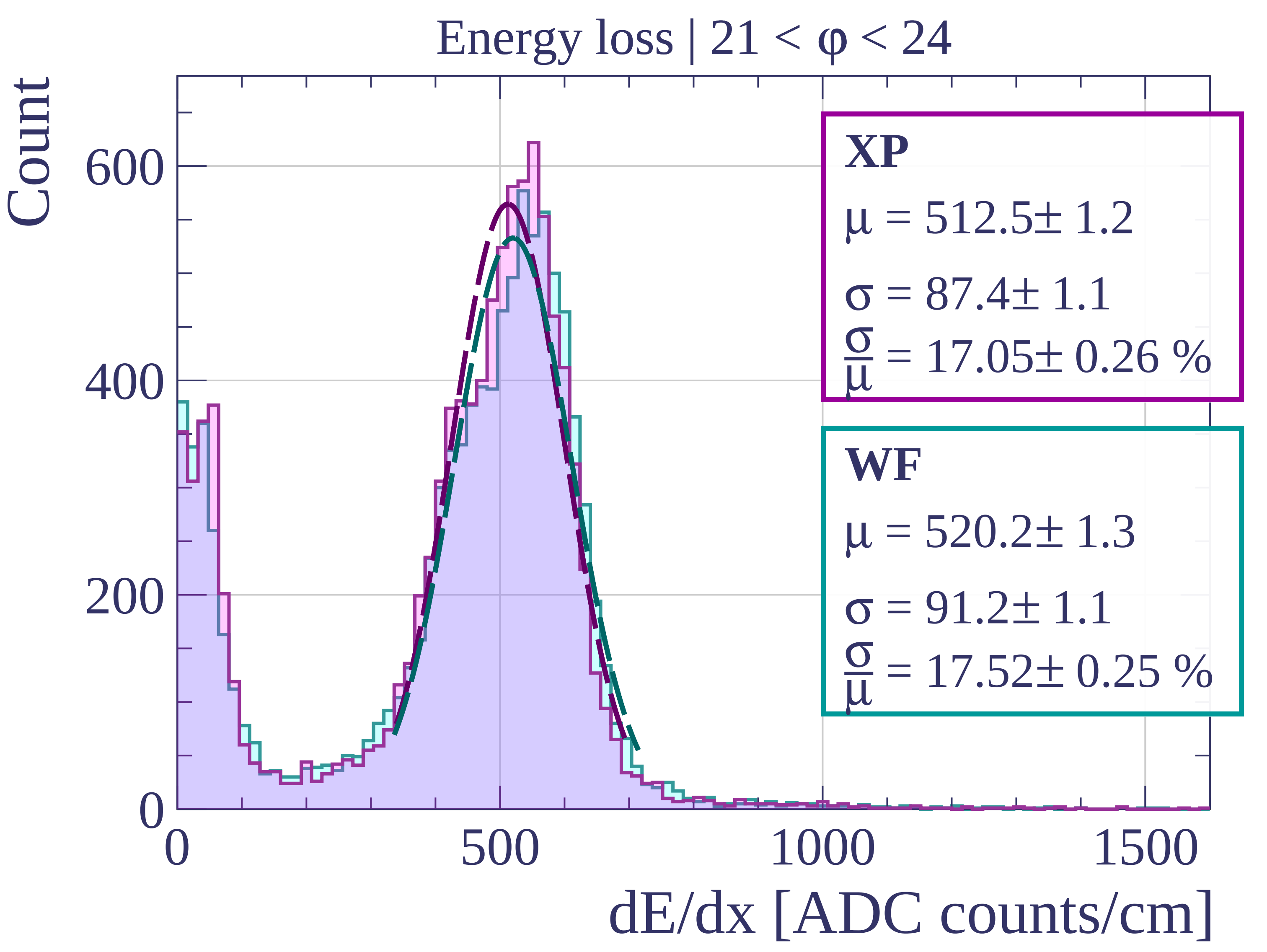
Energy loss | 12 < φ < 15
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = -1678.5± 66.9
σ = 350.9± 10.9
σ/μ = -20.90± 0.18 %
WF
μ = -1657.5± 56.2
σ = 342.9± 10.6
σ/μ = -20.69± 0.06 %



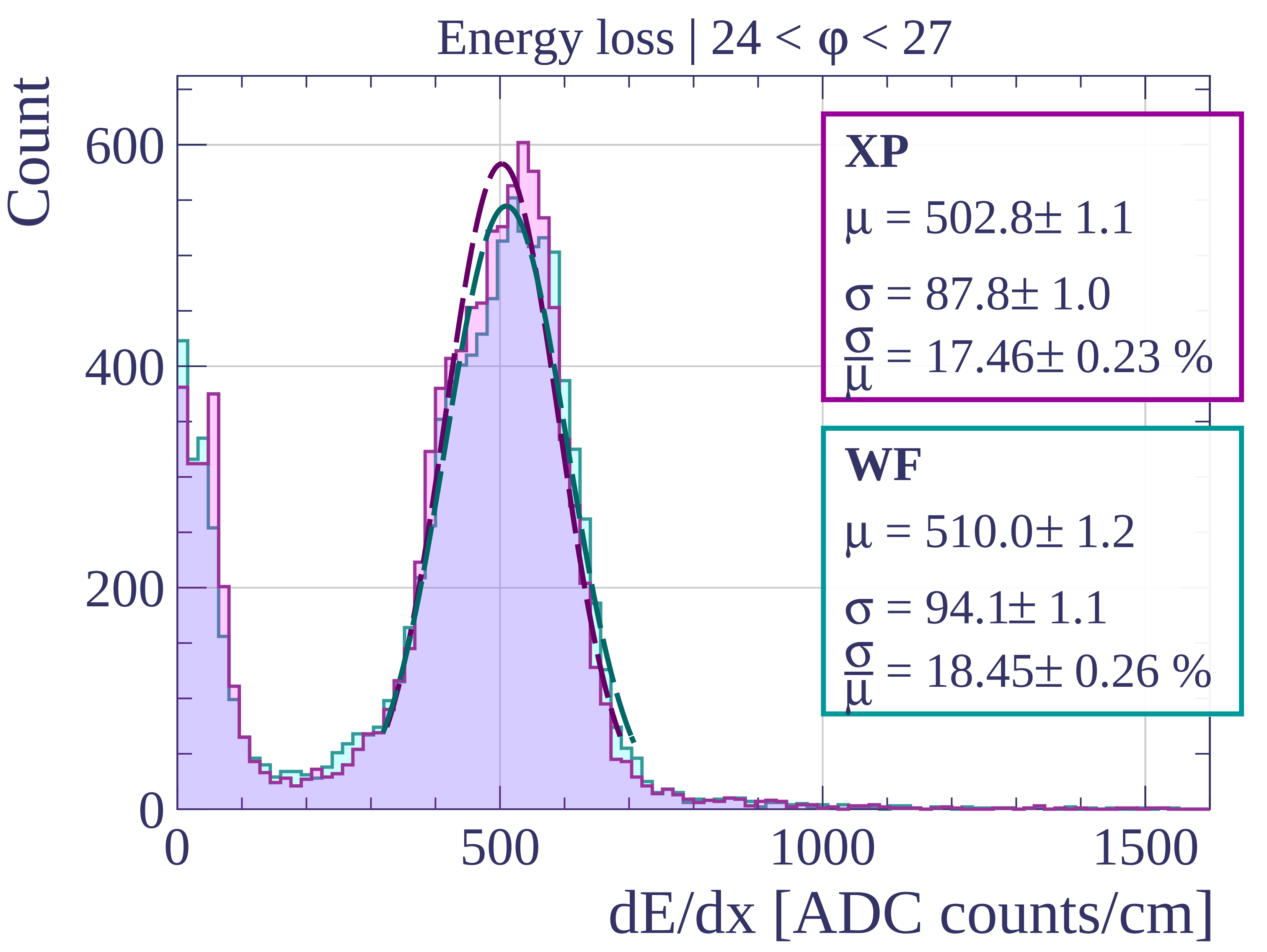
Energy loss | 15 < φ < 18
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = -1566.5± 65.4
σ = 331.3± 10.8
σ/μ = -21.15± 0.20 %
WF
μ = -1838.7± 92.8
σ = 380.5± 17.0
σ/μ = -20.69± 0.12 %

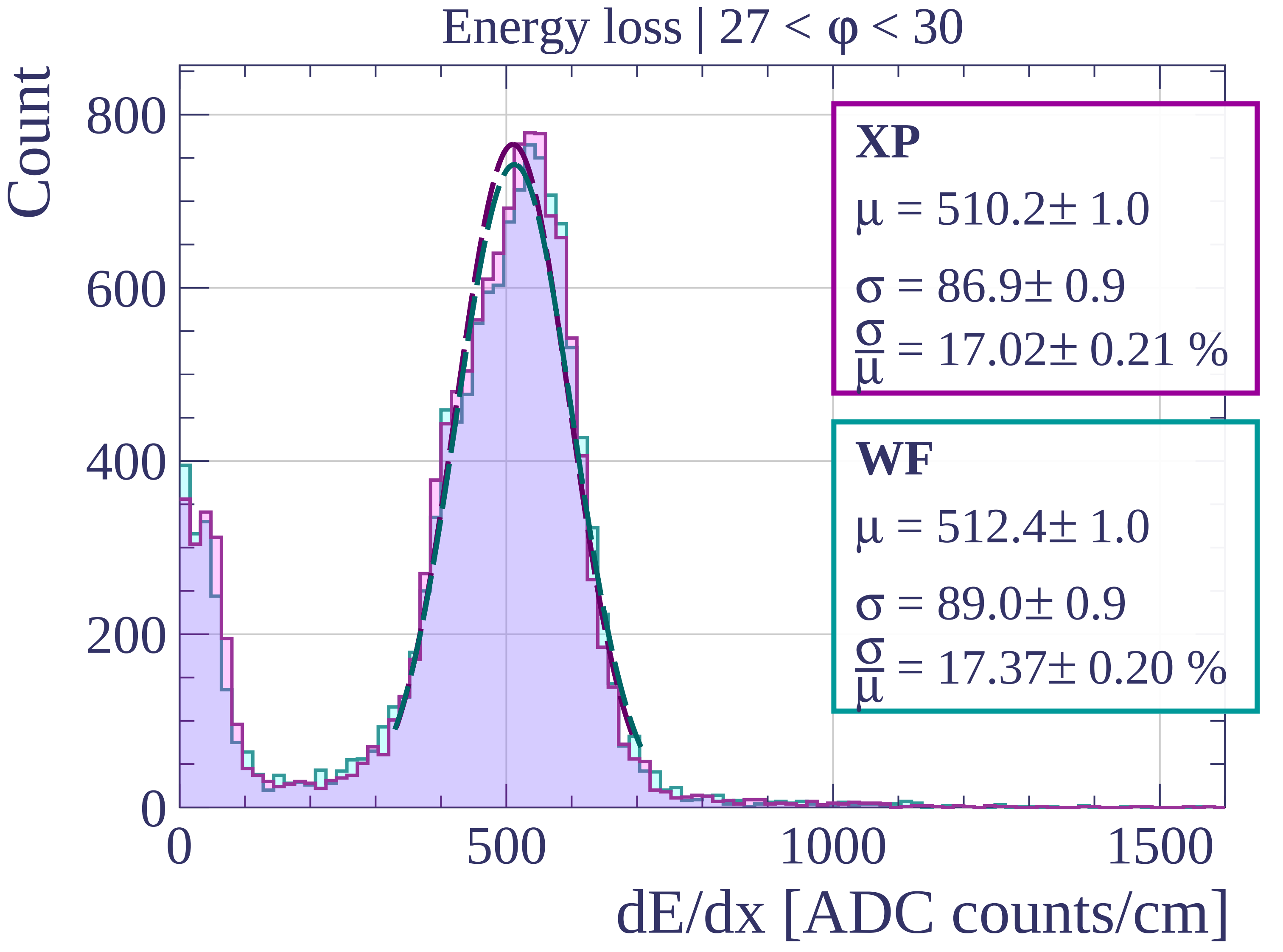


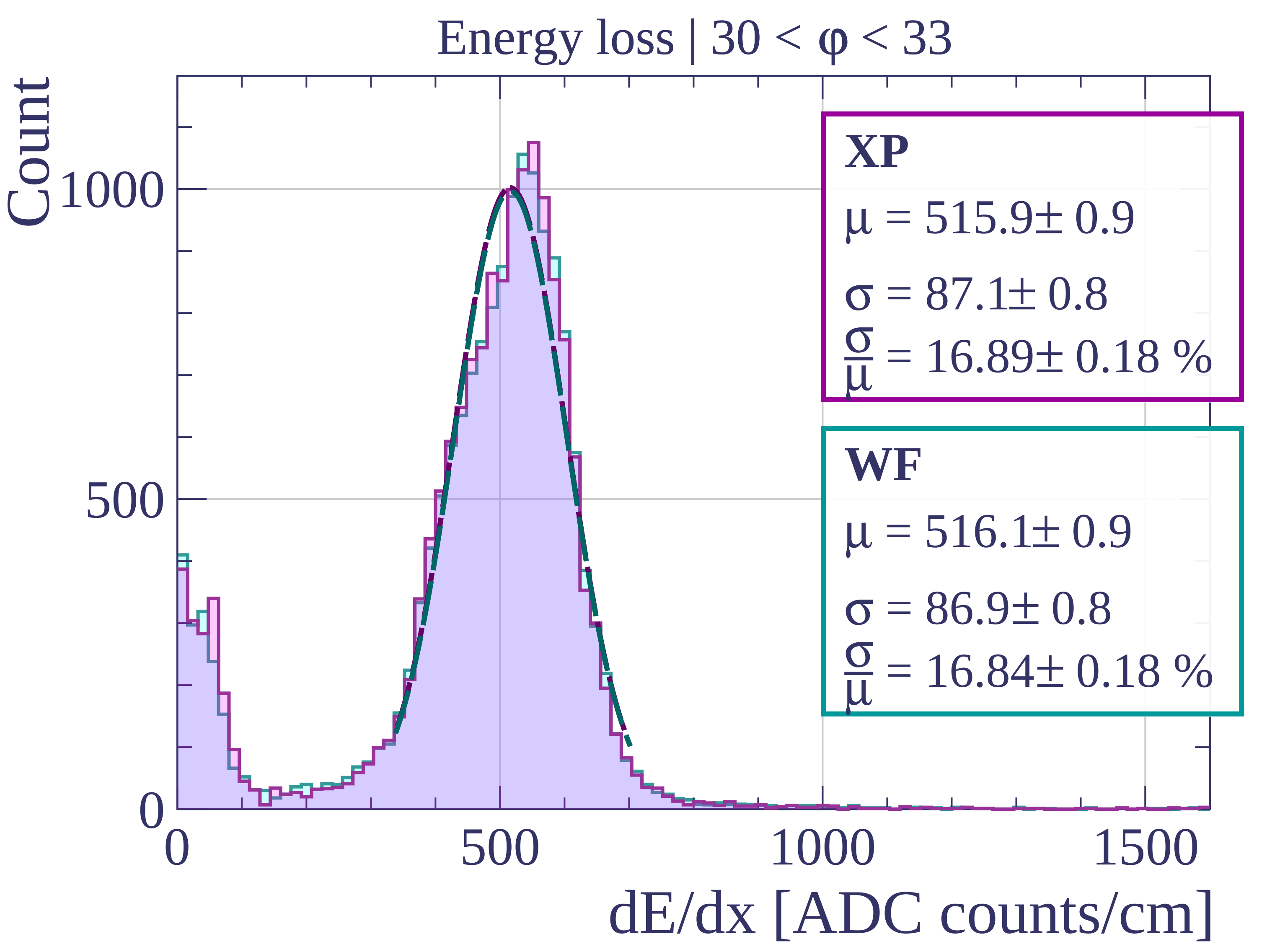
Energy loss | 18 < φ < 21
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
μ = 515.1± 1.3
σ = 88.5± 1.1
σ/μ = 17.18± 0.26 %
WF
μ = 522.3± 1.4
σ = 93.0± 1.2
σ/μ = 17.80± 0.27 %



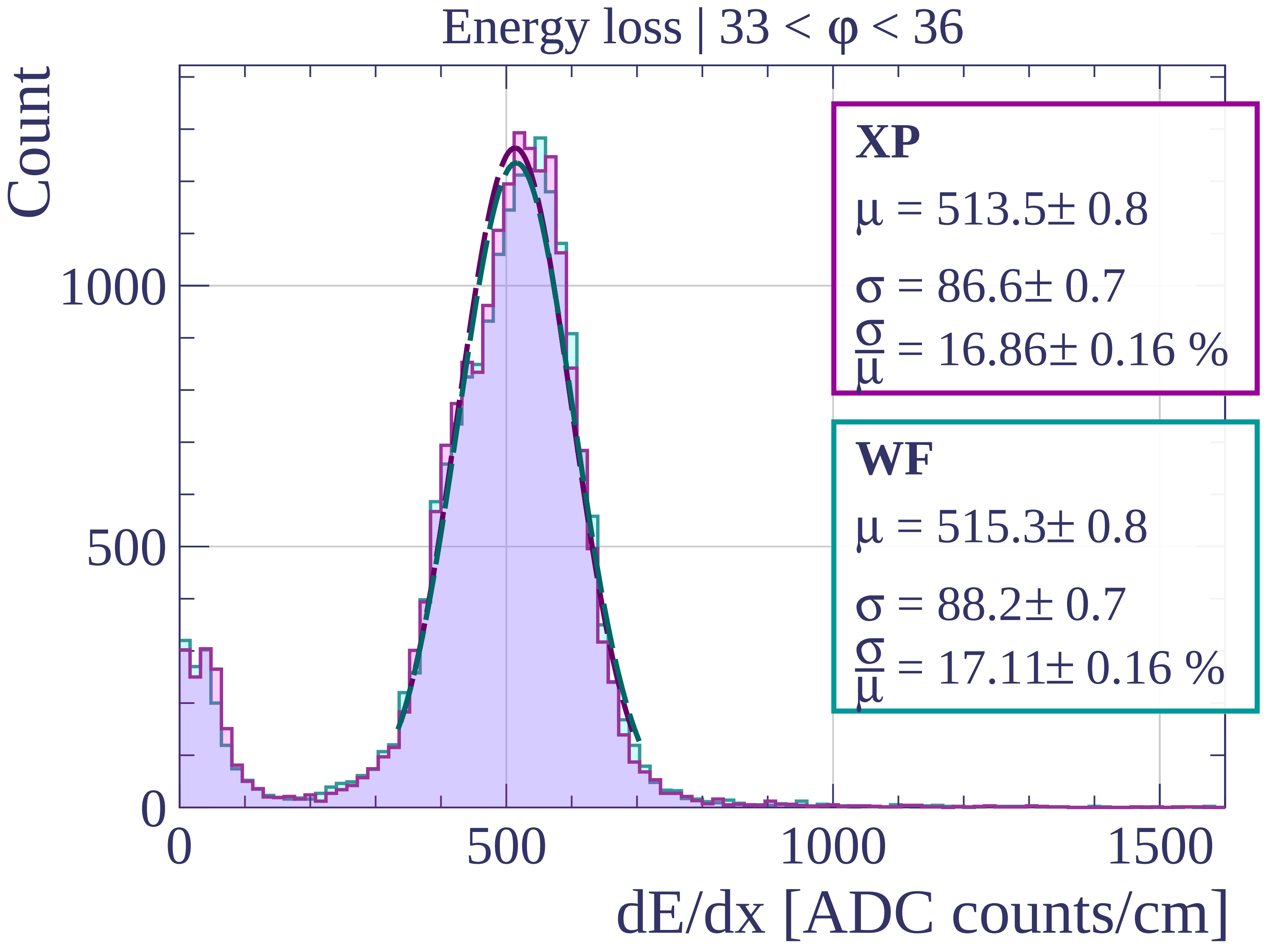
Energy loss | 21 < φ < 24
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 512.5± 1.2
σ = 87.4± 1.1
σ/μ = 17.05± 0.26 %
WF
μ = 520.2± 1.3
σ = 91.2± 1.1
σ/μ = 17.52± 0.25 %


Energy loss | 24 < φ < 27
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 502.8± 1.1
σ = 87.8± 1.0
σ/μ = 17.46± 0.23 %
WF
μ = 510.0± 1.2
σ = 94.1± 1.1
σ/μ = 18.45± 0.26 %

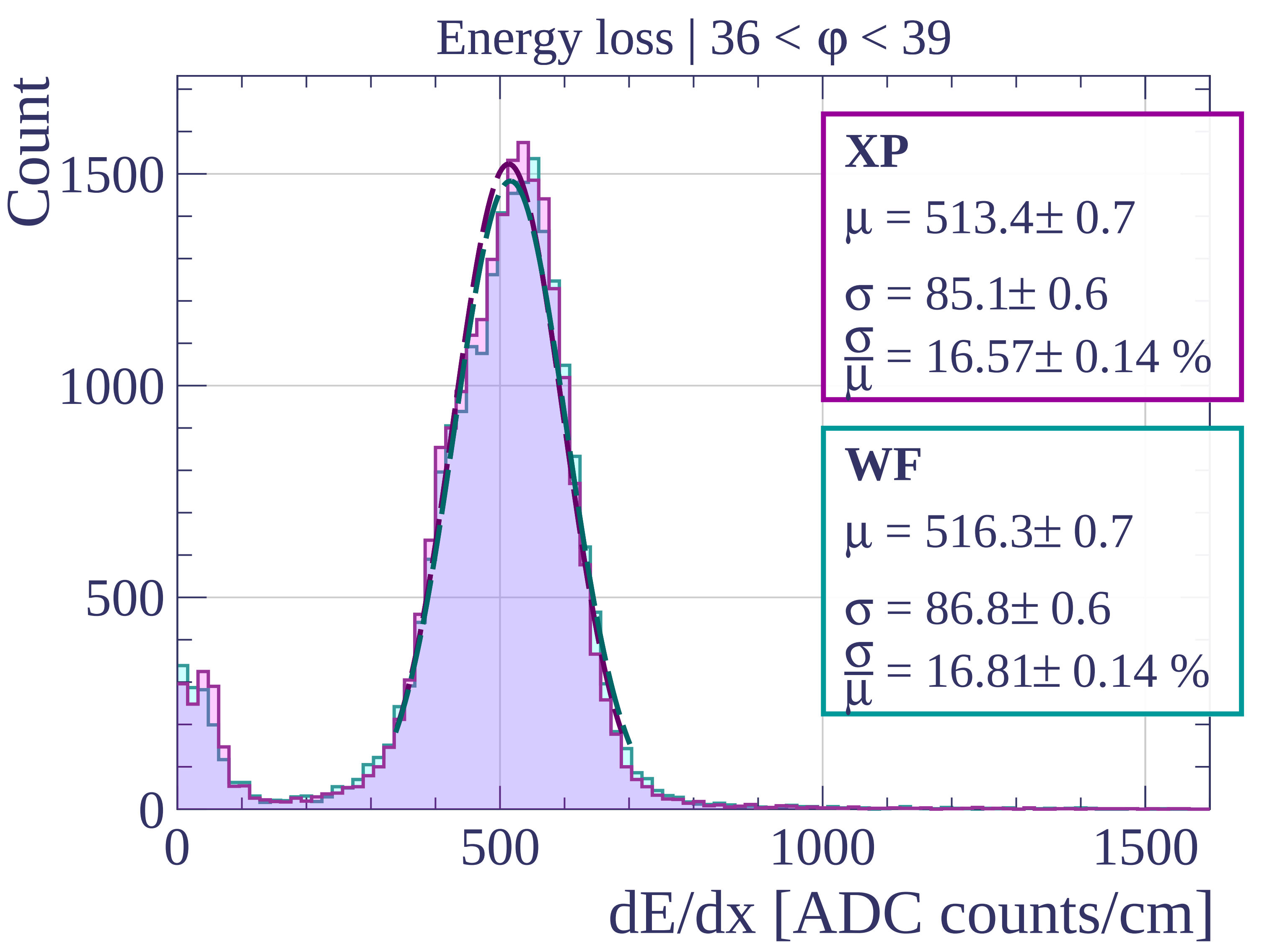

Energy loss | 27 < φ < 30
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 510.2± 1.0
σ = 86.9± 0.9
σ/μ = 17.02± 0.21 %
WF
μ = 512.4± 1.0
σ = 89.0± 0.9
σ/μ = 17.37± 0.20 %



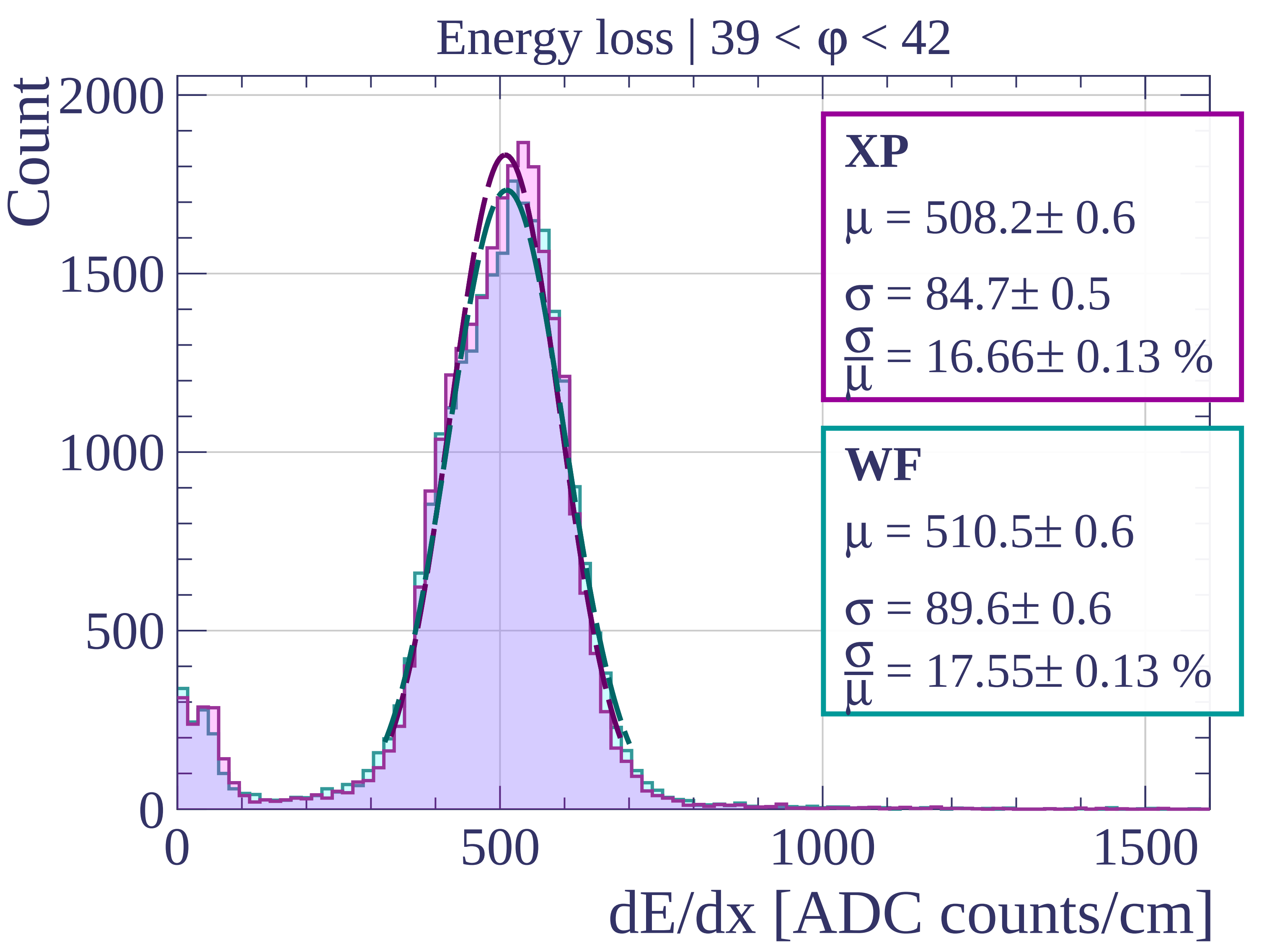
Energy loss | 30 < φ < 33
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 515.9± 0.9
σ = 87.1± 0.8
σ/μ = 16.89± 0.18 %
WF
μ = 516.1± 0.9
σ = 86.9± 0.8
σ/μ = 16.84± 0.18 %

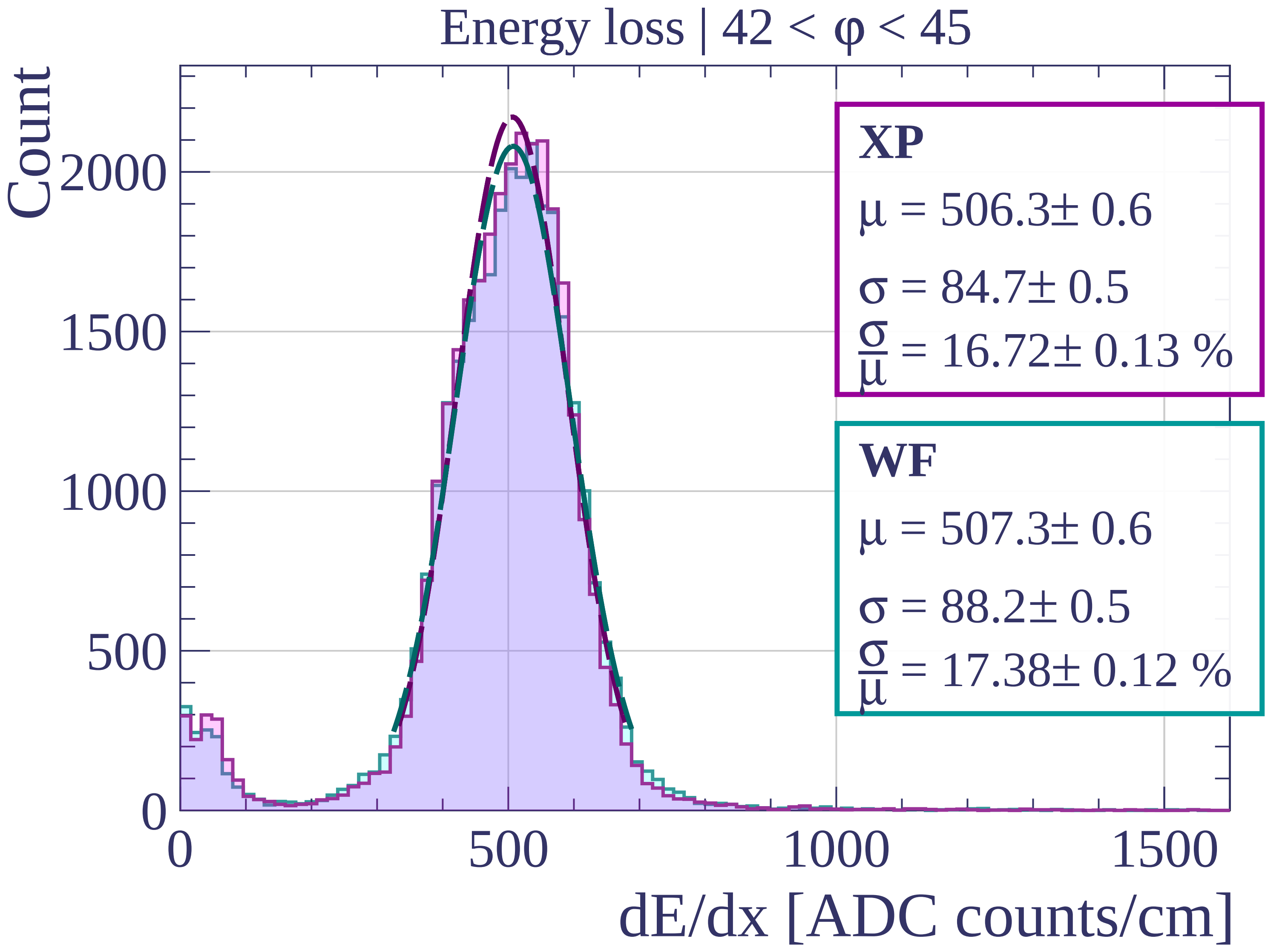


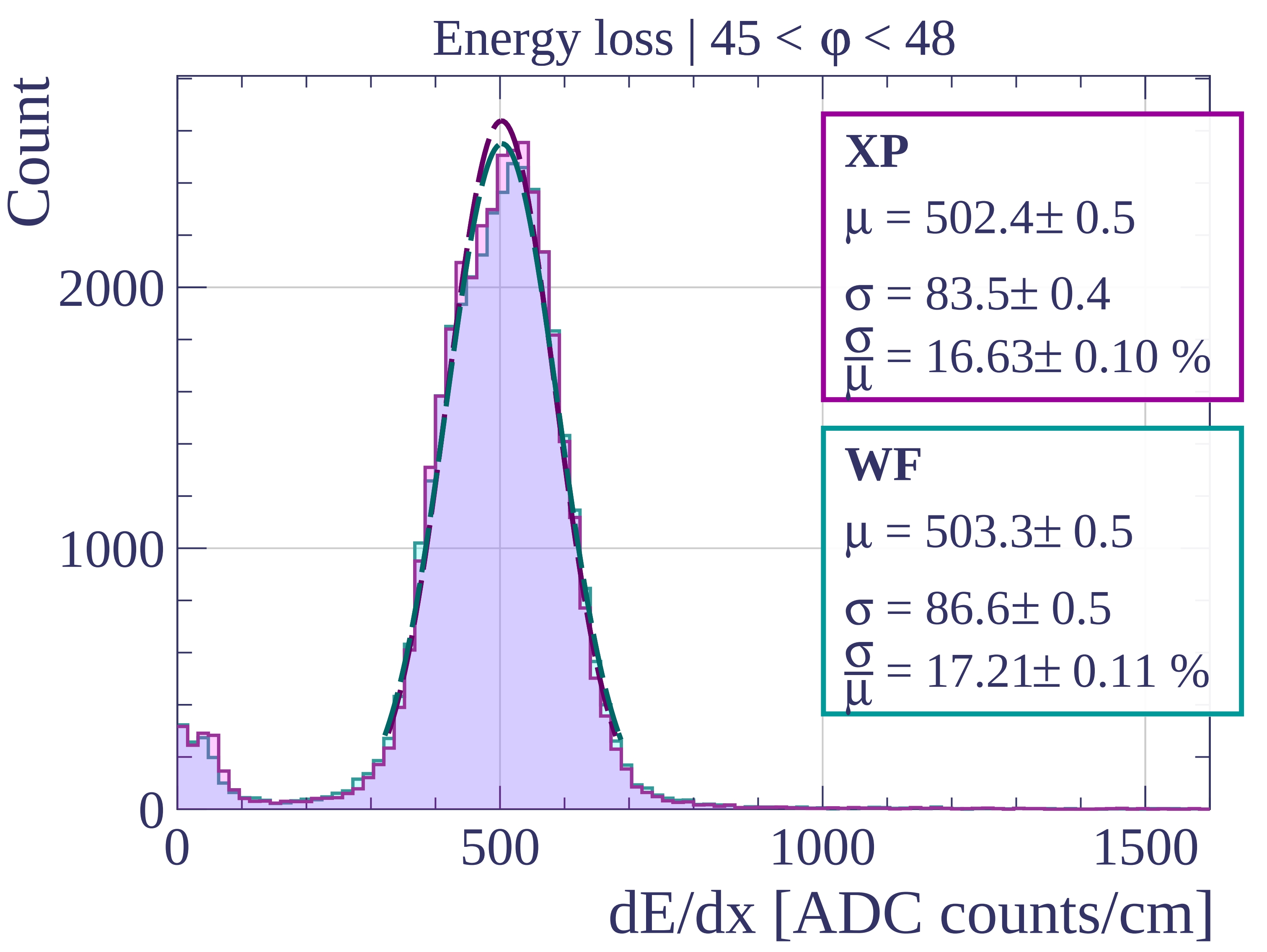
Energy loss | 33 < φ < 36
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
500
1000
XP
μ = 513.5± 0.8
σ = 86.6± 0.7
σ/μ = 16.86± 0.16 %
WF
μ = 515.3± 0.8
σ = 88.2± 0.7
σ/μ = 17.11± 0.16 %



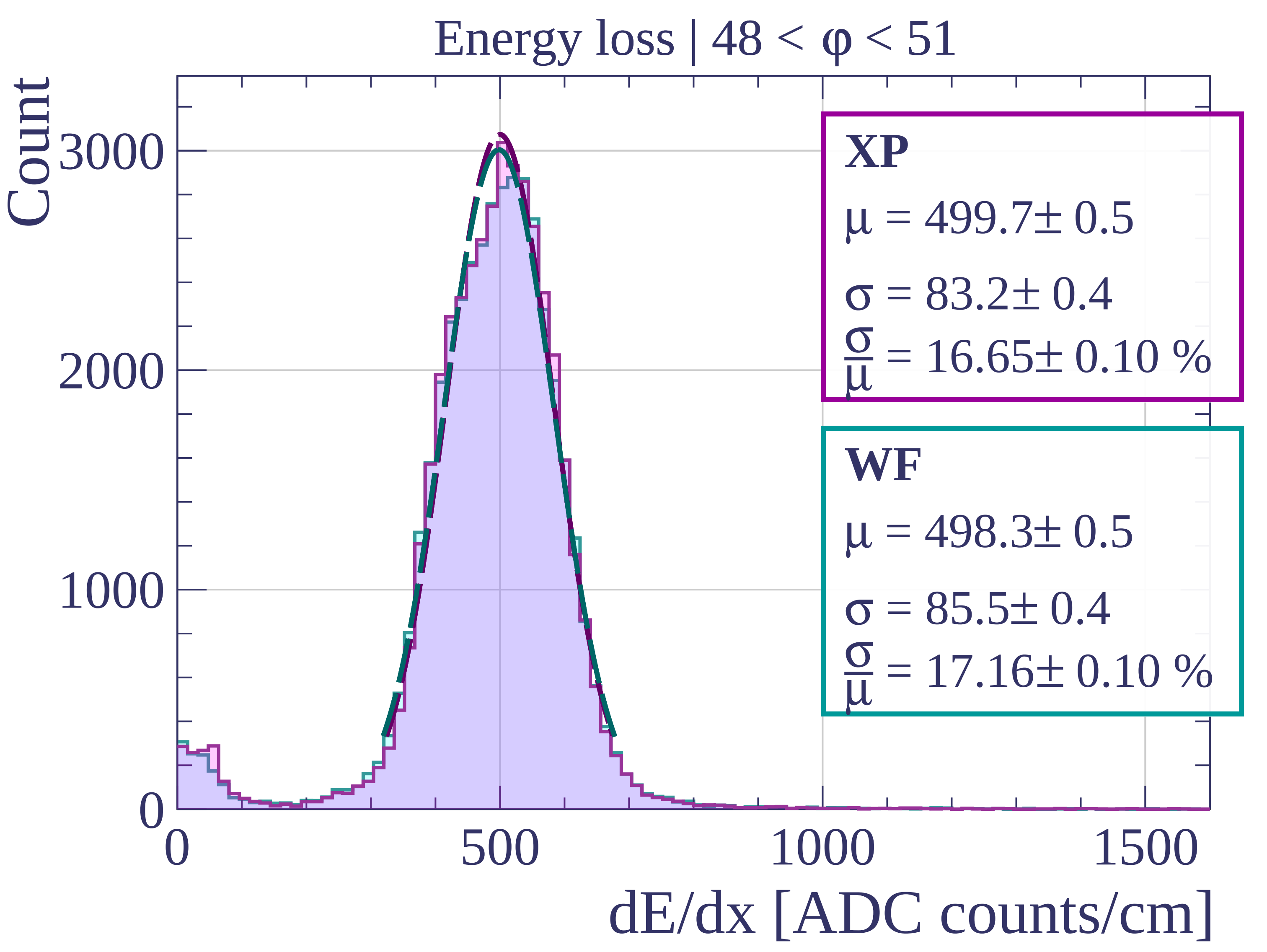
Energy loss | 36 < φ < 39
Count
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 513.4± 0.7
σ = 85.1± 0.6
σ/μ = 16.57± 0.14 %
WF
μ = 516.3± 0.7
σ = 86.8± 0.6
σ/μ = 16.81± 0.14 %

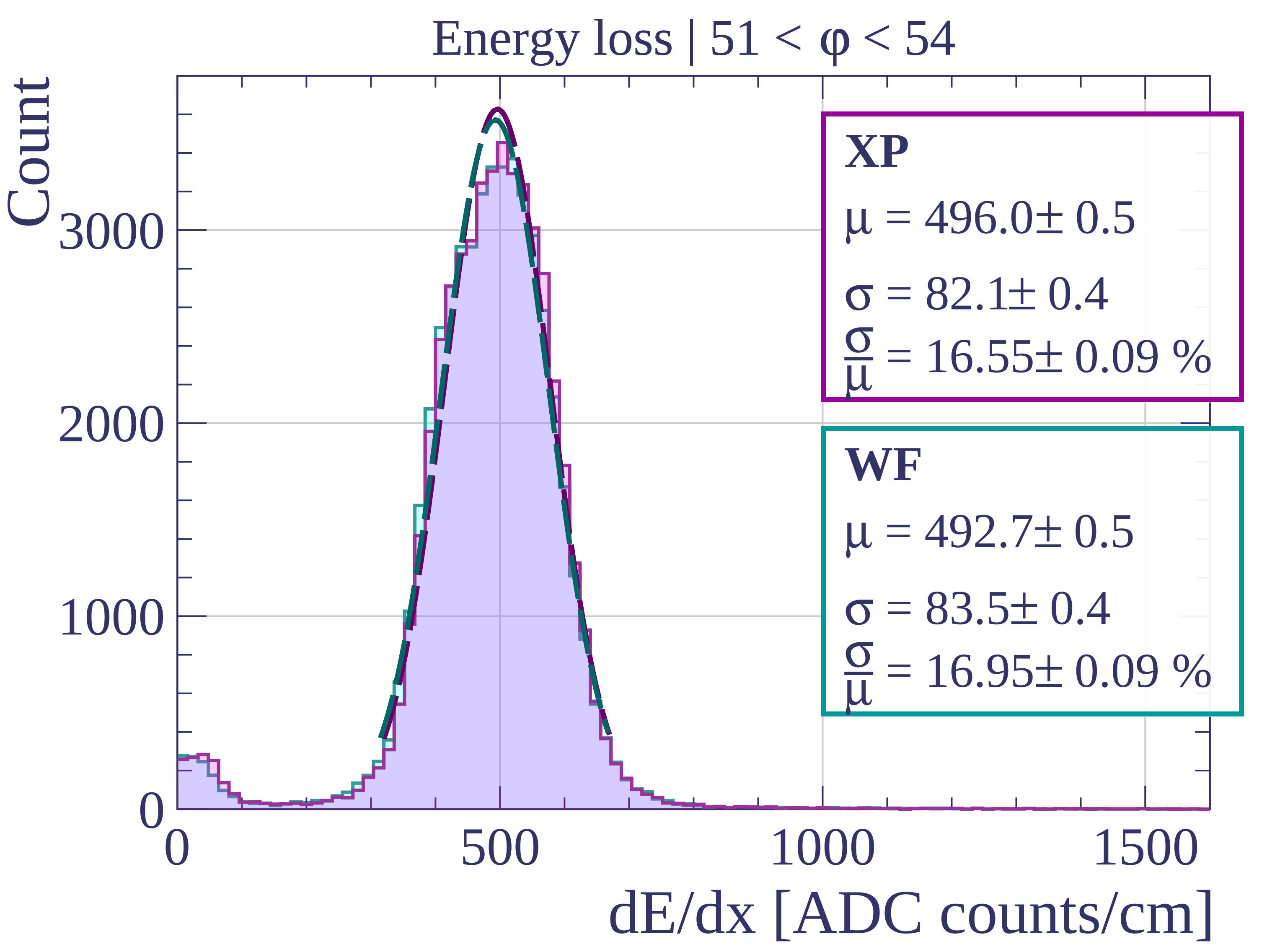

Energy loss | 39 < φ < 42
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 508.2± 0.6
σ = 84.7± 0.5
σ/μ = 16.66± 0.13 %
WF
μ = 510.5± 0.6
σ = 89.6± 0.6
σ/μ = 17.55± 0.13 %

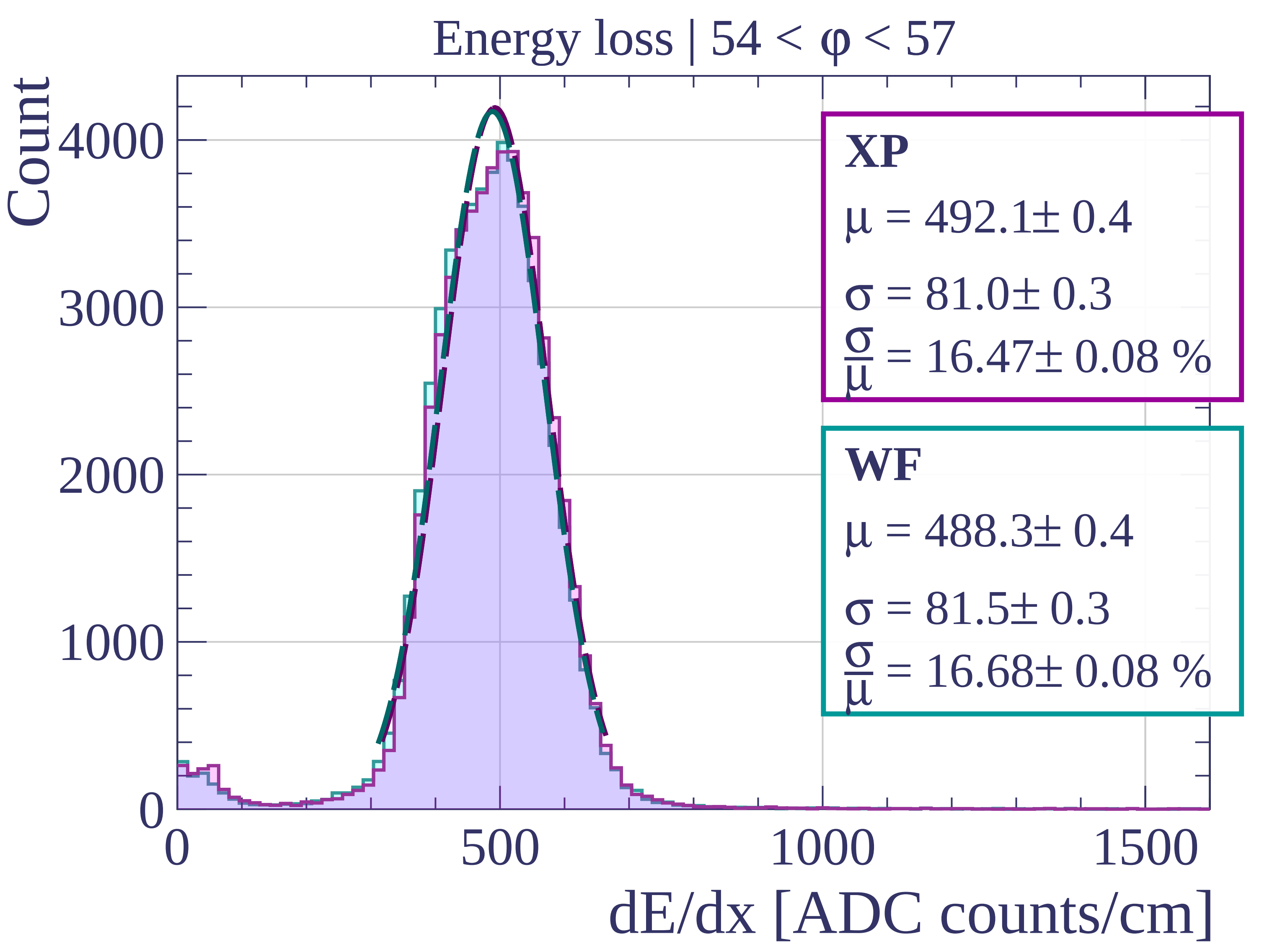

Energy loss | 42 < φ < 45
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
0
500
1000
1500
XP
μ = 506.3± 0.6
σ = 84.7± 0.5
σ/μ = 16.72± 0.13 %
WF
μ = 507.3± 0.6
σ = 88.2± 0.5
σ/μ = 17.38± 0.12 %

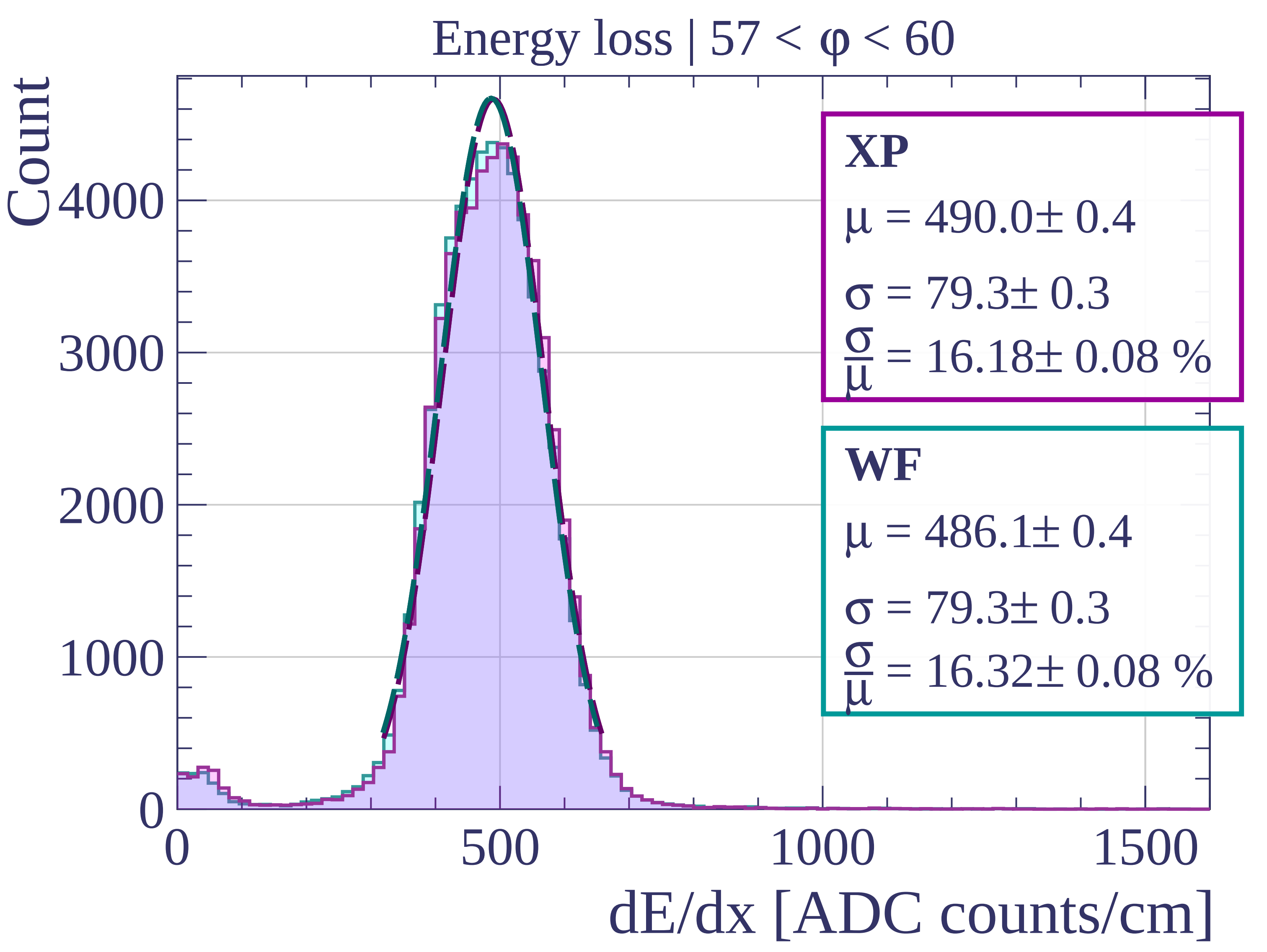


Energy loss | 45 < φ < 48
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 502.4± 0.5
σ = 83.5± 0.4
σ/μ = 16.63± 0.10 %
WF
μ = 503.3± 0.5
σ = 86.6± 0.5
σ/μ = 17.21± 0.11 %

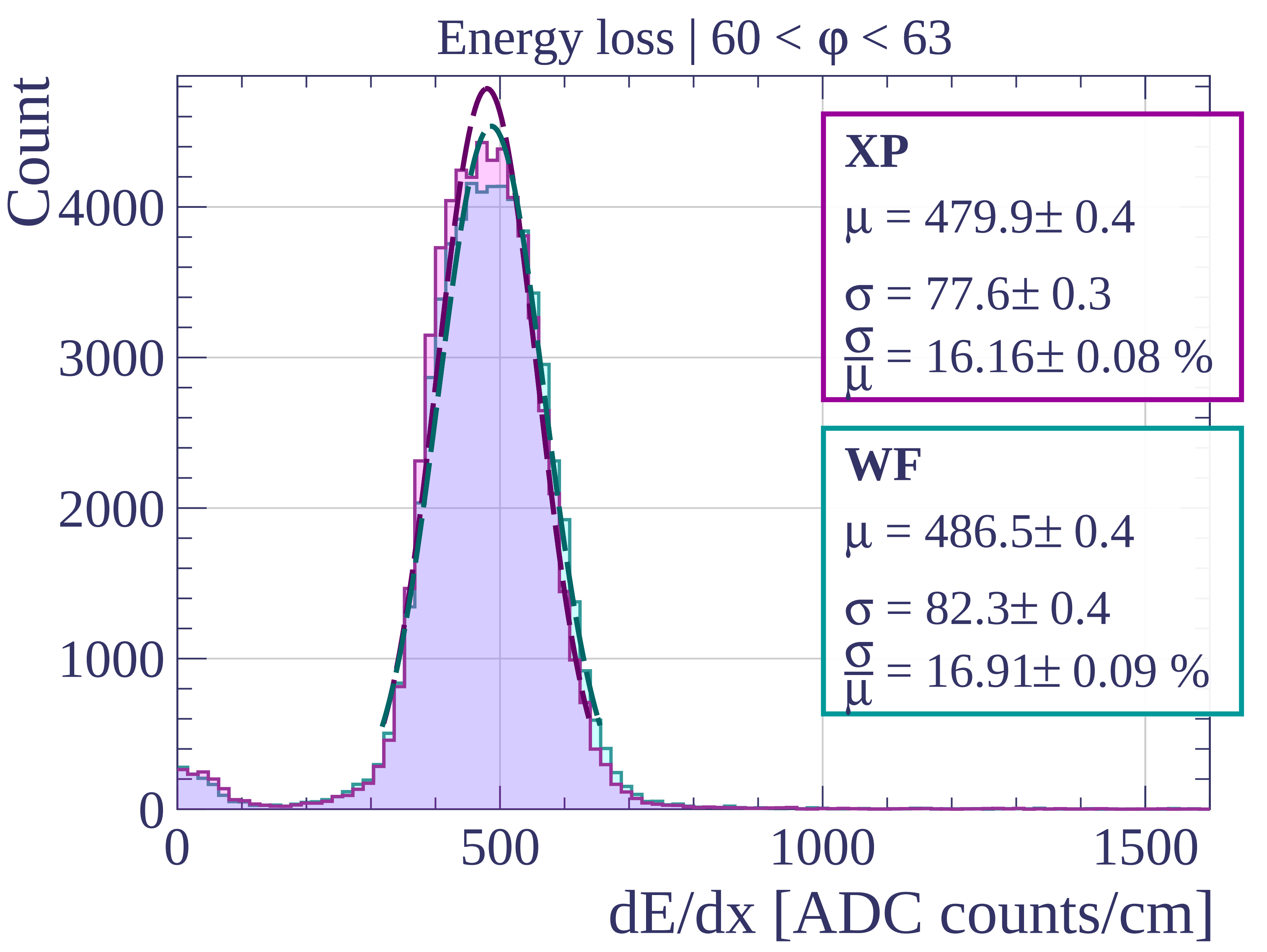

Energy loss | 48 < φ < 51
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 499.7± 0.5
σ = 83.2± 0.4
σ/μ = 16.65± 0.10 %
WF
μ = 498.3± 0.5
σ = 85.5± 0.4
σ/μ = 17.16± 0.10 %

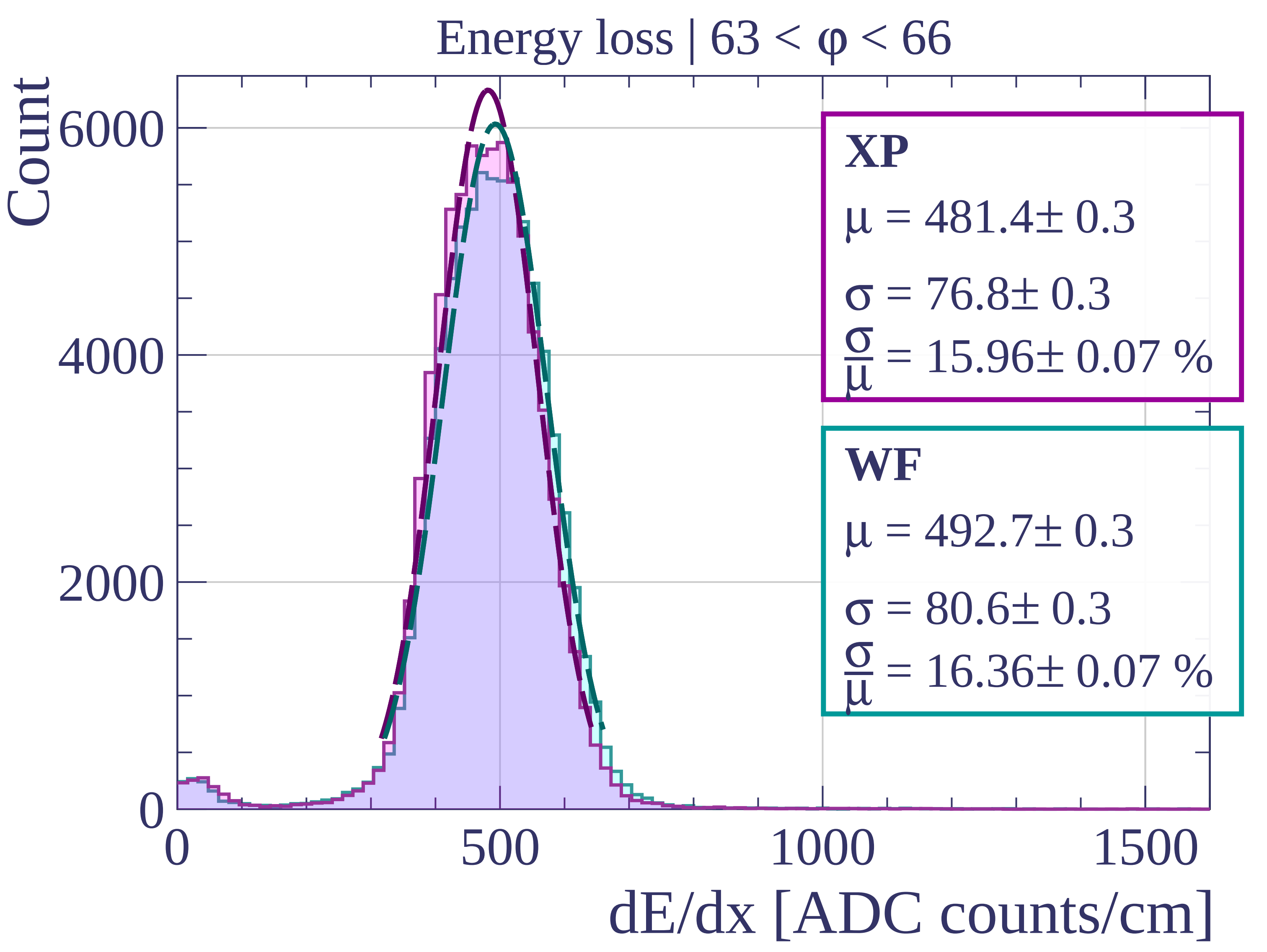

Energy loss | 51 < φ < 54
Count
0
1000
2000
3000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 496.0± 0.5
σ = 82.1± 0.4
σ/μ = 16.55± 0.09 %
WF
μ = 492.7± 0.5
σ = 83.5± 0.4
σ/μ = 16.95± 0.09 %

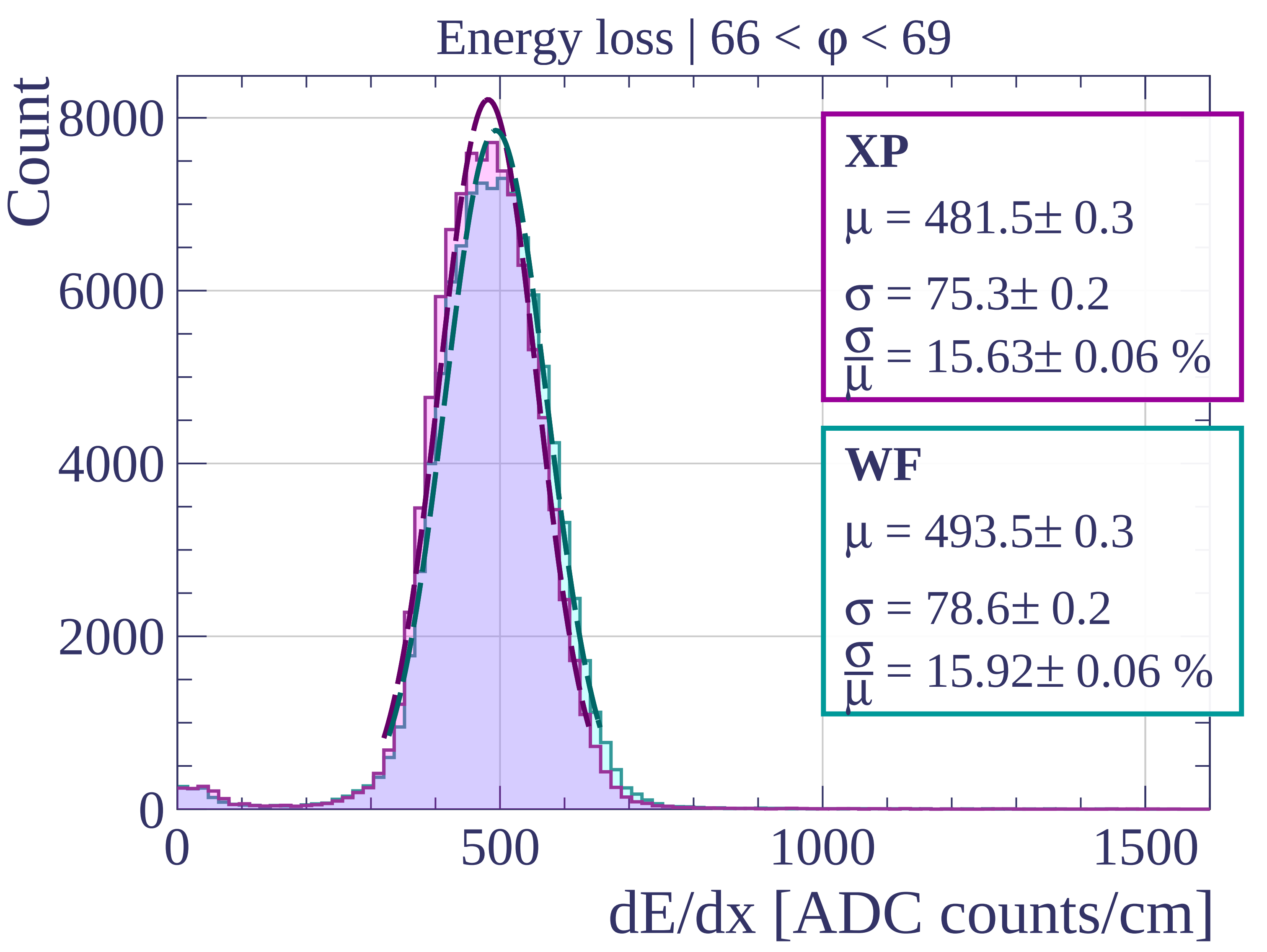

Energy loss | 54 < φ < 57
Count
dE/dx [ADC counts/cm]
0
1000
2000
3000
4000
0
500
1000
1500
XP
μ = 492.1± 0.4
σ = 81.0± 0.3
σ/μ = 16.47± 0.08 %
WF
μ = 488.3± 0.4
σ = 81.5± 0.3
σ/μ = 16.68± 0.08 %

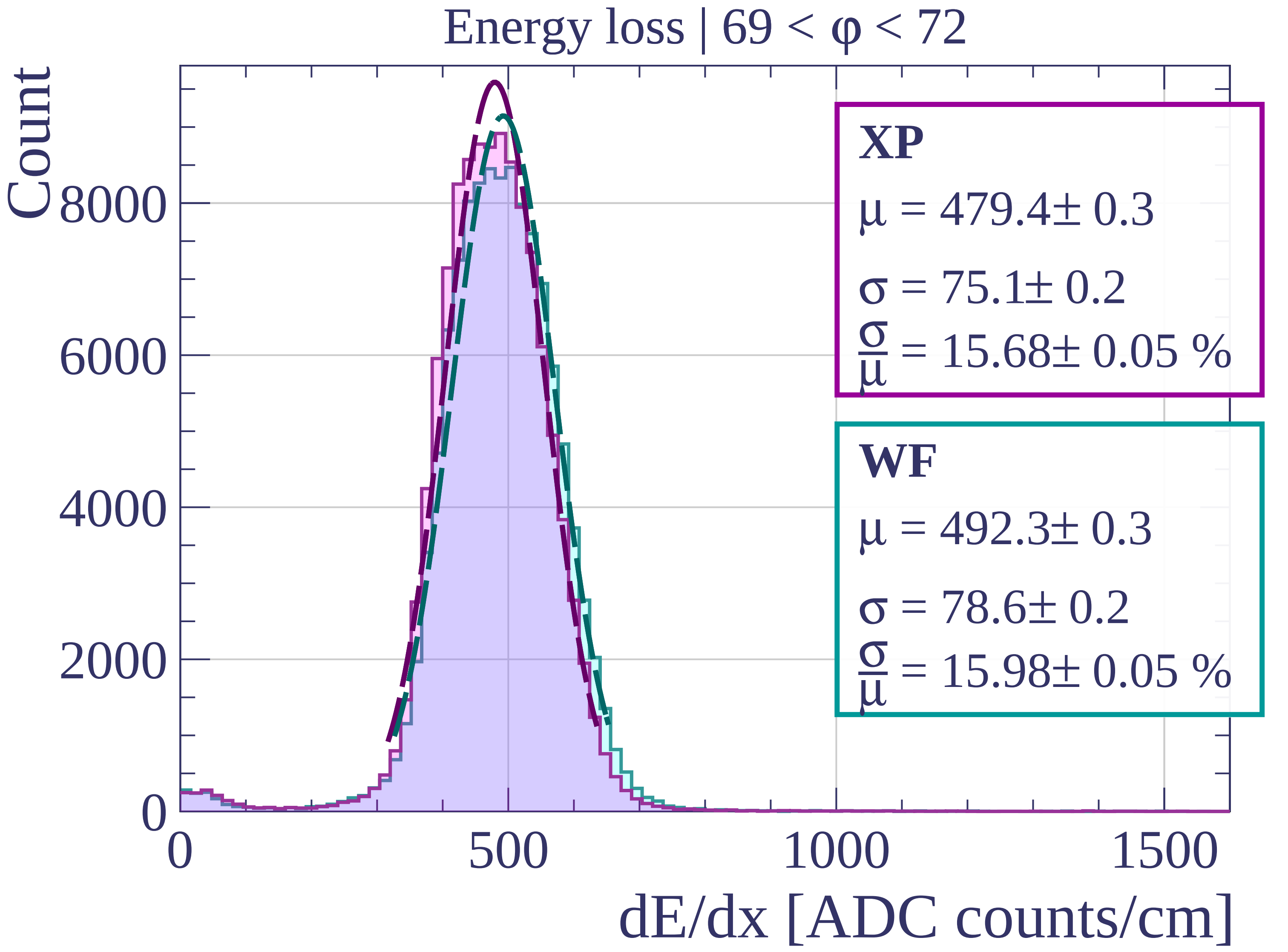


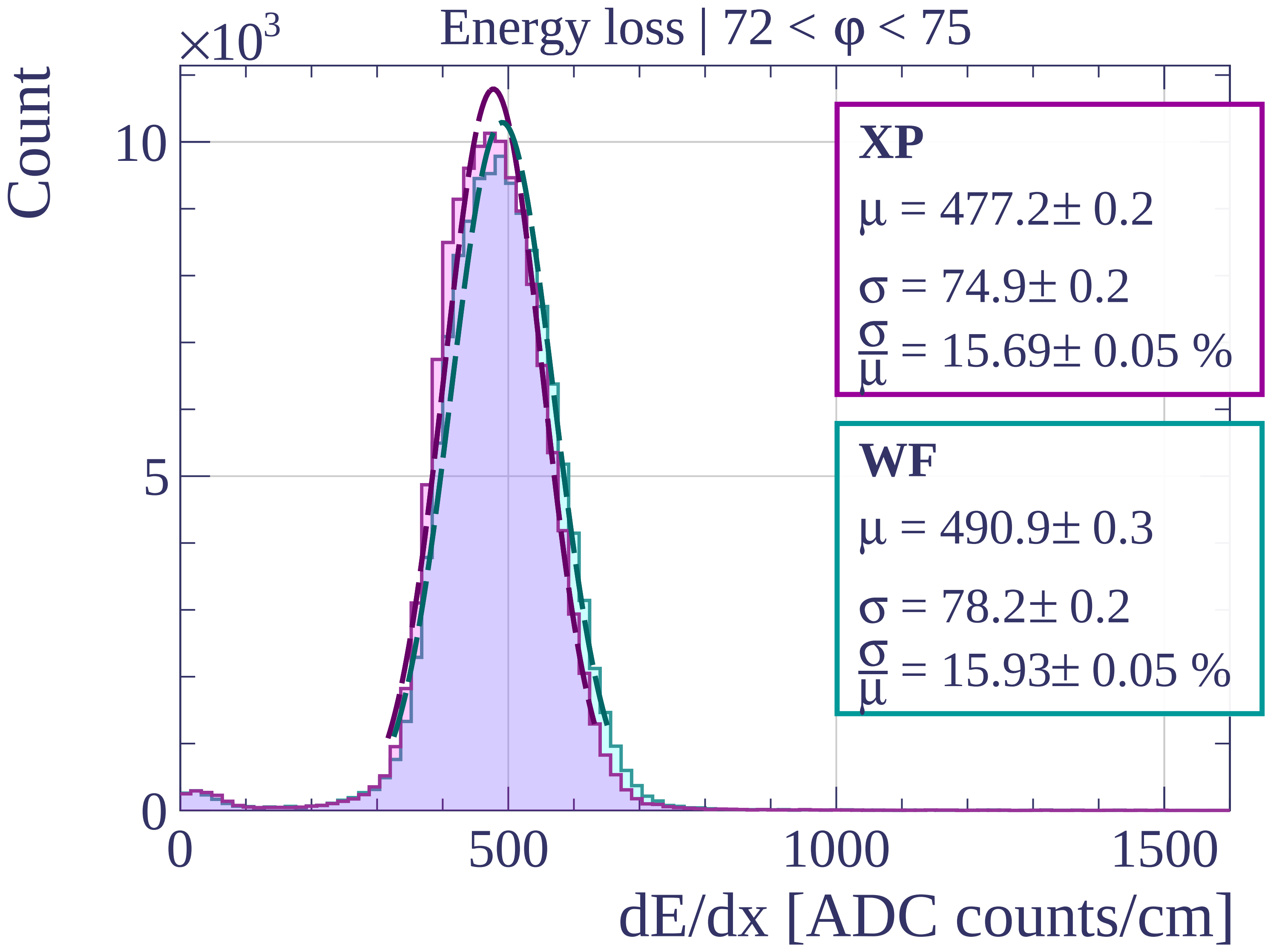
Energy loss | 57 < φ < 60
Count
dE/dx [ADC counts/cm]
0
1000
2000
3000
4000
500
1500
XP
μ = 490.0± 0.4
σ = 79.3± 0.3
σ/μ = 16.18± 0.08 %
WF
μ = 486.1± 0.4
σ = 79.3± 0.3
σ/μ = 16.32± 0.08 %


Energy loss | 60 < φ < 63
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 479.9± 0.4
σ = 77.6± 0.3
σ/μ = 16.16± 0.08 %
WF
μ = 486.5± 0.4
σ = 82.3± 0.4
σ/μ = 16.91± 0.09 %

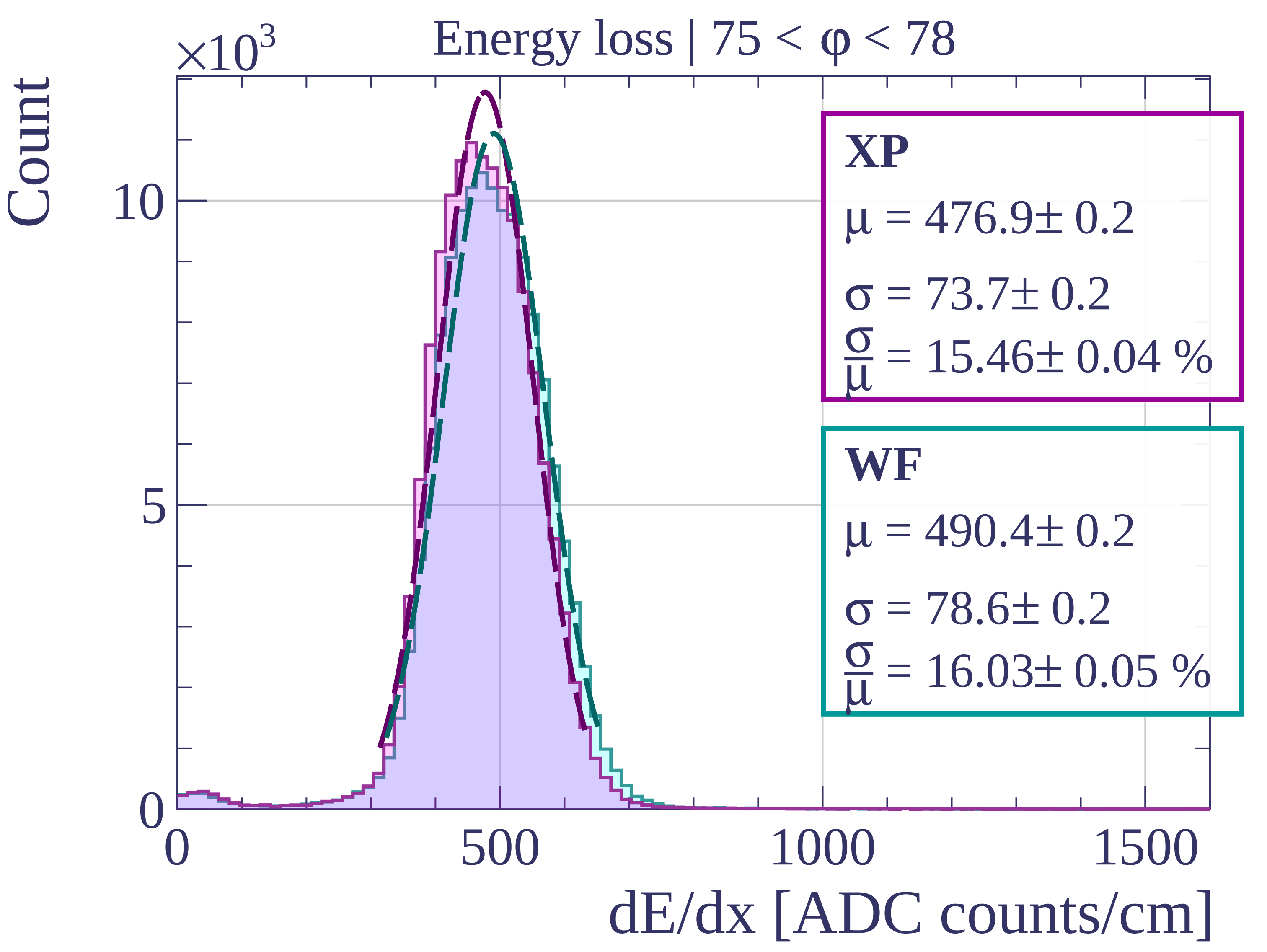

Energy loss | 63 < φ < 66
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 481.4± 0.3
σ = 76.8± 0.3
σ/μ = 15.96± 0.07 %
WF
μ = 492.7± 0.3
σ = 80.6± 0.3
σ/μ = 16.36± 0.07 %


Energy loss | 66 < φ < 69
Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 481.5± 0.3
σ = 75.3± 0.2
σ/μ = 15.63± 0.06 %
WF
μ = 493.5± 0.3
σ = 78.6± 0.2
σ/μ = 15.92± 0.06 %

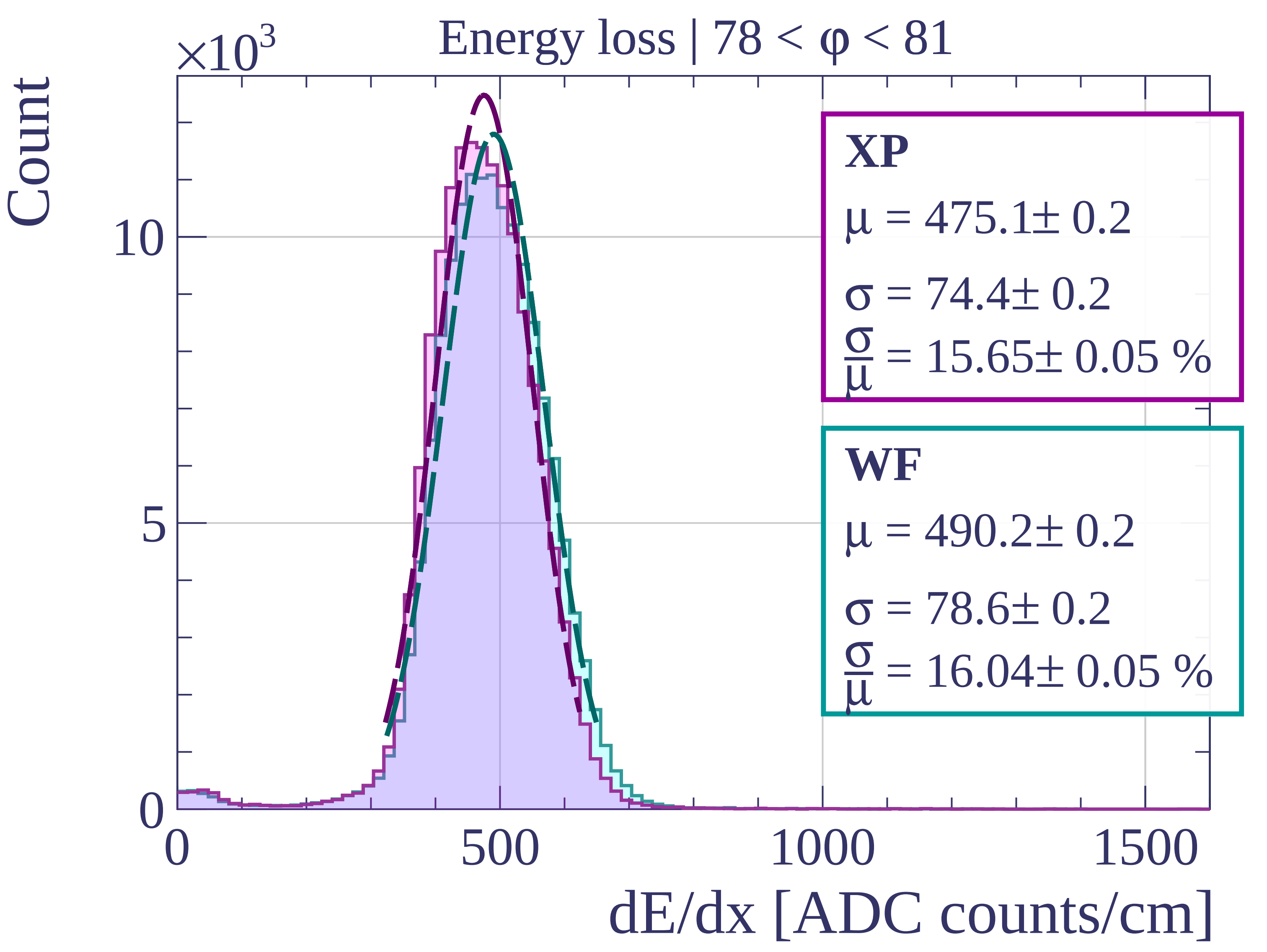

Energy loss | 69 < φ < 72
Count
0
2000
4000
6000
8000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 479.4± 0.3
σ = 75.1± 0.2
σ/μ = 15.68± 0.05 %
WF
μ = 492.3± 0.3
σ = 78.6± 0.2
σ/μ = 15.98± 0.05 %

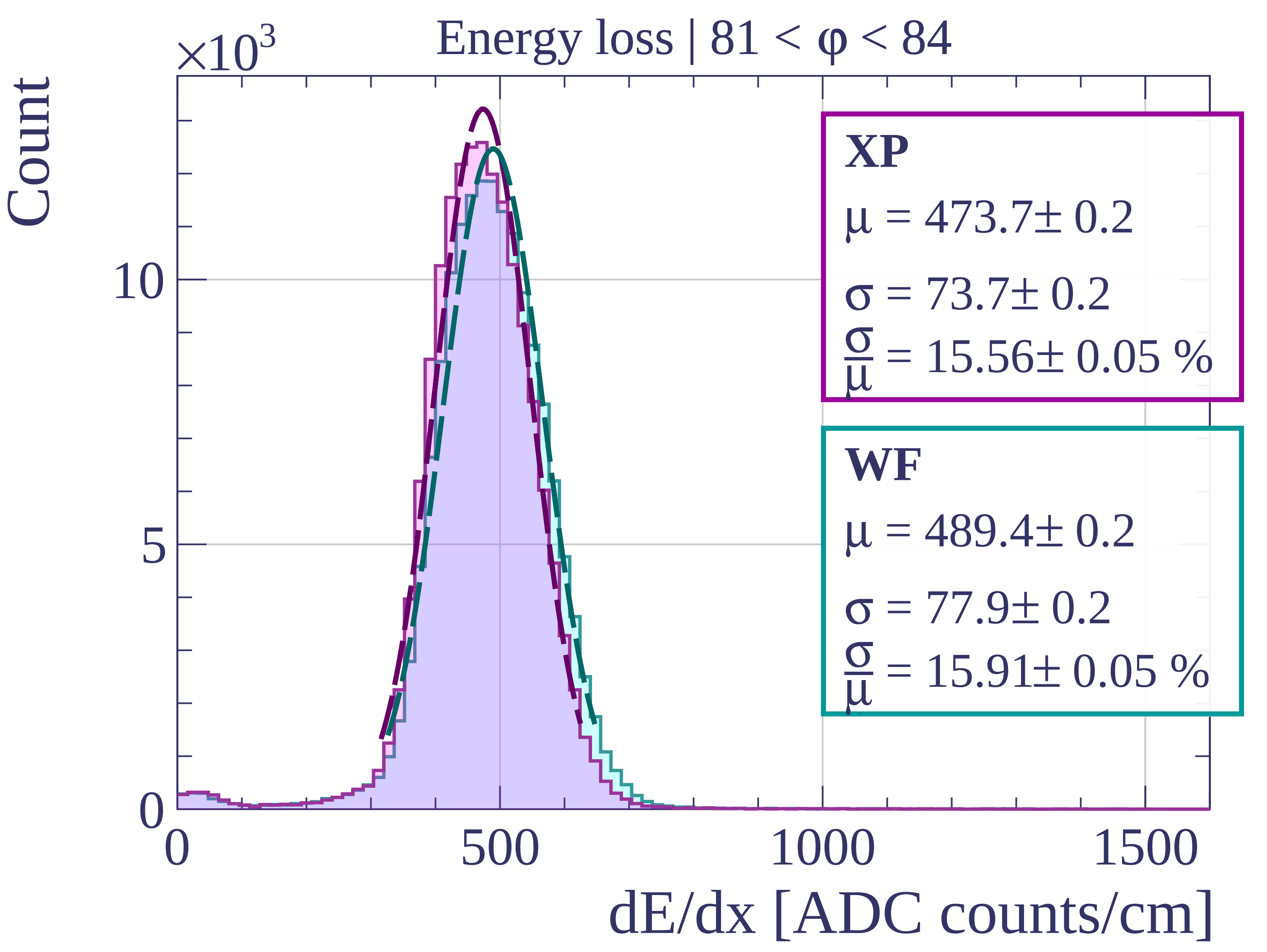


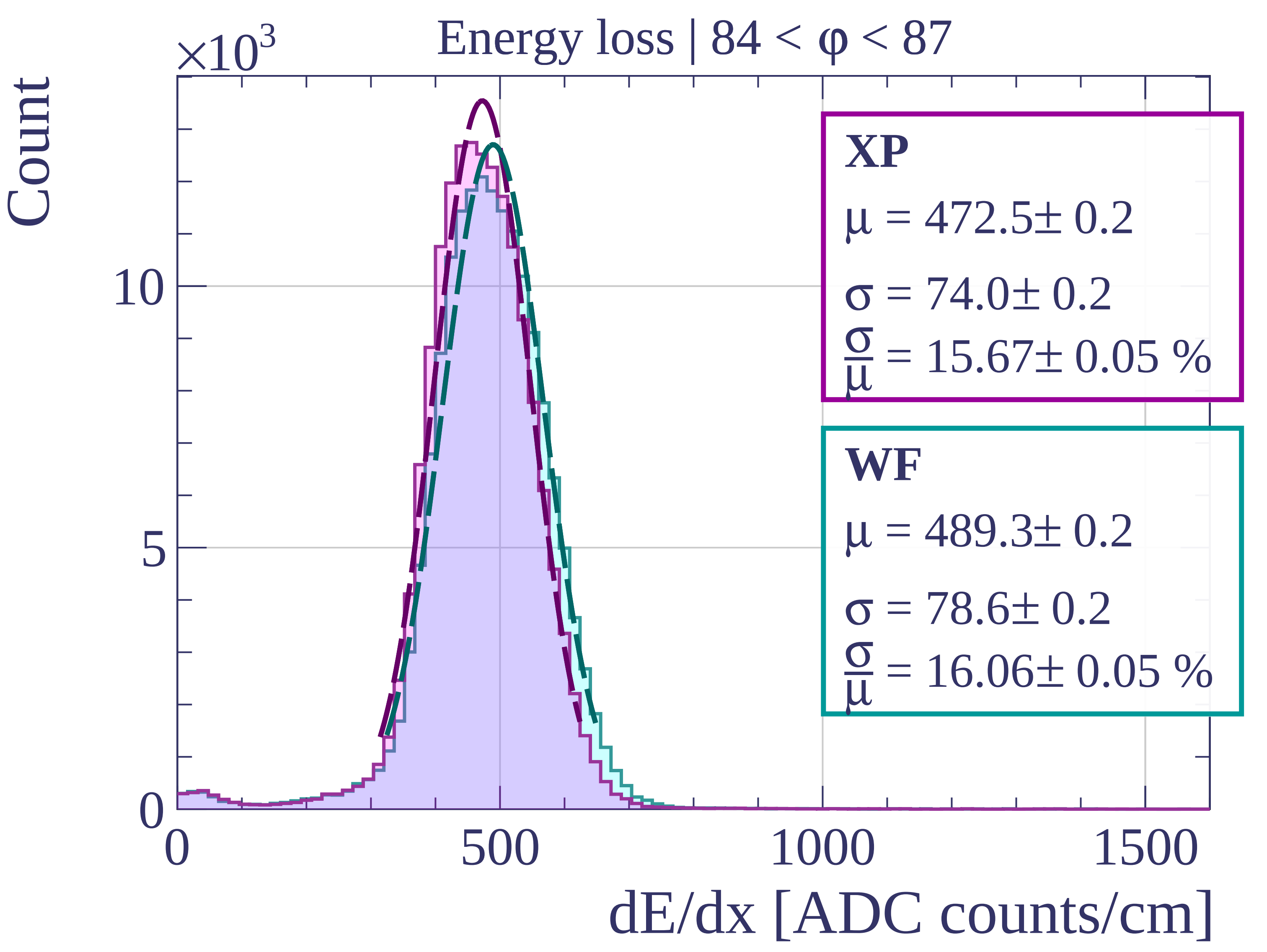
Energy loss | 72 < φ < 75
×10³
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 477.2± 0.2
σ = 74.9± 0.2
σ/μ = 15.69± 0.05 %
WF
μ = 490.9± 0.3
σ = 78.2± 0.2
σ/μ = 15.93± 0.05 %

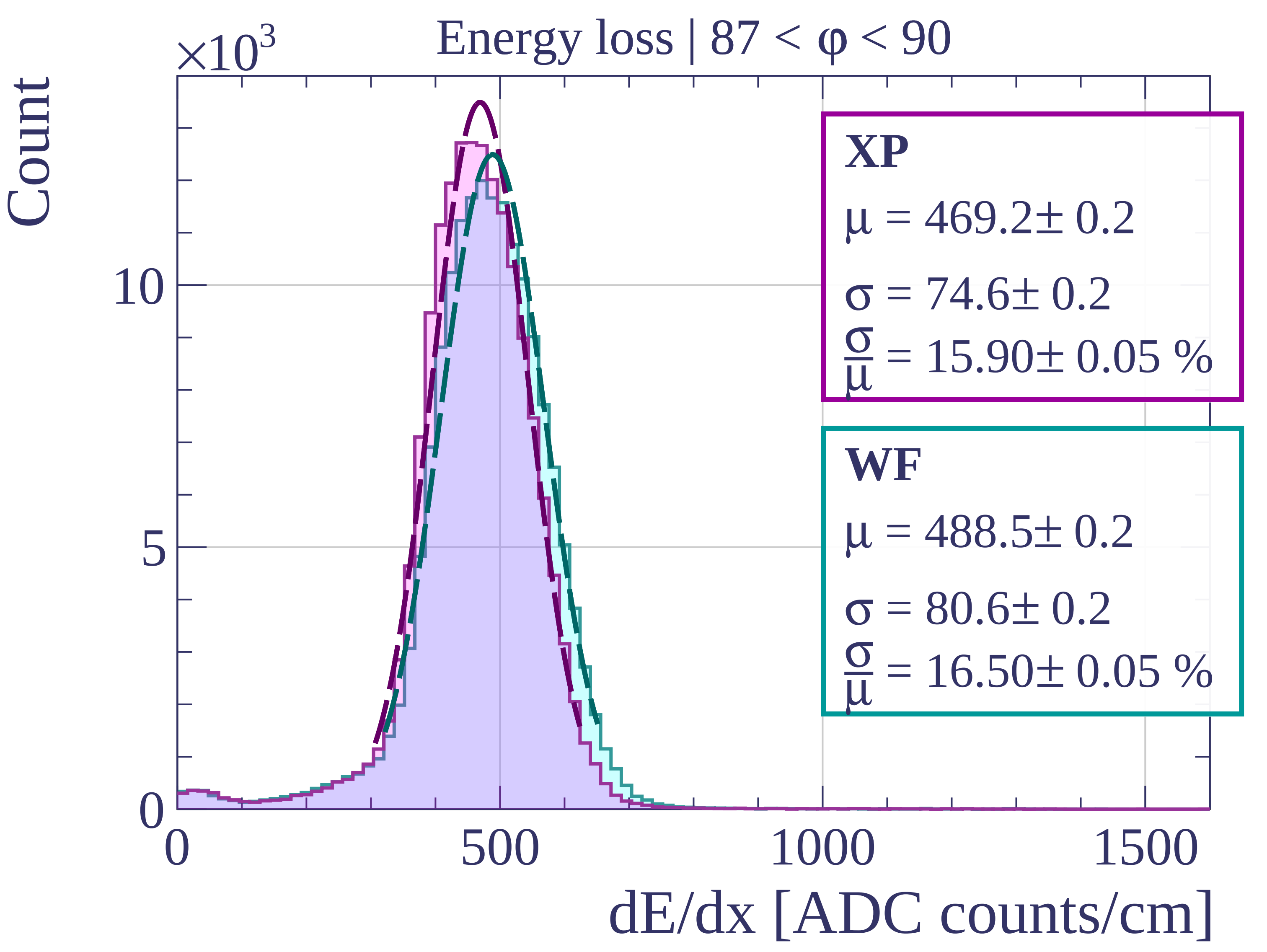


Energy loss | 75 < φ < 78
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 476.9± 0.2
σ = 73.7± 0.2
σ/μ = 15.46± 0.04 %
WF
μ = 490.4± 0.2
σ = 78.6± 0.2
σ/μ = 16.03± 0.05 %

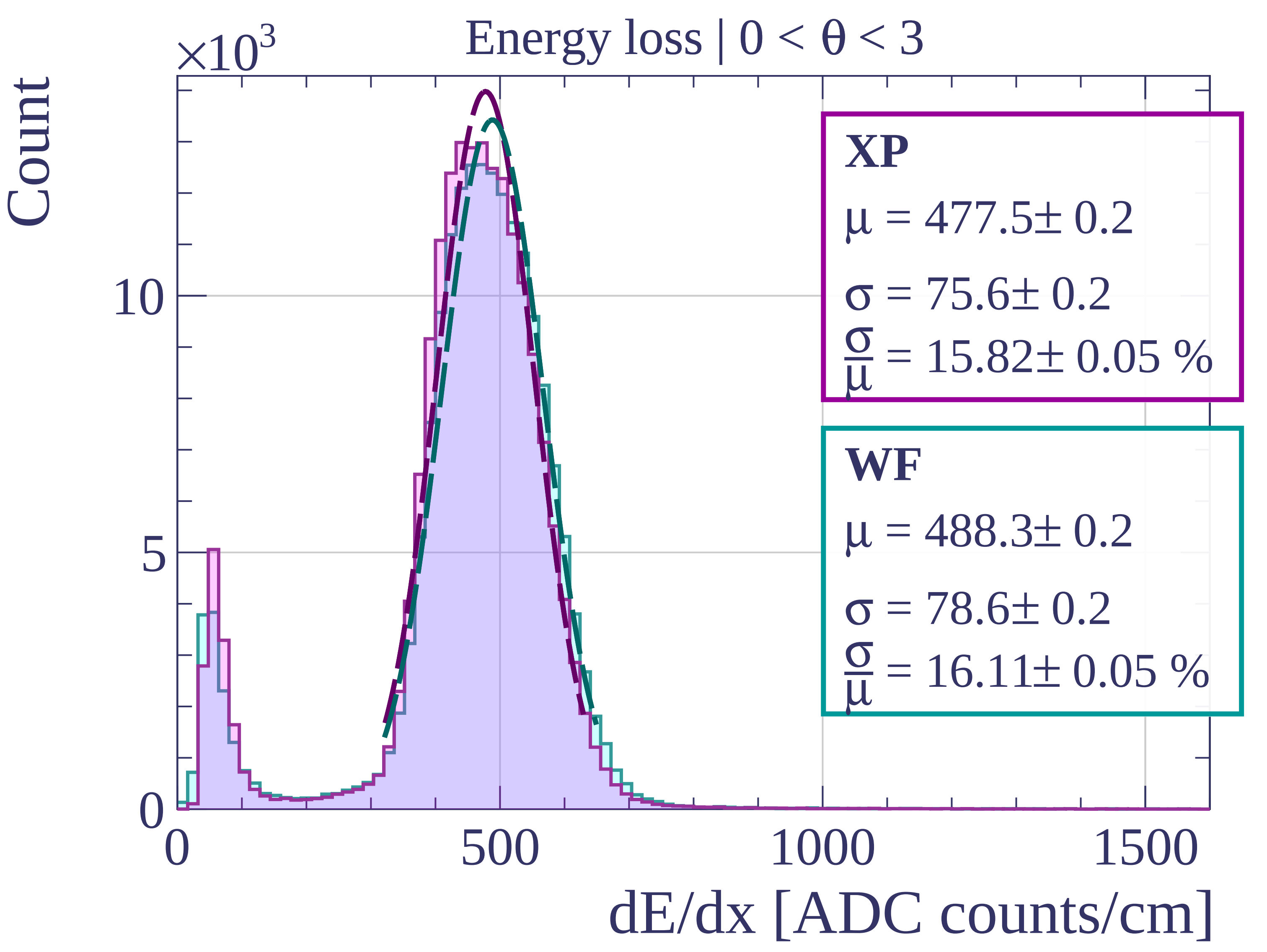

Energy loss | 78 < φ < 81
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 475.1± 0.2
σ = 74.4± 0.2
σ/μ = 15.65± 0.05 %
WF
μ = 490.2± 0.2
σ = 78.6± 0.2
σ/μ = 16.04± 0.05 %


Energy loss | 81 < φ < 84
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 473.7± 0.2
σ = 73.7± 0.2
σ/μ = 15.56± 0.05 %
WF
μ = 489.4± 0.2
σ = 77.9± 0.2
σ/μ = 15.91± 0.05 %

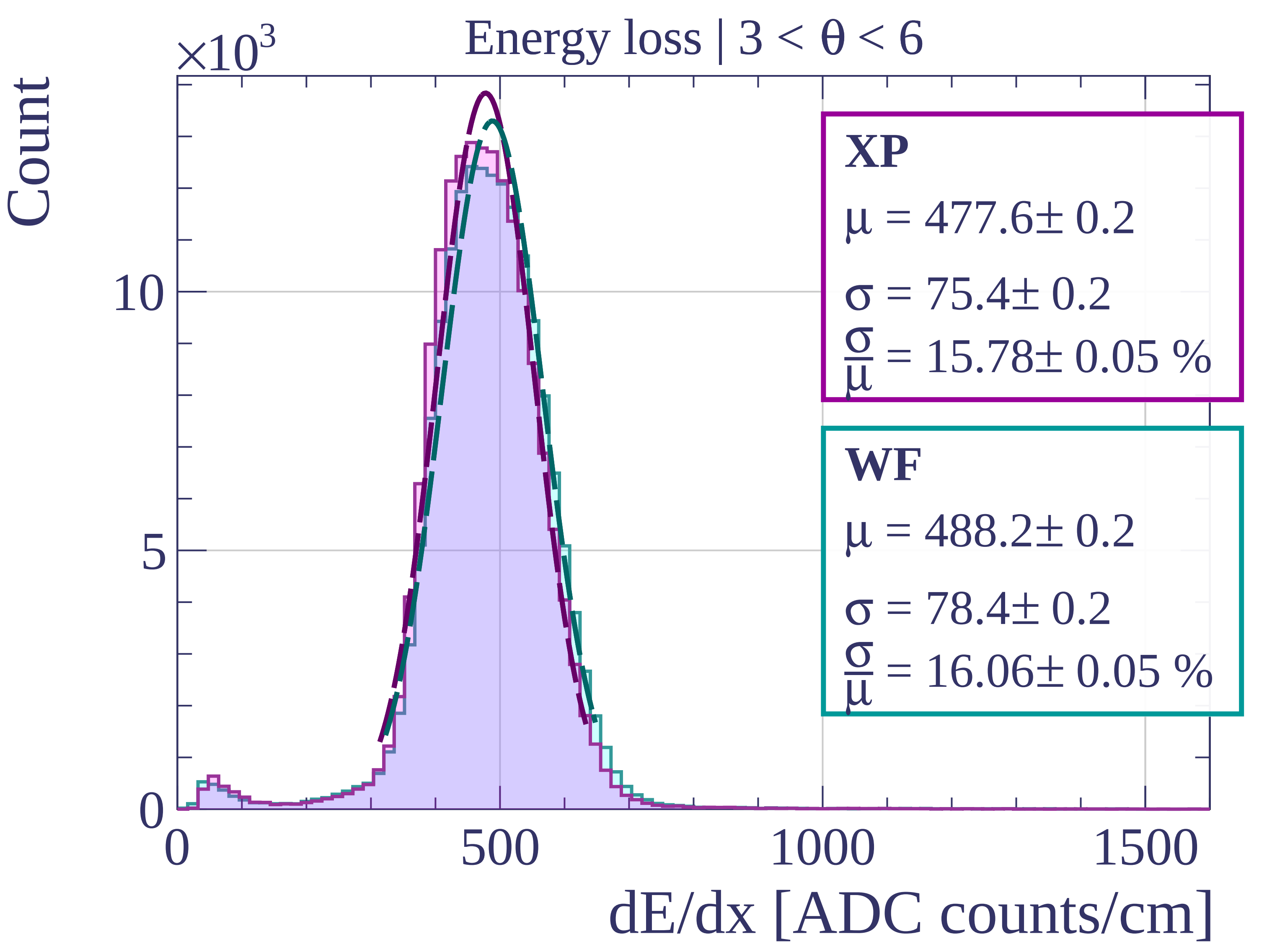

Energy loss | 84 < φ < 87
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 472.5± 0.2
σ = 74.0± 0.2
σ/μ = 15.67± 0.05 %
WF
μ = 489.3± 0.2
σ = 78.6± 0.2
σ/μ = 16.06± 0.05 %



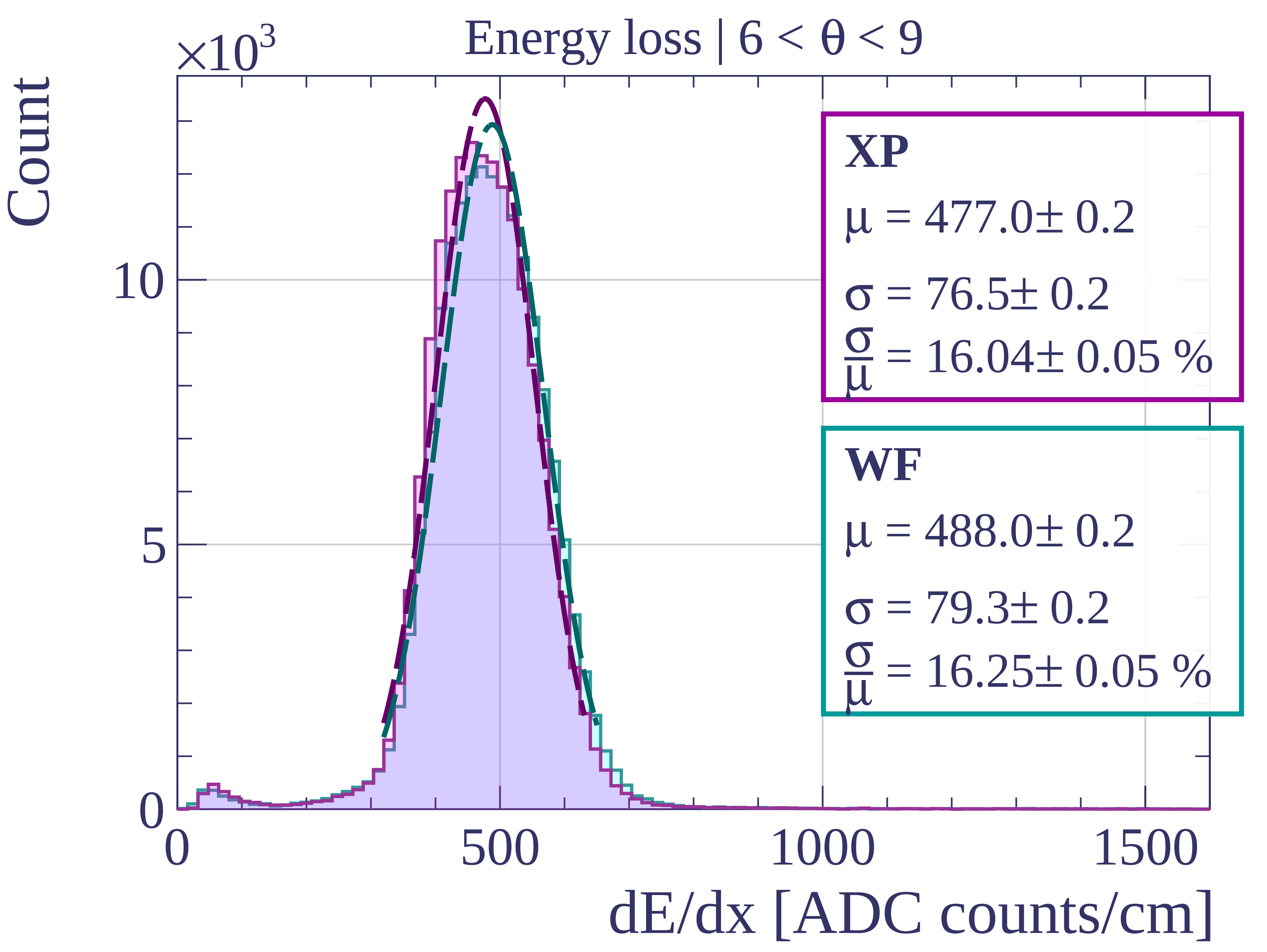
Energy loss | 87 < φ < 90
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 469.2± 0.2
σ = 74.6± 0.2
σ/μ = 15.90± 0.05 %
WF
μ = 488.5± 0.2
σ = 80.6± 0.2
σ/μ = 16.50± 0.05 %

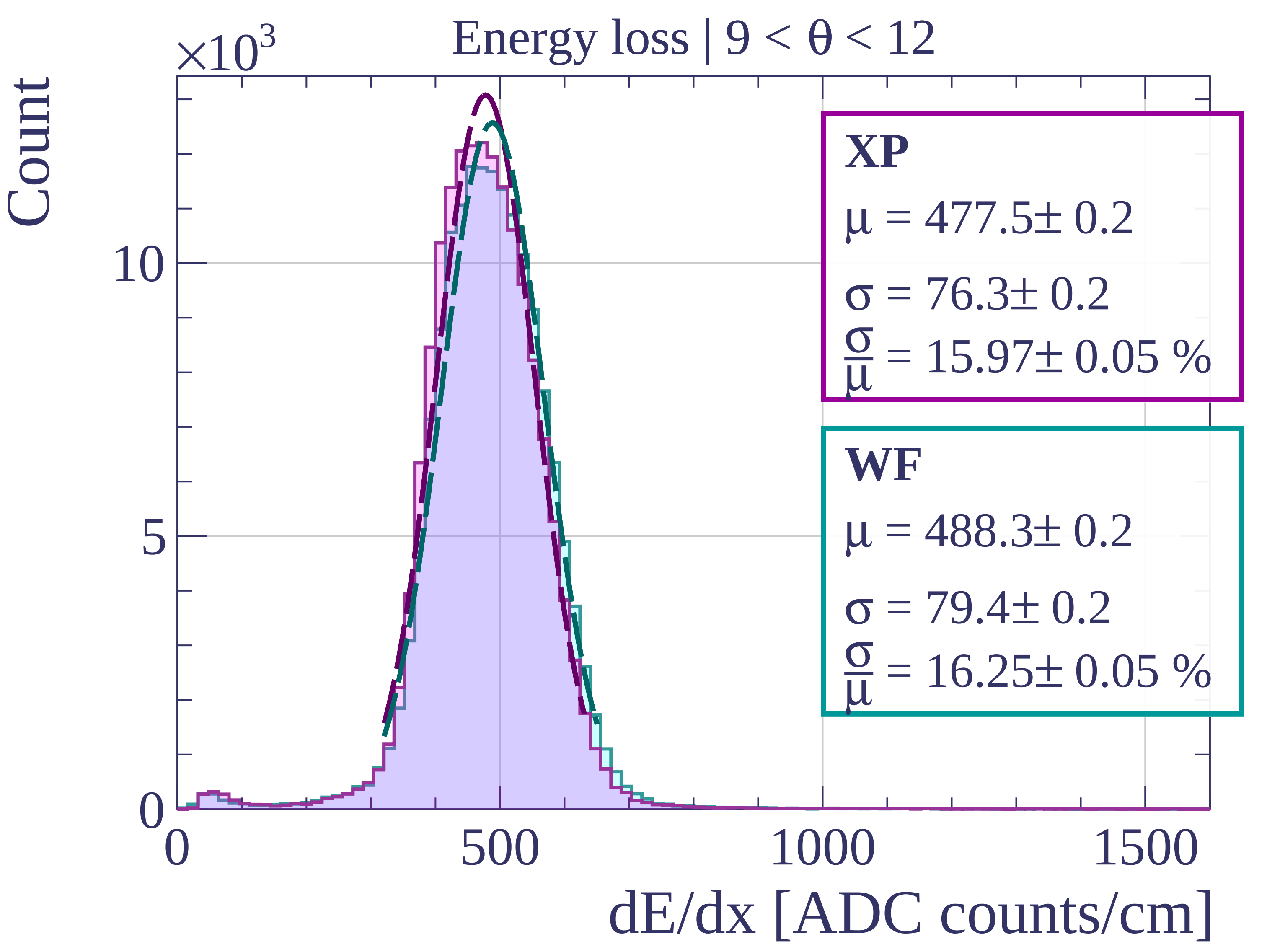


Energy loss | 0 < θ < 3
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 477.5± 0.2
σ = 75.6± 0.2
σ/μ = 15.82± 0.05 %
WF
μ = 488.3± 0.2
σ = 78.6± 0.2
σ/μ = 16.11± 0.05 %

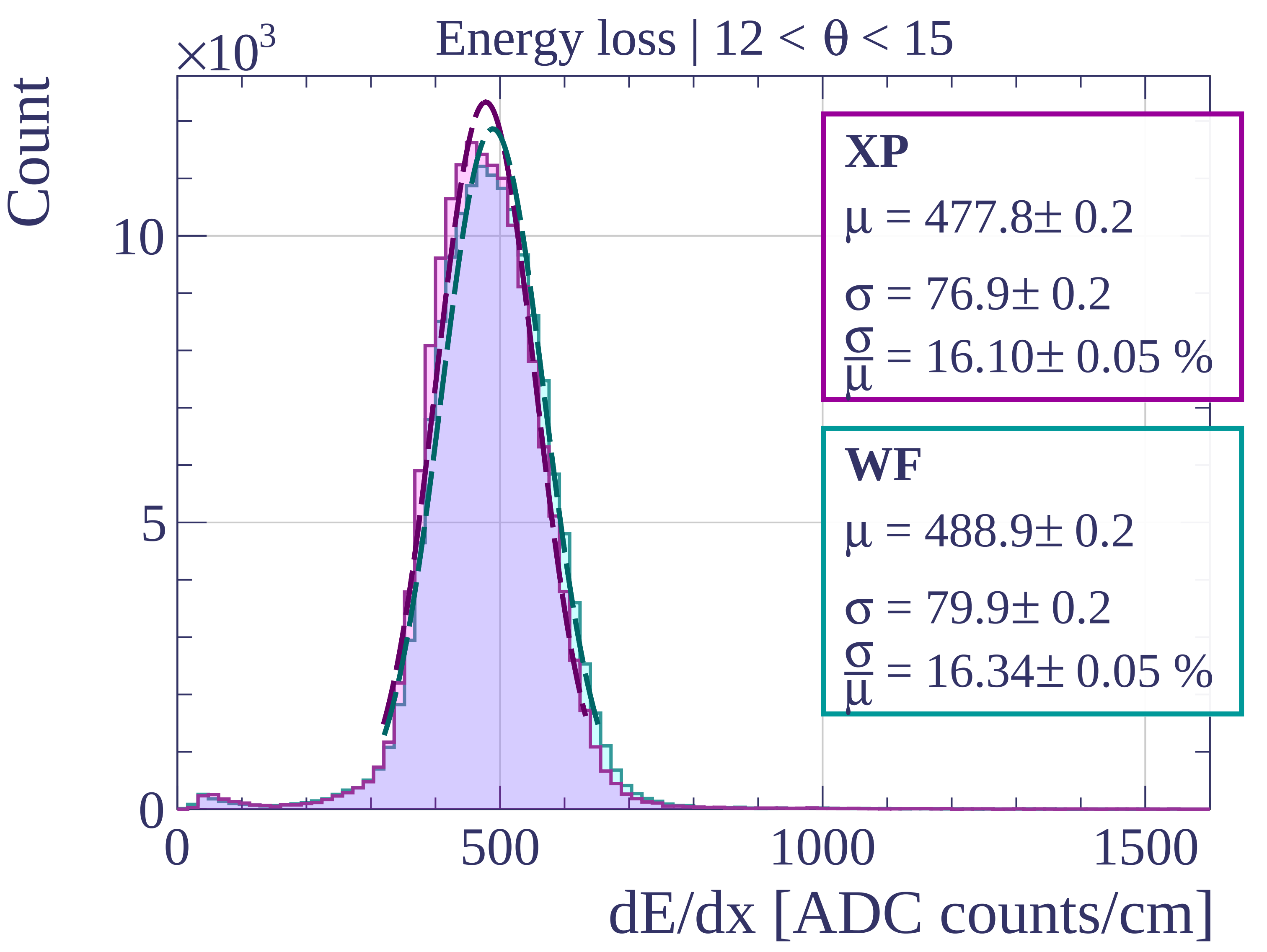


Energy loss | 3 < θ < 6
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 477.6± 0.2
σ = 75.4± 0.2
σ/μ = 15.78± 0.05 %
WF
μ = 488.2± 0.2
σ = 78.4± 0.2
σ/μ = 16.06± 0.05 %

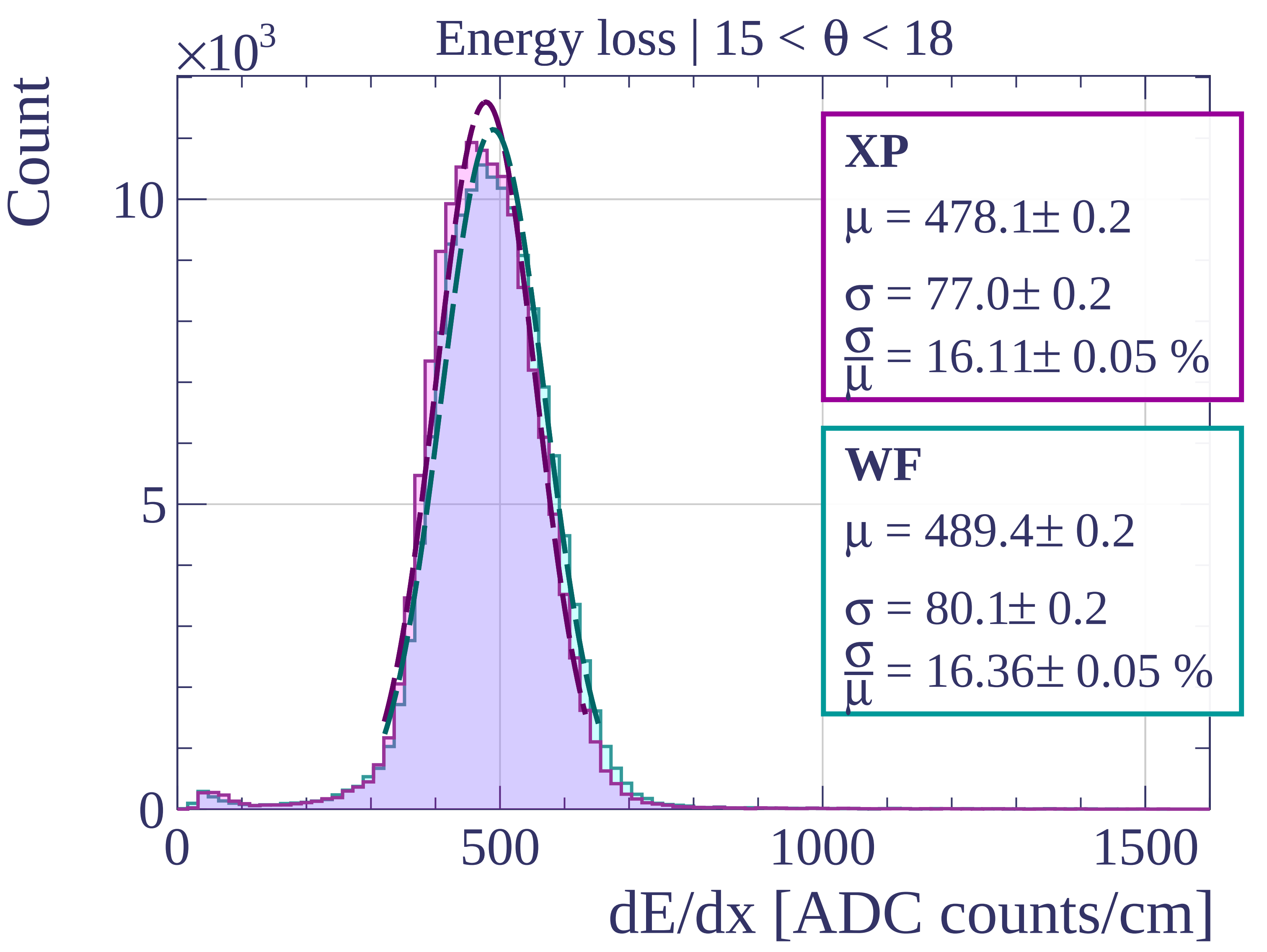

Energy loss | 6 < θ < 9
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 477.0± 0.2
σ = 76.5± 0.2
σ/μ = 16.04± 0.05 %
WF
μ = 488.0± 0.2
σ = 79.3± 0.2
σ/μ = 16.25± 0.05 %



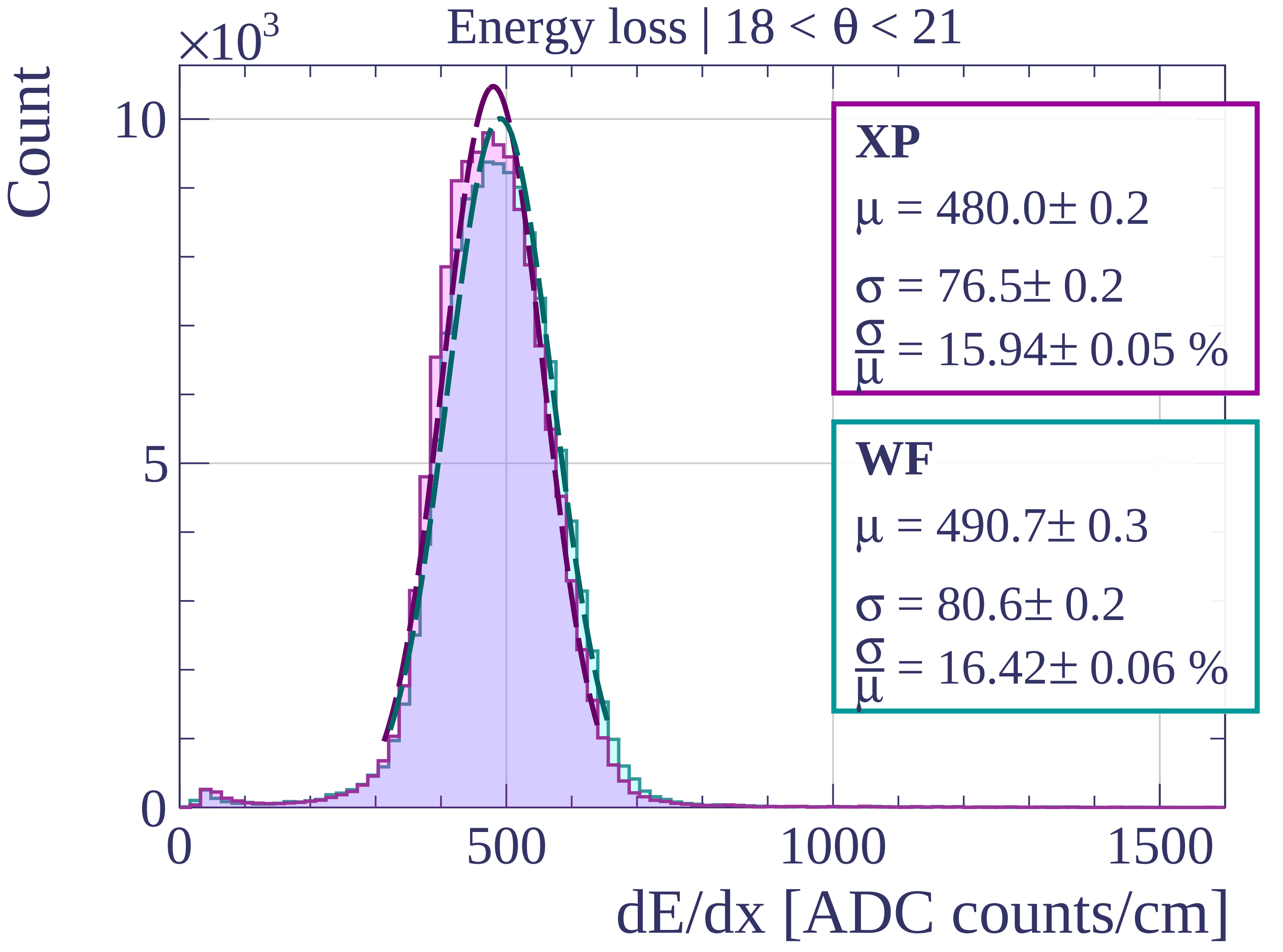
Energy loss | 9 < θ < 12
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 477.5± 0.2
σ = 76.3± 0.2
σ/μ = 15.97± 0.05 %
WF
μ = 488.3± 0.2
σ = 79.4± 0.2
σ/μ = 16.25± 0.05 %



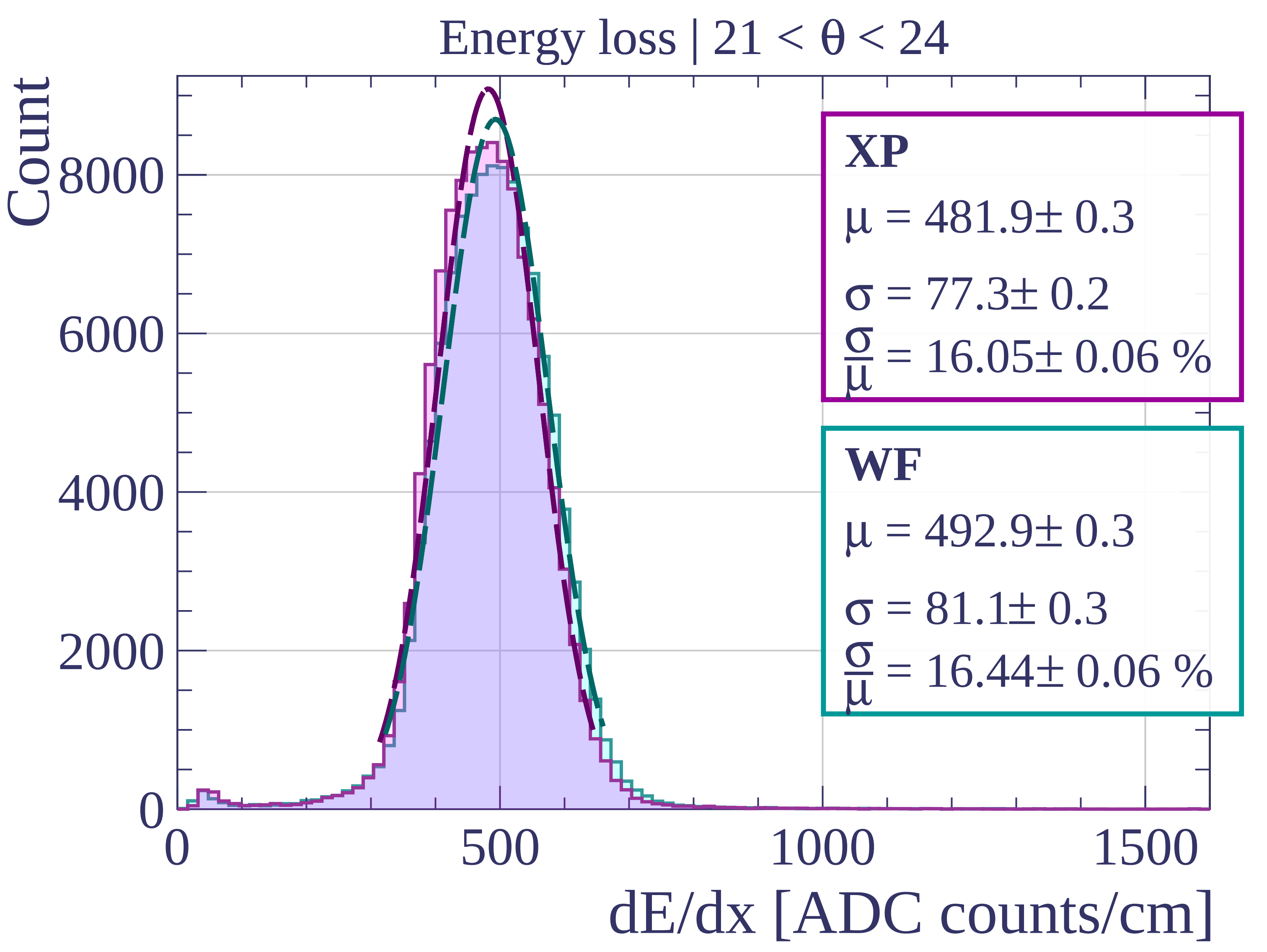
Energy loss | 12 < θ < 15
×10^3
Count
0
5
10
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 477.8± 0.2
σ = 76.9± 0.2
σ/μ = 16.10± 0.05 %
WF
μ = 488.9± 0.2
σ = 79.9± 0.2
σ/μ = 16.34± 0.05 %



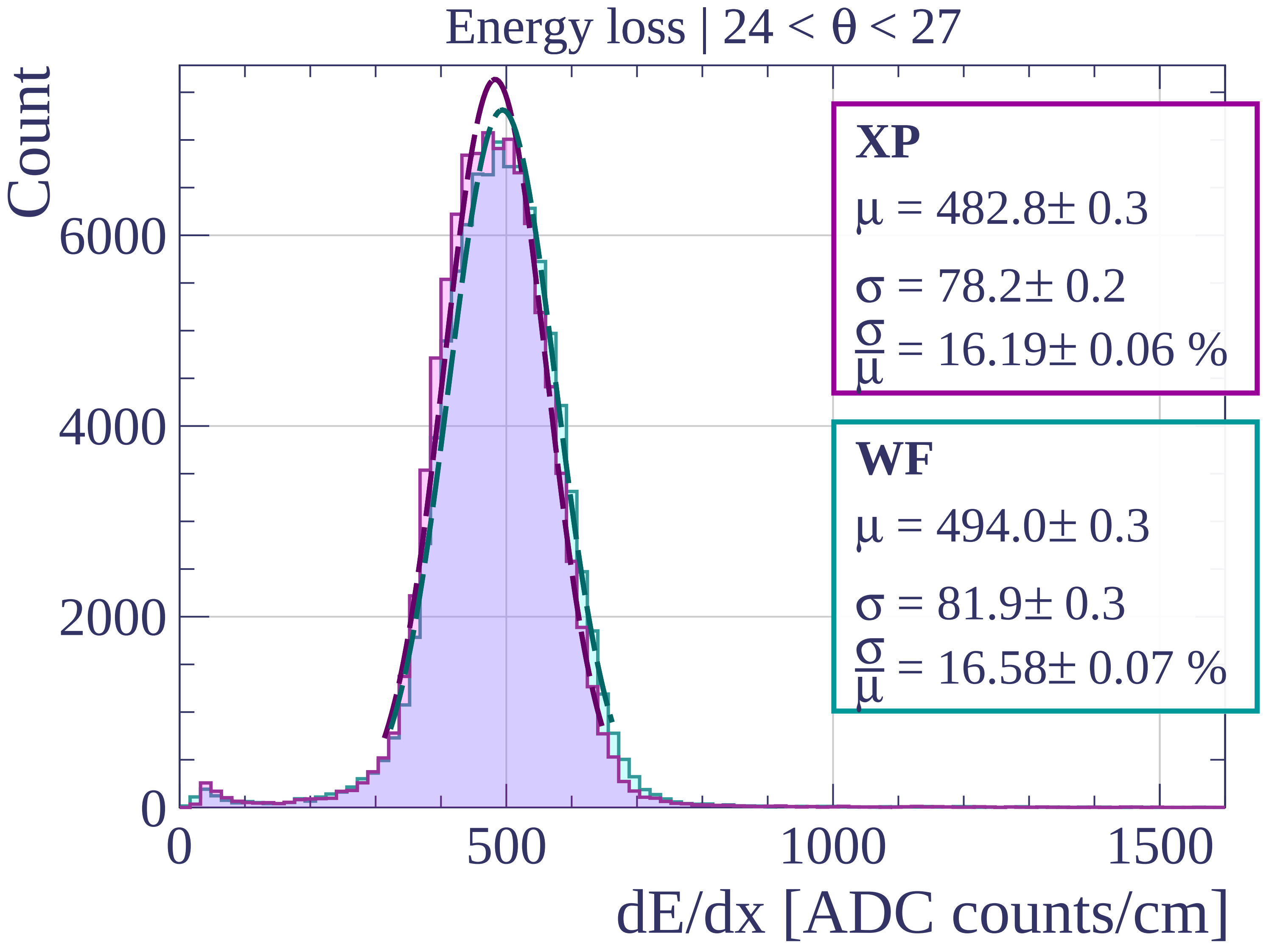
Energy loss | 15 < θ < 18
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 478.1± 0.2
σ = 77.0± 0.2
σ/μ = 16.11± 0.05 %
WF
μ = 489.4± 0.2
σ = 80.1± 0.2
σ/μ = 16.36± 0.05 %



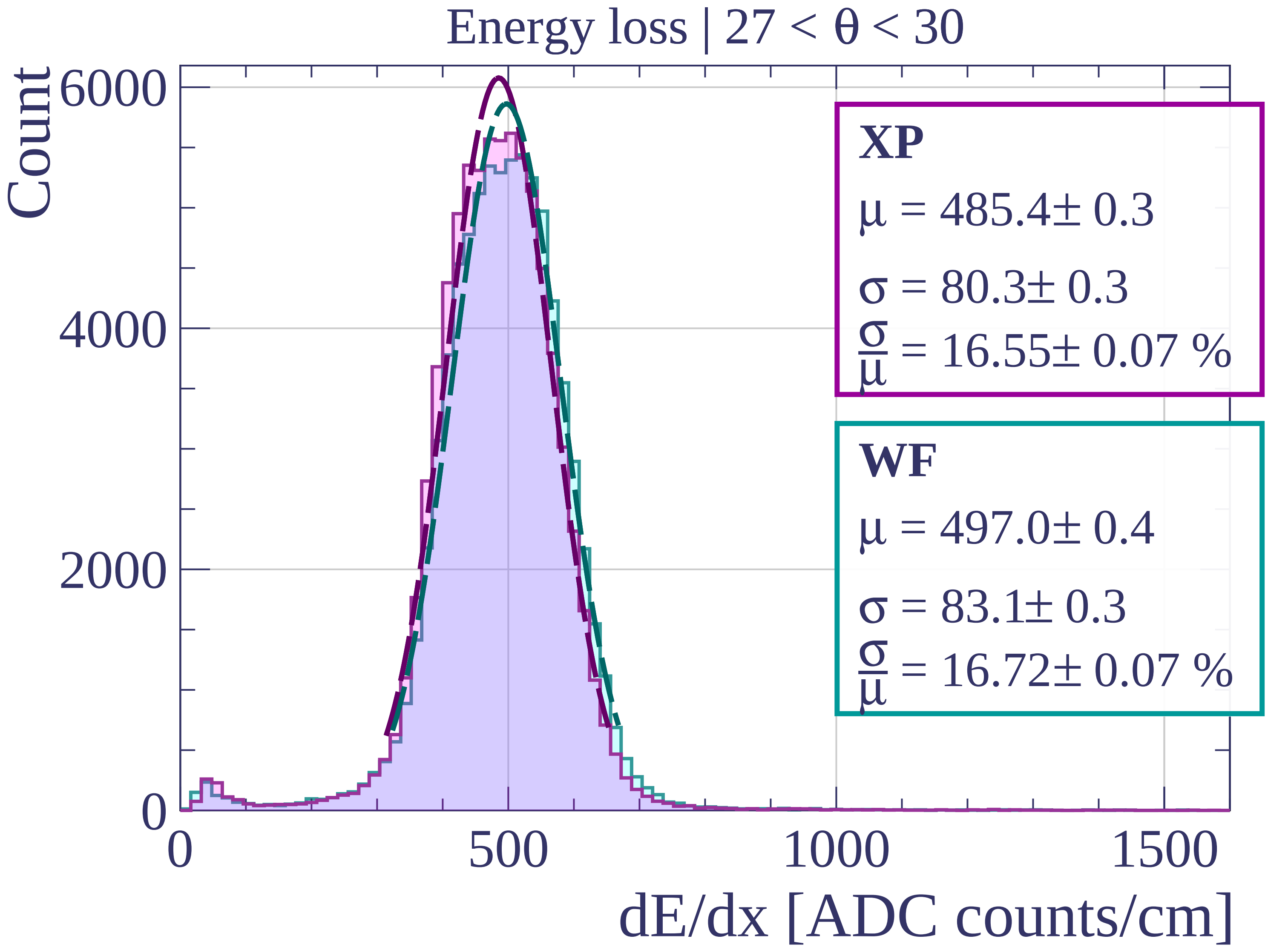
Energy loss | 18 < θ < 21
×10^3
Count
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 480.0± 0.2
σ = 76.5± 0.2
σ/μ = 15.94± 0.05 %
WF
μ = 490.7± 0.3
σ = 80.6± 0.2
σ/μ = 16.42± 0.06 %



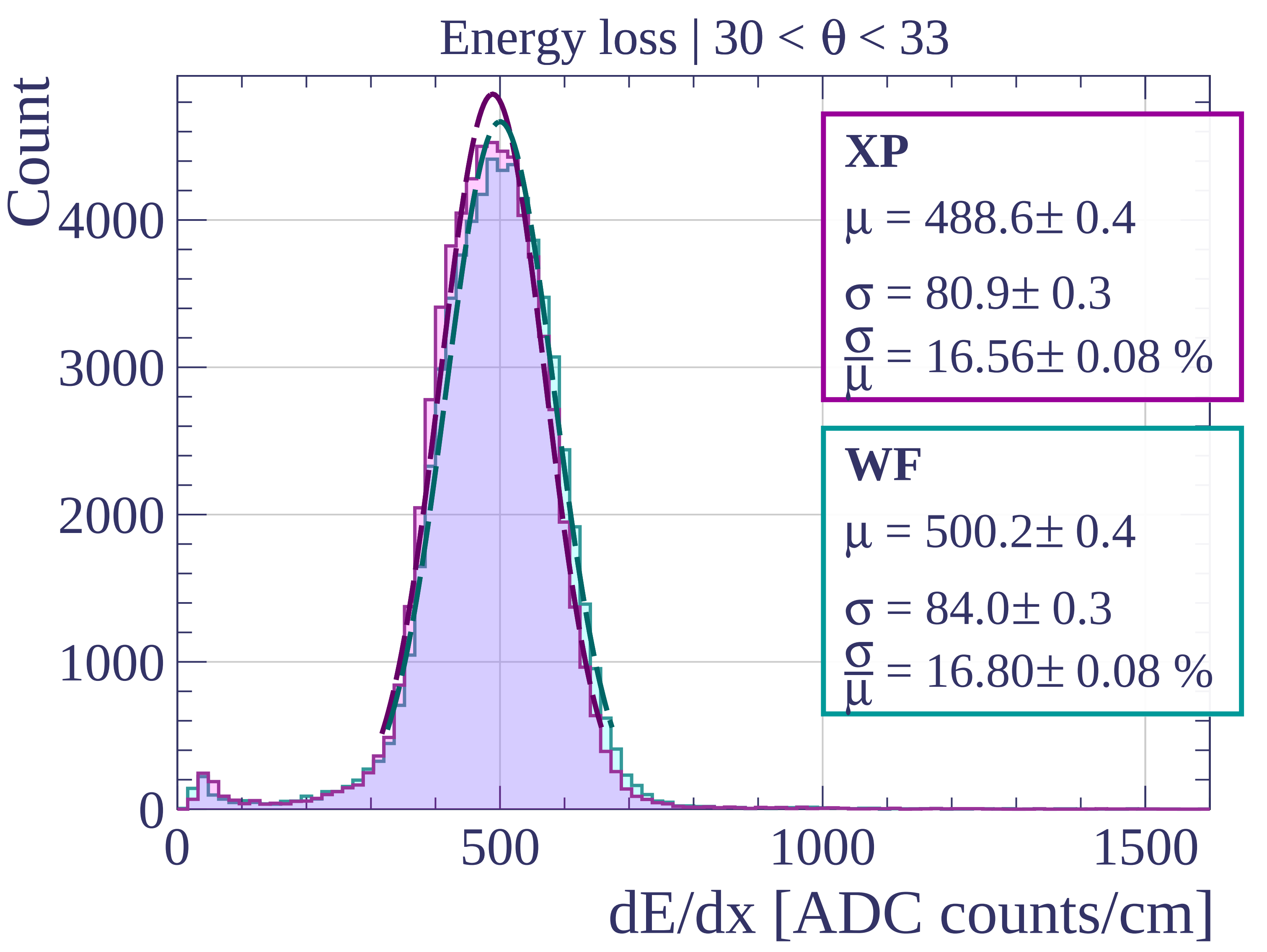
Energy loss | 21 < θ < 24
Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 481.9± 0.3
σ = 77.3± 0.2
σ/μ = 16.05± 0.06 %
WF
μ = 492.9± 0.3
σ = 81.1± 0.3
σ/μ = 16.44± 0.06 %



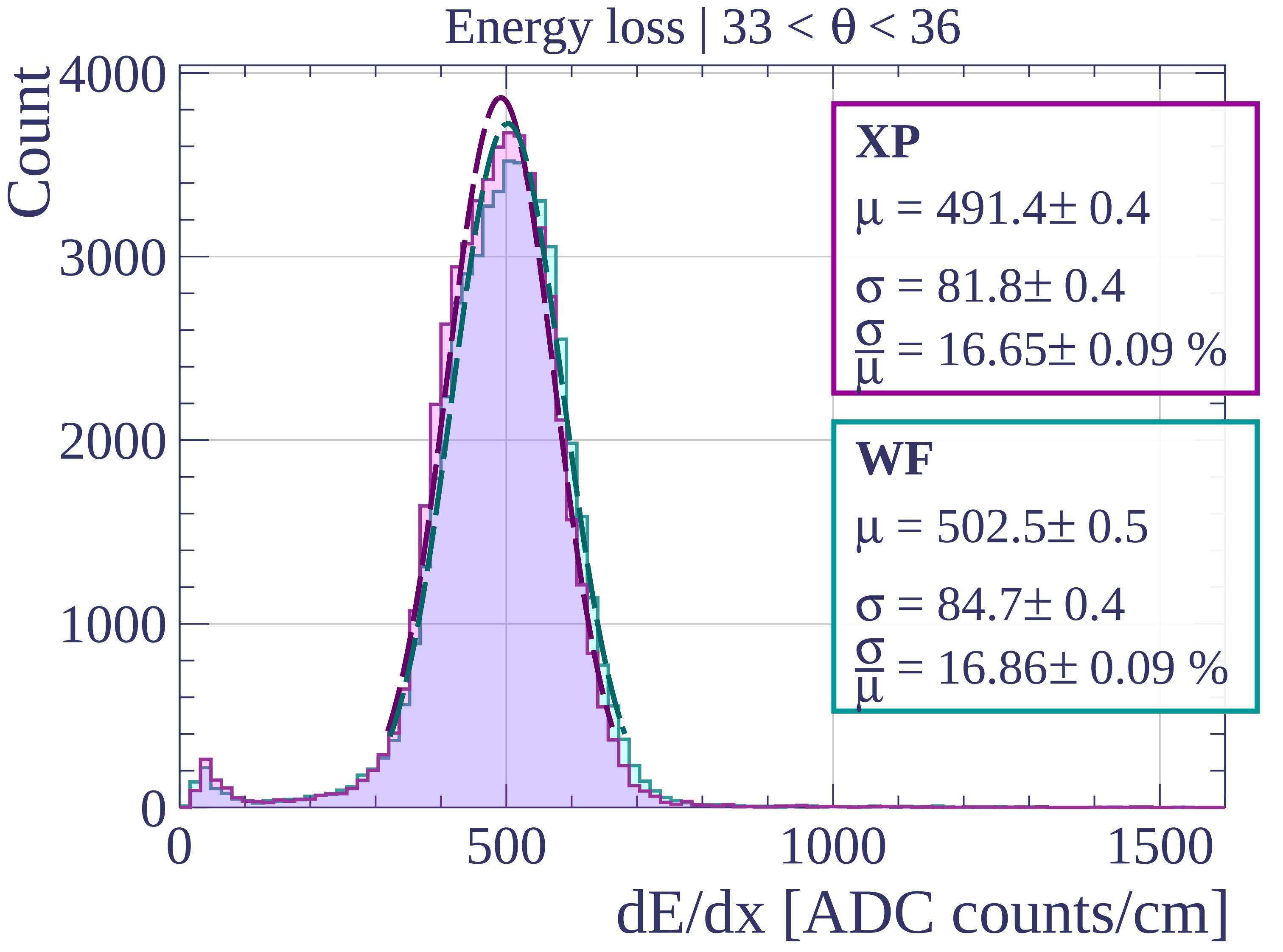
Energy loss | 24 < θ < 27
Count
0
2000
4000
6000
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 482.8± 0.3
σ = 78.2± 0.2
σ/μ = 16.19± 0.06 %
WF
μ = 494.0± 0.3
σ = 81.9± 0.3
σ/μ = 16.58± 0.07 %



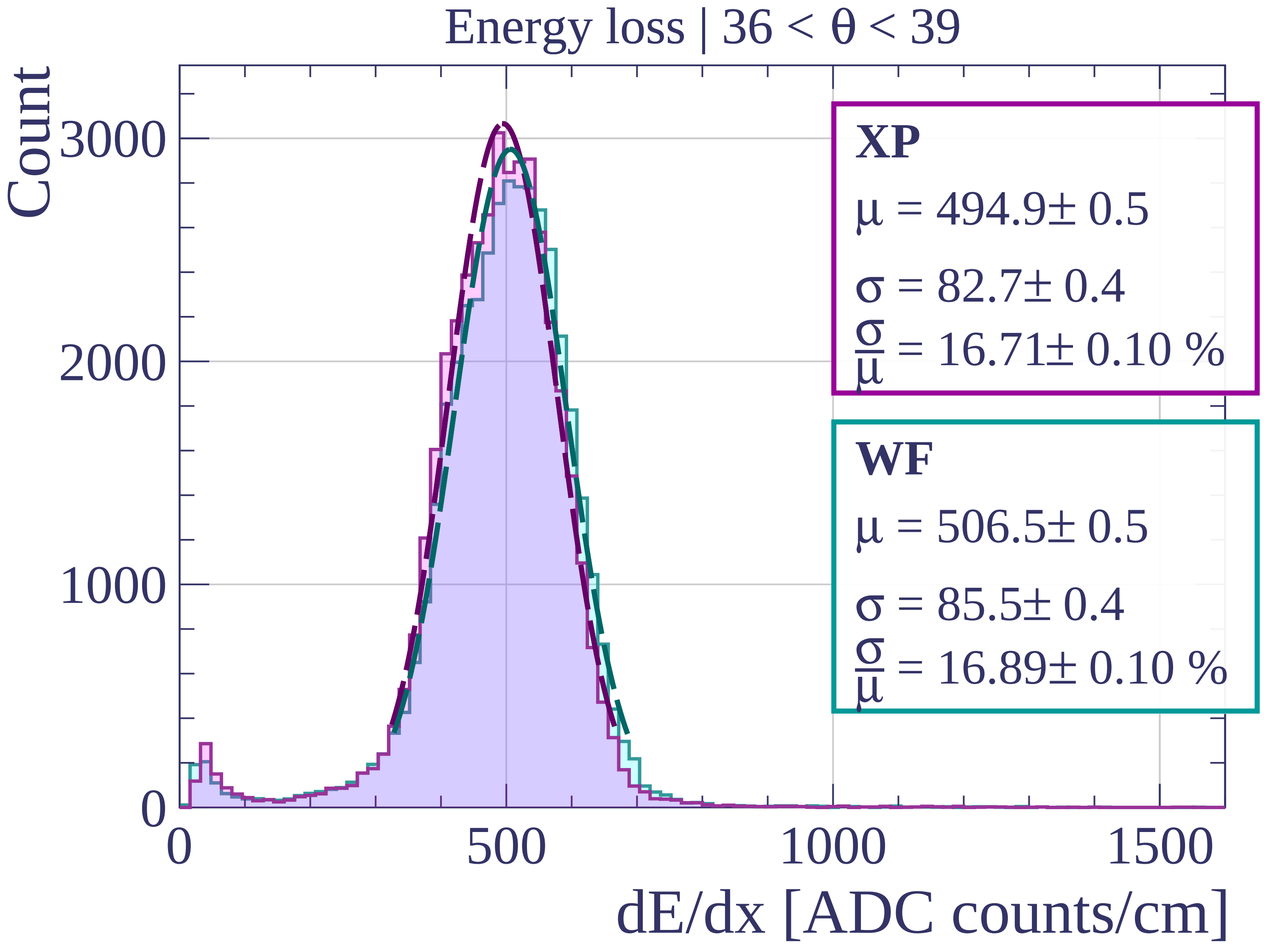
Energy loss | 27 < θ < 30
Count
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 485.4± 0.3
σ = 80.3± 0.3
σ/μ = 16.55± 0.07 %
WF
μ = 497.0± 0.4
σ = 83.1± 0.3
σ/μ = 16.72± 0.07 %

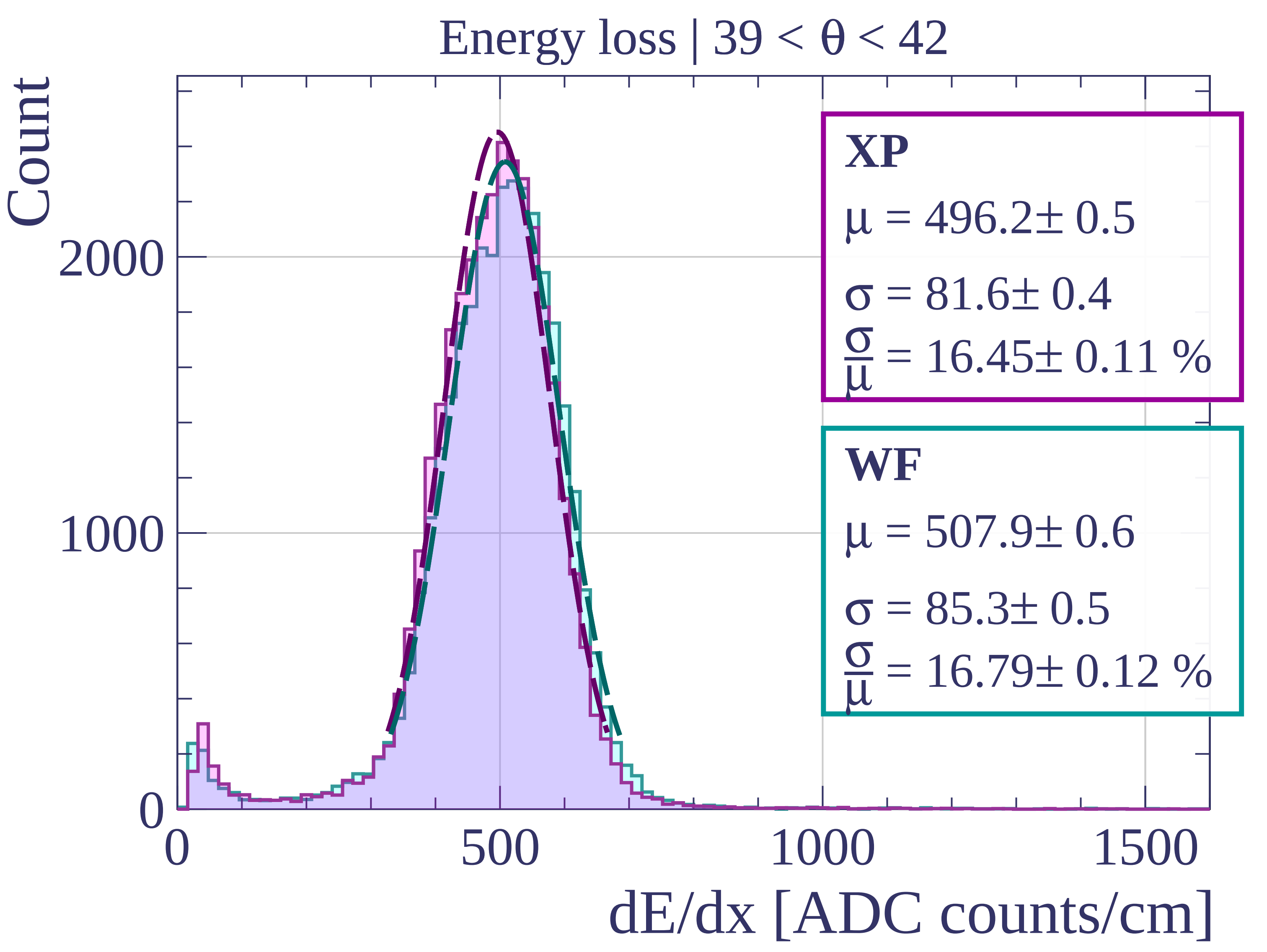


Energy loss | 30 < θ < 33
Count
dE/dx [ADC counts/cm]
0
1000
2000
3000
4000
0
500
1000
1500
XP
μ = 488.6± 0.4
σ = 80.9± 0.3
σ/μ = 16.56± 0.08 %
WF
μ = 500.2± 0.4
σ = 84.0± 0.3
σ/μ = 16.80± 0.08 %

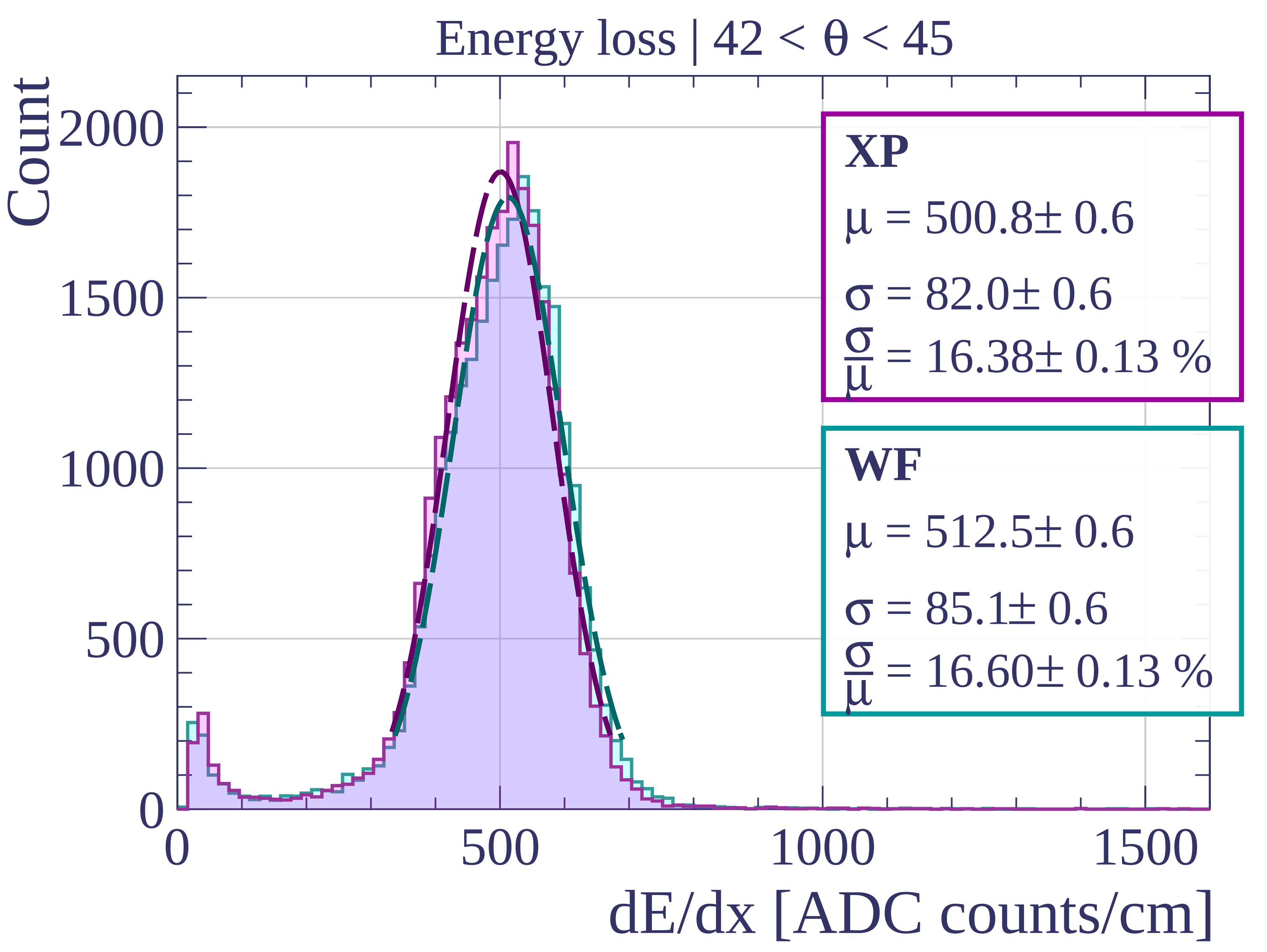


Energy loss | 33 < θ < 36
Count
4000
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 491.4± 0.4
σ = 81.8± 0.4
σ/μ = 16.65± 0.09 %
WF
μ = 502.5± 0.5
σ = 84.7± 0.4
σ/μ = 16.86± 0.09 %

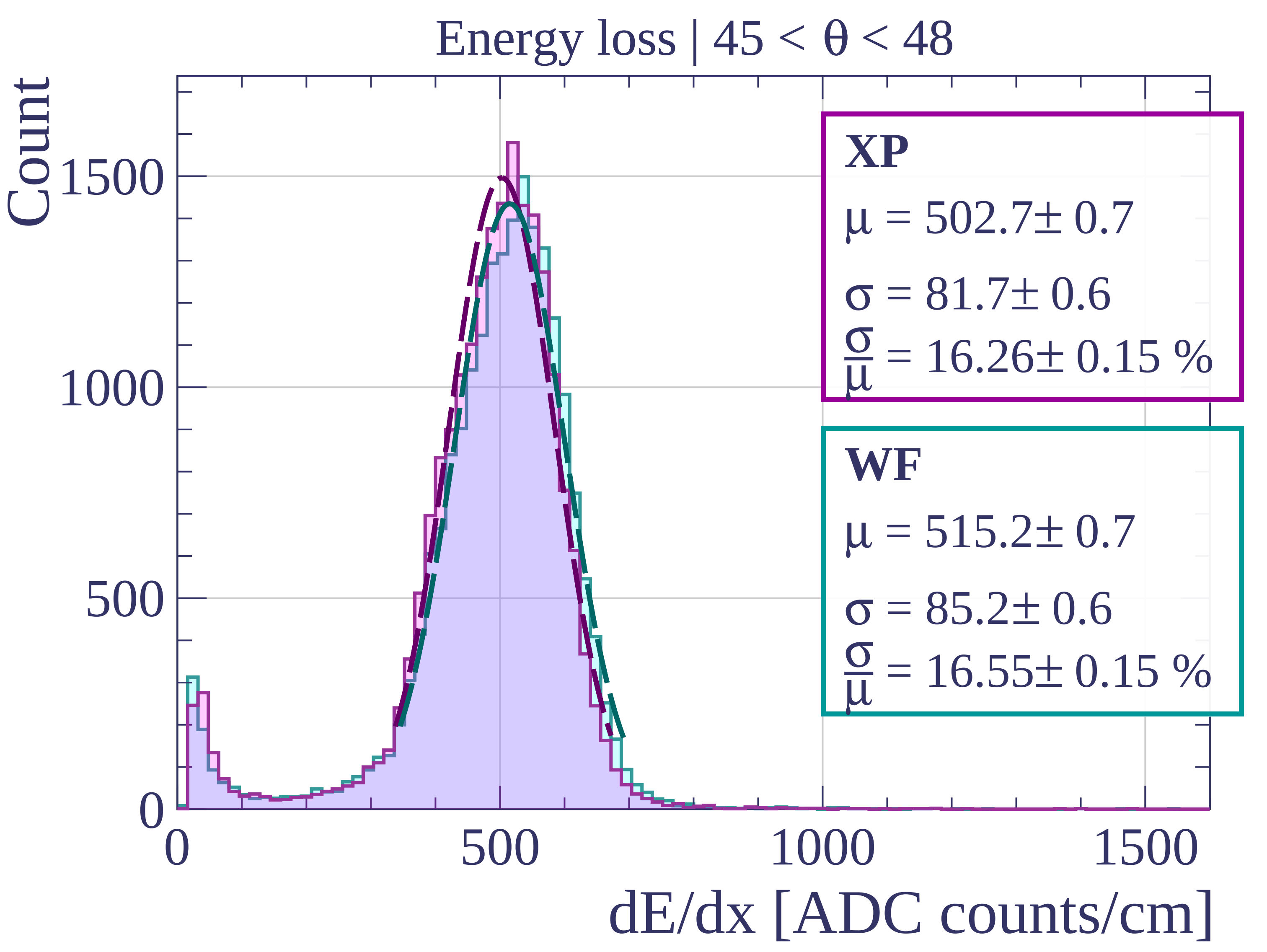

Energy loss | 36 < θ < 39
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 494.9± 0.5
σ = 82.7± 0.4
σ/μ = 16.71± 0.10 %
WF
μ = 506.5± 0.5
σ = 85.5± 0.4
σ/μ = 16.89± 0.10 %



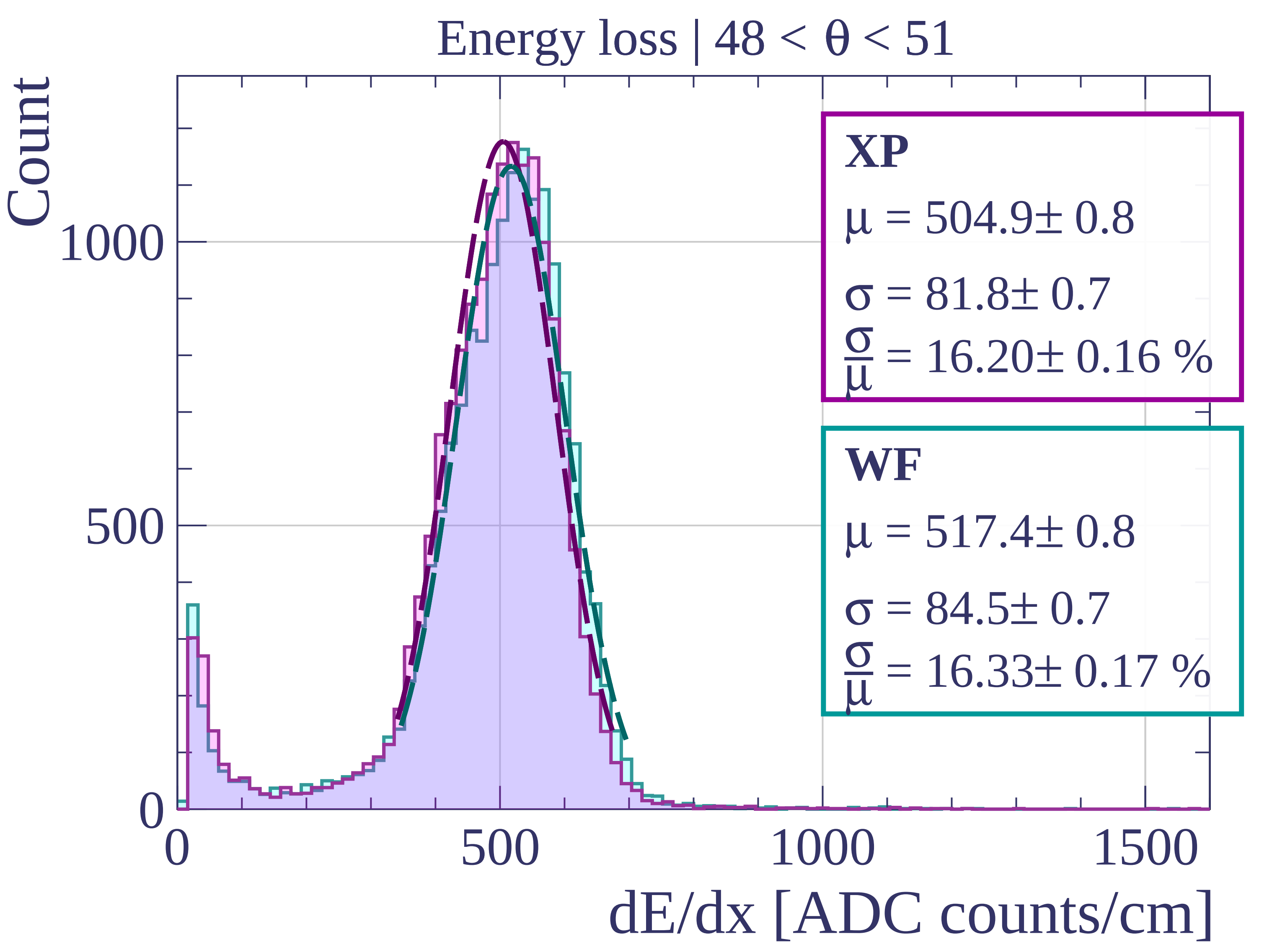
Energy loss | 39 < θ < 42
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 496.2± 0.5
σ = 81.6± 0.4
σ/μ = 16.45± 0.11 %
WF
μ = 507.9± 0.6
σ = 85.3± 0.5
σ/μ = 16.79± 0.12 %



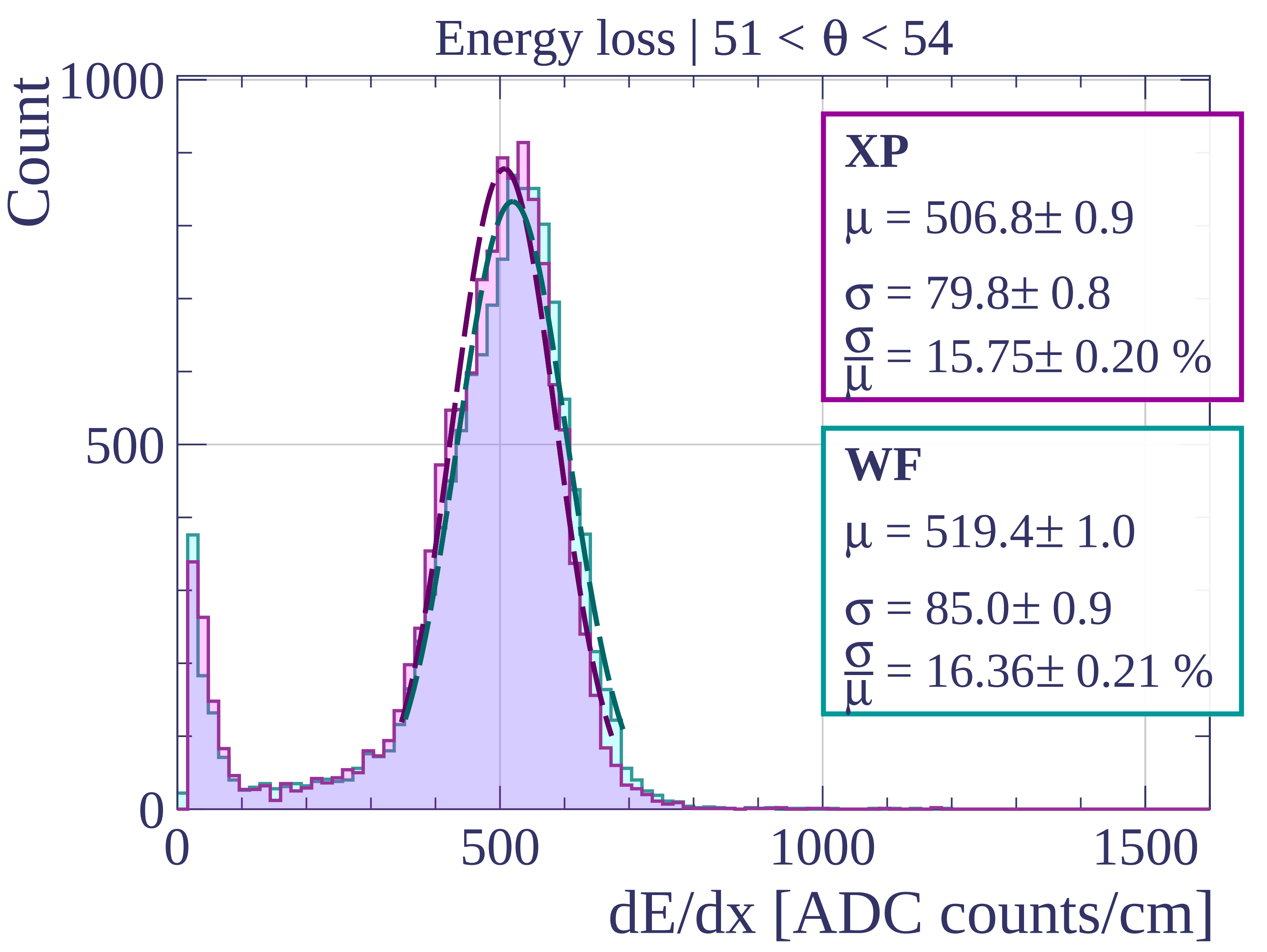
Energy loss | 42 < θ < 45
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 500.8± 0.6
σ = 82.0± 0.6
σ/μ = 16.38± 0.13 %
WF
μ = 512.5± 0.6
σ = 85.1± 0.6
σ/μ = 16.60± 0.13 %

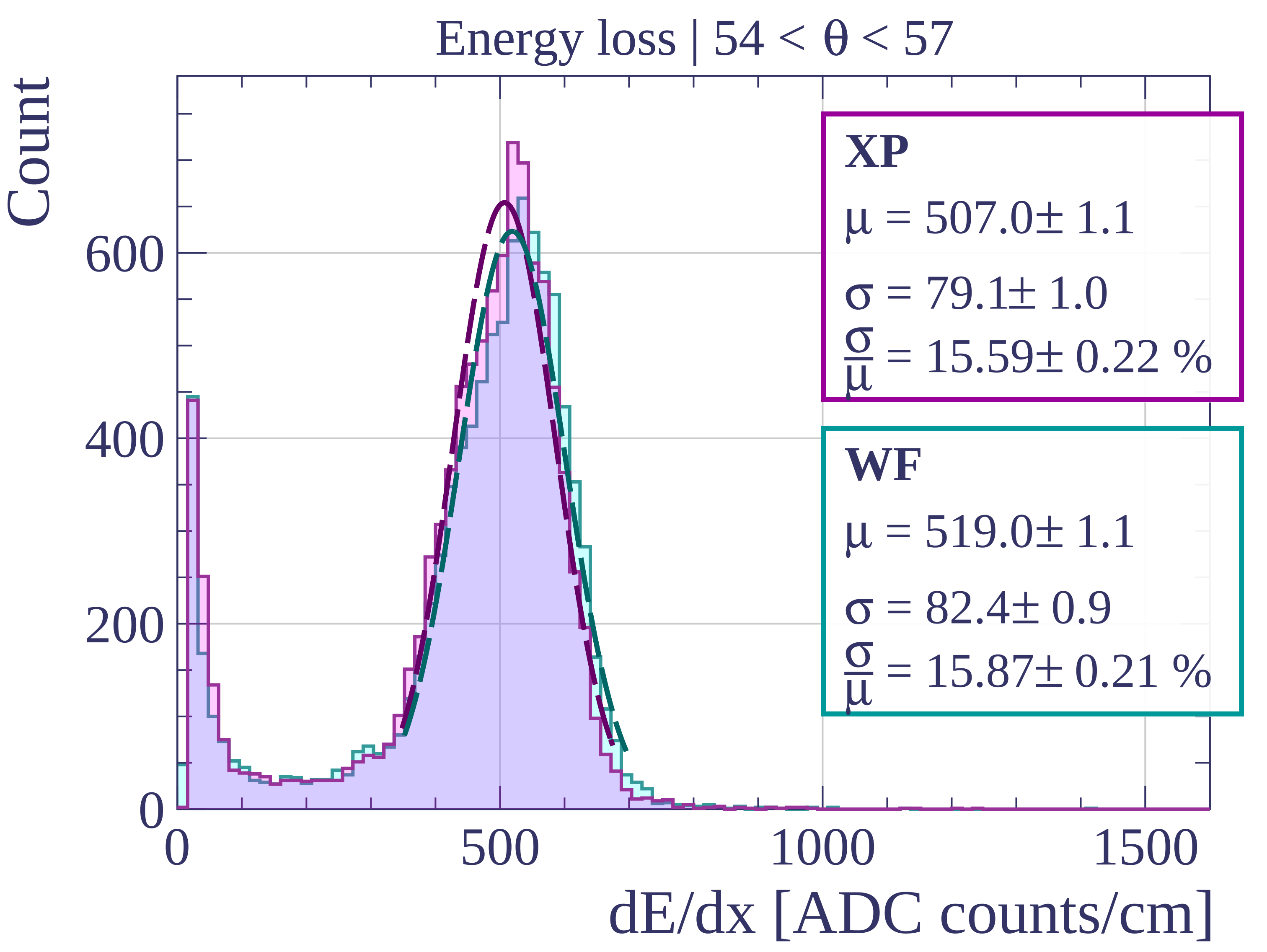


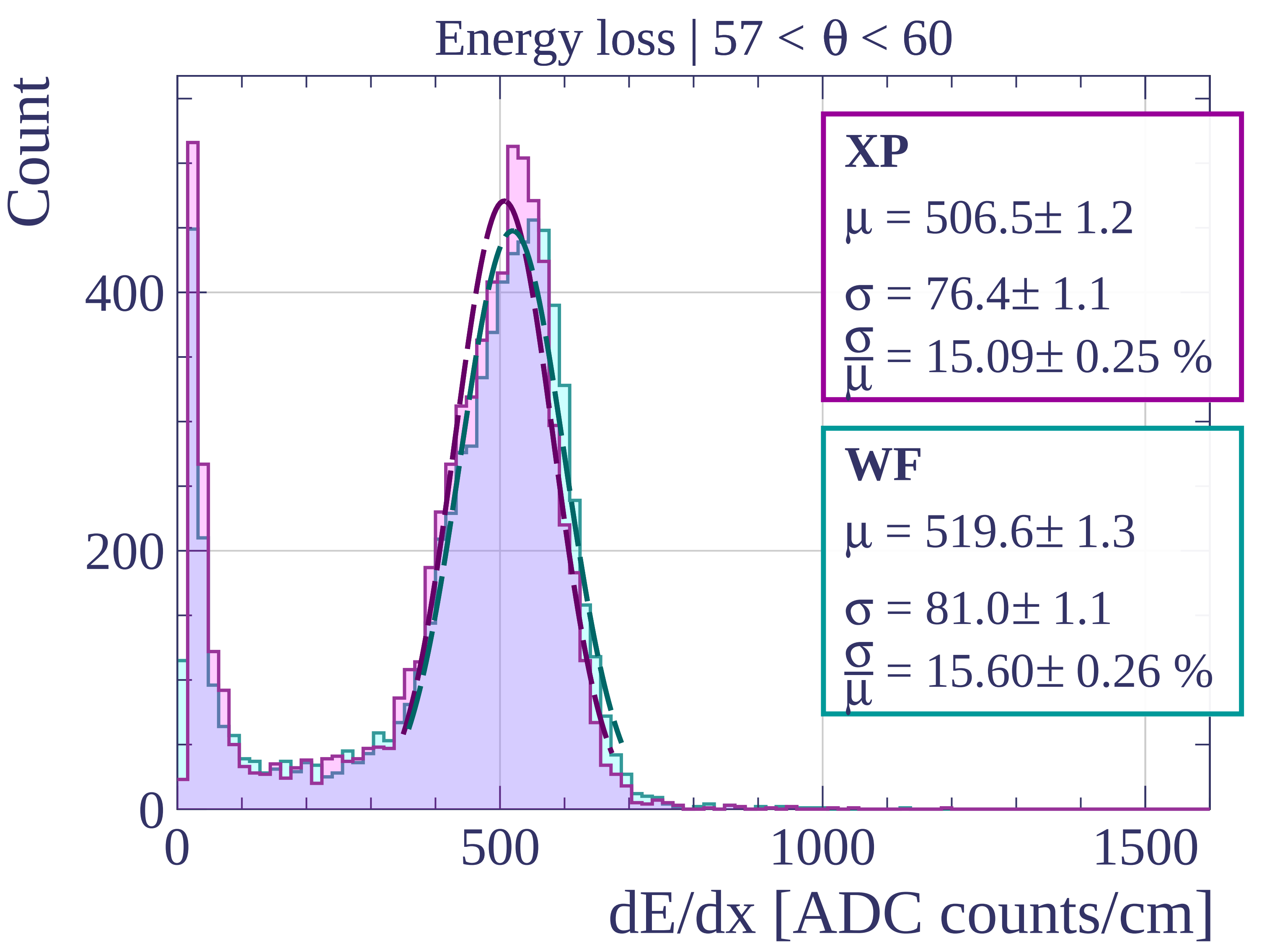
Energy loss | 45 < θ < 48
Count
0
500
1000
1500
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 502.7± 0.7
σ = 81.7± 0.6
σ/μ = 16.26± 0.15 %
WF
μ = 515.2± 0.7
σ = 85.2± 0.6
σ/μ = 16.55± 0.15 %



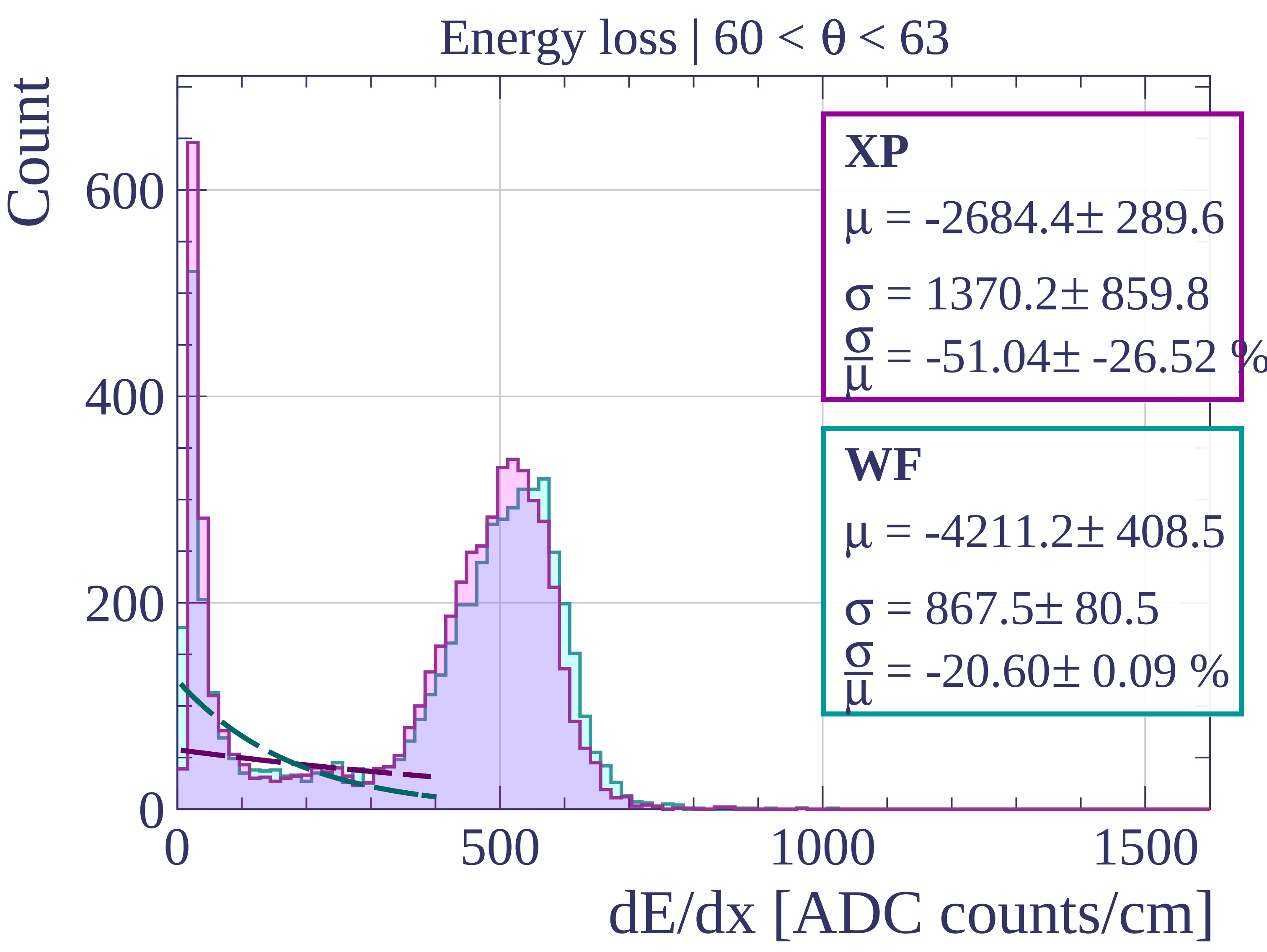
Energy loss | 48 < θ < 51
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = 504.9± 0.8
σ = 81.8± 0.7
σ/μ = 16.20± 0.16 %
WF
μ = 517.4± 0.8
σ = 84.5± 0.7
σ/μ = 16.33± 0.17 %


Energy loss | 51 < θ < 54
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = 506.8± 0.9
σ = 79.8± 0.8
σ/μ = 15.75± 0.20 %
WF
μ = 519.4± 1.0
σ = 85.0± 0.9
σ/μ = 16.36± 0.21 %

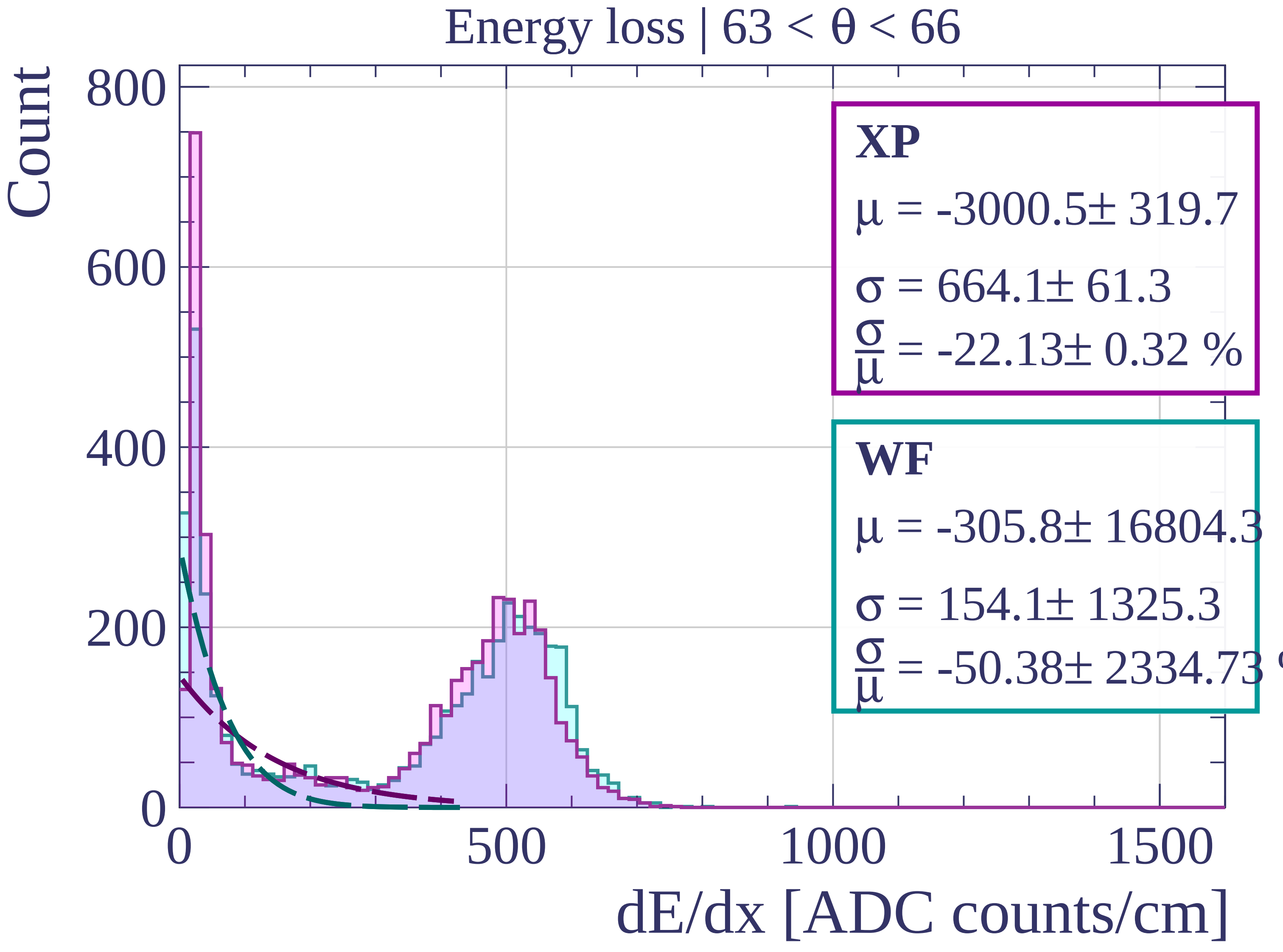

Energy loss | 54 < θ < 57
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
XP
μ = 507.0± 1.1
σ = 79.1± 1.0
σ/μ = 15.59± 0.22 %
WF
μ = 519.0± 1.1
σ = 82.4± 0.9
σ/μ = 15.87± 0.21 %



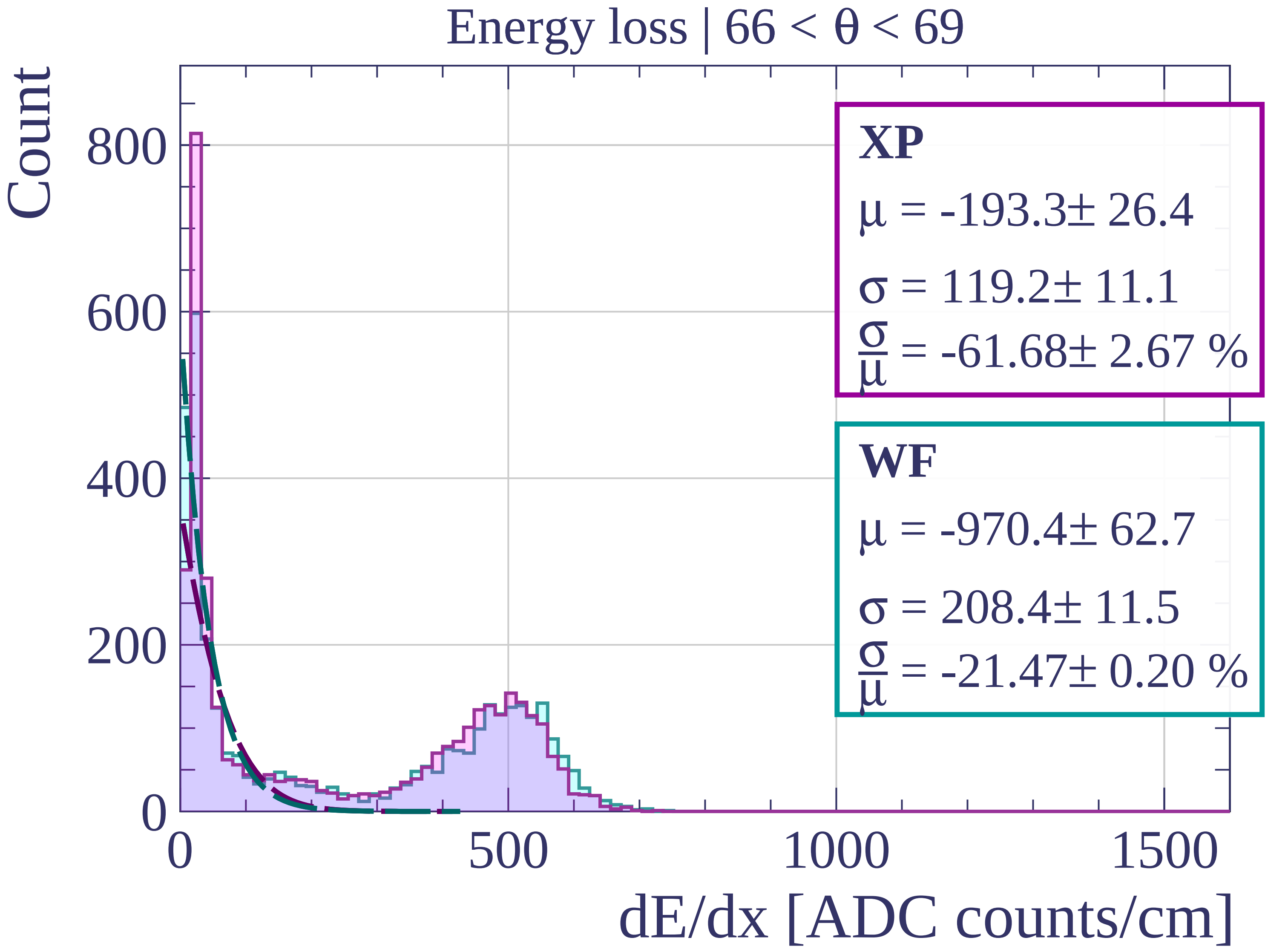
Energy loss | 57 < θ < 60
Count
0
200
400
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 506.5± 1.2
σ = 76.4± 1.1
σ/μ = 15.09± 0.25 %
WF
μ = 519.6± 1.3
σ = 81.0± 1.1
σ/μ = 15.60± 0.26 %



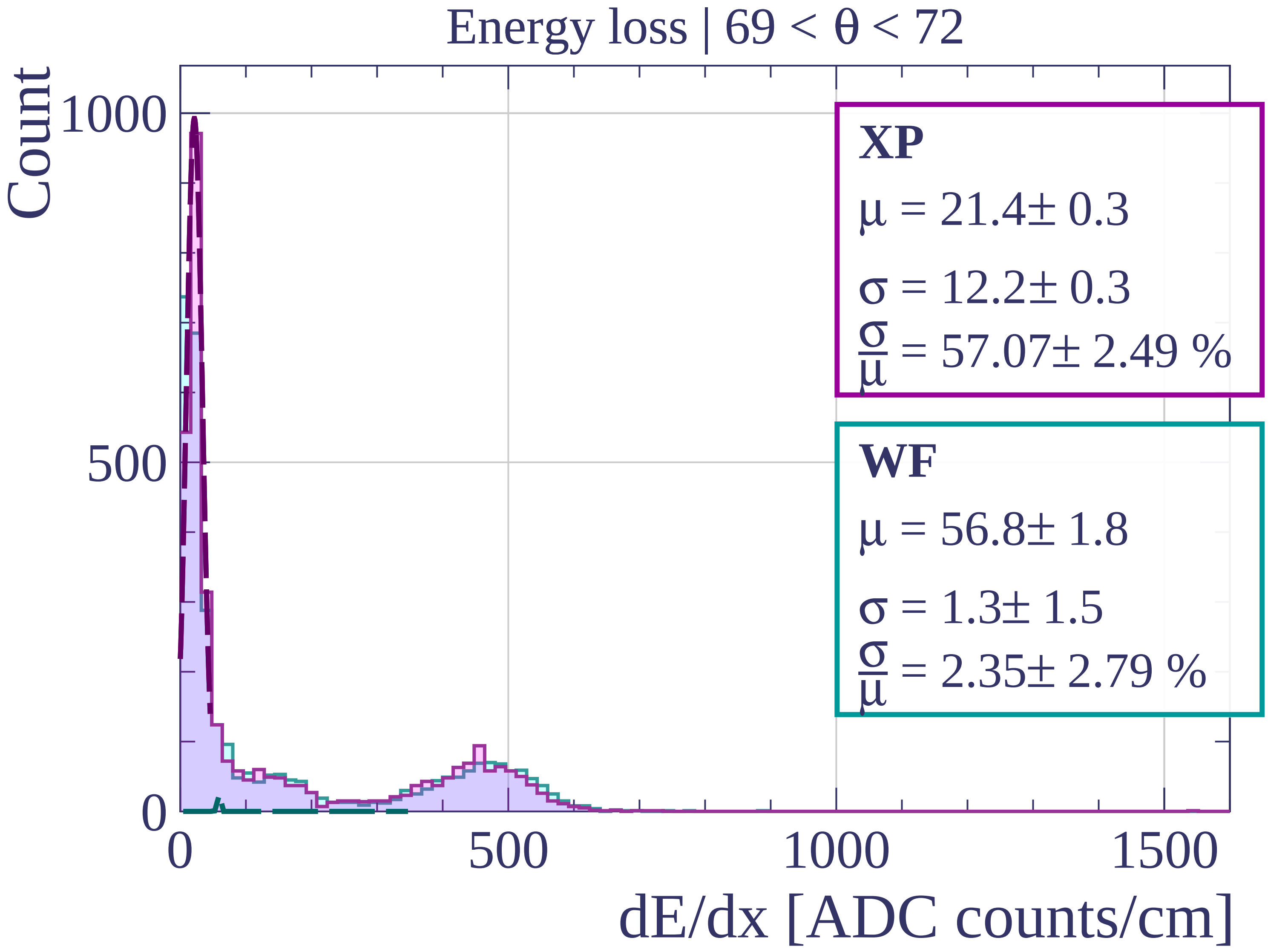
Energy loss | 60 < θ < 63
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = -2684.4± 289.6
σ = 1370.2± 859.8
σ/μ = -51.04± -26.52 %
WF
μ = -4211.2± 408.5
σ = 867.5± 80.5
σ/μ = -20.60± 0.09 %

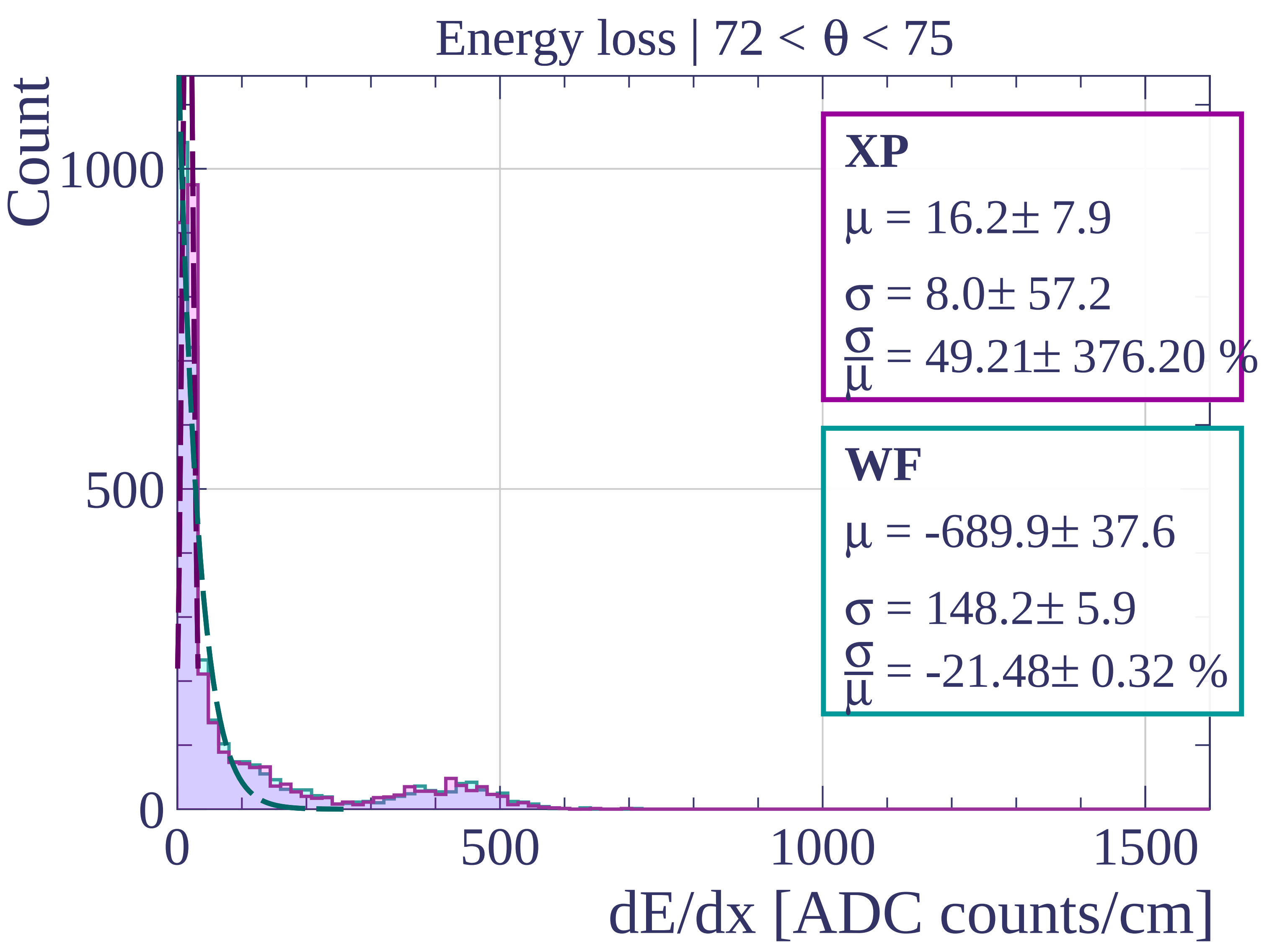


Energy loss | 63 < θ < 66
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = -3000.5± 319.7
σ = 664.1± 61.3
σ/μ = -22.13± 0.32 %
WF
μ = -305.8± 16804.3
σ = 154.1± 1325.3
σ/μ = -50.38± 2334.73

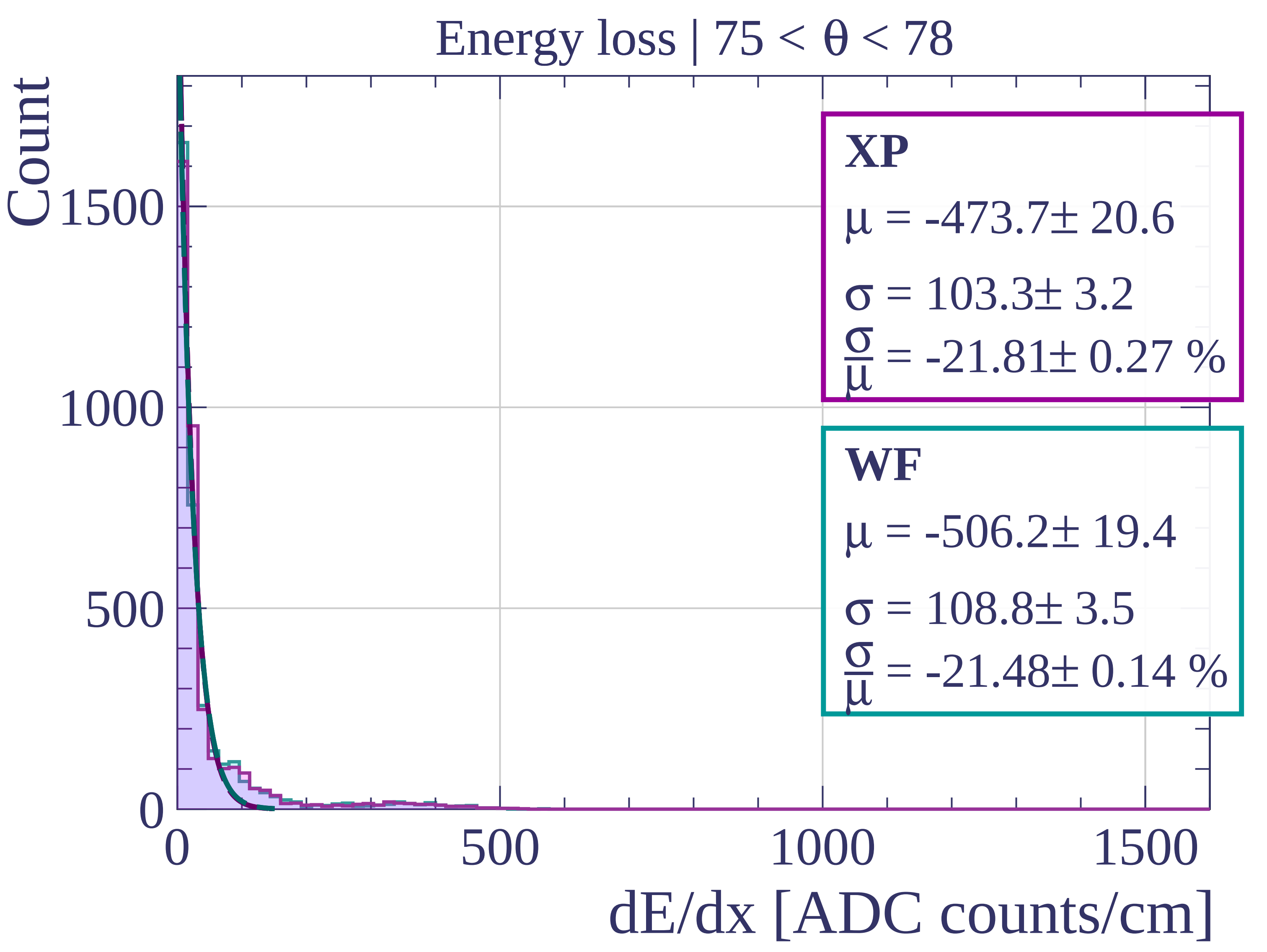


Energy loss | 66 < θ < 69
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = -193.3± 26.4
σ = 119.2± 11.1
σ/μ = -61.68± 2.67 %
WF
μ = -970.4± 62.7
σ = 208.4± 11.5
σ/μ = -21.47± 0.20 %

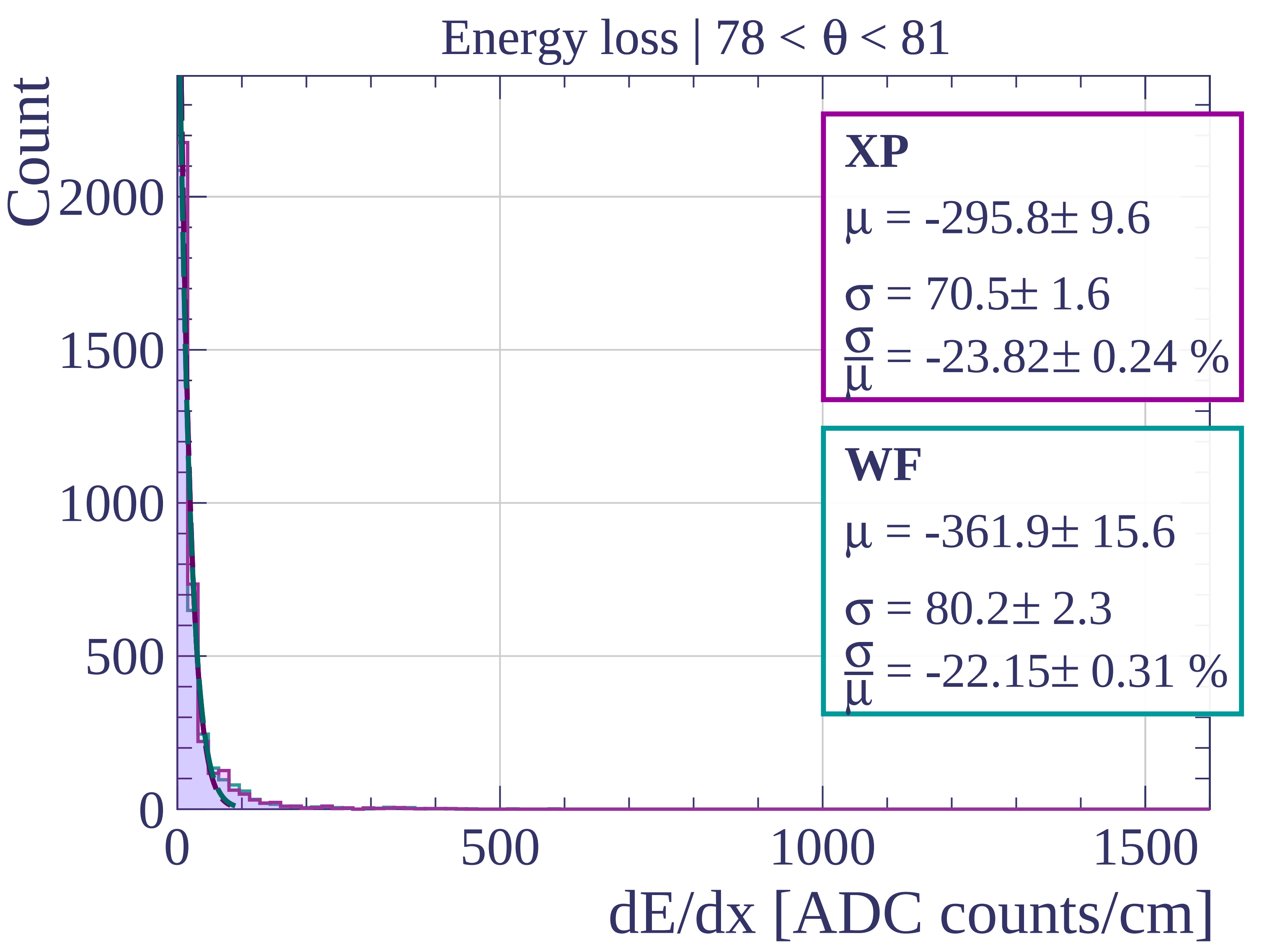

Energy loss | 69 < θ < 72
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 21.4± 0.3
σ = 12.2± 0.3
σ/μ = 57.07± 2.49 %
WF
μ = 56.8± 1.8
σ = 1.3± 1.5
σ/μ = 2.35± 2.79 %

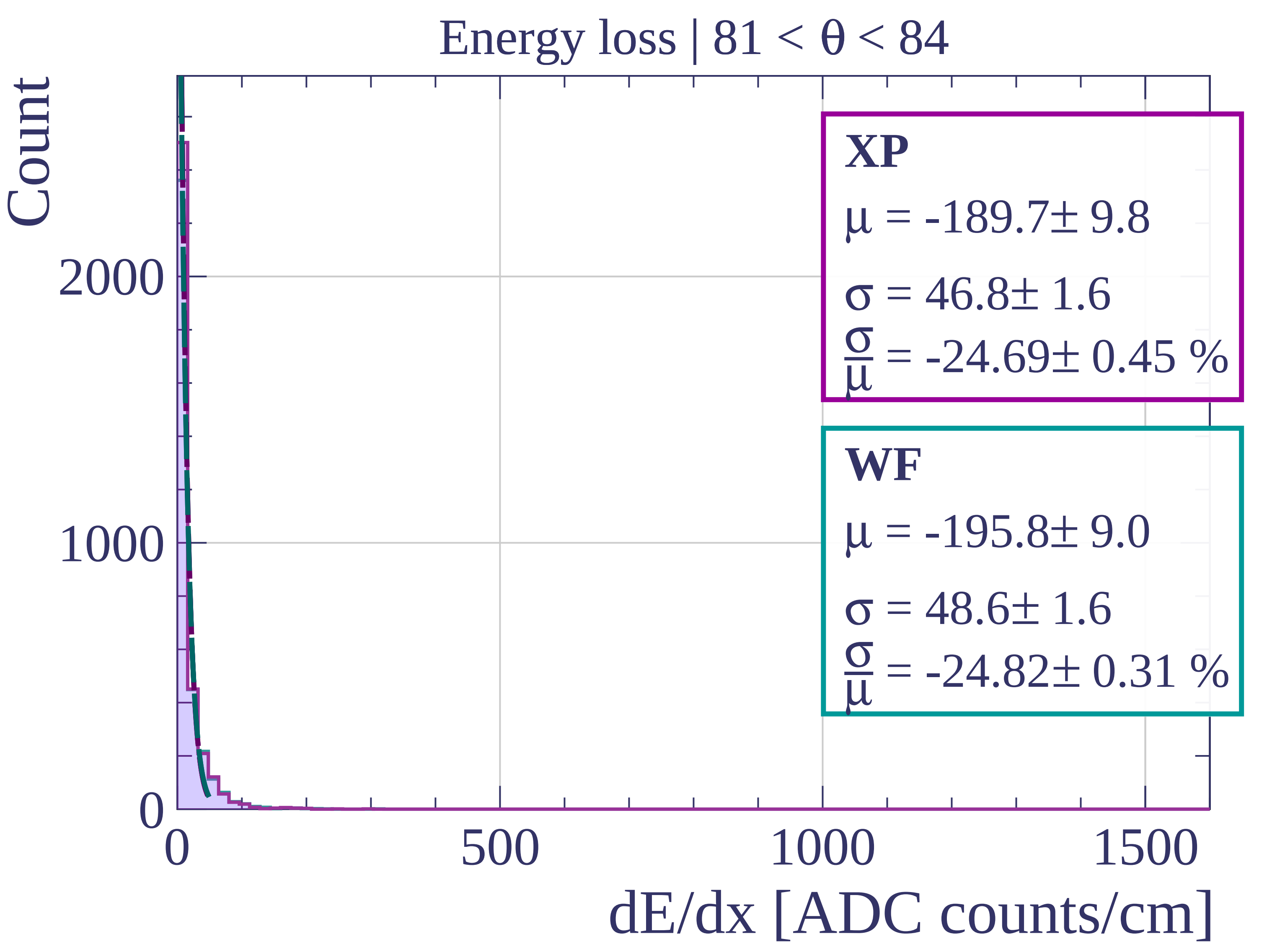


Energy loss | 72 < θ < 75
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 16.2± 7.9
σ = 8.0± 57.2
σ/μ = 49.21± 376.20 %
WF
μ = -689.9± 37.6
σ = 148.2± 5.9
σ/μ = -21.48± 0.32 %

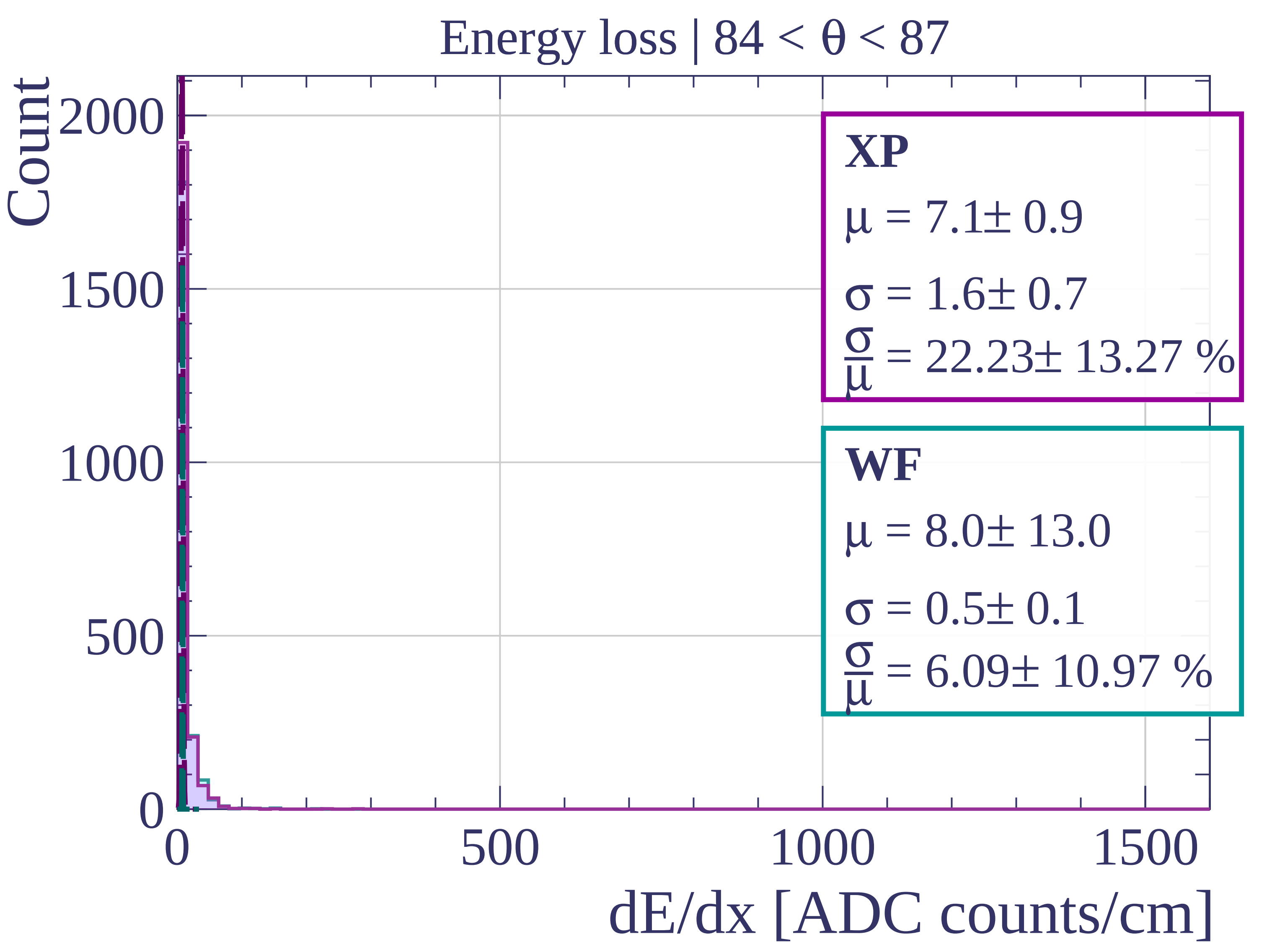


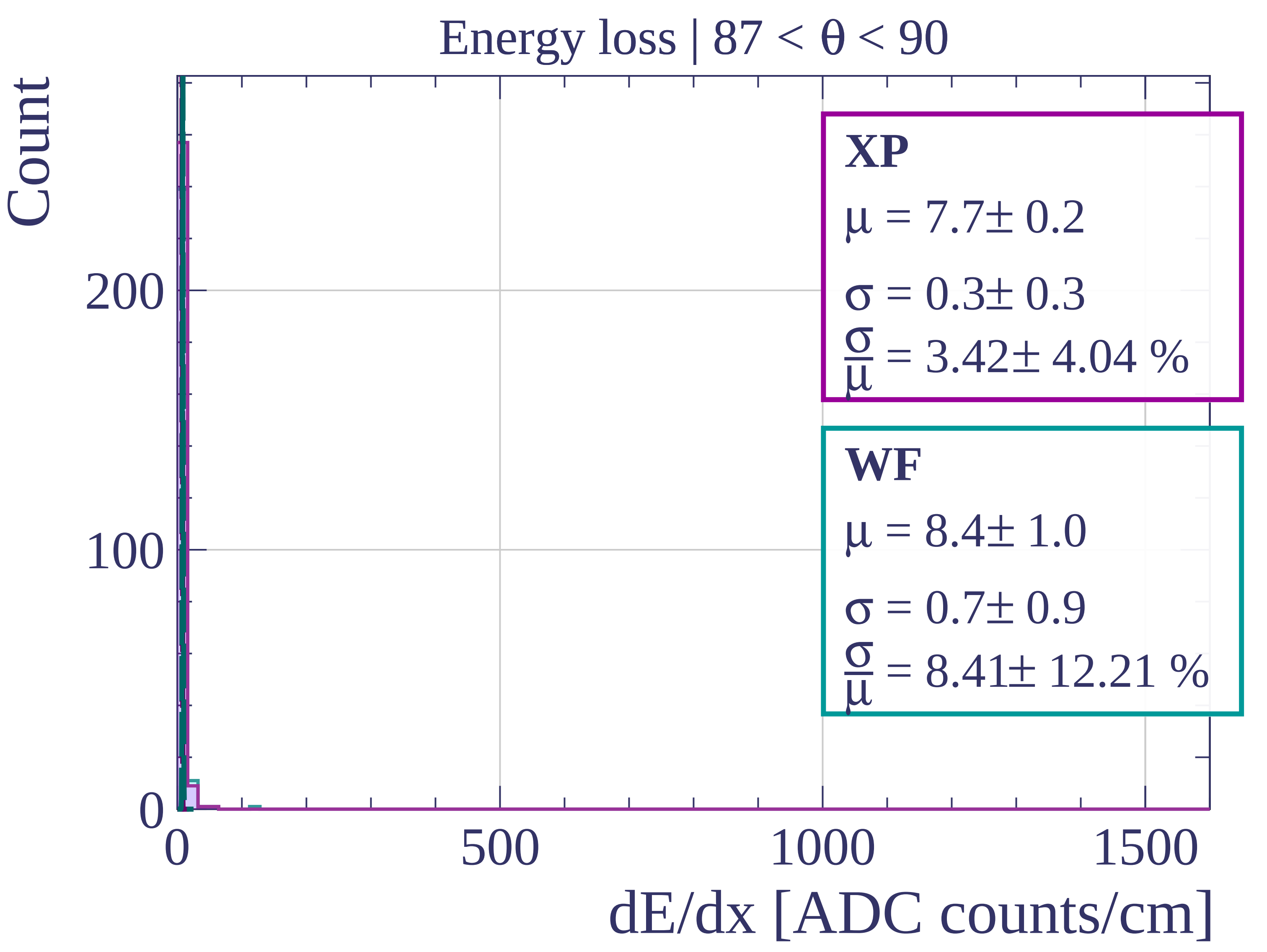
Energy loss | 75 < θ < 78
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
μ = -473.7± 20.6
σ = 103.3± 3.2
σ/μ = -21.81± 0.27 %
WF
μ = -506.2± 19.4
σ = 108.8± 3.5
σ/μ = -21.48± 0.14 %

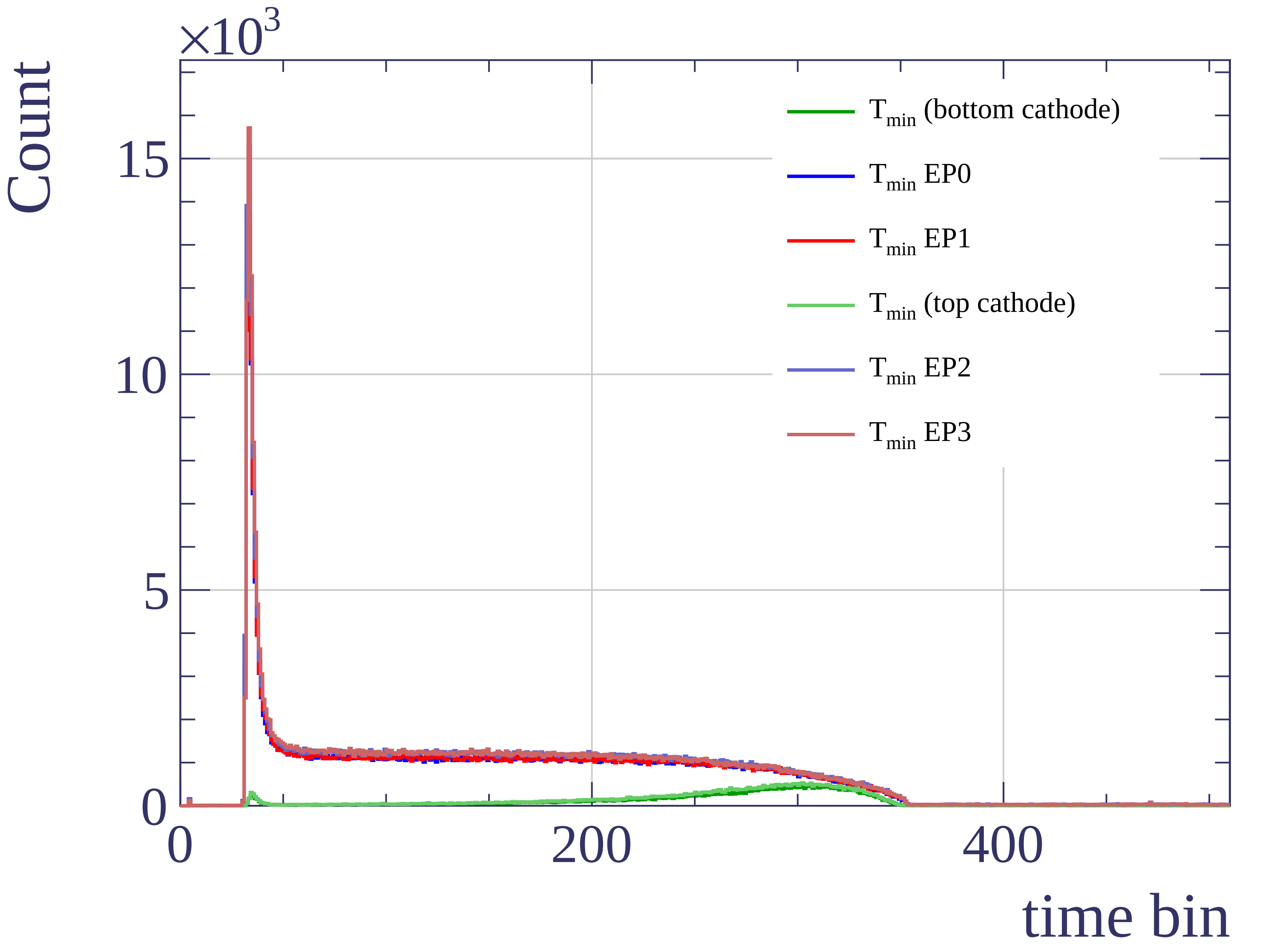

Energy loss | 78 < θ < 81
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
0
500
1000
1500
XP
μ = -295.8± 9.6
σ = 70.5± 1.6
σ/μ = -23.82± 0.24 %
WF
μ = -361.9± 15.6
σ = 80.2± 2.3
σ/μ = -22.15± 0.31 %



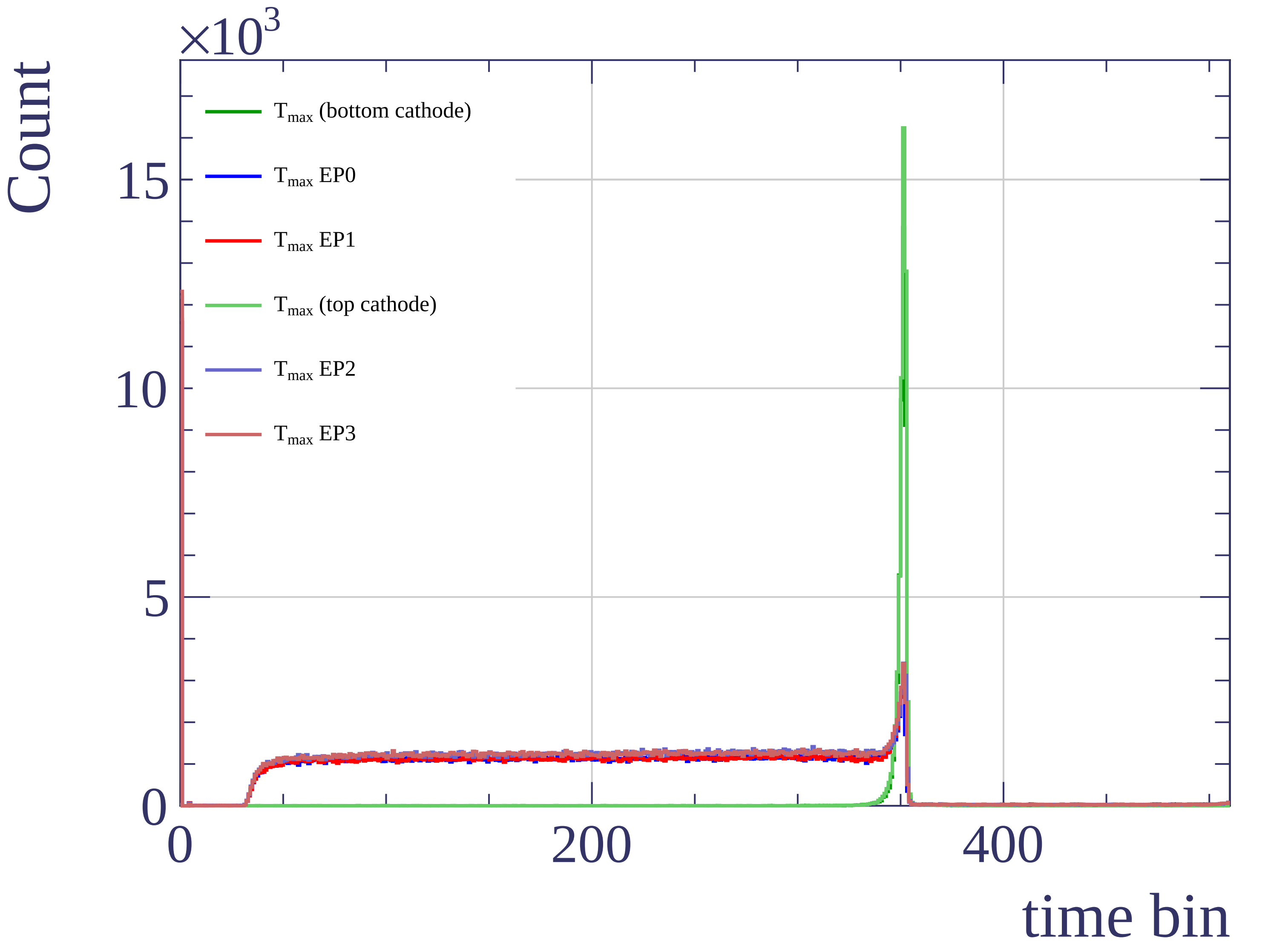
Energy loss | 81 < θ < 84
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = -189.7± 9.8
σ = 46.8± 1.6
σ/μ = -24.69± 0.45 %
WF
μ = -195.8± 9.0
σ = 48.6± 1.6
σ/μ = -24.82± 0.31 %


Energy loss | 84 < θ < 87
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
μ = 7.1± 0.9
σ = 1.6± 0.7
σ/μ = 22.23± 13.27 %
WF
μ = 8.0± 13.0
σ = 0.5± 0.1
σ/μ = 6.09± 10.97 %


Energy loss | 87 < θ < 90
Count
dE/dx [ADC counts/cm]
0
100
200
0
500
1000
1500
XP
μ = 7.7± 0.2
σ = 0.3± 0.3
σ/μ = 3.42± 4.04 %
WF
μ = 8.4± 1.0
σ = 0.7± 0.9
σ/μ = 8.41± 12.21 %



×10^3
Count
15
10
5
0
0
200
400
time bin
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



×10^3
Count
time bin
T_max (bottom cathode)
T_max EP0
T_max EP1
T_max (top cathode)
T_max EP2
T_max EP3
0
5
10
15
0
200
400